Panasonic

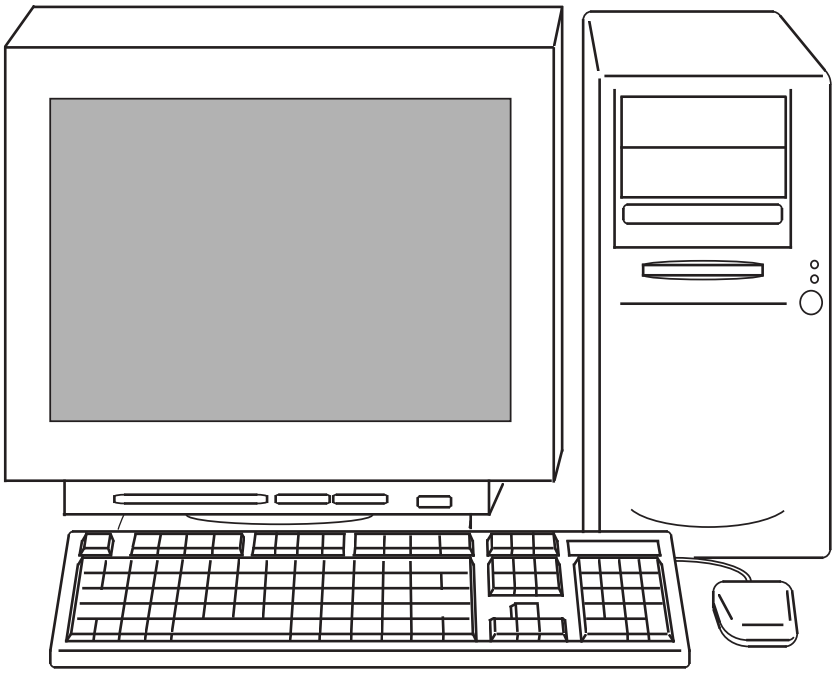
Operating Instructions

Material Control

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A523E11 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1.OVERVIEW

This chapter explains the overview of PanaCIM-EE/Material Control.

1.1 Overview

Material Control tracks component and feeder inventories in production zones and supporting areas.
It also controls the life limit for MSD components.

■ Main Features *1

Material Control - Base (License)

1. Managing the reel ID
2. Monitoring the remaining component quantities
3. Creating the work orders

Material Control – Material Allocation (License) *2

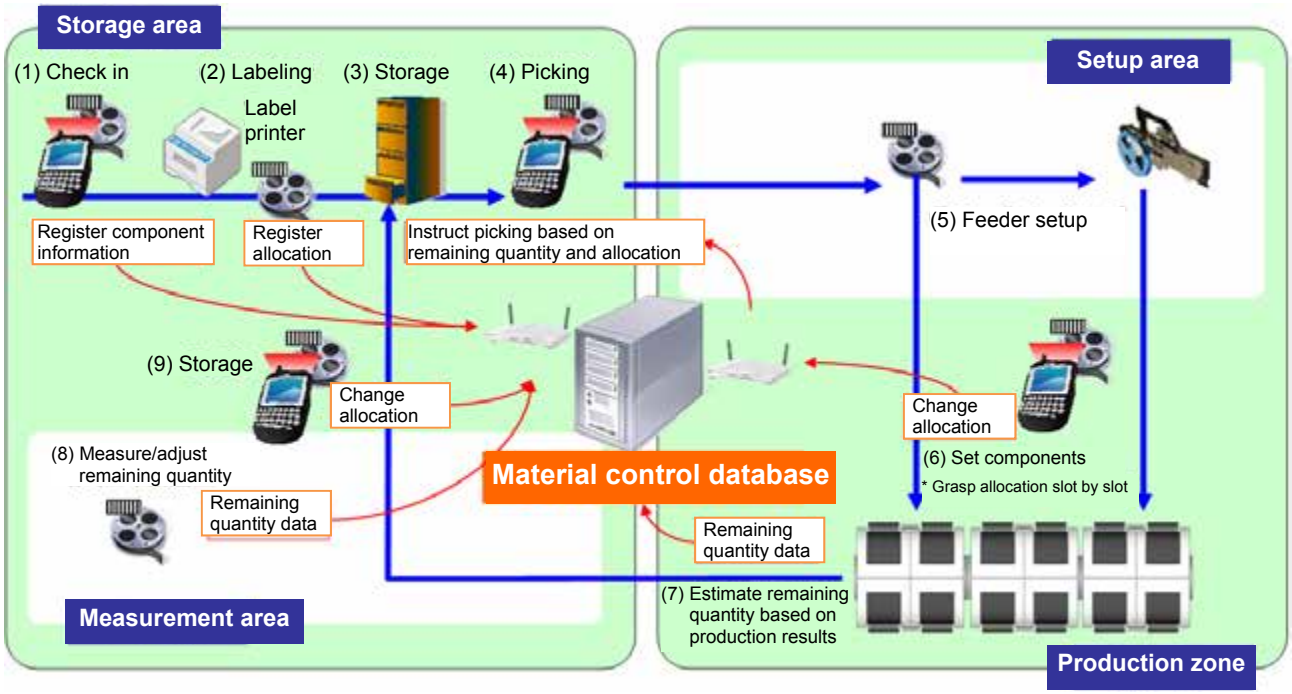
1. Controlling the component inventories and allocations
2. Work orders (Distributing components)
3. Work orders (Creating reservation/low material list)

Material Control – MSD (License) *2

1. Controlling the life limit of MSD components

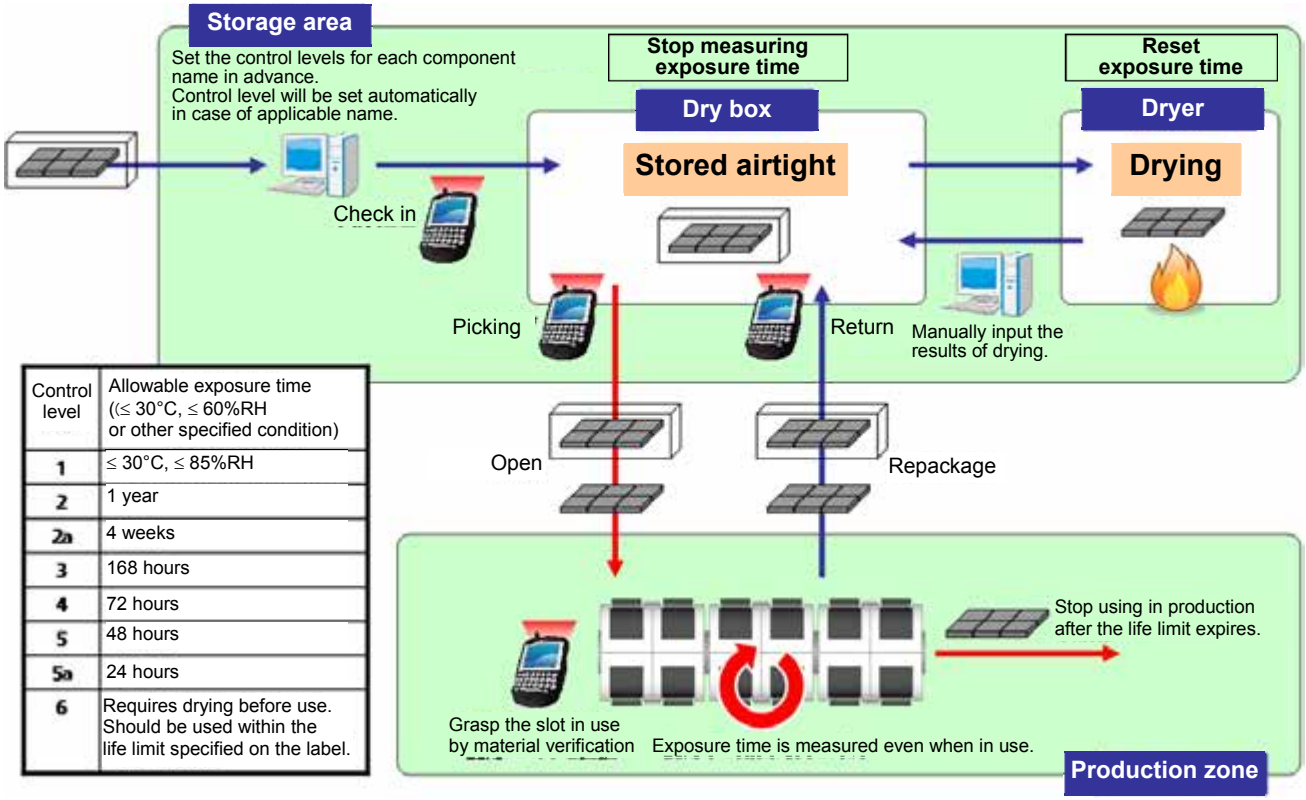
*1.To use the Material Control module, the Material Verification module (license) is required.
*2. Material Control – Base license is required to use this function.

■ Overview of Base and Material Allocation



*3. Barcode labels to be provided by the customer.

■ Overview of MSD



Control level	Allowable exposure time ((≤ 30°C, ≤ 60%RH or other specified condition)
1	≤ 30°C, ≤ 85%RH
2	1 year
2a	4 weeks
3	168 hours
4	72 hours
5	48 hours
5a	24 hours
6	Requires drying before use. Should be used within the life limit specified on the label.

*Depending on the control level, measurement of exposure time may continue, even when the MSD component is repackaged.

■ Material Control module functions

Main features	Detailed functions		Description	Required license *		
	Large category	Small category		Base	Material Allocation	MSD
Remaining quantity monitoring	Remaining quantity data		Estimates the remaining quantity based on the production results.	○	—	—
Work orders	Creating work orders		Instructs the operator to input production plan.	○	—	—
	Reports	Production plan	Shows the entered production plan			
	Component distribution	Distribution	Enables component reservation per panel.	○	○	—
		Reserved list	Creates list of reserved components per panel.			
		Low material list	Creates list of low materials per panel.			
Material inventory and allocation control	Inventories		Controls component inventory.	○	○	—
	Material Allocation		Controls material allocation and reel IDs.			
MSD life limit control	MSD life limit control		Controls the life limit of MSD components. ・Stops measuring exposure time in the dry box. ・Resets the exposure time in the dryer.	○	—	○

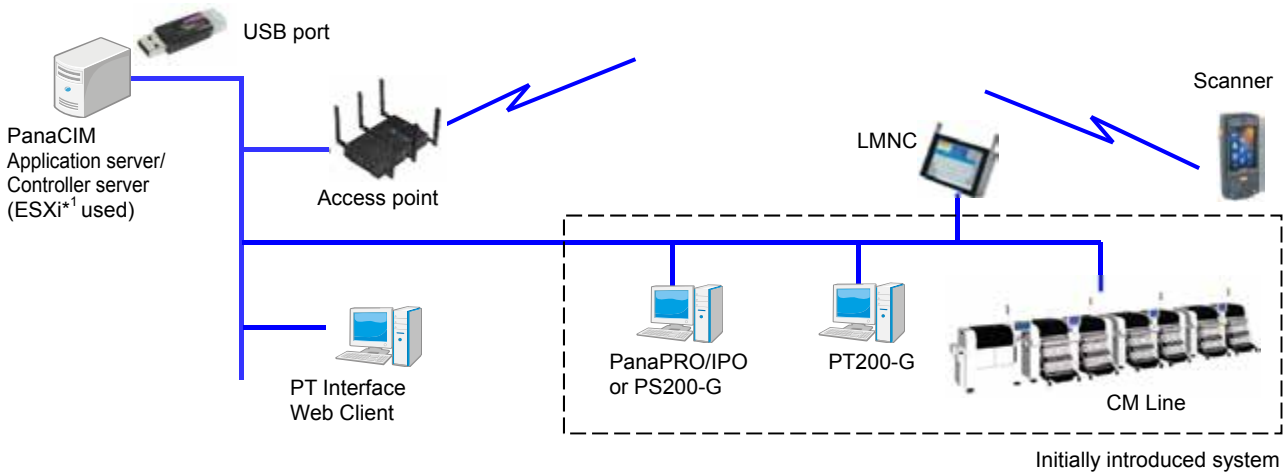
* Material Verification module (license) is also required.

1.2 System Configuration

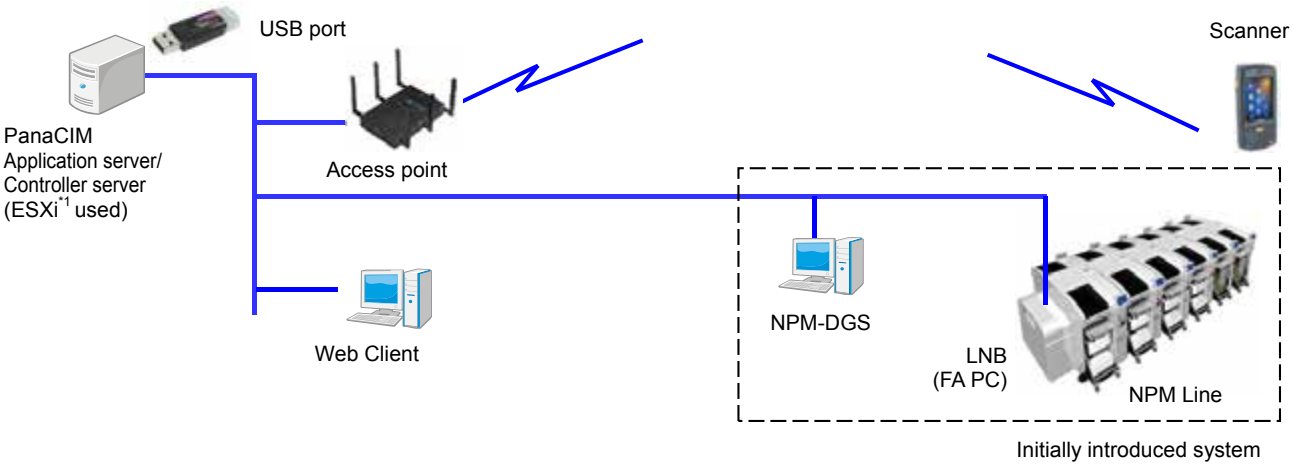
This section shows the schematics of systems with PanaCIM for small-scale and large-scale production.

■ System with PanaCIM (for small-scale production)

CM Line



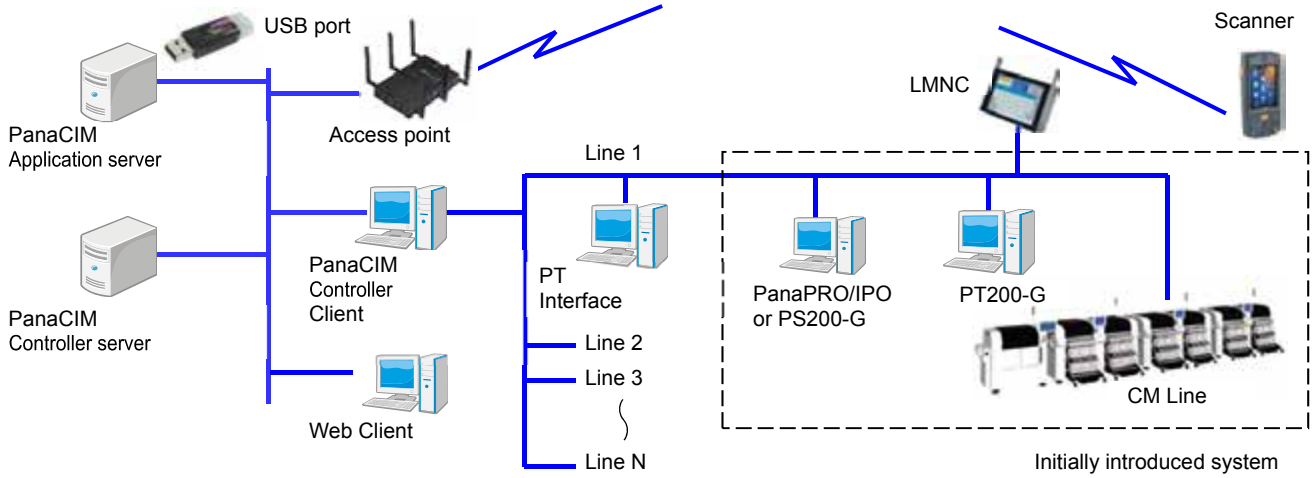
NPM Line*2



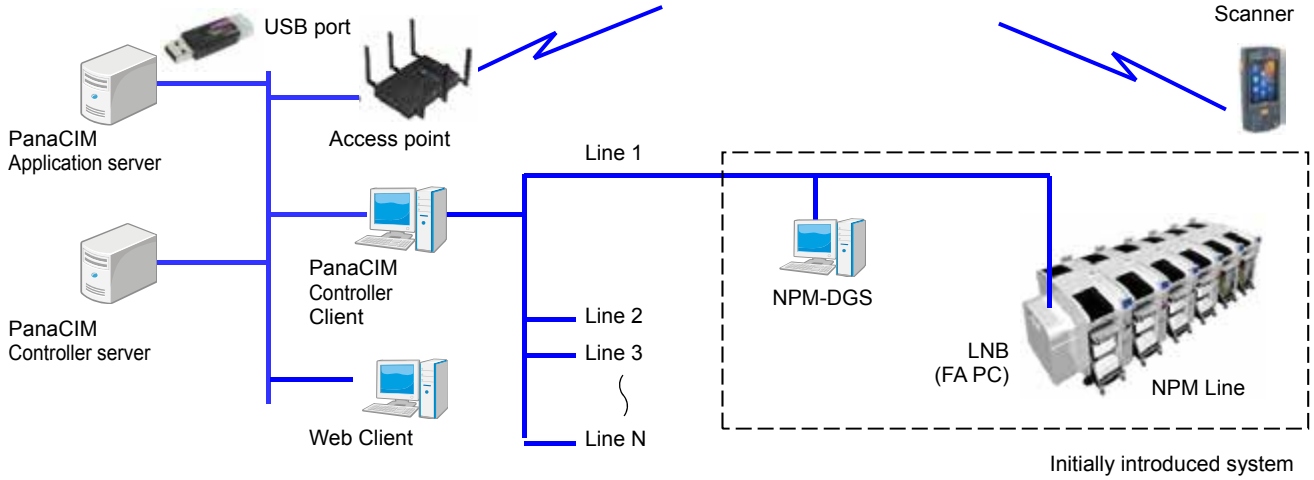
*1. ESXi: A virtualization tool that allows multiple applications to run on a single server.
*2. NPM line can also include the models such as AM100, SPD, SPG, and SPV as NPM-related machines.

■ System with PanaCIM (for large-scale production)

CM Line



NPM Line[*1]



*1. NPM line can also include the models such as AM100, SPD, SPG and SPV as NPM-related machines.

1.3 Functional Overview

■ List of Material Verification functions

Function		Specification
Base	**Configuration**	
	Configuration	Materials that can be controlled by the PanaCIM / Material Control are components and feeders.
		Each component has property information as shown in Tables 1-1 and 1-2.
		Each feeder has property information as shown in Tables 2-1 and 2-2.
		As material storage areas (supporting areas) other than the production line, material areas and maintenance areas (for PanaCIM/maintenance) can be defined.
		Material areas can be defined as Active, Limited, or Inactive.
		Material areas can be registered either in the production route or in other route.
		Multiple movable units for storing materials (shelves: storage units) can be registered in each material area.
		Each storage unit has a unique ID (name) consisting of 20 alphanumeric characters.
		Storage units can be defined as follows: Storage unit - L1: Collection & delivery box: Can be controlled as a single unit. - L2: Shelf: Unit itself and the number of shelves can be controlled. Fixed type: - Only the materials in specific status (Unloaded/Loaded) can be controlled. - Selecting None allows all materials to be checked in without limitation. * Maximum entries of materials per storage unit: Unlimited
		Storage units can be defined as Active or Inactive.
		Status of the registered materials is defined as follows: - Unloaded: Neither mounted on the machine nor loaded on a feeder. - Loaded: Loaded on (data written to) a feeder via material verification or material control operation - Mounted: Mounted on the machine via material verification or material control operation - Spliced: Spliced to an existing reel via material verification or material control operation
	Remain Monitoring	
	Remain Monitor	2 warning levels can be set based on time or number of panels. * Warning level 2 value must be less than the warning level 1 value.
		Any desired machine can be selected (Multiple choice possible).
		Selects the field to be displayed.
		Selects the field to be sorted.
		Resizes the fonts displayed on the remain monitor.
		Enables the setting for sending mails according to the remain warning.
		Color of each component is updated in real time according to the following: Alarm level 1: Yellow Alarm level 2: Orange Part empty: Red Picked up: Green
		Can be changed manually between Picked up (green) and Reset (colorless).
		Shows the list of the selected component.

Function		Specification
Base	**Life limit control**	
	Normal Part Life Limit Control	Controls life limit of normal parts Registers the expiration date of the registered parts by group.
	Expiration Warning for Life Limit Control Parts	Displays a warning message for the life limit control parts according to the setting.
	Restriction of Expired Life Limit ControlParts (During Material Verification)	Displays an error message according to the setting and it is prohibited to use the expired parts for production when the expired life limit control part is verified.
	Interlock Cauased By Expired LifeLimit Control Parts (During Production)	If the life limit control part is expired during production, production is temporarily suspended according to the setting, and the machine does not resume production until new (not expired) part is verified.
	Material inventory control - Scanner -	
	Material Movement	Material is moved from one location to another by scanning each items barcode unique ID. * Scanning: Check in, Material Verification, Move Storage Unit * Feeder Load/Unload Function is for the BM series only, and therefore not supported.
		Current location and movement history can be viewed via the Web Application.
	Estimated Remaining Quantity Tracking	Updates the number of parts used for production and the adjusted parts remain in real time.
		Shows the updated component remaining quantities in "Material status" in the Scanner menu, and in the “Inventory List” report of web application.
		When the Allocation Control module is active, updated remaining quantity is displayed when reserving the components of work order.
	Automatic Material Consumption (Exhaust and Splice)	Clears the remaining quantity of the target reel when parts exhaust has occurred.
		Clears the remaining quantity of the target reel when switching to the spliced reel.
	Restriction on Used Reels	Reels whose remaining quantity cleared to zero cannot be used in the reservation/verification processes.
	Verify Material ID	Completes verification by scanning the material ID just once (without scanning part name, etc.).
		By scanning a material ID, the information registered in advance, including part name, lot, vendor, and quantity can be loaded.
	Adjust Material Quantity	Remaining quantity can be changed by manually inputting a value to be added to or reduced from that of the target component.
	Consume Material Function	Clears the remaining quantity of the target component to zero manually.
	Check In Function	When a new material is checked in, initial information such as Check in Area, Storage Unit, Material ID, Component Name, Lot, Vendor, Quantity, etc., can be registered.
		Registered materials can be moved to different areas/storage units.
	Move Storage Unit Function	Storage units can be moved to different areas.
		Locations of the materials on the target storage units are also moved.
	Split Function	Divides the components on a reel having a material ID to two or more reels.
		When splitting a reel, a new material ID is issued and attribute information such as component are taken over from the source material ID.
	Feeder Load/Unload Function (for BM series only)	Material ID can be given to or removed from the target feeder.
		In case of double feeder, two material IDs can be given to or removed from the left and right lanes.

Function		Specification
Base	Material Status Check (location, quantity, etc.)	Shows the property information registered under the entered material ID.
		Displayed property information as per Tables 1-1, 1-2, 2-1 and 2-2.
	Storage Unit Inventory Check	Inputting the ID of a storage unit shows the list of materials stored in the unit.
	Material Search	Component search: Inputting the target part name shows the list of locations of reels containing the component concerned.
		Feeder search: Inputting the width/pitch of the target feeder shows the list of locations of feeders concerned. * Material Control for feeders is not supported, therefore excluded.
	Prevent Material Rescanning during Exhaust and Splice	Prevents the active material ID from being rescanned in case of parts exhaust or splicing. * Rescanning the material ID is prohibited not only in the same slot, but also in different slots, tables, machines and lines.
	Prevent Stealing of Mounted/Spliced Material	Prevents the parts mounted on the machine and the spliced parts from being removed during production.
	Report - Web Application -	
	Inventory/Allocation Control	Shows the list of component inventories and allocations.
	Material Quantity Report	Shows the report of component remaining quantities.
	Material Adjustment Report	Shows the history of changes to the remaining quantity.
	Material Usage Report	Shows the list of the parts used.
	Material Position History Report	Shows the history of the positions of the parts used.
	Feeder Position History Report	Shows the feeder position information when feeder control is made. * Material control for feeders is not supported, and therefore excluded.
	Material Movement Report	Shows the report of history of materials moved.

Function		Specification
Material Allocation	**Configuration**	
	Configuration	Request for input of Operator ID should be able to set in work order editing.
		Buffer (parts required for production) for all reserved components should be able to set.
		Buffer (%) for each reserved component name should be able to set.
		Buffer (%) for each reserved component name should be able to set for each product program.
	Work Order - Web Application -	
	Detailed Production Schedule	Work order can be scheduled according to the production plan.
		Users that can browse work orders in the read-only mode can be defined.
		Created work orders are color-coded according to the reservation/ production status.
		Week/day, time interval, and target route can be set.
		Work order can be created, edited and deleted.
		Work order can be imported from CSV files.
	Create Work Order	Work order can be created by setting the following information: Work Order ID, Route (production line), Panel Name, Side, Mix Name, Tact, Patterns, Produced Patterns, Started/Ended Time, Changeover Time
		Work order ID must be unique. However, the same ID may be used for two work orders if their Side (top/bottom) is different from each other.
		A work order ID can have up to 20 characters.
		Number of patterns is acquired from the product program and cannot be edited.
		Tact Time multiplied by Produced Panels is set as the default for Ended Time.
		Changeover time can take on any value between 0 and 9999 (minutes).
		If Tact Time, Produced Patterns, and Started Time have been changed, Ended Time will be automatically changed accordingly.
		After setting the product program information on the Create Work Order screen, if the Line Tact Time, Patterns/Panel, Patterns and Setup Time are input or updated, Ended Time is calculated based on the Started Time and the work order will be updated automatically.
		“Patterns/Panel” is acquired from the product program information on the database, and cannot be edited.
		After a work order is saved, the created work order appears on the main page (PanaCIM-EE Detailed Production Schedule).
		Work order with a past date/time cannot be created.
		If saved with some items yet to be input, a message prompting the operator to input the mandatory items. Mandatory items: Work Order Name, Line Tact Time, Patterns Quantity * Route Name, Product Name, Side, and Setup Mix Name can be selected from the drop-down list.
		“Line Tact Time” shows the last edited/set value as default.

Function		Specification
Material Allocation	Reservation and Procurement	The Component Reservation screen for created work orders can be accessed from the PanaCIM-EE Detailed Production Schedule.
		On the Component Reservation screen, the location, priority and buffer by which the component is searched can be set.
		The Reserved Components List screen can be accessed from the Component Reservation screen.
		The Reserved Components List screen shows whether or not the reserved components are sufficient for the quantities of the components to be used.
		Reservation of components is enabled only for one work order/line within the same time.
	Material Preparation - Scanner -	
	Replenish Low Component	Replenishment of low components can be done using the mobile scanner.
	Picking	Reserved components can be collected by scanning according to the work order.
		Shows the situation of the collected components reserved for each work order.
	Transfer Work Order	At the start of production, transfer the work order name along with the product program name via “Program Transfer”. The work order name is displayed at the Route Status on the Route Editor.
		When changeover has occurred, the current work order ends and switches to the next work order.
		Work order can be ended manually.
	Report - Web Application -	
	Material of Report	Shows the list of low materials among the components reserved in the work order.
	Material Reservation Report	Shows the list of the reserved parts in the work order.
	Work Order History	Shows the history of work orders.
	Material Movement History	Shows the history of materials moved among locations.
	Changeover Setup Report	Shows the setup report at product changeover.
	Configuration/Registration	
	Configuration	The following levels for the MSD components can be set: Level 1: Unlimited (Condition: 30°C and 85% or above) Level 2: 1 year Level 2a: 4 weeks Level 3: 168 hours Level 4: 72 hours Level 5: 48hours Level 5a: 24 hours Level 6: Requires drying before use. After drying, use the MSD within the life limit specified on the label.
		MSD can be registered for existing components according to the above levels.
		MSD can be registered using the scanner when registering new components.
		Dry boxes can be registered.
		Warning message for MSD can be set at 2 levels according to the remaining time.

Function		Specification
MSD	**System level**	
	MSD Shelf Life Time Control (Unopened MSD)	When the package of the MSD is not opened, the life time is controlled by shelf time.
	MSD Floor Life Time Control (Opened MSD)	After the package of the MSD is opened, the life time is controlled by floor time.
	MSD Inventory Control	In addition to remaining quantity and location, the attribute information (level, lifetime) of MSD can be managed.
	MSD Status Monitor	Shows the status of MSD components in real time.
		Display is updated in real time according to the time used and warning setting.
	MSD Expiration Warning	Shows warning messages for MSD at 2 levels according to the setting.
	Restriction on Expired MSD (Material Verification)	If an expired MSD is collated, an error will be returned and use of it will not be allowed.
	Expired MSD Interlock (during production)	If the life limit of MSD has expired during production, production will be stopped and will not be restarted until a new MSD (within life time) is collated.
	Operation - Scanner -	
	Unpackage	This process takes place when the package of the MSD is opened, and countdown of the life time starts at the same time.
	Package	This process takes place when the package of the MSD is sealed, and countdown of the life time stops at the same time (according to the MSD level).
	Remove MSD from Dry Box	This process takes place when the MSD is removed from the dry box, and countdown of the life time starts at the same time.
	Check in MSD into Dry Box	This process takes place when the MSD is put back into the dry box, and countdown of the life time stops at the same time (according to the MSD level in Table 3-1).
	Report - Web Application -	
	MSD Status Report	Shows the status of MSD components.
	MSD History Report	Shows the history of MSD components.

Table 1-1: Component - Static Properties

Type	Property	Description
Unique ID (Material ID)	Date/time*	Date and time the part was registered in the PanaCIM
	Component number*	Registered number (name) of the component
	Type*	Category: Component
	Vendor	Manufacturer
	Lot	Manufacturing lot
	Initial quantity*	Initial value of the component quantity
	User definition #1	User-defined property
	Component class	Brightness/color rank of the component * Generally used for LED parts.

Note: *: Mandatory items

Table 1-2: Component - Dynamic Properties

Type	Property	Description
Unique ID (Material ID)	Status	(Usable/Reserved/Inactive)
	Last updated	The date and time remaining quantity or MSD attribute was last changed or static attribute was last added.
	Expected remain	Expected number of components remaining on the reel
	Status	(Unloaded/Loaded/Mounted, etc.)
	Location	Machine or support area on which the material is currently located/stored
	Feeder position	Location of the feeder on the machine

Table 2-1: Feeder - Static Properties

Type	Property	Description
Unique ID (Feeder ID)	Type*	Category: Feeder
	Loadable*	Number of reels that can be loaded on the selected feeder
	Machine type*	Type of the machine on which the selected feeder can be used
	Table size	Sizes of the XY table on which the selected feeder can be used
	Carriage type	Carriage type of the selected feeder
	Handle color	Color of the feeder handle
	Package type	Package type of the part
	Supply method	Component supply method defined by manufacturer
	Tape width	Width of the tape feeder
	Main pitch	Tape feed amount (full)
	Sub pitch	Tape feed amount (partial)
	Reel dia.	Maximum diameter of the reel that can be used on the selected feeder
	Cutter location	Position of the cutter on the selected feeder
	Registered date*	Date the feeder was used (registered) on the PanaCIM
	Vendor	Vendor information of the selected feeder
	Serial number	Serial number of the selected feeder
	Model number	Model number of the selected feeder
	Component number	Component number of the selected feeder

Table 2-2: Feeder - Dynamic Properties

Type	Property	Description
Unique ID (Feeder ID)	Status	(Active/Inactive)
	Location	Zone/area on which the feeder was located
	Feeder position	Location of the feeder on the machine
	Status	(Unloaded/Loaded/Mounted, etc.)

Note: *: Mandatory items

2. FUNCTION - MATERIAL CONTROL - BASE

This chapter explains a setup and operation of PanaCIM-EE / Material Control-Base.

2.1 Overview

2.1.1 Preparation

Ensure that you have satisfied the items in the checklist below before configuring Material Control.

- **Barcodes**: A barcode is a machine-readable representation of data used to identify objects and collect data about them. Ensure that all materials, areas, and zones have been barcoded with the appropriate label types (listed below).
  - Material ID
    - Part Number (Mandatory)
    - Initial Quantity (Mandatory)
    - Vendor (Arbitrary)
    - Lot (Arbitrary)
    - User data (Arbitrary)
  - Feeder ID
  - Material Area
    - L1 Storage Unit (per unit without control of shelves)
    - L2 Storage Unit (per unit with control of shelves)
    - L2 Storage Sub Unit (per shelf)
  - Production Zone (Production area/Machine)

* Please reference the Base document for more information regarding barcodes.

- **Areas**: Determine what areas will be included in the configuration of your factory. Take some time to diagram how your factory is laid out, and how your materials will be stored. Remember to include all of your zones, routes, storage units, worktables, and scanners in your layout. For guidance in configuring Areas, please refer to the PanaCIM Base document.

2.2 Settings

The following details configurations necessary to use the Material Control Base functionality.

The following table highlights what scanner platform is required for implementing operations at various locations:

Location	Equipment Type	Required Hardware
	CM/DT/NPM series	Mobile Scanner
Material Areas	N/A	Mobile Scanner
Maintenance Areas	N/A	Mobile Scanner
Hand Place Areas	N/A	Mobile Scanner

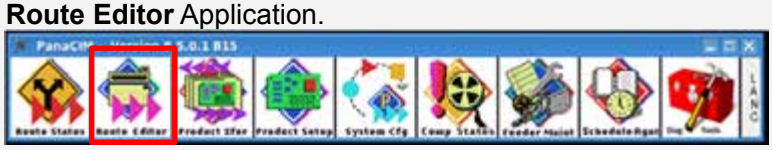
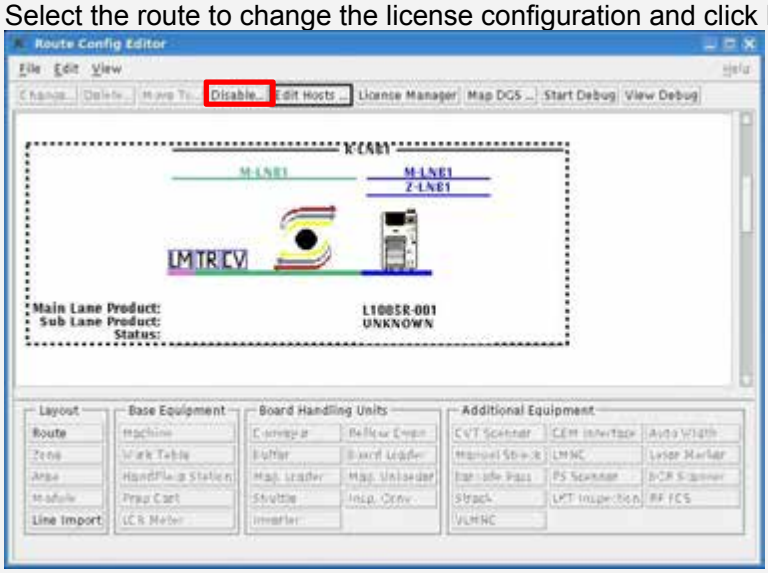
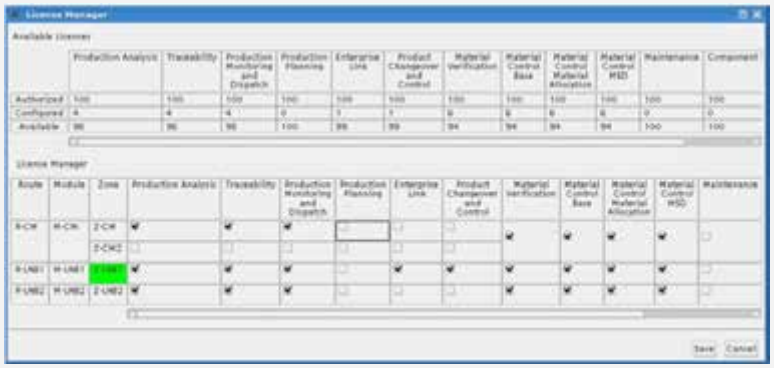
Reference the below table for a brief overview of what Applications will be used to fulfill various configuration settings.

Application	Overview
System Configuration	The System Configuration Application is used to configure the site-dependent aspects of the PanaCIM system. Use the System Configuration Application to configure the following settings for the Material Control Base Module: • Configure Mobile Scanner Settings • Configure Mobile Scanner Operator Settings • Create Storage Units* • Configure Material Control Options • Configure Attributes • A Material Area must be created in Route Editor before a Storage Unit can be created. Please refer to the PanaCIM Base documentation for guidance in creating a Material Area. • Material area must also be created before the Material Handling icon  will be displayed at the bottom of the Mobile scanner.
Route Editor	Route Editor Settings are specific to a particular production line. The settings can be observed while the object is running, but the object *must be disabled or stopped* in order to make any changes. Use the Route Editor Application to configure the following settings for the Material Control Base Module: ● Configure Licenses ● Configure Monitoring by Machine ● Configure Verification Mode
Product Program	OPTI Flag: For Product Import to work with Material Control, the OPTI flag must be set inside the original MAI file. When OPTI flag is set to “Yes”, the Optimized data is used and when set to “No”, the machine will use its own optimized data.

2.2.1 Route Editor

■ License Configuration

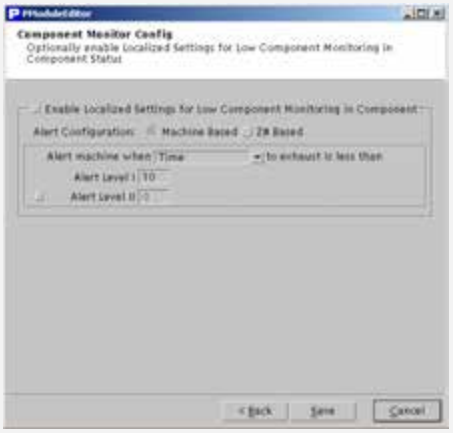
Before using the PanaCIM/Material Control - Base, the license needs to be enabled via the Route Editor of the PanaCIM controller server.

Step	Procedure
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.
4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Material Control - Base, place a check mark in the following checkboxes: • **Material Verify:** Enables the selected license module by module. • **Material Control - Base:** Enables the selected license module by module.  **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After configuring the license, click **Save** and exit the License Manager.

■ Configure Monitoring by Machine

Use the Route Editor Application to make configurations by Machine rather than globally. Choose to configure Time and Panel, or by Z# based to configure by table slot location.

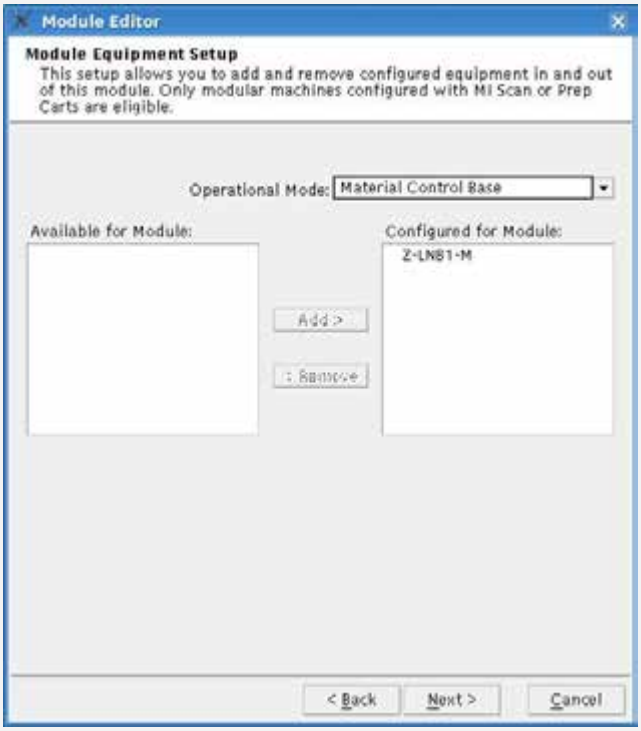
Refer to the action table below to configure monitoring by Machine using the Route Editor Application. (It assumes that the production line has been configured beforehand.)

Step	Procedure
1	Choose the 'Route Editor' icon from the main menu.
2	Select the **Module** to invoke the Module Editor.
3	Click **Next** through the screens until the **Component Monitor Config** screen is reached.
4	Check the **Enable Localized Settings…..**checkbox.
5	Select the **Alert Configuration**. • **Machine Based**: Select your alert level type by machine. • **Z# Based**: Select your alert level type by product for each Z# or table slot location.
6	Set **Alert Options** as appropriate to Machine Based or Z# Based. *Machine Based*  *Z# Based* 
7	Click **Save**.

■ Verification Mode

To enable verification with the material ID, it is necessary to select ‘Material Control Base’ as the verification mode in the Rote editor - Module editor setting.

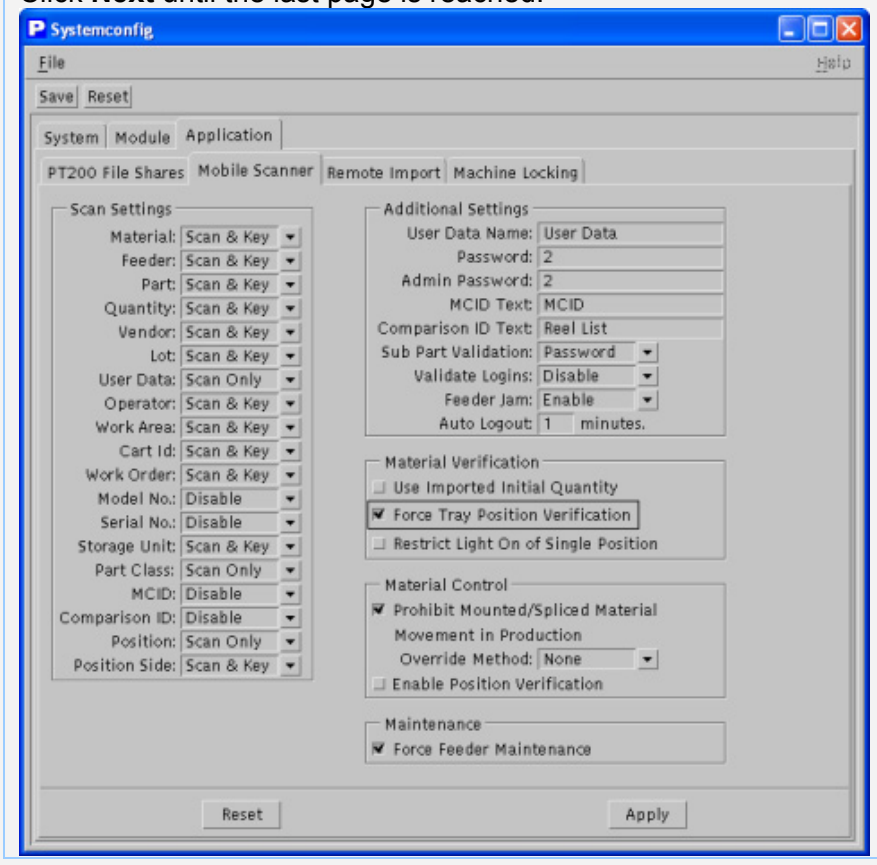
The following shows the steps to set the route editor for selecting the verification mode.

Step	Procedure
1	Choose the ‘Route Editor’ icon from the main menu.
2	Choose the module to be set up and open the module editor.
3	Click **Next** until the **Module setup** screen is reached.
4	The following is chosen by operation mode. • **Material Control Base:** Verification by Material ID is performed. Modular machine shows CM, DT, and the NPM series after CM402 here. 

2.2.2 System Configuration

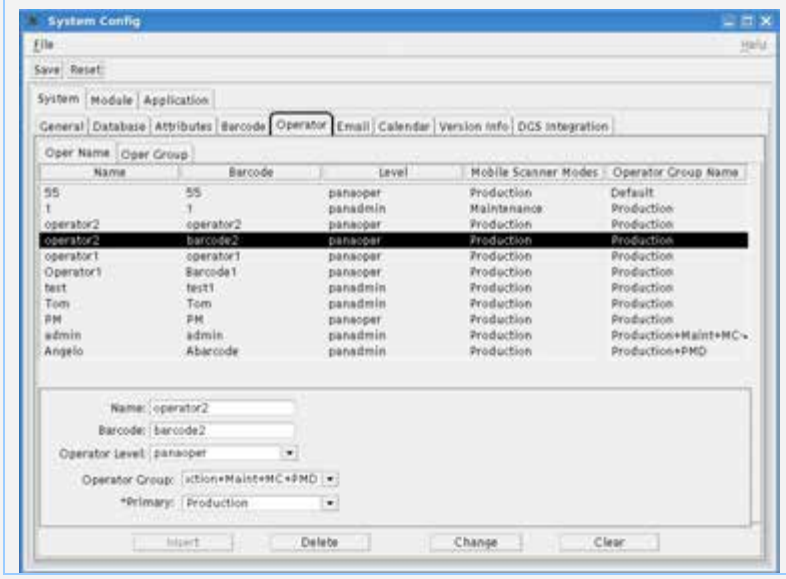
■ Configure Mobile Scanner Settings

Refer to the action table below to configure how your Mobile Scanner for CM/NPM series machines will operate by applying settings in the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **Application** - **Mobile Scanner** tab.
3	Configure the following Scan Settings by choosing from the corresponding drop-down lists: • **Material:** Scan&Key or Scan Only • **Quantity:** Scan&Key or Scan Only • **Feeder:** Scan&Key or Scan Only • **Storage Unit:** Scan&Key or Scan Only
4	Enable the following functions as necessary. • Select the **Prohibit Mounted/Spliced Material Movement in Production** checkbox to ensure that a Material ID currently Mounted or Spliced in another feeder position/machine cannot be used in Part Exhaust, Splice Warning, Splice Anytime, Generic Part Scan, or PreSetup operations. • Select an **Override Method**. If you select either Yes/No or Password, the Material ID will be acceptable only after a Yes/No prompt or Password entry. Select **None** if an override is prohibited.
5	Click **Next** until the last page is reached. 
6	Click **Apply** and **Save**.

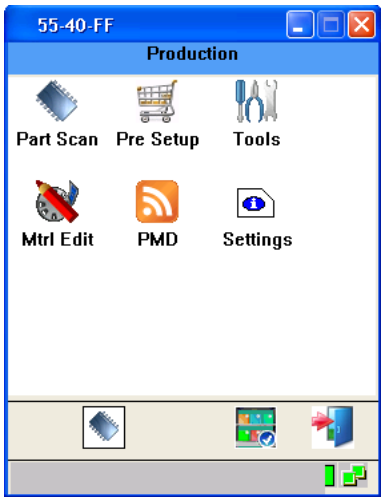
■ Configure Mobile Scanner Operator Settings

Refer to the action table below to configure operator(s) Mobile Scanner settings using the System Configuration Application.

Step	Action
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **Application** tab.
3	Click the **Operator** tab. 
4	Enter the operator **Name**, **Barcode** (Operator ID), and select the **Operator Level** from the drop-down list.
5	If the operator will be using more than one function mode, you may select alternate function modes from the **Operator Group**. *Please note that only those operations selected can be performed by that particular operator*.
6	Next, select the settings the operator will require. Select the **Primary** function mode that the operator will be performing (Production, Maintenance, or Material).
7	Select **Insert** after you have satisfied your selections.
8	Repeat these steps for each operator and then click **Save**.

In the CCS mode, if the operator set in the Material (Material Control) is logged in, the following menu screen appears. Select the desired material handling icon to access the Mtl Handling screen.

[Production] menu



[Production] menu options
- ■ Used for material verification
  - Setup: Enables verification.
  - Routine check: Component names can be checked periodically.
  - Pre Setup: Materials can be set on the offline cart in advance.
- ■Used for material control
  - Mtrl Edit: Allows the materials to be consumed or split.
- ■ Common to the scanners
  - Tools: Enables barcode tests, etc.
  - Settings: Allows default location andr sound option to be set.

[Mtl Handling] menu

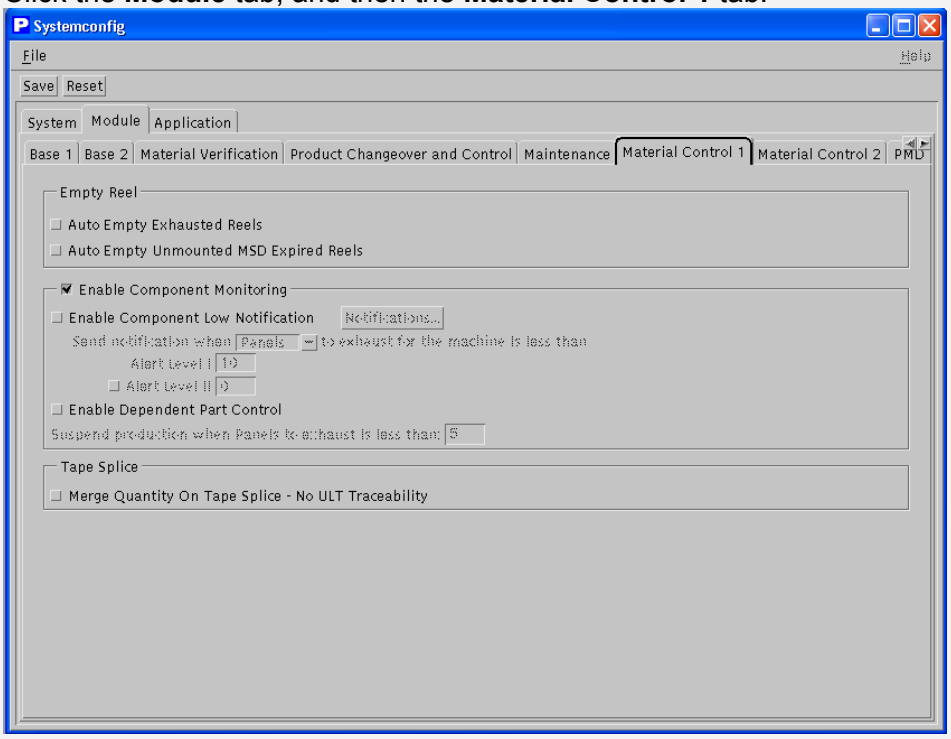


[Mtl Handling] menu options
- Check In: Registers the materials.
- Move StorUnit: Moves the storage unit.
- Mtrl Edit: Materials can be consumed or divided.
- Mtrl Allocation: Materials can be allocated or picked up.
- MSD/Paste: MSDs can be sealed or unsealed.
- Mtrl Prep: Feeders can be allocated or removed.
- Tools: Enables barcode tests and inventory checks.
- Settings: Allows software version to be checked and sound option to be set.

■ Configure Material Control Options

Refer to the action table below to configure Material Control options using the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **Module** tab, and then the **Material Control 1** tab.



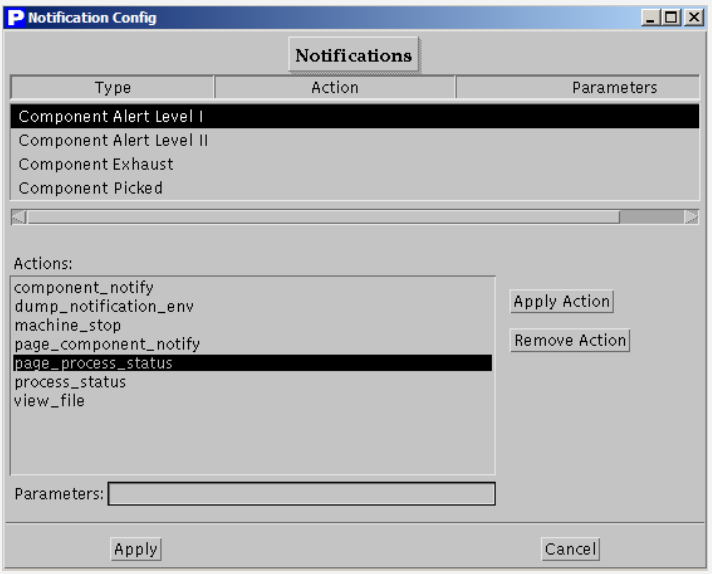
<Empty Reel> field

- **Auto Empty Exhausted Reels**: If exhausted component or spliced tape has been detected and a feeder with a new unique ID has been mounted, the estimated remaining quantity of the previous unique ID mounted in this same location will automatically be set to 0 (empty).
- **Auto Empty Unmounted MSD Expired Reels**: Check this option to set the MSD that has been checked in but unused until the expiration date as “Expired SMD”.
- **Reuse Empty Material Barcodes**: Check this option to reuse a unique ID after the item has been consumed. If this option is not checked, the unique ID will be locked out for use. (only for RF CVT scanner)
  Note: This option is not supported.

<Enable Component Monitoring> field

- **Enable Component Monitoring**: Activates the component monitor. Check this option to monitor the remaining component quantities via the PanaCIM Component Status Application.
- **Enable Component Low Notification**: Check this option to monitor and alert for low components on a global scale for all machines. If enabled, the rules are applicable at all times, even if no individual Component Status Applications are open.
  1. Select the notification **Type**. See Appendix for a listing of notifications.
  2. Select the desired action from the **Actions** list.
  3. Enter additional data into the **Parameters** field.
  4. Either click:

- **Apply Action**. Update the notification item.
- **Remove Action**. Removes the notification item



- **Enable Dependent Part Control:** The “Enable Dependent Part Control” is used with LED Binning. This is used to allow creation of dependent parts for primary parts.
  Note: This option is not supported.

<Predefined reel> field

- **Use Predefined Reels Only**: This is a customizable feature. If selected, material cannot be scanned if it does not exist in your factory. Contact Panasonic to use this option.
  Note: This option is not supported.

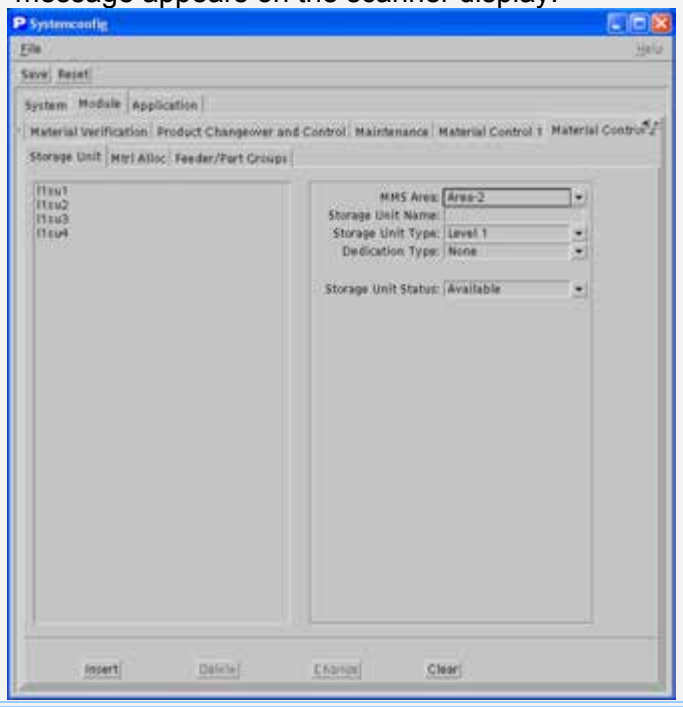
3	Click **Apply** and then **Save**.

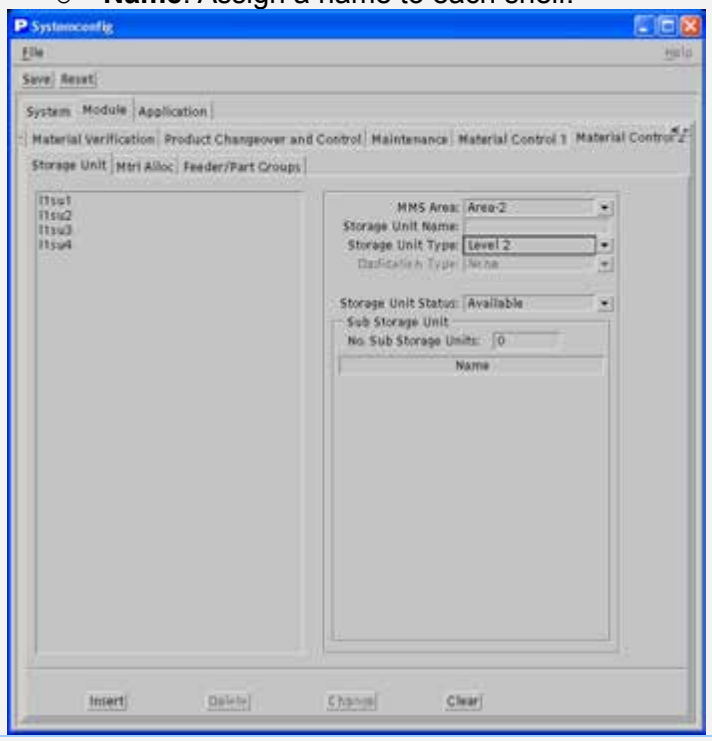
■ Create Storage Units

Use the System Configuration Application to create Storage Units and assign them to a Material area in PanaCIM. This configuration is only applicable if areas have been created.

* This is an optional configuration and can only be done if a Material Area has been created first. Please refer to the PanaCIM Base documentation for guidance in adding a Material Area.

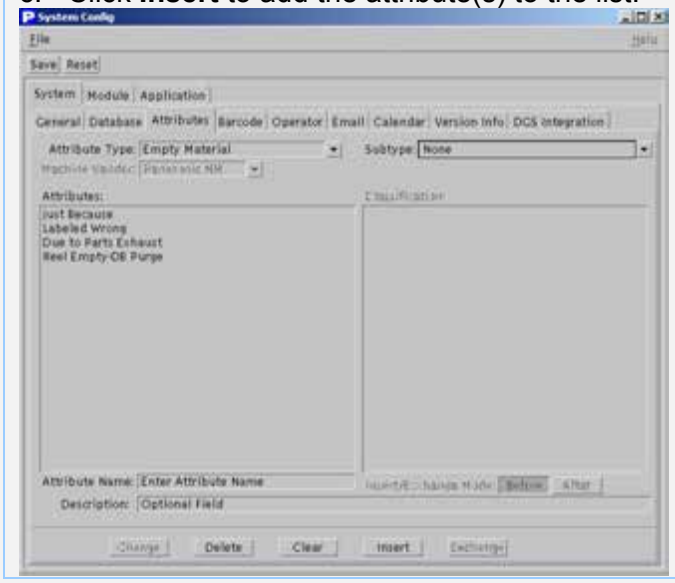
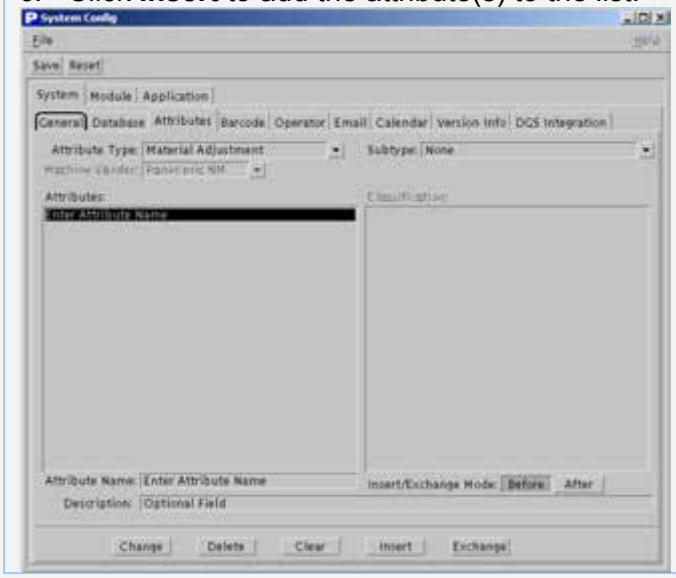
Refer to the action table below to create Storage Units using the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **Module** tab, the **Material Control 2** tab, and then the **Storage Unit** tab to view the configuration options.
3	Click the drop-down next to the **MMS Area** field, and enter a particular name in the **Storage Unit Name** field. **Note:** MMS Area values are defined in the Route Editor window.
4	Determine the control level of the storage unit from the Storage Unit Type drop-down list. • **Level 1**: Choose this level for the units (collection & delivery box, etc.) involving shelves that do not require controlling. ○ **Dedication Type**: This option is available only when the Level 1 is selected. The sub storage unit described later will not be included. Set this option when to control the feeders by discriminating the presence of the reels. **Loaded Feeder**: Reels are loaded. **Unloaded Feeder**: Reels are not loaded. **None**: Unconcerned ○ **Machine Family**: This option is available only when Loaded/Unloaded is selected from **Dedication Type**. Select the applicable machine type. **Note:** If different materials are checked in to the storage unit, an error message appears on the scanner display. 

	• **Level 2**: Choose this level for the units (with multiple shelves) involving shelves that require controlling. Populate the following options in the Sub Storage Unit field that is enabled only when the Level 2 is selected. ○ **No. Sub Storage Units**: Set the number of shelves. ○ **Name**: Assign a name to each shelf. 
5	Choose a **status** for your storage unit. Selecting Available from the drop-down mean that items can be checked in to it. Selecting Unavailable means that items cannot be checked in to it.
6	Click **Insert** to create and display the new Storage Unit in the list on the left side of the screen.
7	To make a change to any of the created Storage Units, select the Storage Unit from the list. Changes can be made to the **Storage Unit Status** and **No. Sub Storage Units** (if a Level 2 Storage Unit). Click **Change** to complete the changes.
8	Click **Save**.

■ Configure Attributes

Refer to the action table below to configure attributes for use with the Consume Material and Adjust Material Quantity functions using the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **System** tab, and then the **Attributes** tab.
3	To apply attributes to your *Consume Material* functionality: a. Select **Empty Material** from the drop-down list as your Attribute Type. b. Next, select an **Attribute Name**, and an optional **Description**. c. Click **Insert** to add the attribute(s) to the list. 
4	To apply attributes to your *Adjust Material Quantity* functionality: a. Select **Material Adjustment** from the drop-down list as your Attribute Type. b. Next, select an **Attribute Name**, and an optional **Description**. c. Click **Insert** to add the attribute(s) to the list. 
5	Click **Save**.

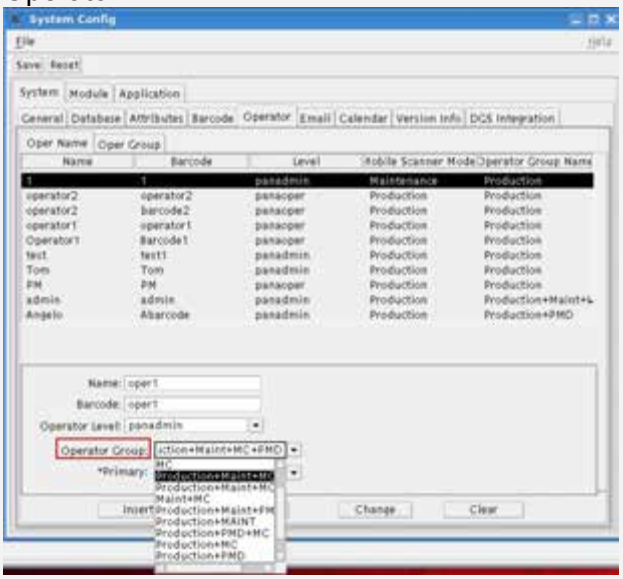
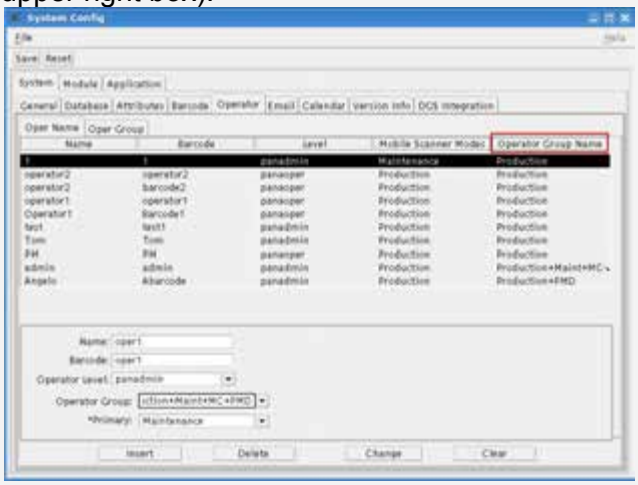
■ Scanner Group Security

The **Operator** Tab under Systems Config in CIMC 8.3.3.0 is updated with following two tabs:

1. **Oper Name**
2. **Oper Group**

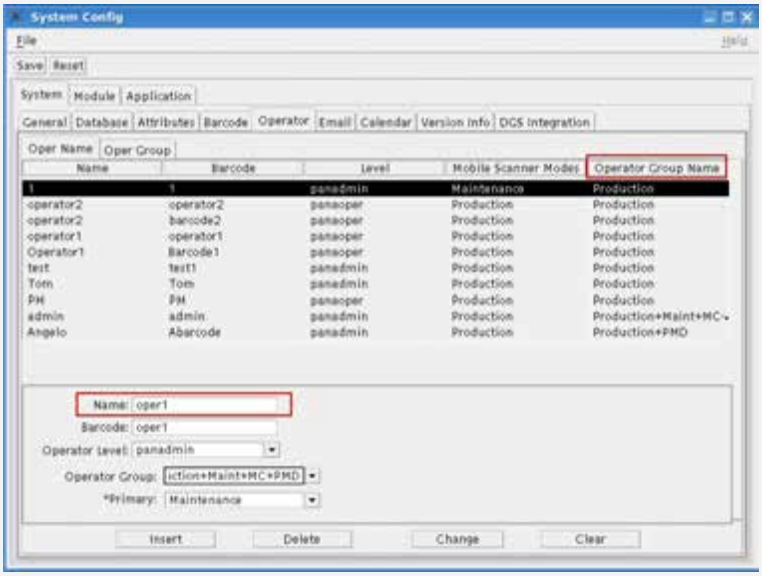
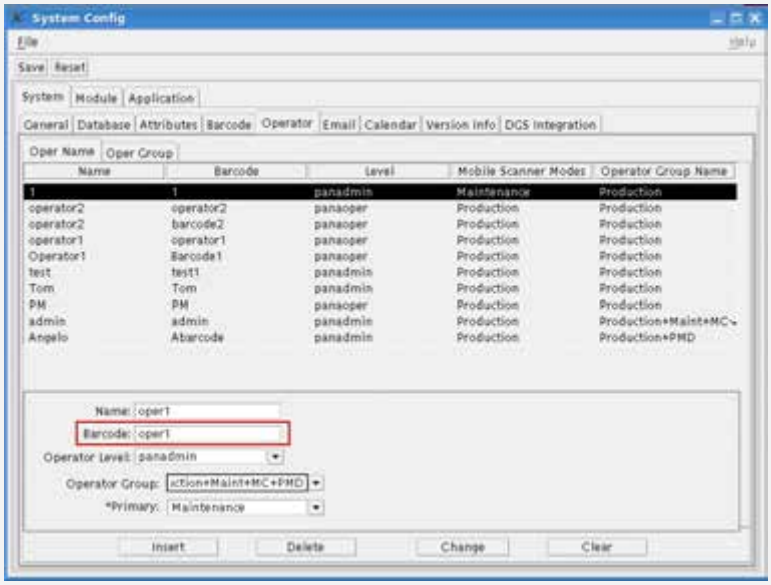
● Oper Name

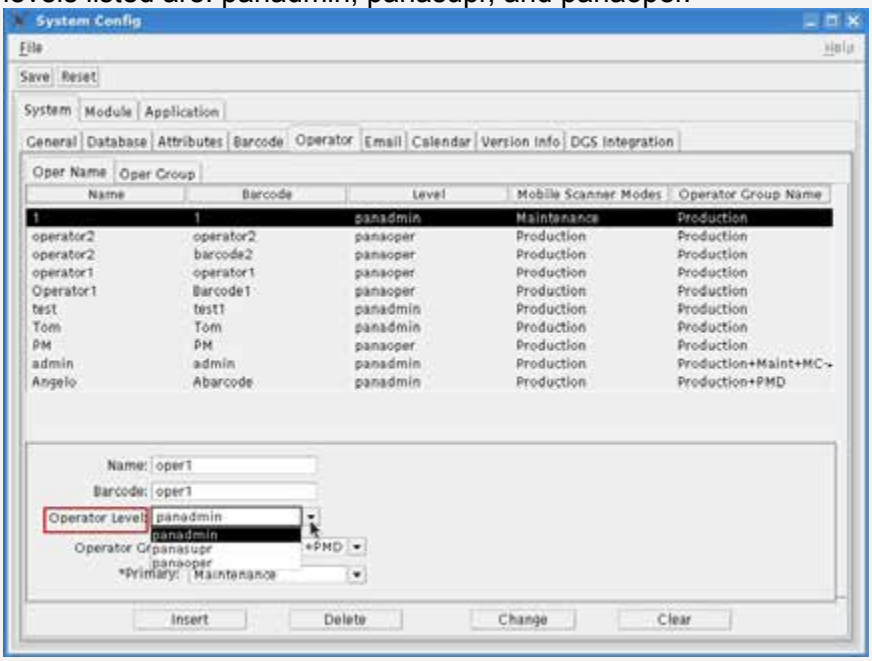
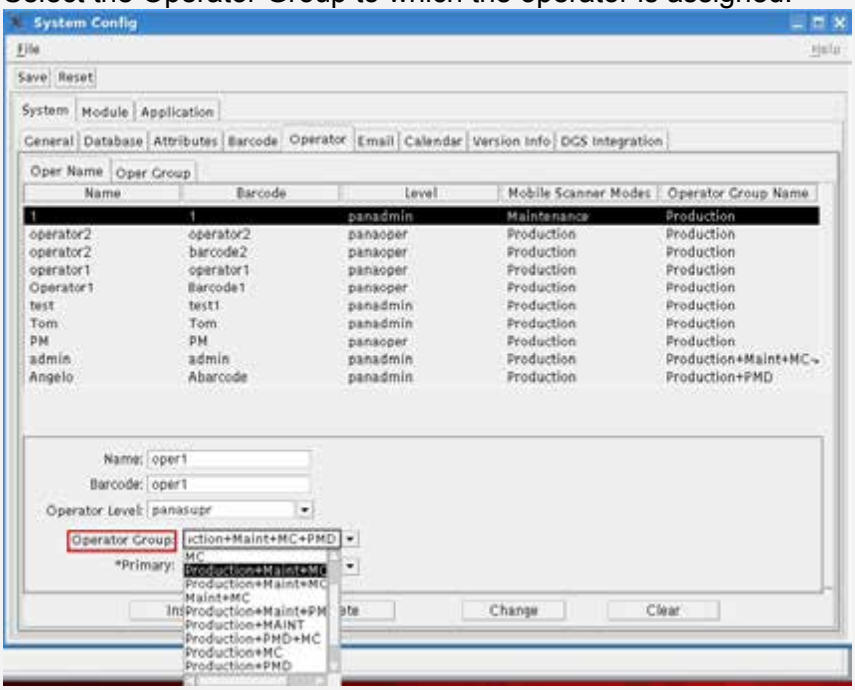
The Oper Name tab allows the user to assign a predefined “Oper Group” to an operator.

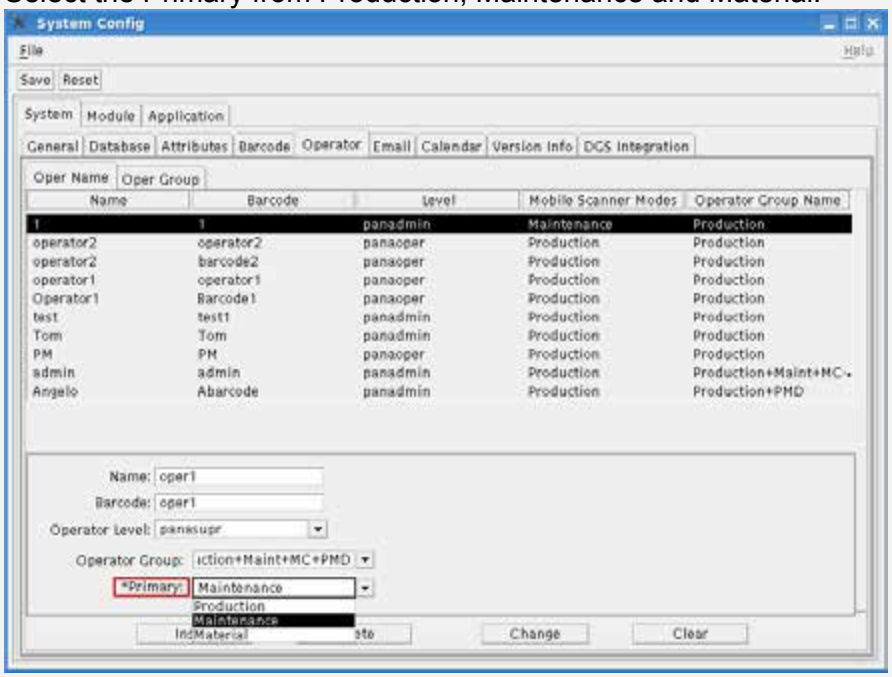
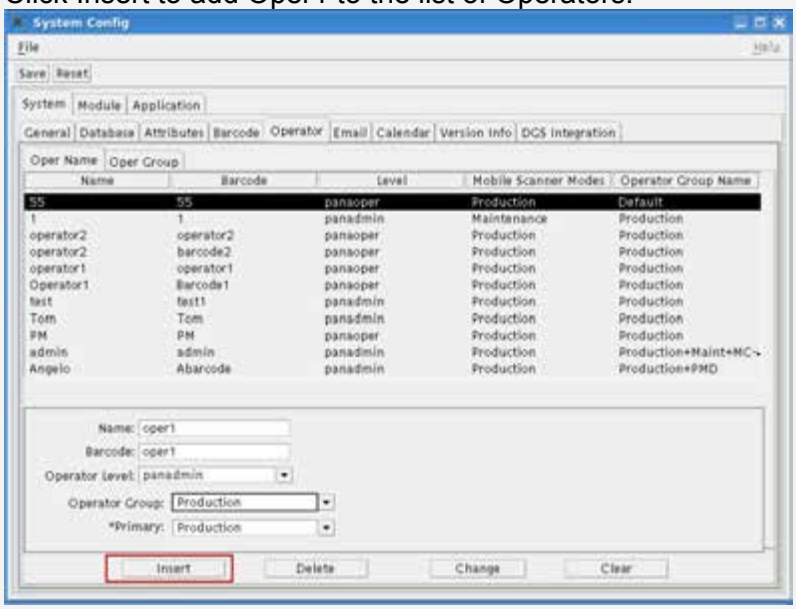
Step	Procedure
1	In **Oper Name** a new field called “**Operator Group**” has been introduced (lower left corner). The user can assign a predefined Oper Group to the selected Operator. 
2	A list of “**Operator Group Names**” assigned to operator appears (upper right box). 

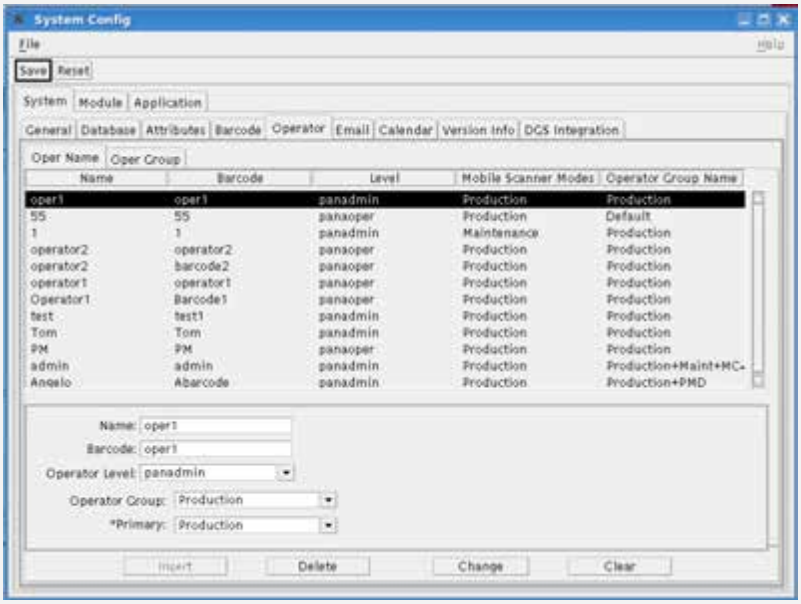
- **Adding Operator**

The following is the procedure to add operator:

Step	Procedure
1	Enter the name of the operator in Systems Config > System > Operator > Oper Name. 
2	Enter the Barcode name. 

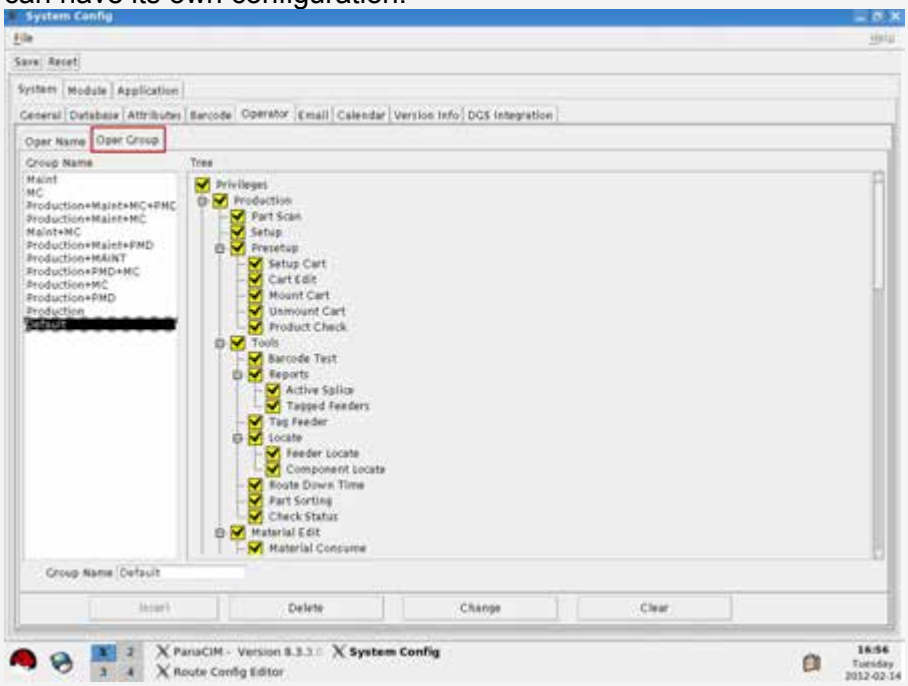
3	Select the Operator Level to which the operator is assigned. The operator levels listed are: panadmin, panasupr, and panaoper. 
4	Select the Operator Group to which the operator is assigned. 

5	Select the Primary from Production, Maintenance and Material. 
6	Click Insert to add Oper1 to the list of Operators. 

7	Click **Save**. Oper1 gets added to the list displaying the level, operator group name it is member of. 

Oper Group

The Oper Group tab allows the user to define the Operator Groups and assign the privileges to see corresponding menu (with exception) in the CCS scanner that will be visible to that operator.

Step	Procedure
1	In **Oper Group** the user has ability to define any number of groups. Each group can have its own configuration. 
2	In order to accommodate existing releases with existing operators configured the following **Group Name Templates** are supplied with PanaCIM and have the following attributes: ➢ **Default** - This group name template cannot be deleted and by default have all the functions and their sub functions enabled (checked). This template can be assigned in the previous screen to any operator. It can also be customized which means the user can check or uncheck any of the functions. This template applies to all scanners in the event where Operator IDs are not enabled in PanaCIM. If the Default Template is assigned to an Operator and the Operator logs into the scanner, only the licensed functions will be visible on the scanner. ➢ **Production** - This template includes all the production functions and sub functions enabled by default. ➢ **Production+PMD** - This template includes the following functions in enabled state: ▪ all the production functions and sub functions ▪ PMD functions ➢ **Production+MC** - This template includes the following functions in enabled state: ▪ all the production functions and sub functions ▪ all Material Control functions and sub functions ➢ **Production+PMD+MC** - This template includes the following

functions in enabled state:
- all the production functions and sub functions
- PMD functions
- All Material Control Functions sub functions

- **Production+Maint** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions

- **Production+Maint+PMD** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - PMD functions

- **Maint+MC** - This template includes the following functions in enabled state:
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions

- **Production+Maint+MC** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions

- **Production+Maint+MC+PMD** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions
  - PMD functions

- **MaterialControl** - This template includes the following functions in enabled state:
  - all Material Control functions and sub functions

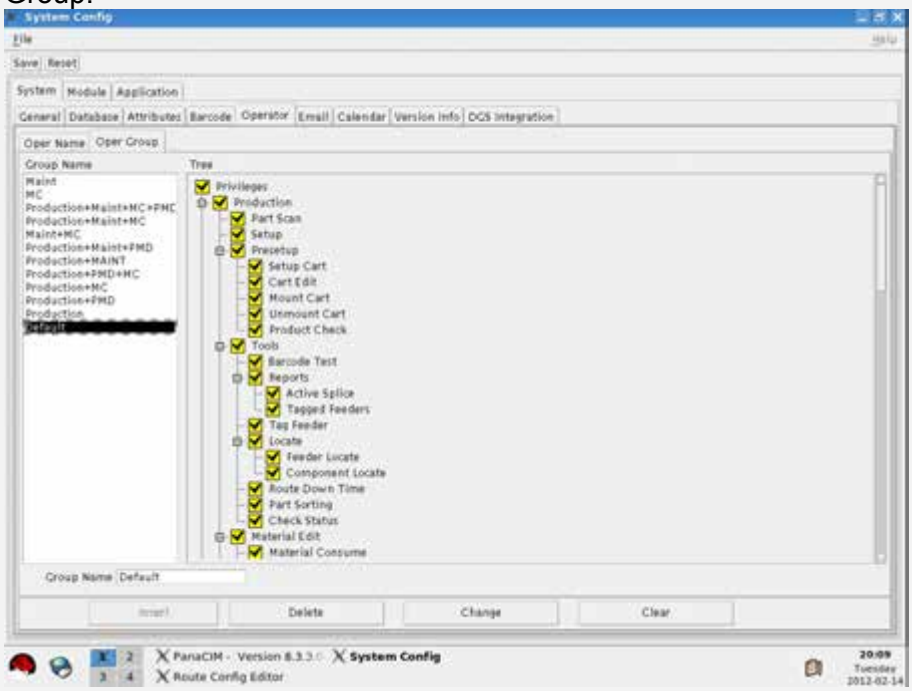
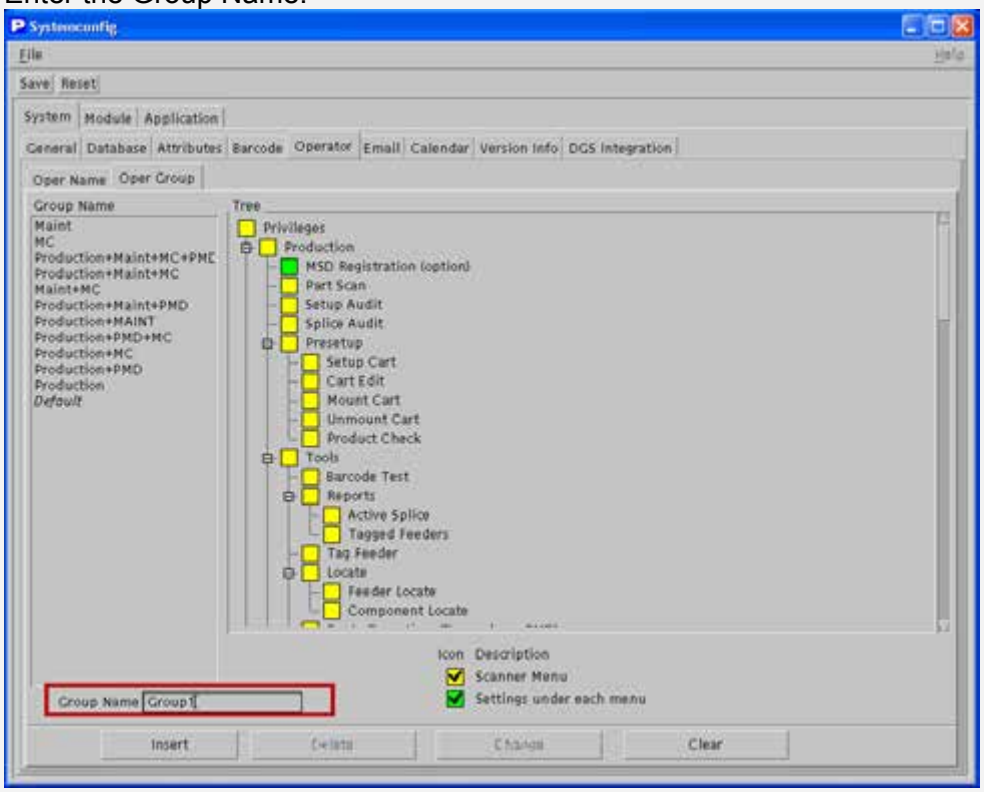
- **Maintenance** - This template includes the following functions in enabled state:
  - all Maintenance functions and sub functions

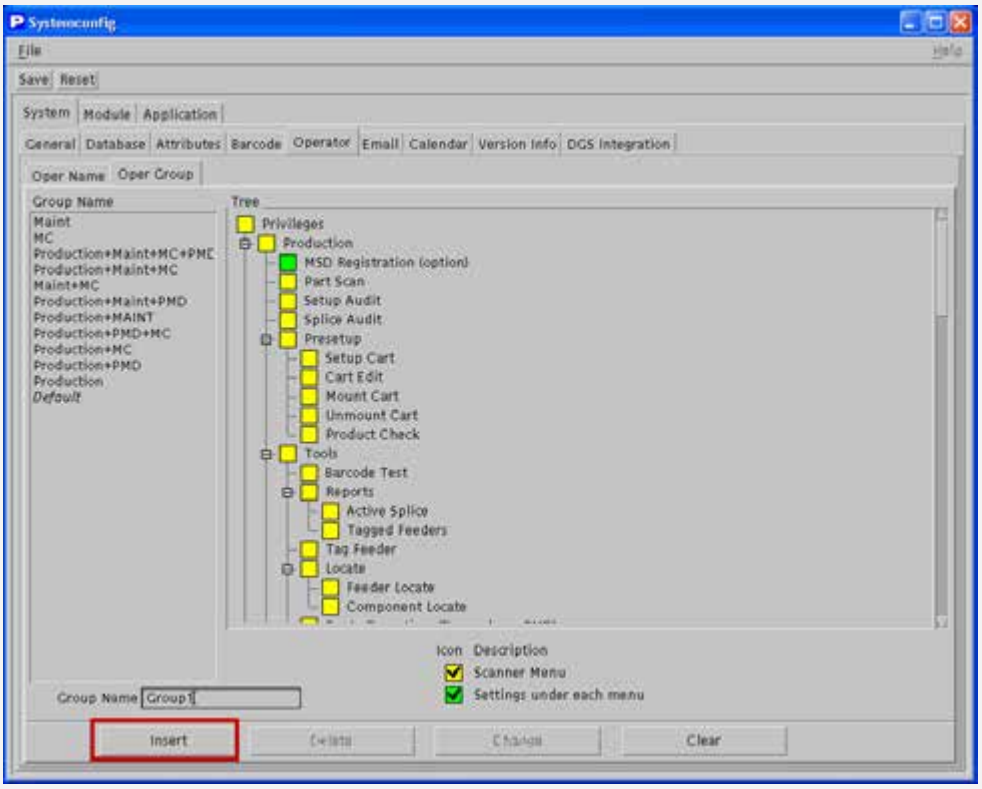
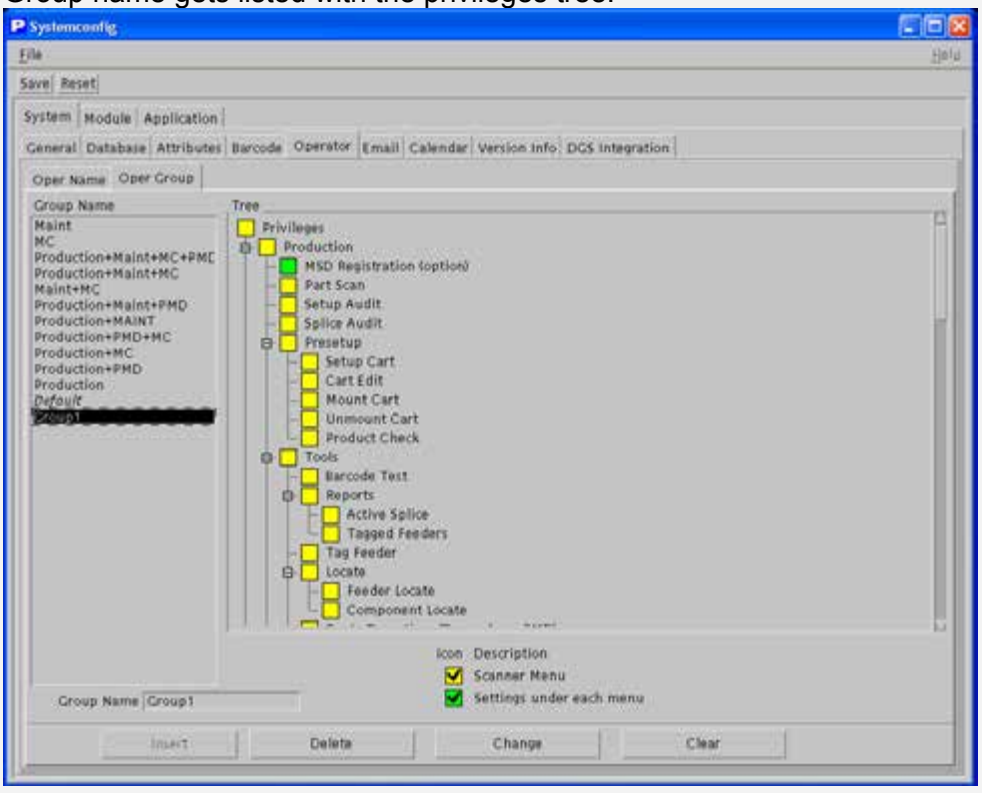
All above Templates, can be edited, renamed or even deleted by the user except for the “Default” Template. The Default Template cannot be renamed or deleted. It can however be customized to display any combination of the existing licensed functions.

Licensing supersede the above configuration, which means that even if a function is checked in the above configuration, it will not be displayed unless the factory has the corresponding licenses.

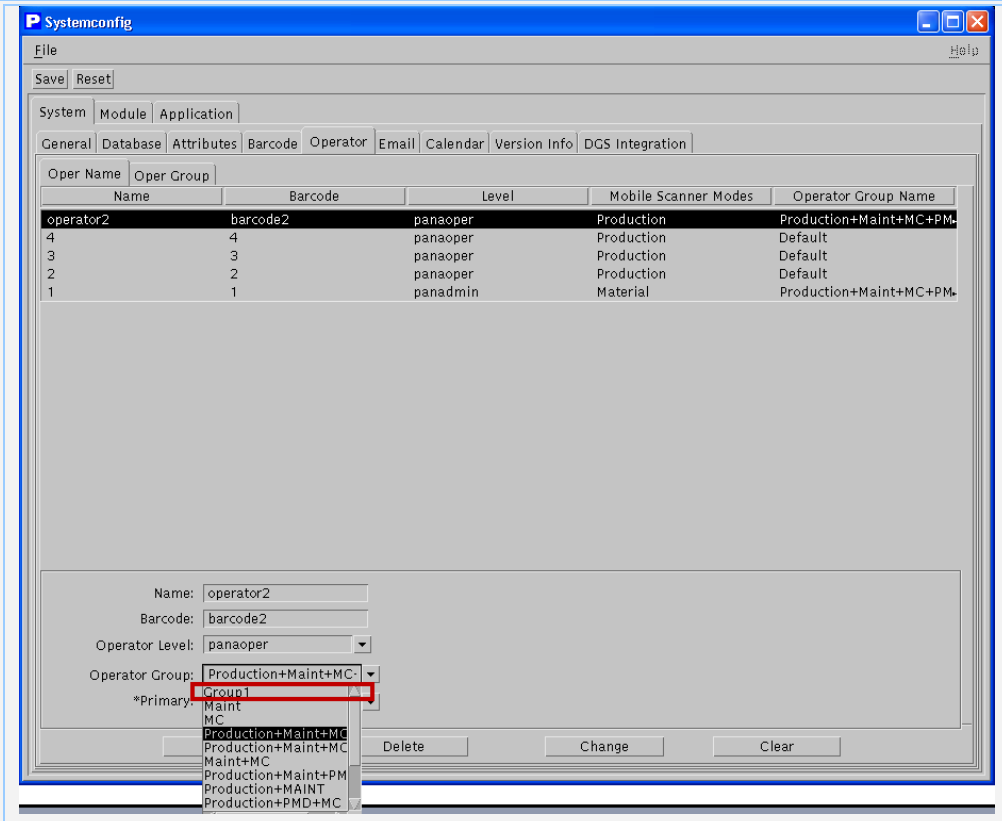
- **Adding Group**

The following is the procedure to add group:

Step	Procedure
1	Enter the name of the operator in systems Config>System>Operator>Oper Group. 
2	Enter the Group Name. 

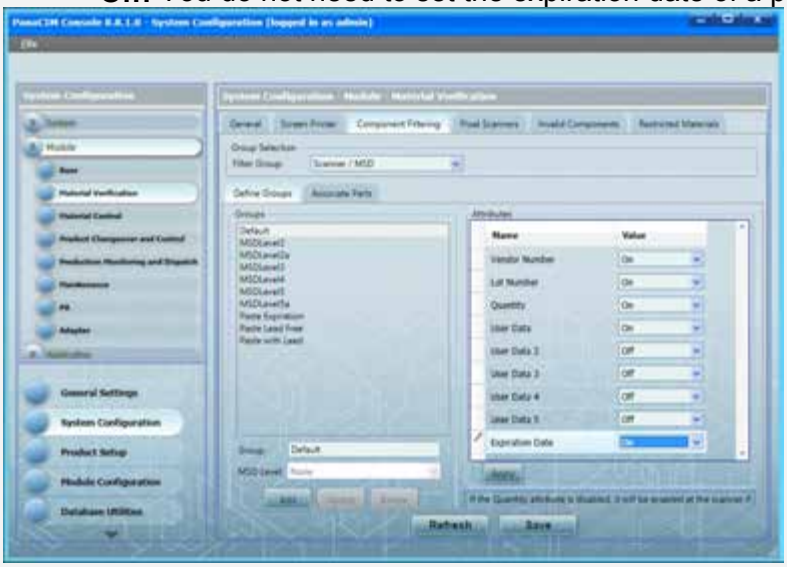
3	Click Insert. 
4	Group name gets listed with the privileges tree. 
5	Select the privileges desired to be assigned to the group and click **Save**. The privileges will be reflected in Operator Group.
6	Close the Systemconfig screen and open it again. Make sure that the added group name is included in the **Operator Group** drop-down list in the **Oper Name** page.

2.2 Settings


Systemconfig
File
Help
Save
Reset
System
Module
Application
General
Database
Attributes
Barcode
Operator
Email
Calendar
Version Info
DGS Integration
Oper Name
Oper Group
Name
Barcode
Level
Mobile Scanner Modes
Operator Group Name
operator2 barcode2 panaoper Production Production+Maint+MC+PM
4 4 panaoper Production Default
3 3 panaoper Production Default
2 2 panaoper Production Default
1 1 panadmin Material Production+Maint+MC+PM
Name: operator2
Barcode: barcode2
Operator Level: panaoper
Operator Group: Production+Maint+MC
*Primary
Group1
Maint
MC
Production+Maint+MC
Production+Maint+MC
Maint+MC
Production+Maint+PM
Production+MAINT
Production+PMD+MC
Delete
Change
Clear

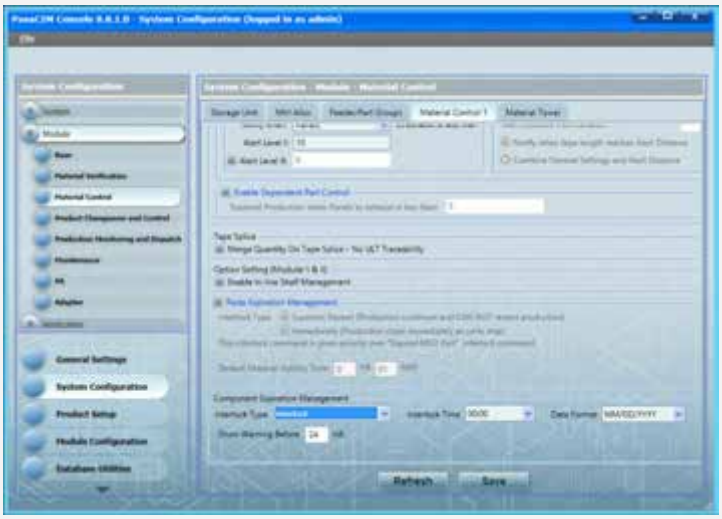
■ Enable Expiration Date

Refer to the action table below to configure the expiration date of parts on the PanaCIM console.

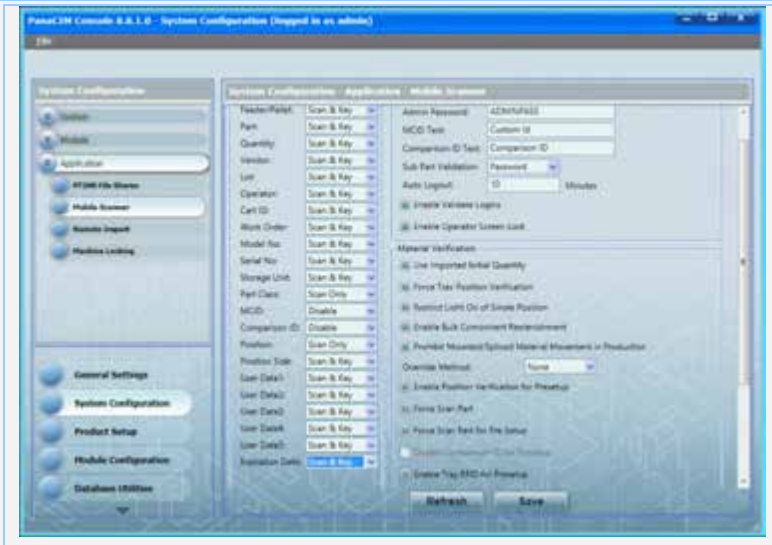
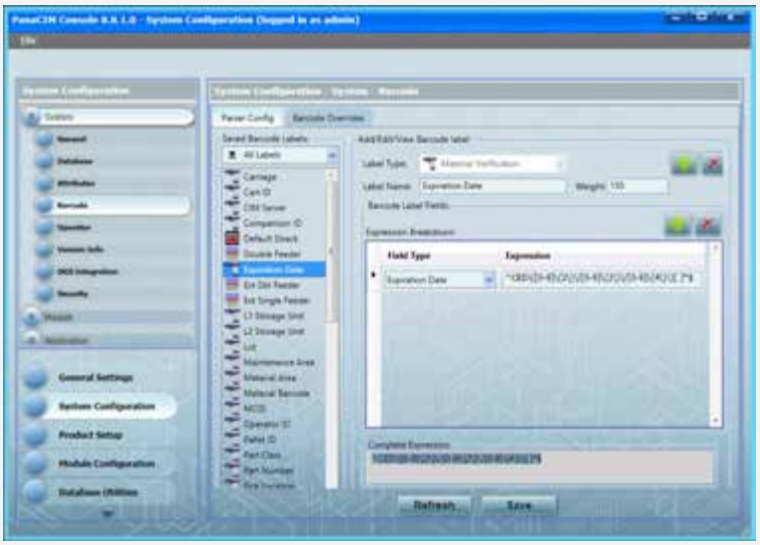
Step	Procedure
1	Open the **PanaCIM console** from the **PanaCIM application** server.
2	Select System Configuration > Module > Material Verification > Component Filtering.
3	In **Define Groups**, select a group that you want to enter the expiration date. Set the <**Expiration Date**> item in Attributes. • **On**: You must set the expiration date of a part when it is registered. • **Any:** You may or may not set the expiration date of a part when it is registered. • **Off:** You do not need to set the expiration date of a part.  **Note)** When you want to control the expiration date of all parts, select <**Default**> in Group. Also for MSD parts, their expiration dates are controlled according to the MSD part standard. Therefore, they are not included here.
4	After the expiration date setting is completed, click **Apply** and then click **Save**.
5	Select System Configuration > Module > Material Control > Material Control1.
6	Configure the following settings in **Component Expiration management**. <Interlock Type> Set whether a warning message is displayed and interlock is activated during production when an expired control part is verified. • **Interlock:** A warning message is displayed and interlock is activated. • **Warning:** Warning display only. • **No interlock/No warning:** Interlock is not activated and a warning message is not displayed. <Interlock Time> Set the time of the expiration when a part is registered.

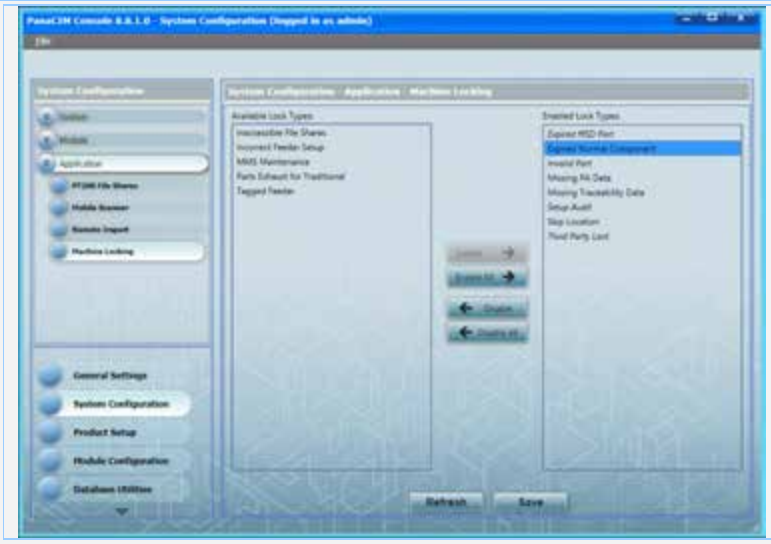
<Date Format>
Set a date format when the expiration date is entered.
If you use barcode, follow the setting below to set parser shown in step 12.
YYYY/MM/DD: Year, Month, Day format. (e.g. 2017/01/31)
MM/DD/YYYY: month, Day, Year format. (e.g. 01/31/2017)
DD/MM/YYYY: Day, Month, Year format. (e.g. 31/01/2017)

<Show Warning Before>
To go back to the interlock time, set the time to display a warning message.
e.g.) Expiration date: 2017/12/31
Interlock time: 13:00
Show warning before: 8 hours
Warning display time: After 2017/12/31 AM05:00



7	Click **Save**.
8	Select System Configuration > Application > Mobile Scanner.
9	Set an <Expiration Date> item by a scanning operation. • **Scan & Key**: Scan barcode of the expiration date or manually enter the expiration date when a material is registered. • **Scan only:** Scan barcode of the expiration date when a material is registered. • **Invalid:** The expiration date is not set when a material is registered.

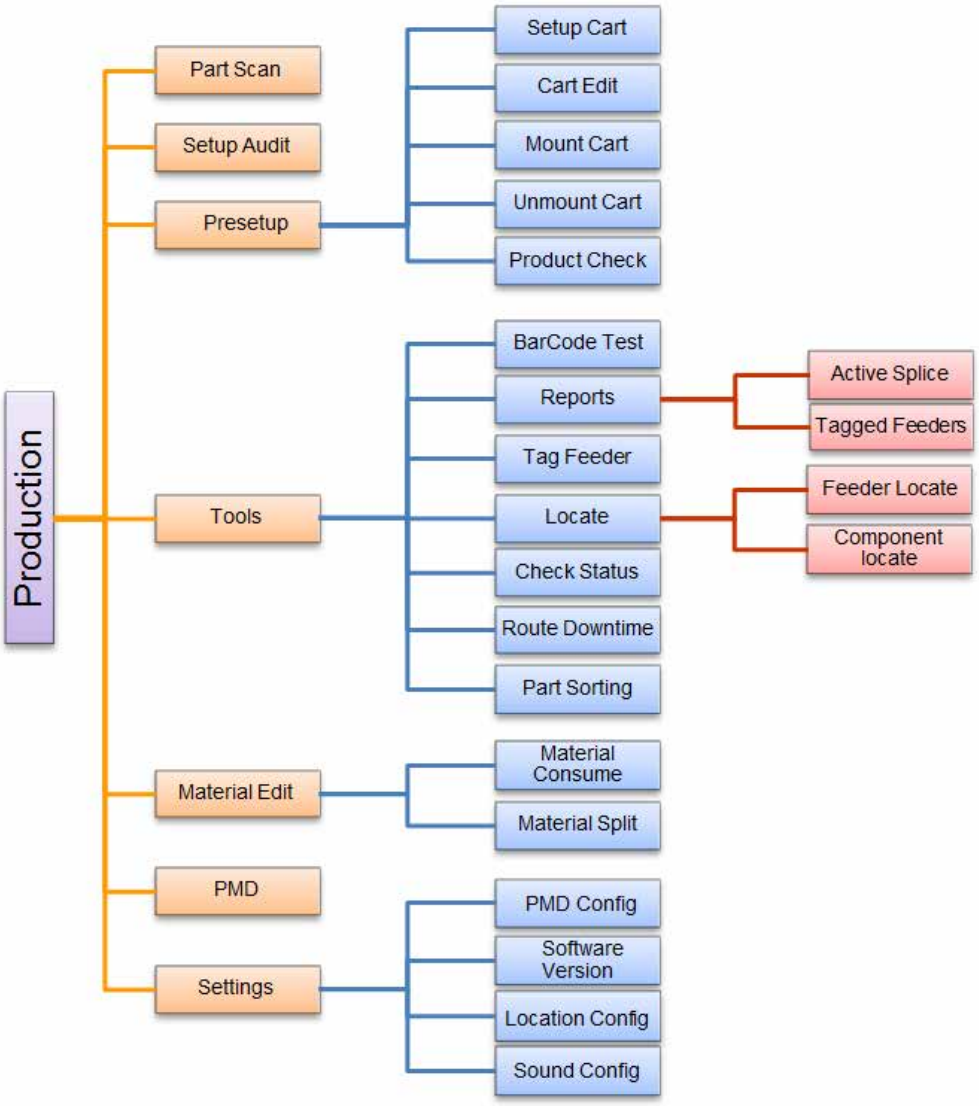
	
10	Click **Save**.
11	Select System Configuration > System > Barcode > Parser Config.
12	Select <Expiration Date> using the saved barcode label. Set label information in Add/Edit/View Barcode Label. 
13	Click **Save**.
14	Select System Configuration > Application > Machine Locking.
15	Set E**xpired Normal Component** when the part in use is expired during production and you want to lock the machine. • Enable: Locked. • Disable: Not locked.

	
16	Click **Save** to finish.

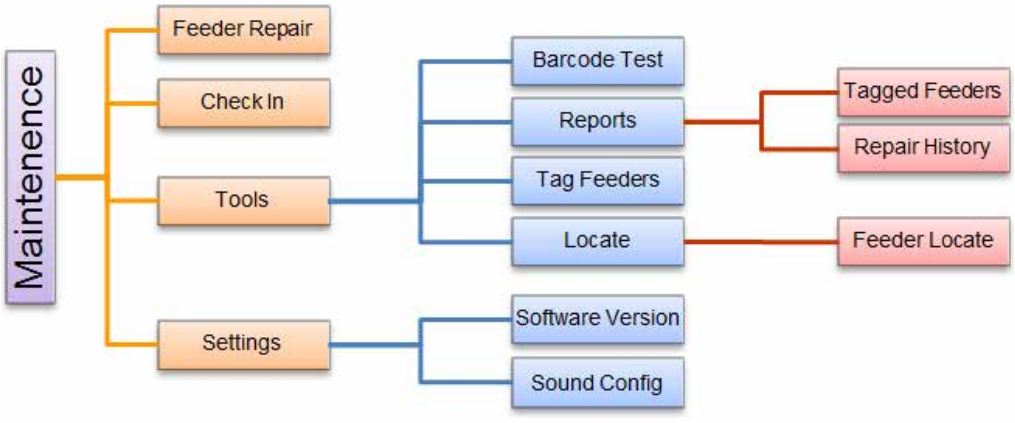
- **Operator functions**

The following is the list of functions available to operators on the CCS scanners.

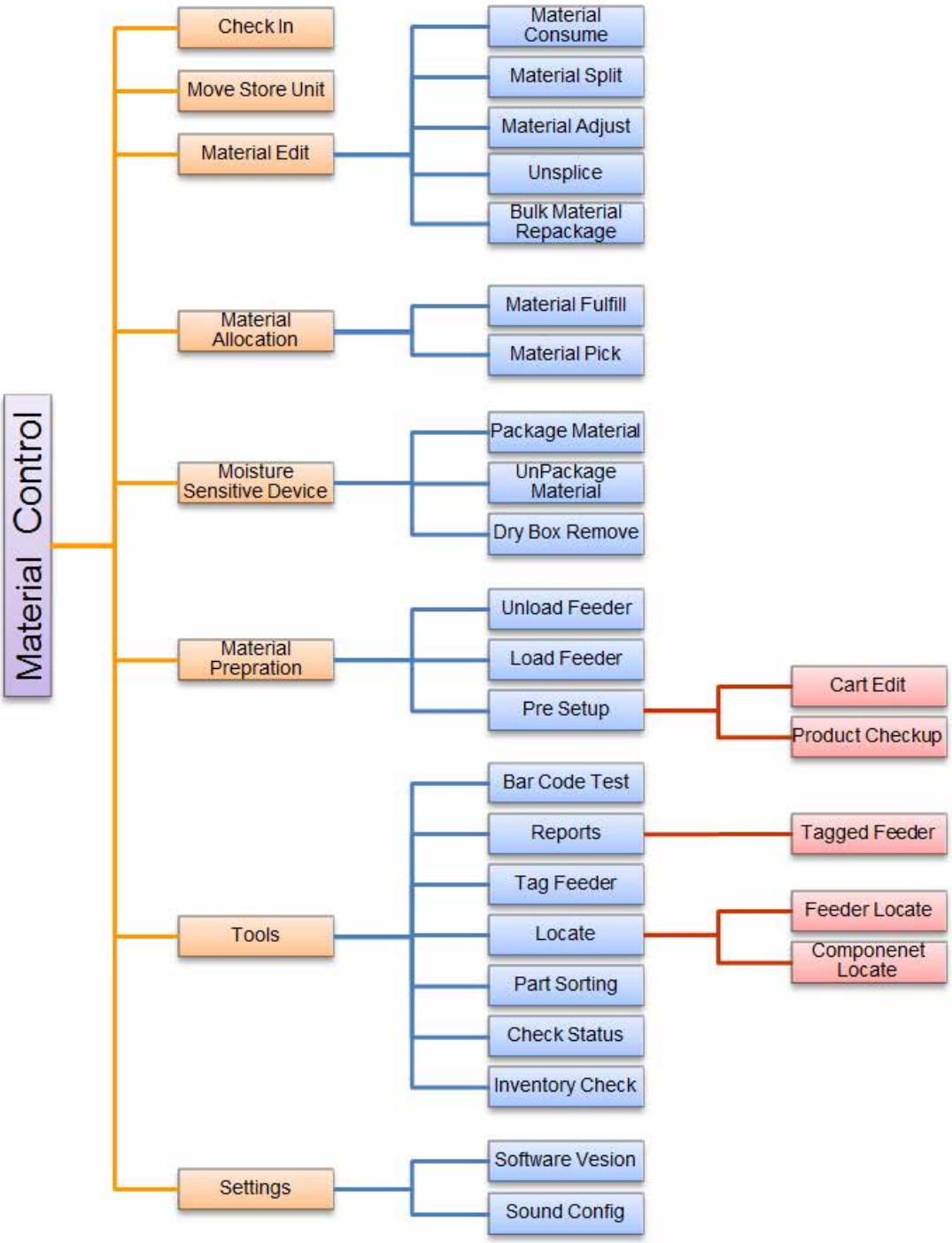
Production



Maintenance

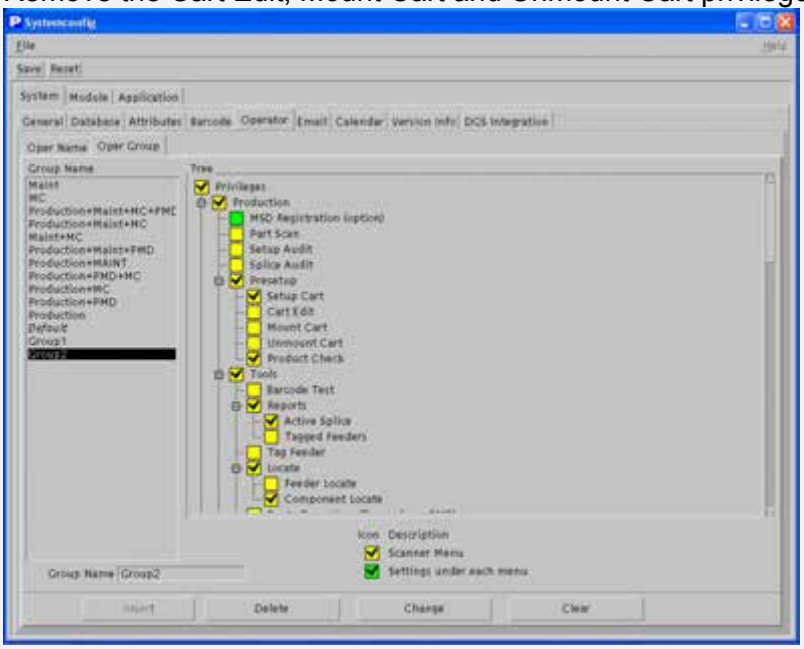
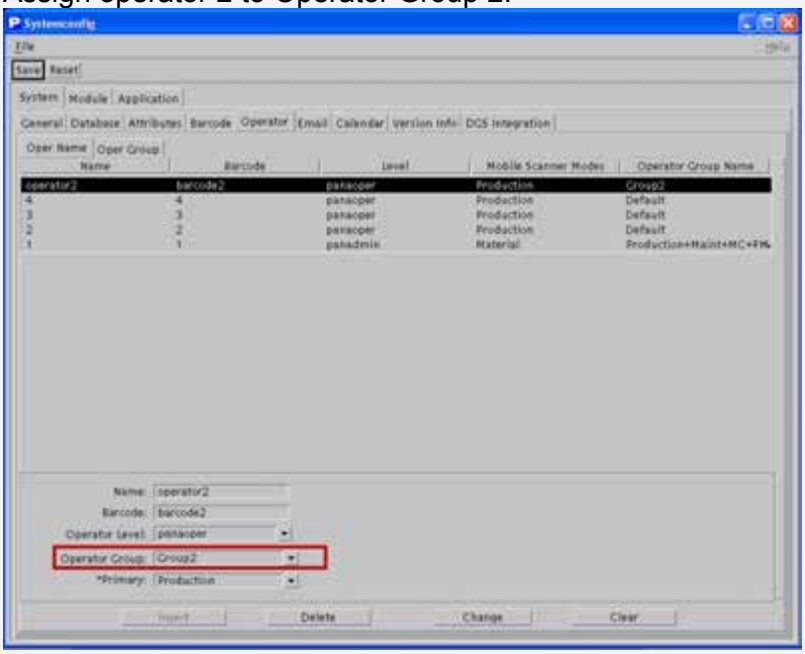


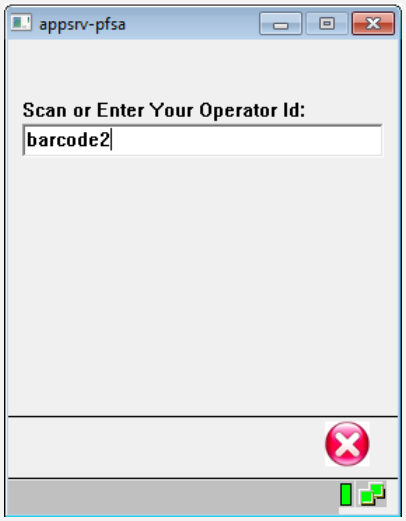
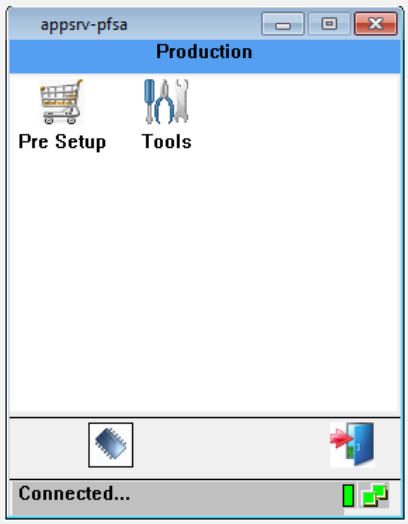
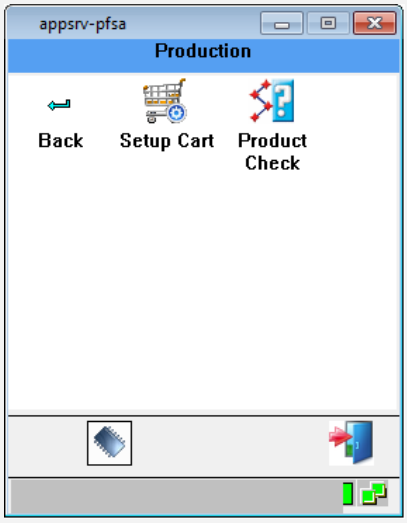
Material Control



Sample Configuration Settings

In this section we will assign some selected privileges to the operator 2 under Group 2 and will validate the privileges to the operator on the scanner.

Step	Procedure
1	Assign the privileges for Presetup >Setup Cart and Product Check to Group2.
2	Remove the Cart Edit, Mount Cart and Unmount Cart privileges from Group2. 
3	Assign operator 2 to Operator Group 2. 
4	Save the changes.

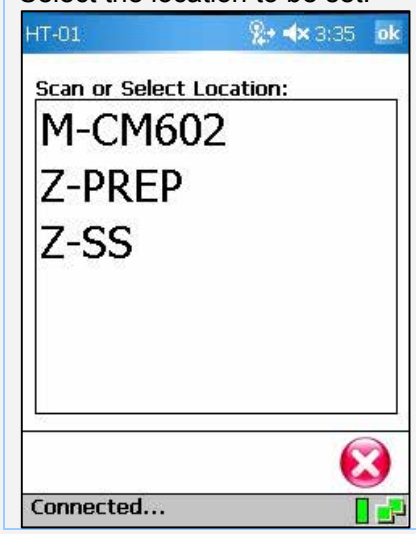
5	On the CCS Scanner login with barcode2. 
6	Once Logged on the Scanner, the operator2 will see the following options. 
7	Under Presetup, the operator 2 will be able to see only Setup Cart and Product Check. 

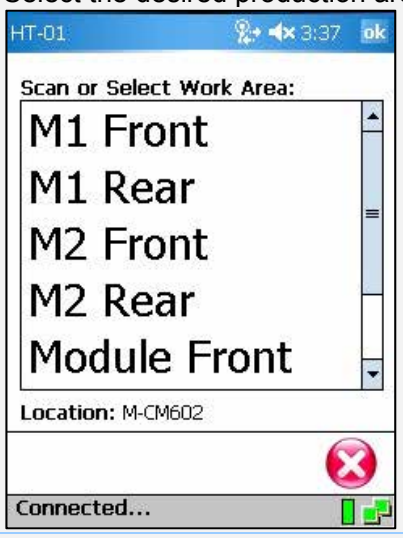
8	The operator 2 will not have access to the Cart Edit, Mount Cart and Unmount Cart privileges. This is because these privileges were removed from to Group2.
9	In this way the groups can be created and assiged different privileges based on the requirements.

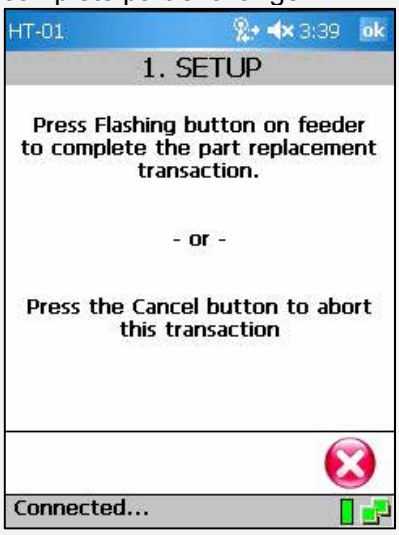
2.3 Operation

2.3.1 Material Handling - Verification

By scanning the registered material ID during verification, the attribute information such as the part number, vendor, lot and quantity associated with the material ID will be applied as verified information. (If unregistered ID is scanned, part number and attribute information need to be entered.)

Step	Mobile Scanner
1	Select **Setup** from the Production main menu. 
2	Select the location to be set. 

3	Select the desired production area from the list. 
4	The Setup screen shown below appears. 
5	Scan or enter the material ID with the part number registered in the list.  **Note:** If the part number associated with the material ID does not agree with the one in the setup list, the scanner displays a message indicating that there is no applicable part number in the list.

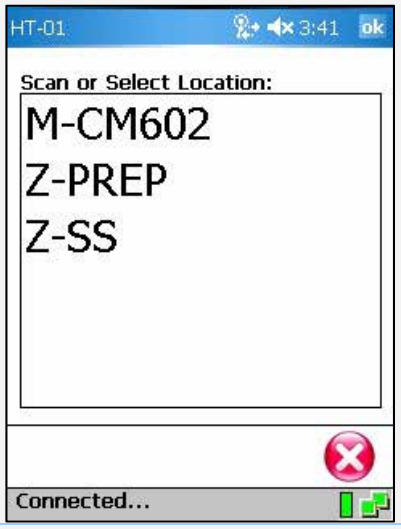
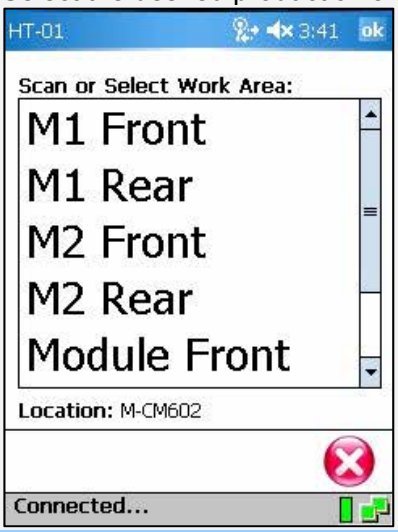
6	A message appears prompting you to press the blinking button of the feeder to complete part exchange..  After pressing the button, the part information will be written into the feeder memory.

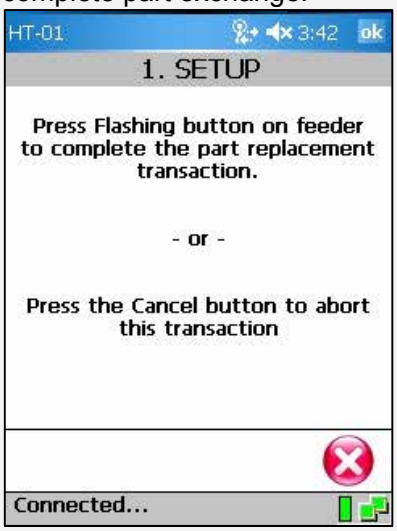
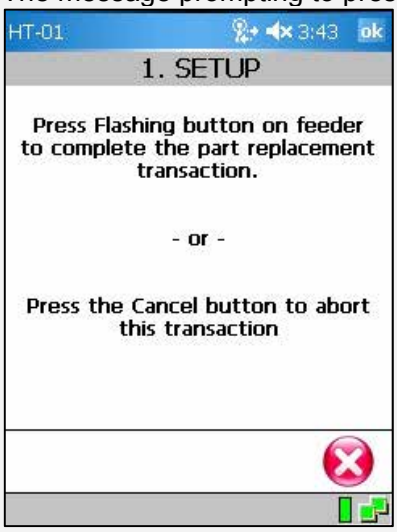
■ Splice Reel Management in the Context of Verification

The reel information obtained by scanning the material ID of splice reel differs between the NPM series and CM series machines.

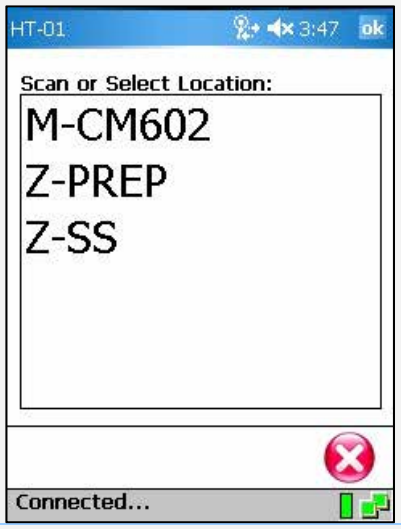
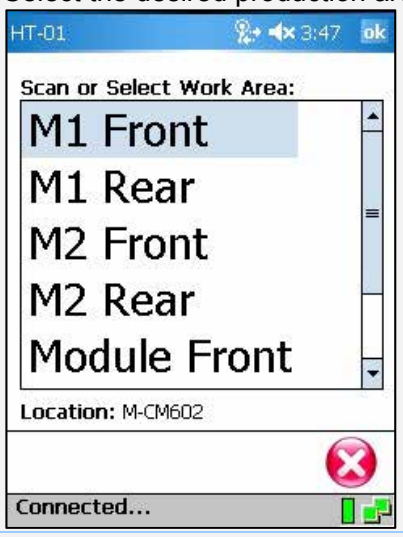
In case of the NPM series machines, if the material ID of the spliced reel is scanned, the reel information before and after the splicing will be written into the feeder memory.
The following shows the procedure for verification with the NPM series machines.

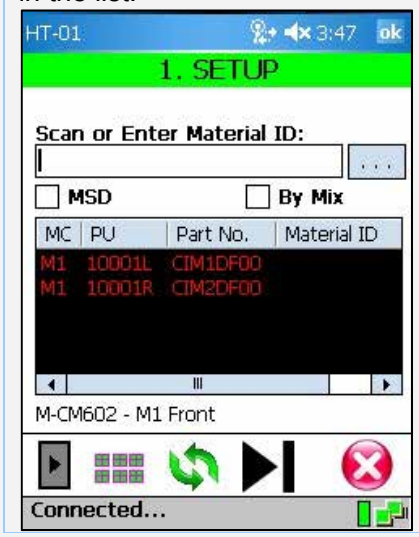
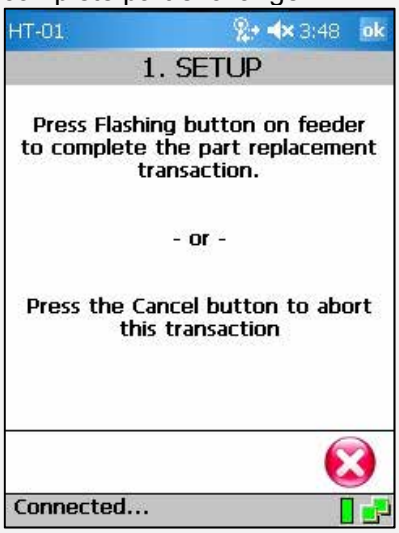
Step	Mobile Scanner
1	Select **Setup** from the Production main menu. 

2	Select the location to be set. 
3	Select the desired production area from the list. 
4	Scan or enter the spliced material ID with the part number registered in the list. 

6	A message appears prompting you to press the blinking button of the feeder to complete part exchange.  After pressing the button, the part information of the reel prior to splicing will be written into the feeder memory.
7	The message prompting to press the blinking feeder appears again.  After pressing the button, the part information of the spliced reel will be written into the feeder memory.

In case of the CM series machines, if the material ID of the spliced reel is scanned by the reel information after the splicing will be written into the feeder memory. (The reel information before the splicing will not be written.)
The following presents the procedure for verification with the CM series machines.

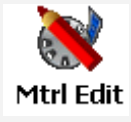
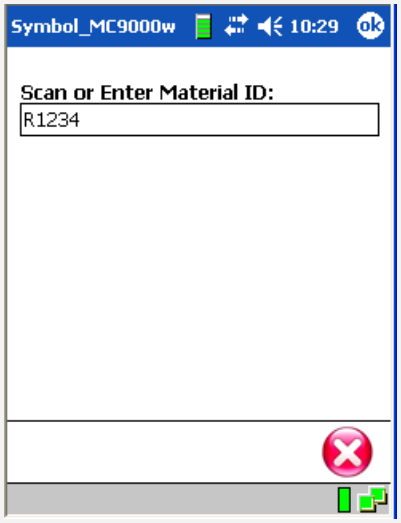
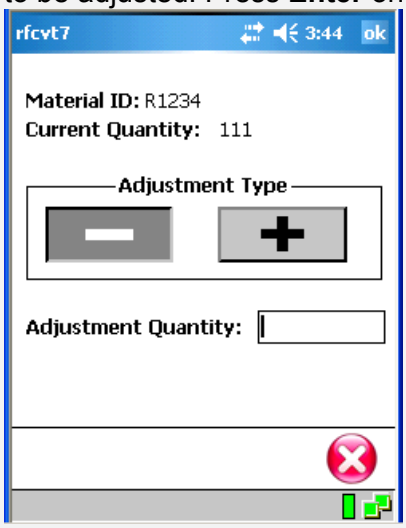
Step	Mobile Scanner
1	Select **Setup** from the Production main menu. 
2	Select the location to be set. 
3	Select the desired production area from the list. 

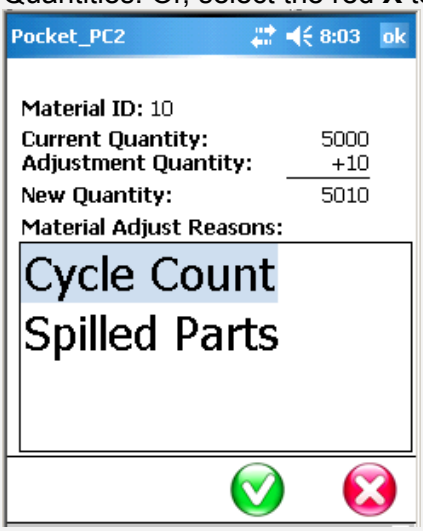
4	Scan or enter the spliced material ID with the part number registered in the list. 
5	A message appears prompting you to press the blinking button of the feeder to complete part exchange.  After pressing the button, the part information of the reel containing a splice joint will be written into the feeder memory. **Note:** If a splice check error occurs due to detection of the spliced joint before splicing, select 'No splice check' from the scanner verification menu and scan the material ID of the spliced reel via the No splice check screen.

2.3.2 Material Handling - Adjust Material Quantity

Using the Scanner, scan the component's unique ID to indicate when quantities have been added or removed. The estimated remaining quantity is then adjusted in the database.

Refer to the action table below for instructions for adjusting material quantity using the Scanner.

Step	Mobile Scanner
1	Select the Material Handling  icon to begin.
2	Select  and then  .
3	Scan or enter your **Material ID** and click Enter. 
4	Select a positive or negative Adjustment Type and then enter the **Quantity** to be adjusted. Press **Enter** on the keypad. 

5	The scanner displays a summary of the adjustments made and prompts you to adjust the quantity of more materials. To continue, select a Material Adjust Reason and click the **green** checkmark to continue Adjusting Quantities. Or, select the red **X** to quit. 

■ Splice Reel Management in the Context of Adjust Material Quantity

The Material Handling operation of Mtrl Adjust has been enhanced to prompt users for which reel to adjust (Main or Splice) if either Material ID is scanned of a joint reel.

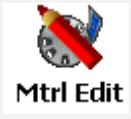
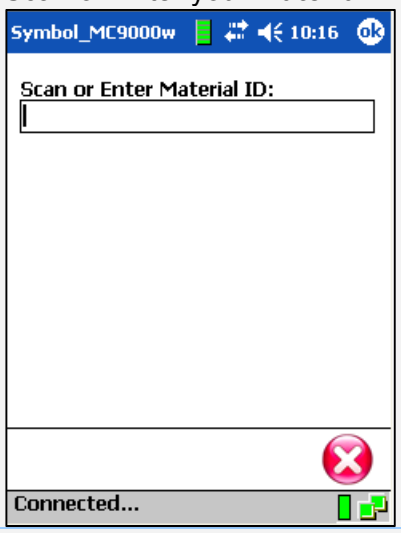
The mobile scanner maintains the relationship of the Main Reel (parent) to the Splice Reel (child) both on and off a machine. If an error is present you will be notified via scanner display.

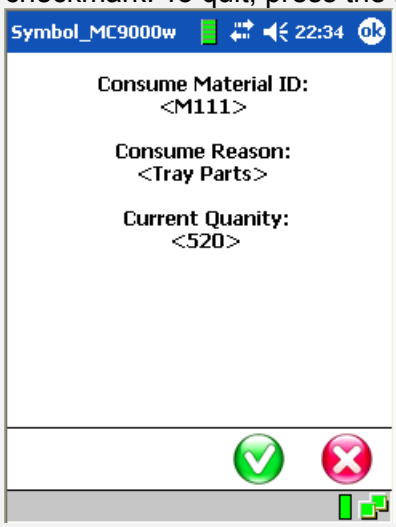
The following table illustrates what the scanner will instruct you to do when performing the Material Adjust action for a spliced reel.

Screen	Description
	The scanner will instruct you to select a reel where the quantity is to be adjusted. Select a Material ID from the window to continue.

2.3.3 Material Handling - Material Consume

Refer to the action table below for instructions for consuming/emptying materials using the Scanner.

Step	Mobile Scanner
1	Select the Material Handling  icon to begin. **Note:** These procedures are the same when in Production  mode.
2	Select  and then  .
3	Scan or Enter your **Material ID** and press **Enter** on the keypad. 
4	Select a **Consume Reason** and press the **green** checkmark. 

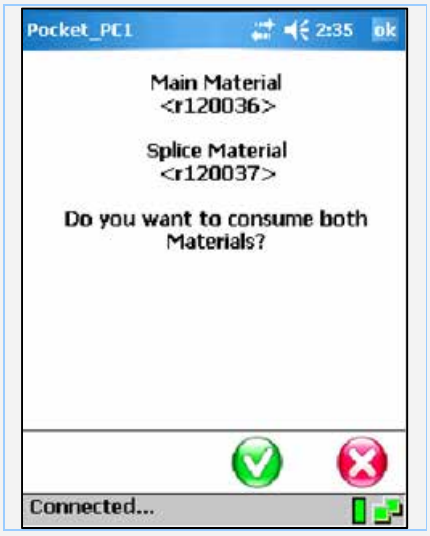
5	The scanner displays a summary of the material consumed and prompts you to consume more Material ID's. To continue, press the **green** checkmark. To quit, press the red **X**. 

- **Splice Reel Management in the Context of Material Consume**

The Material Handling Operation of Material Consume has been enhanced to consume both the Main Reel and Splice Reel simultaneously if either Material ID is scanned on a joint reel.

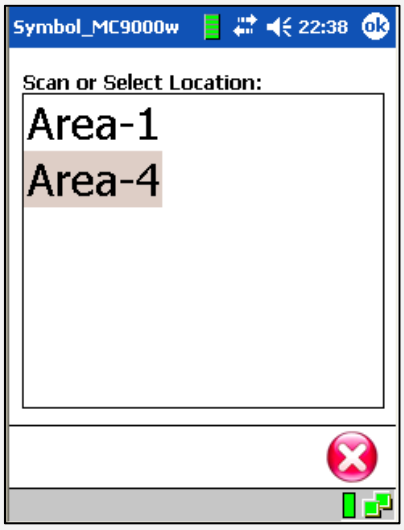
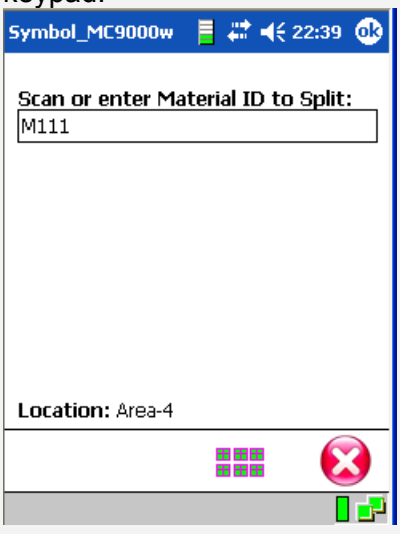
The mobile scanner maintains the relationship of the Main Reel (parent) to the Splice Reel (child) both on and off a machine. If an error is present you will be notified via scanner display.

The following table illustrates common errors that may occur when performing a Material Consume for a spliced reel.

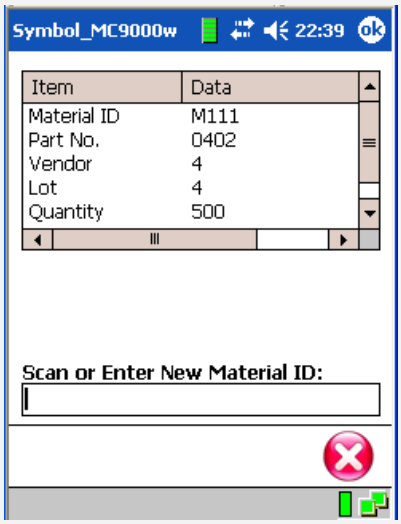
Screen	Description
	The scanner will ask if you would like to consume both the main and spliced reel if detected. Click the green checkmark to continue, or the red **X** to quit.

2.3.4 Material Handling - Split Material

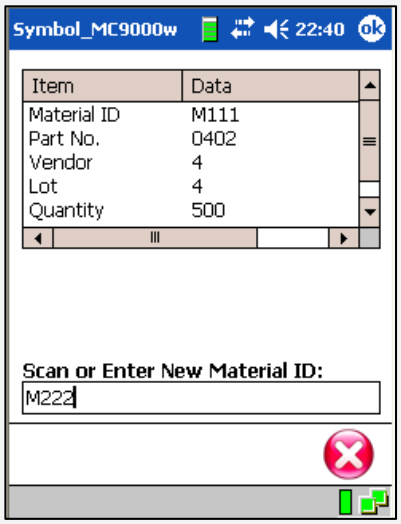
Refer to the action table below for instructions on how to split up the quantity of a material, and assign it to one or multiple materials using the Mobile Scanner.
Splitting the quantity of a component causes an error in the remaining quantity. Care should be taken when the components have been exhausted.

Step	Procedure
1	Select the Material Handling  icon to begin. **Note:** These procedures are the same when in Production  mode.
2	Select  and then  .
3	Barcode scan or select an Area from the scanner display. **Note:** The current Area will be highlighted on the scanner display. 
4	Scan or manually enter the new **Material ID** and press **Enter** on the keypad. 
5	The scanner displays a summary of the Material ID to Split and prompts you

to enter new material ID.

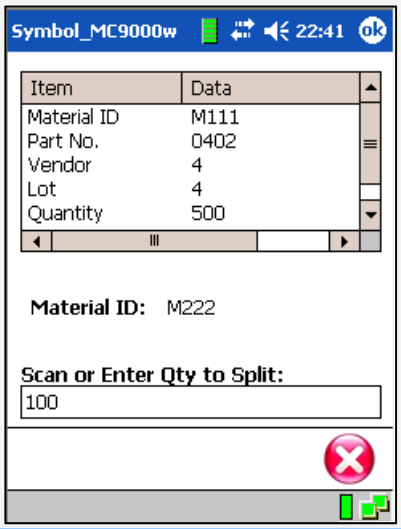


6 Scan or enter new Material ID.



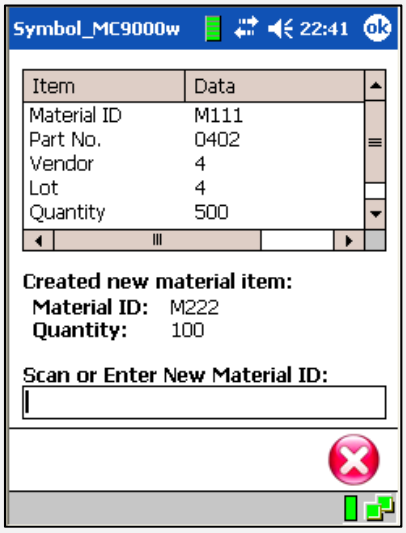
As indicated in the above screen, Material ID M111 is being split to create new Material ID M222.

7 After new material ID is scanned/entered, following screen appears to enter quantity for the split:



8 After scanning/entering quantity to split, new Material ID is created with

quantity split from original material id assigned to the newly created Material ID.



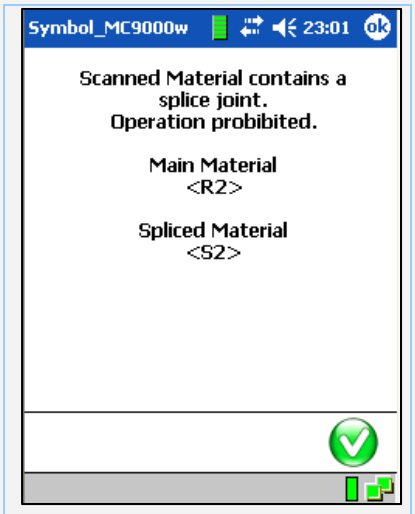
Repeat steps 6 and 7 to create new Material IDs from same Material ID via material split function.

Click the **X** button to close the screen after required number of Material IDs has been created.

Splice Reel Management in the Context of Split Material

The Material Handling operation of Material Split has been blocked from use on a joint reel. The mobile scanner maintains the relationship of the Main Reel (parent) to the Splice Reel (child) both on and off a machine. If an error is present you will be notified via scanner display.

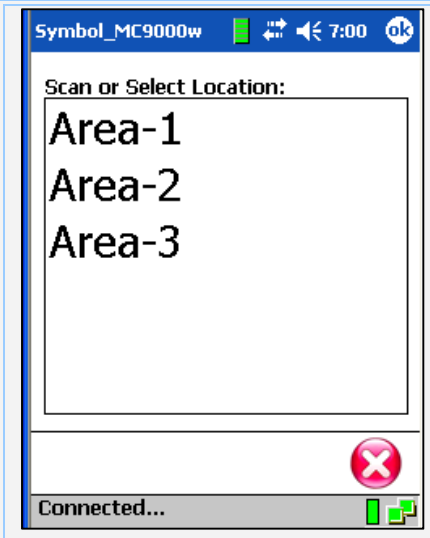
The following table illustrates a common error that may occur when performing a Material Split for a spliced reel.

Screen	Description
	If the scanner detects that the reel contains a splice joint, the following error message appears. You must un-splice the reel in order to continue. Then you may split. Correct this error and click the green arrow to continue.

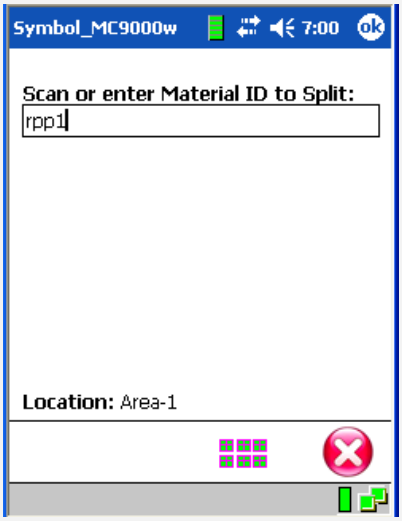
Allow Material split of a Reserved reel only up to the Unreserved Qty

If a reserved reel contains part quantity over and above the reserved quantity, only this additional quantity can be split. Refer to the action table below that explains this concept.

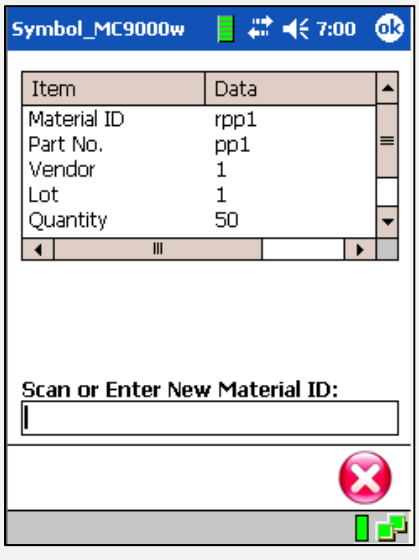
Step	Procedure
1	In the **Material Handling** screen, click **Material Split**. This is shown below: 
2	After clicking Material Split, following screen appears to scan or select a material area contained reserved reels.

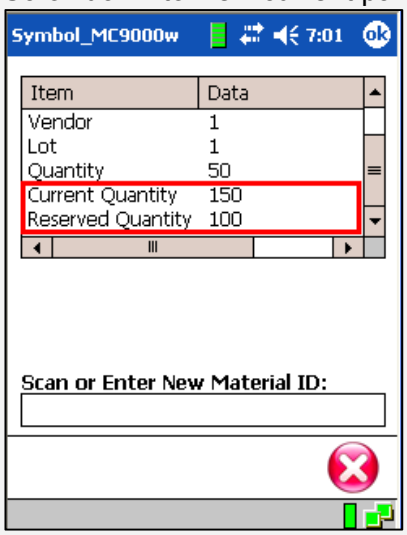
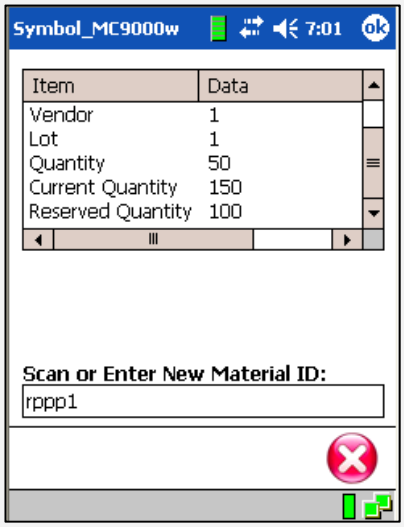
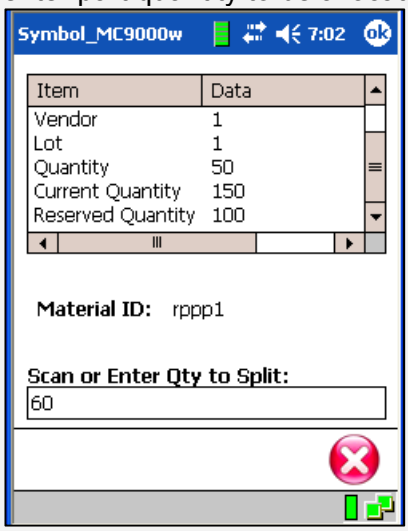


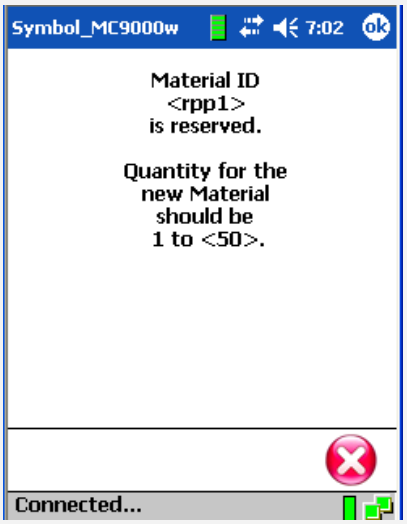
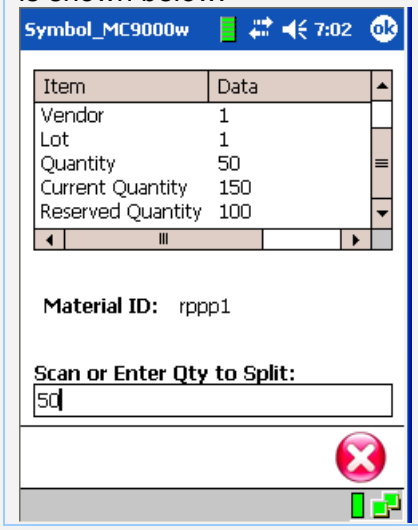
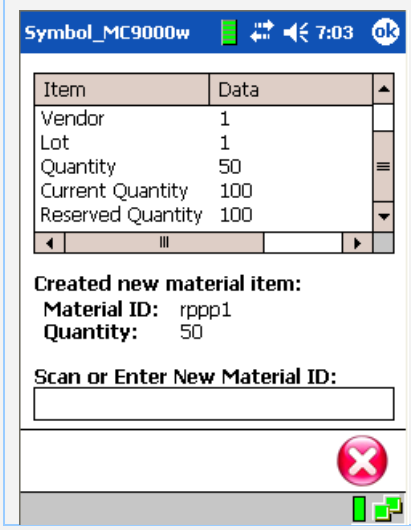
3 After scanning or selecting a material area, following screen appears to enter material ID to split. Enter the reel barcode of the reserved reel that needs to be split.



4 After scanning or entering material ID to split, following screen appears to enter a new material ID that will hold the quantity split from the reserved reel.

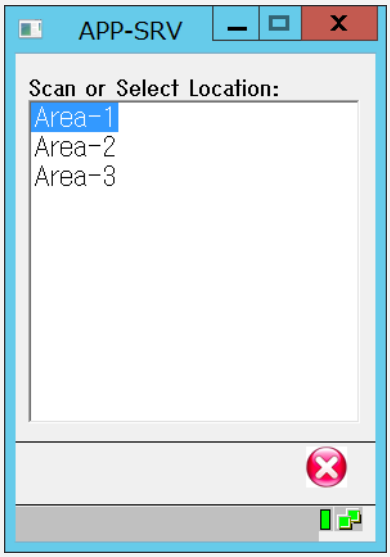
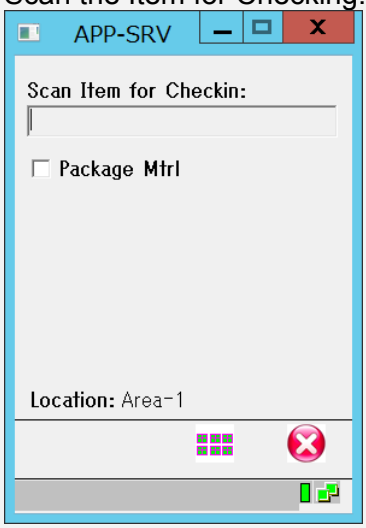


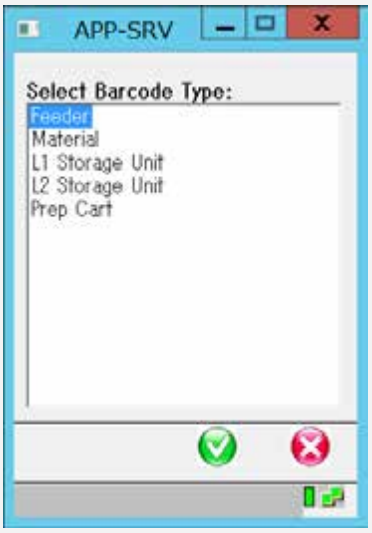
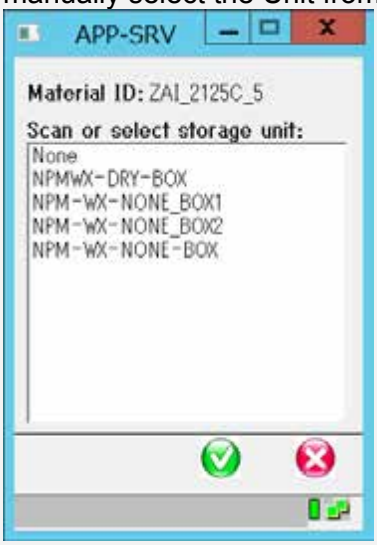
5	Scroll down to view current part quantity and reserved quantity. This is shown below:  **Note:** As shown in the screen above, current part quantity is 150 while reserved reel quantity is 100. The buffer quantity of 50 can be split to create additional reels.
6	Scan or enter material ID of the new reel that will hold the buffer quantity split from the reserved reel. 
7	After scanning or entering material ID of the new reel, following screen appears to enter part quantity to be allocated for the newly entered material barcode. 

8	If part quantity entered for the new reel exceeds the buffer quantity (Current quantity - reserved quantity), following message appears:  **Note:** Buffer quantity in the resereved reel with barcode MC0001A is 50 in the scenario considered here.
9	Enter part quantity in the new reel which is less than or equal to the buffer quantity. This is shown below: 
10	After entering the correct part quantity, reserved reel is split to create a new reel. 

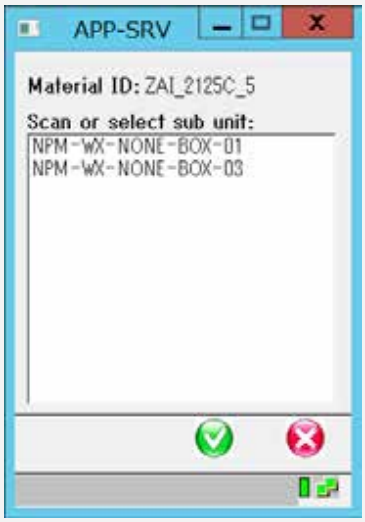
2.3.5 Material Handling - Check In Materials

Refer to the action table below for instructions on how to check in materials using the Mobile Scanner. This function is used to register new material ID or to move existing material ID.

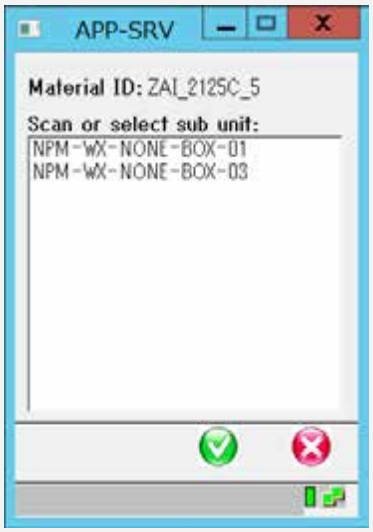
Step	Procedure
1	Select the Material Handling  icon to begin.
2	Select  .
3	Scan or Select an **Area**. 
4	Scan the Item for Checking.  **Note 1 - New ID:** If this ID is new, the scanner prompts you to enter part details. Scan the Part No. barcode, or key it in and press enter on the keypad. The scanner displays the Material ID and Part No., and requests you to scan or manually enter: • Vendor • Lot • Quantity

	Note 2 - Inactive Area: If the Item resides in an Inactive Area, the scanner displays the message “Material ID *xxx* is from an inactive area. Ok to move?”. Press **Apply** to continue with the Check In.
5	After entering the item for check in, following screen appears to select the barcode type of material being checked in:  **Note:** Above screen appears, when operators enter the item for checking. When operator scans the item for check in, scanner already knows the scanned barcode type and hence the above screen to select barcode type will not appear in this case. After selecting appropriate barcode type (for reels, barcode type is material), a message appears (shown in step 7) that the item has been checked in.
6	In case material area contains a storage unit, following screen appears to select the storage unit. If applicable, barcode scan a **Storage Unit**, or manually select the Unit from the scanner display and press **Apply**. 

Note: If you scanned a Storage Unit, and it is a **Storage Unit Level 2**, scan or select the **Unit** as shown below:



After name of the L2 storage unit is scanned /selected, following screen appears to scan or select storage sub unit:



7 The scanner summarizes what has been checked in.

Press the **Check Status**  on the scanner screen.



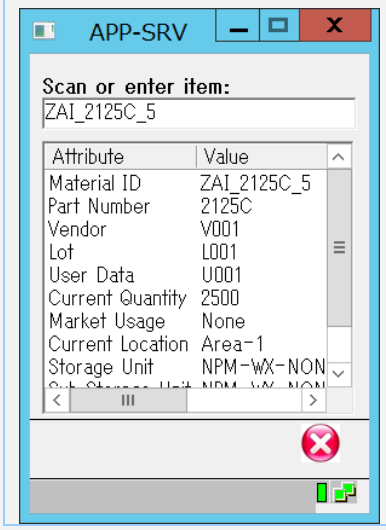
The **Check Status** icon on the scanner screen is available only when you checked in under the following conditions.

1. Normal component are checked in.
2. A splicing reel is checked n
   *The splicing unit is only displayed.
3. Invalid component/restricted component is checked in.
4. Substitute component is checked in.
5. Humidity control component is checked in.

8

The status of the target material ID is displayed.

Scan/enter another material ID to perform the operation, or select  to return to the information display screen for the material ID you checked in.



Splice Reel Management in the context of Material Checking

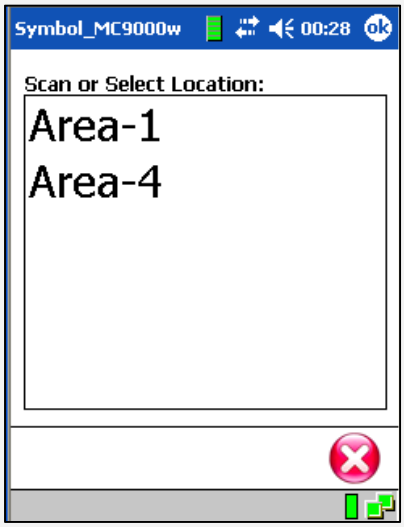
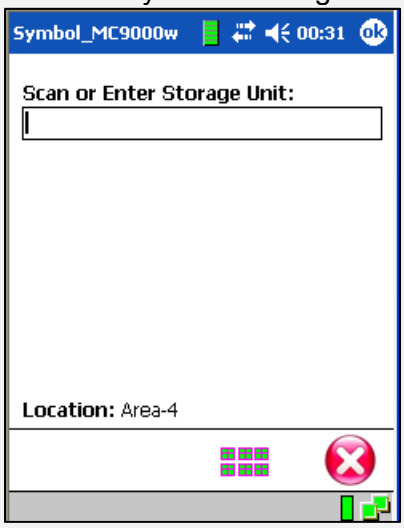
The Material Handling operation of Check In has been enhanced to move both the Main Reel and the Splice Reel simultaneously if either Material ID is scanned. The mobile scanner maintains the relationship of the Main Reel (parent) to the Splice Reel (child) both on and off a machine. If an error is present you will be notified via scanner display.

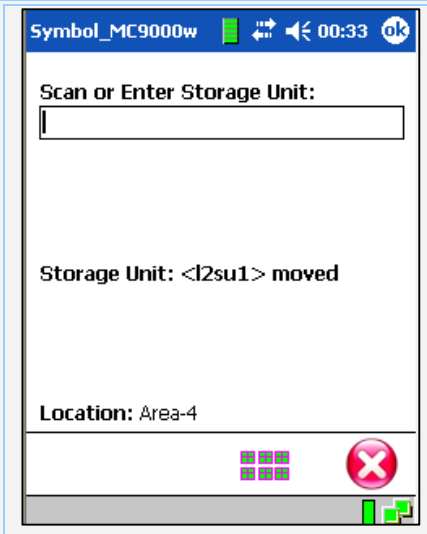
The following table illustrates common errors that may occur when performing a Check In.

Screen	Description
	If the scanner detects an invalid component at the main reel (or 'parent' reel), an error message appears on the scanner. To continue, press the **green** checkmark. To quit, press the red **X**.
	If the scanner detects an invalid component at the spliced reel (or 'child' reel), an error message appears on the scanner. To continue, press the **green** checkmark. To quit, press the red **X**.
	If the scanner detects an invalid component at the main reel and the spliced reel, an error message appears on the scanner. To continue, press the **green** checkmark. To quit, press the red **X**.

2.3.6 Material Handling - Move Storage Unit

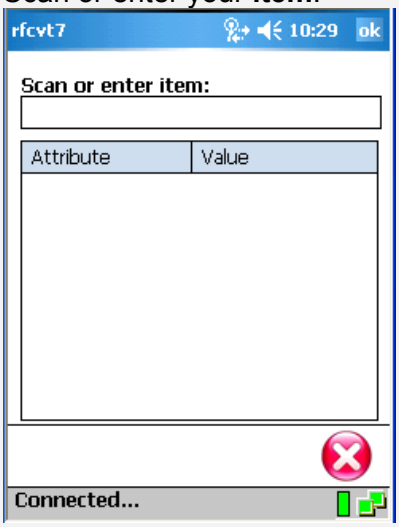
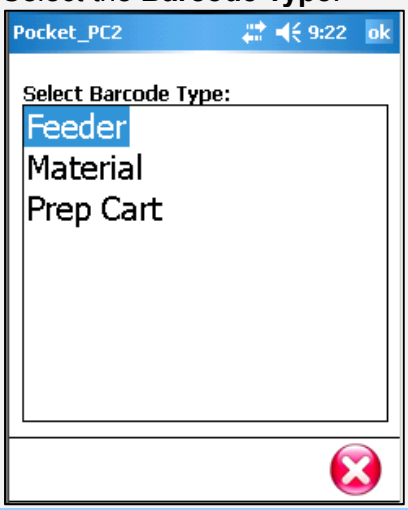
Refer to the action table below for instructions on how to move a Storage Unit to a Material Area using the Mobile Scanner.

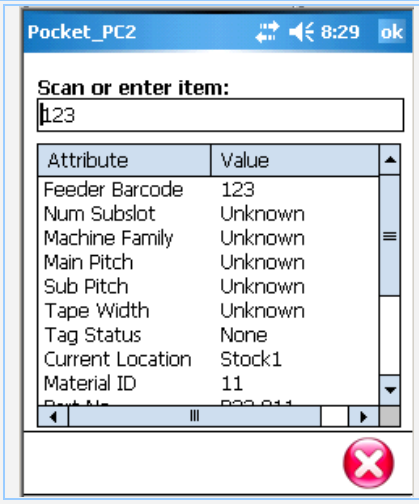
Step	Procedure
1	Select the Material Handling  icon to begin.
2	Select  .
3	Scan or select the desired area. 
4	Scan or key in the Storage Unit. 
5	The scanner will display a summary of the Storage Unit moved and will prompt you to move more Storage Units.


Symbol_MC9000w
00:33
ok
Scan or Enter Storage Unit:
Storage Unit: <l2su1> moved
Location: Area-4

2.3.7 Material Handling - Material Status Check

Refer to the action table below for instructions on how to display the status of an item using the Mobile Scanner.

Step	Procedure
1	Select the Material Handling  icon to begin.
2	Select  and then  .
3	Scan or enter your **item**. 
4	Select the **Barcode Type**. 
5	The status of the item displays. Scan or enter a new item, or select  .

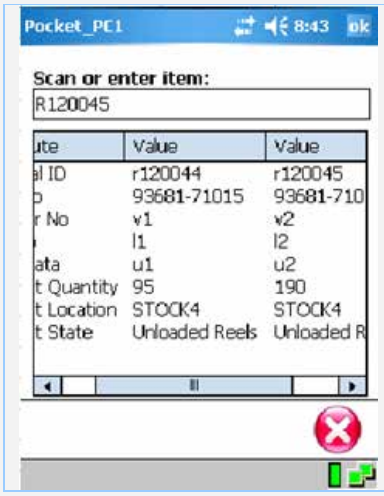


- **Splice Reel Management in the Context of Material Status Check**

The Material Handling operation of Status check has been enhanced to display the status of both the main reel and spliced reel when either material id is scanned of a joint reel.

The mobile scanner maintains the relationship of the Main Reel (parent) to the Splice Reel (child) both on and off a machine. If an error is present you will be notified via scanner display.

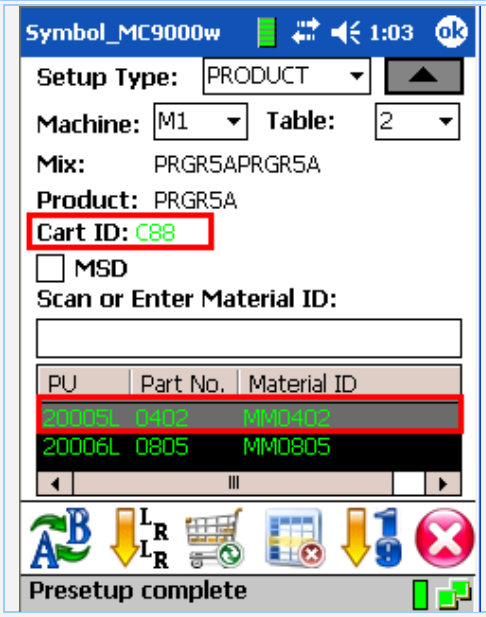
The following table illustrates what the scanner display will look like when scanning a spliced reel.

Screen	Description
	The status appears. Scan another item or click the red **X** to quit.

- **Viewing Presetup Cart ID during Status Check**

When Presetup is completed in Material Control mode, presetup cart ID can be seen at the time of checking the status of reel barcode. Refer to action table below to view presetup Cart ID

Step	Procedure
1	Consider a completed presetup as shown below:



2

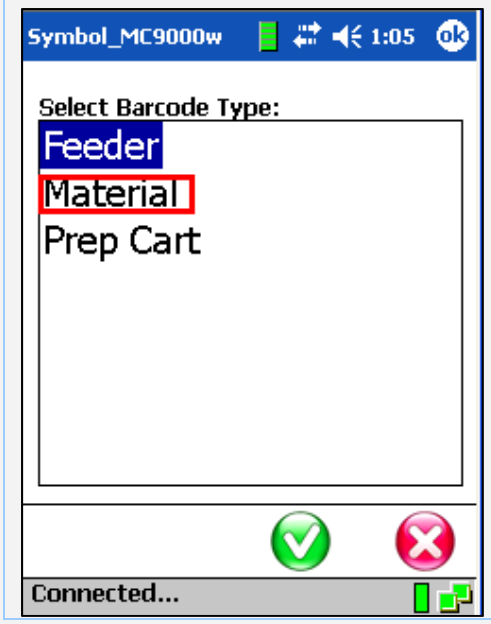
Select 

and then 

.

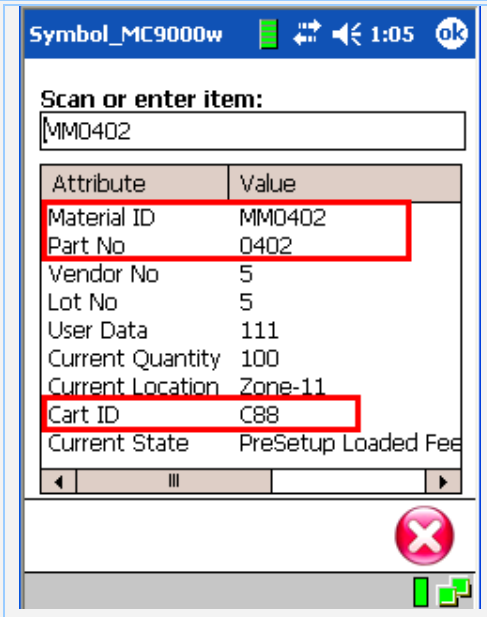
3

Select Barcode Type as **Material** and click 
.



4

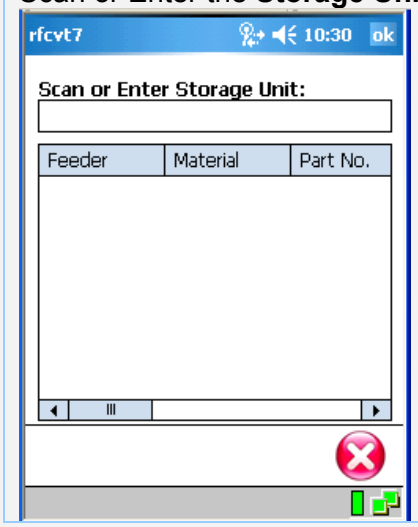
After clicking 
following screen appears to enter reel barcode. Enter reel barcode of the reel used for Presetup. This is shown below:



As shown above, Cart ID is displayed for the reel used for presetup.

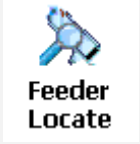
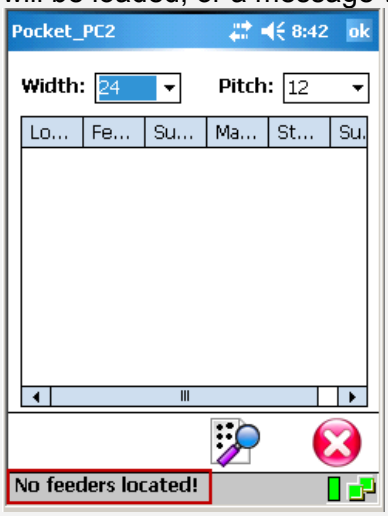
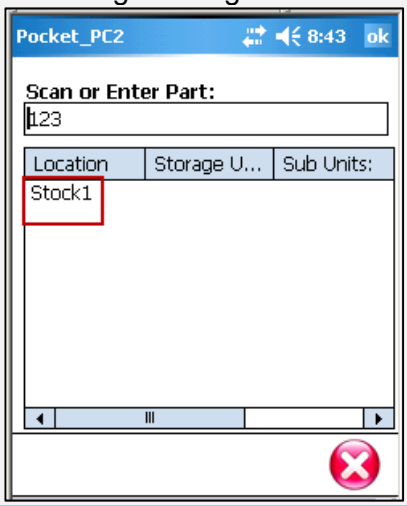
2.3.8 Material Handling - Storage Unit Inventory Check

Refer to the action table below for instructions for how to display the contents of a storage unit using the Mobile Scanner.

Step	Procedure
1	Select the Material Handling  icon to begin.
2	Select  and then  .
3	Scan or Enter the **Storage Unit**. 
4	Contents of the storage unit will be displayed.

2.3.9 Material Handling - Locate Material

Refer to the action table below for instructions for how to display locations of a particular part number using the Mobile Scanner.

Step	Procedure
1	Select the Material Handling  icon to begin.
2	Select  and then  .
3	Select  or  .
4	If you chose **Feeder Locate**: Select the **Width** and **Pitch**, then select the **magnifying glass** icon. Feeders will be loaded, or a message will appear stating that No feeders located.  If you chose **Comp Locate**: Enter or Scan the **Part**. All part locations will be displayed, or you will receive a message stating that there are no part locations. 

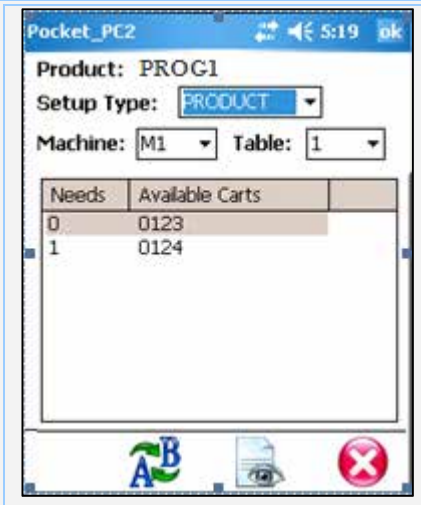
Note: Storage Unit and **Sub Unit** columns are located right after location column.

2.3.10 Material Handling - Perform a Product Check

Refer to the action table below for instructions for how to display the Cart ID number of PreSetups that best represent cart contents of the selected Product.

Note: The cart with the closest match is displayed first. The current feeder setup on the cart can be displayed if desired.

Step	Procedure
1	Select the Material Handling  icon to begin. **Note:** These procedures are the same when in Production  mode.
2	Select  and then  .
3	Select  .
4	Follow the steps below to select a product program. a. Select the **Route** b. Select the Module. c. Select the Product. d. Click the **green** checkmark. 
5	The following menu will appear if the product contains both a top side and a bottom side. Select **Top** or **Bottom**.
6	All Cart IDs for Presets that use the selected Product will appear.



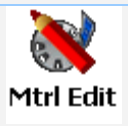
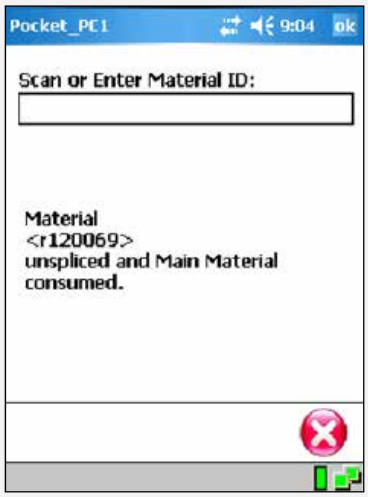
- If Setup Type is set to **PRODUCT or PCB**, the Needs column indicates the number of feeders that require verification in order to complete the selected Product setup.
- If Setup Type is set to **Mix**, the Needs column indicates the number of feeders that require verification in order to complete the setup for all the Products in the Mix containing the selected Product.

7

Perform one of the following:

- Select a **Cart** from the list.
-  Return to Product List menu.
-  List feeders in the setup on the selected cart.
-  Cancel the operation.

2.3.11 Material Handling - Unsplice

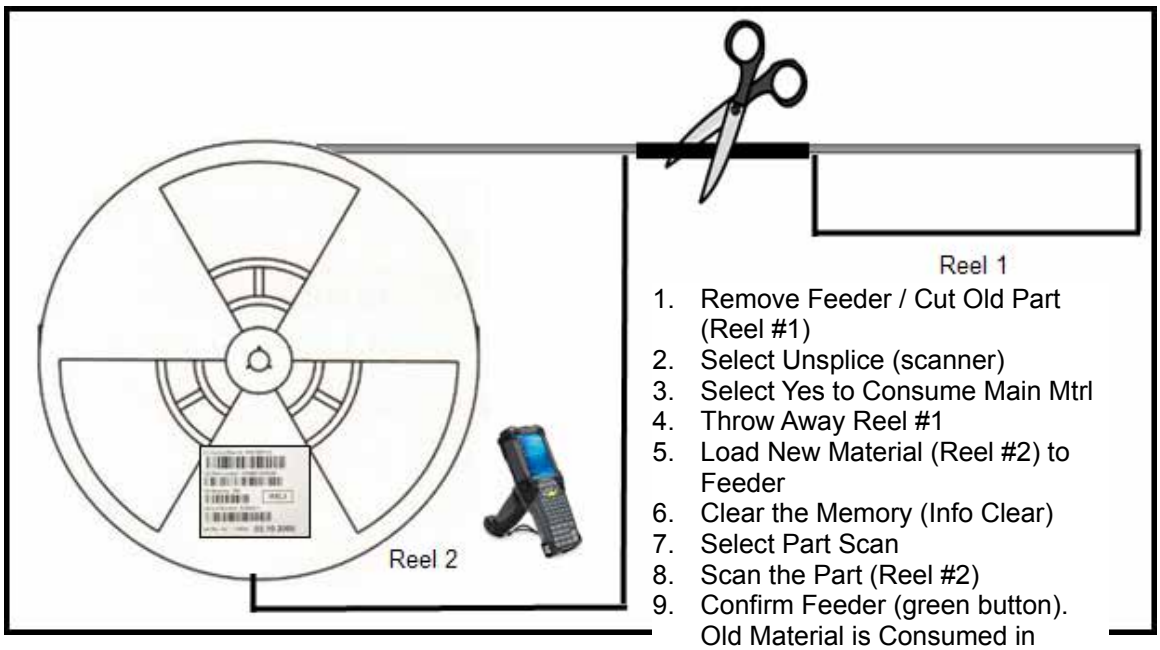
Step	Procedure
1	Select the Material Handling  icon to begin. **Note:** These procedures are the same when in Production  mode.
2	Select  and then 
3	Select the green checkmark  to change the existing reel that splicing has been already done to the consumed state. If you do not want to change the existing reel to the consumed state, select  . To cancel operation, click  . 
4	If you selected yes to consume the Main Material, the scanner display will indicate that the Material was un-spliced and the Main Material was consumed.  If you opted not to consume the Main Material, the scanner will indicate as such:



Note: If you are scanning an MSD part and it has expired, the scanner will display as such. Select the green arrow to un-splice or click the red **X** to cancel the operation.



The following job aid provides an operator quick steps to reference easily when performing unsplice operation on the factory floor.

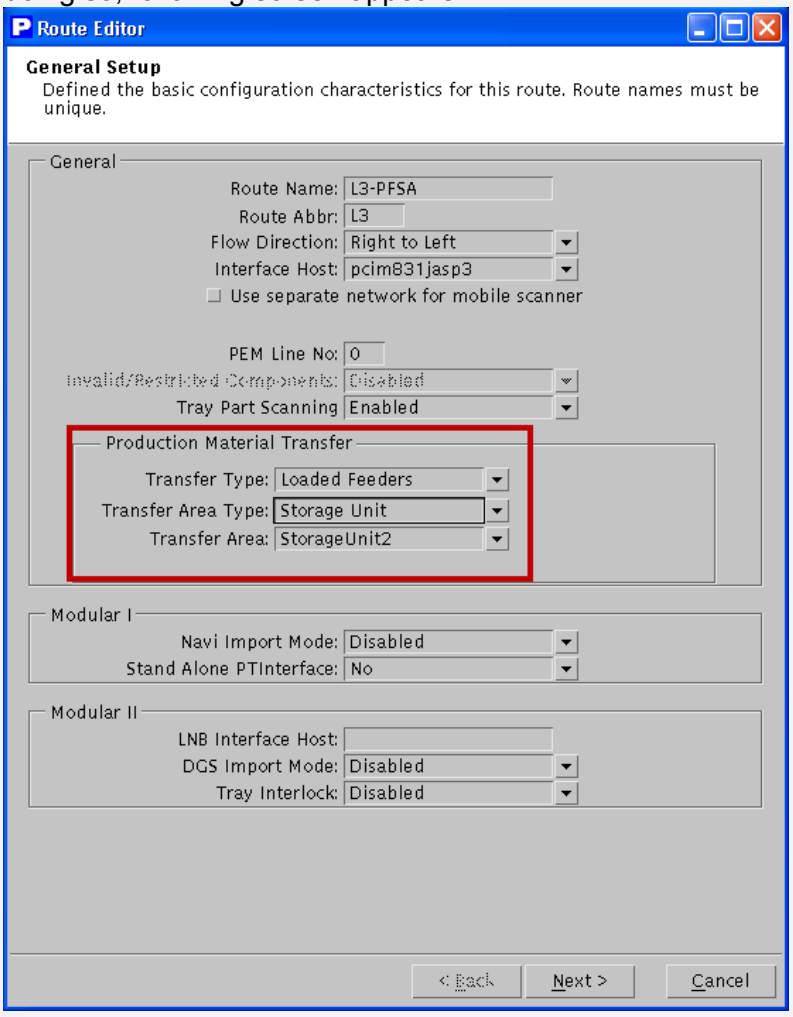


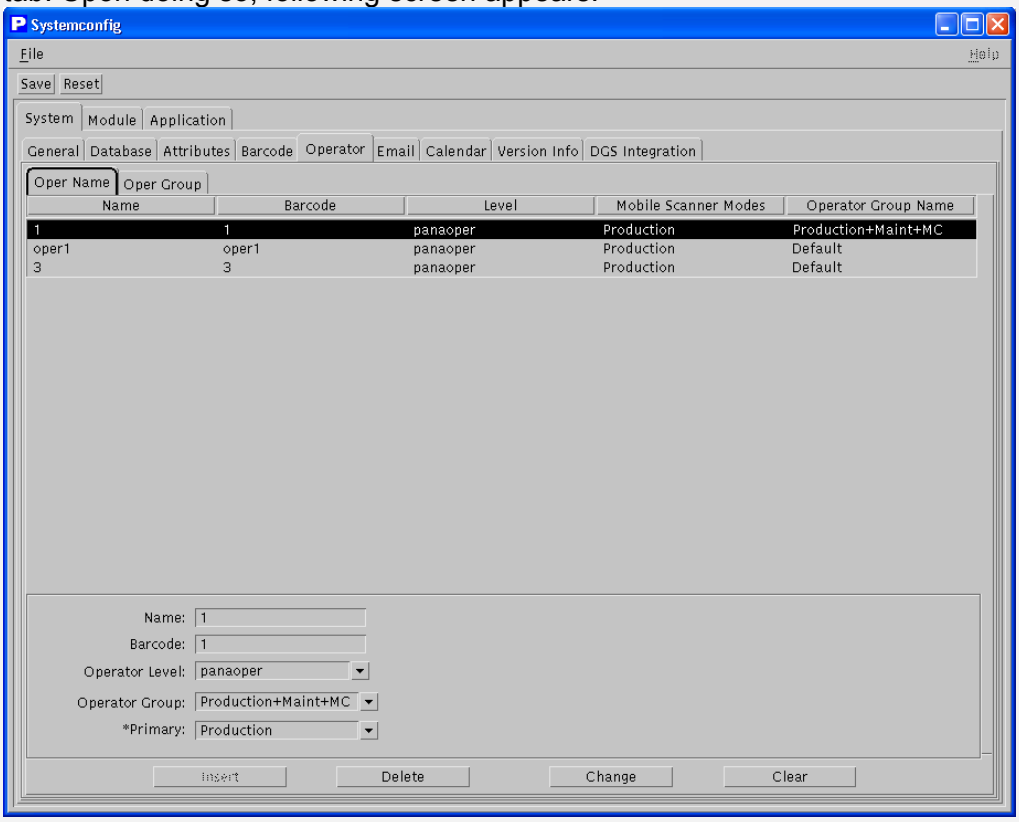
1. Remove Feeder / Cut Old Part (Reel #1)
2. Select Unsplice (scanner)
3. Select Yes to Consume Main Mtrl
4. Throw Away Reel #1
5. Load New Material (Reel #2) to Feeder
6. Clear the Memory (Info Clear)
7. Select Part Scan
8. Scan the Part (Reel #2)
9. Confirm Feeder (green button). Old Material is Consumed in PanaCIM Automatically
10. Main Memory is Programmed w/Reel #2
11. Restart Production
12. Switch [[START]] ON after [[UNLOCK]] ON when restart production

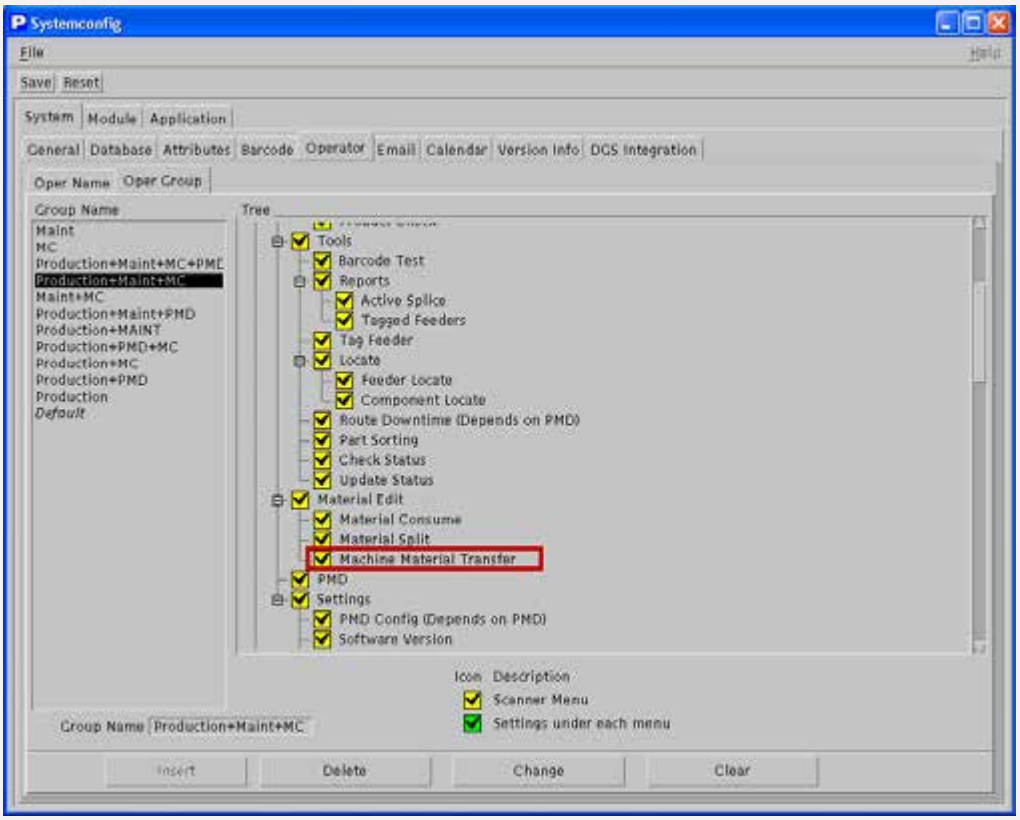
2.3.12 Material Handling - Batch Material Transfer

Previously, operators had to scan each reel barcode separately to move the material from production floor to material area. This process took a long time and hence a batch scan option is introduced to move all the material at once to the designated storage location configured in the route with one scan on the mobile scanner. With this solution, great deal of time is saved in transferring production material from machines to the default material area.

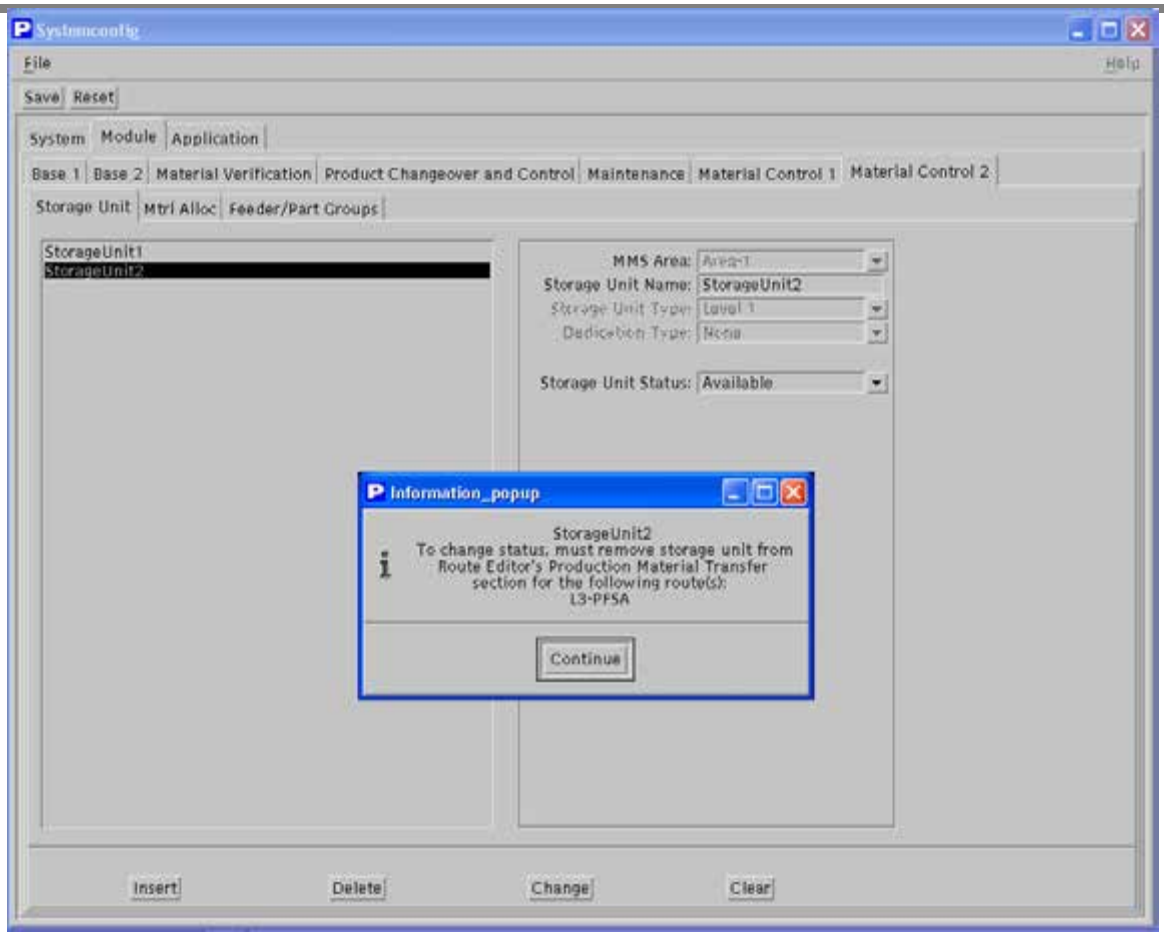
■ Configuration

Step	Procedure
1	Double-click a route for which batch material transfer needs to be set. Upon doing so, following screen appears: 
2	Choose one of the following options from the **Transfer Type** drop-down list. • **Loaded Feeders:** Indicates that feeders loaded with reels have to be checked in to transfer area. • **Unloaded Material:** Indicates that unloaded material have to be checked in to transfer area. Set **Transfer Area Type** as **Storage Area** or **Storage Unit** by selecting respective options from the drop-down list. ・**Storage Area:**

	Select this option to specify the storage area as the storage location. ・ **Storage Unit:** Select this option to specify the storage unit in the storage area as the storage location. Select name of material area or storage unit from the **Transfer Area** drop-down list depending upon the option selected in Transfer Area Type drop-down list. ・ When **Storage Area** is selected, set to storage area name. ・ When **Storage Unit** is selected, set to storage unit name.
3	Keep on clicking the **Next** button until **Save** button appears at the bottom of Route Editor window. Click the **Save** button to save the configuration.
4	Launch **System Config** and then select **System->Operator->Oper Name** tab. Upon doing so, following screen appears: 

5	Select an operator from **Oper Name** tab that needs to be granted Machine Material transfer privileges, click **Oper Group** tab, select Group name to which selected operator belongs and finally check the **Machine Material Transfer** checkbox underneath **Material Edit**. 
6	Click Save button on the top left hand side of **Systemconfig** window to save the changes.

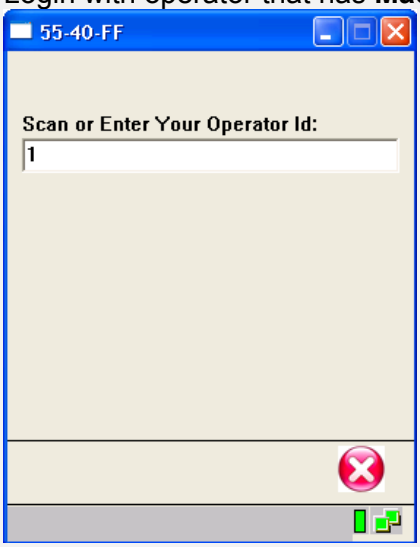
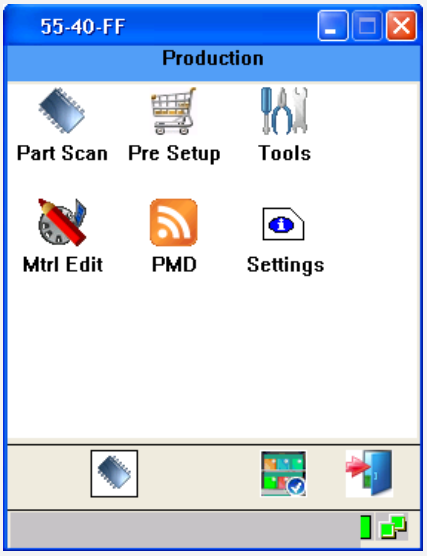
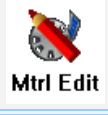
Note: Any attempt to change storage unit status for which production material transfer is configured at route level results in following message being displayed:

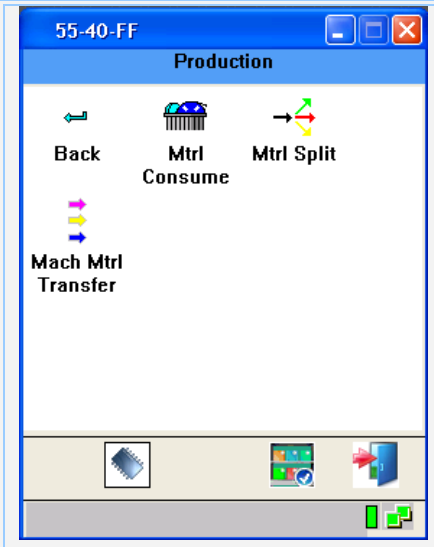


Any attempt to delete the storage unit which is configured for production material transfer at route level results in following message being displayed:



■ Batch Scan

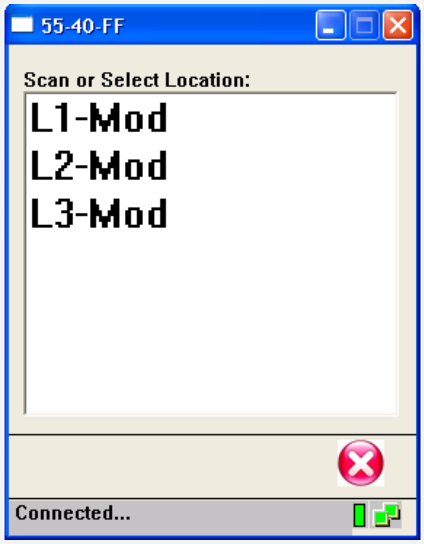
Step	Procedure
1	Login with operator that has **Machine Material Transfer** privileges. 
2	Click  in Production main menu screen. 
3	After clicking  , following screen appears:



4

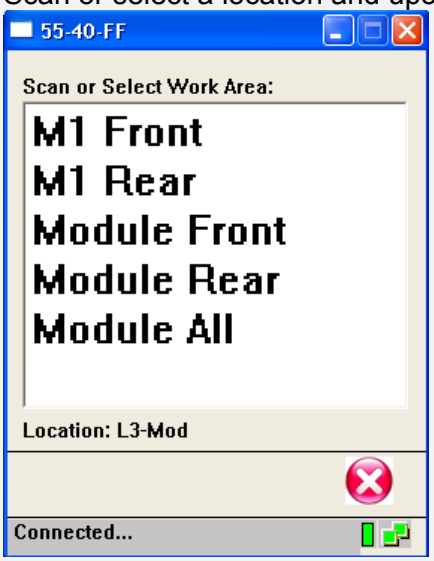
Click 

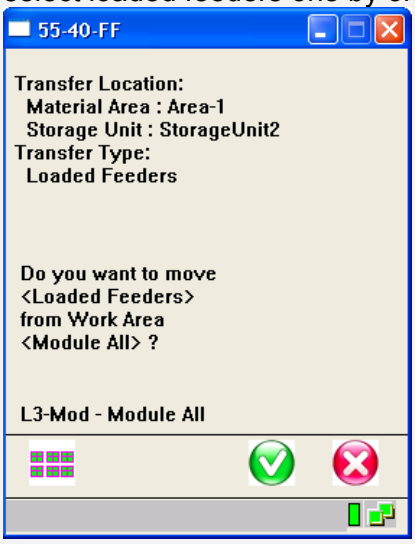
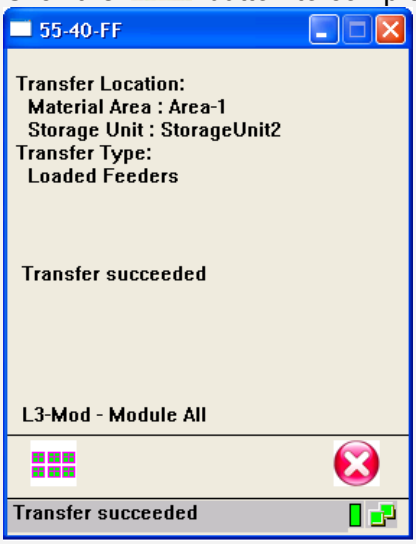
. Upon doing so, following screen appears:

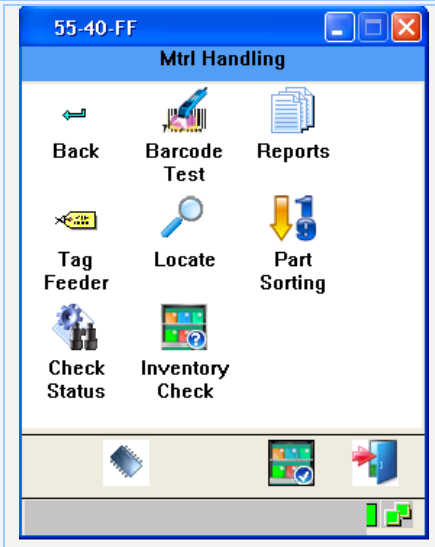


5

Scan or select a location and upon doing so, following screen appears:



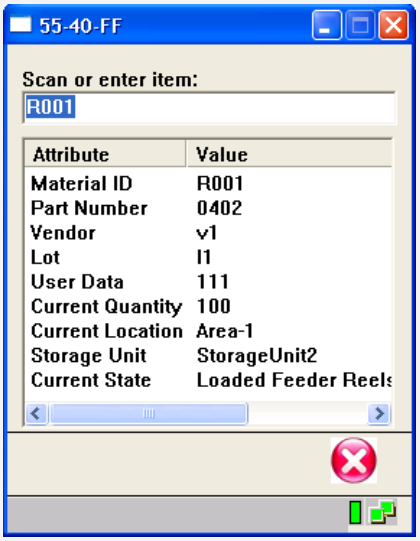
6	Scan or select a work area and upon doing so, following screen appears indicating that all the loaded feeders from selected work area associated with a route will be moved to storage unit. This prevents the operator from having to select loaded feeders one by one. 
7	Click the  button to complete the transfer of loaded materials. 
8	In the Material Handling screen, click  . Upon doing so, following screen appears:

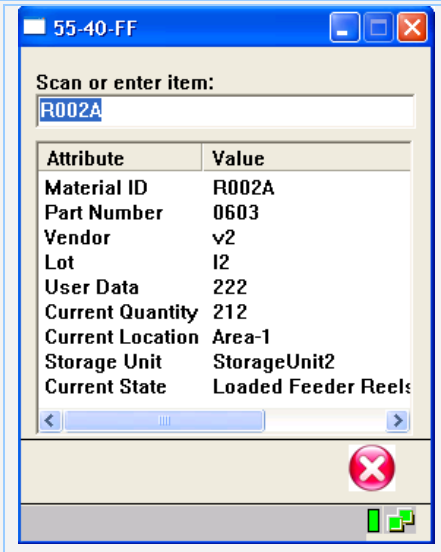


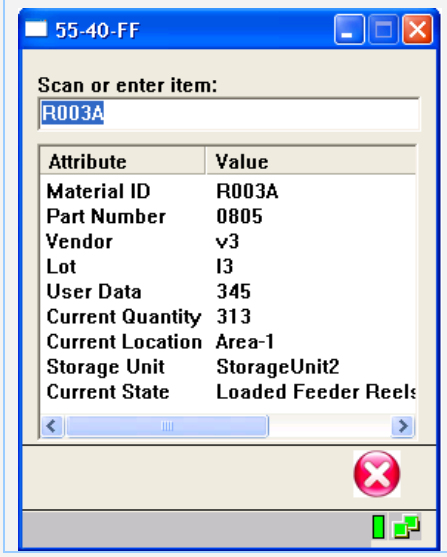
9

Click 

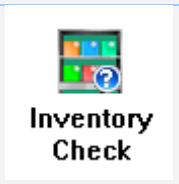
. Upon doing so, screen appears in which you can enter reel barcode to confirm that it has indeed been checked into the designated location configured in the route which can either be material area or storage unit.



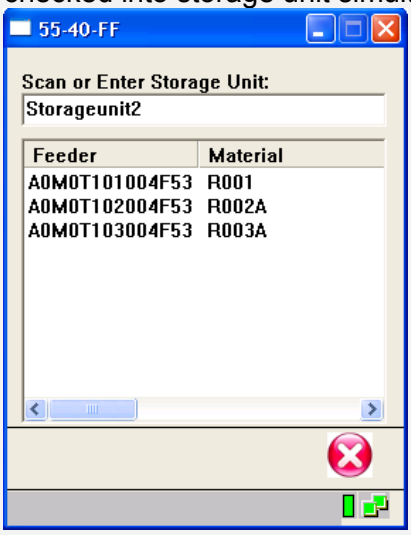


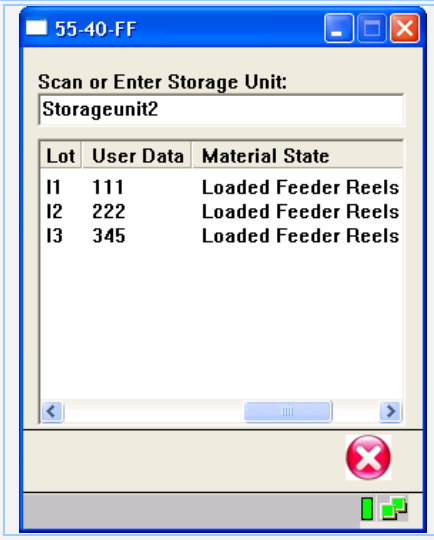


10

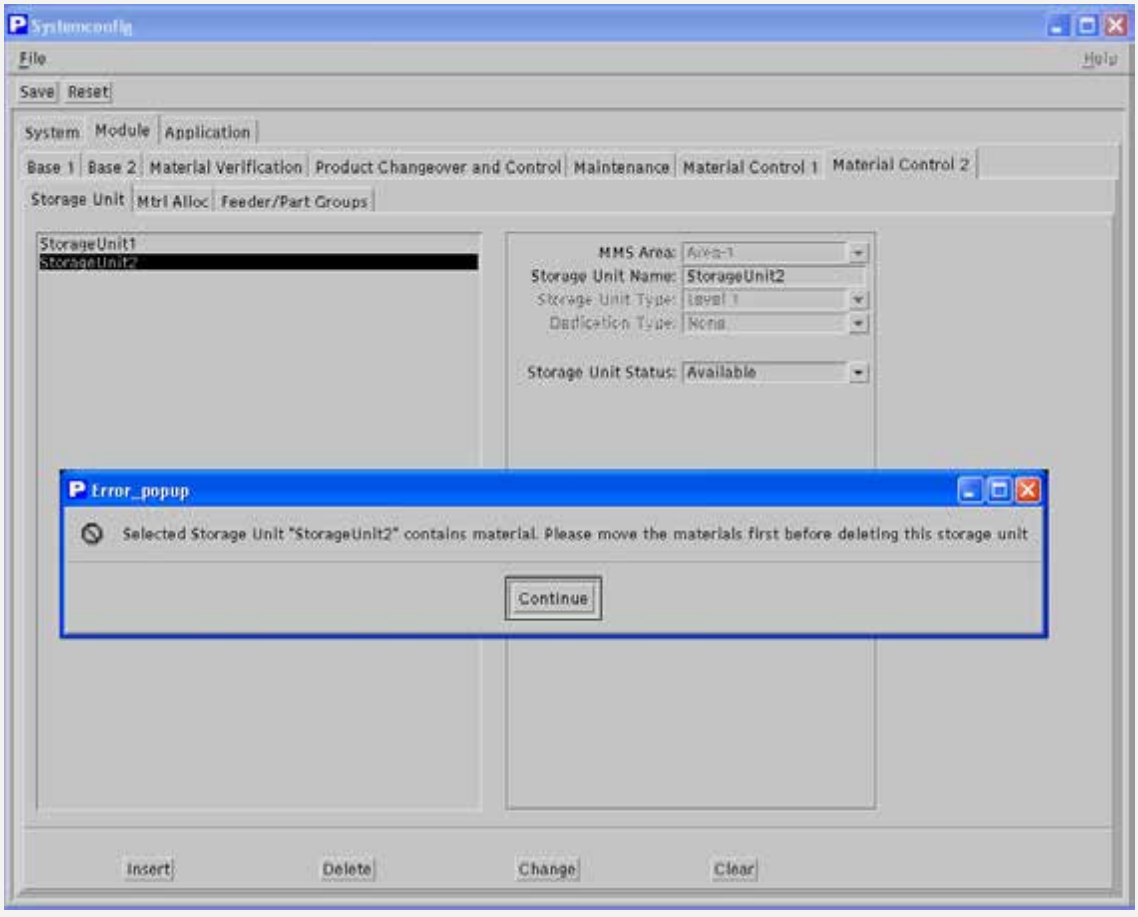


You can click in the Material Handling screen and then scan/enter storage unit to confirm that all the loaded feeders have been checked into storage unit simultaneously.



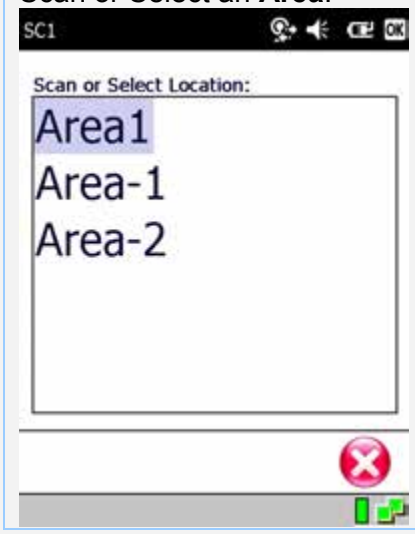


Note: Once the production material is moved to storage unit, it cannot be deleted until the material is moved out of it and any attempt to do so results in following message being displayed



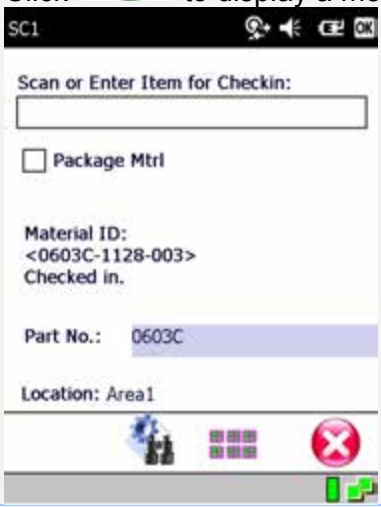
2.3.13 Material Handling –Check in Expiration Date Control Parts

Refer to the action table below for instruction of how to check in expiration date control parts to the material area using the mobile scanner.

Step	Procedure
1	Select 
2	Scan or Select an **Area**. 
3	Scan or enter the material ID. 

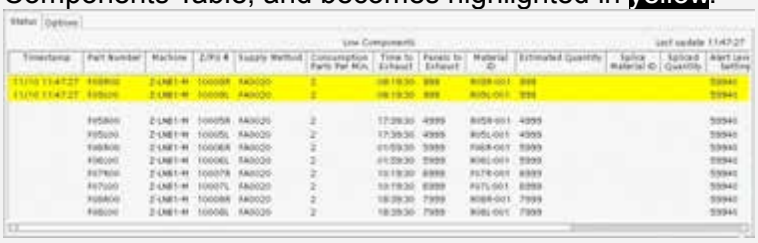
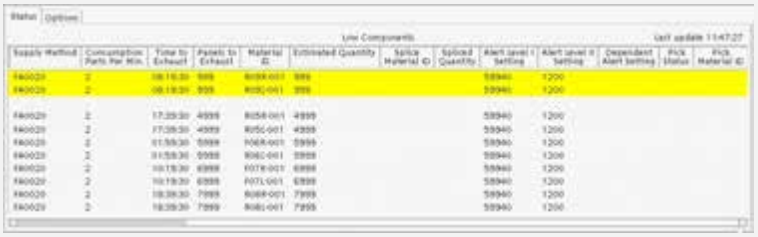
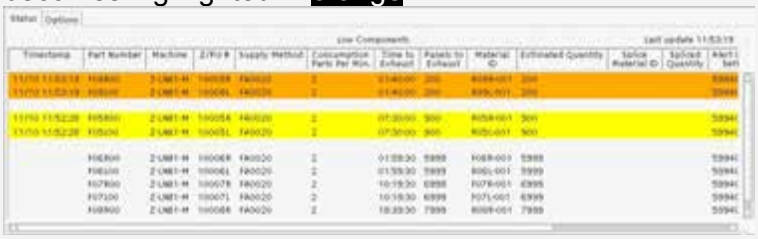
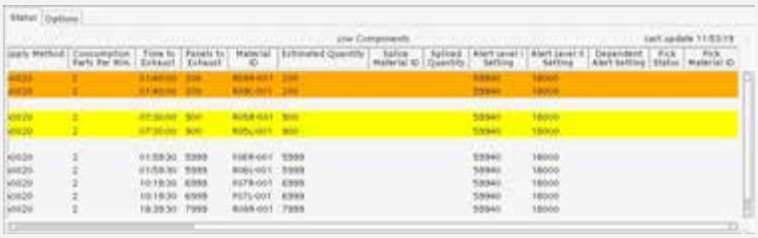
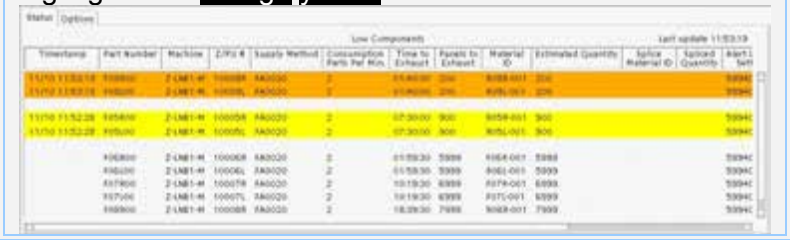
4	When a check-in item is entered, the following screen appears for selecting the barcode type.  Select a material to check in a reel including the expiration date control parts.
5	If it is a new material ID, scan the part number using the mobile scanner or manually enter the number. 

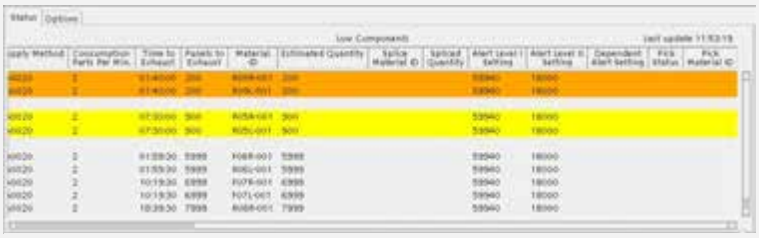
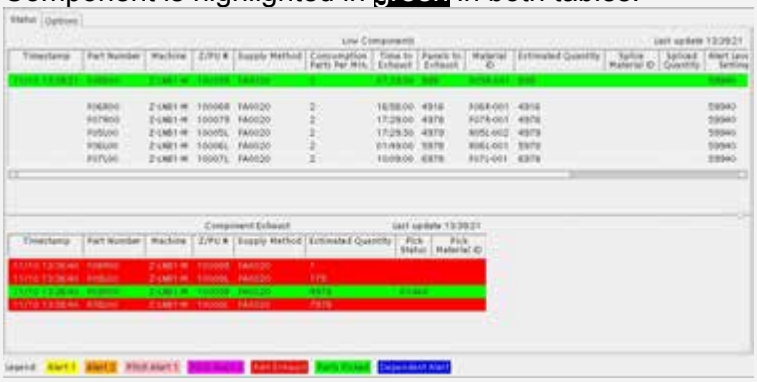
6	Enter detailed part information. 
7	Scan or enter the expiration date.  If you set **Expiration Date** to **Any**,  appears at the bottom left of the screen. Clicking this button allows you to **skip entry of the expiration date**. Note) If the entered date is the same day of **Expiration Date** or **Show Warning Before,** a message appears according to the setting.

8	Select **Storage Unit** to check in.  If the selected storage unit has a sub unit, also select a sub unit.  Click  to display a message screen for material ID check in. 

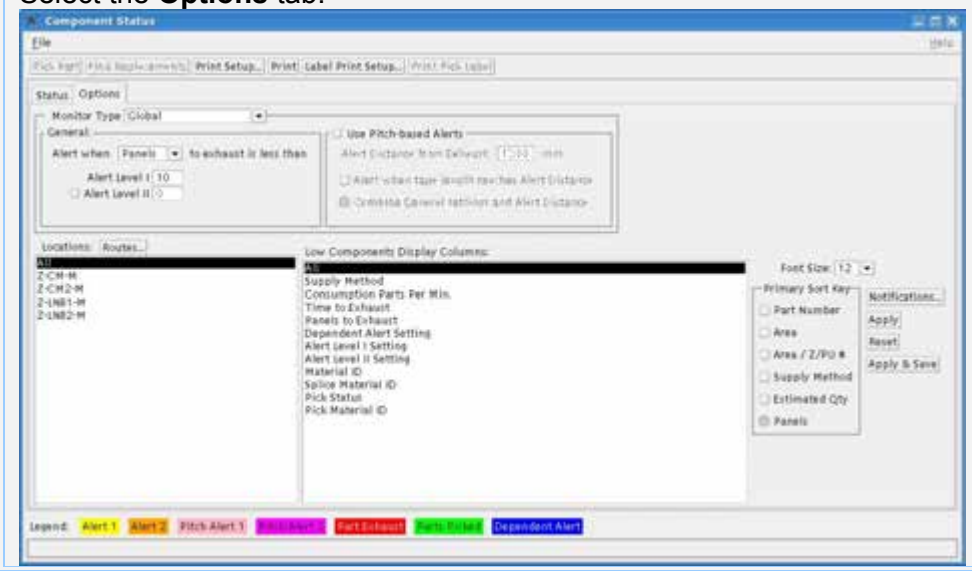
2.3.14 Production - Component Status Tracking

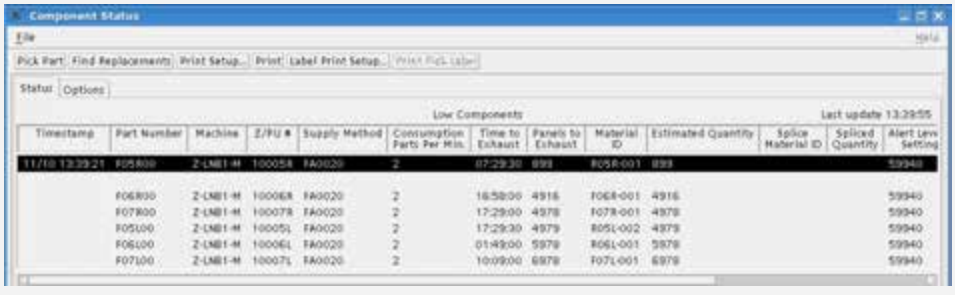
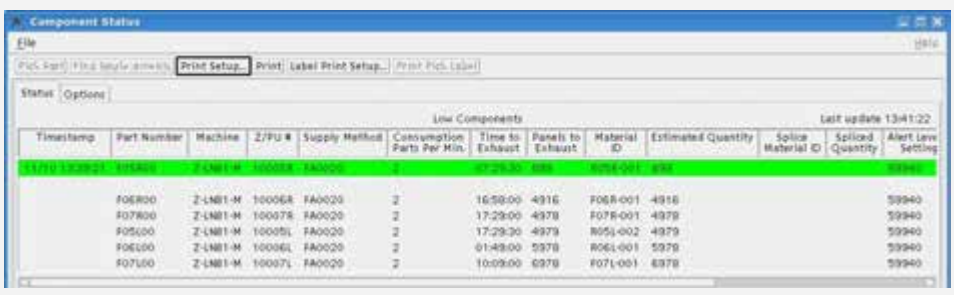
Use the Component Status Application to track component inventories and component related production statistics within your factory. The output of the Component Status Application is displayed in the Low Components Table and the Component Exhaust Table. Use this display to flag a component as ‘picked’ to show intent of action for the component. It typically shows intent to resupply parts for a component that is close to, or is in exhaust.

Component Action	Application Response
Drops below Alert Level II value	Component listing is moved to the top of the Low Components Table, and becomes highlighted in yellow.  
Drops below the Alert Level I value	Component listing is moved to the top of the Low Components Table *after* the Alert Level II entries, and becomes highlighted in orange.  
Does NOT drop below Alert Level I and Alert Level II values	Sorts by threshold value and sort order in the Low Components Table *beneath* those components highlighted in orange/yellow. 

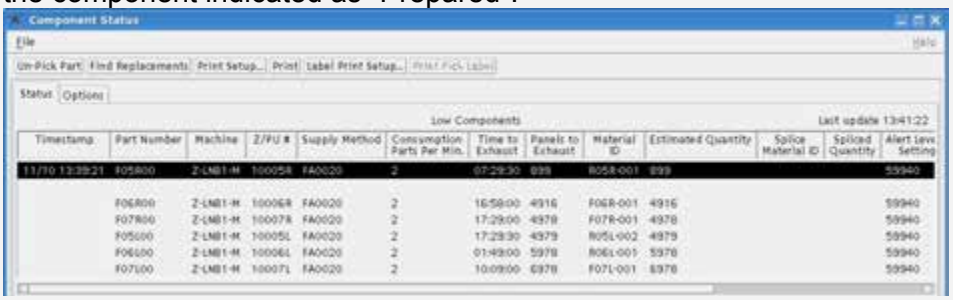
	
Exhausts completely	Component listing moves to the Component Exhaust Table, and is highlighted in red. 
Flagged as picked	Component is highlighted in green in both tables. 

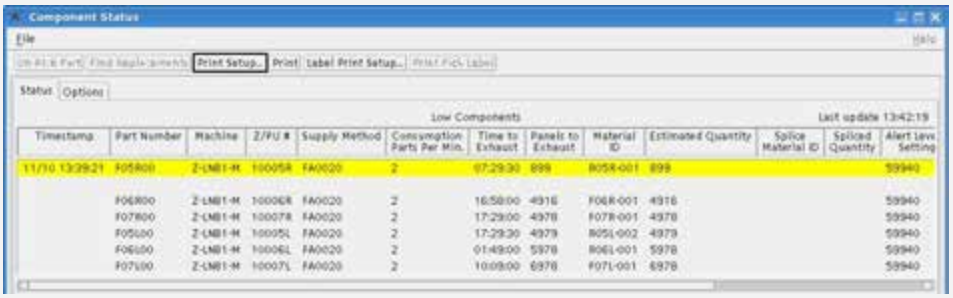
Refer to the action table below to configure the Component Status Application.

Step	Procedure
1	Open the **Component Status** Application from the PanaCIM Main Menu.
2	Select the **Options** tab. 

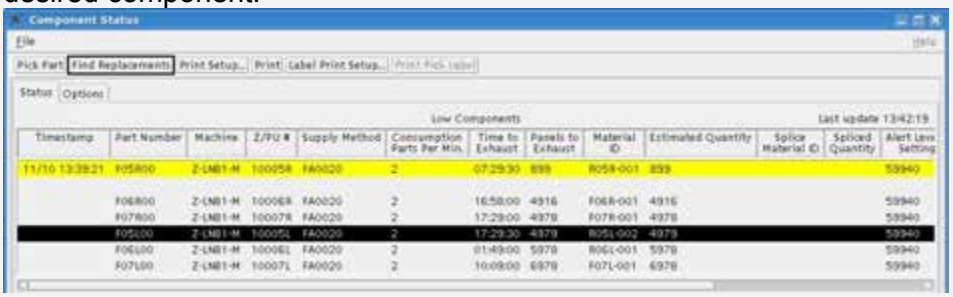
3	Select **Panels** (recommended option) or **Time** from the Alert when drop-down list.
4	Enter a value in the **Alert Level I** field. A warning will generate for each component estimated to exhaust within this set value. Alert Level I items will be displayed on the status page in yellow.
5	Optionally, enter a value into the **Alert Level II** field. A critical warning will be generated for each component estimated to exhaust within this set value. This value must be less than the value determined for Alert Level I. Alert Level II items will be displayed on the status page in orange.
6	Select an **area** (or areas) to monitor from the Locations list.
7	Select a **column** to display in the Low Components list. Select All to display all of the columns. Material ID and Splice Material ID are also added to the Low Component Display Columns.
8	Select a **Font Size** from the drop-down list for the text used in the Low Components Table and Component Exhaust Table.
9	Components are sorted by the configured limit type. Define a secondary sort order in the **Sort By** column. To order by limit type only (no alternate column defined), select None.
10	Select: • **Notifications** to configure notification settings. • **Apply** to activate the settings. • **Apply & Save** to active and save the settings. • **Reset** to reset the settings to their original state. Applicable notifications include: component_notify, dump_notification_env, page_component_notify andview_file. The dialog that appears is also available for configuration in System Configuration. If you choose to configure notification settings, see the Appendix for more detail.
11	Select the **Status** tab. **Pick Part:** Allows a component with which a part exhaustion warning occurred to be flagged that parts have been resupplied. Select a component not labeled as Pick Part.  

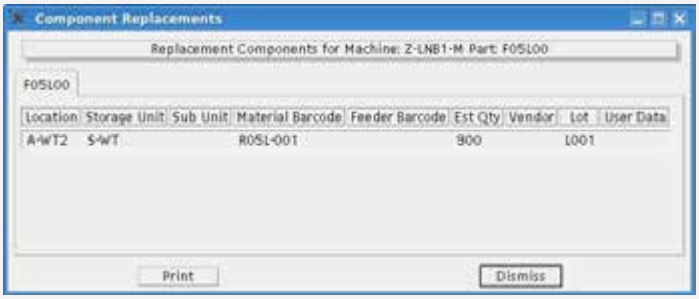
Un-Pick Part: Reverts the picked component back to unpicked state. Select the component indicated as “Prepared”.





12 **Find Replacements:** Used to locate the replacement reel. Select the desired component.





2.3.15 Production - Material Edit (CM/DT/NPM Only)

Material Edit operations include:

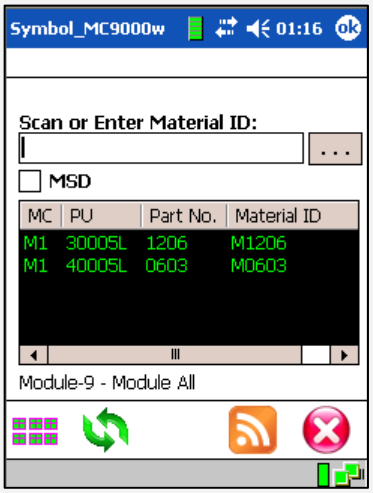
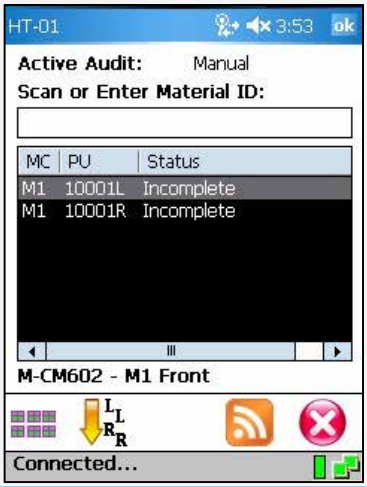
- Material Consume
- Material Split

Material Consume and Material Split operations can be performed in either Material Handling  mode or Production  mode using the Mobile Scanner. The actions for performing these functions are identical for both modes. Therefore, please reference the procedures documented in the **Material Handling – Material Consume and Split Material** sections above.

2.3.16 Production - Setup and Verification under Production (CM/DT/NPM Only)

Procedures for performing these operations in Material Control mode are very similar to how they are performed in Material Verification mode. Consequently, please reference the Material Verification document for procedures on the above items while keeping the following variations in mind. When the *Material Control - Base* option is enabled the following variations to the procedures will occur:

Production operations

Procedure	Variation
Setup	Before scanning a part you will be prompted to scan the Material ID. After scanning the Material ID, you will be prompted to input Part Details if the Material ID is *unknown*.  *Part Scan - Material ID (mobile scanner)*
Routine check	When performing Routine check operation via the mobile scanner, you will be prompted to scan the material ID instead of Part No. After scanning the Material ID, you will be prompted to input Part Details if the Material ID is *unknown* 

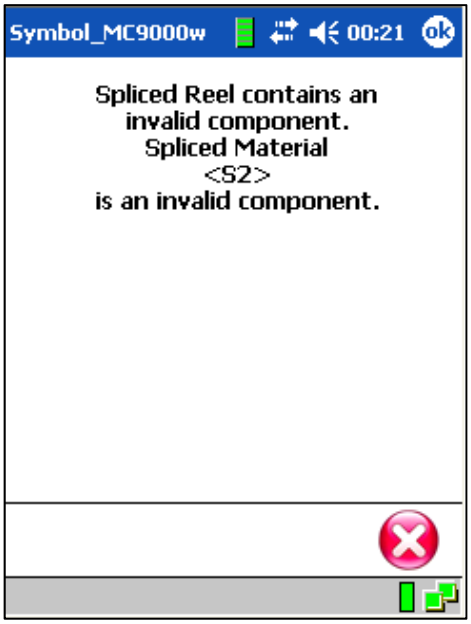
■ Splice Reel Management in the Context of Part Scan

When a reel with a pending splice is later remounted to a feeder using the Part Scan operation, the system recognizes that the reel contains a pending splice point and uses the data from the Main Reel to program the feeder Main memory for the location at which part scan operation is performed.

However, if part scan operation is performed using joint reel with pending splice containing an invalid substitute for original part in both the reels, following screen is displayed:



If Invalid Substitute part is present in the reel joined to the main reel, following message is displayed:



If Invalid substitute part is present in main reel only, following message is displayed:



As evident from above screens, operator has to click on the **X** button and correct the error by using a join reel containing a valid substitute for original part during part scan operation.

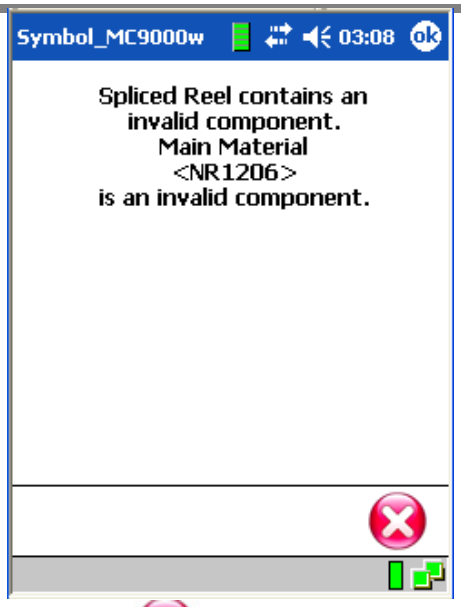
■ Splice Reel Management in the Context of Part Replenish Operation

If a Material ID that is already spliced to a mounted reel is scanned, a message will display stating that this action is prohibited. It is because multiple splice joints are not supported.

For example, say a reel with Material ID S011 is already spliced to an existing reel with Material ID M013 in order to replenish the part 0402. Splice Joint is present at Feeder location 20005L. If the Material ID S011 is now scanned to replenish the part 002 at another location say 20007L, following message will be displayed.



During part replenish operations, when a spliced reel is scanned that contains an invalid substitute, the following message will appear.

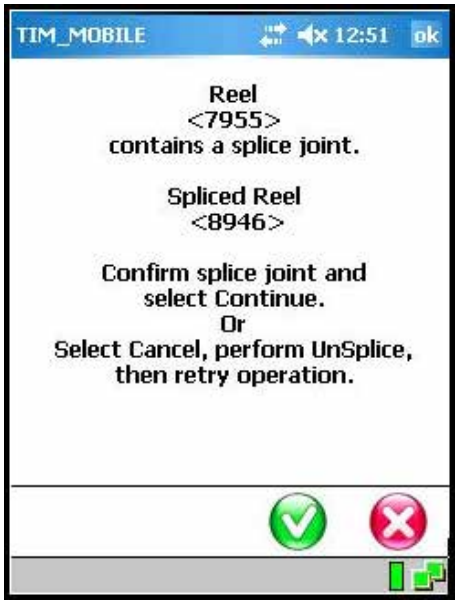


Click the button and correct the error.

Note: In the scenario considered above, the mounted reel contains the original part but the reel being spliced (reel barcode NR1206) to this mounted reel contains invalid substitute of original part.

■ Splice Reel Management in the Context of Presetup

When performing a PreSetup using the Mobile Scanner, you will receive the following error message if the reel contains a splice joint.



2.3.17 Production - PreSetup(CM/DT/NPM Only)

Following operations can be performed when 

icon is selected from Production main menu of the scanner:

- Cart Edit
- Product Check
- Part Scan
- PreSetup
- Part Sort

* Splice verification is not available with the Pre Setup regardless of the machine type (NPM or CM).

For details of performing above mentioned operations, please refer to **5. PRESETUP** in Material Verification user manual.

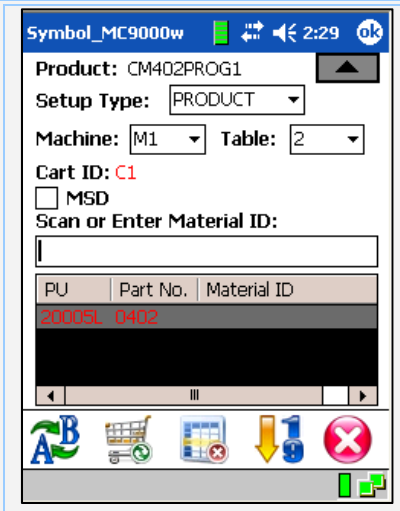
■ Cart Edit and Product Check

Cart Edit and Product Check operations can be performed in either Material Handling  mode or Production  mode of the Mobile Scanner. The actions for performing these functions are identical for both modes. Therefore, please reference the procedures documented in the corresponding sections above.

■ Presetup Operation

Procedures for performing Presetup operation in Material Control mode is very similar to how they are performed in Material Verification mode. Consequently, please reference the Material Verification document for procedures on the above items while keeping the following variations in mind. When the ***Material Control – Base*** option is enabled the following variations to the procedure for performing presetup will occur.

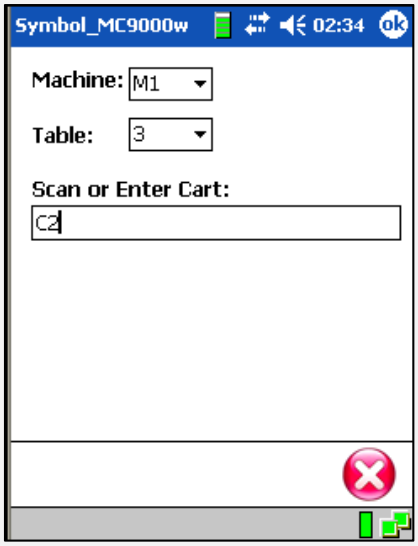
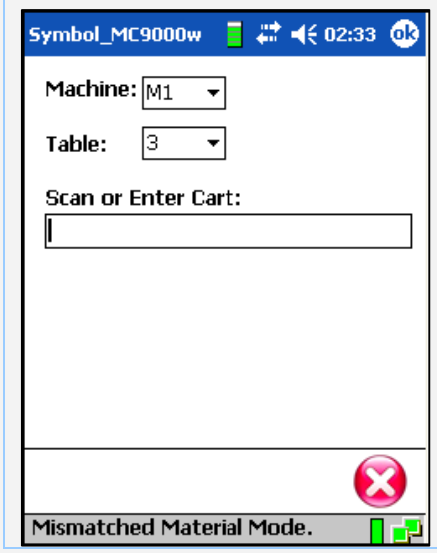
Procedure	Variation
PreSetup	In mobile scanner, after scanning or entering cart ID, you will be prompted to scan or enter material ID as shown below

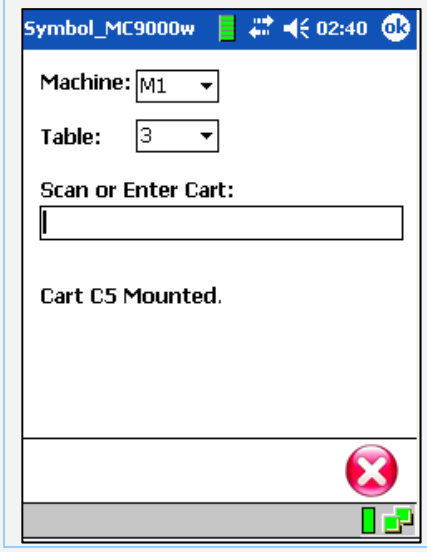


Mobile Scanner

2.3.18 Production - Modular Presetup (CM/DT/NPM Only)

New enhancement has been added to facilitate preparation of cart on ready to use feeder in material control mode. The feeder should be prepared in Material control mode before adding to the machine. If the feeder is prepared in Material verification mode and then added to the machine you will see a mismatch error.

Step	Procedure
1	Setup Cart Operation on the scanner to create a cart ID C2 in Material Verification Mode. 
2	Add feeder in Material Control mode to another module of a different route.
3	You will see a Mismatched Material Mode message as below. 
4	Setup Cart Operation on the scanner to create a cart ID C5 in Material Control Mode.
5	Add feeder in Material Control mode to another module of a different route.

6	Machine will get added in Material Control mode to another module of a different route. 
7	The cart gets mounted successfully.

2.3.19 Report

In PanaCIM Material Control, you can create various reports. The reports can be created from the PanaCIM Web application that starts up by accessing the PanaCIM application server from browser (Internet Explorer 11 or later).

For details of the PanaCIM Web application, refer to PanaCIM-EE [Basic Operations (Web application)]



■ List of reports

The table below is the list of available reports.

● Material Control - Base

Report	Description
Material Quantity Report	Shows a report for remaining parts.
Printer Material Usage Count Report	Shows a list of the number of times that current material is used for each material barcode for the screen printer and the threshold values (warning count, maximum use count)
Material Adjustment Report	Shows the quantity adjusted against material barcode. Reason for adjustment is shown together with addition or subtraction to the current parts corresponding to material barcode.
Material Usage Report	Shows a list of details of all reel used during production.
Material Position History Report	It is a trace record of a material location. When to bring a material into the location and when to bring out it from the location are recorded.
Feeder Position History Report	It is a trace record of a feeder location. When to bring a feeder into the location and when to bring out it from the location are recorded.
Material Movement Report	Provides information related to movement of a material. If the location of materials changes, quantity of the materials re traced
Material Edit Report	Shows the total remaining materials per part number.
Material Registration Report	Shows details of registered material ID.
Materials Summary Report	Shows the total remaining materials per part number.
Part Position Report	Shows the current position of a material.
Material Management Report	Shows detailed information of each material.

■ Filter criteria

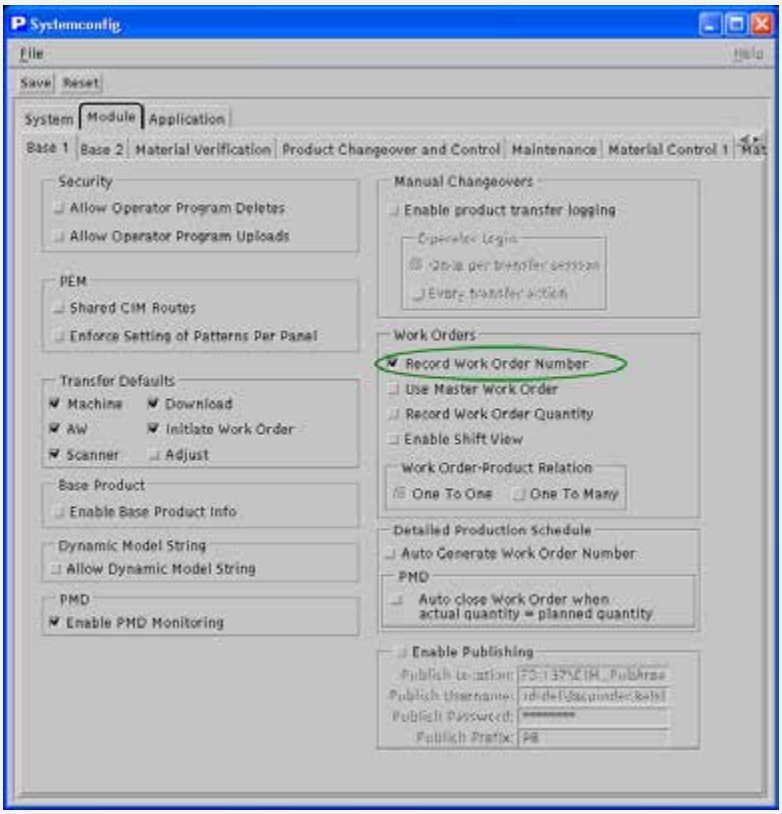
For filter criterial of a report, refer to Basic Operations (Web application 3.2.2 [Filter Criteria].

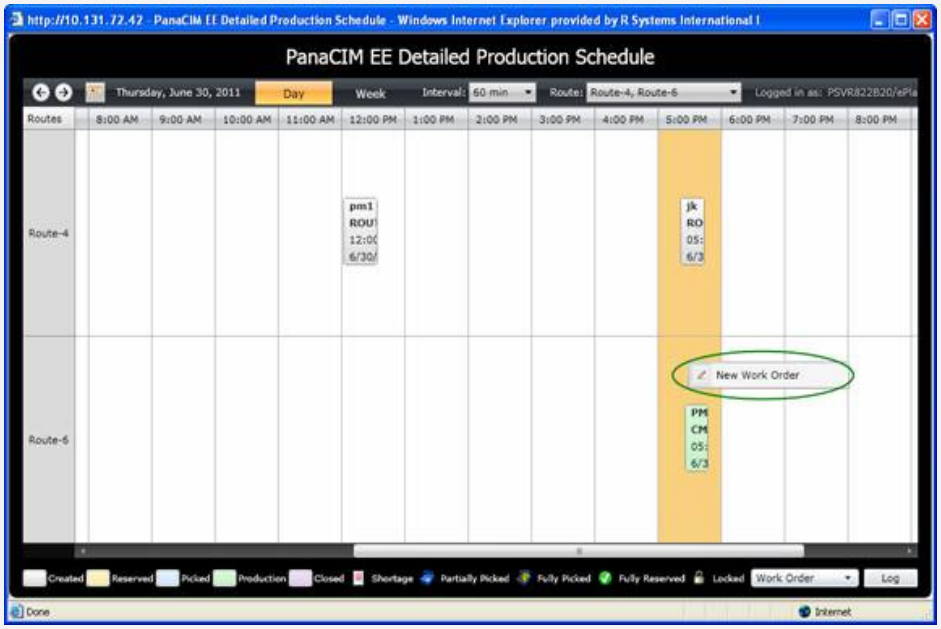
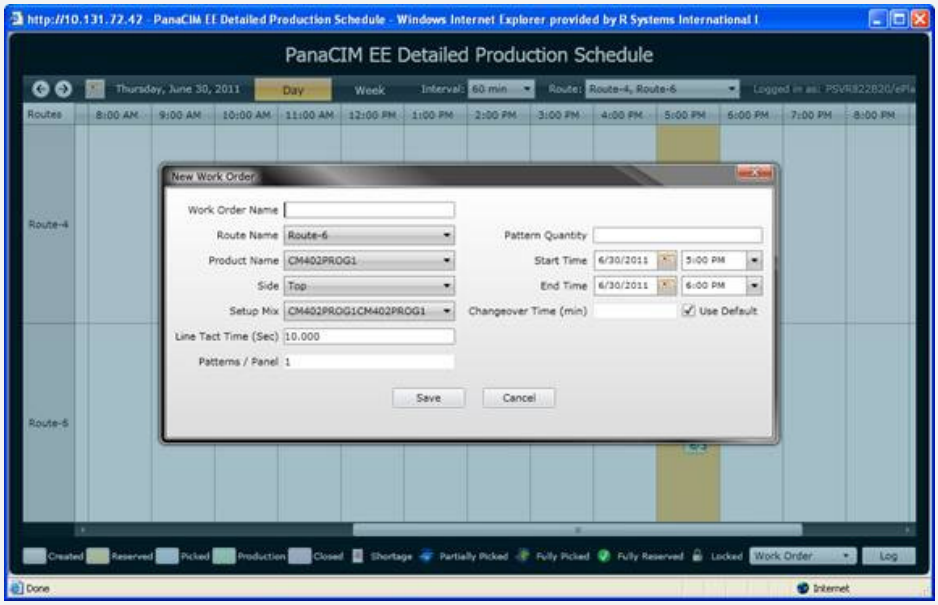
■ Operation procedure

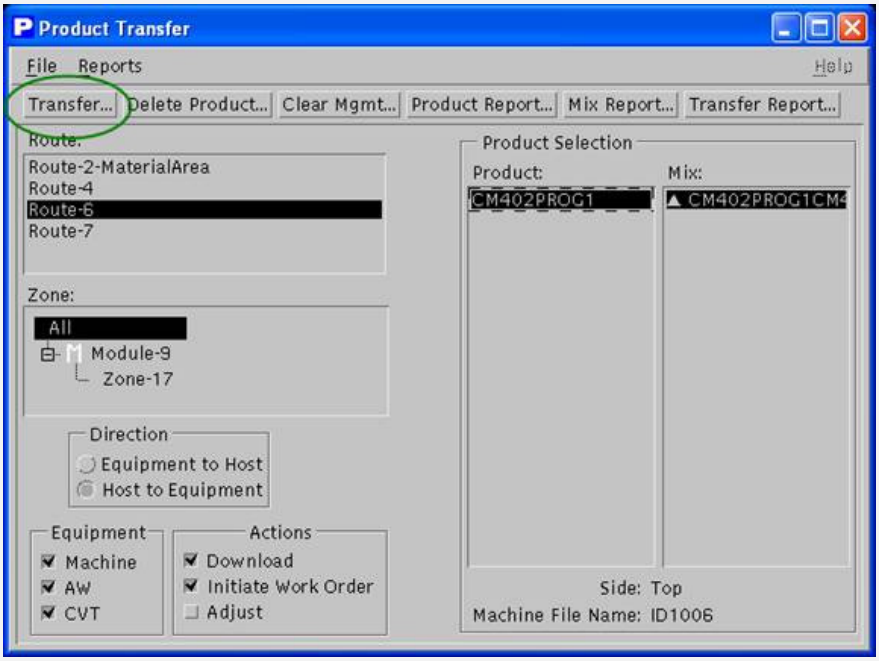
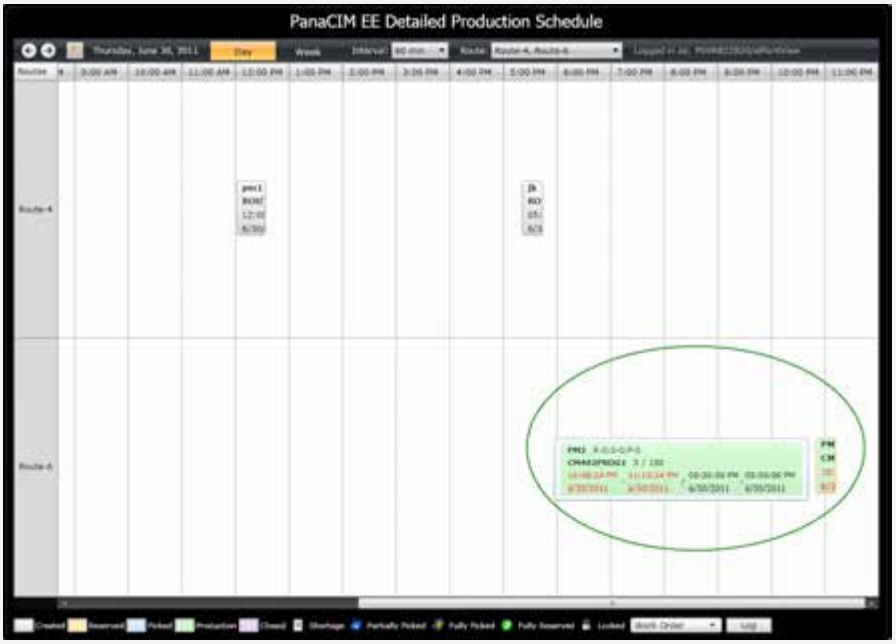
For how to output a report, refer to refer to Basic Operations (Web application) 3.3.1 [Operation Procedure].

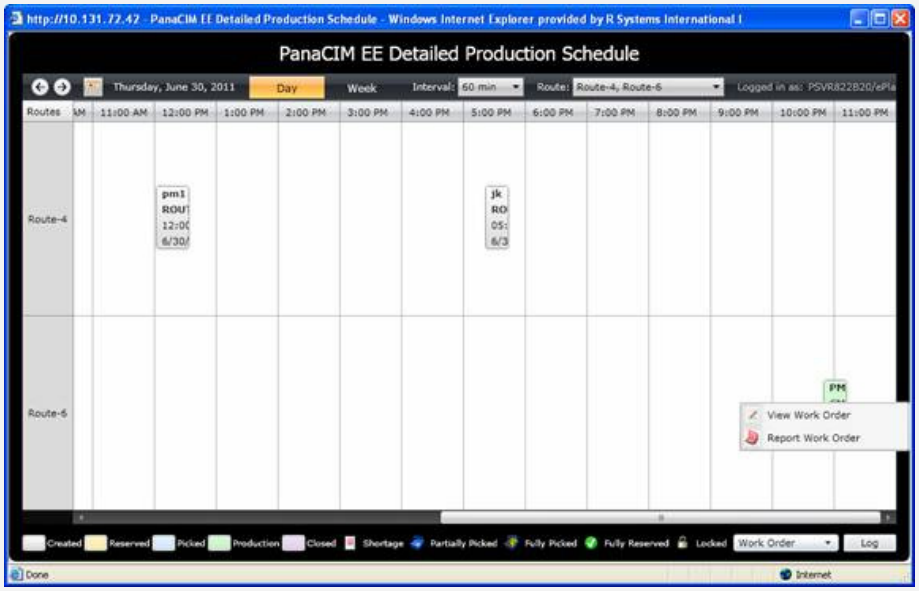
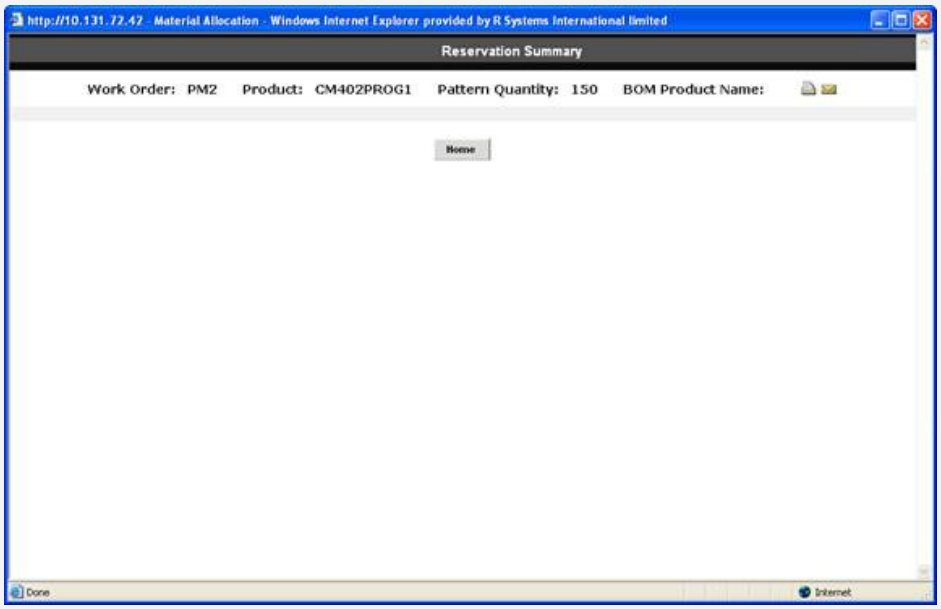
■ Report for Work Order in production from Detailed Production Schedule

New enhancement has been added to facilitate generation of **"Report Work Order"** from Detailed Production Schedule User Interface while the work order is in production.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu. Go to **Module > Base 1**> and check **Record Work Order Number**. 
2	From the PanaCIM Web Application, choose **Charts and Reports** > **Manage Work Orders**. 

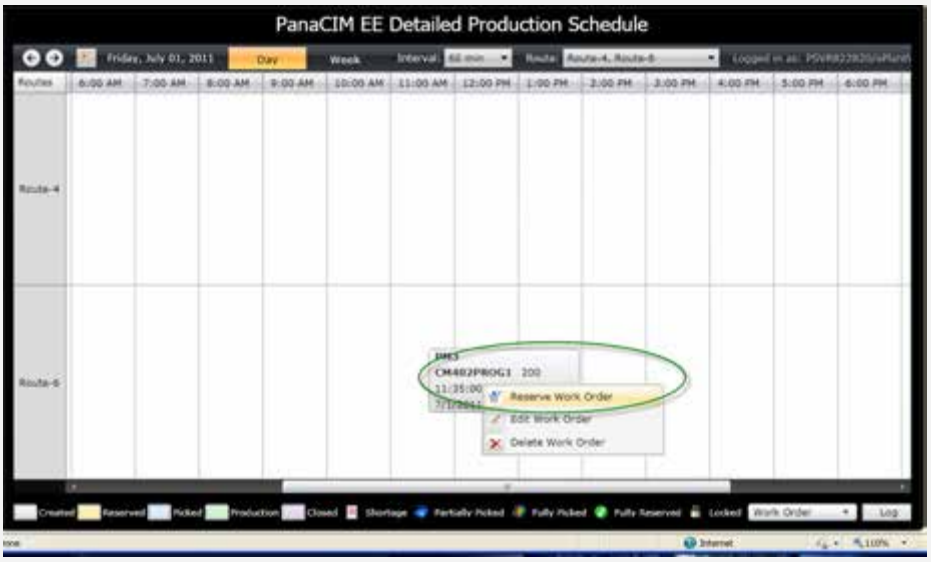
3	Right click on DPS and click on **New Work Order** to create a work order. (For details on how to reserve the components on DPS, refer to 3.3.) 
4	Create a new work order for the required time and schedule it. 
5	On App Launcher, click Product Transfer. 

6	At **Route** select the Route to schedule and at **Product Selection** select the product and click **Transfer**. 
7	At the scheduled work order time you will see the **schedule turns green**. 

8	Right Click on the green schedule and click **Report Work Order**. 
9	You will see the details of the work order as below while the work order is still running. 

■ Search Location with Pre Setup Zone on Material Reservation

New enhancement has been added to search by Pre Setup zone in the list of Searchable Location on Material Reservation. This is done to include the CM/NPM Pre Setup Zones in the list of Searchable areas for CM/NPM series Work Orders.

Step	Procedure
1	From the PanaCIM Web Application, choose **Charts and Reports** > **Manage Work Order**. 
2	On Detail Production Schedule right click the schedule and click **Reserve Work Order**. (For details of the detailed production schedule, refer to 3.3.) 

3	You will see the searchable locations with **Area, Pre Setup Zone**, **Production Zone**. 

3. FUNCTION - MATERIAL CONTROL - MATERIAL ALLOCATION

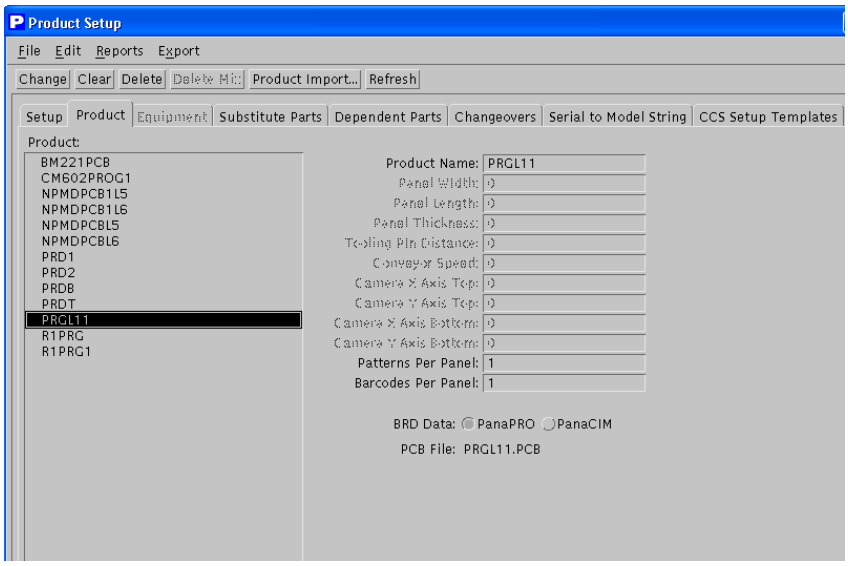
This chapter explains a setup and operation of PanaCIM-EE / Material Control-Allocation.

3.1 Overview

3.1.1 Preparation

In order to support Material Allocation operations, a new Area must be created using the special type “Limited”. (Refer to the PanaCIM Base document for details on how to create areas.) This Area is dedicated to registering incoming picked material from the warehouse based on the Work Order shortage list. The material picked to fulfill the Work Order shortages should be checked in to this special area. The material must remain in this area until the Material Handler fulfills the Work Order shortages with this material. If the material is not stored in this Area, it is considered available for anyone to use and could inadvertently be reserved to a Work Order other than the intended Work Order. Once the fulfillment process is completed, the Material can be moved from this Area to another Area.

Ensure each product in Product Setup is configured with the proper value for Patterns Per Panel (Product tab in Product Setup).

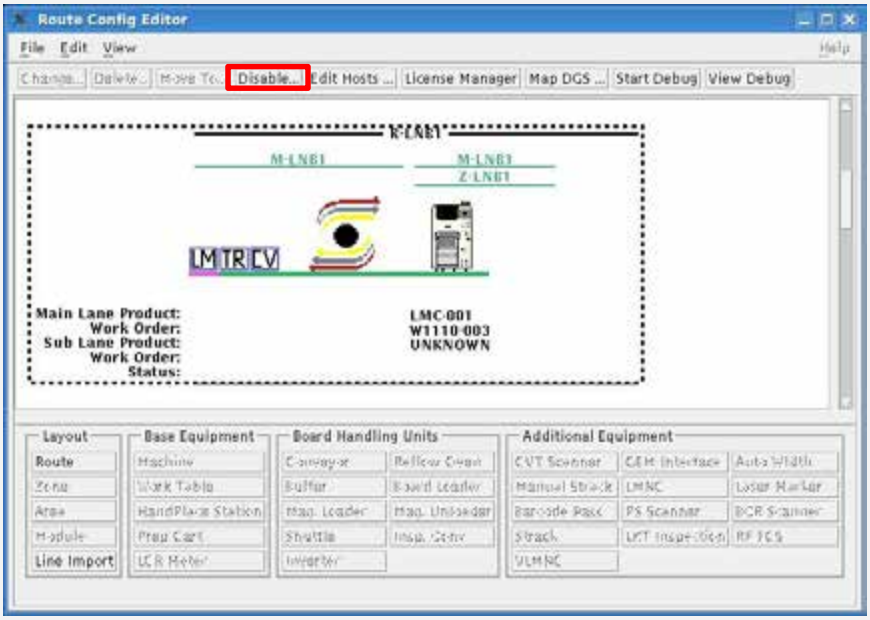


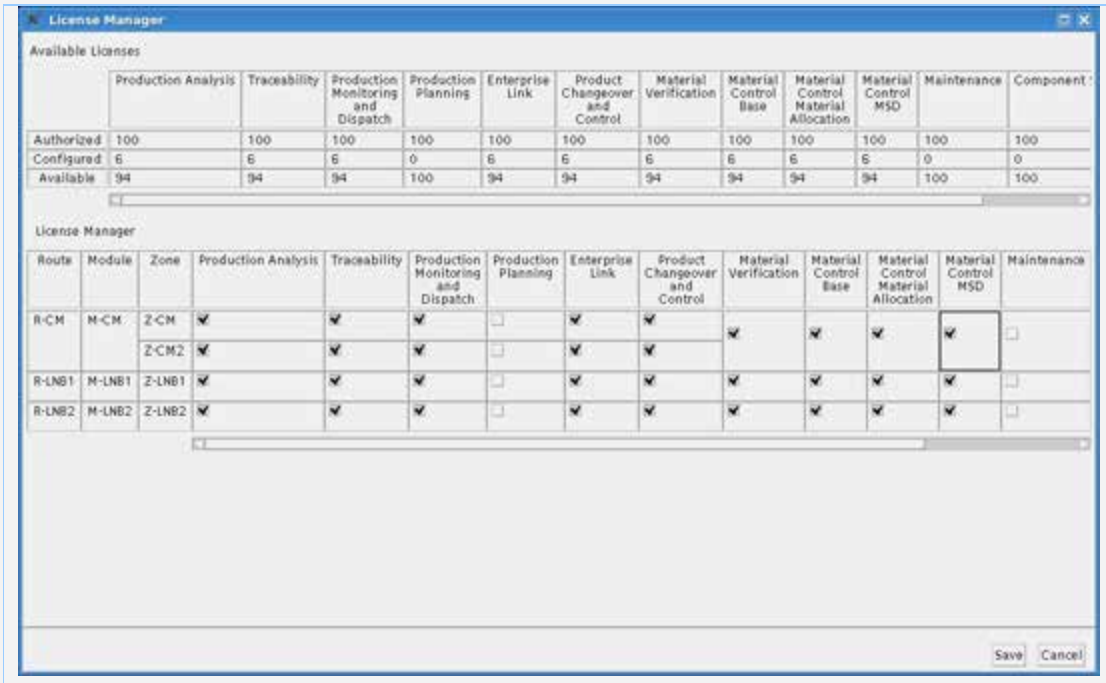
3.2 Setting

3.2.1 Route Editor

■ License Configuration

Before using the PanaCIM/Material Control – Material Allocation, the license needs to be enabled via the Route Editor of the PanaCIM controller server.

Step	Procedure
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.
4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Material Control - Base, place a check mark in the following checkboxes: • **Material Verification:** Enables the selected license module by module. • **Material Control - Base:** Enables the selected license module by module. • **Material Control - Material Allocation:** Enables the selected license module by module.

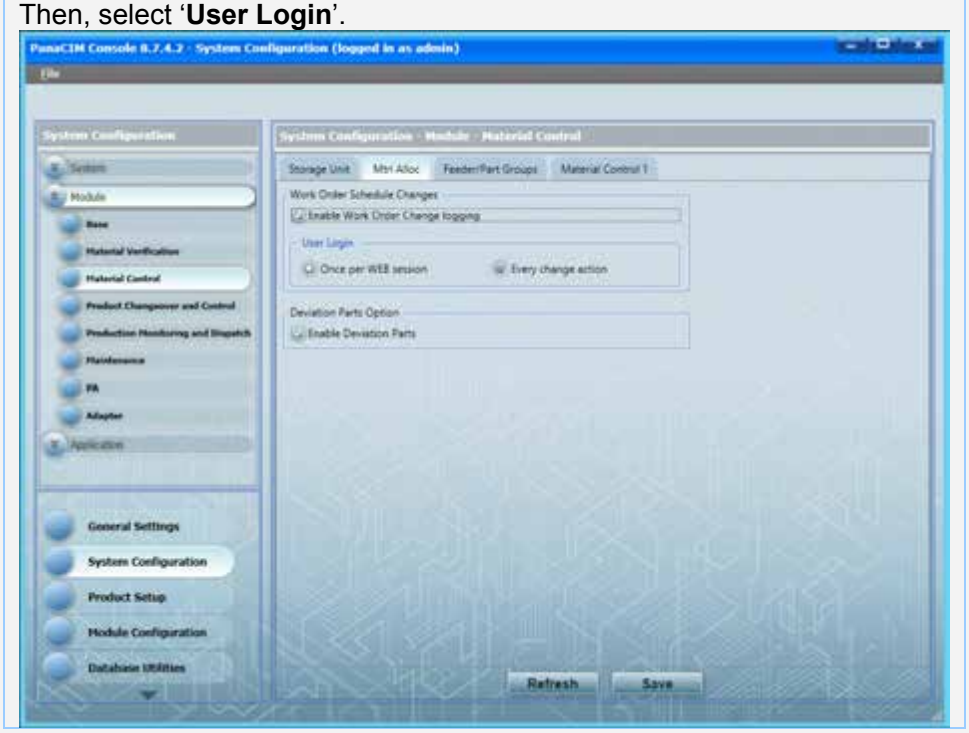
Step	Procedure
	 **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After configuring the license, click **Save** and exit the License Manager.

3.2.2 System Config

■ Enable Work Order Change Logging

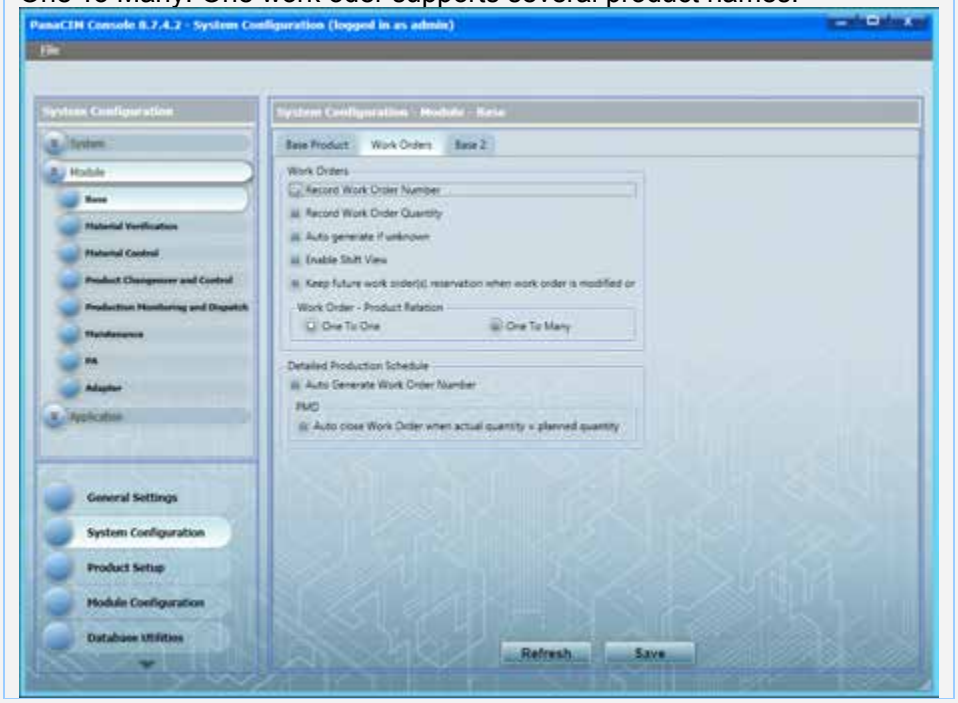
Refer to the action table below to configure if the User Login should be accepted when creating, editing, or deleting a work order using the System Configuration Application.

Refer to the action table below to enable work order change logging using the System Configuration Application.

Step	Procedure
1	Select **PanaCIM Console** -> **System Configuration** -> **Module** -> **Material Control**.
2	Click the **Mtrl Alloc** tab.
3	Check '**Enable Work Order Change logging**' checkbox in **Work Order Schedule Changes**. Then, select '**User Login**'. 
4	Click **Save**.

■ Configure Work Order Mapping

Refer to the action table below to configure the Work Order mapping relationship.

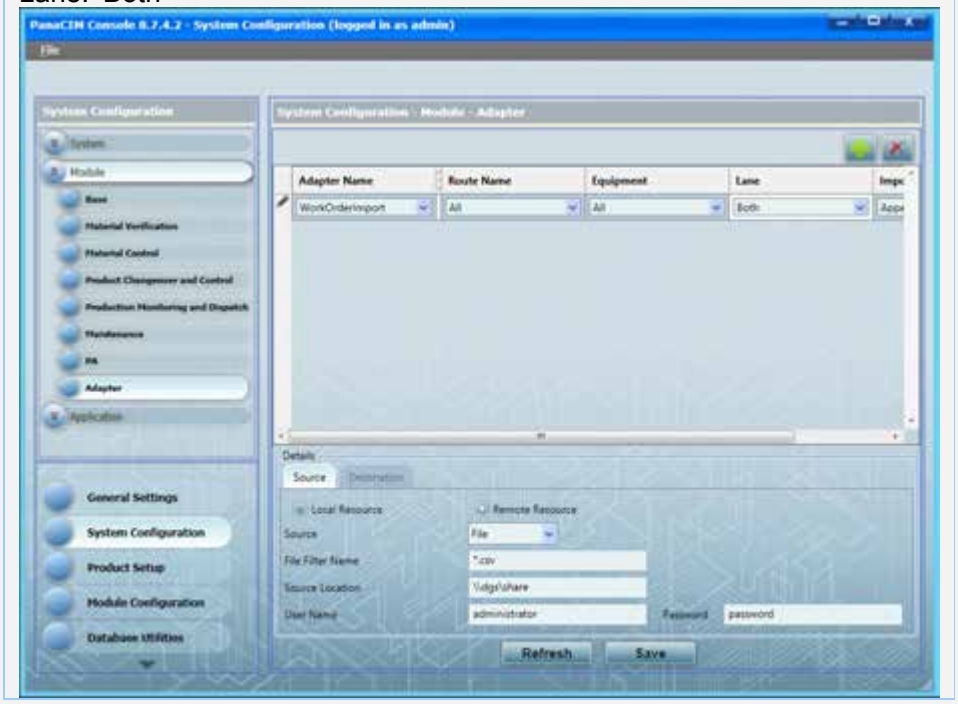
Step	Procedure
1	Select **PanaCIM Console** -> **System Configuration** -> **Module** -> **Base**.
2	Click the **Work Order** tab.
3	Check the '**Record Work Order Number**' checkbox in **Work Orders** to enable. Select either one of **Work Order** – **Product Relation**. One To One: One work order supports one product name. One To Many: One work oder supports several product names. 
4	Click **Save**.

When you select **One To Many** in **Work Order** – **Product Relation**, there are the following restrictions in material reservation / picking using a scanner described later.

- Operations over several routes becomes impossible. Therefore, **All** will not be displayed in Route Name.
- When **All** is selected in Prodcut, all materials for work orders included in the selected work order are displayed.

■ Automatic import of work order

The following shows how to set an adopter when automatic import is enabled in work order import.

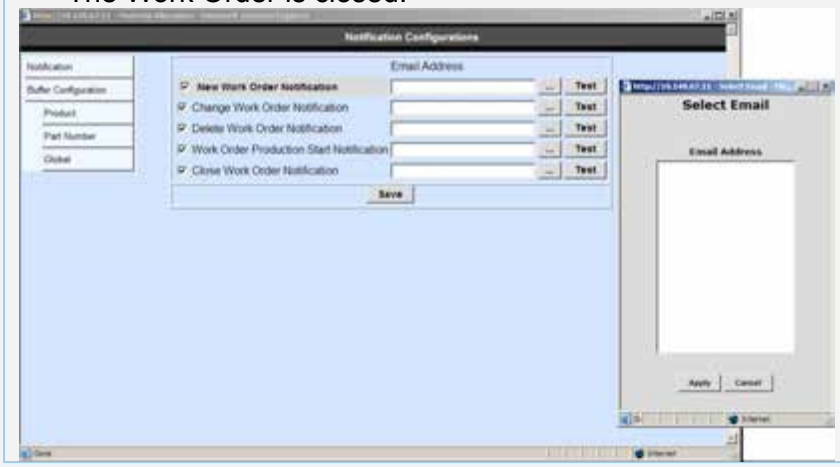
Step	Procedure
1	Select **PanaCIM Console** -> **System Configuration** -> **Module** -> **Adapter**.
2	Click **+** on the upper right of the window to add a row.
3	Select the followings from drop-down list in each item. Adapter name: 'WorkOrderImport' Route Name: 'All' Equipment: 'All' Lane: 'Both' 
4	Select the **Source** tab in **Details**.
5	Enter / select each item. ・Select **Remote Resource**. ・Source: Select **File**. ・File Filter Name Enter '*.csv'. ・Source Location: Specify a folder of file allocation desination. ・User Name: Enter a user name of PC that a file is allocated. ・Password: Enter a password of PC that a file is allocated.
6	Click **Save.**

3.2.3 Web Application

■ Configure Email Notifications

Access the Material Control Application via PanaCIM EE Launchpad to configure email notifications to be sent when a work order is added, changed, deleted or completed.

Refer to the action table below to configure email notifications.

Step	Procedure
1	Open a web browser and enter the appropriate URL in to the **Address** field. a. using the server: http://<localhost>/eplantview/ b. using a PC other than the server: http://<webservername>/eplantview
2	Enter your **User name** and **Password**. **Note:** When logging in for the first time, log in as an Administrator with a local user ID (not a domain ID). By default, local Administrators have access to all lines and reports, and can provide access to all other users.
3	Click **OK**.
4	Select **Material Control**, and click the **Configuration** link.
5	Enable email notifications by placing a checkmark next to one or more actions. You can configure notifications for the following events: • A new Work Order is created. • An existing Work Order is created. • An existing Work Order is deleted. • The production of the Work Order is started. • The Work Order is closed. 
6	Check-marked events become active. Enter an email address (or click  to select an email), and then click **Apply**.
7	Click **Save**.

■ Global Buffer Configuration

Access the Material Control Application via PanaCIM EE Launchpad to configure a global buffer percentage to be used for all material reservations. However, if a Part number based buffer or product–part number based buffer is configured for a part, it will override this buffer.

Example: Reel Parts Global Buffer = 5% Tray Parts Global Buffer = 10%
Product A Estimated Use ⇒Reel Part A - 1000 pcs., Reel Part B - 2000 pcs.
Product A Req'd Reserve ⇒Reel Part A - 1050 pcs., Reel Part B - 2100 pcs.
Product A Estimated Use ⇒Tray Part E - 1000 pcs., Tray Part F - 2000 pcs.
Product A Req'd Reserve ⇒Tray Part E - 1100 pcs., Tray Part F - 2200 pcs.
Product B Estimated Use ⇒Reel Part C - 4000 pcs., Reel Part D - 5000 pcs.
Product B Req'd Reserve ⇒Reel Part C - 4200 pcs., Reel Part H - 5500 pcs.
Product B Estimated Use ⇒Tray Part G - 4000 pcs., Tray Part H - 5000 pcs.
Product B Req'd Reserve ⇒Tray Part G - 4400 pcs., Tray Part H - 5500 pcs.

Refer to the action table below to configure a global buffer percentage for material reservations.

Step	Procedure
1	Open a web browser and enter the appropriate URL in to the **Address** field. a. using the server: http://<localhost>/eplantview/ b. using a PC other than the server: http://<webservername>/eplantview
2	Enter your **User name** and **Password**. **Note:** When logging in for the first time, log in as an Administrator with a local user ID (not a domain ID). By default, local Administrators have access to all lines and reports, and can provide access to all other users.
3	Click **OK**.
4	Click the **Material Control** link and then select **Buffer Configuration**.
5	Click **Global underneath "Buffer Configuration"**. Enter a Buffer percentage that is *greater* than 0 and click **Save**. The buffer percentage can be set for Reel Parts and Tray Parts separately. 

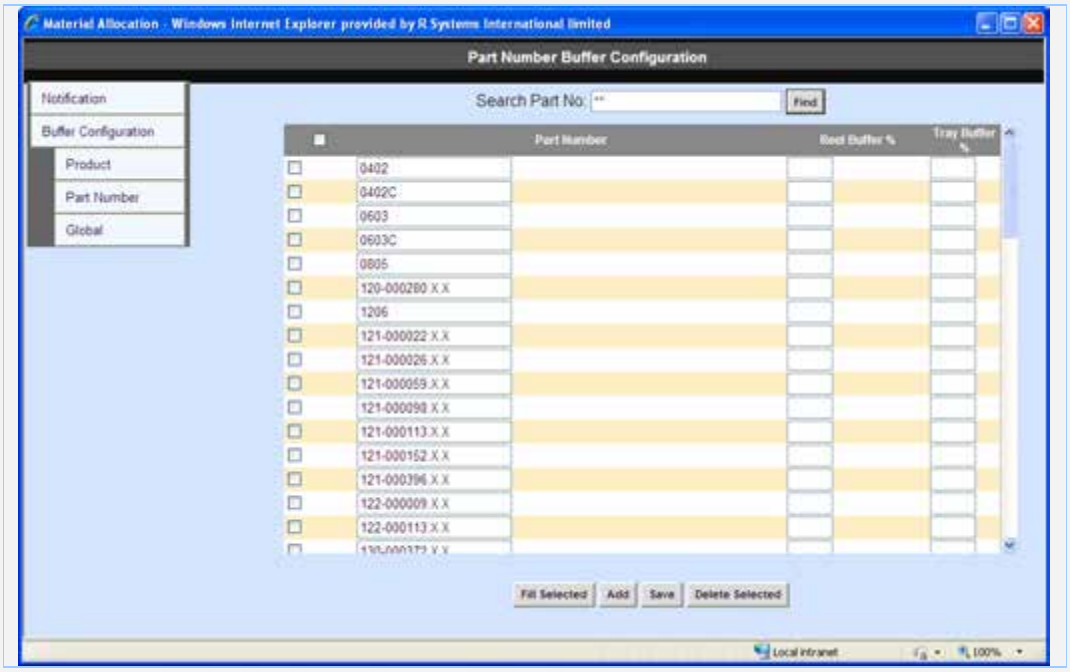
■ Part Number Buffer Configuration

Access the Material Control Application via PanaCIM EE Launchpad to configure a buffer percentage for an individual reel part number or Tray part number that will be used for all material reservations. If a product-part number based buffer is configured for a part, it will override this buffer.

Example:
Reel Part# Buffer Part A = 2%, Part B = 5%, Part C = 8%, Part D = 10%
Tray Part# Buffer Part E = 5%, Part F = 10%, Part G = 15%, Part H = 20%
Product A Estimated Use ⇒ Reel Part A - 1000 pcs., Reel Part B - 2000 pcs.
Product A Req'd Reserve ⇒ Reel Part A - 1020 pcs., Reel Part B - 2100 pcs.
Product A Estimated Use ⇒ Tray Part E - 1000 pcs., Tray Part F - 2000 pcs.
Product A Req'd Reserve ⇒ Tray Part E - 1050 pcs., Tray Part F - 2200 pcs.
Product B Estimated Use ⇒ Reel Part C - 4000 pcs., Reel Part D - 5000 pcs.
Product B Req'd Reserve ⇒ Reel Part C - 4320 pcs., Reel Part D - 5500 pcs.
Product B Estimated Use ⇒ Tray Part G - 4000 pcs., Tray Part H - 5000 pcs.
Product B Req'd Reserve ⇒ Tray Part G - 4600 pcs., Tray Part H - 6000 pcs.

Refer to the action table below to configure a buffer percentage for an individual part number for material reservations.

Step	Procedure
1	Open a web browser and enter the appropriate URL in to the **Address** field. a. using the server: http://<localhost>/ePlantView/ b. using a PC other than the server: http://<webservername>/ePlantView/
2	Enter your **User name** and **Password**. **Note:** When logging in for the first time, log in as an Administrator with a local user ID (not a domain ID). By default, local Administrators have access to all lines and reports, and can provide access to all other users.
3	Click **OK**.
4	Click the **Material Control** link and then select **Configuration**.
5	Click **Part Number**. • Search for the part number by clicking **Find**. • Select a Part Number (or multiple part numbers) • Reel Buffer % or Tray Buffer % must be greater than 0. • Click **Save** when complete.

Step	Procedure
	

■ Product-Part Number Buffer Configuration

Access the Material Control Application via PanaCIM EE Launchpad to configure a Reel /Tray buffer percentage for product-part number combination that will be used for all material reservations.

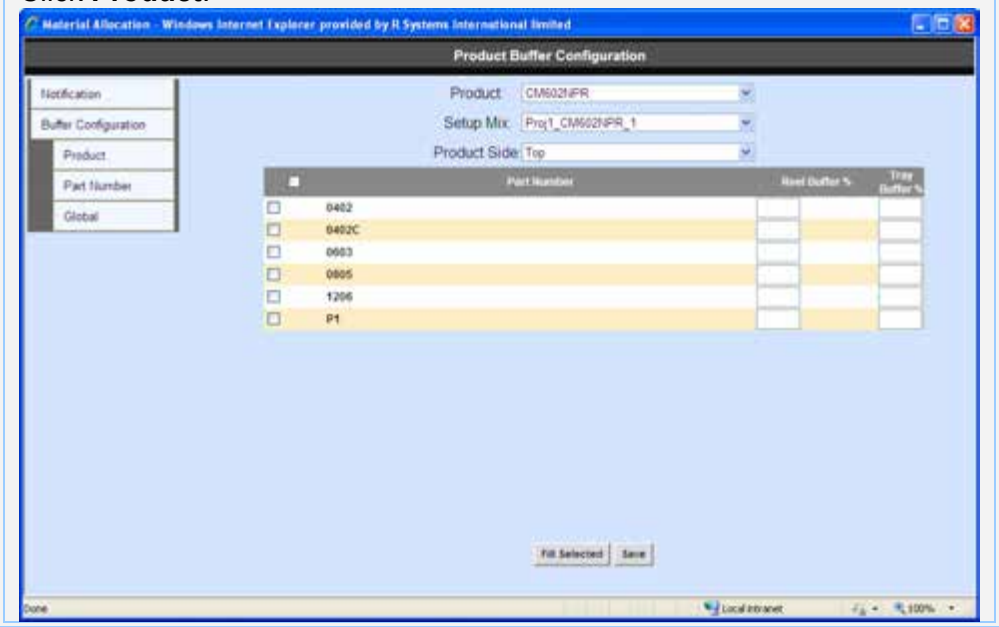
Example:

Product Name (A)
- Reel Part# Buffer — Part A = 2%, Part B = 5%
- Tray Part# Buffer — Part E = 5%, Part F = 10%

Product Name (B)
- Reel Part# Buffer — Part C = 5%, Part D = 10%
- Tray Part# Buffer — Part G = 10%, Part H = 20%

Product A Estimated Use ⇒ Reel Part A - 1000 pcs., Reel Part B - 2000 pcs.
Product A Req'd Reserve ⇒ Reel Part A - 1020 pcs., Reel Part B - 2100 pcs.
Product A Estimated Use ⇒ Tray Part E - 1000 pcs., Tray Part F - 2000 pcs.
Product A Req'd Reserve ⇒ Tray Part E - 1050 pcs., Tray Part F - 2200 pcs.
Product B Estimated Use ⇒ Reel Part C - 4000 pcs., Reel Part D - 5000 pcs.
Product B Req'd Reserve ⇒ Reel Part C - 4200 pcs., Reel Part D - 5500 pcs.
Product B Estimated Use ⇒ Tray Part G - 4000 pcs., Tray Part H - 5000 pcs.
Product B Req'd Reserve ⇒ Tray Part G - 4400 pcs., Tray Part H - 6000 pcs.

Refer to the action table below to configure a buffer percentage for product-part number combinations for material reservations.

Step	Procedure
1	Open a web browser and enter the appropriate URL into the **Address** field. a. using the server: http://<localhost>/eplantview/ b. using a PC other than the server: http://<webservername>/eplantview
2	Enter your **User name** and **Password**. **Note:** When logging in for the first time, log in as an Administrator with a local user ID (not a domain ID). By default, local Administrators have access to all lines and reports, and can provide access to all other users.
3	Click **OK**.
4	Click the **Material Control** link and then select **Configuration**.
5	Click **Product**. 

Step	Procedure
6	If there are multiple mixes for the product, choose a mix name from the **Setup Mix** drop-down. Doing this will display the all parts for the product mix.
7	Select a **Product Side** from the drop-down. This screen will refresh to display the part number for the product mix and side.
8	**Check** the part for which you want to assign a buffer percentage, and then enter a buffer amount (in %) for each part in the **Reel Buffer %** or **Tray Buffer %** box. This amount must be greater than zero. For reel parts, **Reel Buffer %** is assigned and for tray parts, **Tray Buffer %** is assigned.
9	Click **Save** after completing your fields and selections

3.2.4 Work Order

■ Detailed Production Schedule - Work Order Reservation for Reels

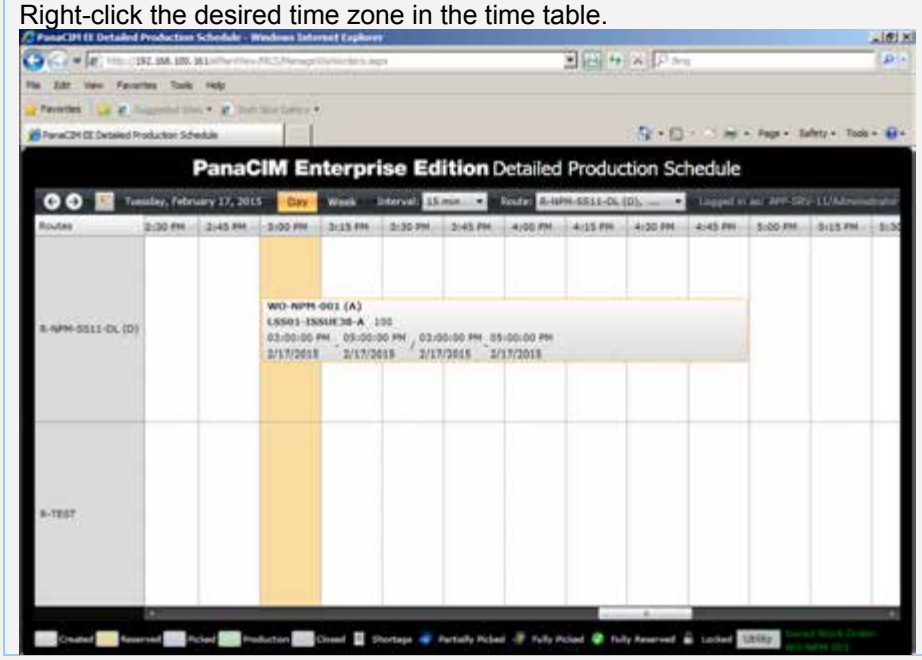
Search Order in Material Reservation to show by default the Unloaded Partial Material before Unloaded Full Material Reels in PanaCIM Enterprise Edition ePlantview.

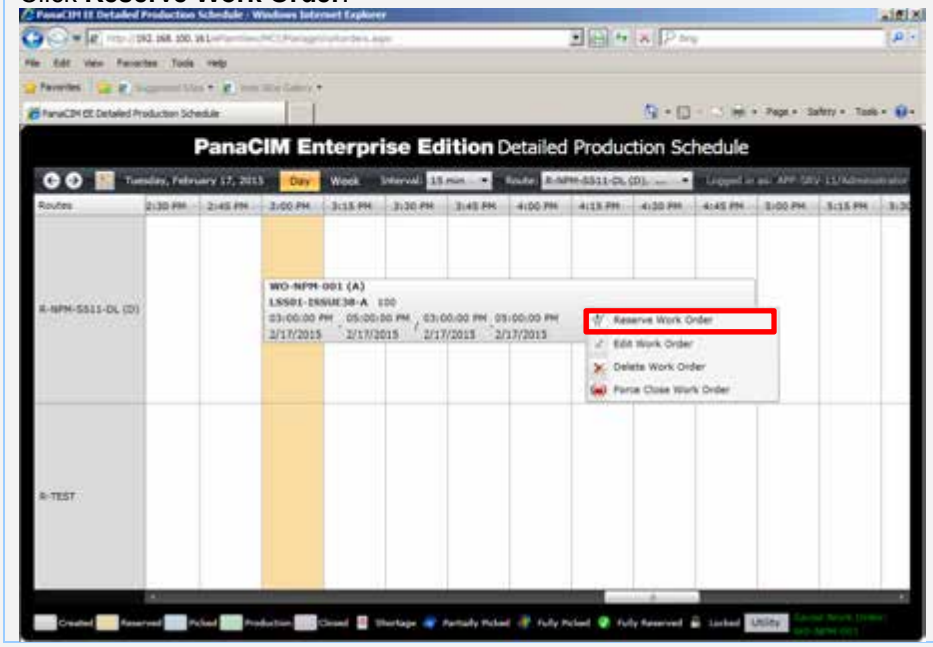
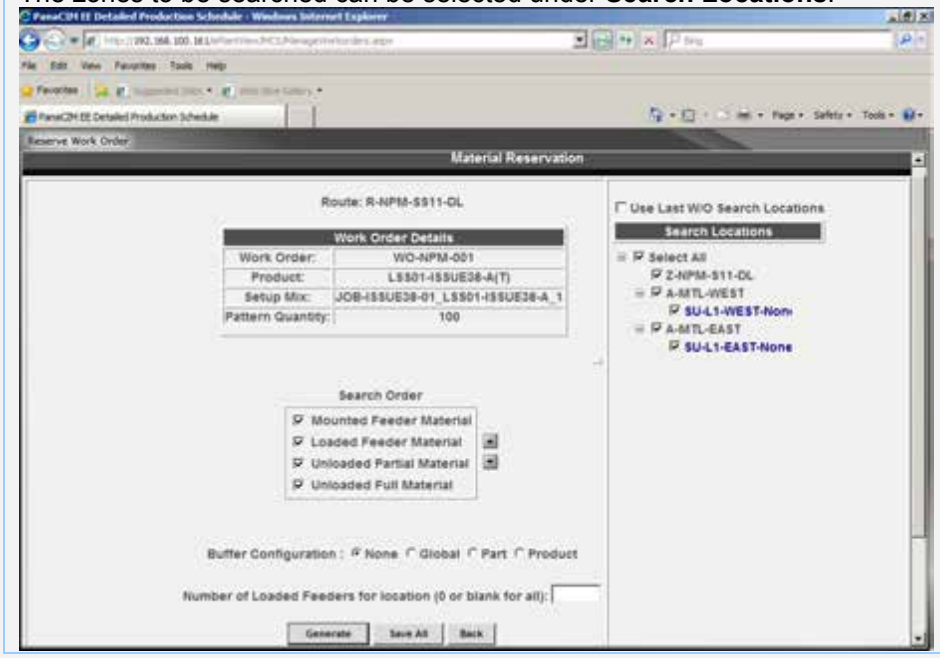
Step	Procedure
1	Open the Web application.
2	Click **Charts and Reports**. 
3	Click **Manage Work Orders**.
4	Right Click the work order in DPS whose reservation details you want to modify. 

5	Click Reserve Work Order. 
6	Select the part status for which you want to change the priority order under **Search Order**. The search order can be set for each product program and the checked materials will be searched for. 
7	You can change the order as desired by using the up and down arrow buttons.

■ Detailed Production Schedule - Storage Unit to Search

Use of Search Location by Zone in the PanaCIM Web application makes it easy to specify the zones to be searched when reserving materials. This position search enables the storage unit to be selected as a part of zone.

Step	Operation
1	Open the Web application.
2	Click **Charts & Reports** in the menu. 
3	Click **Manage Work Orders**.
4	Right-click the desired time zone in the time table. 

5	Click **Reserve Work Order**. 
6	The zones to be searched can be selected under **Search Locations**. 

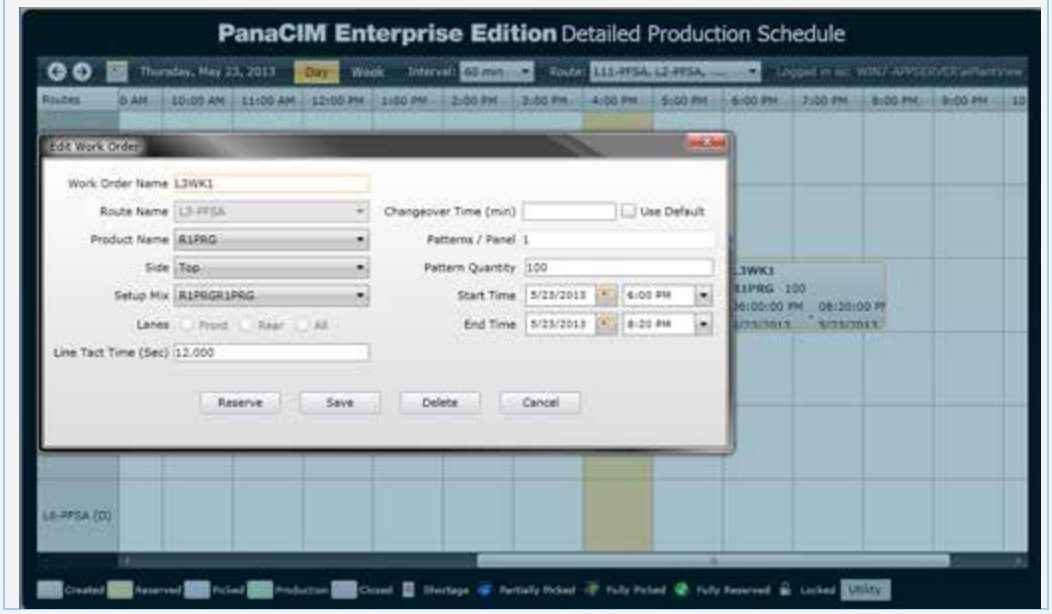
Detailed Production Schedule - Add view only access to the Detailed Production Schedule from Web Browser

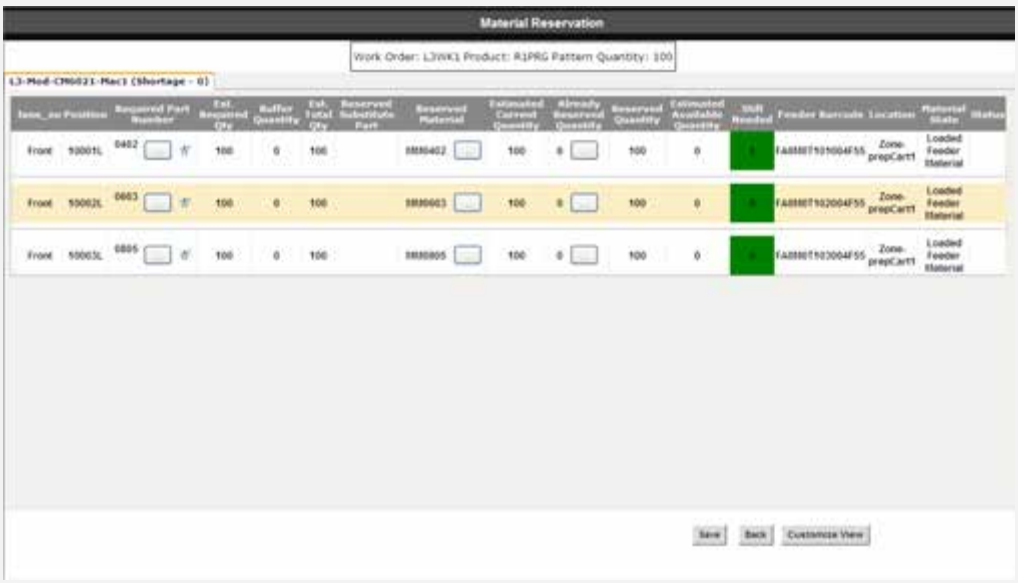
To add view only access to the detailed Production schedule from a web browser follow the procedure below. This is done to allow operators to view the schedules. The operators will not be able to make any changes to the detailed production schedule.

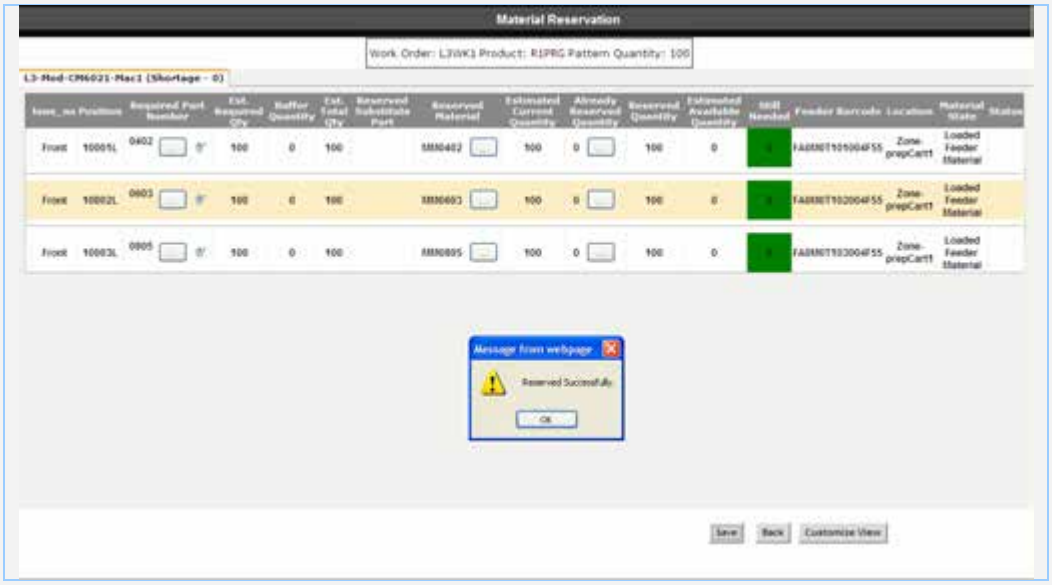
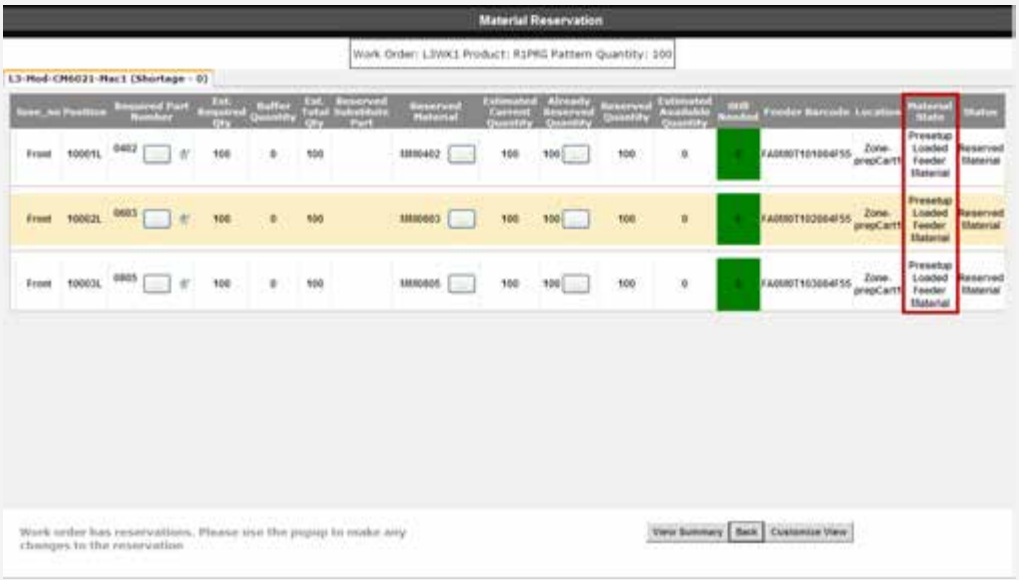
Step	Procedure
1	Create new user account and put into epvOperators group;
2	From the PanaCIM Web Application, choose **Administration** > **User Administration**. Display 'epvOperators' under GROUPNAME and place a check in the 'Manage Work Orders' checkbox under Action list. Then click **Update** to add it to the epvOperators group.
3	Enter following entry in epv_admin.SPC_ACTION table: Field Name Values Action SPC_CANRESERVE Action Type Work Order Reservation File Path NULL
4	Login in as ePlantView and new operator account to see the difference.

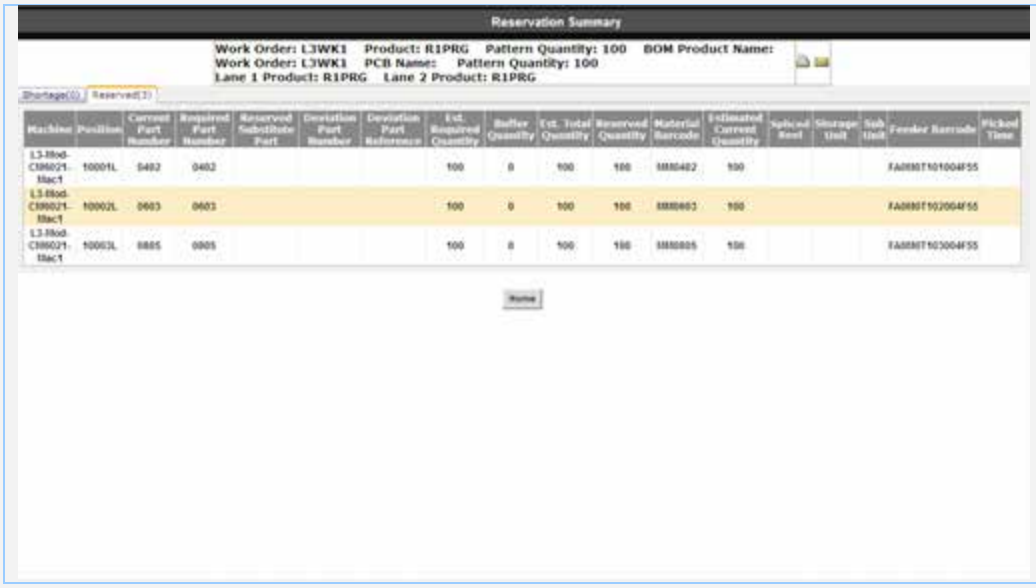
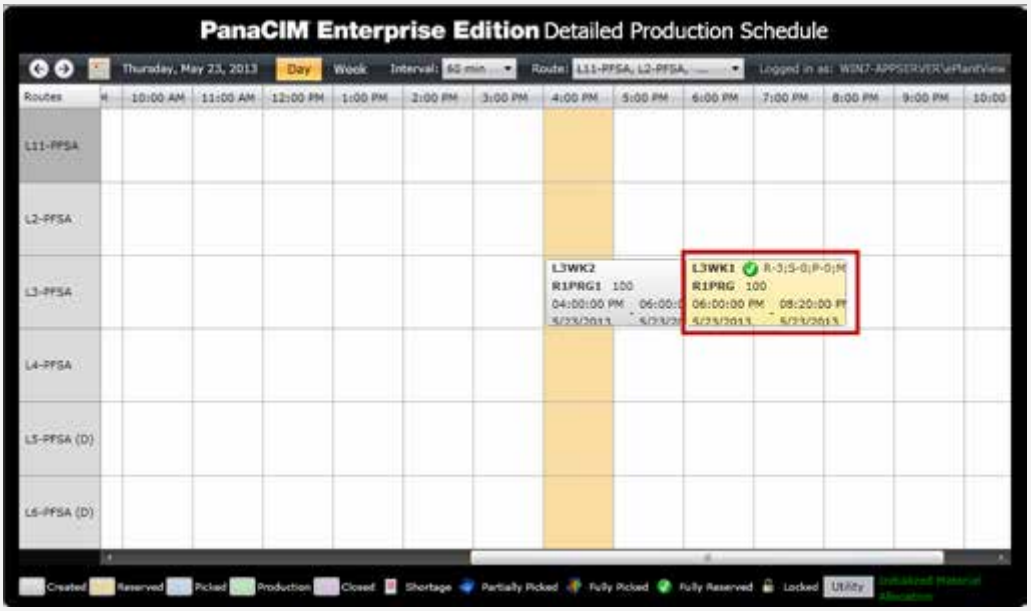
■ Detailed Production Schedule - Work Order Reservation for reels existing on presetup cart

Reels loaded on a presetup cart can be reserved for a work order. While selecting material for reservation on material reservation page, presetup carts are represented by green color while storage units are represented by blue color. Refer to the action table below detailing the steps.

Step	Procedure
1	Right-click the newly created work order and select **Edit Work Order** option. Following screen appears: 
2	Click the **Reserve** button in the **Edit Work Order** dialog. **Material Reservation** screen appears as shown below:  As shown above, material to be reserved can be selected from Presetup cart as well in addition to storage units. Presetup carts are colored green while storage units are represented by blue color.

Step	Procedure
3	Click the **Generate** button in the Material Reservation screen. Following screen appears:  **Note:** If the parts being reserved are also present in another product for which work order is created, then instead of above shown screen, an error message might appear. 
4	Click the **Save** button in the Material Reservation page. A message **"Reserved successfully"** appears as shown below:

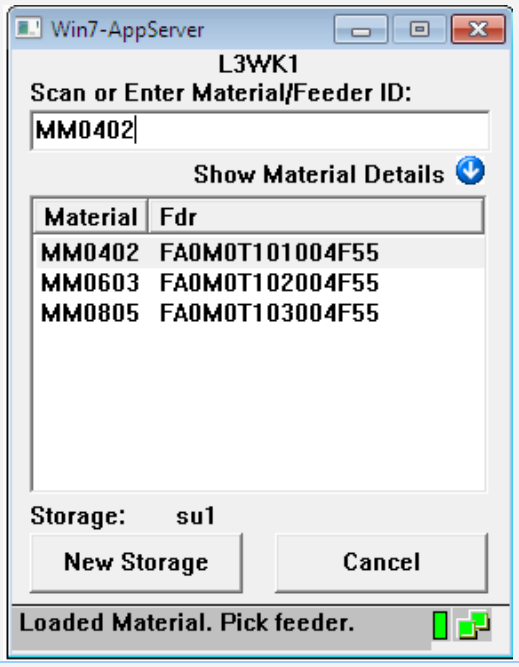
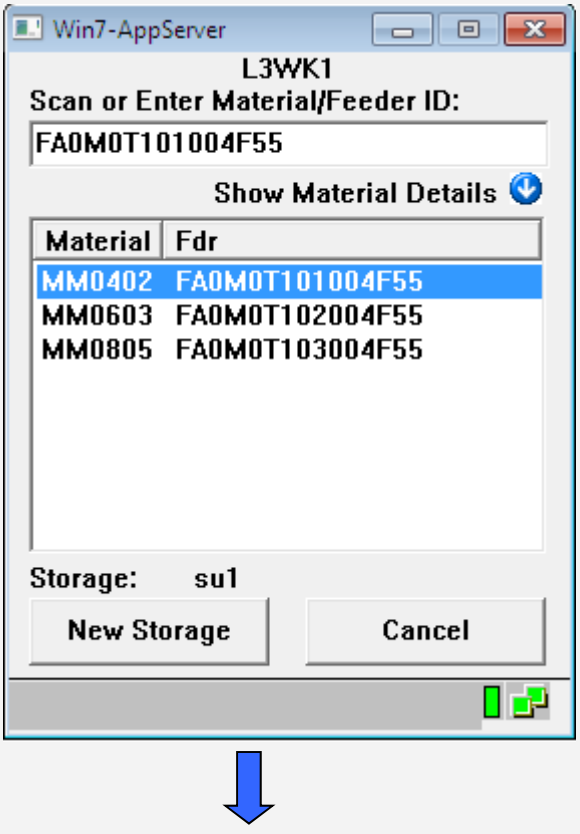
Step	Procedure
	
5	After closing the **"Reserved Successfully"** message box, following screen appears:  As indicated in the **Material State** column, reels loaded on presetup cart are successfully reserved.
6	Click the **View Summary** button. Following screen appears:

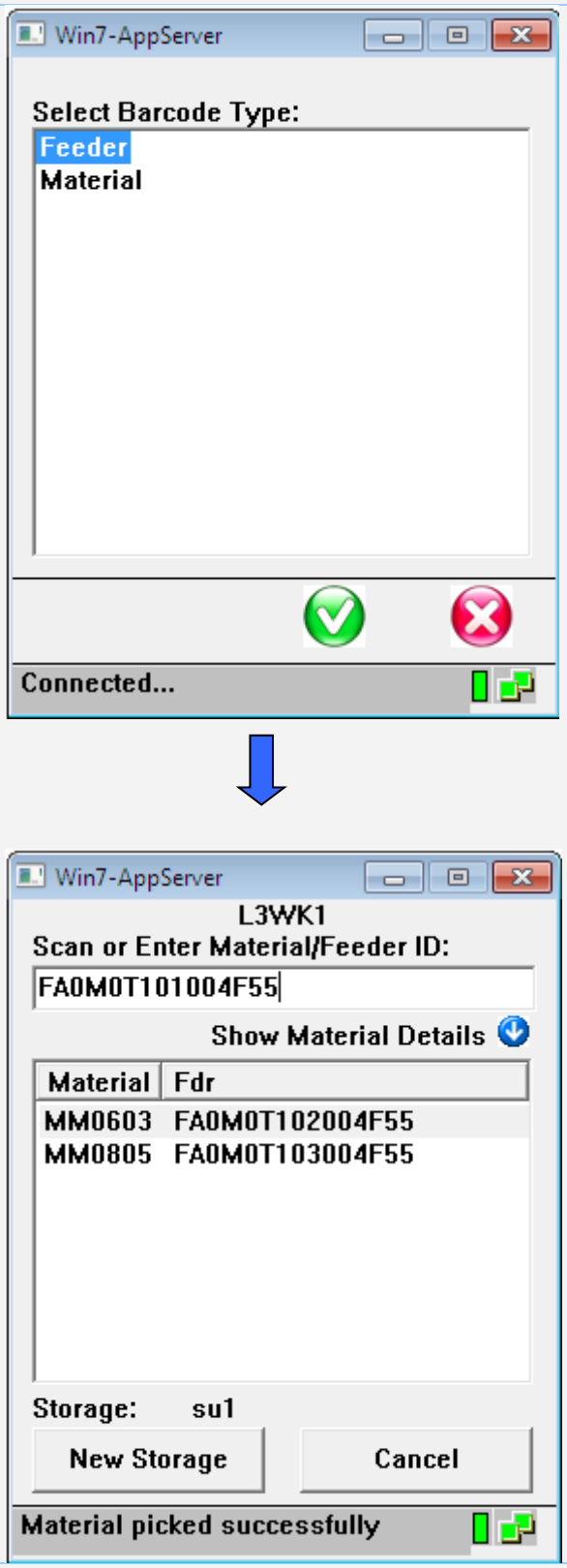
Step	Procedure
	
7	In the **Detailed Production Schedule (DPS)** screen, work order color is changed to reflect successful reservation:  **Note:** 'X' next to the following letters in the work order represents the following: P: Number of reels reserved in work order S: Number of reels running short in work order. M: Number of reels mounted on the machine

● Material Pick for Reels Loaded on Presetup Cart

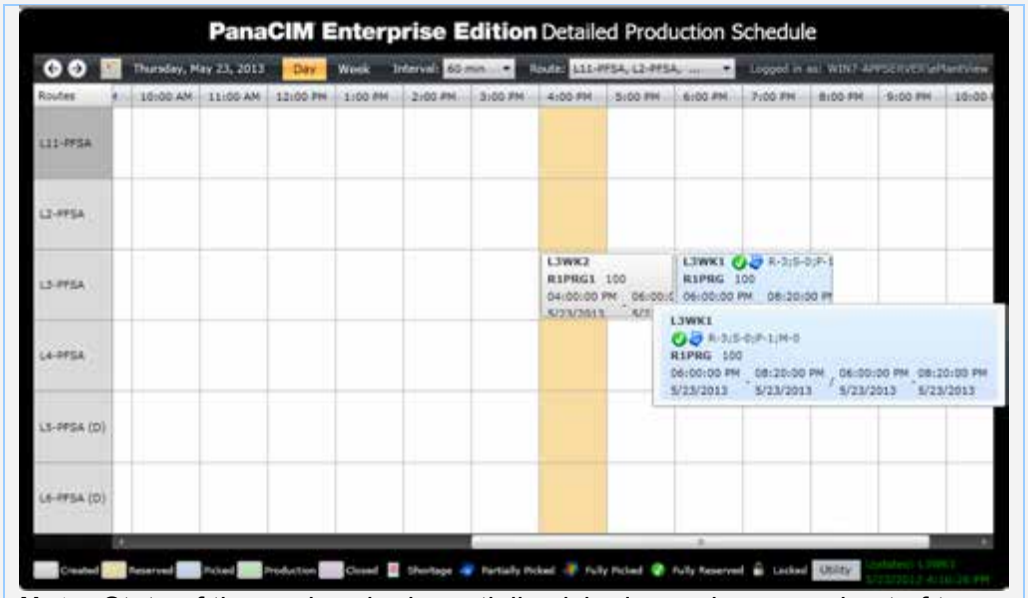
When material pick operation is performed from the mobile scanner for reserved reels existing on Presetup carts, Feeder barcode needs to be entered instead of reel barcode. This is illustrated below:

Step	Procedure
1	Reel barcode of the reel mounted on a presetup cart is not accepted at the time of material pick as shown below:

Step	Procedure
	
2	Feeder barcode is accepted at the time of material pick as shown below: 

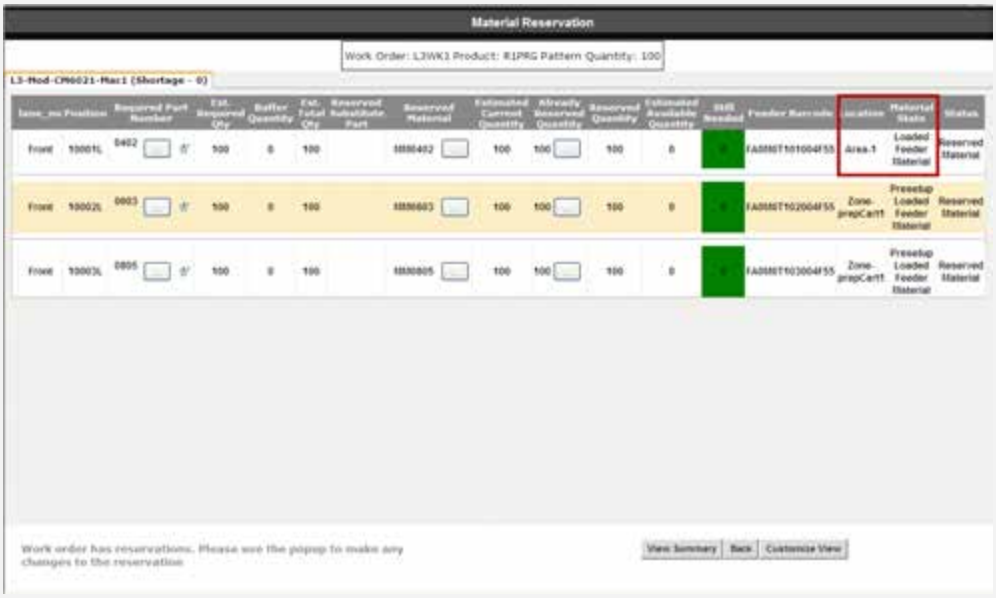
Step	Procedure
	
3	Material Pick is represented on the DPS screen as shown below:

Step	Procedure

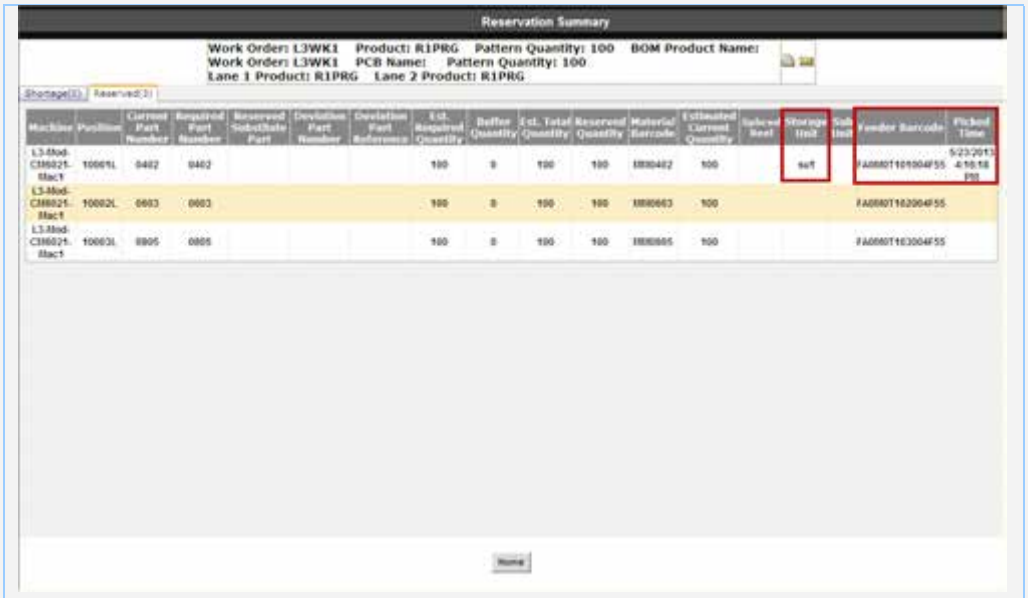
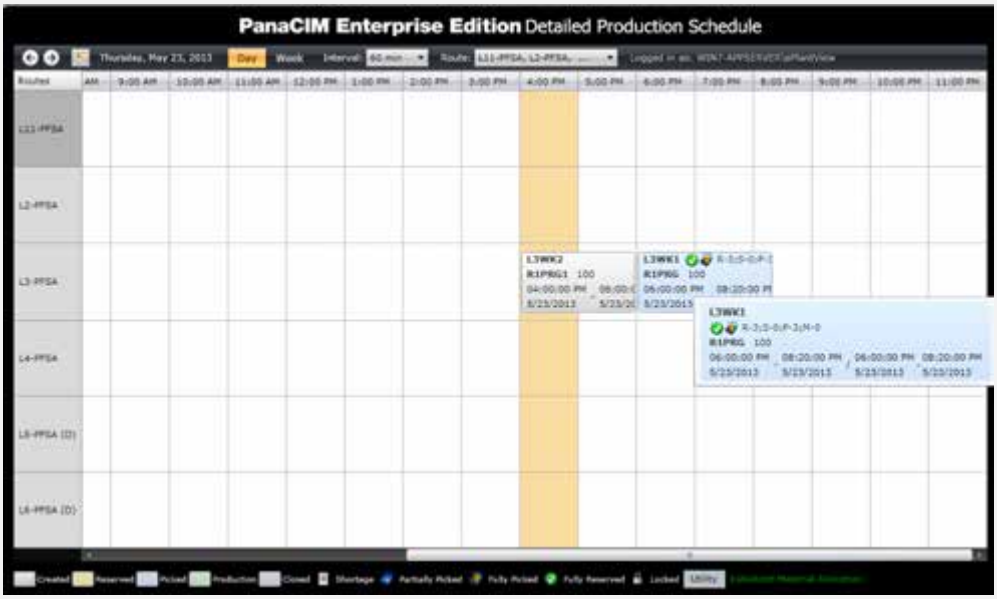


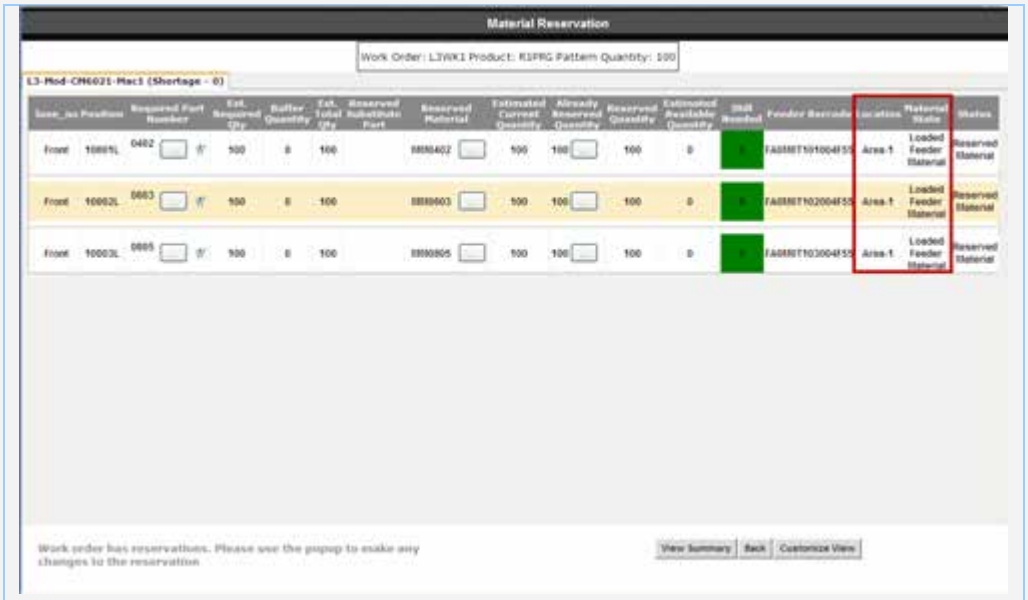
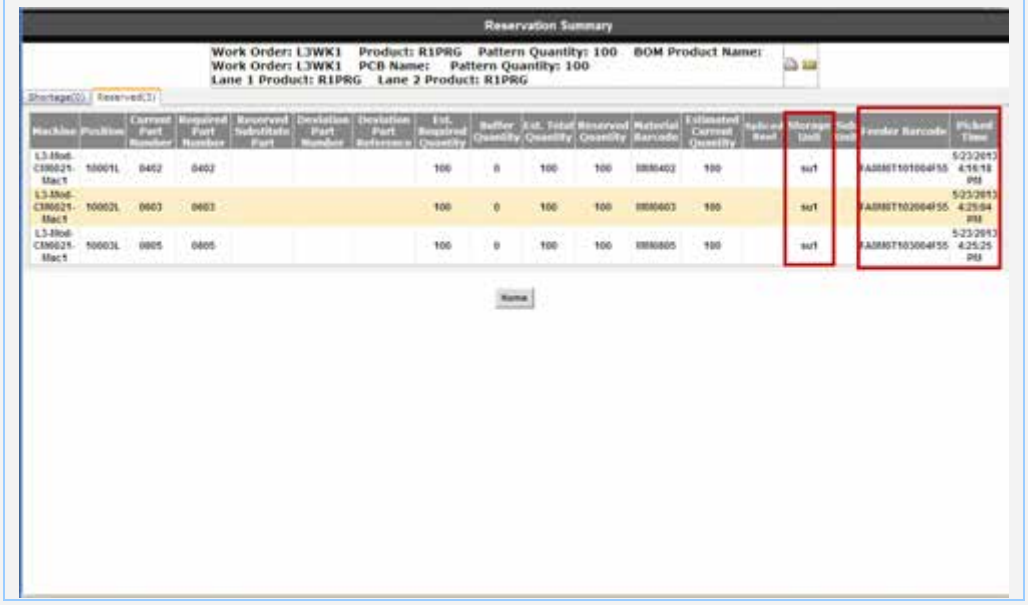
Note: State of the work order is partially picked as only one reel out of two reels has been picked.

Material Reservation page reflects the reel pick as shown below:



Work Order summary page shows the pick time as shown below:

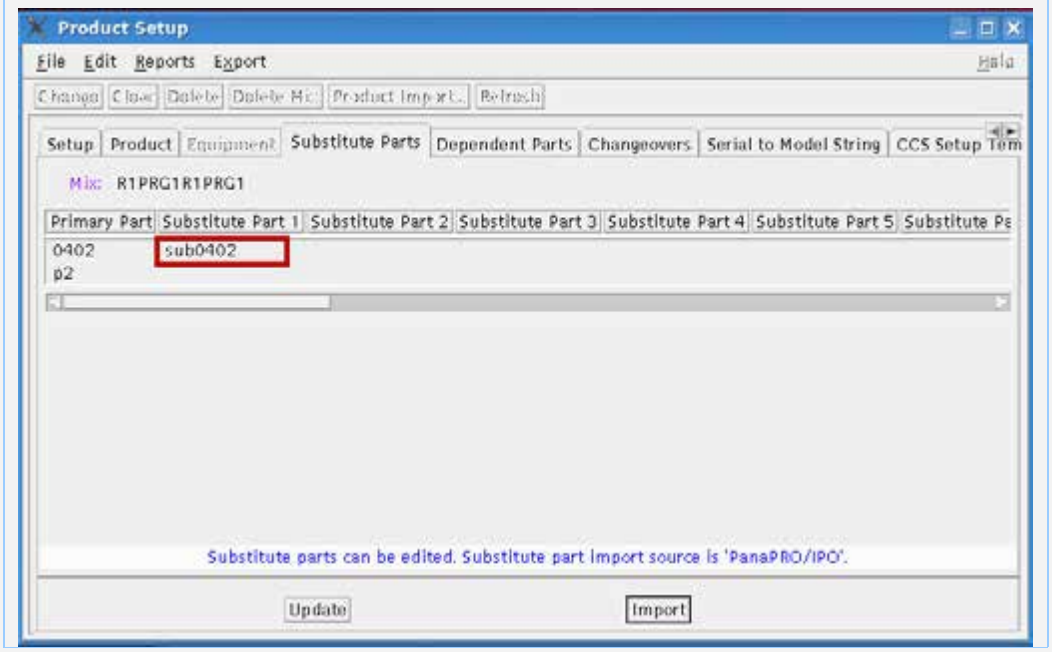
Step	Procedure
	
4	After all the reels loaded on presetup carts are picked (in this case 2), state of DPS screen is shown below:  **Material Reservation** page and **Reservation Summary** page also reflect material pick as shown below:

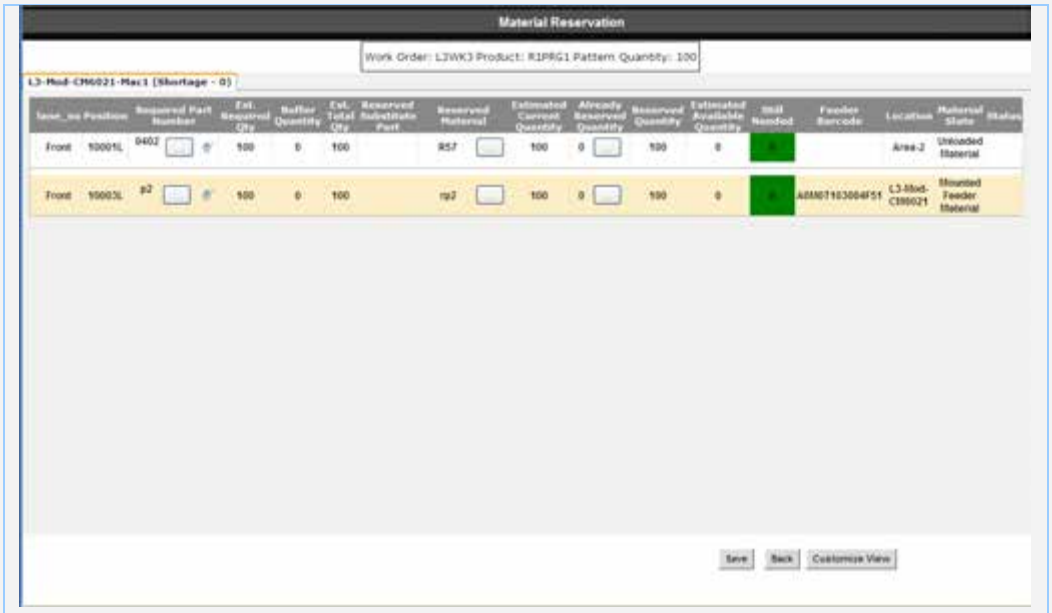
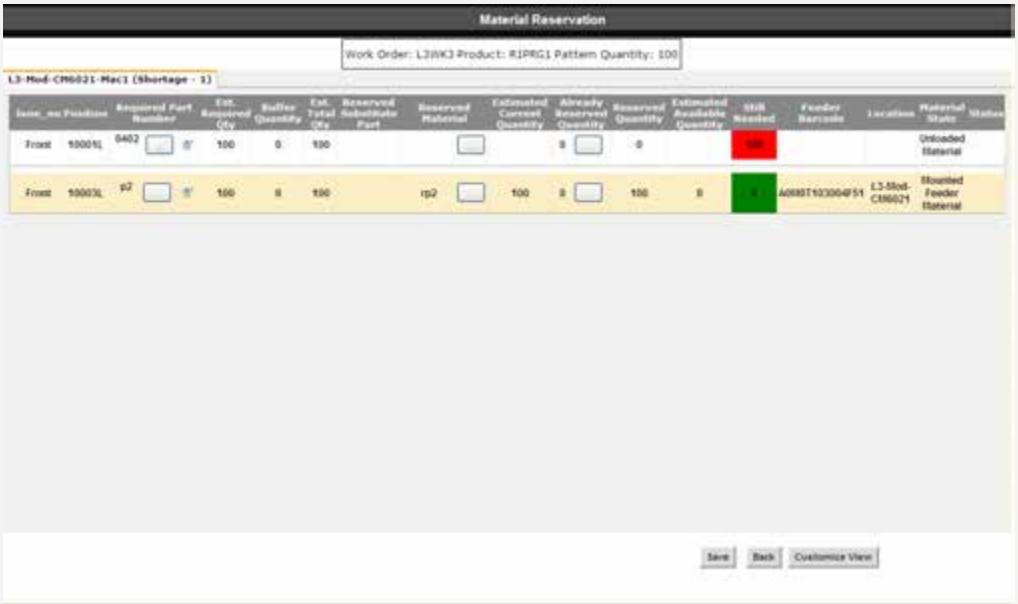
Step	Procedure
	
	

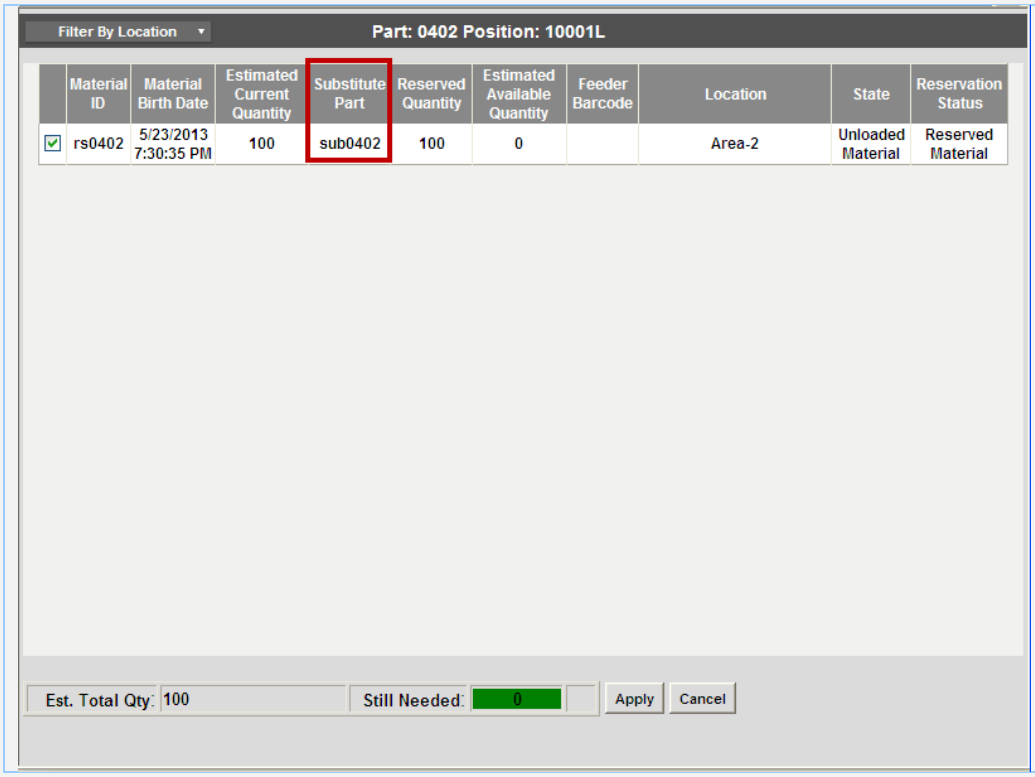
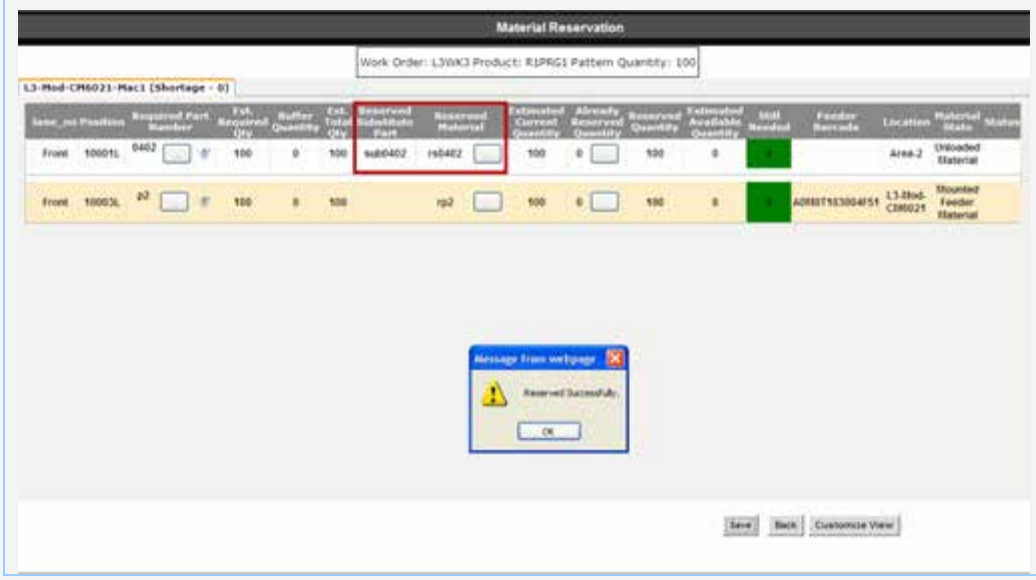
■ Detailed Production Schedule - Reserving Substitute Parts for a Work Order

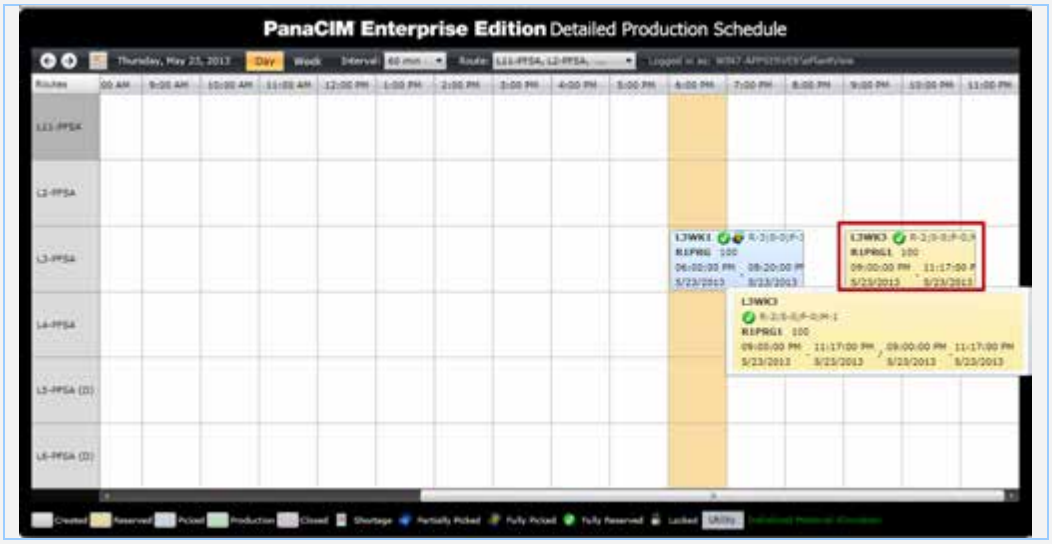
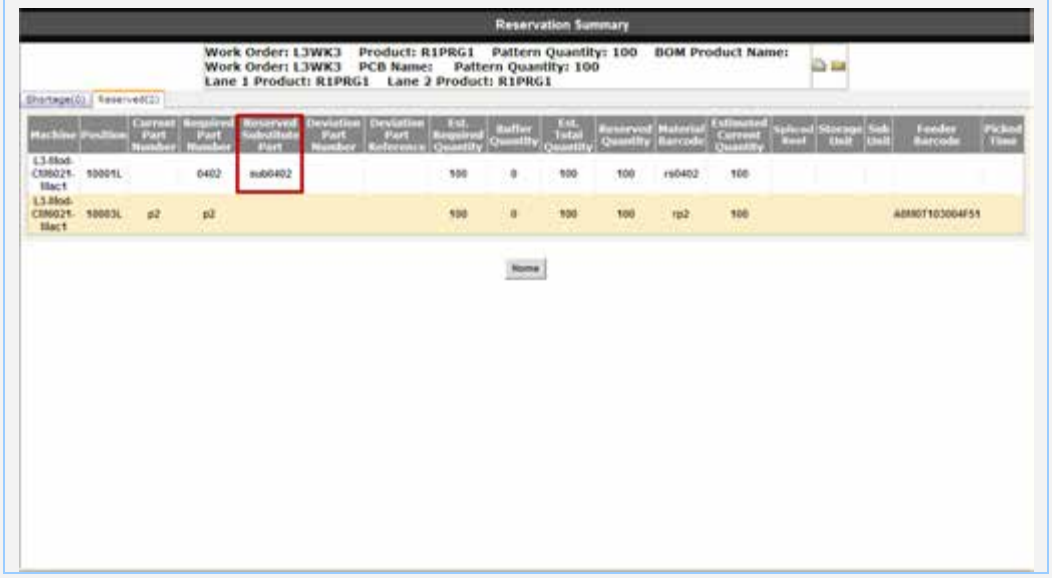
Substitute parts can be reserved for a work order. Reservation is first attempted using primary part, followed by substitute part. Therefore, reel containing primary part should exist in PanaCIM database if reservation is to be done for substitute part. For example, if the reel containing primary part is empty and no other reel containing primary part exists on the production floor, then instead of waiting for another reel to arrive from warehouse, reel containing substitute part (which exists in production floor) can be used for reservation.

Refer to the action table below that shows steps showing reservation when one or more substitute parts exist for a primary part.

Step	Procedure
1	Consider the following product for which substitute part is defined in the product setup as shown below: 
2	Open the material reservation screen in the DPS as shown below:

Step	Procedure
	 **Note:** If a reel containing primary parts does not exist in PanaCIM database but only reel containing substitute parts exists, state of material reservation screen is the one shown below:  Reserved material column is populated for part **p2** because its reel ID contains primary part only while for part 0402, only reel that exists is the one for substitute part (sub0402).
3	When the Material reservation screen opens, **Reserved Material** column shows the reel consisting of primary parts only. To reserve a reel containing substitute parts, click on ... button adjacent to the reel barcode shown in the **Reserved Material** column. Upon clicking ... button, following screen appears. In this screen, check the material ID (in this case rs0402) for substitute part (sub0402) and finally click the **Apply** button.

Step	Procedure
	
4	Click the **Save** button in Material Reservation screen after selecting a reel containing substitute part. 
5	Work order is reserved successfully when reel consisting of substitute part is used as shown below:

Step	Procedure
	
6	Reservation Summary page also shows the substitute part as shown below: 

■ Detailed Production Schedule - Material Reservation Support for Loaded Feeders

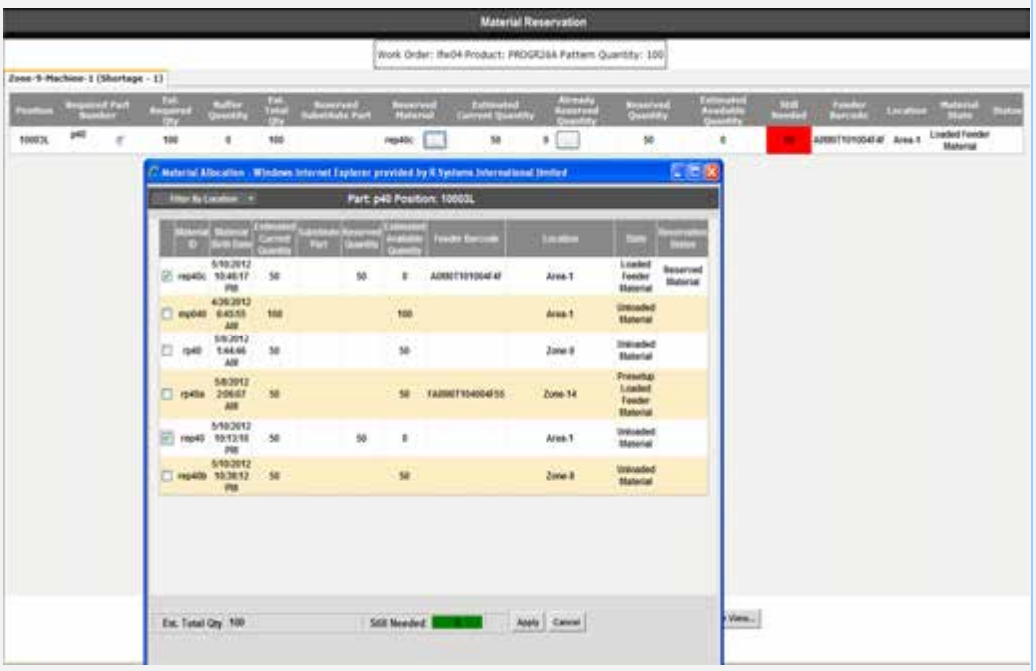
You can specify number of loaded feeders to be used for material reservation. This helps to regulate the use of loaded feeders so that they can be allocated to other work orders rather than loading feeder again and again for new work orders. For instance, if you have unloaded material available, it can be used for reservation thereby conserving the loaded feeders existing on production floors. Refer to the action table below that covers this scenario.

Step	Procedure
1	Material Reservation screen contains an option to specify number of loaded feeders to use. This is shown below:  As shown in the screen above, the number of loaded feeders to be used for reservation from the selected locations are specified. Whatever value is entered in the **Number of Loaded Feeders for location** textbox, the corresponding number of loaded feeders are automatically used for reservation. However, if no value is entered in the textbox, all the loaded feeders existing in the selected locations are automatically used for reservation.
2	As shown in the screen above, one loaded feeder is to be used. After clicking **Generate** button, following screen appears:

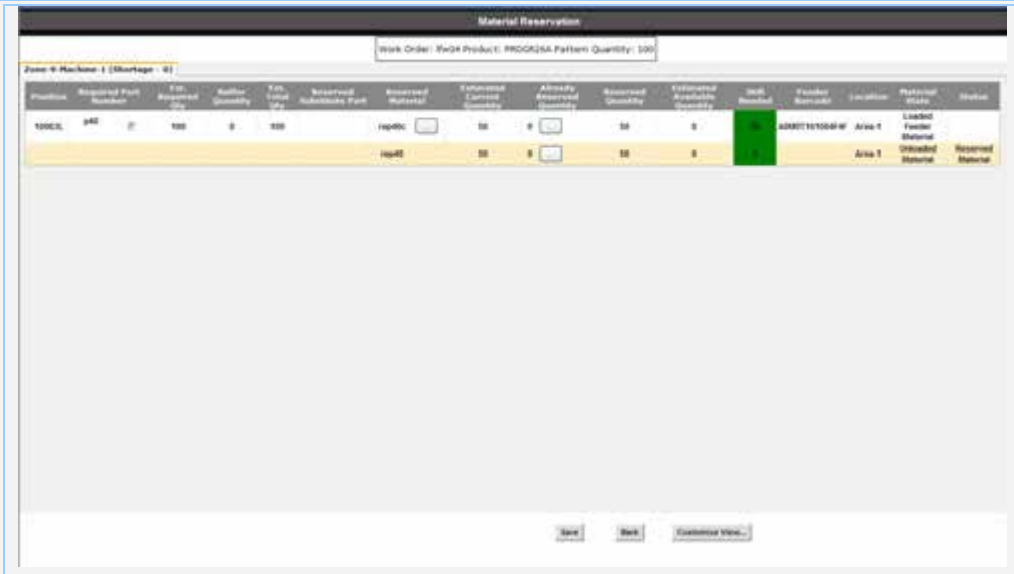


Note: Since one loaded feeder that is automatically suggested for reservation is not sufficient to meet the required quantity, you can manually choose additional feeders. This is shown below:

a. Click the ... button adjacent to the reel barcode in the **Reserved Material** column. Following screen appears:



b. Choose additional reels for reservation till the required quantity is fulfilled and click the **Apply** button. After doing so, following screen appears:

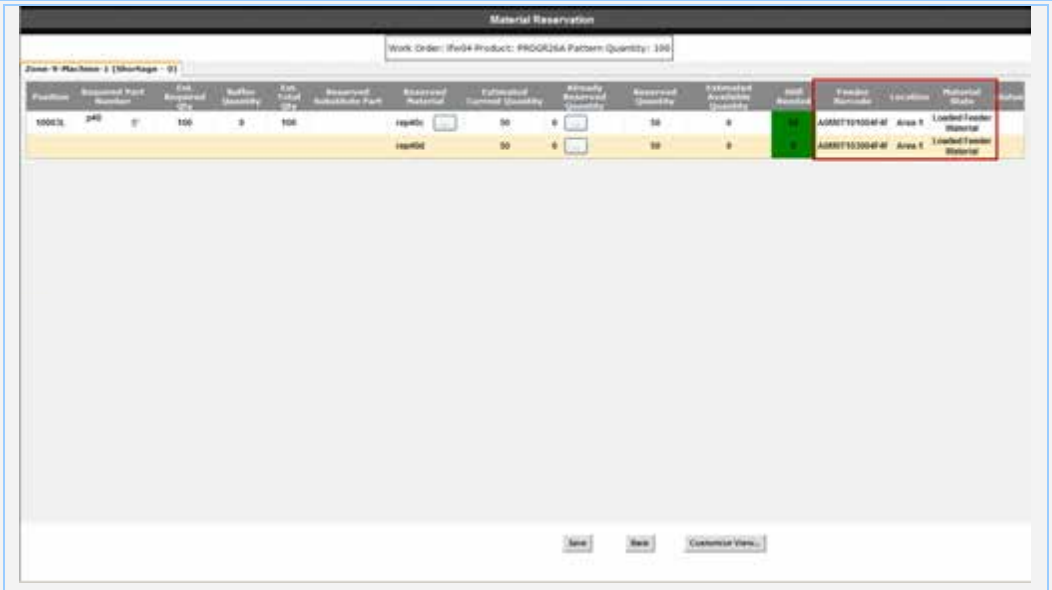


If you have additional mounted feeders / unloaded material available, you can use them to fulfill shortage for a work order instead of loading new feeders as it saves time.

3 You can specify two loaded feeders to be used as shown below:



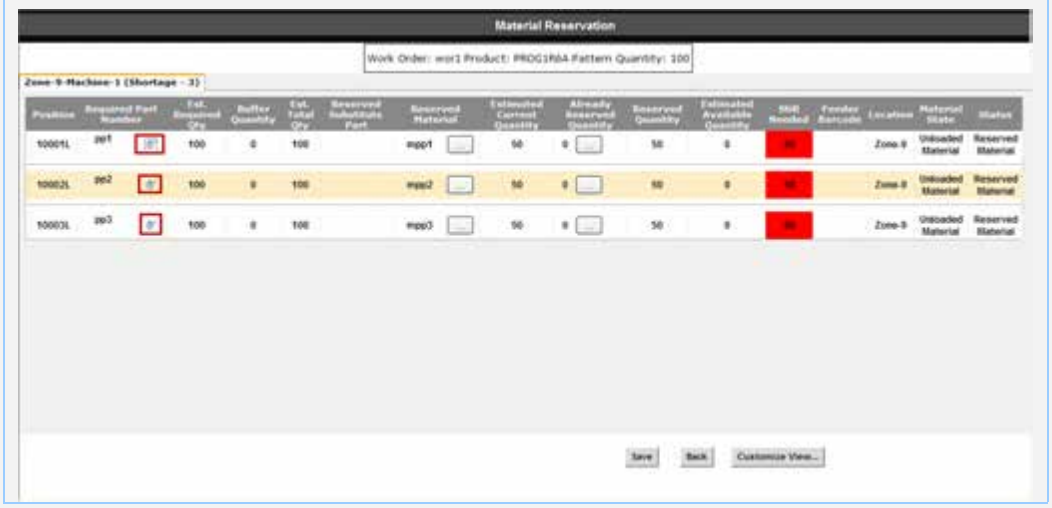
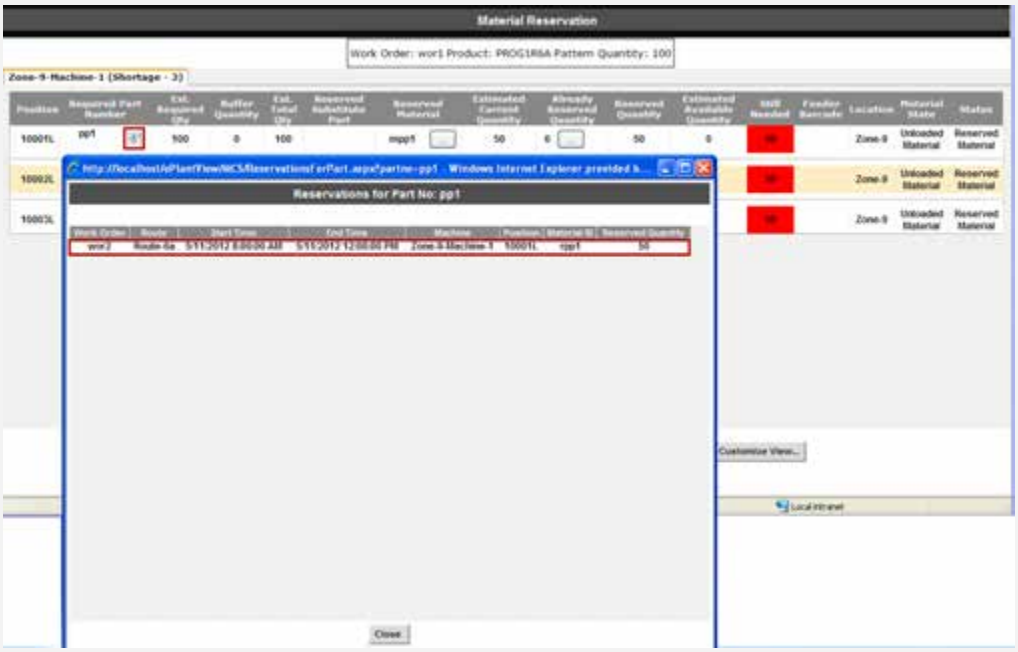
4 Upon clicking the **Generate** button, two loaded feeders are automatically suggested for reservation as shown below:

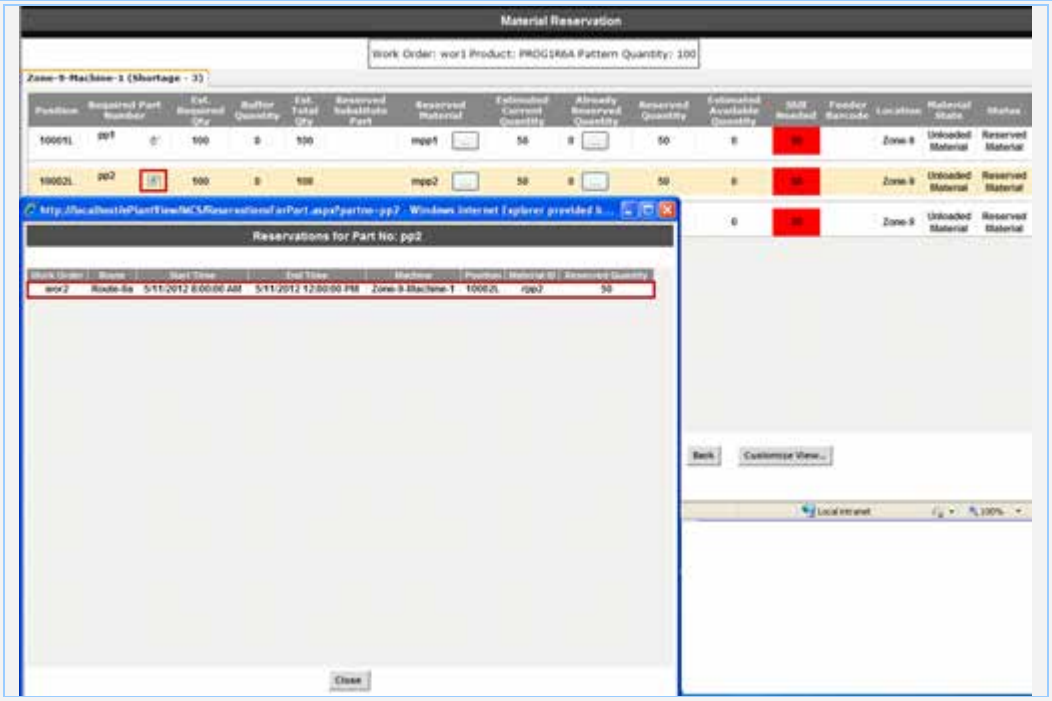


Note: In the scenario considered, only two loaded feeders exist for part p40 in Area-1. Hence, in the screen shown in step 3, if **Number of loaded feeders** is left blank, above screen shown in step 4 appears because all the loaded feeder are automatically suggested for reservation.

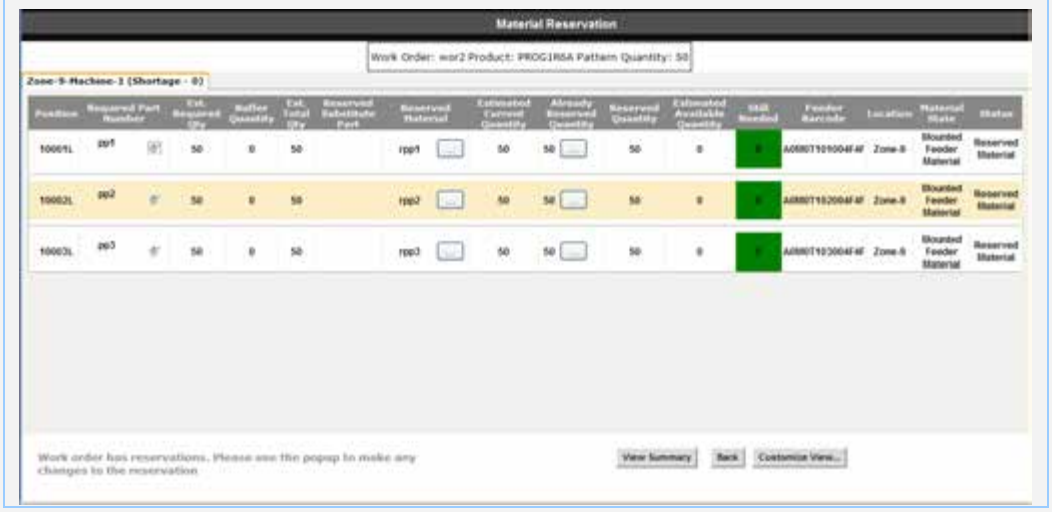
■ Detailed Production Schedule - Displaying Existing Reservations for a Part at the time of New Reservation

While reserving a part for a particular work order, you can view existing reservation of same part for different work orders. This is to decide if part needs to be released from an existing reservation and reallocated to another reservation to fulfill a shortage.

Step	Procedure
1	Consider the scenario shown below in which shortages exist at the time of reserving parts for a work order: 
2	Click the icon adjacent to part number in the **Required Part Number** column. icon is displayed irrespective of whether part has shortages or not. If the part number is also reserved for other work orders, following screen appears showing reservations for this part number as shown below: 

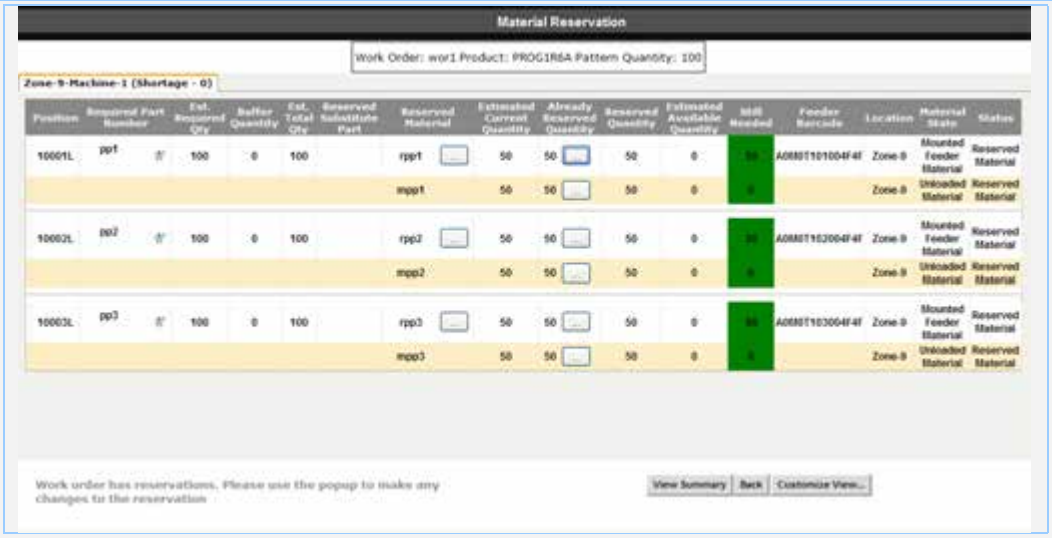


3 Work order to which a part is already reserved can be viewed by right-clicking it in the DPS screen. After doing so, following screen appears:



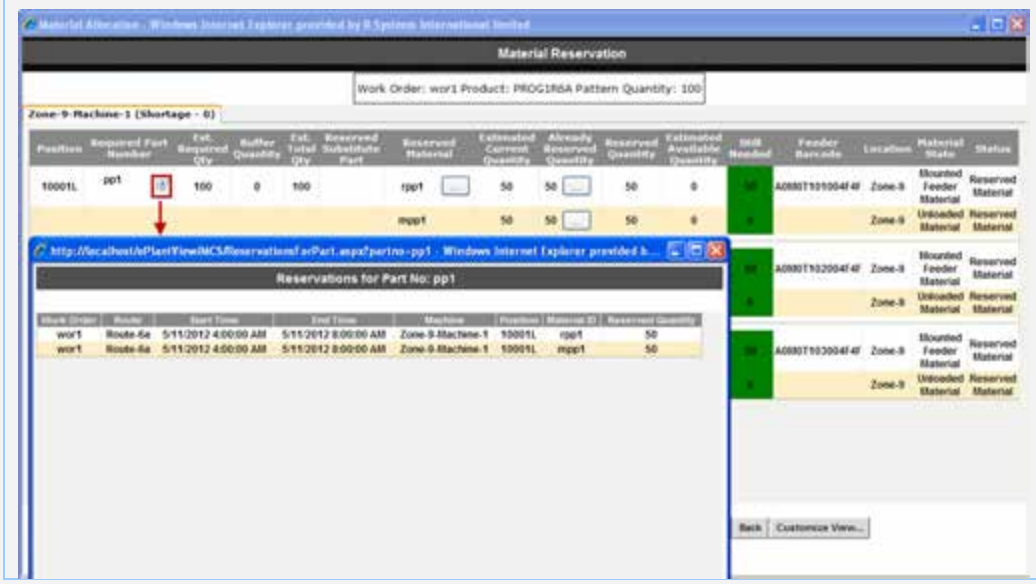
4 To fulfill the shortage, you can release the part reservation from existing work order shown in step 3 above and reallocate them to work order with shortages.

5 Once the reservation for the work order (which earlier had shortages) is completed after reallocation, details can be seen in the screen below:



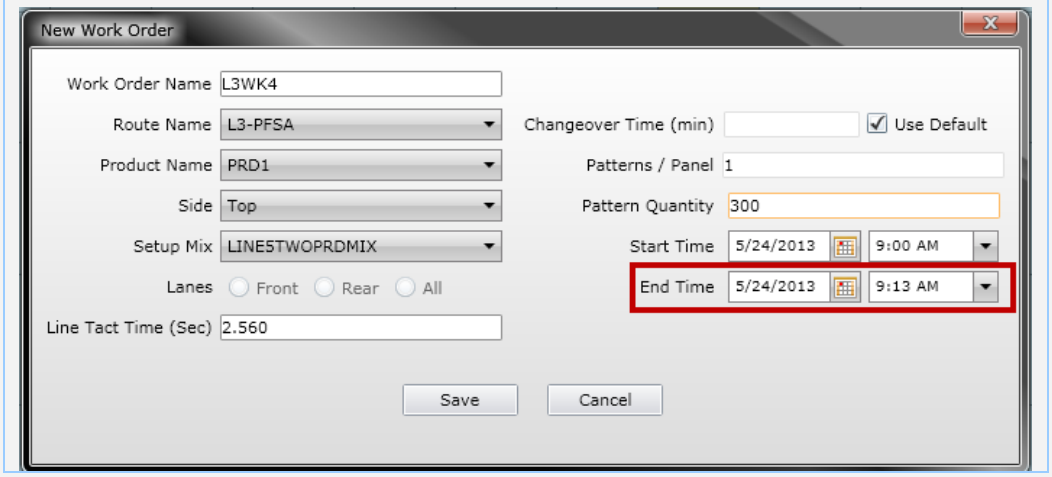
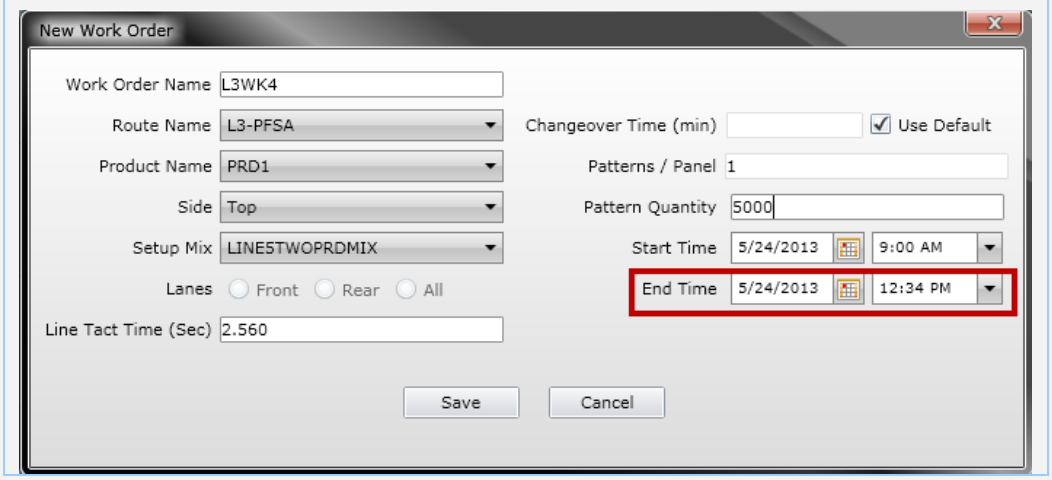
6

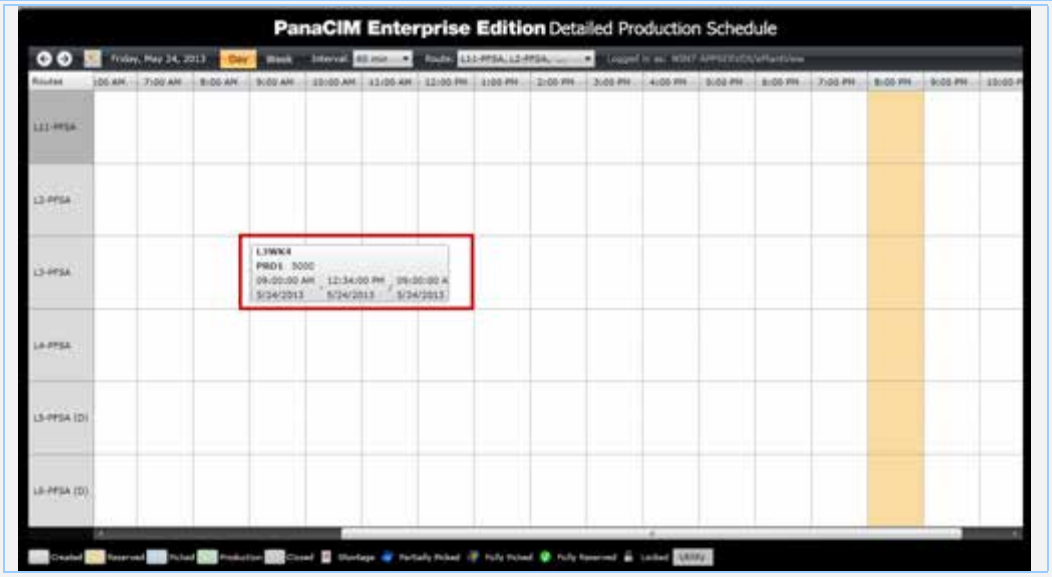
Now, if you click the icon, reservations exist only for one work order (in this case wor1) as shown below:



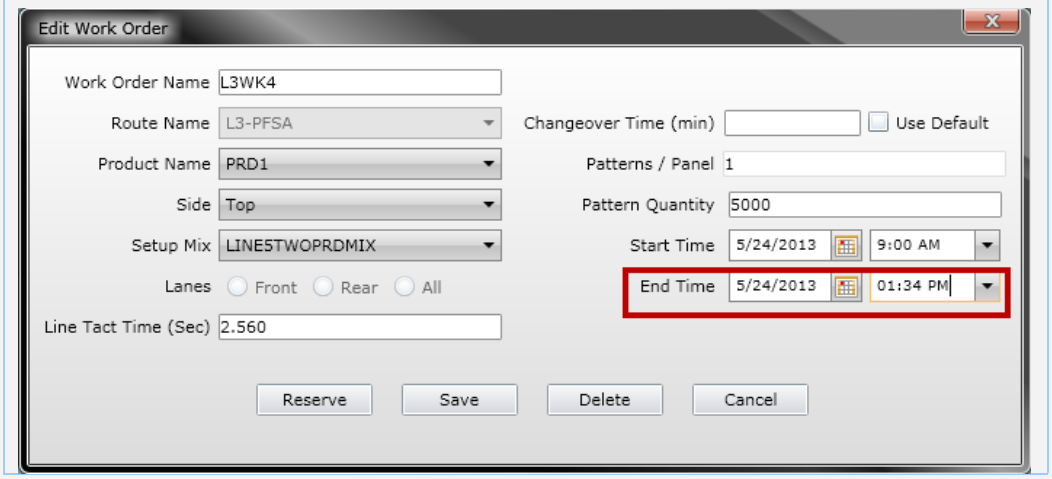
■ Detailed Production Schedule - Include Unplanned Downtime for Work Orders

Users are now allowed to include unplanned down time in their work orders. This is accomplished by allowing the user to adjust the end time of the Work order time range. After the user creates a Work order with all the details including Line Tact time, Patterns/Panel and Pattern Quantity, the system automatically calculates the time range based on the user selected Start time and work order details. Users can now modify this time range for unplanned downtime by changing the End time in the Work order.

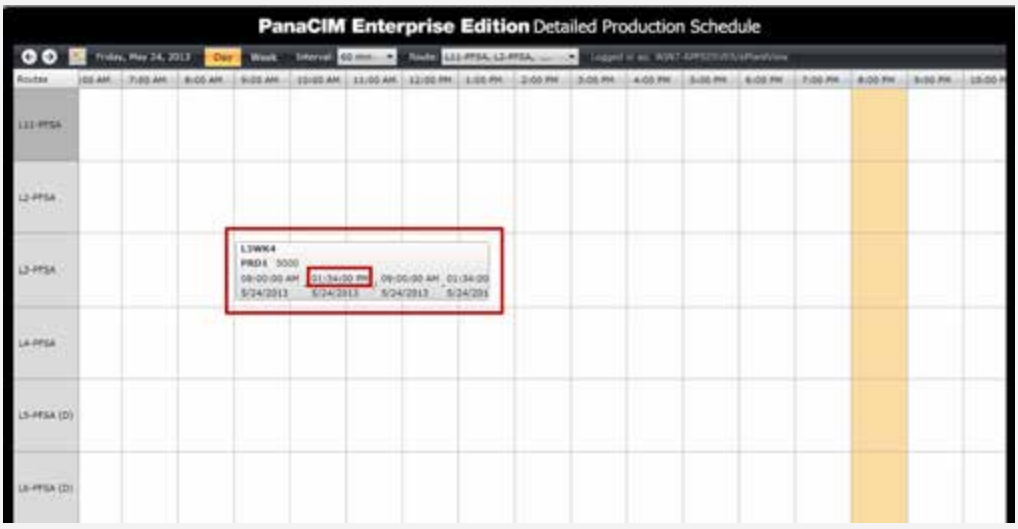
Step	Procedure
1	At the time of creating work order, end time is automatically calculated based upon Line Tact time, Patterns/Panel and Pattern Quantity. 
2	Upon changing the pattern quantity, end time of work order automatically changes as shown below: 
3	Click the **Save** button to save the work order. Saved work order appears in the DPS screen as shown below:



4 You can change the automatically calculated end time of the newly created work order as shown below. This is to incorporate any unplanned downtime.



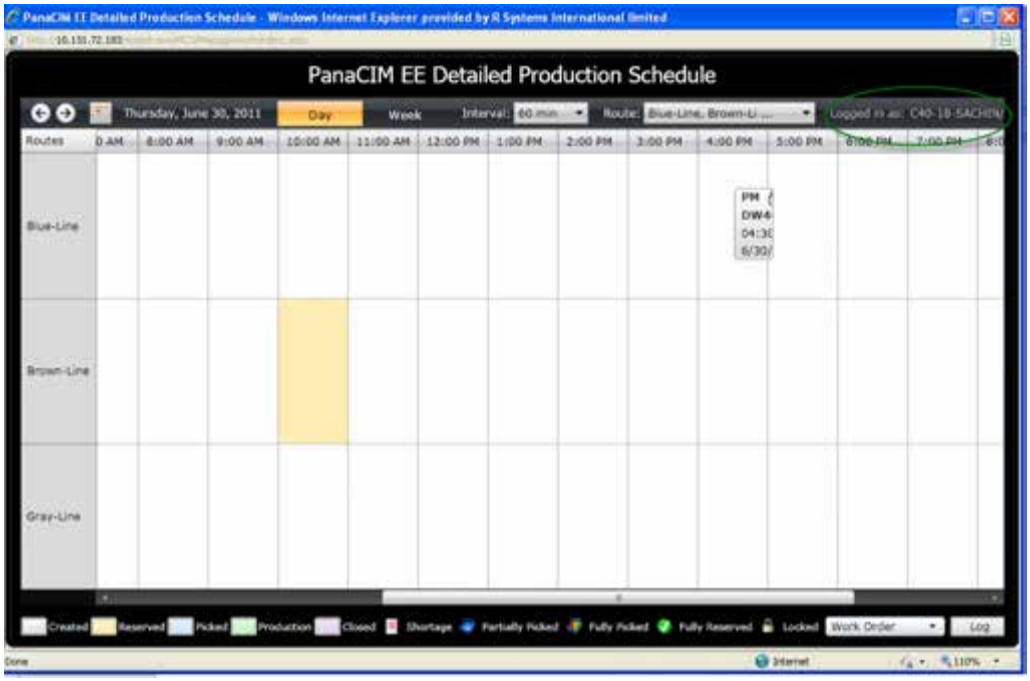
5 Manually changed end time is reflected in the work order upon clicking **Save** button:



3.3 Operation

3.3.1 Material Planning - Detailed Production Schedule - Login ID on Detailed Production Schedule

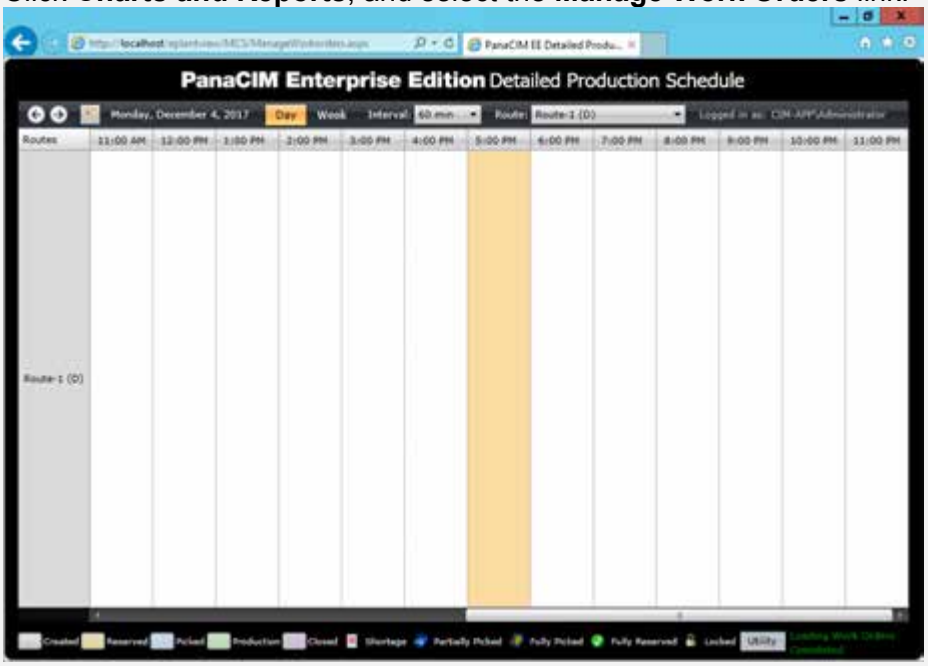
The login ID information is made available on the Top- Right side of the Detailed Production Schedule screen. This is indicative of the status of the user.

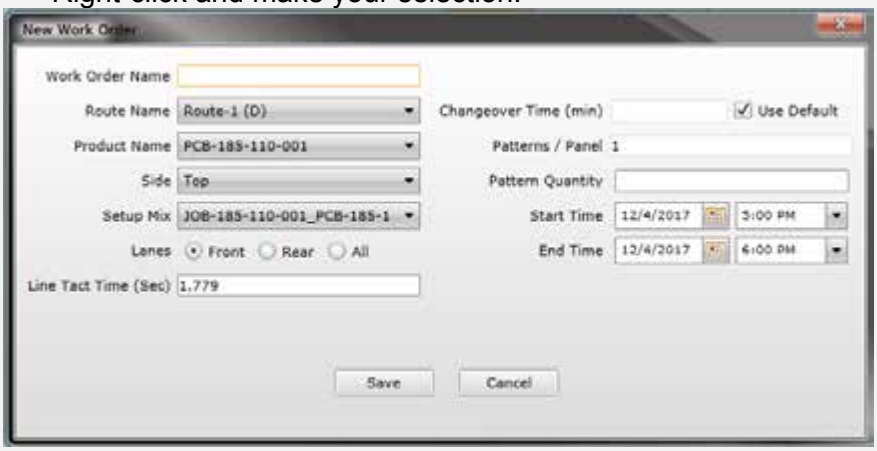


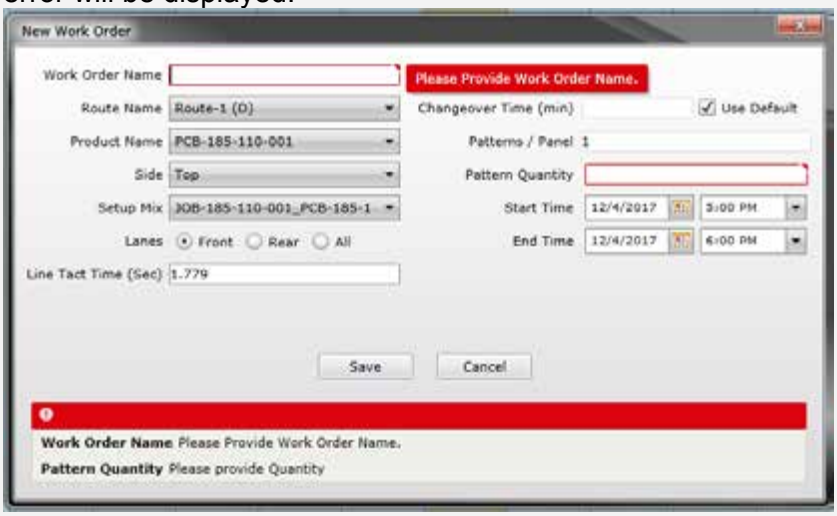
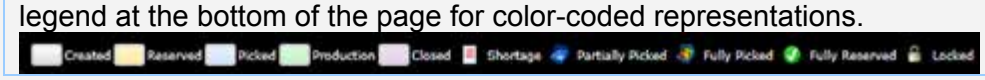
3.3.2 Material Planning - Detailed Production Schedule - Work Order Management

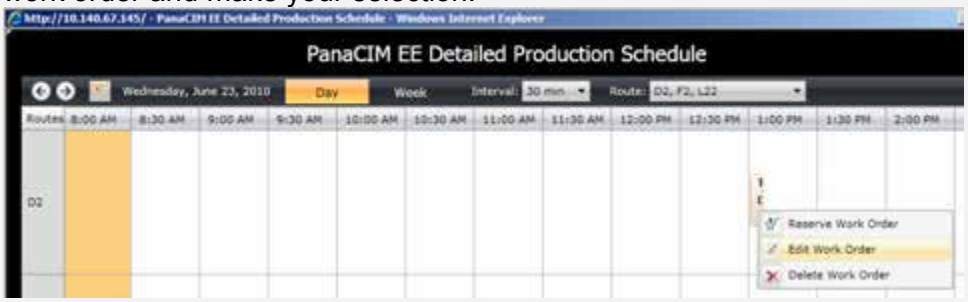
Work orders must be created and reservations and shortages must be configured for each Route during the Material Planning Phase. Please note that a work order should be created only if it has been confirmed for production. After work order creation, materials must be reserved to ensure they are dedicated to your work order.

Refer to the action table below for guidance in using the Detailed Production Application to manage your work orders.

Step	Procedure
1	Access the **PanaCIM EE Launchpad**. a. Open a web browser and enter the appropriate URL in to the **Address** field. • using the server: http://<localhost>/ePlantView/ • using a PC other than the server: http://<webservername>/ePlantView b. Enter your **User name** and **Password**. When logging in to the Launchpad for the first time, log in as Administrator with a local user ID (not a domain ID). By default, local Administrators have access to all lines and reports, and can provide access to all other users. c. Click **OK**. The PanaCIM EE Launchpad appears. **Notes**: The fonts and layouts will be displayed differently depending on the browser type used. The PanaCIM EE Launchpad is compatible with the following browser types: • Internet Explorer 5.0 to 7.0 (Windows) **Important:** Using browser buttons or navigating to another web address can result in loss of data or Application errors. Also note that Explorer menu selections (right-click) **Back** and **Forward** are not compatible with this version.
2	Click **Charts and Reports**, and select the **Manage Work Orders** link. 

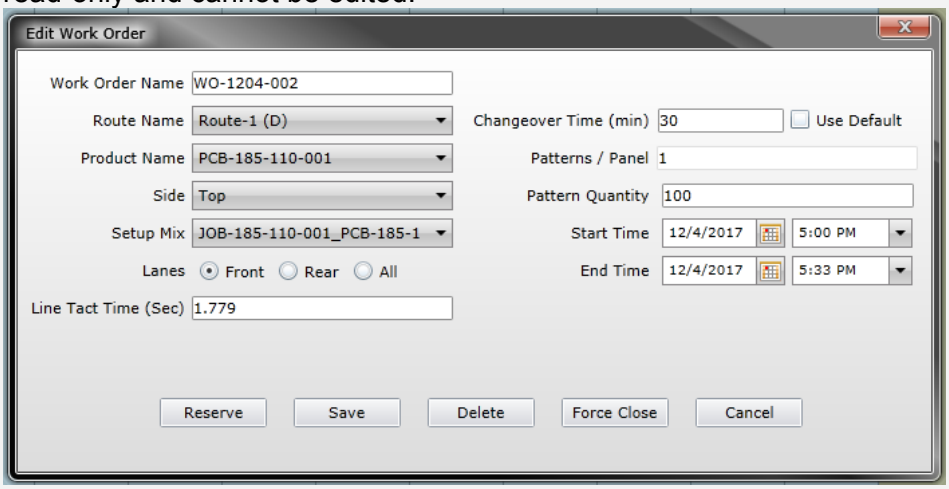
3	Invoke the **New Work Order** dialog by doing one of the following: • Click the Schedule page. • Right-click and make your selection.  **Fields** • **Work Order Name:** Enter a name for the work order. • **Route Name:** Use the drop-down selection to choose a route. The default displays the first route in the list. • **Product Name:** Use this drop-down list to choose a product associated with the route. If no products are associated with the route, the list will remain empty. • **Side:** Use the drop-down to select the board side (top or bottom) based on the product selected. The default displays side 'Top'. • **Setup Mix:** Use the drop-down to select the mix related to the product. The default displays the first mix of the product. • **Line Tact Time (Sec):** This can be edited and automatically populated as long as the data is held in the database. • **Patterns / Panel:** This can be edited and automatically populated as long as the data is held in the database. • **Pattern Quantity:** Enter the number of patterns to be manufactured per board. • **Start Time:** Enter the time to start production. Time already past is not allowed. The default displays the best available time for the selected route based on Quantity and Cycle Time. • **End Time:** Enter a production end time. User can change the end time to adjust for down times amount in each Work Order Time range • **Changeover Time (min):** This optional field is populated with the value from the product setup table if you click **Use Default**. If nothing is available, the value will be 0. However, you have the ability to change the value specific to your work order. If you opt to enter a value, the minimum value can be 0 minutes and the maximum value can be up to 99999 minutes.

3	**Notes:** • To minimize error when completing the work order page, dependent controls are available only after the predecessor value is valid. If you happen to miss a field, an error will be displayed.  • When Line Tact Time, Pattern Quantity, and Start Time are entered or modified, the end time will be calculated automatically. However, you may modify the End Time if desired.
4	After completing the fields, click **Save**. The Work Order is now saved to the main interface. A snapshot representation of the work orders for each route will appear on this interface. To view the entire snapshot, hover your mouse over it to invoke a popup view, or double-click the snapshot to display the entire work order.  This snapshot responds to real time messages and updates accordingly. If there is a delay in start time and/or the end time differs from the expected end time, the snapshot displays the altered time in red font. The state of the work order is also represented in the snapshot by color. Refer to the legend at the bottom of the page for color-coded representations. 

5	You have the ability to modify your display by selecting from daily or weekly views, as well as the time interval you would like to view for each route. Please note that: • The default time period is for the current day for the selected route, and the starting point is an hour prior to the current time. • You have the option to use a calendar to aid in scheduling. Click the calendar icon to invoke this feature. • From the Route drop-down list at the top of the page, select only the lines (routes) you have interest in displaying. This application only has the ability to display what will fit in the production schedule window (scrolling is not permitted in this application). 
6	If for some reason you want to modify or delete your work order, simply right-click the work order and make your selection.  **Note:** You may also invoke the Edit Work Order dialog by double-clicking an existing work order.

6 The following guidelines apply:

- **Edit Work Order:** Special guidelines apply for editing work orders based on the state of the work order. If the work order status is anything other than 'Created' it will be read-only and cannot be edited.

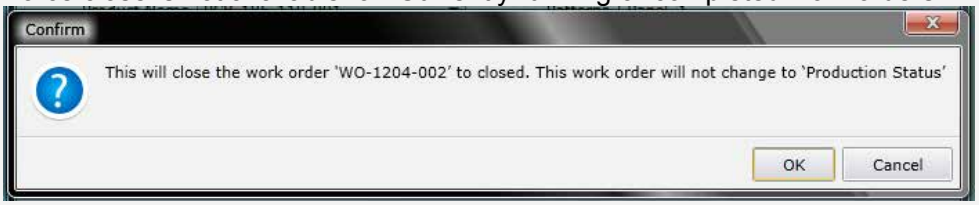


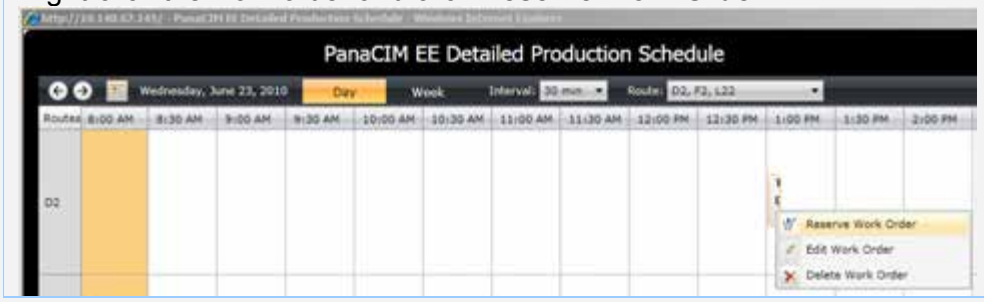
- **Delete Work Order:** Click **Delete** to remove the work order. *A currently running, or completed work order cannot be deleted.* Work orders can be deleted only if they are not running or not completed. Keep in mind that if you choose to delete a work order:
  - All prepared materials are released, and associated reservations are cleared.
  - All prepared materials will be kept in the last known locations.
  - All material reservations are cleared, and these materials are available for new reservations.

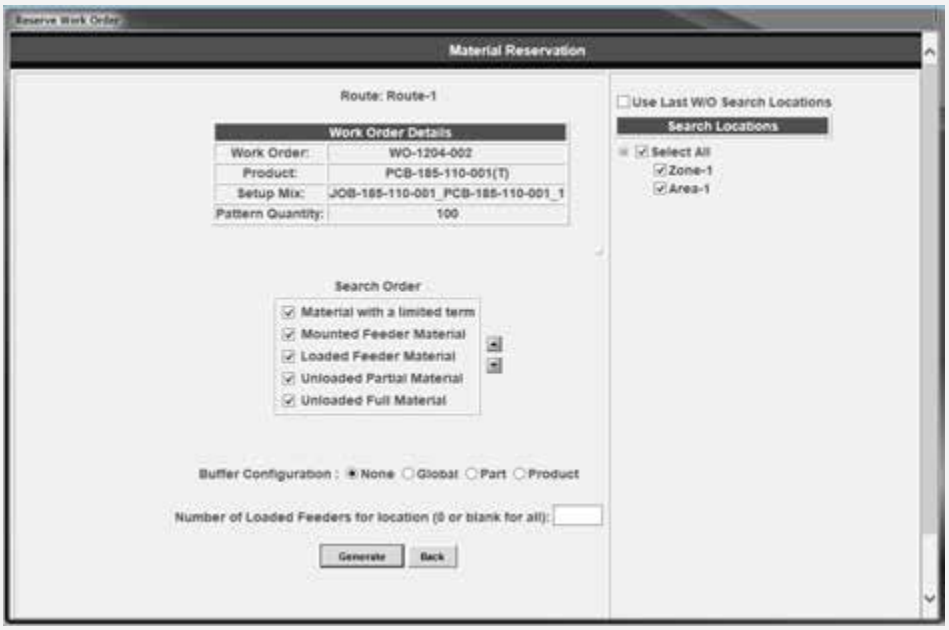
  The following operations require you to manually delete a work order and then create a new work order. During these operations the work order name can be reused for the newly created work order.
  - Splitting the work order into 2 or more work orders.
  - Move a work order within a route.
  - Move a work order from one route to another.

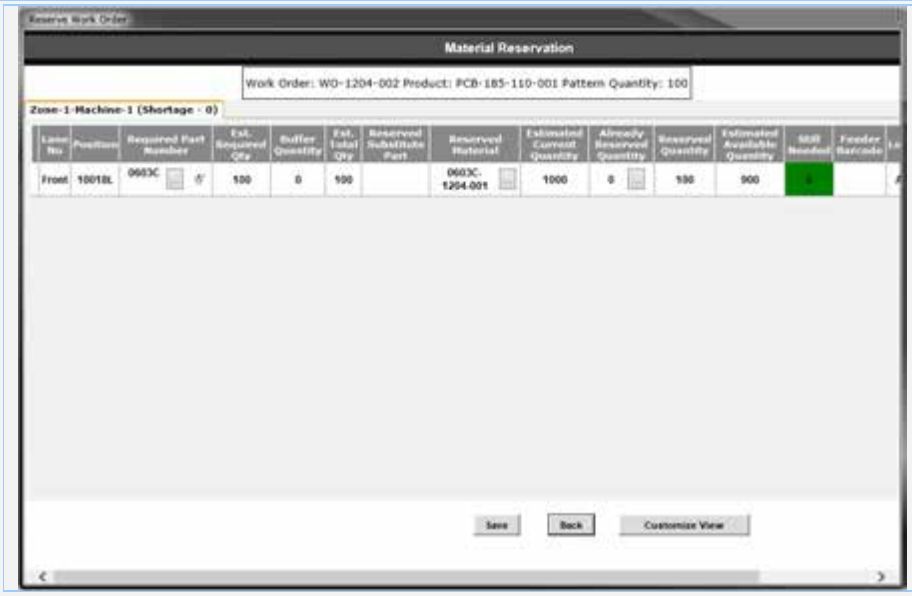
- **Force Close Work Order:** Choose **Force Close Work Order** from the right-click menu. The dialog shown below appears. Clicking **OK** closes the work order forcibly.

  Force close is available for: Created, reserved or collected work orders
  Force close is not available for: Currently running or completed work orders



7	After you have satisfied your new work order selections, materials must be allocated/reserved in order to dedicate them to your work order. Right-click the work order and click **Reserve Work Order**. 

8	The system automatically displays the previous work order based on the schedule unless this is the first time the changeover has been generated. In that case, the system will not consider any material on the machine. The Product, Setup Mix and Quantity appear for the previous and current work order. a. **Search Order**：You can set search priorities for target material conditions (positions). Select your search criteria by placing a checkmark next to the item. Modify the search order by clicking the arrows to the right of the search box. *If more than one material exists under the same condition, a material registered earlier has propriety to be reserved. **Note:** Differences between a Full and Partial Material: • **Unloaded Full Material:** Reels for which the current quantity is the same as the initial quantity. • **Unloaded Partial Material:** Reels for which the current quantity is not the same as the initial quantity. *This is for BM series machines only*. • Select an **Exchange Mode**. *This is for BM series machines only*. b. **Buffer Configuration**：Select when buffer is used for the reserved parts. c. **Search Location**：Select a zone/area of search target. Selecting '**Select All**' will search all locations. d. Click **Generate**. 
9	Place a **checkmark** next to your selections and click **Reserve**. Click **OK** at the dialog prompt. A summary of your shortages and reservations appears.


Material Reservation
Work Order: WO-1204-002 Product: PCB-185-110-001 Pattern Quantity: 100
Zone-1-Machine-1 (Shortage - 0)
Front
10018L
0603C
500
0
100
0603C-1204-001
1000
0
500
900
Save
Back
Customize View

The table below outlines the main State transitions for a Work Order in the Detailed Production Schedule. For some main States, there are applicable Sub-states.

From	Trigger	To	Description
None	1. Manual Entry 2. File Import	Created	1. User manually enters and commits a work order. 2. User imports a work order from an external .CSV file.
Created	At least one Reservation is committed	Reserved	User commits a component reservation for at least 1 and up to all locations.
		Partial	Reservation(s) exist for at least 1 location but less than all locations.
		Full	Reservation(s) exist for all locations.
Reserved	All reservations deleted	Created	User has deleted every reservation for all locations of the work order.
Reserved	Pick operation started	Picked	User has begun to pick reserved components for the work order.
		Partial	At least one component is picked, but not all.
		Full	System detects all components are picked.
Picked	All components un-picked	Created	User has unpicked all components for every location. No reservations existed before the picking operation started.
Created	Pick operation started	Picked	User has begun picking components without any previous reservations.
Created, Reserved, Picked, PreSetup	Work order activated in Route	Production	User has selected the work order from Product Transfer and activated it in a Route.
Production	1. New work order activated in the Route. 2. Manual close.	Closed	User has selected the next work order from Product Transfer and activated it in a Route. User manually closes the work order from Route Status.

Sub-states follow the below guidelines:

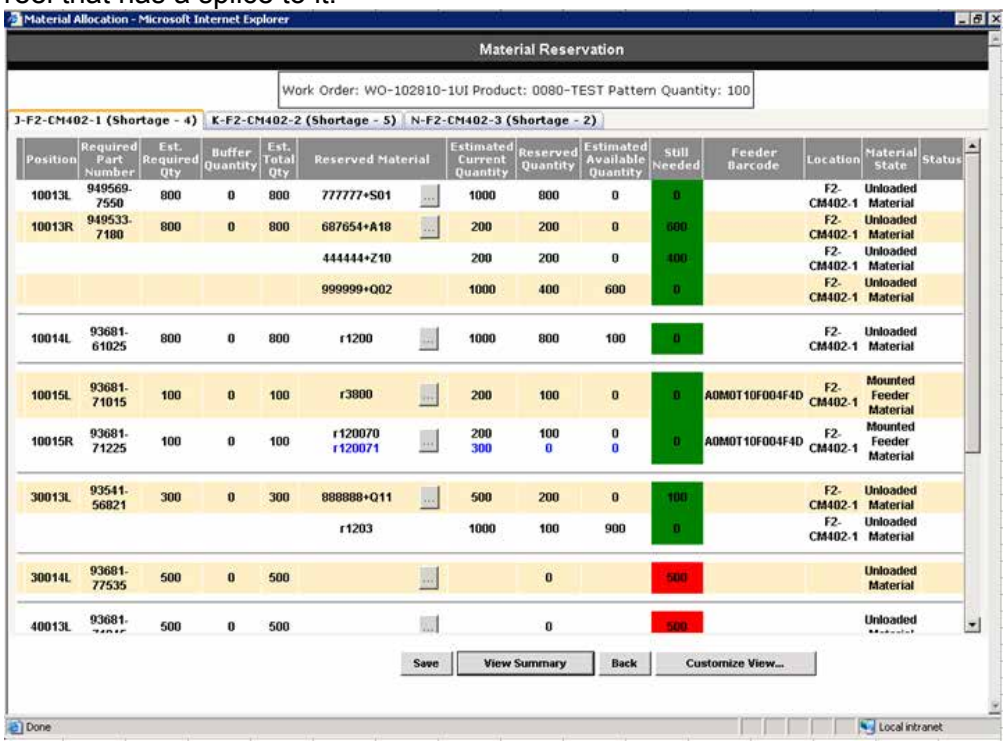
State	Sub-States	Reserved	Picked	PreSetup
Reserved	Partial / Full	Manage Reserve Sub-State value	Retain / Manage last known sub-state if Reserved state was previously active	Retain / Manage last known sub-state if Reserved state was previously active

Picked	Alert / Partial / Full / Incomplete		Manage Picked sub-state value	Retain / Manage last known sub-state if Picked state was previously active.
PreSetup	Started / Completed			Manage PreSetup sub-state value

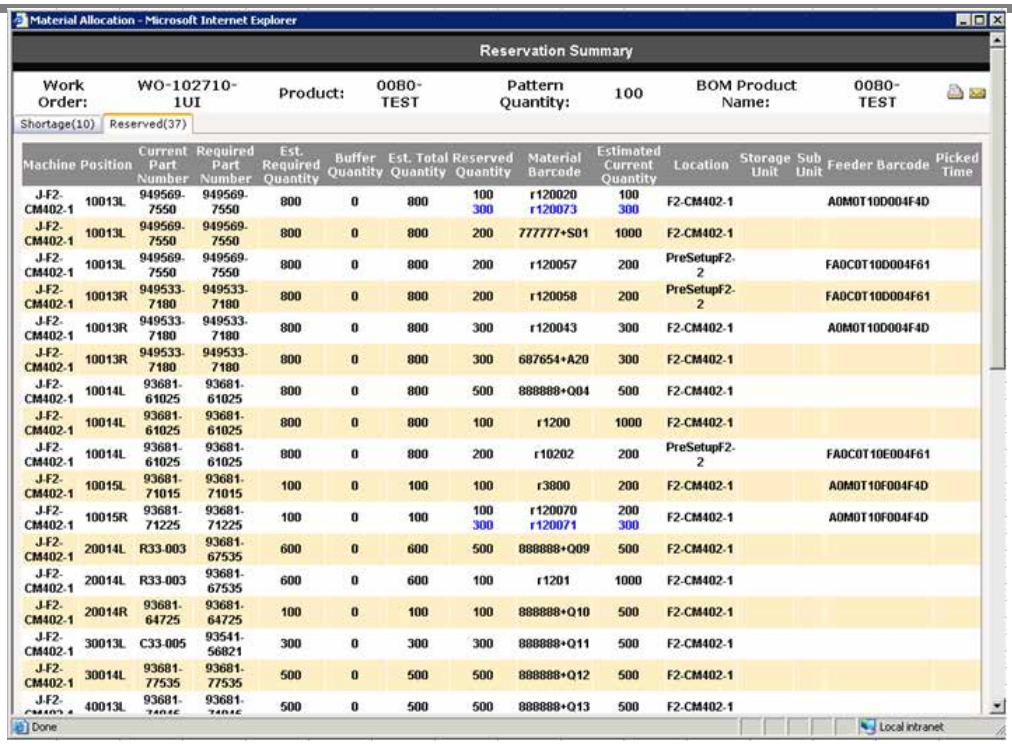
- **Splice Reel Management in the Context of Material Reservation**

The Reservation operation has been updated to display the Splice Reel Material ID of any reel that has a reel spliced to it. It has also been updated to mark both the Main Reel and Splice Reel as Reserved, if either is selected for reservation.

If the Reservation includes a spliced reel, the Material ID of the reel that has the reel spliced to it appears on the Material Reservation page. The Main Reel and Splice Reel will show as Reserved if either reel is selected for reservation. The Reservation Summary Report will display the Splice Reel Material ID of any reel that has a splice to it.



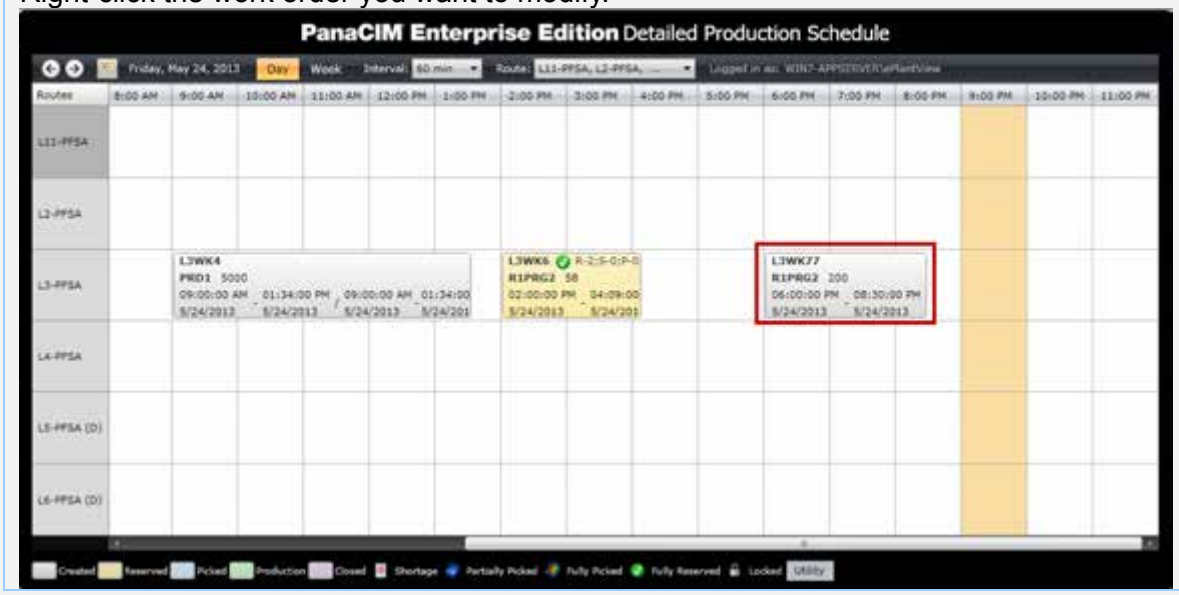
3.3 Operation

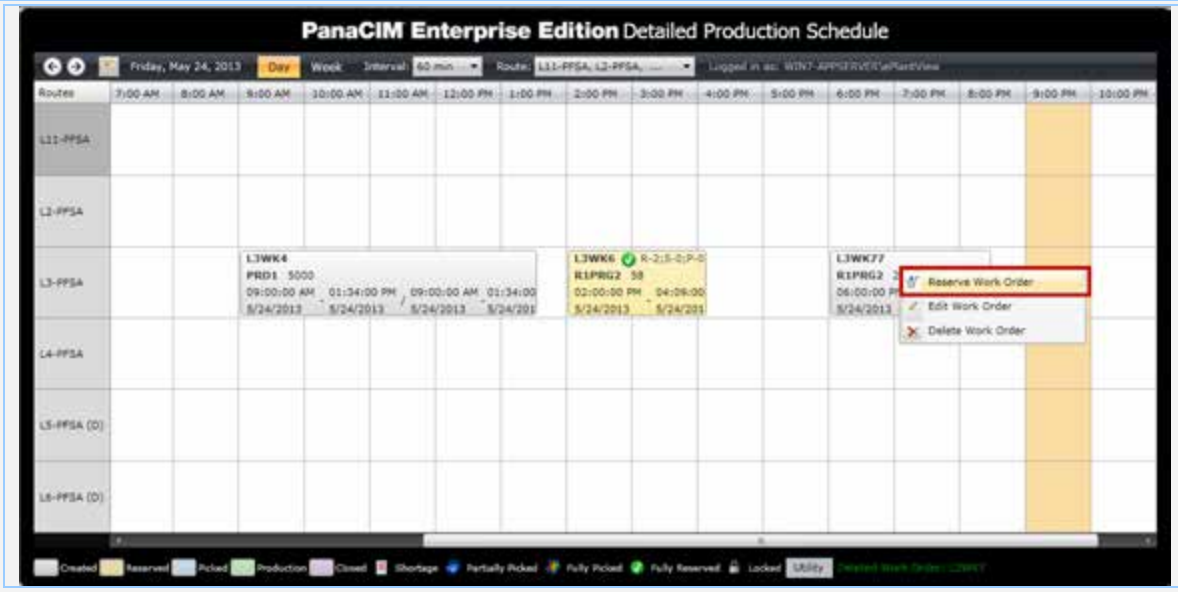


Machine	Position	Current Part Number	Required Part Number	Est. Required Quantity	Buffer Quantity	Est. Total Quantity	Reserved Quantity	Material Barcode	Estimated Current Quantity	Location	Storage Unit	Sub Unit	Feeder Barcode	Picked Time
J-F2-CM402-1	10013L	949569-7550	949569-7550	800	0	800	100 300	r120020 r120073	100 300	F2-CM402-1			A0M0T10D004F4D	
J-F2-CM402-1	10013L	949569-7550	949569-7550	800	0	800	200	777777+S01	1000	F2-CM402-1				
J-F2-CM402-1	10013L	949569-7550	949569-7550	800	0	800	200	r120057	200	PreSetupF2-2			FA0C0T10D004F61	
J-F2-CM402-1	10013R	949533-7180	949533-7180	800	0	800	200	r120058	200	PreSetupF2-2			FA0C0T10D004F61	
J-F2-CM402-1	10013R	949533-7180	949533-7180	800	0	800	300	r120043	300	F2-CM402-1			A0M0T10D004F4D	
J-F2-CM402-1	10013R	949533-7180	949533-7180	800	0	800	300	687654+A20	300	F2-CM402-1				
J-F2-CM402-1	10014L	93681-61025	93681-61025	800	0	800	500	888888+Q04	500	F2-CM402-1				
J-F2-CM402-1	10014L	93681-61025	93681-61025	800	0	800	100	r1200	1000	F2-CM402-1				
J-F2-CM402-1	10014L	93681-61025	93681-61025	800	0	800	200	r10202	200	PreSetupF2-2			FA0C0T10E004F61	
J-F2-CM402-1	10015L	93681-71015	93681-71015	100	0	100	100	r3800	200	F2-CM402-1			A0M0T10F004F4D	
J-F2-CM402-1	10015R	93681-71225	93681-71225	100	0	100	100 300	r120070 r120071	200 300	F2-CM402-1			A0M0T10F004F4D	
J-F2-CM402-1	20014L	R33-003	93681-67535	600	0	600	500	888888+Q09	500	F2-CM402-1				
J-F2-CM402-1	20014L	R33-003	93681-67535	600	0	600	100	r1201	1000	F2-CM402-1				
J-F2-CM402-1	20014R	93681-64725	93681-64725	100	0	100	100	888888+Q10	500	F2-CM402-1				
J-F2-CM402-1	30013L	C33-005	93541-56821	300	0	300	300	888888+Q11	500	F2-CM402-1				
J-F2-CM402-1	30014L	93681-77535	93681-77535	500	0	500	500	888888+Q12	500	F2-CM402-1				
J-F2-CM402-1	40013L	93681-	93681-	500	0	500	500	888888+Q13	500	F2-CM402-1				

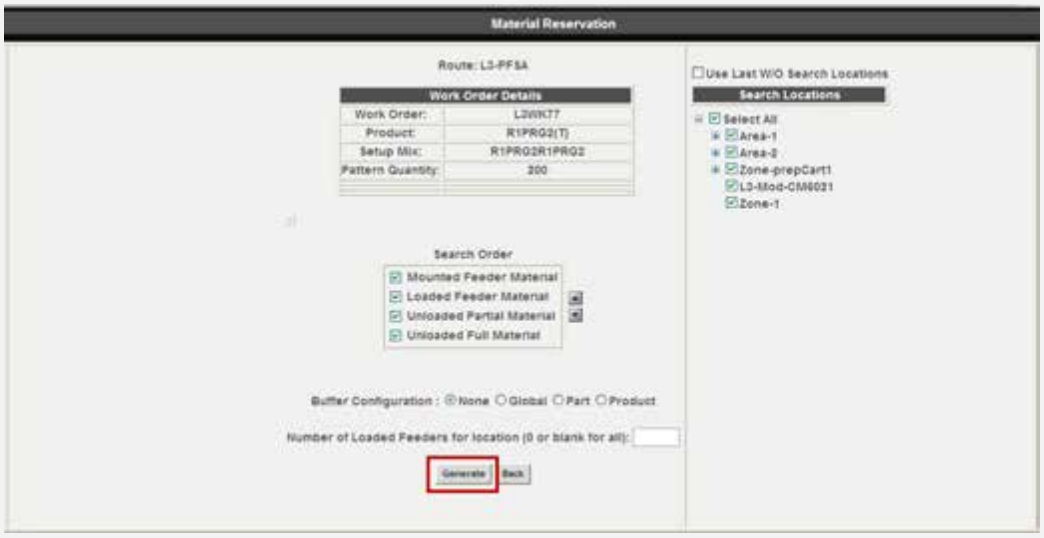
3.3.3 Material Planning - Detailed Production Schedule - The display of the amount of reserved components

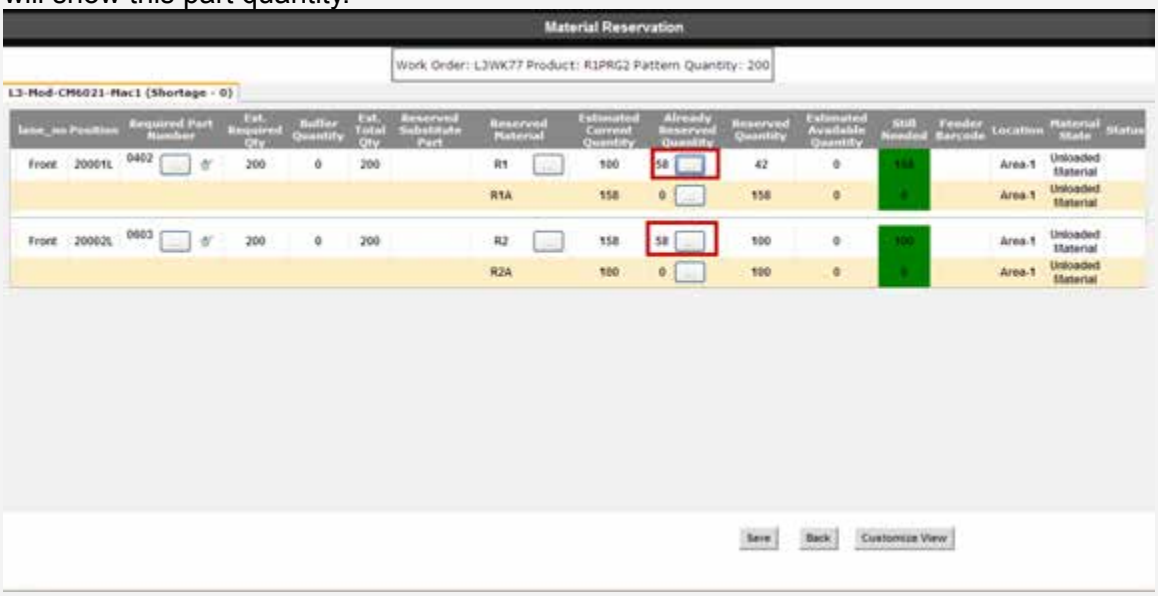
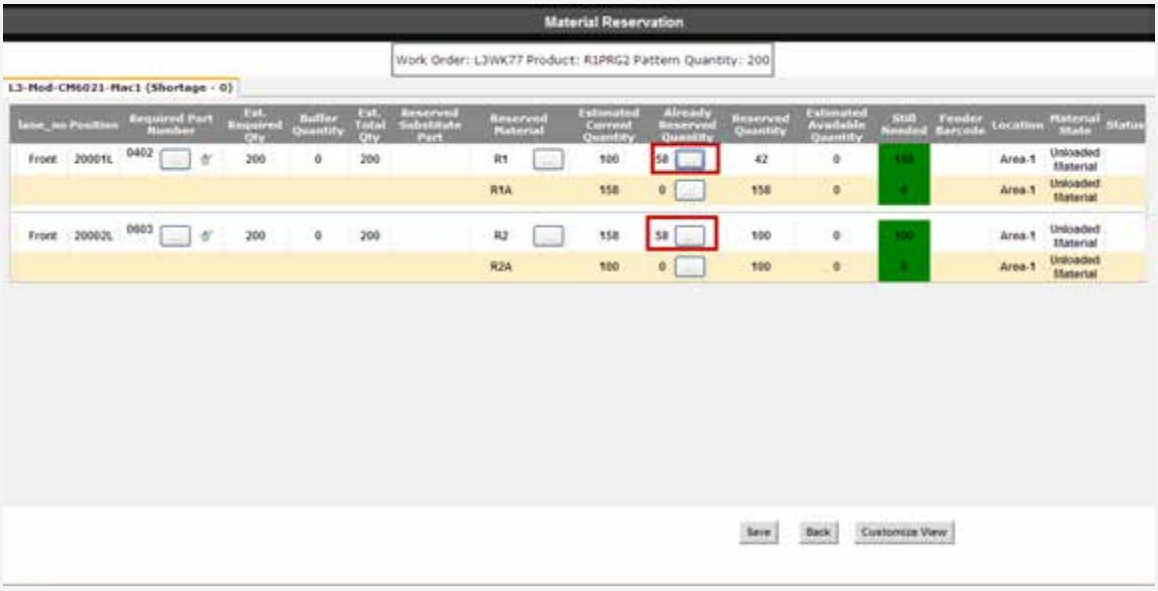
An additional column for Already Reserved Quantity is made to Material Reservation in PanaCIM Enterprise Edition ePlantview.

Step	Procedure
1	Open the PanaCIM Web application.
2	Click on **Charts and Reports**. 
3	Click **Manage Work Orders** to display DPS screen.
4	Right-click the work order you want to modify. 
5	Click **Reserve Work Order**.

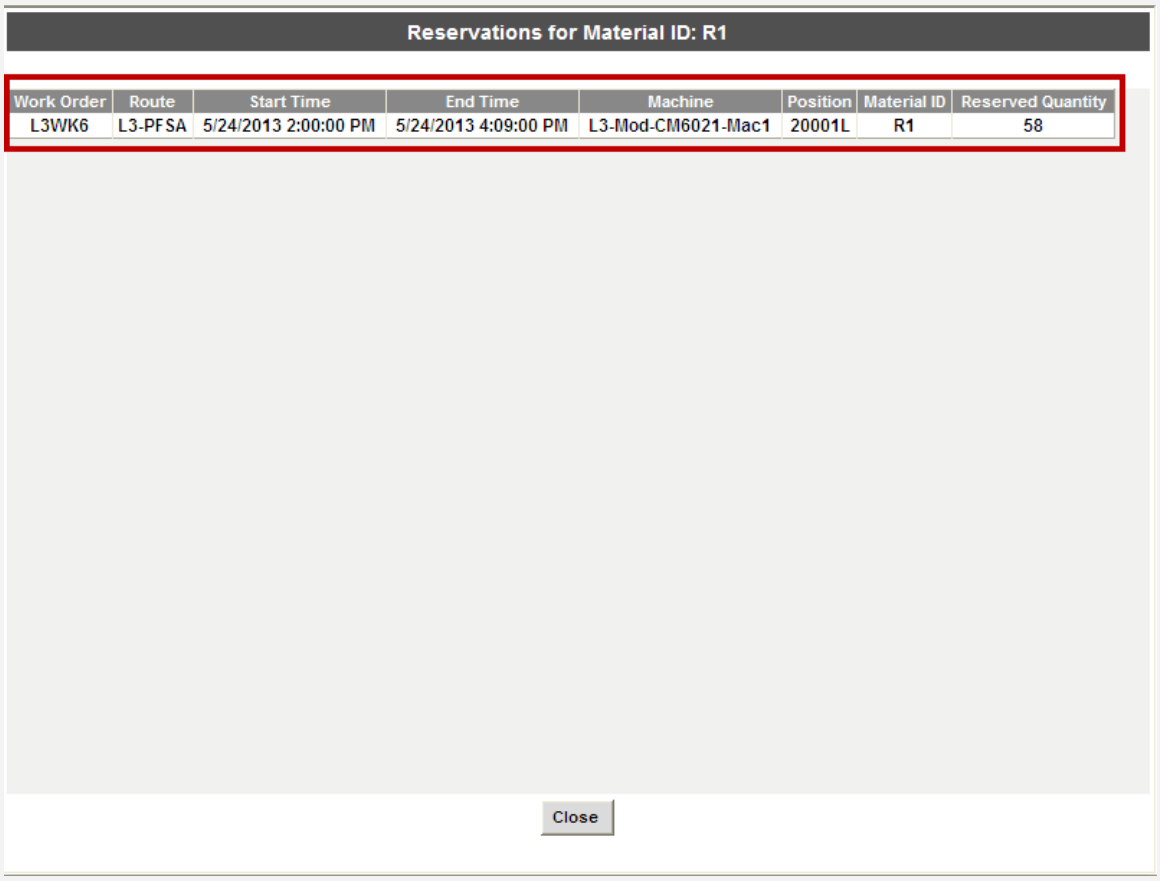


6 Click on **Generate** to proceed with work order reservation.



7	Material Reservation screen appears showing **Already Reserved Quantity** column along with other columns. This column will contain 0 if the reel is not reserved for any work order. If some part quantity in the reel is reserved for earlier work order, this column will show this part quantity. 
8	Click on the ... button in **Already Reserved Quantity** column to view part quantity that is alreay reserved for earlier work order. A report of **Reservation for Material ID** with Required Part Number will be displayed. 

9	Reserved quantity will be shown in a report of reservation for the material ID. One feeder reel can be reserved for one work order in one line at a time.

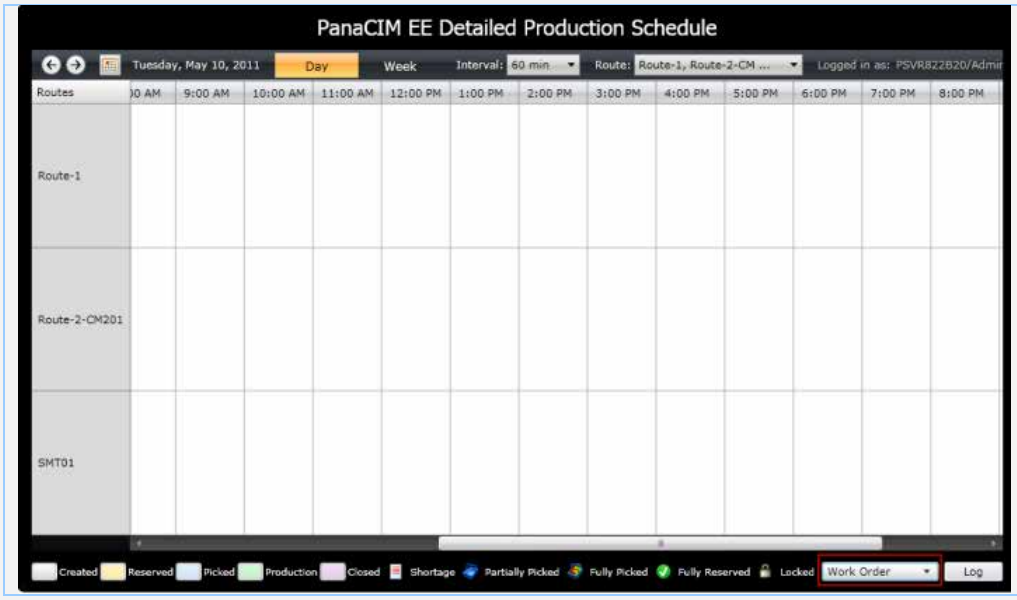
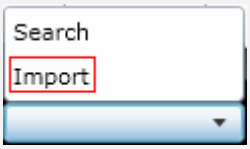
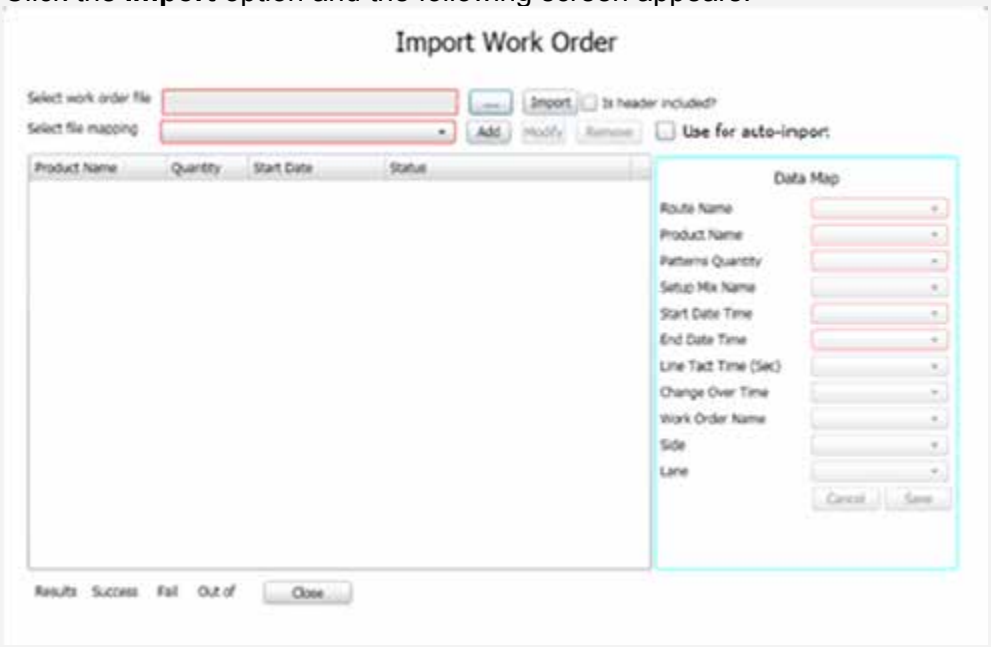


3.3.4 Material Planning - Detailed Production Schedule - Import Work Orders from a csv file

Work orders can be created in bulk using the import option through Production Analysis manage work orders functionality. It can create the work order from input CSV file. A work order contains important information, such as route name, product name, quantity, start date, end date, and status related to the production schedule of a product.

Usually production details required to manufacture a product are existing in the csv file. This csv file can then be imported using newly created **Import Work Order** page existing in the Production Analysis application.

Step	Procedure
1	Open **Web application**.
2	Click **Chart and Reports** from the menu. 
3	Click **Manage Work Orders**.
4	<Detailed Production Schedule> screen appears

Step	Procedure
	
5	Click the **Work Order** button at the bottom right corner. An Option appears to import work orders as shown below: 
6	Click the **Import** option and the following screen appears:  The fields displayed on the **Import Work Order** screen are described in the following table:

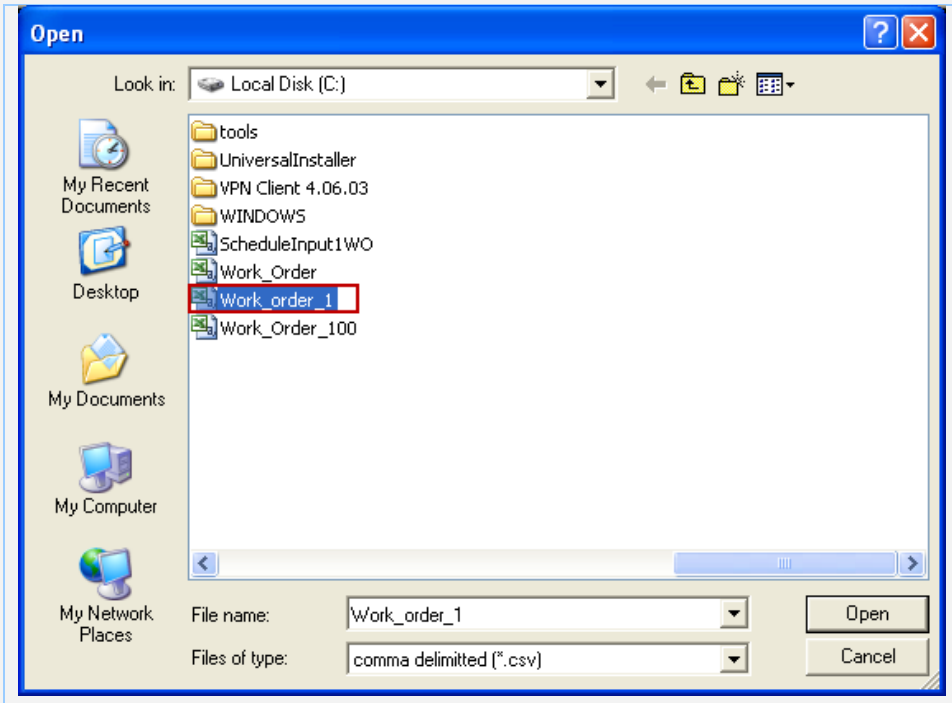
Field Name	Description
Select work order file	Refers to the name of CSV file containing work order details.
….	Launches File Open dialog to select files from local or network folder.
Import	Imports work orders from the selected file after data mapping is established in **Data Map** section of the page.

Step	Procedure

Is header included?	Check the checkbox if a csv file has column header. If this is enabled, the column header of the selected csv file is displayed in a data map. If this is disabled, column 1, column 2 are displayed.
Select file mapping	Used to select a user-defined mapping for the selected work order file. To add, enter a definition name.
Add	Enables **Data Map** section in order to define a new mapping between production parameters on the left hand side and the individual columns of the selected csv file using the columns name listed on the right-hand side.
Modify	Used to modify an existing user-defined mapping for a csv file.
Remove	Deletes an existing user-defined mapping for a csv file.
Use for auto-import	If this option is checked, automatic import is enabled using the selected file maping.
Route Name	Refers user to map column containing route name in the selected csv file.
Product Name	Refers user to map column containing product name in the selected csv file.
Patterns Quantity	Assigns the number of patterns in the selected csv file.
Setup Mix Name	Assigns a column of setup mix name in the selected file.
Start Date Time	Assgins a column of the production start date and time in the selected file.
End Date Time	Assgins a column of the production end date and time in the selected file.
Line Tact Time (Sec)	Assigns a column of the line tact time in the selected file. Refers the line tact time as the time taken to produce one PCB.
Change Over Time	Assigns a column of the changeover time in the selected file. Refers the change over time as the time taken to carry out changeover between ending of last work order and starting of new work order.
Work Order Name	Assigns a column of the work order name in the selected csv file.
Side	Assigns a column of the PCB side (Top or Bottom) in the selected csv file.
Lane	Assigns a column of the placement machine lane number in the selected csv file.

7	Click the ellipsis button ( ) adjacent to **Select work order file** textbox. **File open** dialog appears to select a csv file as shown below:

Step	Procedure



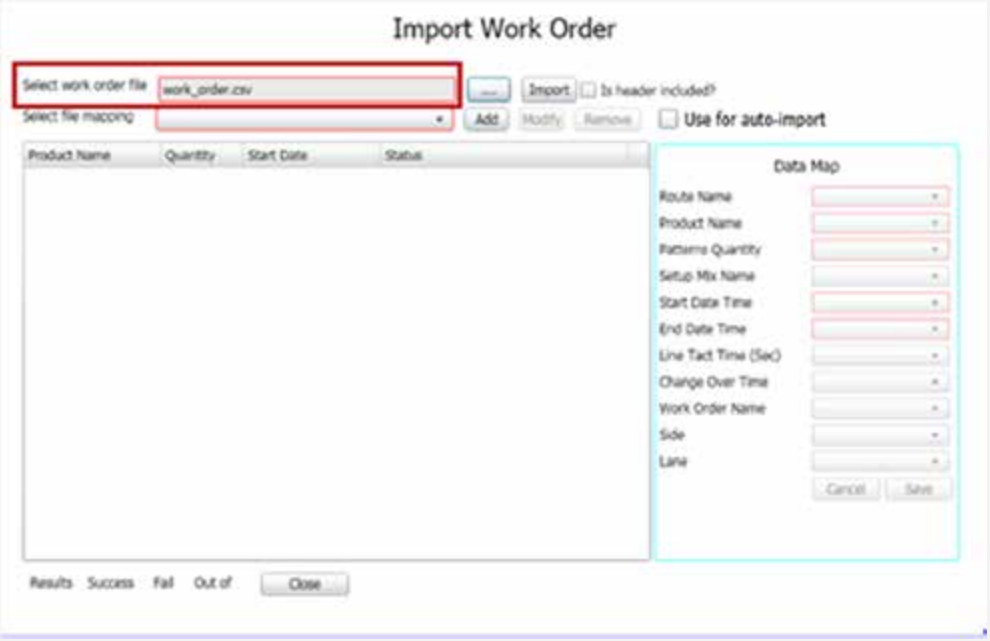
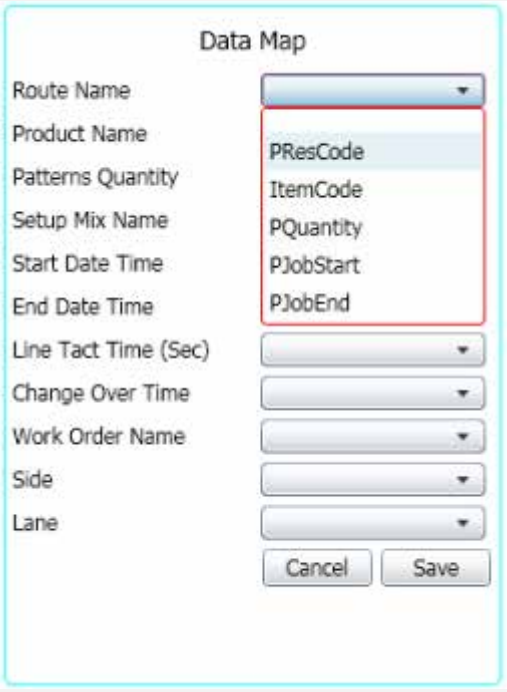
Select the name of the csv file and click the **Open** button.

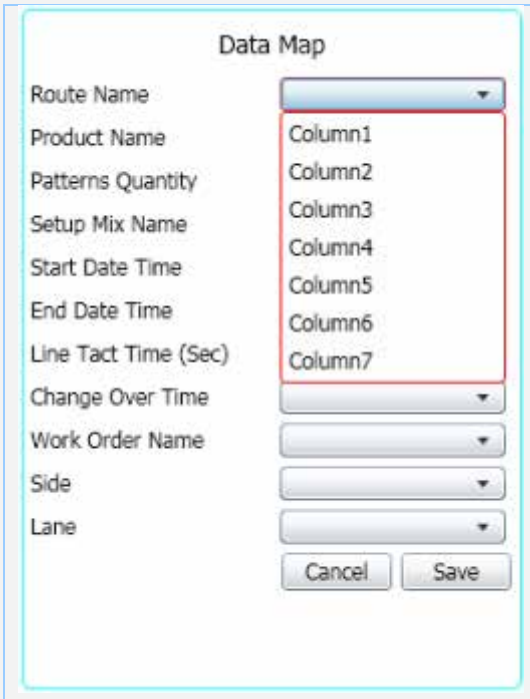
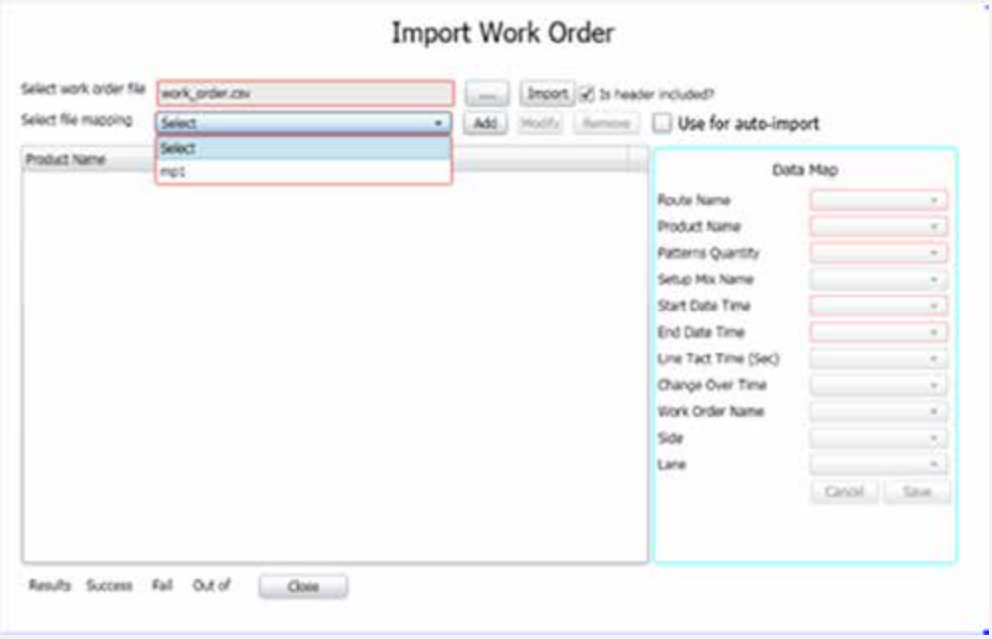
A csv file to be used must be created with the contents of the followings.

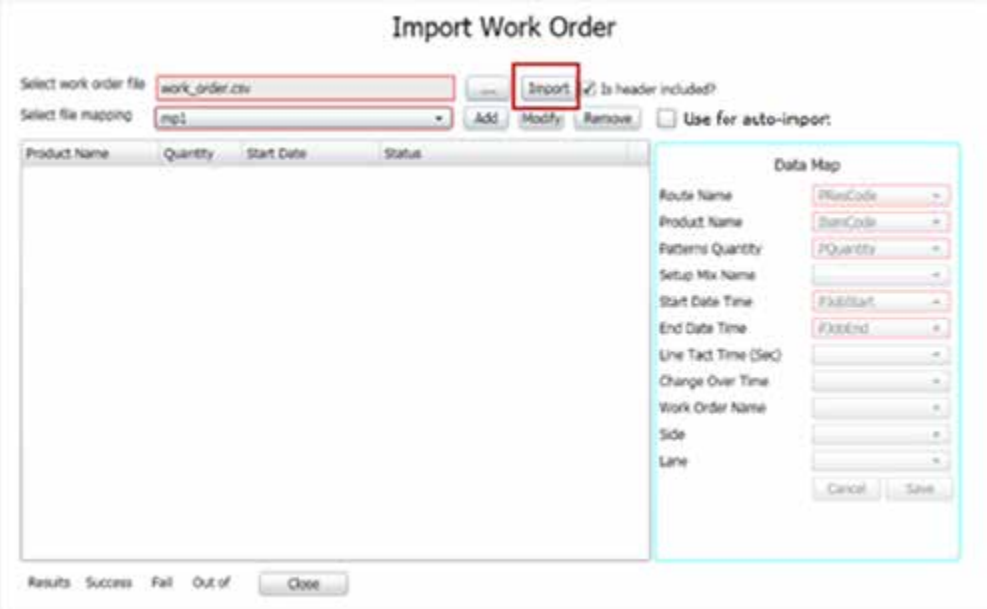
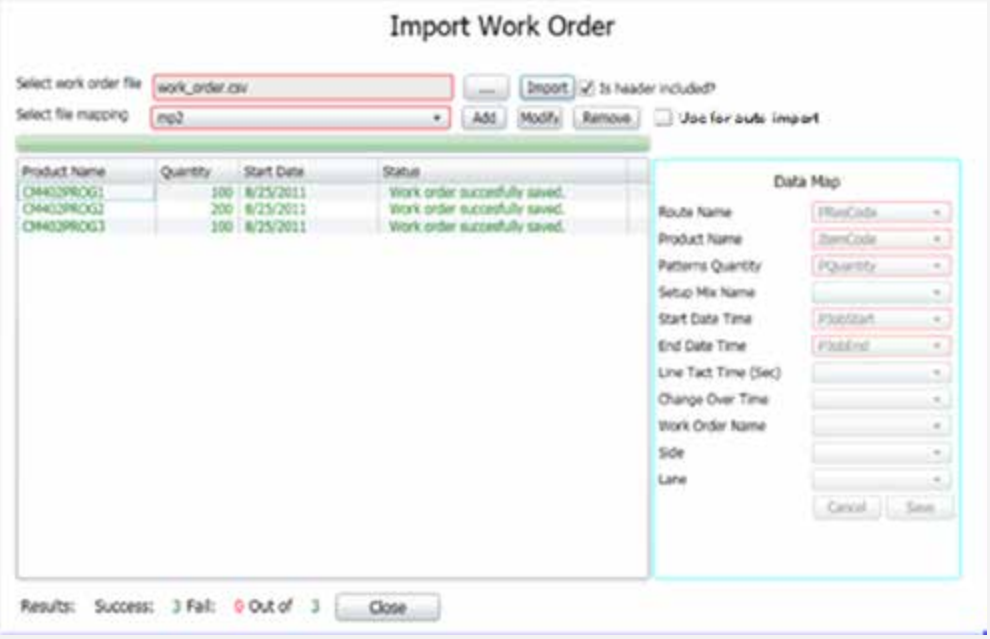
route	product	qty	mix	start	end	tact	change	order	side	lane
Route-1	Product01	500	JOB01_PRODUCT01	04/08/2017 05:00	04/08/2017 06:00	10	5	Order01	T	0
Route-1	Product02	500	JOB02_PRODUCT02	04/08/2017 06:30	04/08/2017 07:30	10	5	Order01	B	1
Route-1	Product03	500	JOB03_PRODUCT03	04/08/2017 08:00	04/08/2017 09:00	10	5	Order01	T	2
Route-1	Product04	500	JOB04_PRODUCT04	04/08/2017 09:30	04/08/2017 10:30	10	5	Order01	B	0
①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪

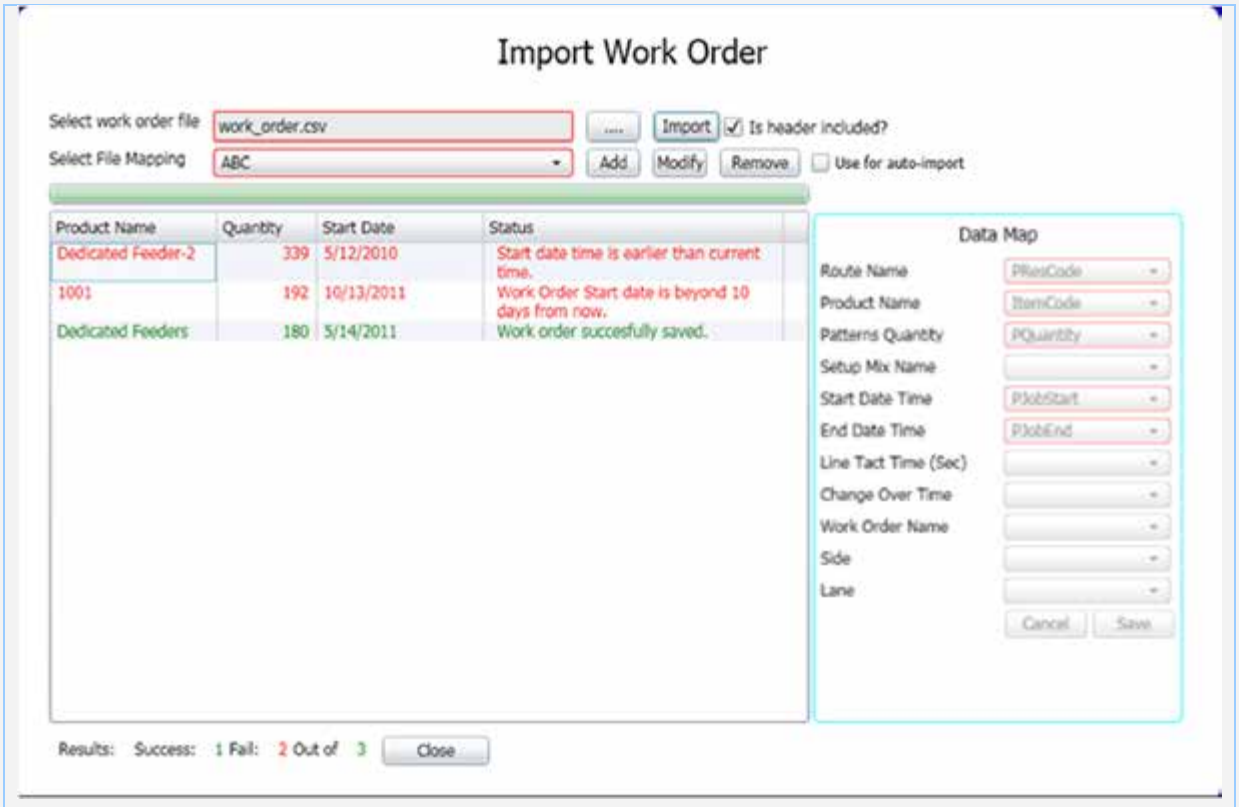
*The following describes points to note of input values when a CSV files is created.

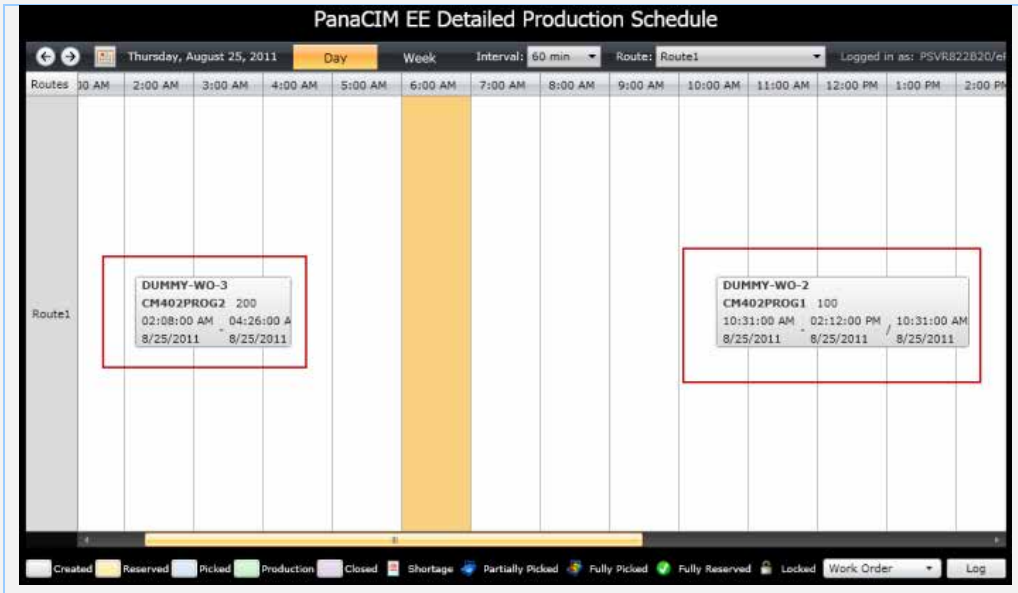
①Route (Route Name)	Even though (D) for dual or (S) for single are assigned to a route name on the production order screen, do not assign when you input the name.
②Proudct (Product Name)	Enter the lot (PCB) name as same as the one displayed on the product setup screen.
③Pattern (Pattern Quantity)	Enter an integer value.
④Mix (Setup Mix name)	Enter a mix name displayed on the product setup screen.
⑤Start (Start Date Time)	Enter the start date in the format “MM/DD/YYYY HH:MM”
⑥End (End Date Time)	Enter the end date in the format “MM/DD/YYYY HH:MM”.
⑦Tact (Line Tact Time (Sec))	Enter an integer value.
⑧Changeover (Changeover Time)	Enter an integer value.
⑨Order name (Work order name)	If an order name under the same work order name and product name has been registered before, it cannot be registered.
⑩Side	Enter the production PCB using “T” for front and “B” for back.
⑪Lane	If production is performed on the front lane, enter “0”, if it is performed on the rear lane, enter “1” and if it is performed on both lanes, enter “2”.

Step	Procedure
8	Click the ellipsis (....) button adjacent to the **Select work order file** textbox and select the CSV file saved on your system. Name of the selected CSV file is populated in the **Select work order file** textbox as shown below: 
9	Click the **Add** button. Now the controls are enabled in the **Data Map** section of the page. If the **Is header included?** option was selected earlier, actual column headers will be presented as a list. Otherwise Column1, Column2 appears as list. **Figure: Option "Is header included ?" selected**  **Figure: Option "Is header included ?" not selected**

Step	Procedure
	 To carry out one-to-one mapping using Data Map, a column of the csv file must be defined for each work parameter from the drop-down list.
10	Enter a mapping name in the **Select file mapping** textbox and click the **Save** button at the bottom of Data Map section to finish defining a mapping. At this stage, you can select the mapping name from the **Select File Mapping** dialog as shown below.  **Note:** Import button is still disabled
11	After selecting the name of the mapping file, **Import** button is enabled as shown below:

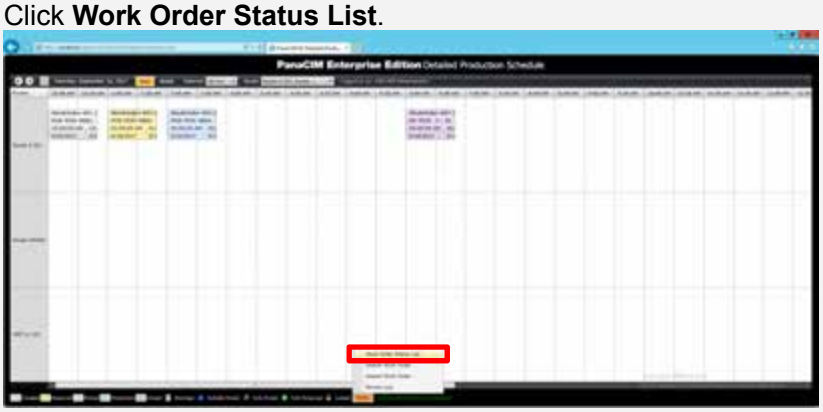
Step	Procedure
	
12	<For manual import> Click the **Import** button. csv file containing work order details is now imported as shown below  **Note:** If there is improper data in the csv file, importing this file results in failures as shown below:

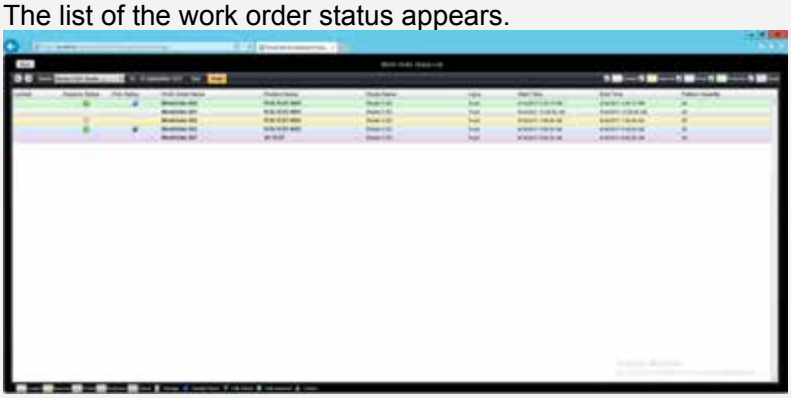
Step	Procedure
	 **Note:** Work orders successfully imported are indicated by green-colored rows while unsuccessful work order imports are indicated by red-colored rows. <For automatic import> Check the **Use for auto-import** checkbox to enable. When a file is arranged to the folder defined in 3.2.2Automatic import of work order, it will automatically be imported. *You do not need to open the import screen for work order. *The order downloaded result is not displayed.
13	Imported work orders are displayed in the **PanaCIM EE Production Schedule screen** as shown below:

Step	Procedure
	

3.3.5 Material Planning -Detailed Production Schedule- Display of Work Order Status List

Use the PanaCIM-EE Web Application to check a list of work order status.

Step	Procedure
1	Open **Web application.**
2	Click **Charts and Reports** from the menu. 
3	Click **Manage Work Orders**.
4	Click **Utility** at the bottom of the schedule table. 
5	Click **Work Order Status List**. 

6	The list of the work order status appears.  The colors of each field and displayed icons are as follows.  You can select which work orders are displayed using the checkbox on the upper right of the screen. The list is automatically updated every five minutes.

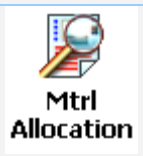
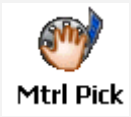
3.3.6 Material Planning - Detailed Production Schedule - The message to Feeder Maintenance (Under Development)

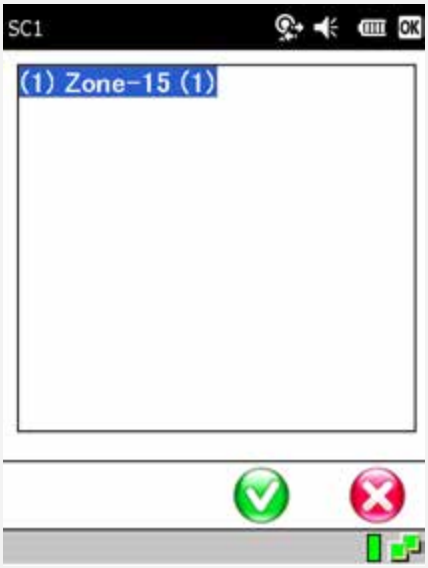
3.3.7 Material Planning - Scanner - Pick Material

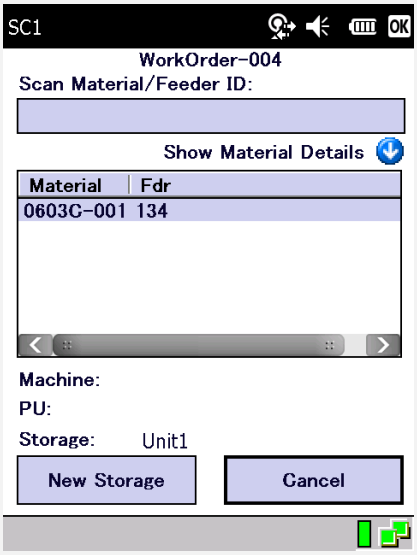
Reserved materials can be picked up (collected) using a mobile scanner for the next setup. The picked up materials are transferred from the scanner to the PanaCIM host and then moved to the production line.

Refer to the action table below to pick up the reserved materials using the mobile scanner.

Note: The following procedures assume that the operator has already had their settings configured in the System Configuration Application.

Step	Procedure
1	Select the Material Handling  mode of the Mobile Scanner.
2	Select  and then  .
3	Select the **Route** and **Product** from the drop-down menus. Then Select a **Work Order** and click the **green** checkmark. 

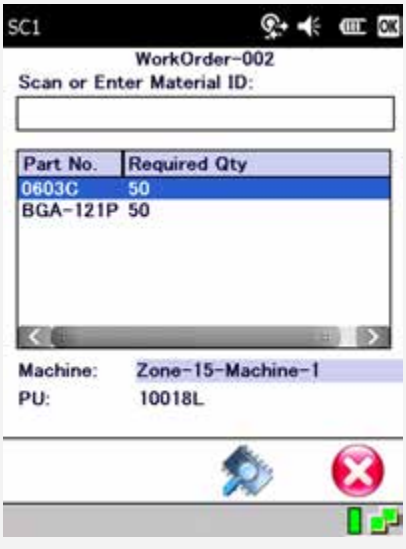
Step	Procedure
4	Scan or select the storage unit containing the materials to be picked up and click the **green** checkmark. 
5	Locations (in alphabetical order) of all materials reserved for the Work Order and the quantity of items in each location to pick are displayed. Select a location to start picking from and click the **green** checkmark. 

Step	Procedure
6	Scan the **Material IDs** of the reserved materials as each one is picked. 
7	After the material ID is entered, the scanner prompts the operator to select the barcode type, as shown below.  The above screen appears only when the material and feeder IDs are entered. If the material and feeder IDs are scanned, the scanner already knows the scanned barcode, and therefore the above screen does not appear.
8	Once all the required materials have been picked, the scanner returns to the Route and Product selection screen. This work order is no longer displayed in the list.

3.3.8 Material Planning - Scanner - Shortage Fulfillment

In addition to allocating materials via the Detailed Production Schedule application, shortages may be fulfilled via a Mobile Scanner. Refer to the action table below to complete backfilling using the Mobile Scanner.

Note: These procedures assume that the operator settings have been configured in the System Configuration Application.

Step	Procedure
1	Select the Material Handling  mode of the Mobile Scanner.
2	Select  and then  .
3	Select the **Route** and **Product** from the drop-down menus. Then Select a **Work Order** and click the **green** checkmark. 
4	The Part No. and Required Quantity appear. Select the **Part Number** and then select the **Material Locate** icon. 

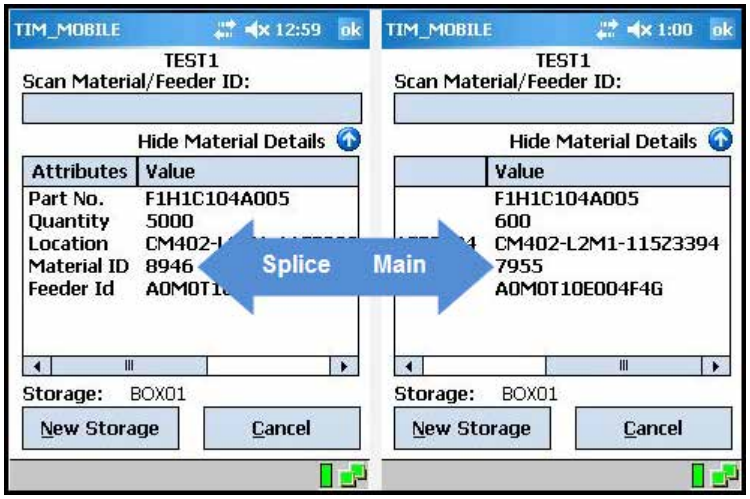
Step	Procedure
5	Detailed information appears. Select the **Material ID** and click the **green** checkmark to fulfill a shortage. (Replenished materials are displayed in the list, if any.)  After shortage of all the parts listed in step 5 is fulfilled, work order is no longer visible in the screen shown in step 4.

- **Splice Reel Management in the Context of Shortage Fulfillment**

The Material Fulfillment Operation has been updated to display attributes of both the Main Reel and the Splice Reel.

In order to view details, click **Show Material Details**. For spliced reels, the first value will be that of the spliced reel. Use the scroll bar to scroll to the right to view the main reel.

In the below example, 8946 is the Material ID of the splice reel and 7955 is the Material ID of the main reel.

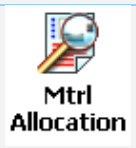


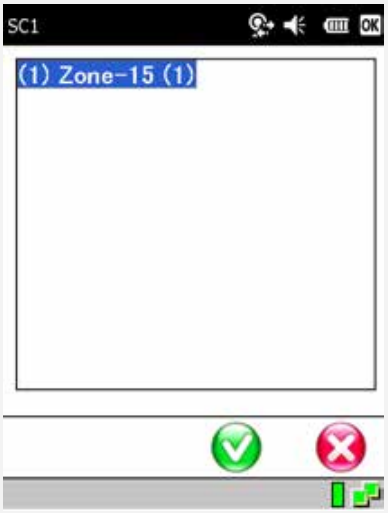
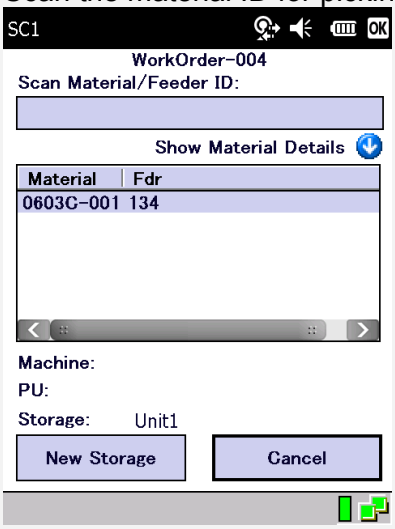
3.3.9 Material Preparation: Scanner/Pickup of Reserved Material

For next presetup, a material to be reserved is picked up by a scanner operation. Data of the picked-up material is transmitted from the scanner to the PanaCIM host, and the material is transfered to the production line.

Refer to the step below for picking up a material using a mobile scanner operation.

Note) The following step is based on the premise that the setup has been completed in the system configuration application.

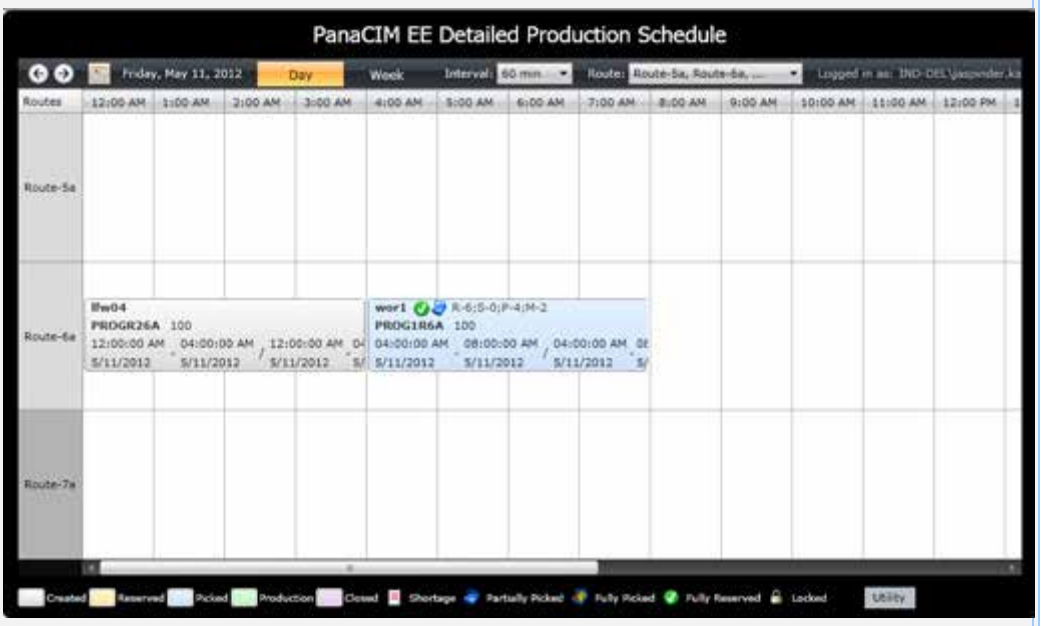
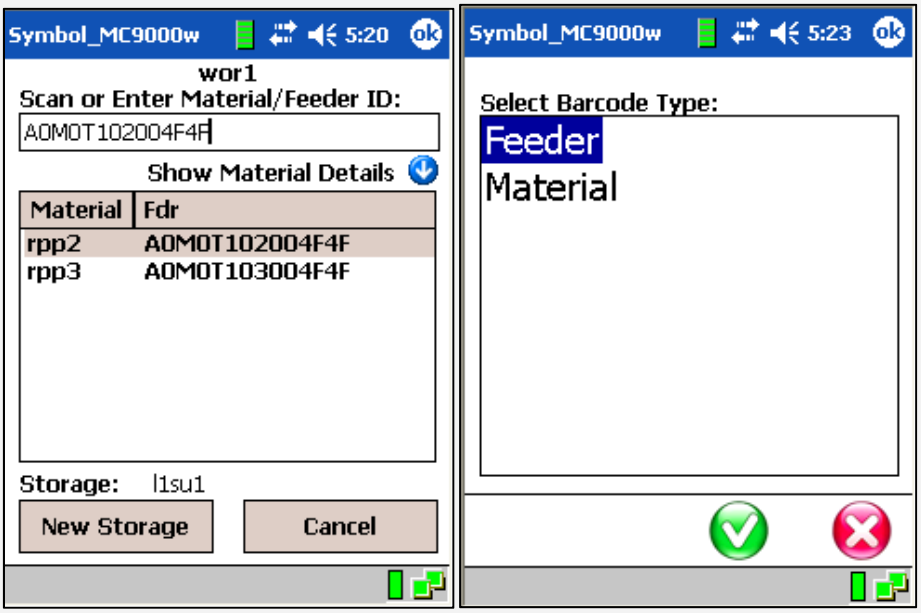
Step	Procedure
1	Select the Material Handling  of the mobile scanner.
2	Select Mtrl Allocation  and then select Mtrl Pick  .
3	From the drop-down menu, select applicable line and production program. Then select an applicable work order and click the green checkmark. 
4	Scan or select the storage unit that a material to be picked up is stored **(move from)** and select the green checkmark.

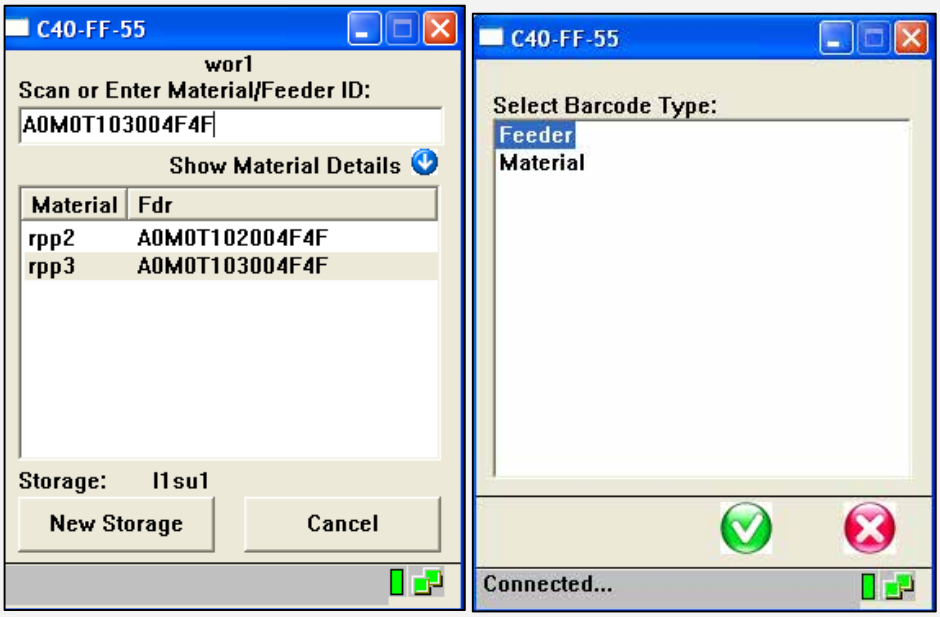
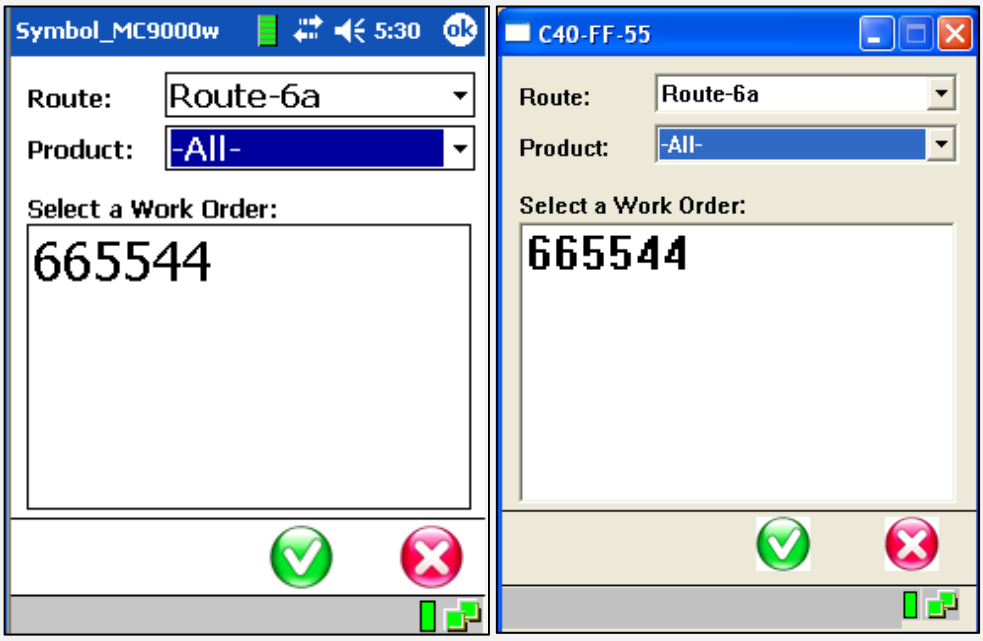
Step	Procedure
	
5	All locations (listed in an alphabetic order) of reserved materials for the work order and their quantities in each location are displayed. Select a location and click the green checkmark. 
6	Scan the material ID for picking up each reserved material, 

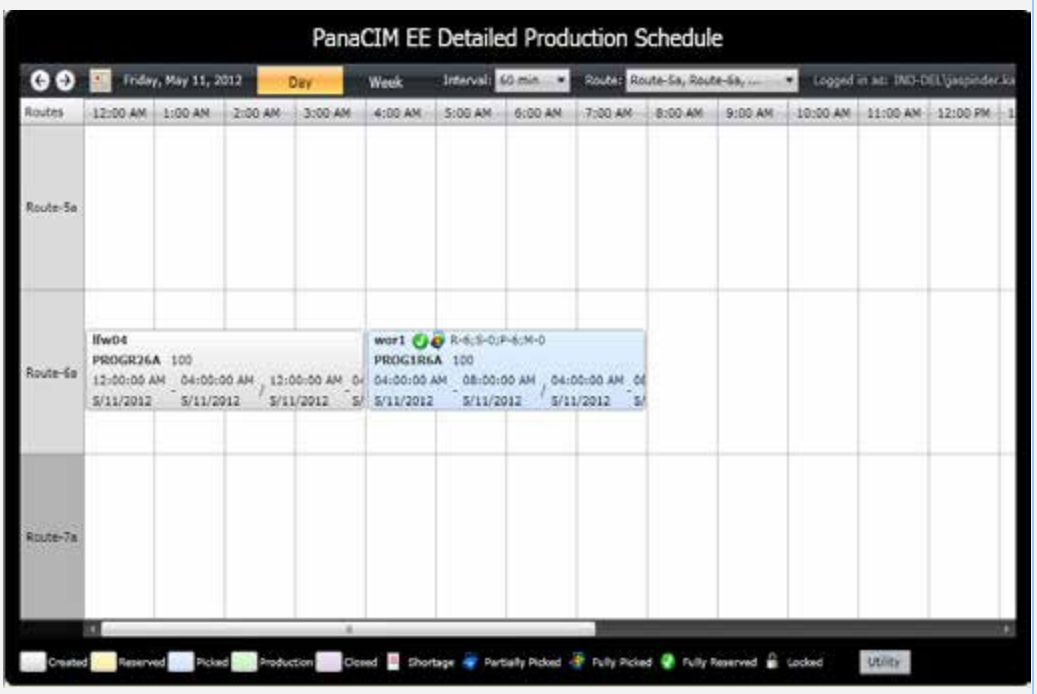
Step	Procedure
7	After typing the material ID, the following screen appears to select the type of the barcode that is keyed in.  If the material/feeder ID is **typed**, the above screen appears. If the material/feeder ID is scanned, the scanner has already determined the barcode type. Therefore the above screen for selecting the barcode type does not appear.
8	After all necessary materials are picked up, the scanner returns to the line and production selection screen. After this, the work order will not be displayed in the list.

3.3.10 Material Preparation - Scanner - Multiple Scanners Performing Pick on Same Work Order

Multiple Scanners can be used to perform material pick operation for the same work order. This is shown below:

Step	Procedure
1	When two scanners start pick operations on the same work order, state of DPS screen is as shown below:  A “Partially Picked” appears in the work order, indicating that the reserved materials are being picked up.
2	In the First scanner, feeder barcode is scanned for picking as shown below: 

Step	Procedure
	At the same time when feeder barcode is being scanned by first scanner, another feeder barcode is being scanned by second scanner. This is shown below: 
3	Instead of one operator scanning two feeder barcodes twice, each of operator has scanned feeder barcode once (from his scanner) assuming that there are two operators picking material for the same work order. Once the two reels are picked, work order (in this case wor1) disappears from both the scanners as shown below: 

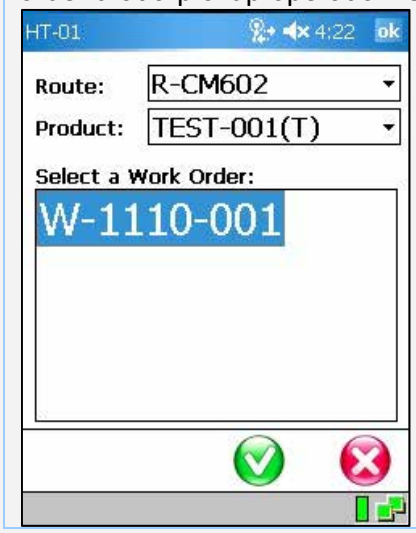
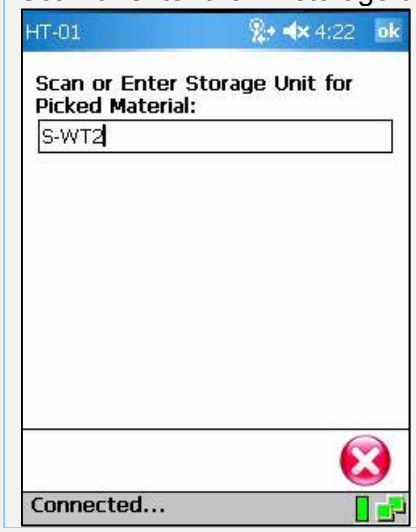
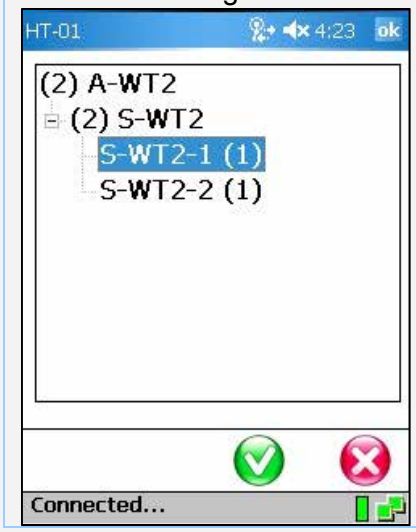
Step	Procedure
4	State of DPS screen after material pick from two scanners simultaneously is shown below:  A “Fully Picked” icon appears in the work order, indicating that all the reserved materials have been picked up (collected)0.

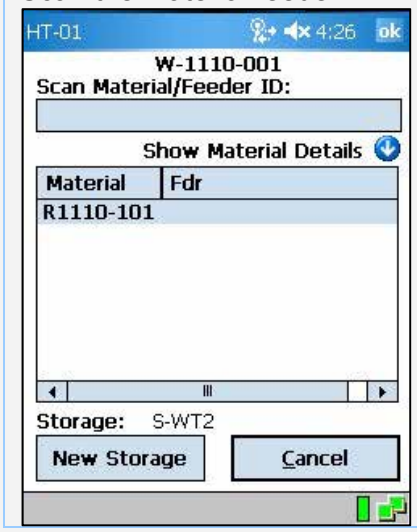
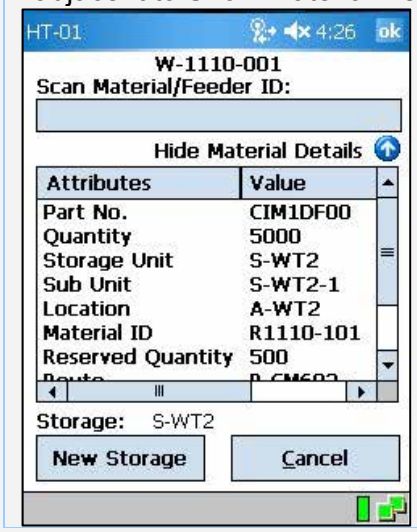
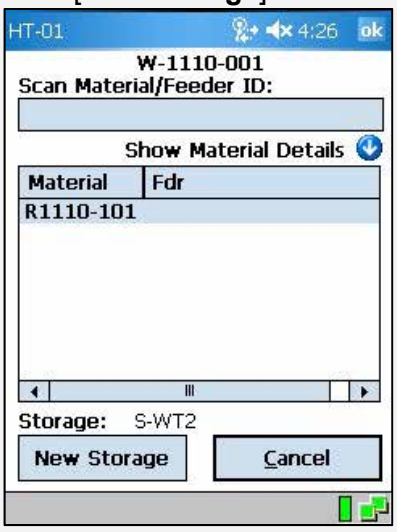
3.3.11 Material Preparation: Scanner/Pickup of Reserved Material (L2 Storage Unit)

The mobile scanner has an extended function related to a pickup operation when materials in the L2 storage unit (Level 2) are picked up.
Note) It is available only for a mobile scanner.

A pickup operation of materials in the L2 storage unit is as follows.

Step	Procedure
1	Using a mobile scanner, move to Production mode. 
2	Click the Material Handling  for opening Mtrl Handling.
3	Click Mtrl Allocation in Mtrl Handling. 

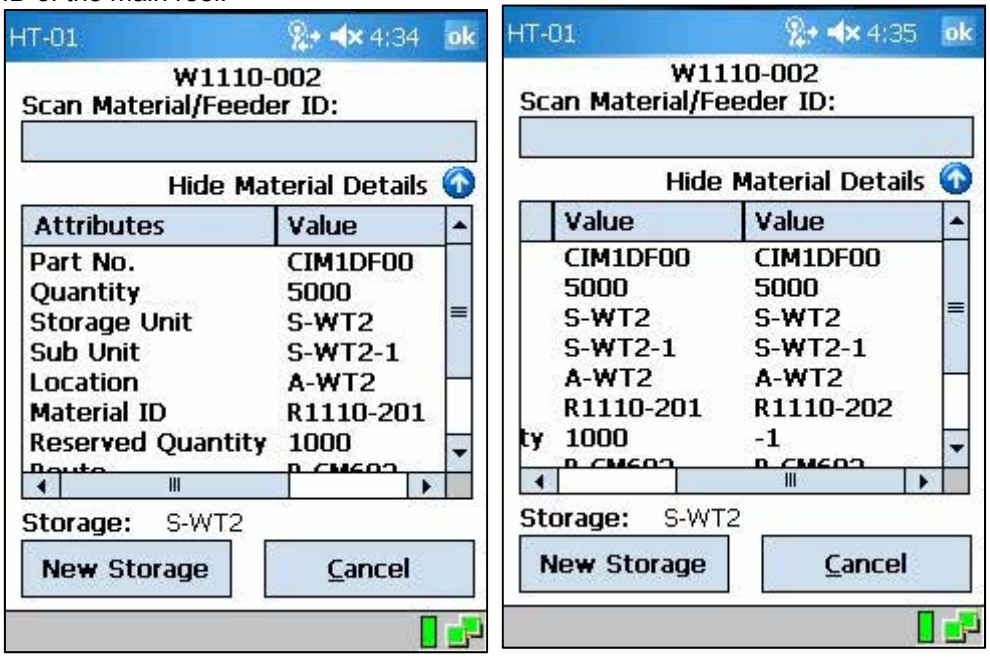
4	Click 'Mtrl Pick' and select **Route, Product** and **Work order**. The work order that a pickup operation is completed is not displayed. 
5	Scan or enter the L2 storage unit of **Move from**. 
6	Select the storage area or zone. 

7	Scan the material/feeder ID. 
8	To view detailed information of the material, select the down arrow icon adjacent to Show Material Details label in step 7. 
9	Click [**New Storage**] to decide **Move to**. 
10	The material is successfully picked up from the L2 storage area.

■ Material pickup for a spliced reel component

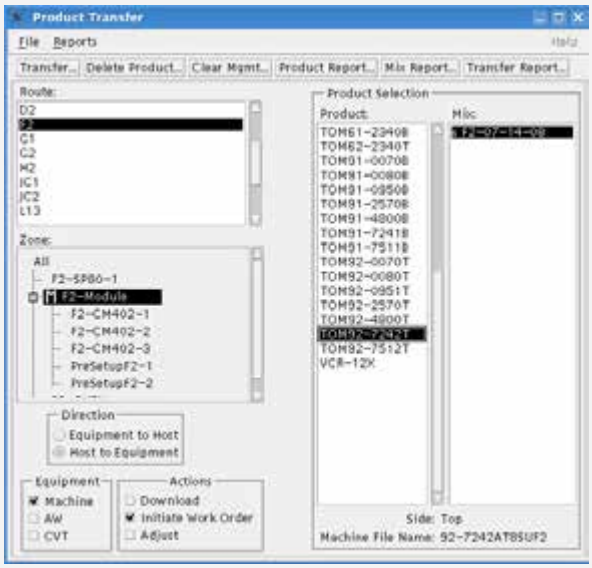
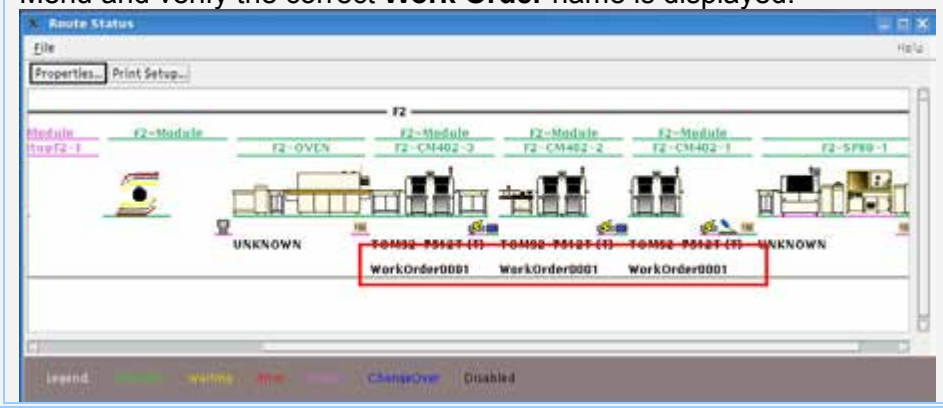
In case a pickup operation is performed for a spliced reel component, attributes of both the main reel and splice reel are allocated.
Details can be displayed in a ‘Pickup' operation described later. For spliced reels, the first value will be that of the spliced reel. Use the scroll bar to scroll to the right to view the main reel.

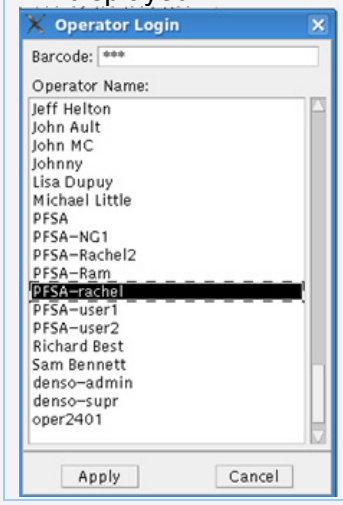
In the below example, R-25L-01 is the Material ID of the splice reel and R-25L-03 is the Material ID of the main reel.



3.3.12 Production - Work Order Logging

Refer to the action table below to configure and use the Work Order logging function.

Step	Procedure
1	Open the **Product Transfer** Application from the PanaCIM Main Menu to initiate a manual Product Changeover.
2	a. Select the **Route**. b. Select the **Module**. c. Select the **Product and Mix**. d. Select **Machine** from the Equipment section. e. Select **Initial Work Order** in the Actions section. f. Click the **Transfer** button.  * Two lanes appear only for the dual lane machines.
3	a. Select the **Work Order**. b. Enter an optional **Reason**. c. Click **Apply**. 
4	The selected Product will be activated at the machine, and the Work Order entered will be opened for use. Next, open the **Route Status** Application from the PanaCIM Main Menu and verify the correct **Work Order** name is displayed. 

5	**Scan all required components** with Material Verification scanners and then run Production.
6	At the end of Production for the current Work Order, you have two options. 1. Automatically close the current Work Order by performing Product Changeover to **Next Work Order** and **Product**. 2. Manually close the current Work Order.
7	If you selected option two, to close the Work Order manually: a. Click the **Machine Zone** from Route Status. A popup will appear. b. Select the **Close Work Order** tab. * This tab is not displayed when you don't check Record Work Order Number in Systems Config > Base 1 > Work Order. c. Enter the **End Time** for the Work Order. d. Click **Apply**. 
8	a. Enter the **Operator Barcode**. This will be the same barcode used during any Material Verification scanning operations. b. Select the **Operator Name** that corresponds to the barcode entered. The Administrator must pre-define the Operator Names to the Operator Barcodes. c. Click **Apply**. If the Barcode matches the Name, the Work Order will be closed. If the Barcode does not match the Name, an error is displayed. 

3.3.13 Master Work Order

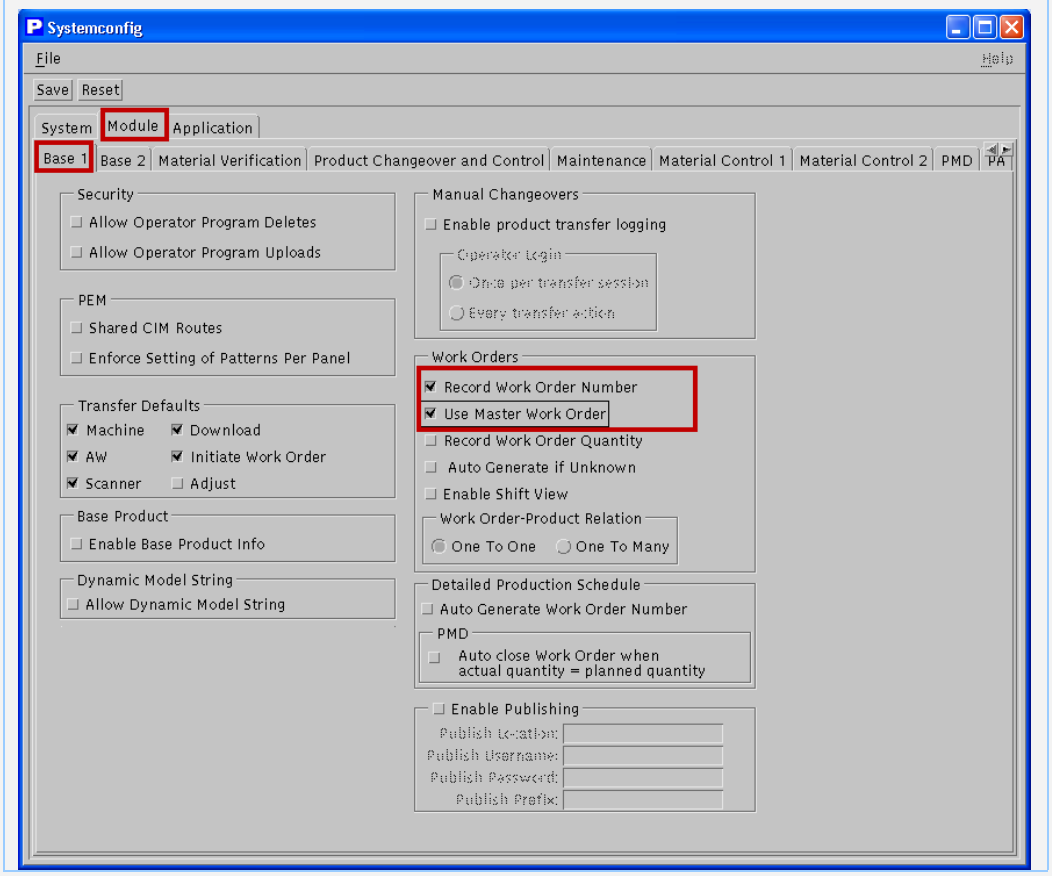
■ Creating a Master Work Order (MWO)

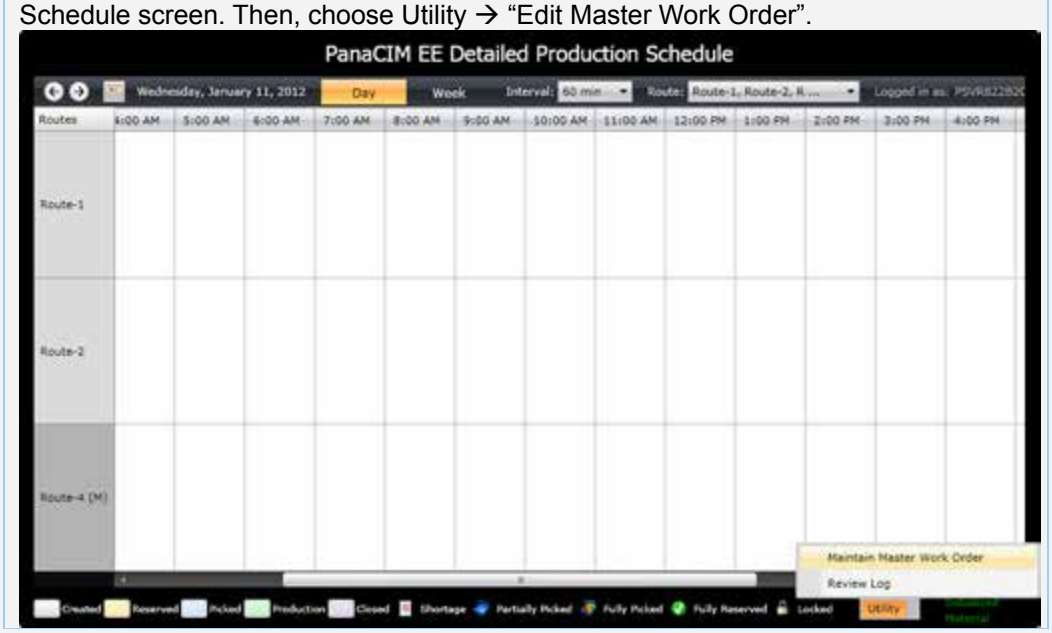
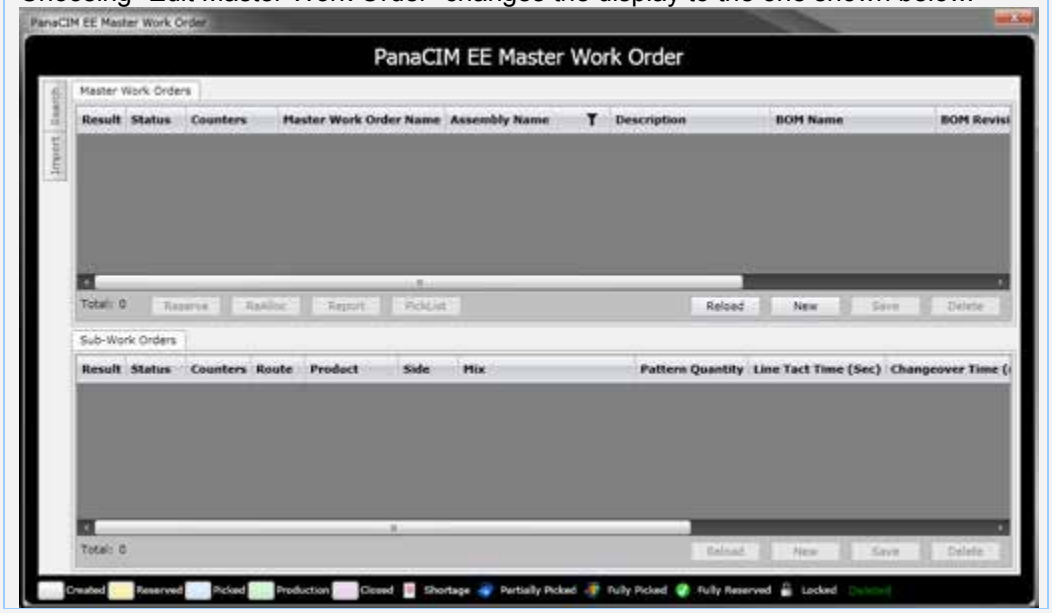
Create a master work order according to the requested work order. In the master work order, register the sub work orders based on the actual production line. In addition, two or more sub work orders can be registered for one master work order. This allows production of a PCB to be carried out on multiple lines at a time, and shortens the production time.

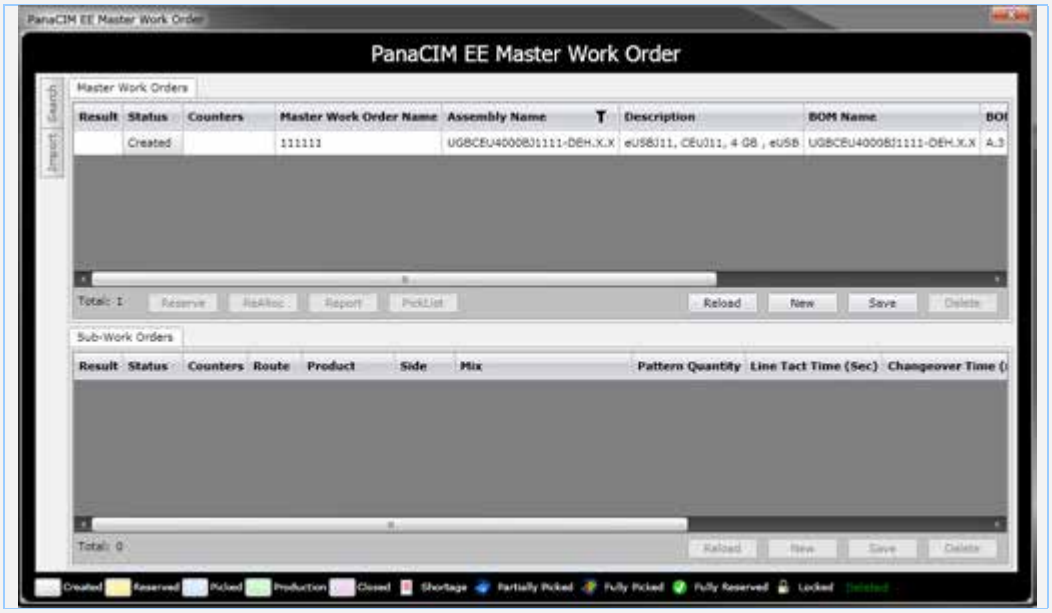
Step	Description
1	Create a master work order according to the requested work order. However, the number of types of PCBs included in one master work order must always be one. In other words, another master work order must be created when producing a different PCB.
2	More than one sub work orders can be created for one master work order. This allows number of PCBs requested as the master work order to be divided to perform production on two or more lines, thus shortening the production time.

Creating a Master Work Order

Take the following steps to create a master work order.

Step	Operation
1	On the System configuration dialog, enable **Record Work Order Number** and **Use Master Work Order**. 
2	From the PanaCIM Web Application, choose **Charts and Reports → Manage Work Orders**. 

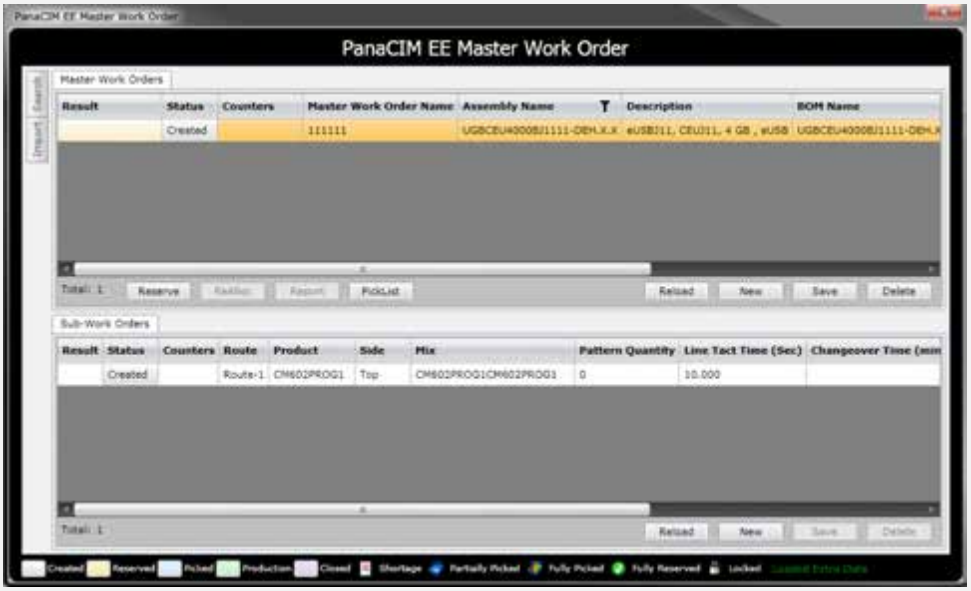
Step	Operation
3	Clicking **Manage Work Orders** displays the PanaCIM EE Detailed Production Schedule screen. Then, choose Utility → "Edit Master Work Order". 
4	Choosing "Edit Master Work Order" changes the display to the one shown below. 
5	Clicking the **New** button adds a new line for registering the master work order under the Master Work Order tab.

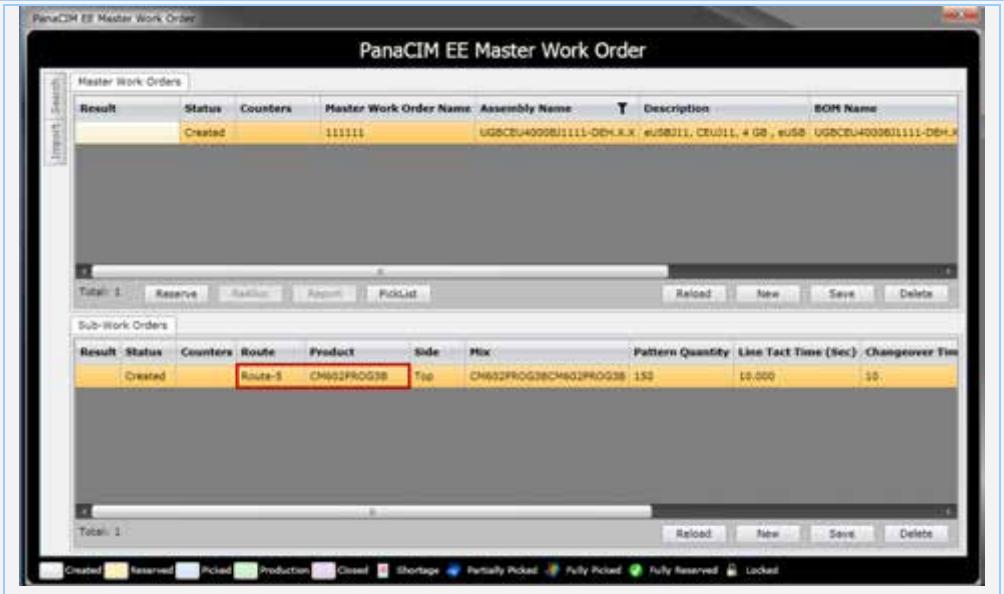
Step	Operation
	
6	For a new master work order, enter the following information and click the **Save** button to register it: <Mandatory information items> The following items are mandatory: - Master Work Order Name: Enter the name of the requested work order. - Assembly Name: Enter the assembly process name of the requested work order. - Description: Enter the description for the requested work order. - Pattern Quantity: Enter the number of patterns to be produced for the requested work order. - Patterns Per Panel: Enter the number of patterns to be produced per PCB. <Optional information items> The following items are optional and can be omitted: - BOM Name: Enter the name of BOM (Bill Of Materials) that lists the parts to be used for the requested work order. - BOM Revision: Enter the revision information for the BOM to be used for the requested work order. - Due Date: Enter the deadline for the requested work order.
7	Click the **Save** button. The Result column shows whether the master work order has been properly saved.

Step	Operation
	 **Note:** The screen above is populated with the following information entered as a master work order: - Master Work Order Name: MWO-PROD-C-001 - Assembly Name:PROD-C-001 - Description: Product C(001) - BOM Name: Omitted - BOM Revision: Omitted - Pattern Quantity: 200 - Patterns/Panel:1 - Due Date: 2012/10/12

Note: If the PanaCIM EE Master Work Order screen is closed and then reopened, sometimes the saved information may not be displayed. In such a case, click the “Reload” button under the Master Work Order tab to display them properly.

Creating Sub Work orders for a Master Work Order

Step	Action
1	Select a master work order and then click the **New** button in the **Sub-Work Orders** tab. A record appears by default as shown below: 
2	Select a route from the Route drop-down list on which Sub work order needs to run. 
3	After selecting a route, list of products imported into that route are displayed in the **Product** drop-down list. This is shown below:



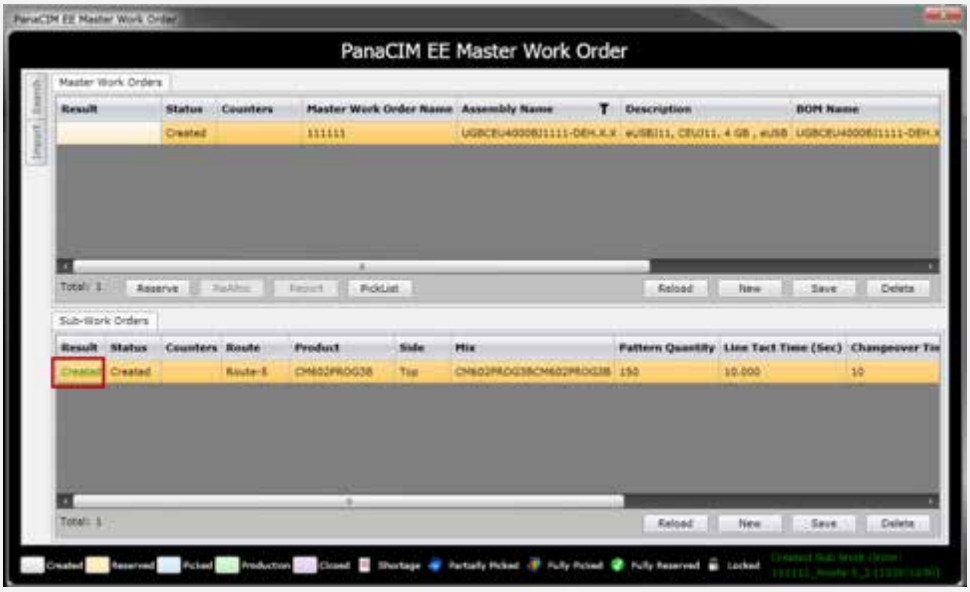
Select a product that needs to be manufactured in some future date. After selecting a product, **Side**, **Mix** and **Line Tact Time (Sec)** columns are also populated.

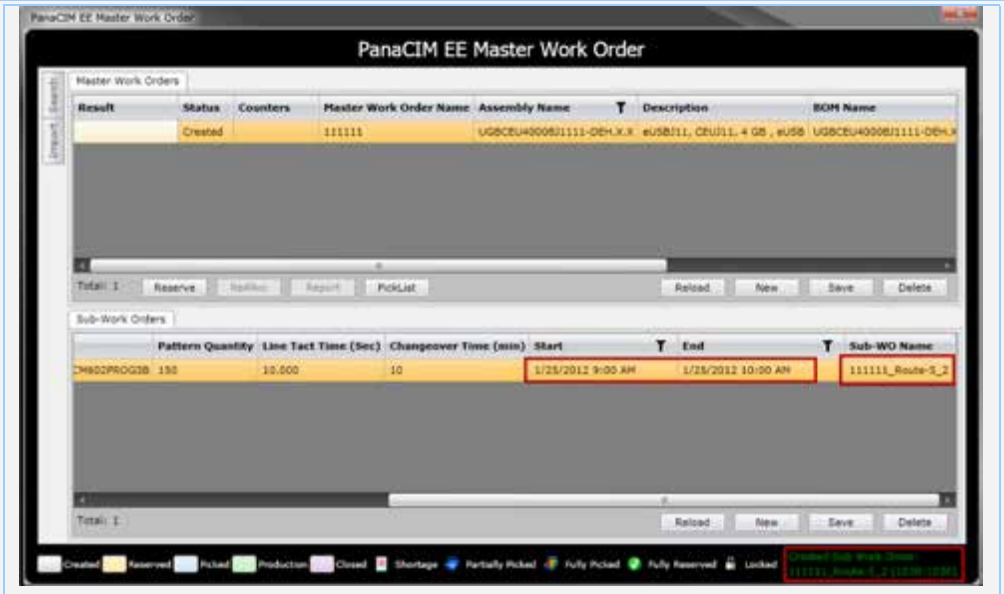
Enter a Pattern Quantity that corresponds to number of boards of that product to be produced out of the total boards specified in the master work order.

Enter Line tact time and Changeover time. Leave Start and End date/time of sub work order as blank to be entered later or it can be entered at the time of creating the sub work order. It is to be noted that start and end date/time of child work order should be less than that of its corresponding Master Work Order.

4

Click the **Save** button. **Sub WO Name** column is populated with name of sub work order.



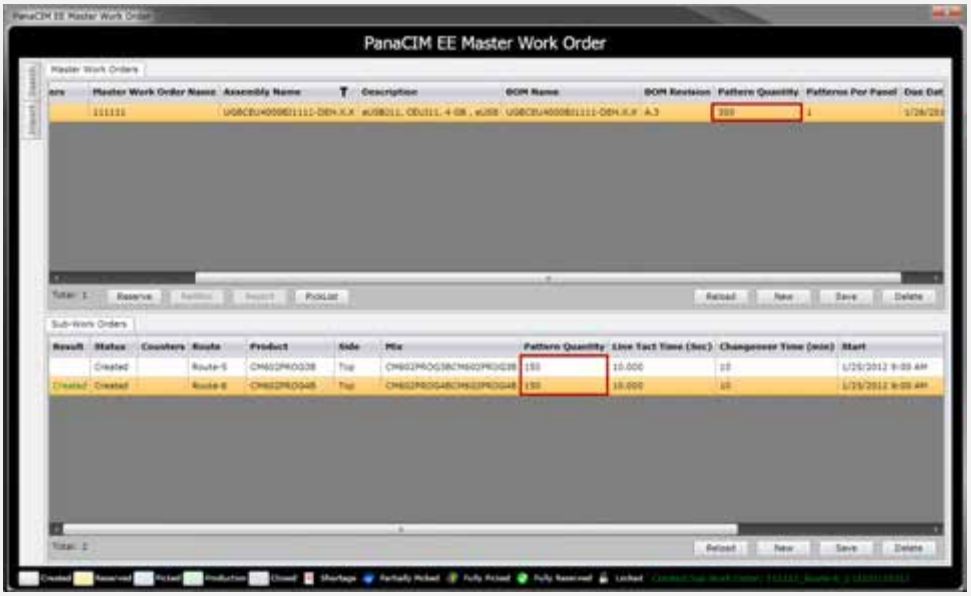


In the example considered above, pattern quantity for master work order is 300 out of which 150 boards are to be produced on Route-5 corresponding to a single sub work order. Due date of Master work order is 26-January-2011 12:00 AM while Start date and end date of sub work order is day earlier than this.

5

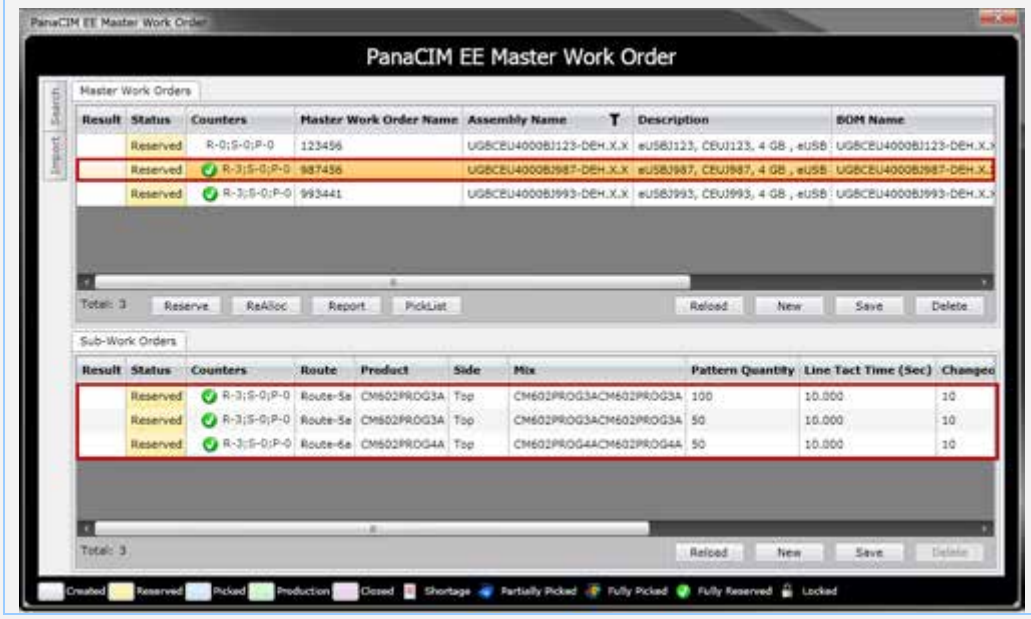
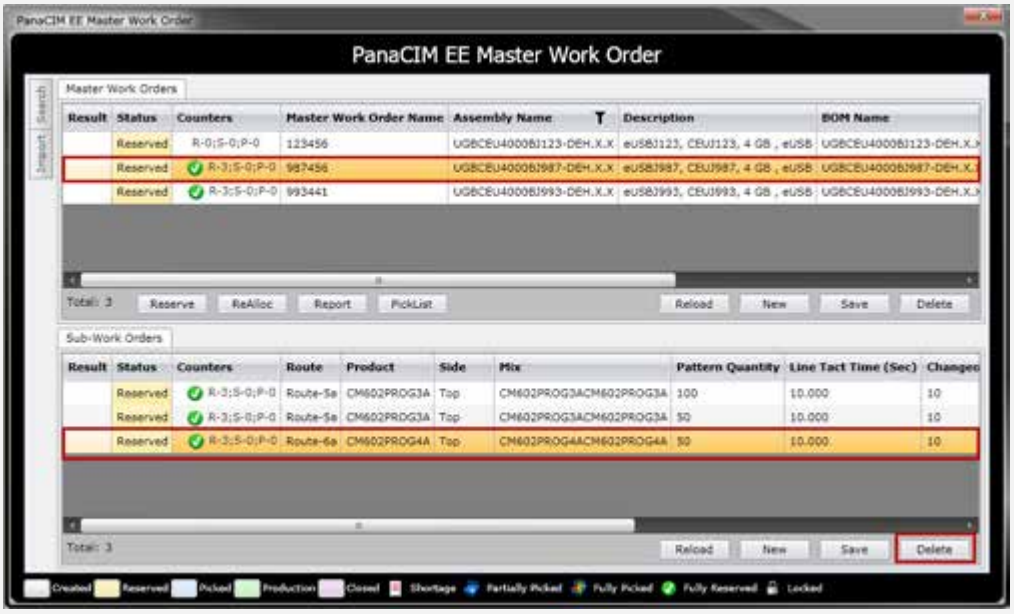
Click the **New** button and repeat the process of creating sub work orders until entire pattern quantity specified in the master work order is distributed among all the sub work orders.

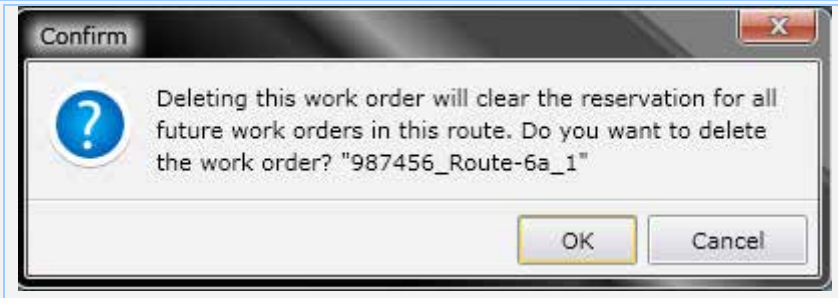
In the example considered above, another sub work order is created to account for remaining Pattern quantity of master work order. As a result, when production is started, 150 boards will be produced in Route-5 while 150 boards will be produced in Route-6.



Deleting Sub Work Orders

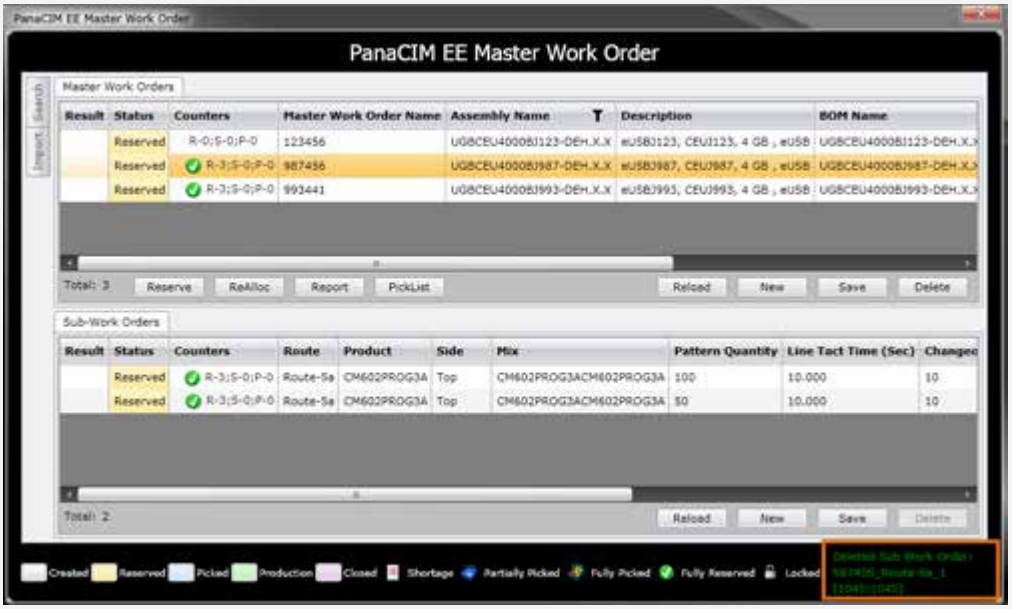
Refer to the action table below for the procedure to delete sub work orders for a master work order.

Step	Procedure
1	Select a Master WorkOrder from the **Master Work Orders** tab. The sub work orders for the selected master work order are displayed in the **Sub-Work Orders** tab as shown below: 
2	Select a Sub work order to delete. **Delete** button is activated as shown below: 
3	Click the **Delete** button.



Click the **OK** button in the above shown dialog.

The Selected sub work order is now deleted as shown below:



Note: If you select a master work order and click the **Delete** button in the **Master Work Orders** tab, all the sub-work orders of master work order are deleted.

- **Master Work order - Field Descriptions**

Field Name	Description
Status	Status of the Master Work Order which can be either be **Created**, **Reserved**, **Picked**, **Production**, and **Closed**.
Master Work Order Name	This is a high level Work order created in Customer's Oracle system to represent demand for a quantity of an end product. It is this work order which is imported into PanaCIM. It can also be created in PanaCIM without Oracle system.
Assembly Name	Name of the end product for which PCBs need to be produced.
Description	Brief explanation of product that needs to be manufactured.
BOM Name	Name of the Product as defined in the BOM (Bill of Materials) or ERP Pick List. This is same as Assembly Name.
BOM Revision	Most recent revision of the Product. Every time a program is changed that contains all the necessary information to assemble a product, revision number is changed to reflect program change.
Pattern Quantity	Total number of PCBs that need to be produced when master work order comes into production.
Patterns Per Panel	Number of PCBs that are assembled on a single panel. For example, if 100 Panels are produced and each panel contains 2 patterns, total number of PCBs produced is 200.
Due Date	Date when master work order comes into production. It is a future date from the current date.

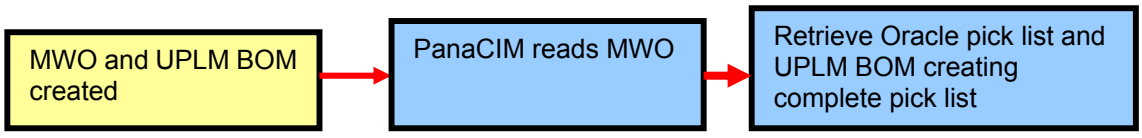
Sub Work Order – Field Descriptions

Field Name	Description
Status	Status of sub work order that corresponds to that of master work order.
Counters	Column that indicates number of reservations (represented by **R**), number of shortages (represented by **S**), number of material picks (represented by **P**).
Route	Name of the route on which production for sub work order will be started when due date of master work order is reached.
Product	Name of the program that runs on the route in order to produce a subset of boards out of total number of boards to be produced when Master work order comes into production.
Side	Side (Top or Bottom) on which components are placed.
Mix	Name of the mix that includes program needed to start production on a route corresponding to a sub work order.
Pattern Quantity	Number of boards that need to be produced on a single route. Production for master work order is distributed among one or more routes corresponding to sub work orders running on these routes. Production can be split in any ratio among sub work orders of master work order. For example, if 300 PCBs need to be produced for a master work order, and there are two sub work orders, you can assign 150 PCBs to each of the two sub work orders or 200 PCBs can be assigned to one sub work order while remaining 100 PCBs can be assigned to other sub work order.
Line Tact Time	Tact time (in seconds) for a route corresponding to production run of sub work order. Tact time refers to average time taken to complete placements on one side of a particular PCB from the moment it enters the route to the moment it leaves it.
Changeover Time	Changeover Time (in minutes) for a route when production for sub work order commences.
Start	Date and time when sub work order comes into production. It can be left blank at the time of creating sub work order. If entered, this date should be less than Due Date of master work order.
End	Date and time when production for sub work order finishes. It can be left blank at the time of creating sub work order. If entered, this date should be less than Due Date of master work order.
Sub-WO Name	Identifier for sub work order. It includes a combination of Master Work Order and Route Name.

Master Work Order Creation (Oracle used)

*** The functions working with the Oracle system are not supported.**

When the work order (master work order) is prepared using the Oracle-based system for production control, take the following steps to register the work order (master work order) in PanaCIM.

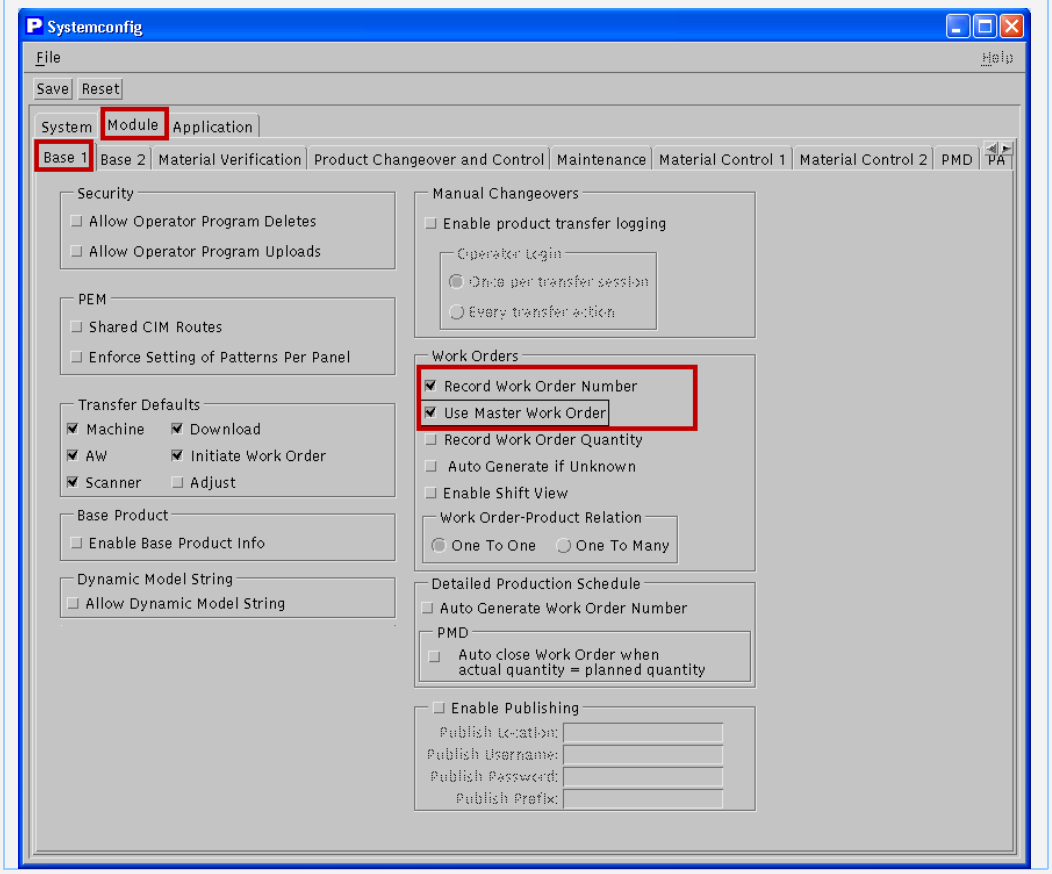


Step	Description
1	Master work order is created in the oracle system. Output of oracle system is in the form of flat file which is imported into PanaCIM via the PanaCIM EE Master Work Order Screen. Master work order will contain at most one PCB order i.e the details needed to manufacture PCB of one particular product. If you need to manufacture PCBs of additional products, additional Master Work Order needs to be created and then imported into PanaCIM. **Note:** Master work order can also be created in PanaCIM.
2	Detailed BOM (Bill of materials) / ERP Pick List is created for an assembly (product). BOM contains essential details like component name, number of components to be placed per panel and total quantity needed to manufacture all the panels.
3	ERP Pick List details are imported into PanaCIM. ERP pick list contains different parts like SMT components, hand place components, and other final assembly components that are kitted and tracked.
4	Programs are created in PanaPRO containing the Parts in the ERP Pick List. These programs are then exported to PanaCIM routes via PEM (Product Exchange Manager).
5	One or more sub work orders for master work order are created. Each Sub work order includes a PanaCIM product whose production runs on a route to create some of the boards out of the total boards that need to be manufactured for a master work order.

Importing Master Work Order (Oracle Used)

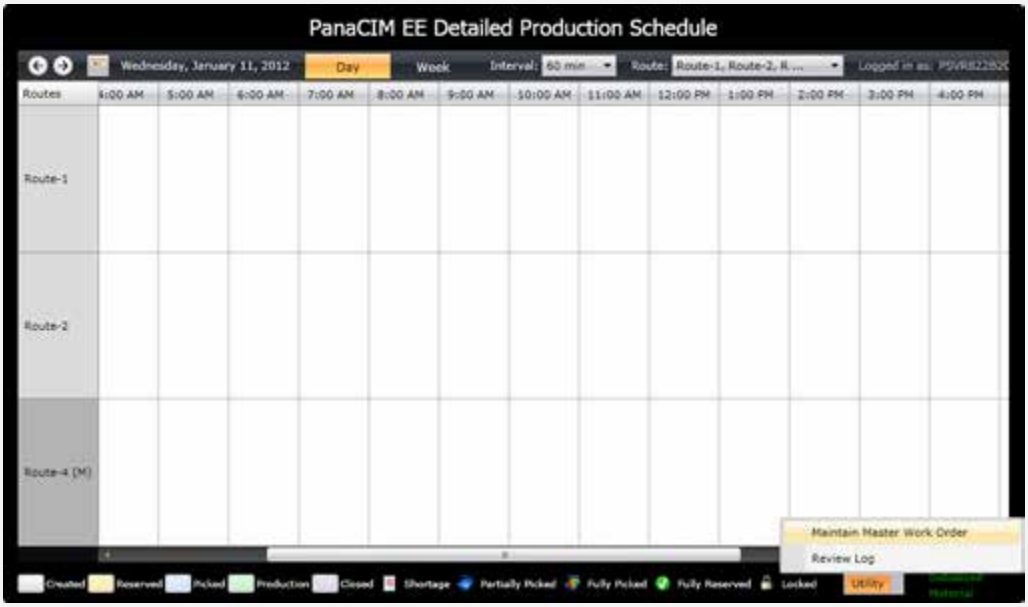
*** The functions working with the Oracle system are not supported.**

Refer to the action table below for enabling tray part exhaust verification in PanaCIM.

Step	Action
1	From the **System Config** dialog, enable the **Record Work Order Number** and **Use Master Work Order** options. 
2	From the PanaCIM Web Application, select **Charts and Reports -> Manage Work Orders** as shown below:

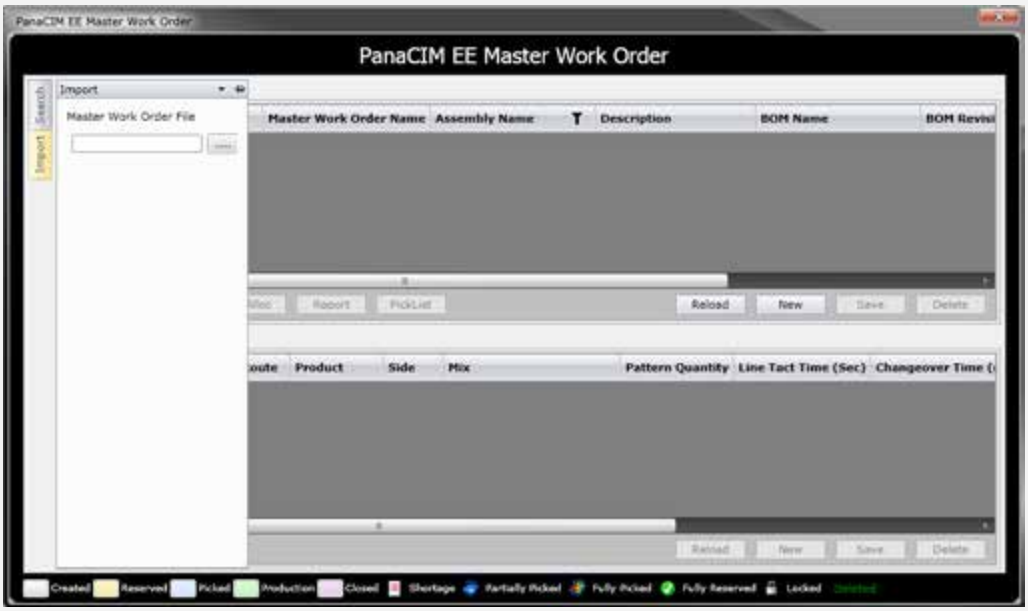


3	After clicking **Manage Work Orders** bubble, **PanaCIM EE Detailed Production Schedule (DPS)** screen appears. At the bottom of the DPS screen, click **Utility -> Maintain Master Work Order** as shown below:
4	After selecting **Maintain Master Work Order**, following screen appears:

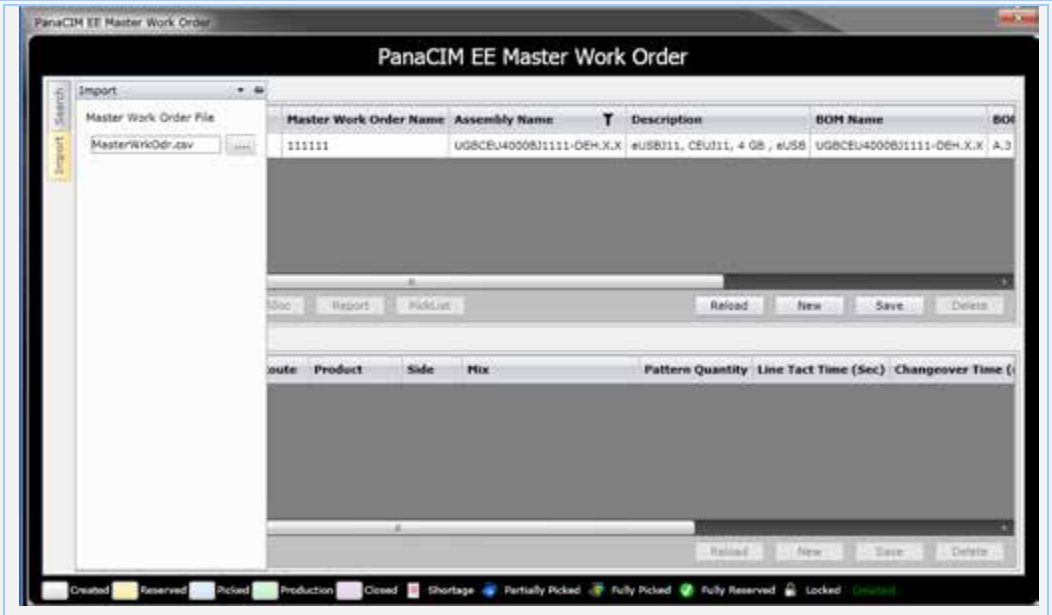




5 Click **Import** to import a master work order file. Textbox along with .... button appears as shown below



6 Click the .... button to open a file open dialog to select the CSV file containing master work order details. After CSV file is imported, **Master Work Order** tab is populated with the details of the work order as shown below:



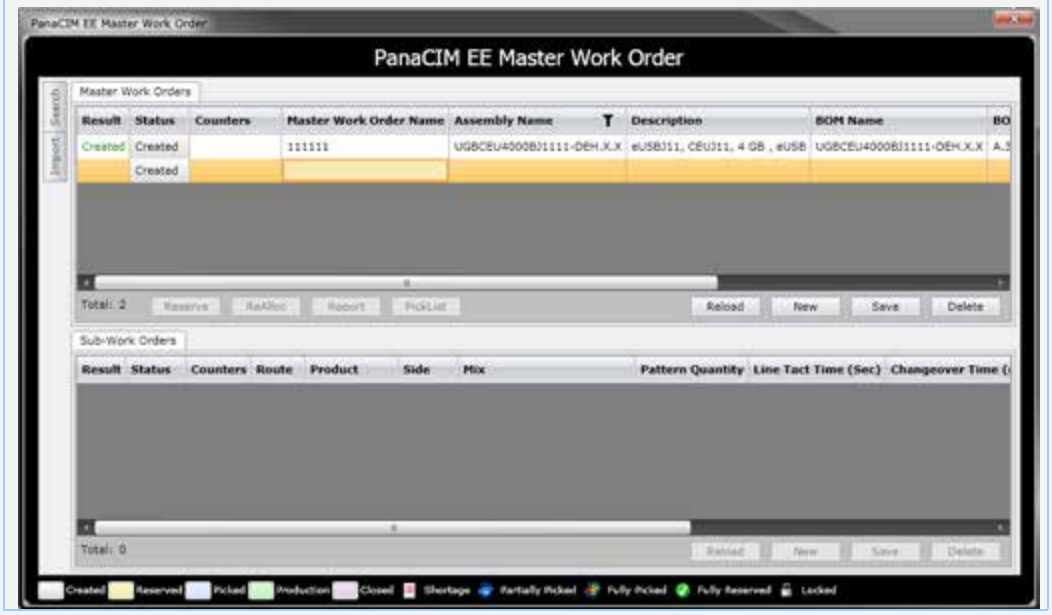
After you click the icon to close the **Import** tab, following screen appears:



7	Click the **Save** button. Master work order is saved as confirmed by the status shown in the **Result** column.



Note: Master work order can also be created by clicking the **New** button. After clicking the New button, a new record appears in which you need to enter Master Work Order Name, Assembly Name, Description, BOM Name, BOM Revision, Pattern Quantity, Patterns Per Panel, and Due Date. Click the **Save** button to save the master work order.



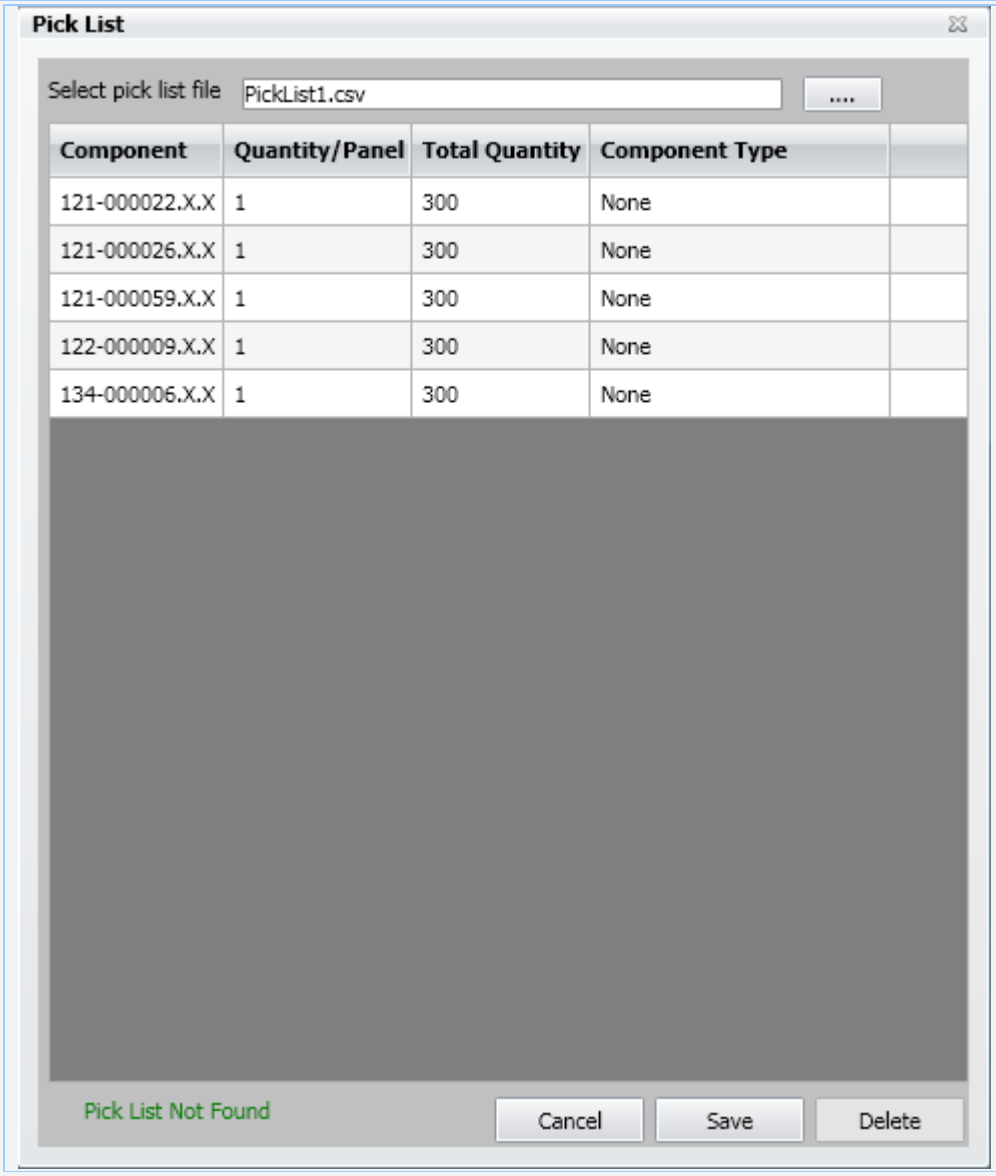
Note: After DPS screen is closed and is reopened, saved master work order may not appear. Hence, click the **Reload** button in the **Master Work Orders** tab to reload the details of saved master work order.

Importing the Pick List for the Master Work order (Oracle Used)

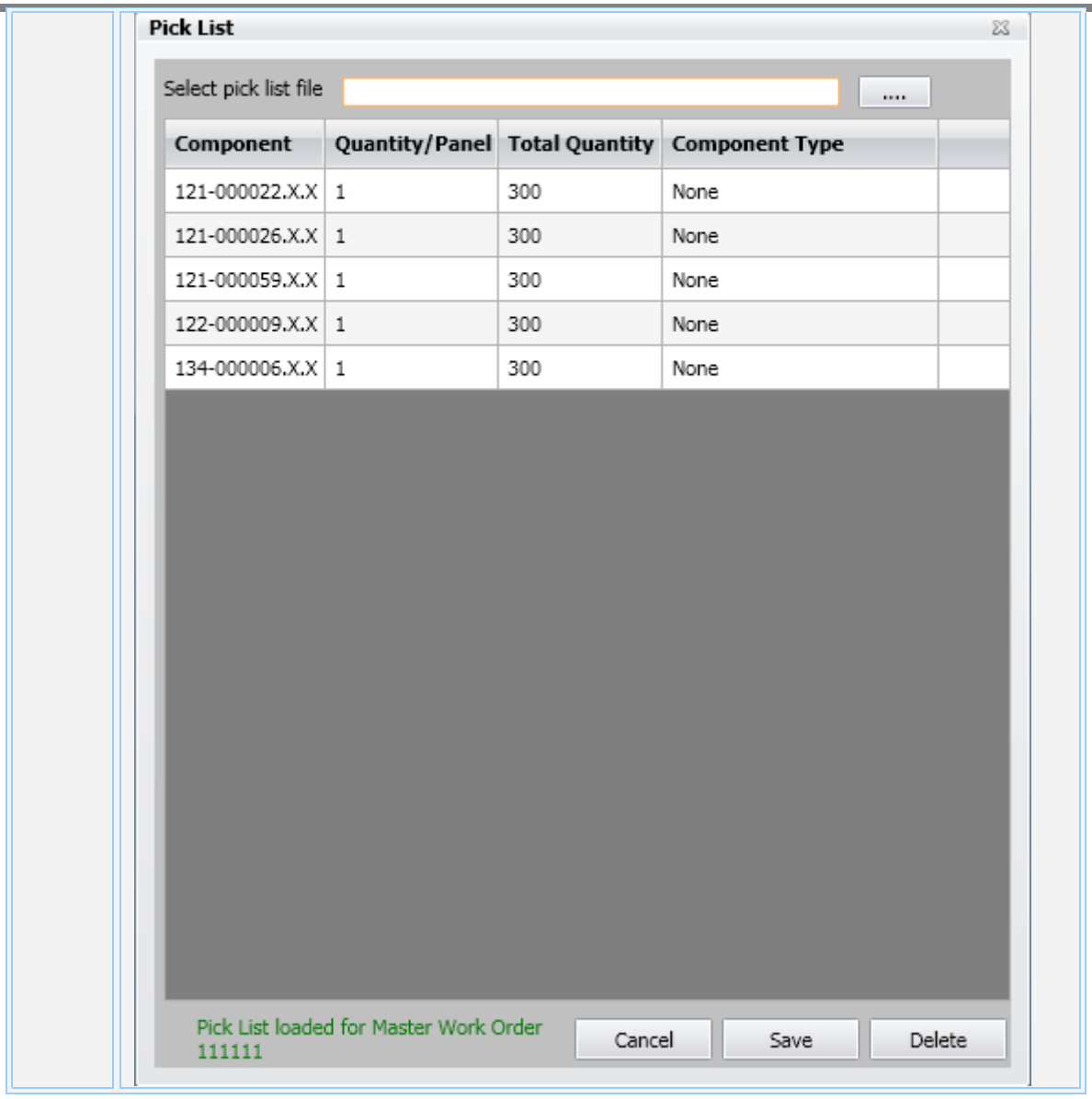
*** The functions working with the Oracle system are not supported.**

Pick List can be created after master work order is created or after its sub work order is created. Refer to the action table below for importing the Pick List for a master work order:

Step	Procedure
1	Select a Master Work Order and click the **Pick List** button. Following screen appears: 
2	Click the **ellipsis** button adjacent to **Select pick list file** textbox. A standard file open dialog appears from which the name of the csv file containing part details can be selected. After selecting pick list file, part details like **Component**, **Quantity/Panel**, **Total Quantity** and **Component Type** are visible.

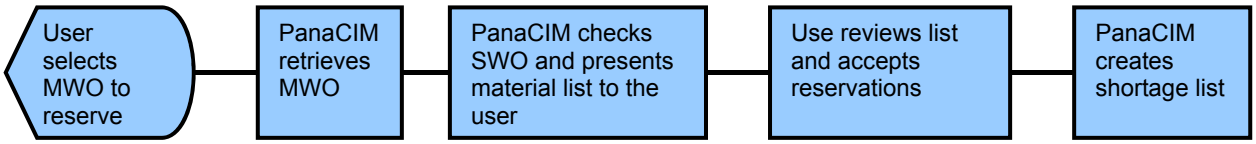


3	Click the **Save** button to save the pick list for the master work order. If you click the **Pick List** button again, saved picked list appears as shown below:


Pick List
Select pick list file
....
Component
Quantity/Panel
Total Quantity
Component Type
121-000022.X.X
1
300
None
121-000026.X.X
1
300
None
121-000059.X.X
1
300
None
122-000009.X.X
1
300
None
134-000006.X.X
1
300
None
Pick List loaded for Master Work Order 111111
Cancel
Save
Delete

■ Master Work Order Reservation

To reserve the components (materials) to be used for production, select a master work order to reserve.
By reserving a master work order, the components (materials) to be used are reserved for each sub order.



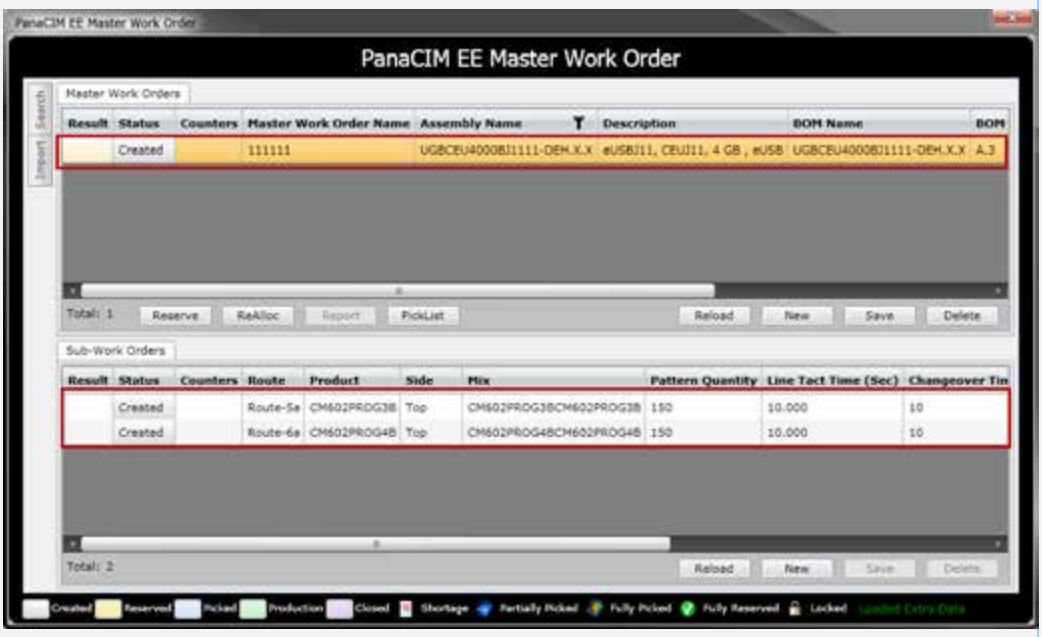
Step	Description
1	Master Work Order to reserve is selected from a list of master work orders. **Note:** Sub Work Order (SWO) does not need to exist to reserve material for MWO.
2	PanaCIM checks master work order and looks for parts that needs to be reserved based upon parts existing in the programs corresponding to sub work orders.
3	Reservation list is presented to the user. The list is displayed for each sub work order. Reservation list contains details of reels that have the parts in the required quantity to fulfill the work order as well as any reels that have partial part quantity. The materials for reservation can be either unloaded reels present in material areas or those mounted on machine slots containing left over part quantity from previous production run.
4	User reviews list and accepts reservations. If there are still shortages after reservation, they are also displayed.
5	In order to fulfill the shortage, user again performs the reservation. This process is repeated until there is no shortage.

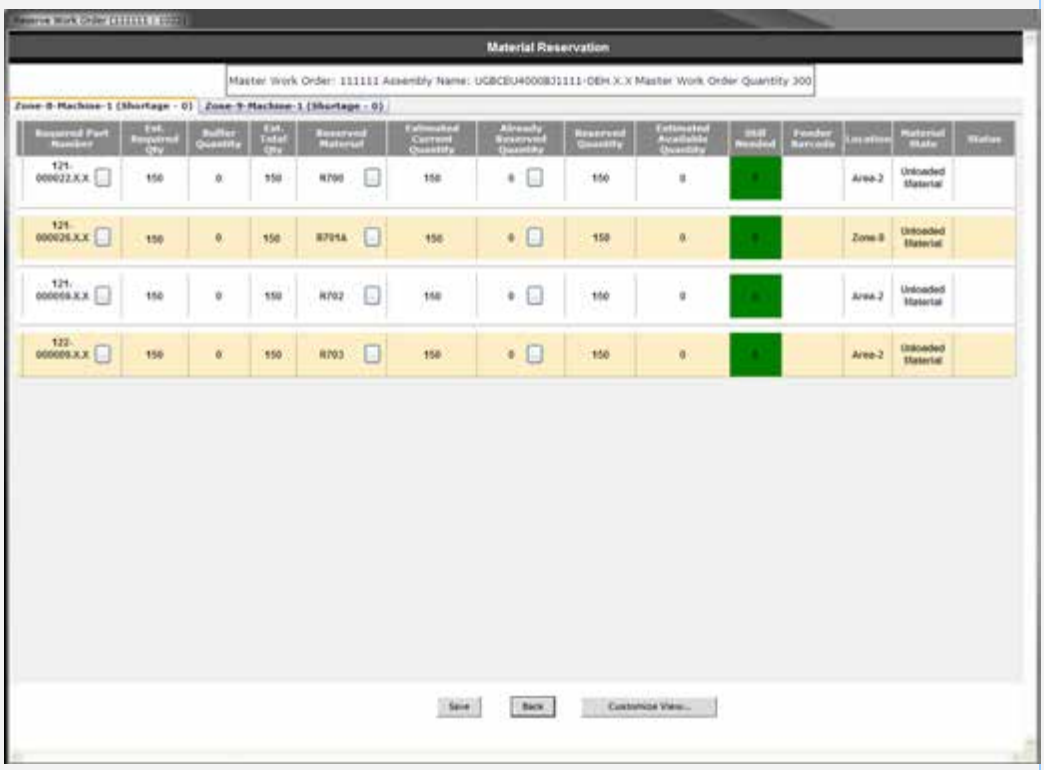
Note: End result of material reservation is to ensure that reels with quantity required to produce all panels as part of master work order production run is available and these reels are committed to a one particular master work order only until production of this master work order is completed.

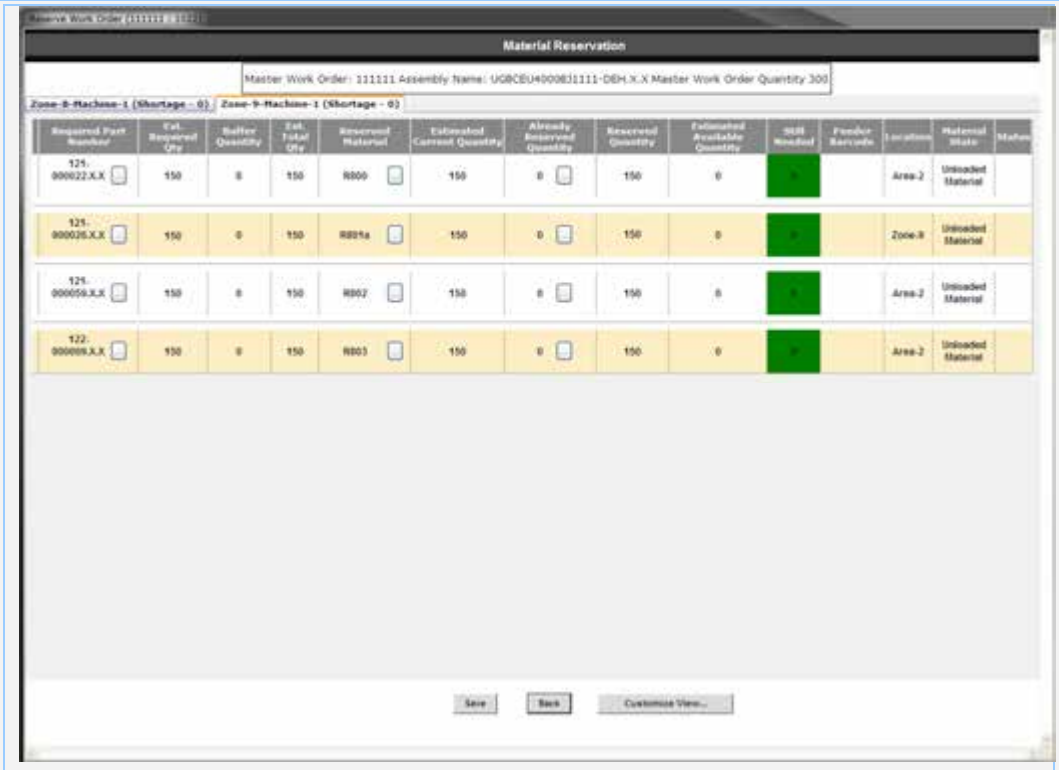
Reserving Master Work Orders

Before reserving reels for master work order, the programs need to be imported into PanaCIM routes.

Note: User does not need to create SWO in order to reserve materials. Reservations can be done against a MWO.

Step	Procedure
1	After creating required sub work orders for the master work order, click the **Reserve** button in the **Master Work Order** tab. 
2	After clicking the **Reserve** button for a master work order (in this case 111111), following screen appears:  As shown in the screen above, **Master Work Order** name, **Assembly Name** and **Master Work Order Quantity** are listed in the top section of **Material Reservation** page. Also, note that material can be reserved from Presetup

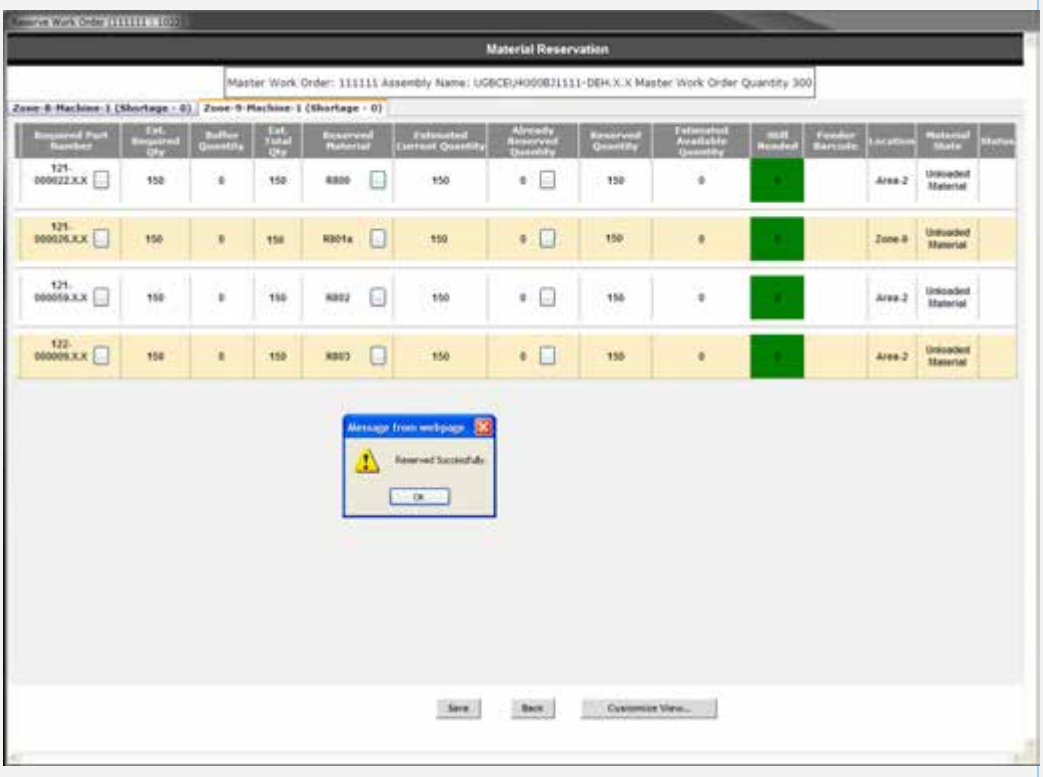
	Carts in addition to storage units.
3	Underneath **Search Order** label on the **Material Reservation** page, **Unloaded Partial Material** and **Unloaded Full Material** are checked by default. This dictates the order in which unloaded reels will be searched in the Search locations listed on the right hand side. Unloaded reels that contain partial part quantity will be considered first for reservation followed by unloaded reels that contain full part quantity as per the work order.
4	Click the **Generate** button. Reels are presented for reservation for each of the PanaCIM routes of child work order. A separate tab appears for each PanaCIM route as shown below: 

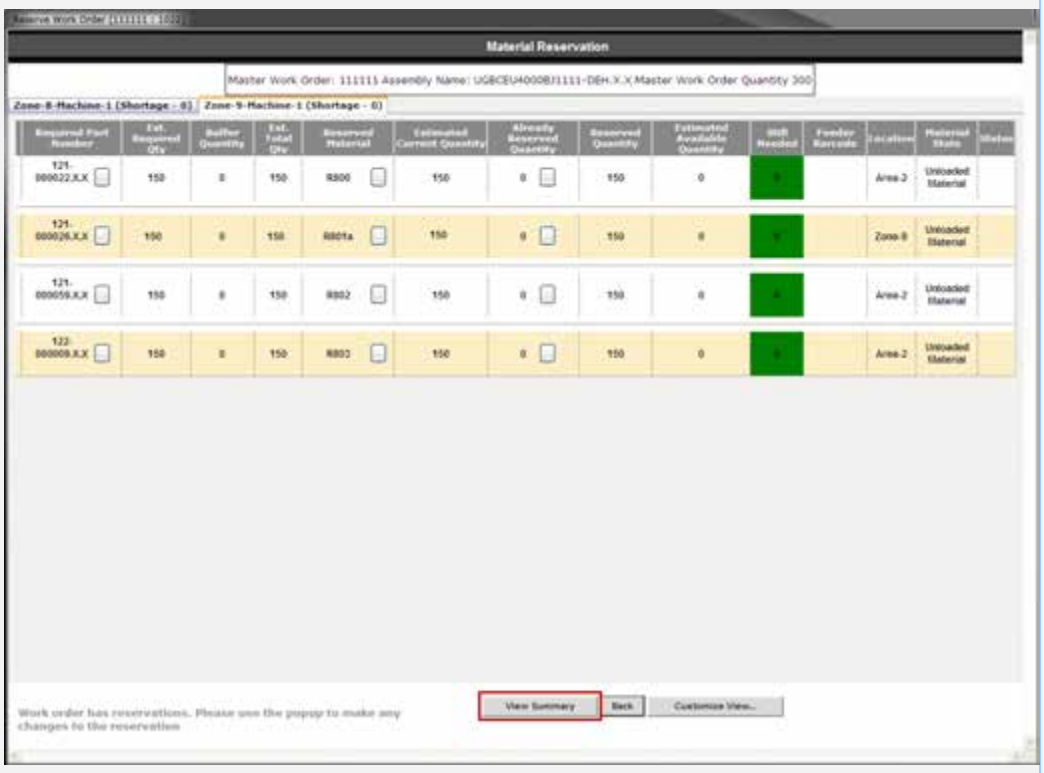
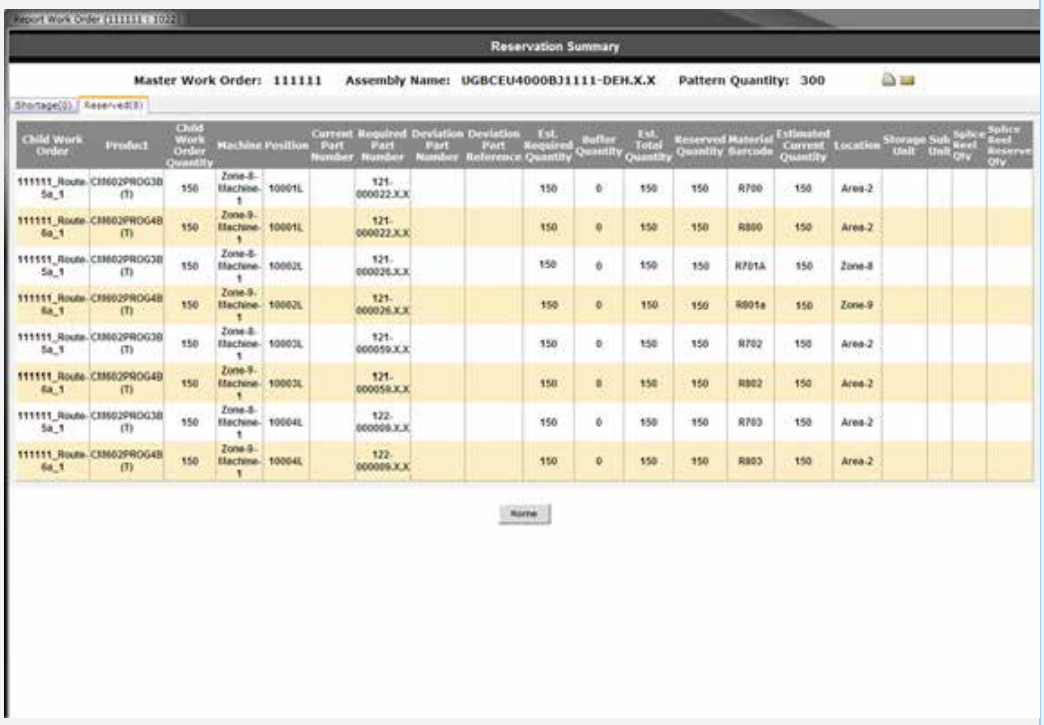


For brief description of columns shown in the above page, please refer to **"Material Reservation - Column Descriptions"**.

Note: Even parts remaining in the mounted feeder cannot be reserved after a product changeover.

5 Click the **Save** button in the **Material Reservation** Screen. A message **"Reserved Successfully"** appears as shown below:



6	Now you can view the reservations by clicking the **View Summary** button at the bottom: 
7	After clicking the **View Summary** button, the **Reservation Summary** page appears containing two tabs - **Shortage** and **Reserved**. Shortage tab lists any materials for which required part quantity is not available for reservation. **Reserved** tab lists all those materials which have been successfully reserved.  Description of most of these columns can be found in the heading **"Material Reservation - Column Description".** Following columns in the **Reservation Summary** page are unique to this page.

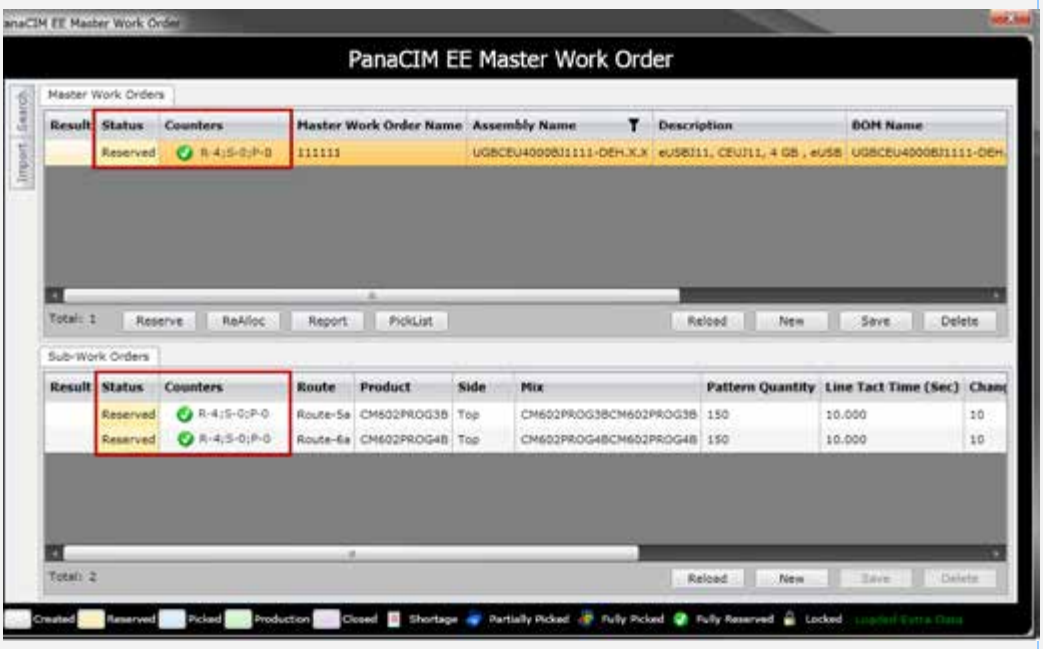
Storage Unit: Name of the storage unit into which picked reel will be placed. It will be empty immediately after material reservation.

Sub Unit: Name of the storage sub unit into which picked reel will be placed. It will be empty immediately after material reservation.

Splice Reel Qty: Part Quantity in the spliced reel used for reservation.

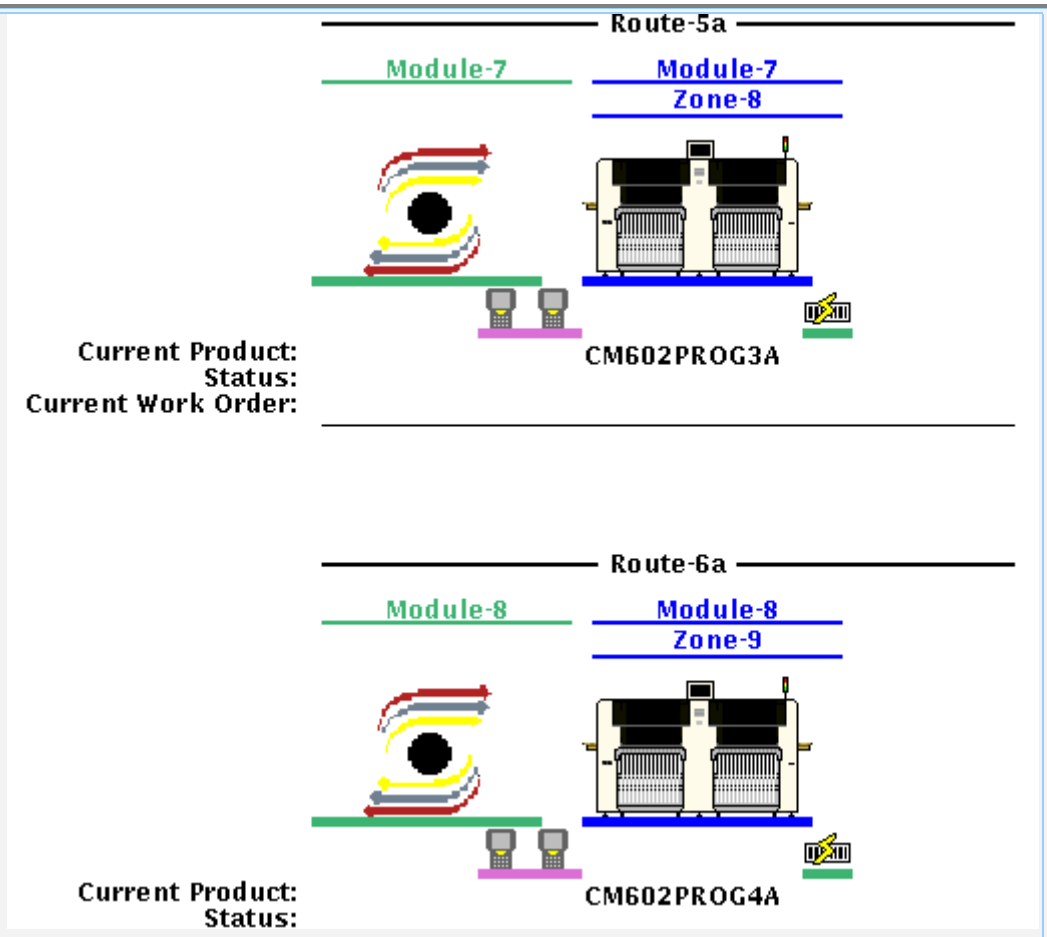
Splice Reel Reserve Qty: Quantity in the spliced reel actually reserved for a sub work order.

8 Click the **Home** button in the **Reservation Summary** page to return to the **PanaCIM EE Master Work Order** Page. **Status** column contains **"Reserved"** for both the Master Work Order and the Child work order as shown below:



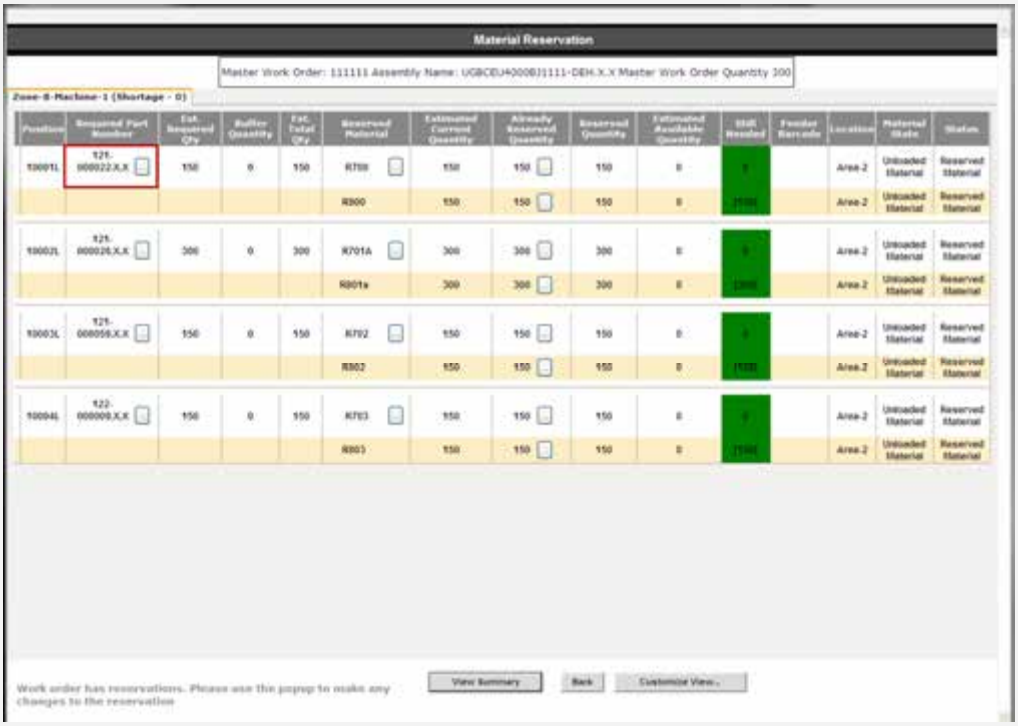
Note:
At the time of reservation, sub work order programs should not be running on the routes.

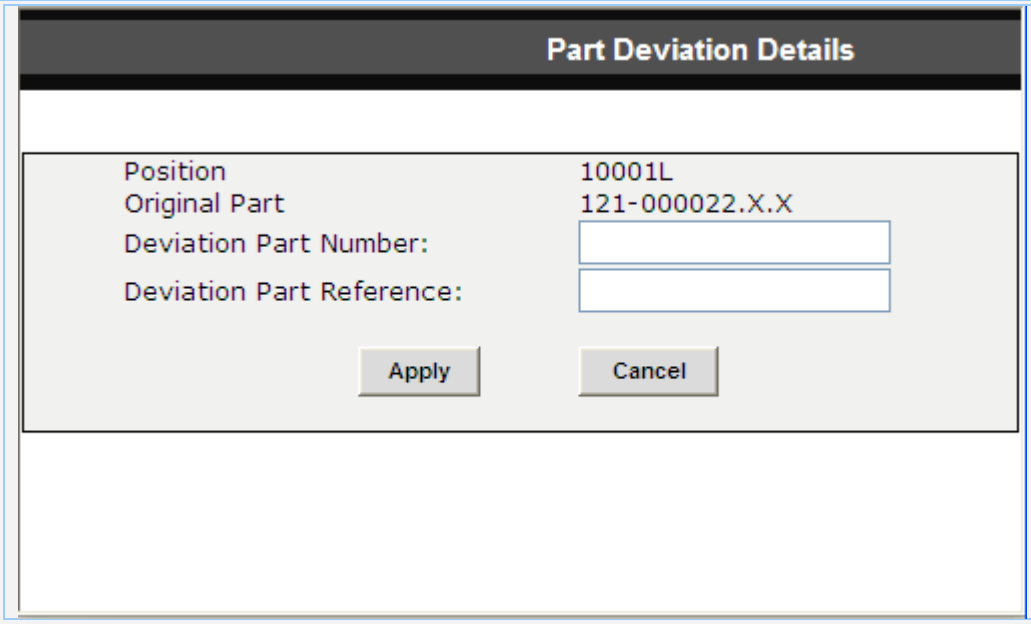
For example, parts are reserved for sub work orders whose production will run on **Route-5a** and **Route-6a** when programs **CM602PROG3B** and **CM602PROG4B** will be activated. But at the time of material reservation, some other programs are activated as shown below:

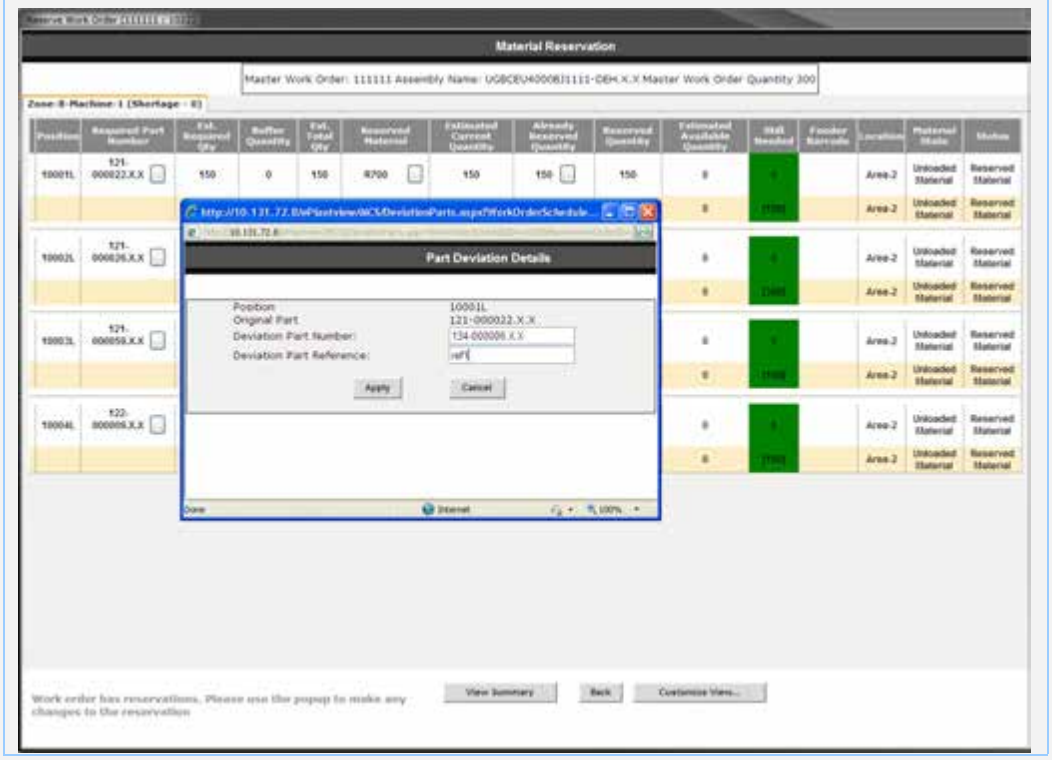

Route-5a
Module-7
Module-7
Zone-8
Current Product:
Status:
Current Work Order:
CM602PROG3A
Route-6a
Module-8
Module-8
Zone-9
Current Product:
Status:
CM602PROG4A

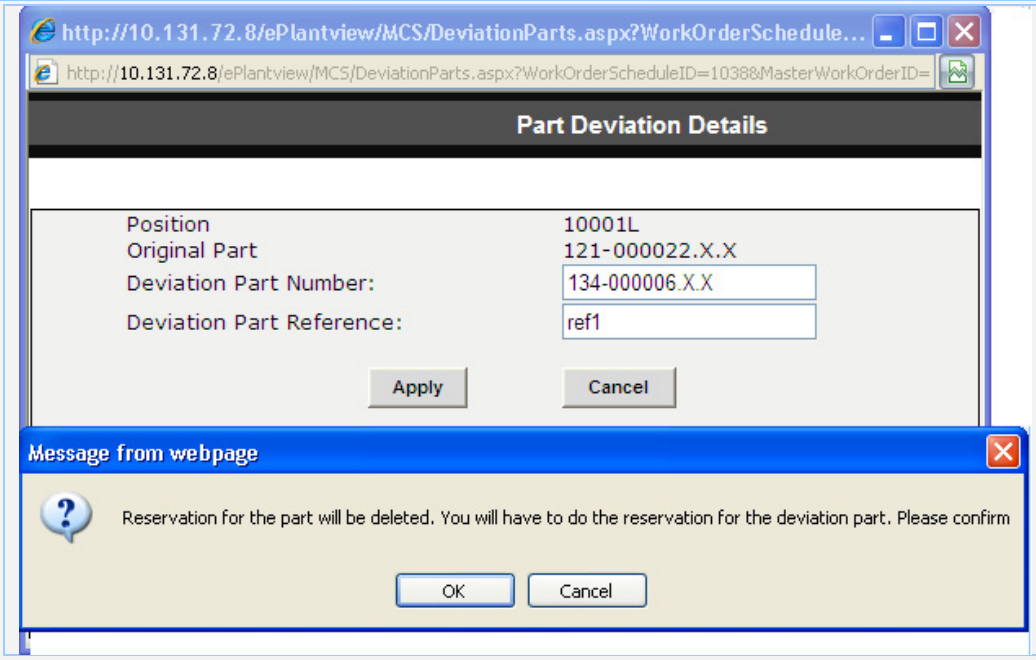
Entering Part Deviation for a Master Work Order

Entering a part deviation is equivalent to specifying a substitute part if primary part is unavailable at a particular feeder location for a reserved master work order. System will remove the current reservation and allow the user to do a manual reservation for the new part number that was entered.

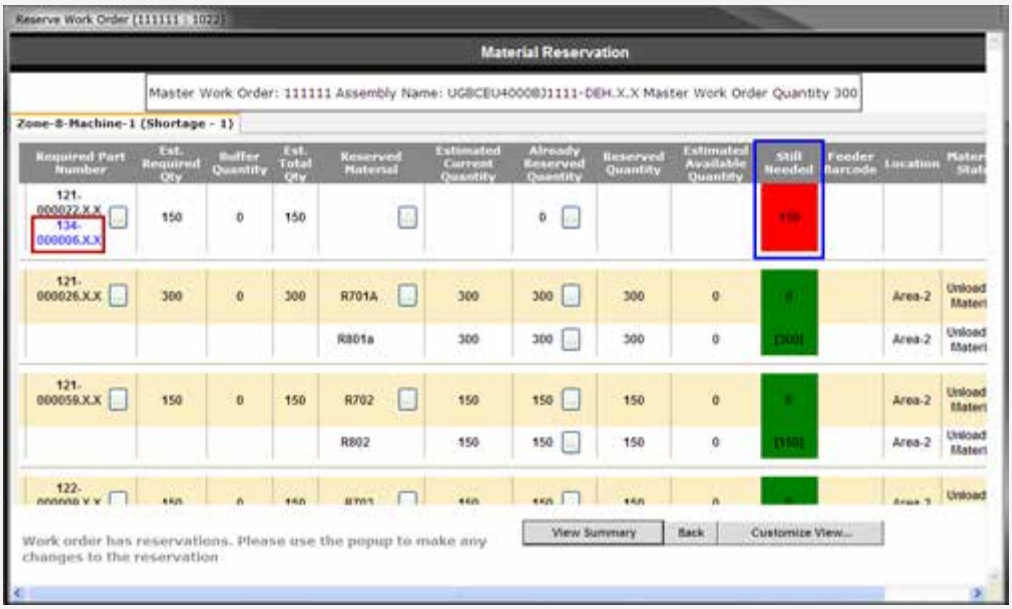
Step	Procedure
1	Launch **Material Reservation** screen from **PanaCIM EE Master Work Order** screen by selecting a Master Work Order and clicking the **Reserve** button. Material reservation screen is shown below: 
2	Click the … button in the **Required Part Number** column for a part number for which alternate part needs to be specified. After clicking the … button, following screen appears:



3	Enter a substitute part number and part reference number in **Deviation Part Number** and **Deviation Part Reference** textboxes respectively. 
4	Click the **Apply** button in the **Part Deviation Details** screen. Following message appears:



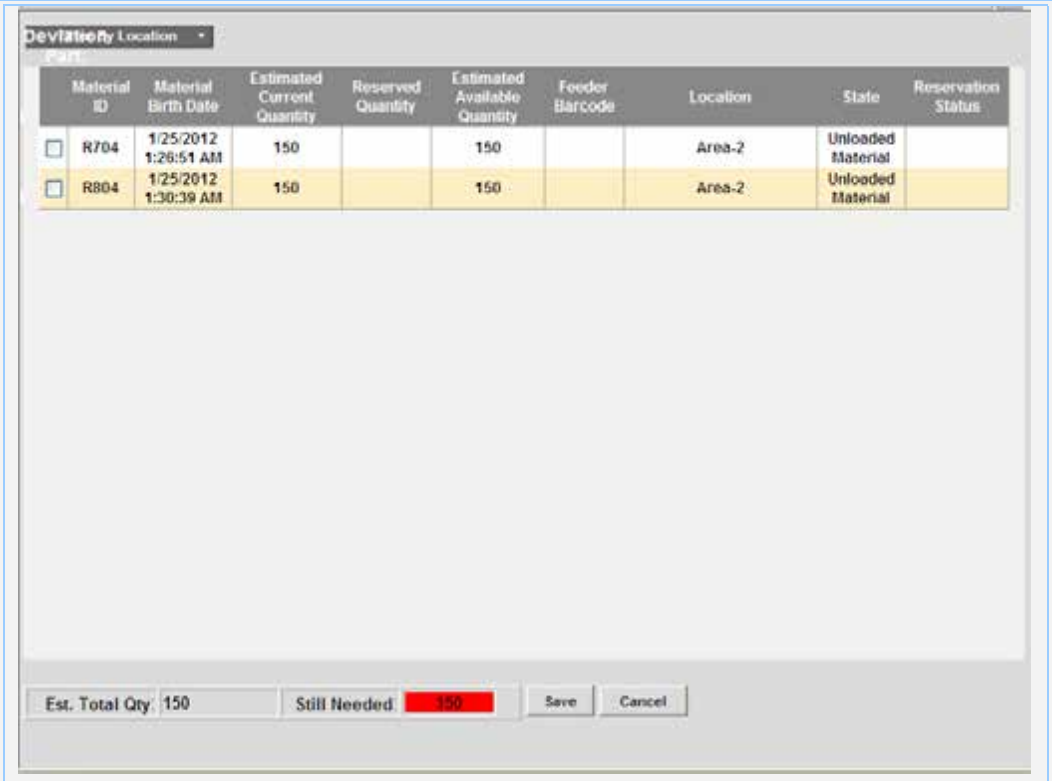
5 Click the **OK** button in the Confirmation message shown in step 4. Following screen appears:



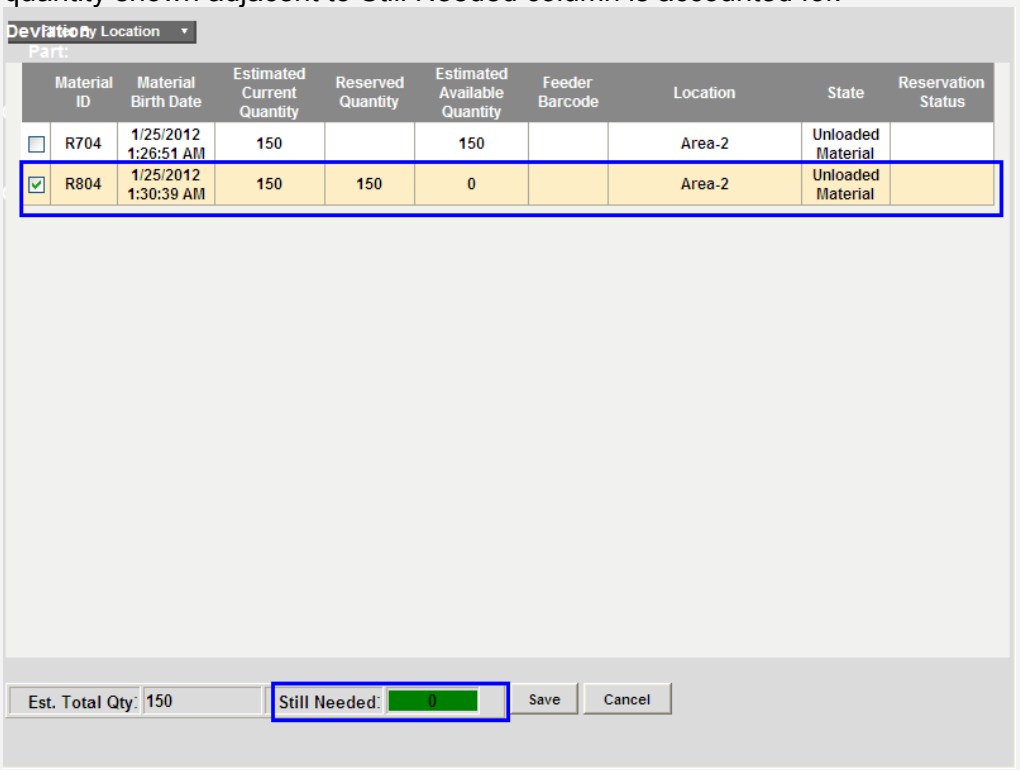
As shown in the screen above, reservation for primary part (in this case **121-000022.X.X**) is deleted and reservation for deviated part (in this case **134-000006.X.X**) has to be done again. This is apparent from the **Still Needed** column which is now highlighted in red color.

6 To reserve the deviated part, click the … button in the **Reserved Material** column to select a reel containing the deviated part.

After clicking the … button in the **Reserved Material** column , following screen appears:

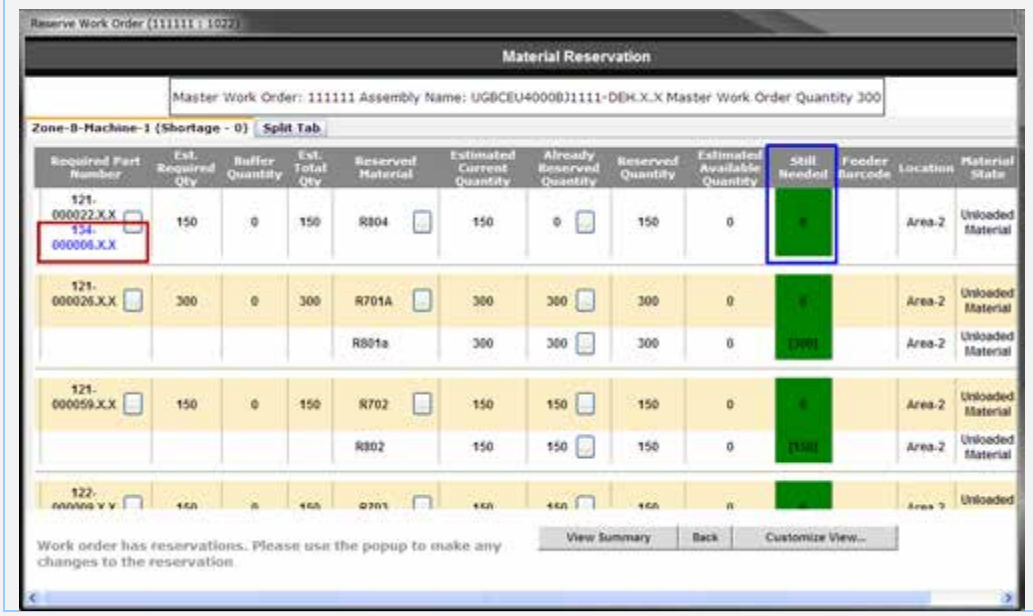


7 Select one or more reels from the screen shown in step 6 until required quantity shown adjacent to Still Needed column is accounted for.

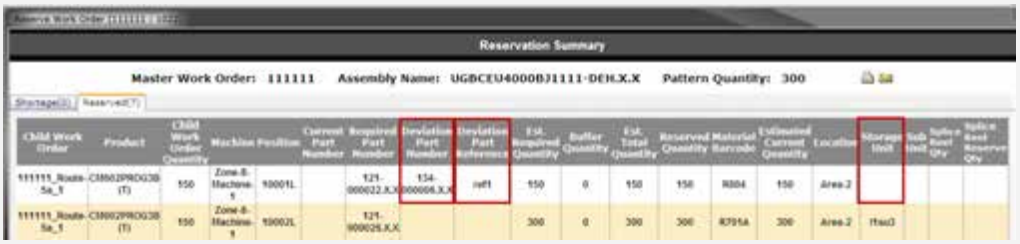


Notice that **Still Needed** quantity is updated to 0 since the required part quantity is available in the selected reel (in this case the one with barcode R804). If one reel is not sufficient to account for **Still Needed** quantity, you can select multiple reels from the above screen until Still Needed quantity is updated to 0.

8 Click the **Save** button adjacent to the **Still Needed** label to reserve the quantity for deviated part.



9 Now if you click the **View Summary** button, the details of deviated part are available as shown below:



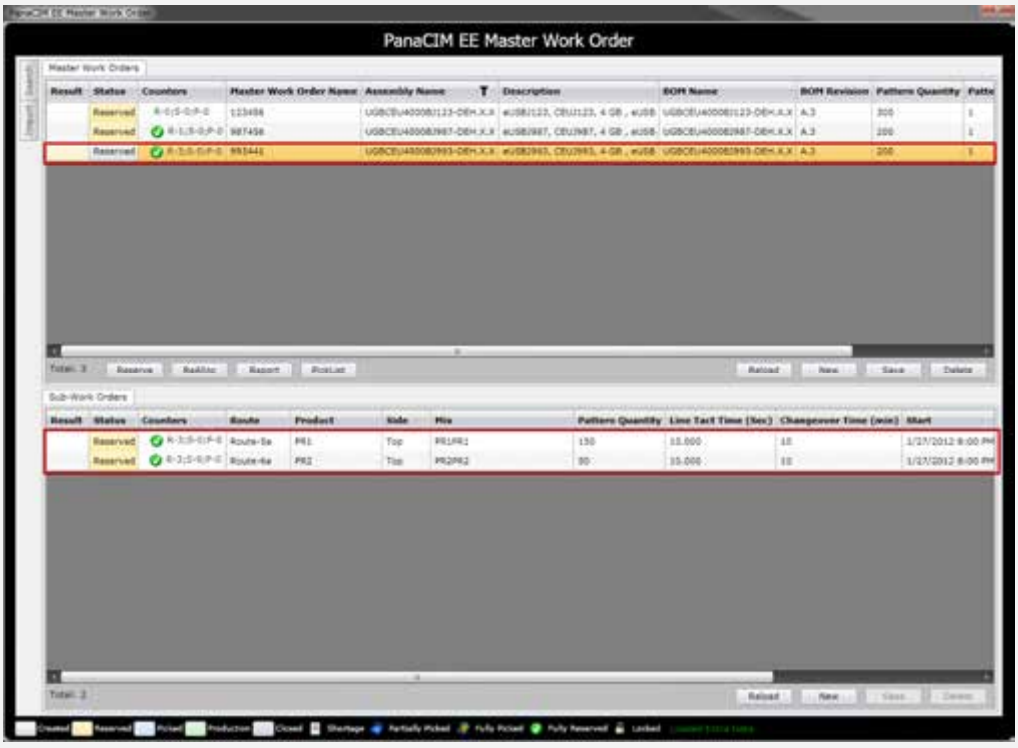
Note: Deviated part still needs to be picked and before running the production, program containing deviated parts need to be re imported into PanaCIM.

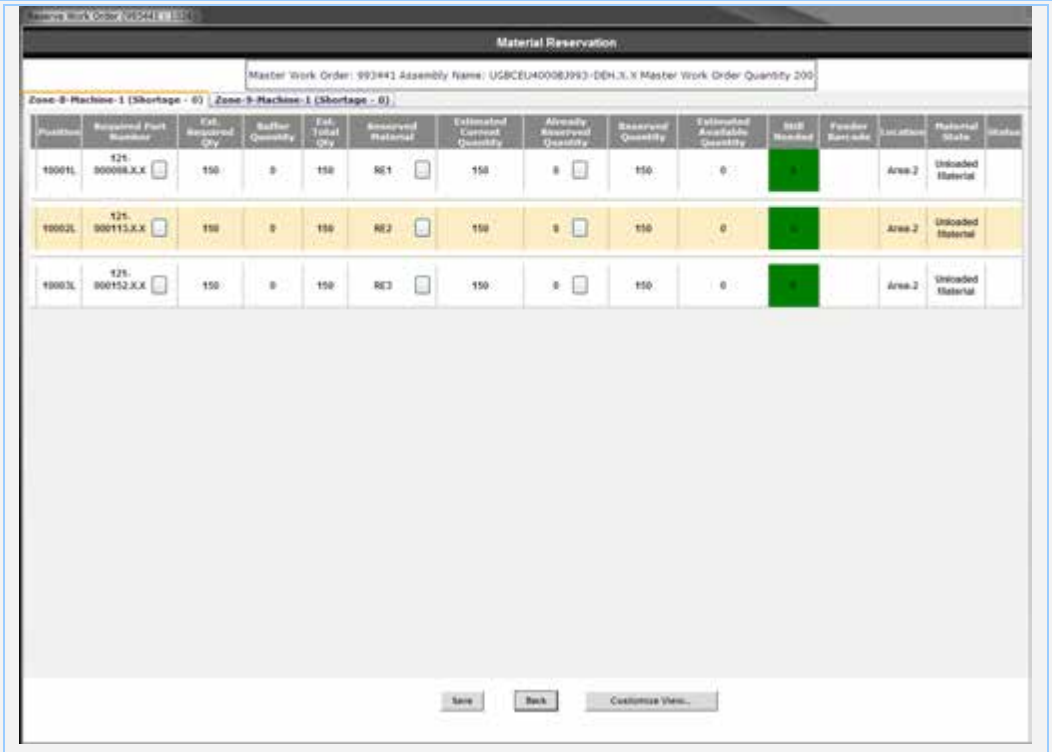
Reallocating a Master Work Order

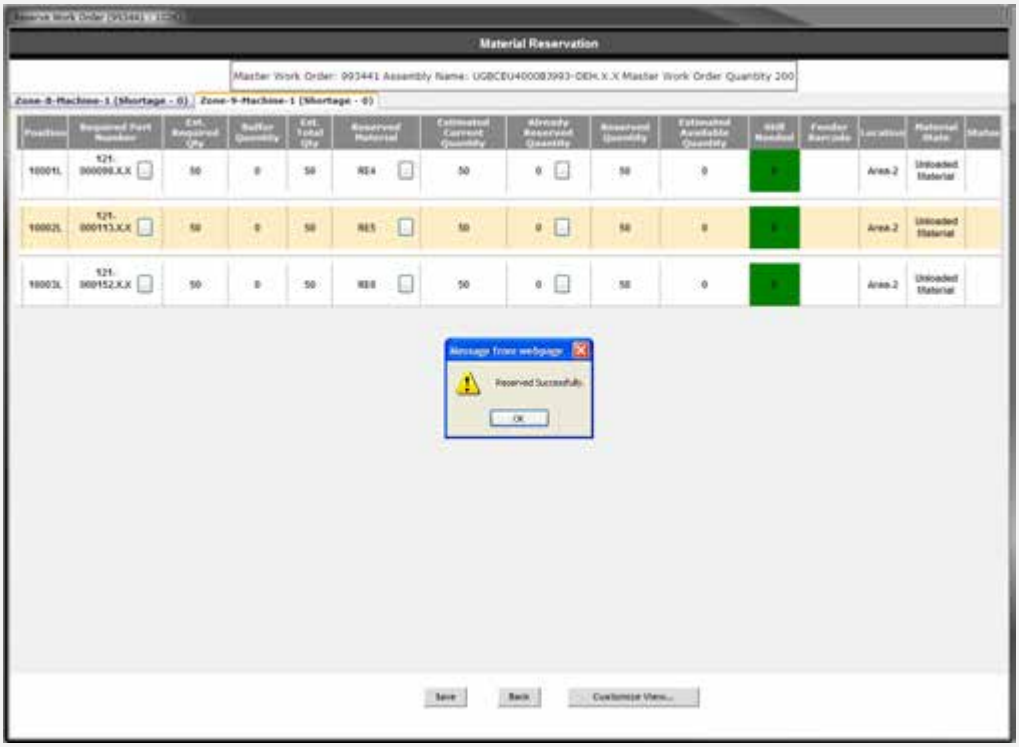
Reallocating a master work order involves changing line assignments so that existing reels used for reservation can be redistributed to fit the needs of new line assignment.

As an example, consider that 200 boards needs to be produced corresponding to a master work order. Out of these 200 boards, 150 boards will be produced corresponding to sub work order 1 and 50 boards will be produced corresponding to sub work order 2. After this master work order is reserved, user decides to change line assignment by adding distributing the production load in such a way that 100 boards will be produced corresponding to sub workorder 1, 50 boards will be produced corresponding to sub work order 2 and 50 boards will be produced corresponding to sub work order 3. In order to accommodate this, user needs to reserve reels for new sub work order and make changes to existing reservations done for previous sub work orders. This is accomplished by means of reallocating an already reserved master work order.

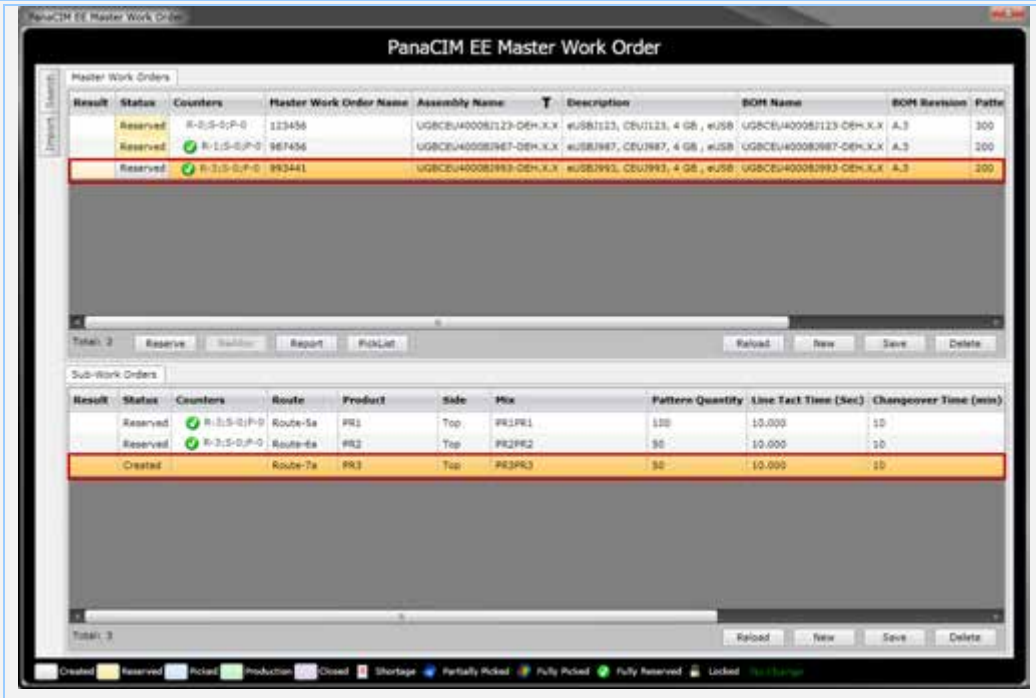
Refer to the action table below for the steps to reallocate an already reserved master work order.

Step	Procedure
1	Consider the following reserved master work order in which 150 boards will be produced on Route-5a corresponding to first sub work order and 50 boards will be produced on Route-6a corresponding to second sub work order.  The reel reservations for this master work order is shown below:



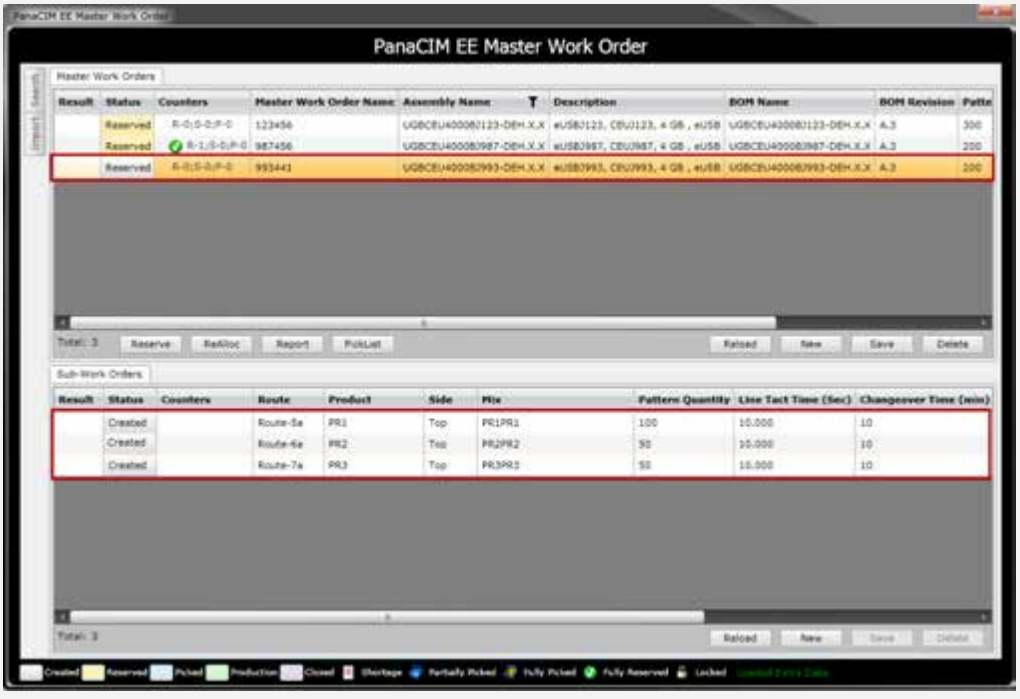


2	To change the line assignment, click the **New** button in the **Sub-Work Orders** tab. After this button is clicked, screen layout changes to the one shown below:



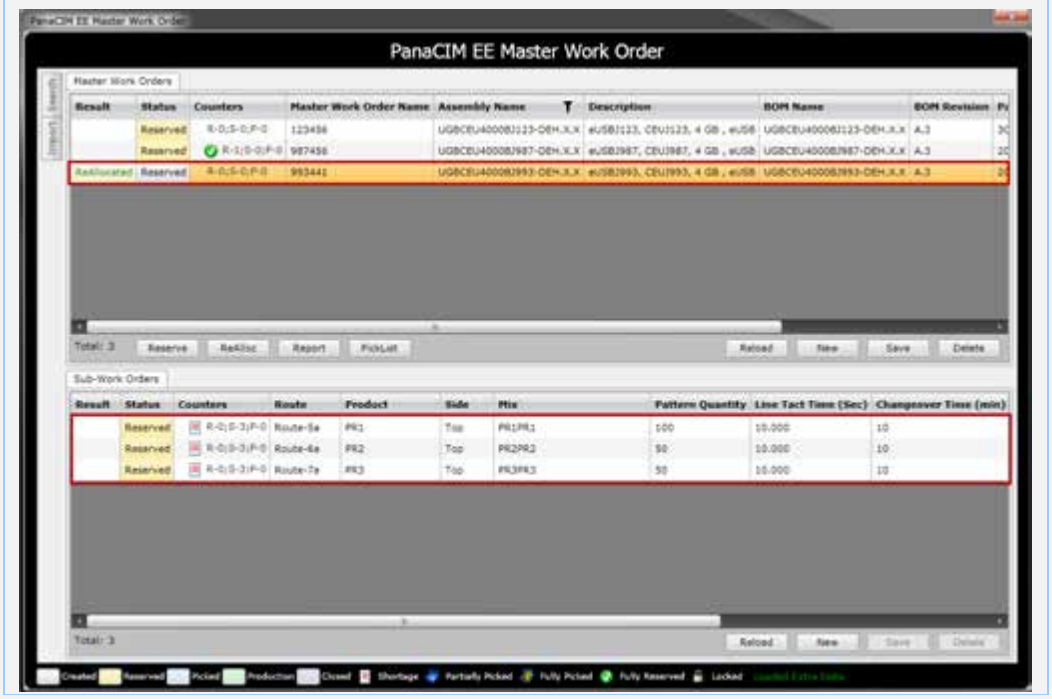
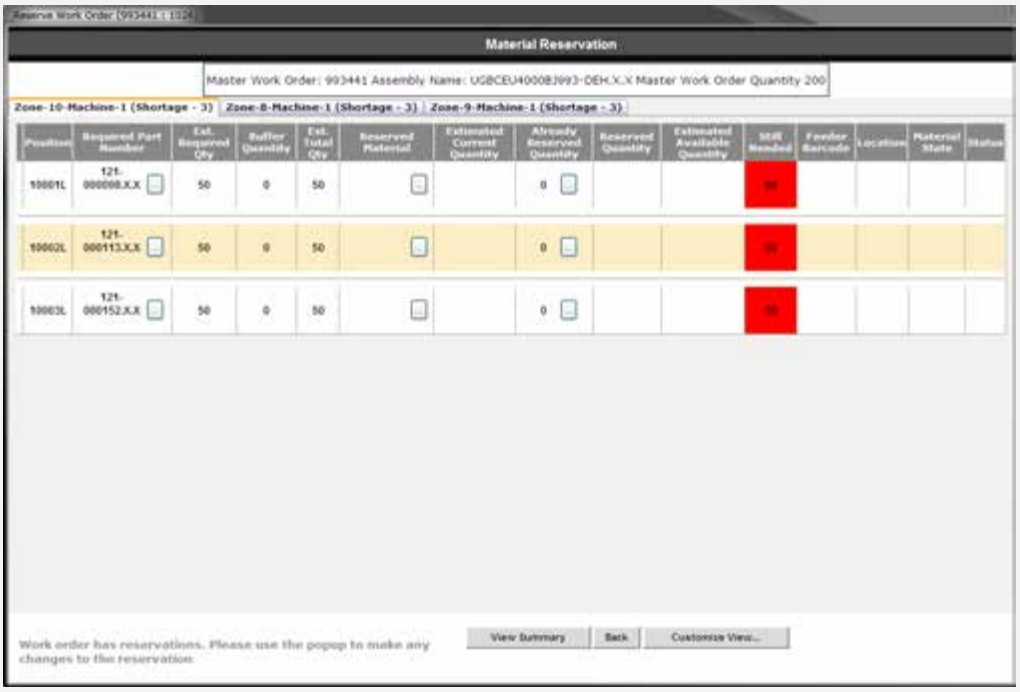
As shown above line assignment is changed such that out of total 200 boards to be produced, 100 boards will be produced on Route-5a (corresponding to sub work order 1), 50 boards will be produced in Route-6a (corresponding to sub work order 2) and another 50 boards will be produced on Route-7a (corresponding to sub work order 3).

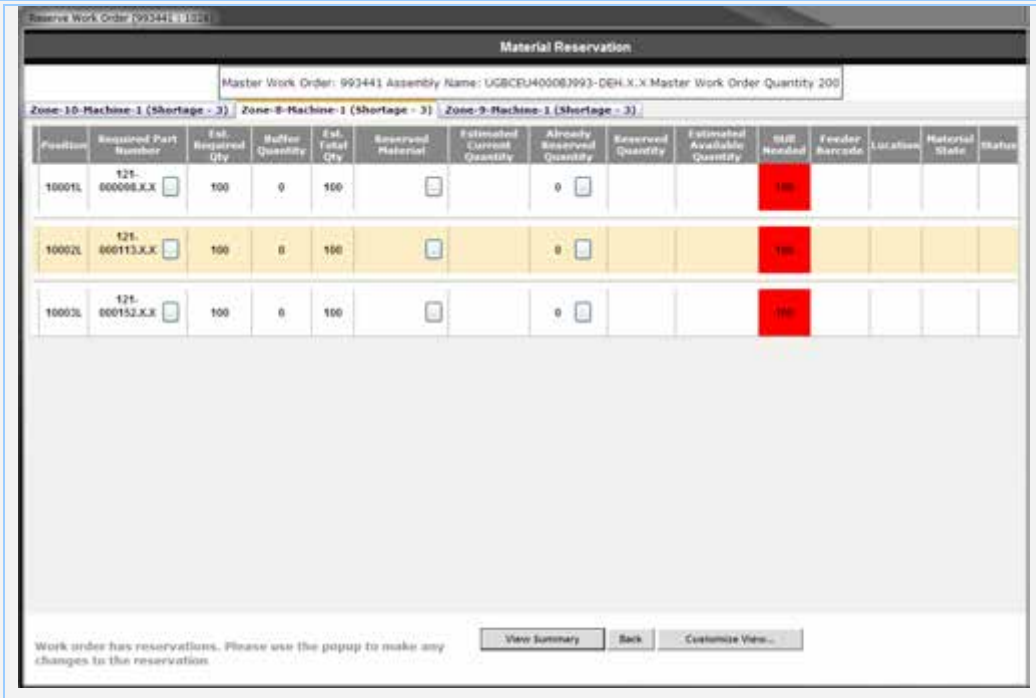
3 Click the **Save** button in the **Sub Work Orders** tab. Layout changes to the one shown below:

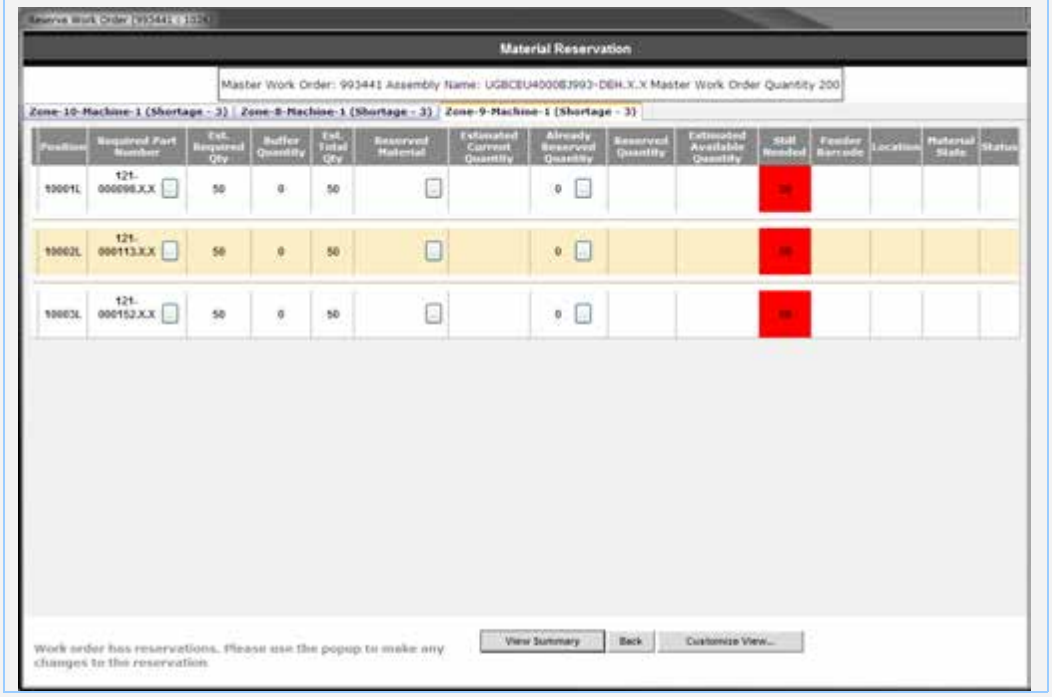


As shown above, status of sub work orders is changed from **"Reserved"** to **"Created"**.

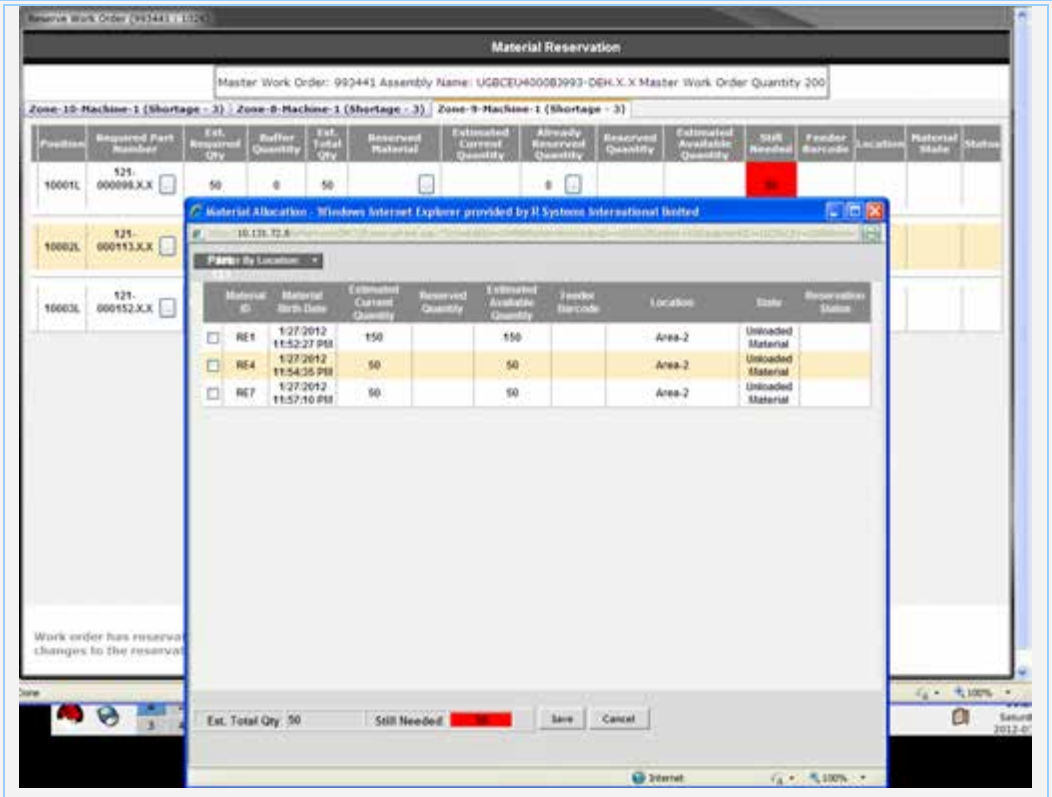
4 Click on the **ReAlloc** button in the **Master Work Orders** tab. After doing so, all reel reservations are deleted from the existing sub work orders as indicated by

	appearance of  icon in the **Counters** column of **Sub Work orders** tab. Also, notice that **Result** column contains keyword **"Reallocated"** for the master work order. 
5	At this point, user has to again reserve the reels for each of the sub work orders by clicking the **Reserve** button. After **Reserve** button is clicked, following screen appears. Note that new tab (in this case **Zone-10-Machine-1**) appears to newly added child sub work order. 



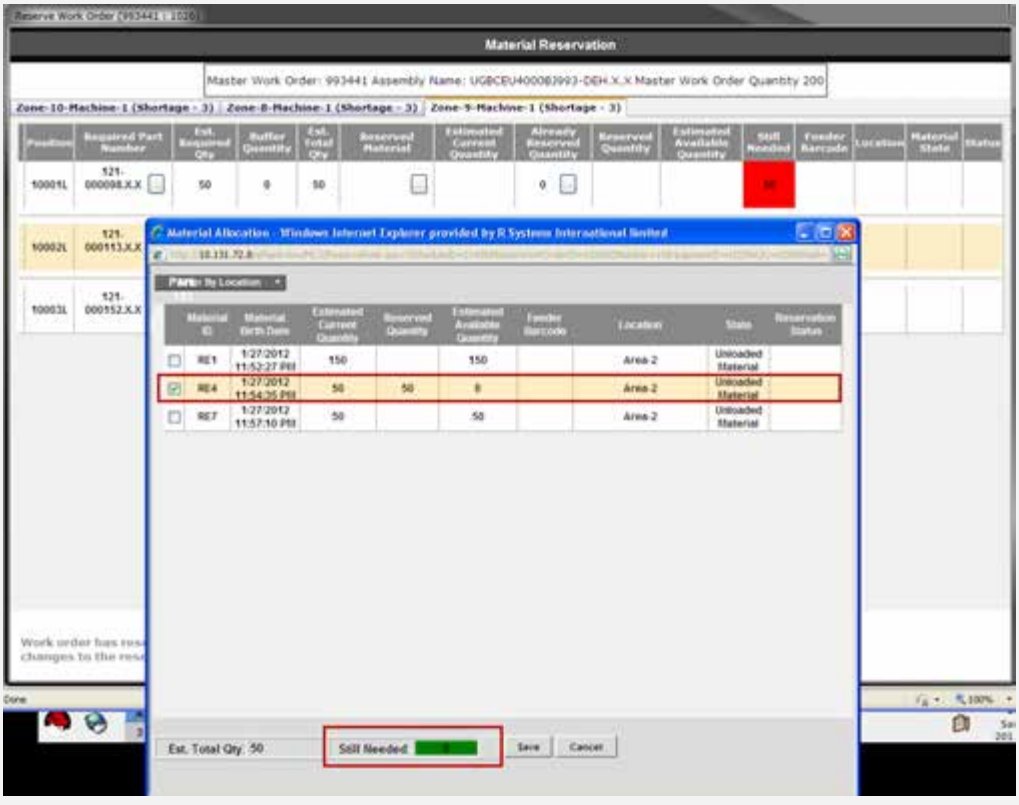


6	At this stage, reels have to be reserved again for each feeder location.
7	Click the … button in the **Reserved Material** column. After doing so, following screen appears:



8 As shown in step 7, all the reels containing part **121-000098.X.X** are displayed. Select one or more reels such that required part quantity shown adjacent to **Still Needed** column is completely accounted for. In this case, selecting reel with barcode RE4 or RE7 fulfills this condition.

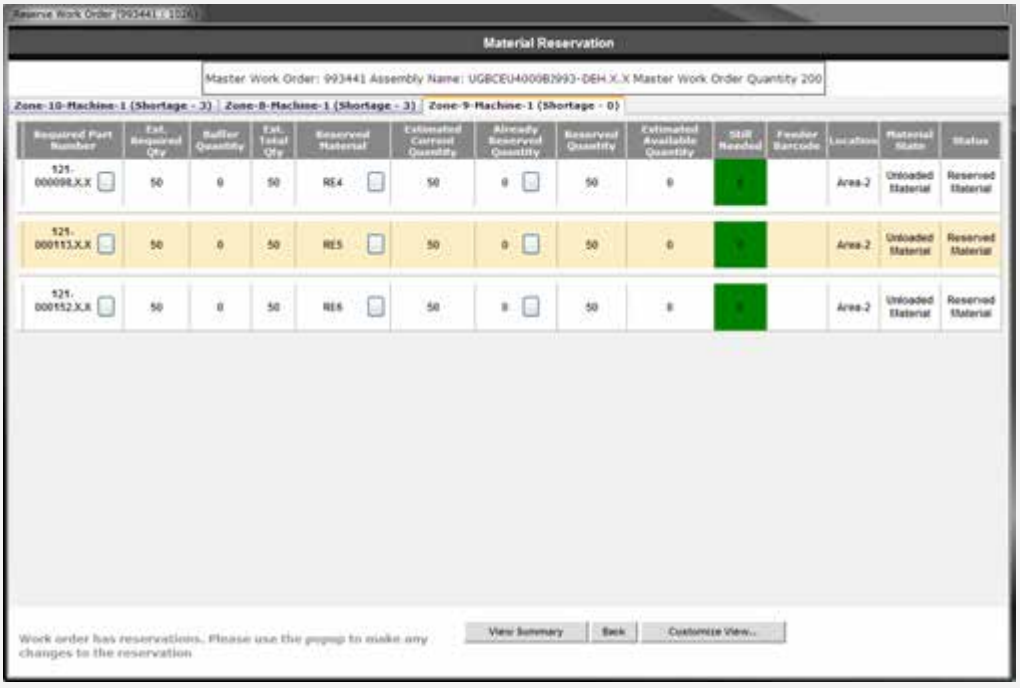
After checkbox corresponding to a reel barcode is clicked, **Still Needed** quantity changes to zero as shown below:

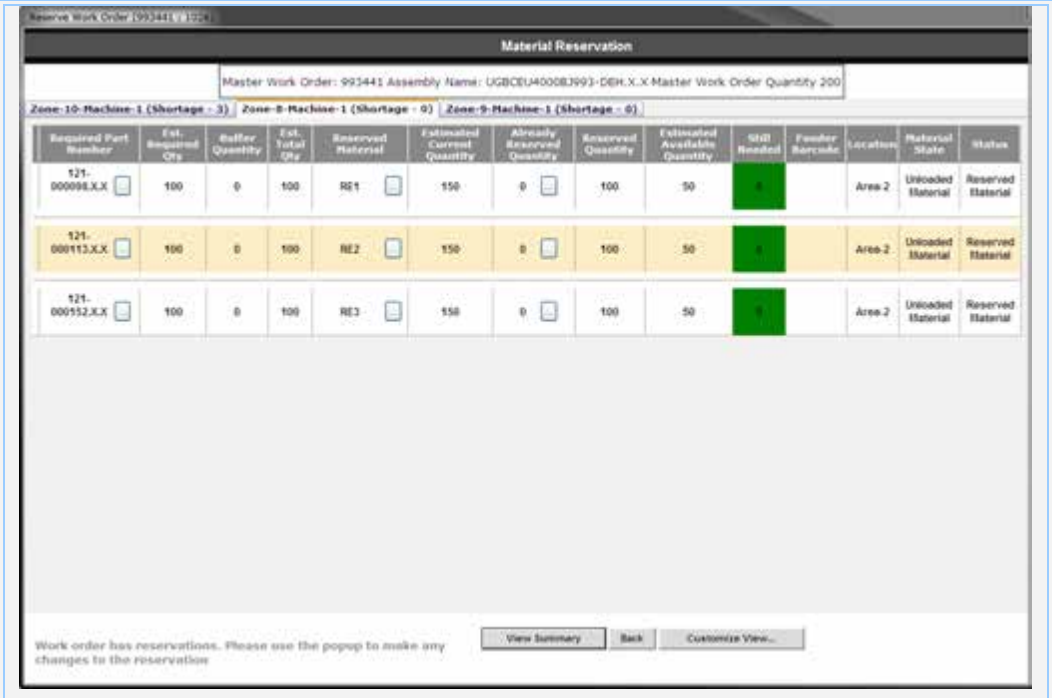


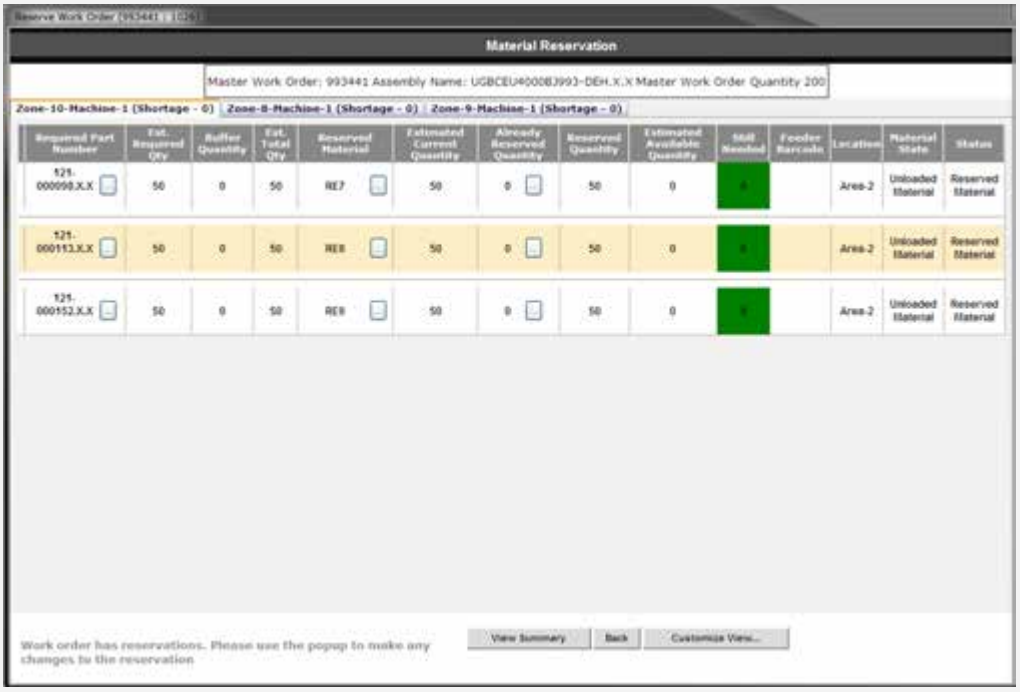
Click the **Save** button to complete the reservation. Now **Material Reservation** page shows the reservation for selected feeder location.



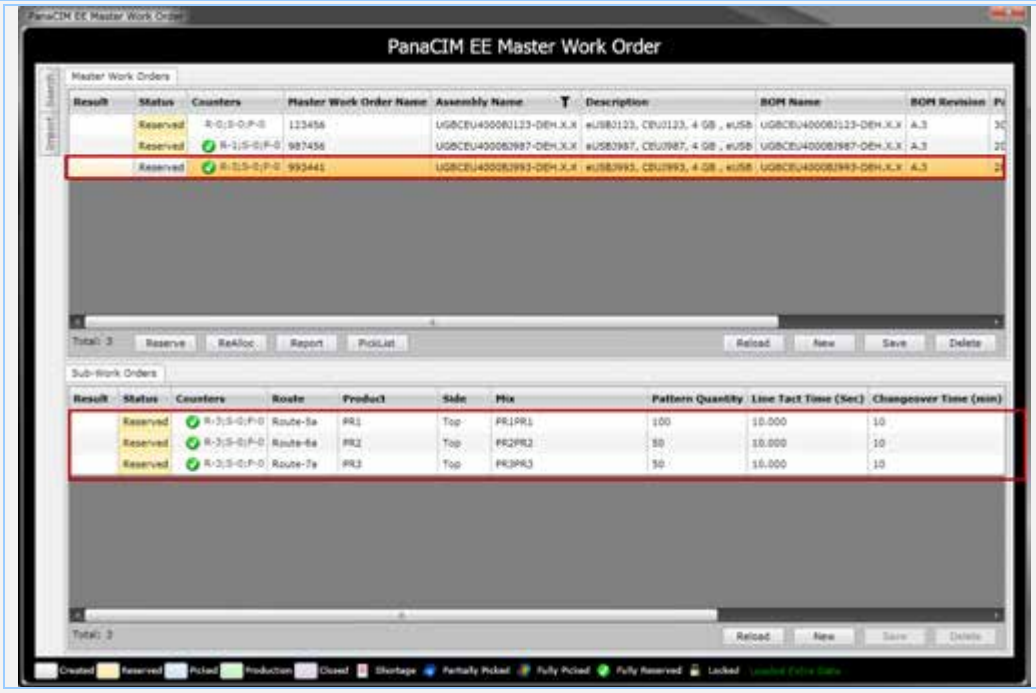
9 Repeat steps 7 and 8, until reservation is completed for all the sub work orders. Reservation details for each sub work order are shown in each individual tab of **Material Reservation** screen.







10	Click the **Back** button in the **Material Reservation** page to return to **PanaCIM EE Master Work Order** page. This page shows the status as fully reserved for each sub work order as shown below:

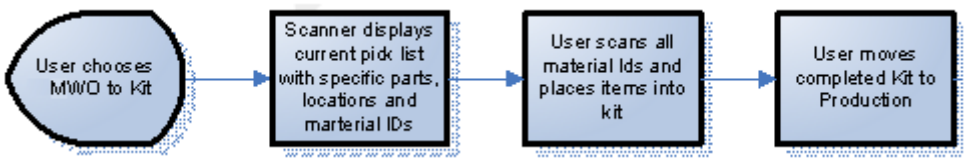

PanaCIM EE Master Work Order
Master Work Orders
Result
Status
Counters
Master Work Order Name
Assembly Name
Description
BOM Name
BOM Revision
Reserved
123456
987456
993441
Total: 3
Reserve
ReAlloc
Report
PickList
Reload
New
Save
Delete
Sub-Work Orders
Route
Product
Side
Mix
Pattern Quantity
Line Tact Time (Sec)
Changeover Time (min)
Route-5a
Route-6a
Route-7a
PR1
PR2
PR3
Top
PR1PR1
PR2PR2
PR3PR3
100
50
10.000
10
Created
Reserved
Picked
Production
Closed
Shortage
Partially Picked
Fully Picked
Fully Reserved
Locked

- **Material Reservation - Column Description**

Column Name	Description
Position	Feeder location at which part needs to be placed as specified in the program file.
Required Part Number	Part Number that needs to be placed at a particular feeder location.
Est. Required Qty	Part quantity required for placement on PCB as per sub work order.
Buffer Quantity	Extra part quantity over and above the required part quantity.
Est. Total Qty	Sum total of Estimated required quantity and Buffer quantity.
Reserved Material	Reel barcode the reel suggested for reservation. After reel reservation is accepted, this reel is reserved for one particular master work order and cannot be allocated to any other work order.
Estimated Current Quantity	Part quantity available in the reserved reel.
Already Reserved Quantity	Quantity that is already reserved by virtue of some earlier reel reservation. Usually it is 0.
Reserved Quantity	Quantity existing in the reel suggested for reservation. This quantity is reserved for one particular master work order and cannot be allocated to any other work order.
Estimated Available Quantity	Difference between Estimated Current Quantity and Reserved Quantity.
Still Needed	If reserved quantity is less than the required part quantity, it is quantity that is still needed to fulfill the shortage. It is calculated as difference between required part quantity and reserved quantity.
Feeder Barcode	Barcode of the feeder containing the reel that is suggested for reservation. This reel contains left over quantity from earlier production run and this left over quantity can be reserved for a sub work order.
Location	Current location of the reel suggested for reservation. For example, it can be unloaded reel present in a material area in which case, name of this material area will be shown in the location column.
Material State	State of the material with regards to whether it is unloaded or still mounted at some feeder location.
Status	Status of the reel with regards to different work order states like reservation, picked e.t.c

Master Work Order Kitting

This feature collects reserved materials from the storage area (storage units) for pickup, and moves them into the storage area (storage unit) as preparations for master work order.

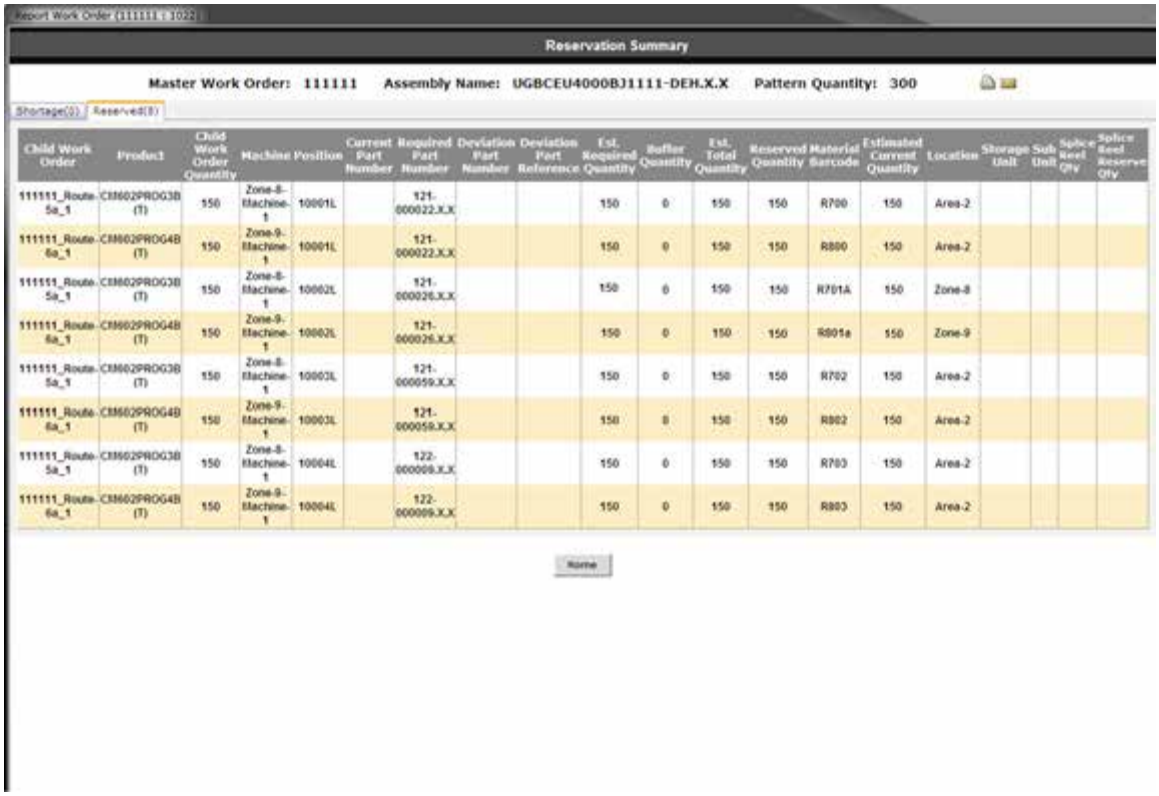


Step	Description
1	User chooses master work order to kit in the scanner. Master work order chosen for kitting is the one for which material reservation is completed.
2	Kitting personnel is prompted to scan or enter the storage unit into which picked material needs to be stored.
3	Scanner shows the locations from which material needs to be picked for a particular master work order. Kitting personnel pick only reserved material for master work order using hand scanner. For example, if four reels have been reserved for a work order out of which three are present in a material handling area and one is present in the production zone, then after selecting master work order, name of this material area and production zone is shown. The number of reels present in each location is indicated in the bracket.
4	A list of materials in the storage unit selected from the pickup list appears. Scan or enter a material ID shown in the material list to pick up the material concerned. If two or more storage units are shown in the pickup list even after all materials in the material list have been picked up, the screen in step 3 returns. Before picking a reel, kitting personnel scan material ID of each reel present in the selected location. Details of each reel can be seen like Part number, part quantity, and location. Scanned material ID is validated to ensure that reel containing correct part is picked.
5	Completed kit is moved to production staging area where it is readied for production.

Material Kitting

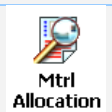
Kitting involves picking material for producing the product and placing it into storage units. These storage units are then moved into production zones for the reels to be loaded to feeders. Once the loaded feeders are mounted to machine slots, production can begin. Material kitting is performed when reels have been successfully reserved for a master work order.

From the **Reservation Summary** page, one can view the reserved reels which need to be picked and their current location. This is shown below:



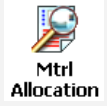
As shown in the above screen, we have to pick eight reels. Out of these eight reels, six are located in Area-1 and one each is located in Zone-8 and Zone-9 respectively.

Refer to the action table below for the procedure to kit a master work order

Step	Procedure
1	Click  icon in the Material Handling screen shown below:



2

After clicking 

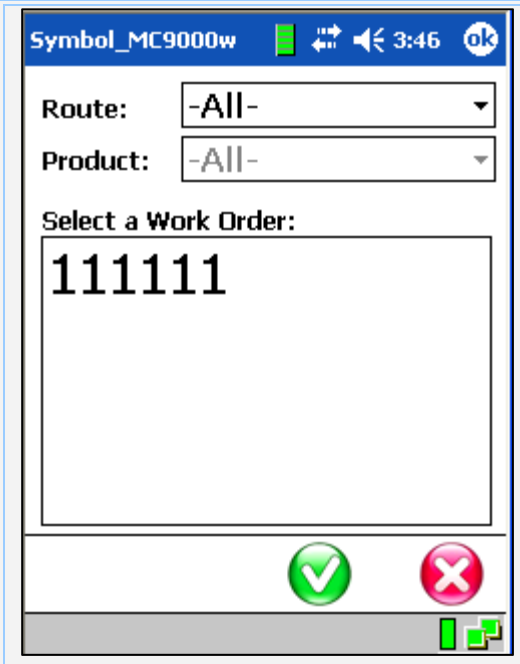
icon, following screen appears.



3

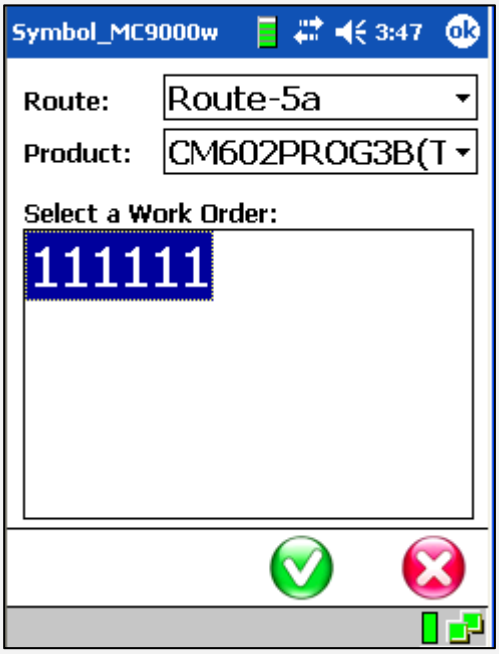
Choose 

from the screen shown in step 2. After doing so, following screen appears showing the Master Work Order (in this case 111111) for which material reservation is successfully completed.



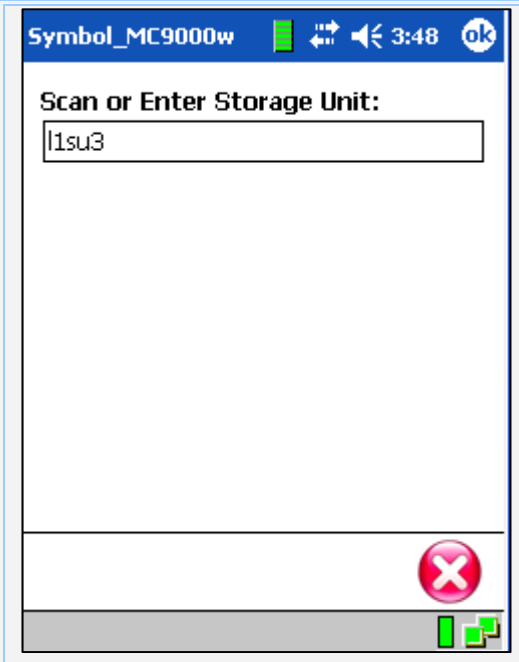
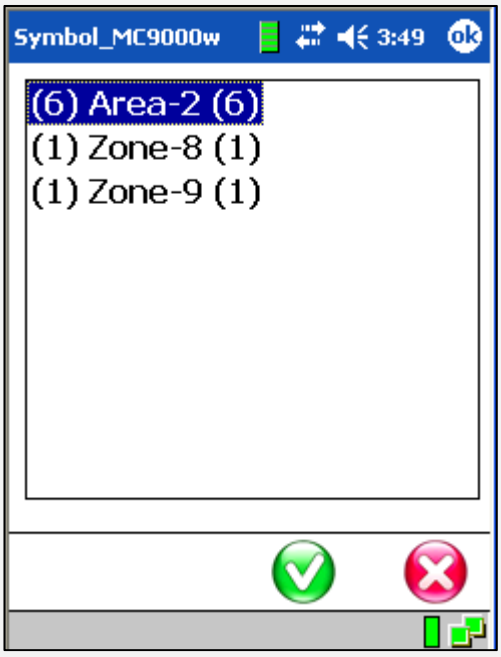
4

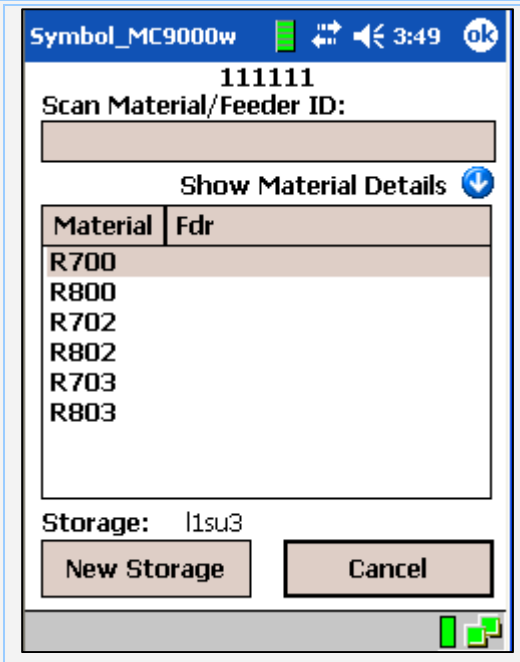
Select the Master work order and then click 



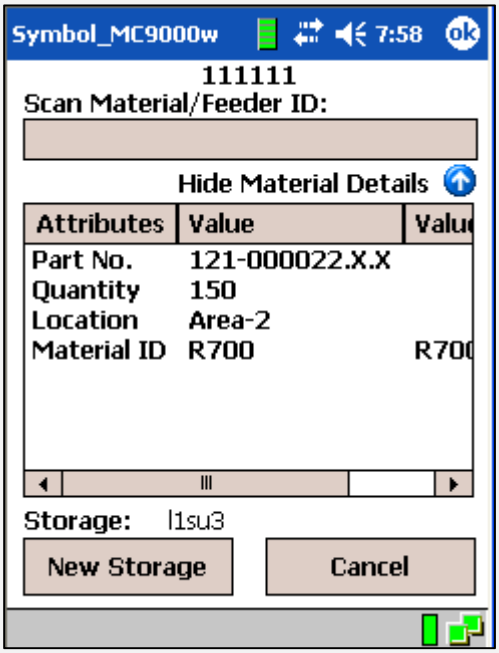
5

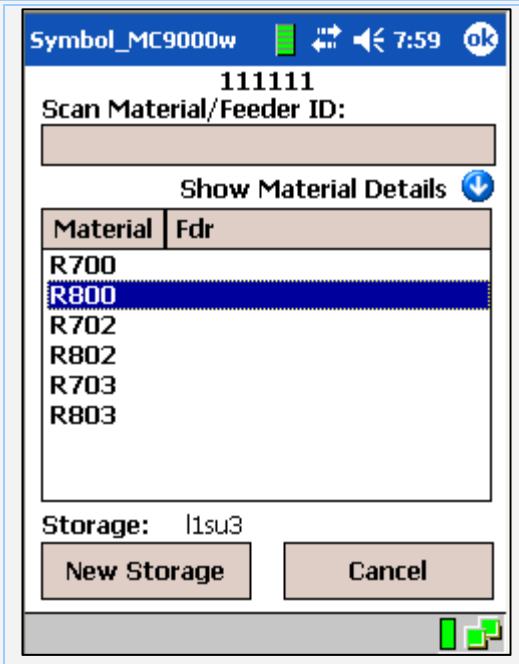
After selecting master work order, following screen appears to scan or enter storage unit into which picked reels will be placed.

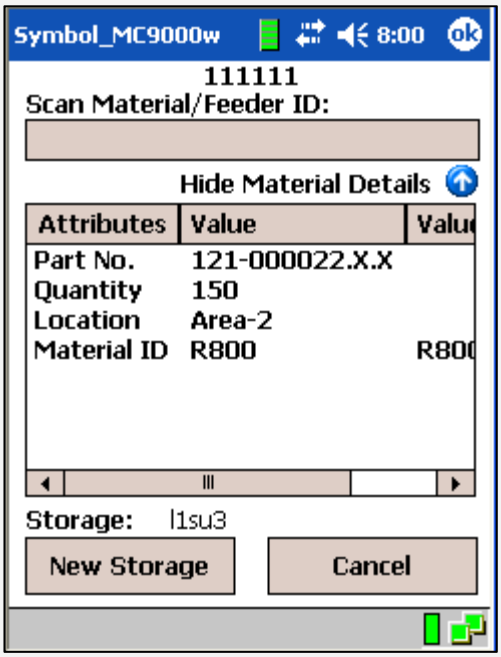
	
6	After scanning or entering storage unit, following screen appears showing the locations in which reserved reels are present for the selected master work order. As shown in the below screen, out of eight reserved reels, six are located in Area-2, one is present in Zone-8 (in unloaded state) and one is located in Zone-9 (also in unloaded state). This is shown below:  Select the location from which reserved reel needs to be picked and click  .
7	After selecting the location in step 6, following screen appears. As shown in this screen, name of master work order (in this case 111111) is shown on the top. Reels present in the selected location are also shown in this screen.

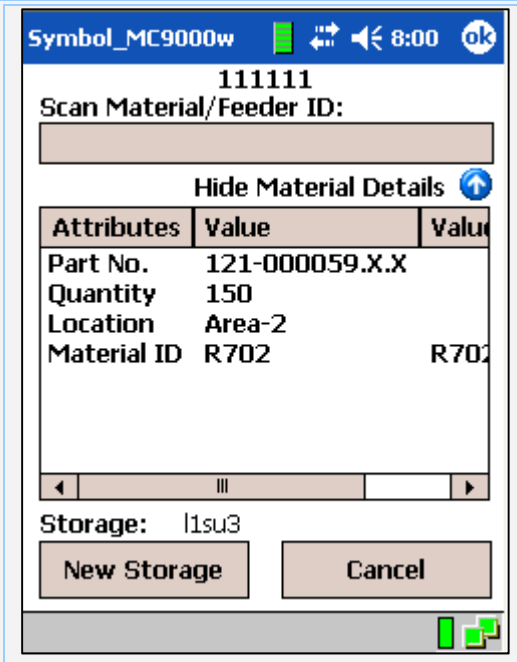


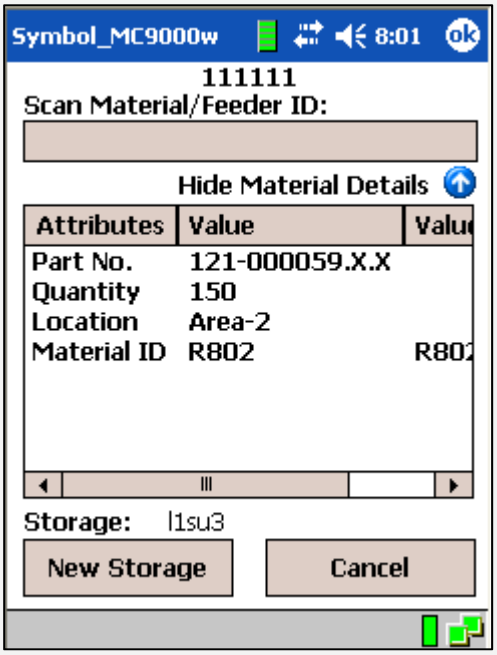
Reel details like its current location, part quantity, part number can be viewed by clicking the button adjacent to **Show Material Details** label. Reel details for the reels located in Area-2 are shown below:

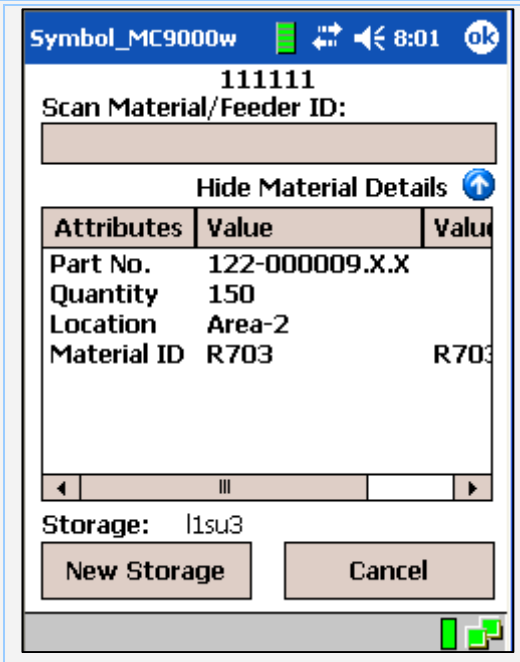


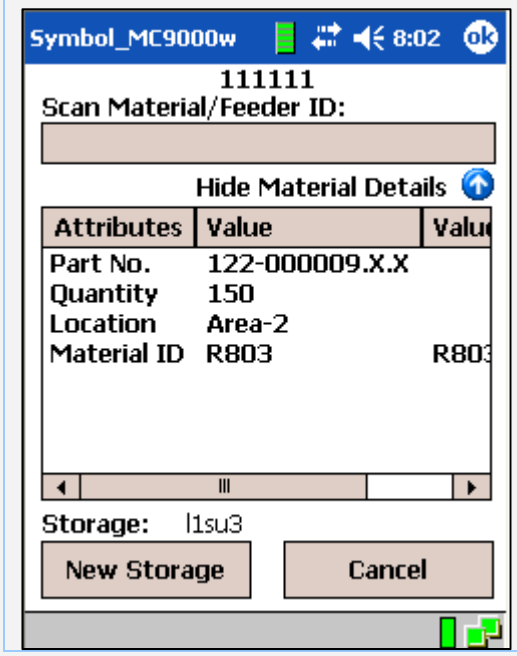

Symbol_MC9000w
7:59
ok
111111
Scan Material/Feeder ID:
Show Material Details
Material
Fdr
R700
R800
R702
R802
R703
R803
Storage: l1su3
New Storage
Cancel


Symbol_MC9000w
8:00
ok
111111
Scan Material/Feeder ID:
Hide Material Details
Attributes
Value
Part No. 121-000022.X.X
Quantity 150
Location Area-2
Material ID R800
Storage: l1su3
New Storage
Cancel

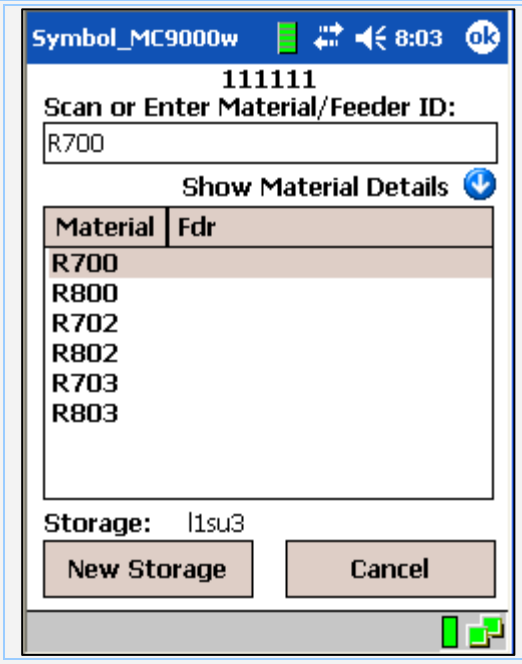

Symbol_MC9000w
8:00
ok
111111
Scan Material/Feeder ID:
Hide Material Details
Attributes
Value
Part No.
121-000059.X.X
Quantity
150
Location
Area-2
Material ID
R702
Storage:
I1su3
New Storage
Cancel

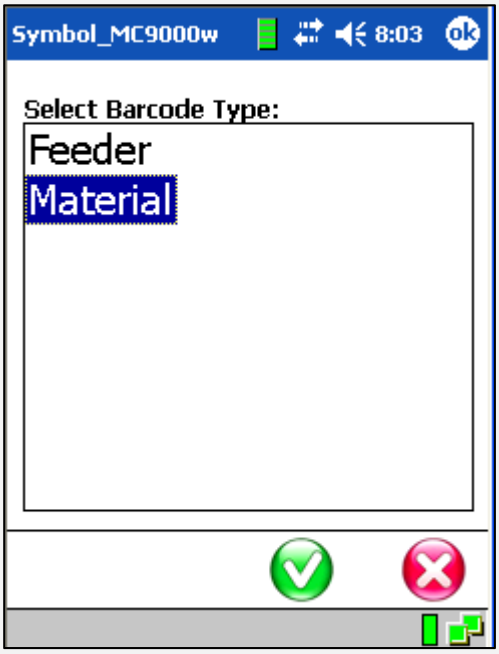

Symbol_MC9000w
8:01
ok
111111
Scan Material/Feeder ID:
Hide Material Details
Attributes
Value
Part No.
121-000059.X.X
Quantity
150
Location
Area-2
Material ID
R802
Storage:
I1su3
New Storage
Cancel

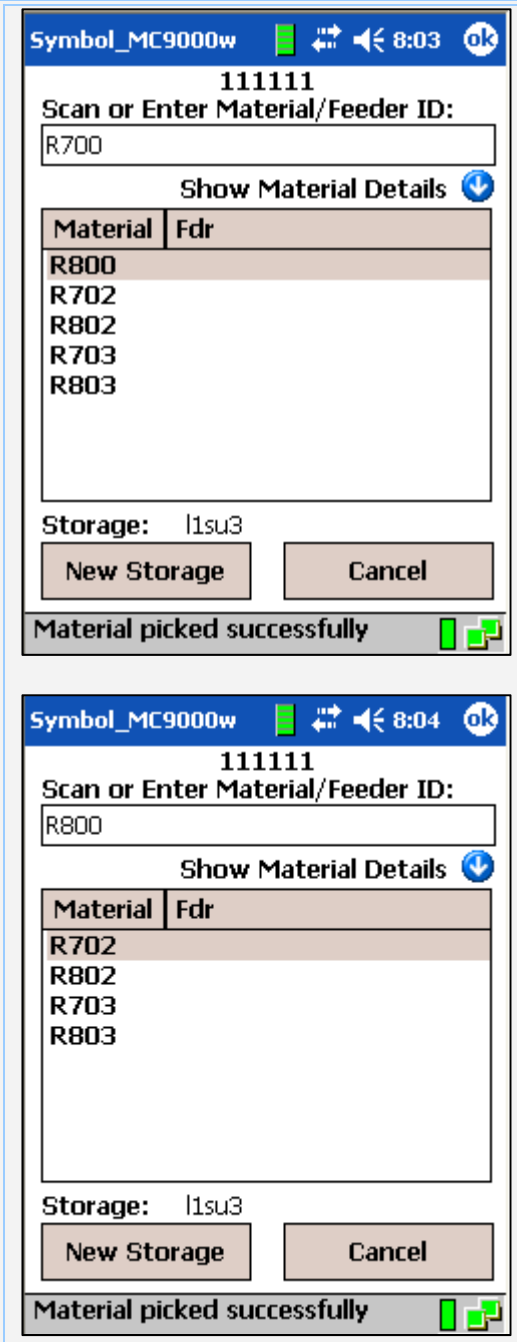


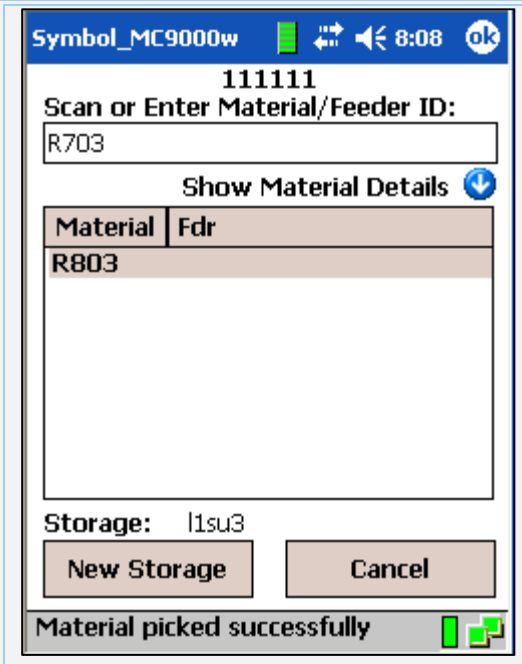


8	Scan or enter reel barcode of the reel that needs to be picked. This is shown below:



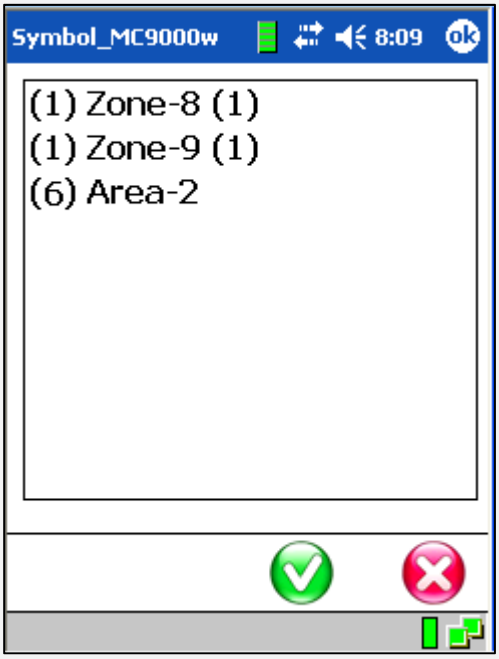
9	After entering reel barcode of the reel, following screen appears to select the type of reel barcode that is keyed in  **Note:** Above screen does not appear when reel barcode is scanned since barcode parser can determine the barcode type.
10	After correct reel barcode is scanned/entered, a message **"Material picked successfully"** appears as shown below. Also this reel is now removed from the list of reels to be picked.


Symbol_MC9000w
8:03
ok
111111
Scan or Enter Material/Feeder ID:
R700
Show Material Details
Material
Fdr
R800
R702
R802
R703
R803
Storage:
l1su3
New Storage
Cancel
Material picked successfully
Symbol_MC9000w
8:04
ok
111111
Scan or Enter Material/Feeder ID:
R800
Show Material Details
Material
Fdr
R702
R802
R703
R803
Storage:
l1su3
New Storage
Cancel
Material picked successfully



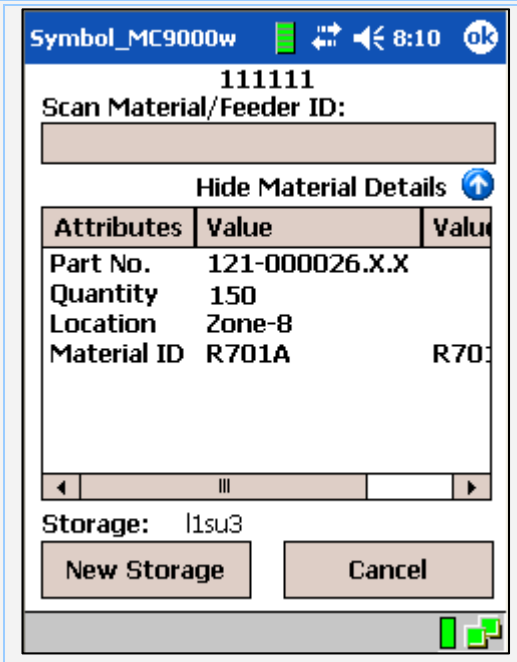
11

When all the reels are picked from a material area (in this case Area-2), brackets to the right of area name containing the number of reels to be picked are removed. This is shown below:

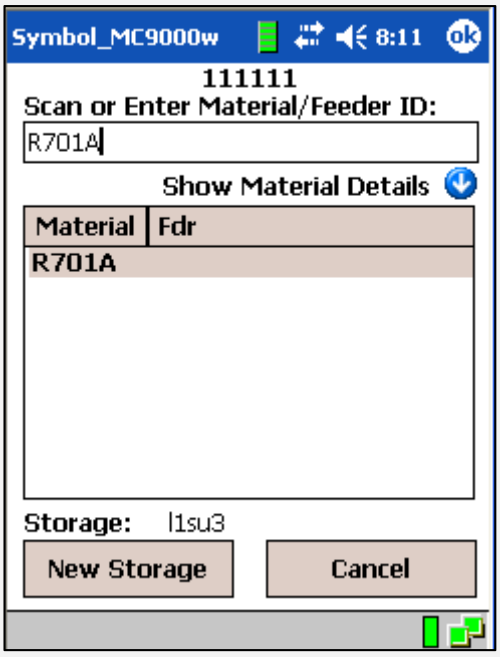


12

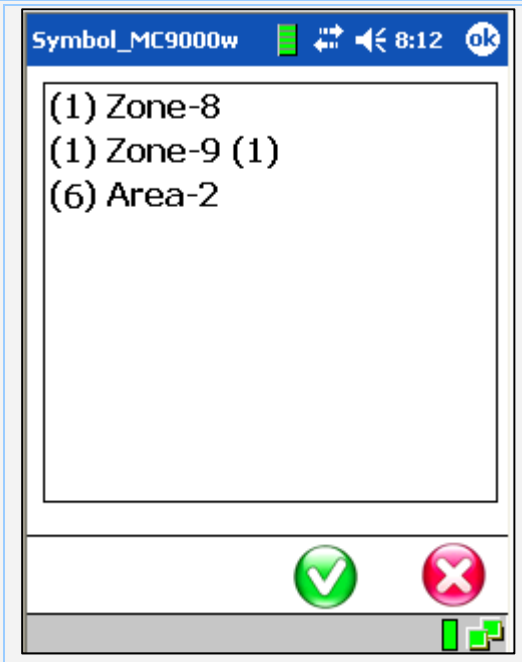
To pick a reel in Zone-8, select it and click  in the screen shown in step 11. You can view the reel details of the reel that needs to be picked from Zone-8 as shown below:



13	Scan/enter the reel barcode of the reel that needs to be picked up from Zone-8 (in this case R701A). This is shown below:



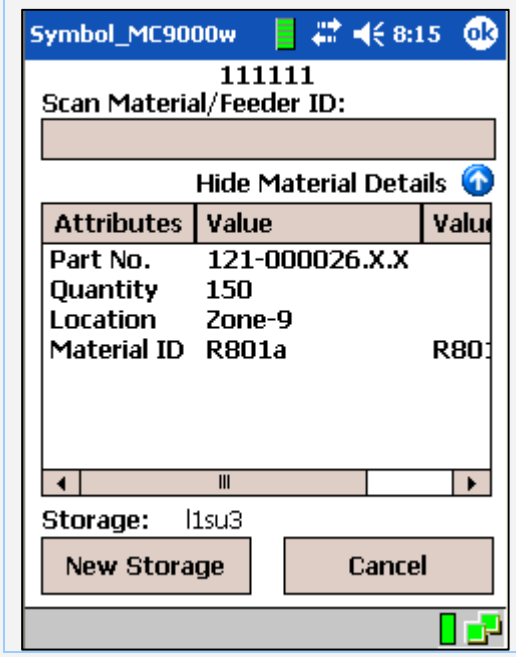
Once this reel is picked, brackets after Zone name disappears as shown in the following screen:



At this stage, all the reserved reels have been picked up from Area-2 and Zone-8 and only one reel remains to be picked from Zone-9 (as indicated by the number in brackets after zone name).

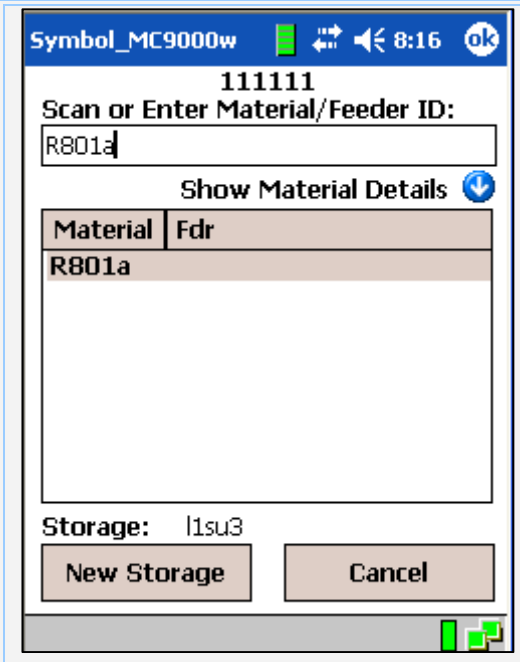
14

Select the Zone name (in this case Zone-9) and click  . After doing so, following screen appears:

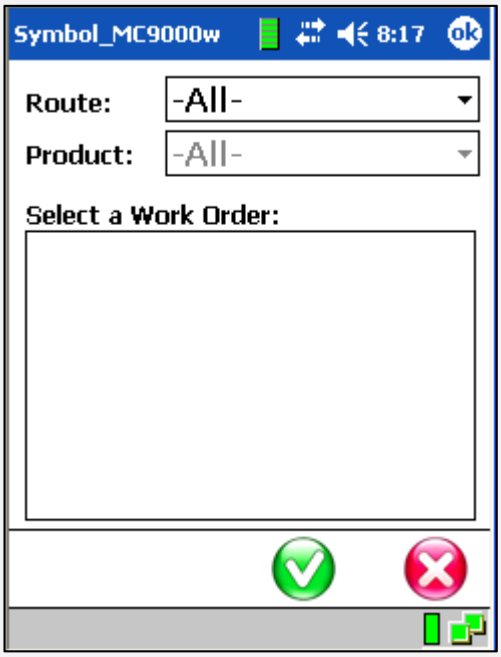


15

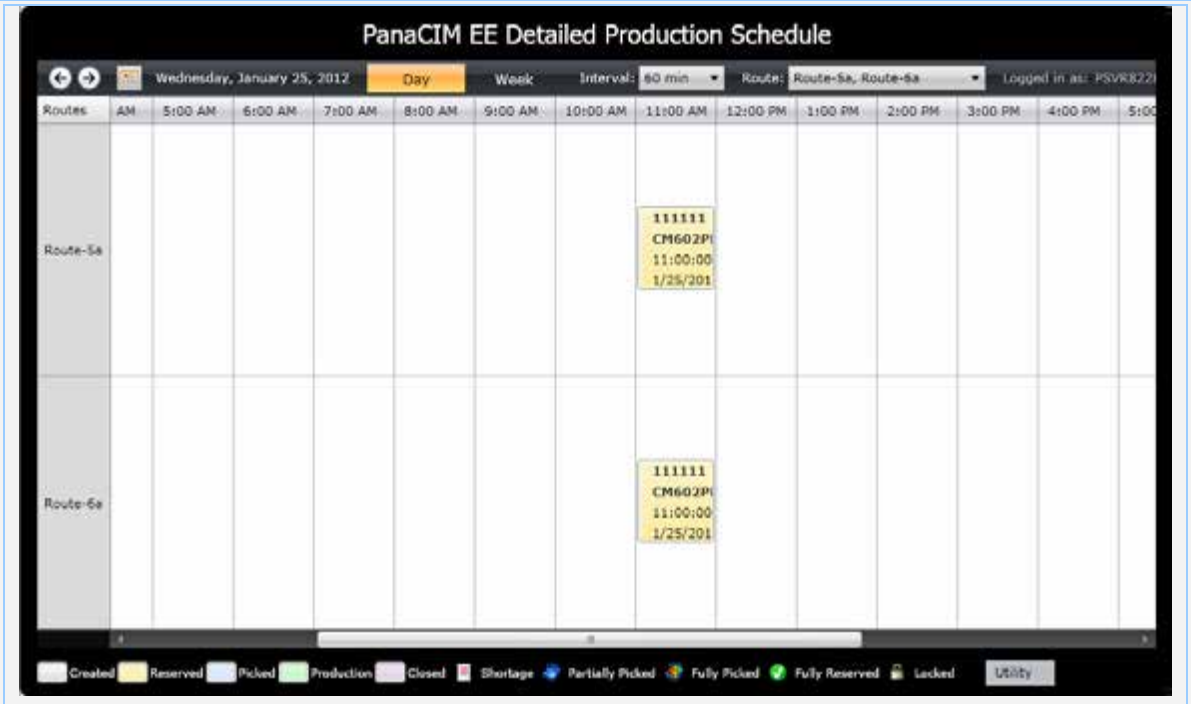
Scan/enter the reel barcode the reel (in this case R801a) to be picked from selected zone (in this case Zone-9). This is shown below:



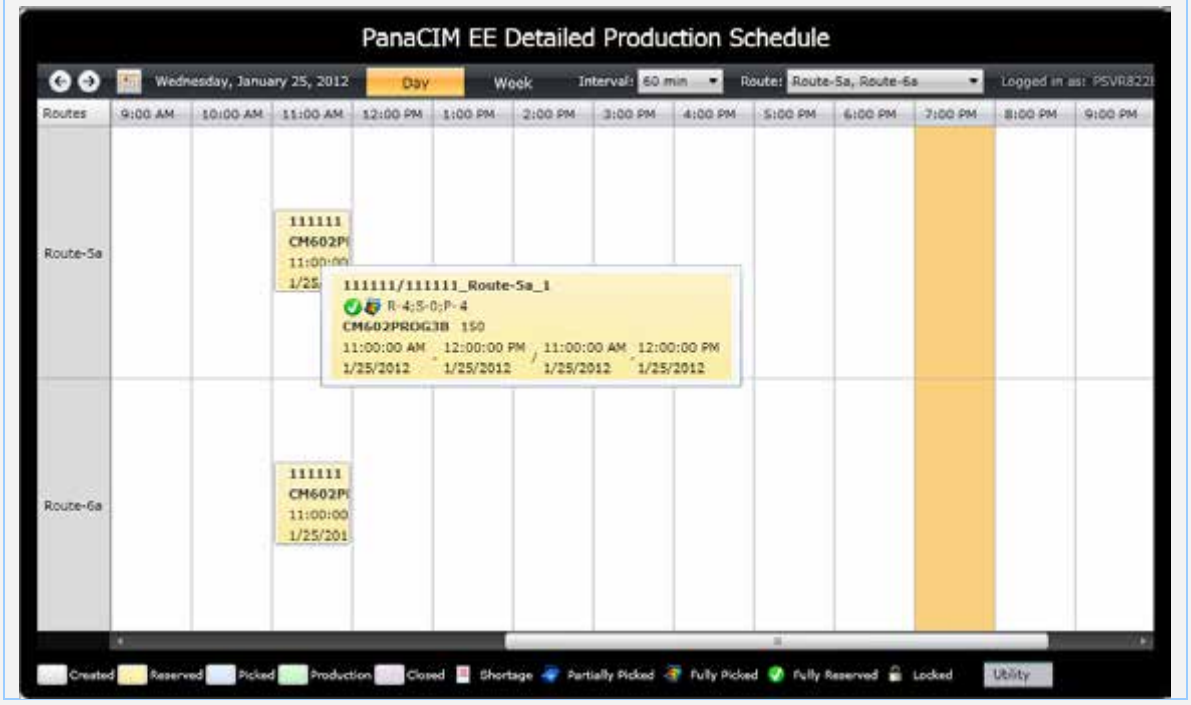
16	Once the only remaining reel is picked, material pick operation is successfully completed for the master work order. This is indicated by disappearance of master work order as shown below:



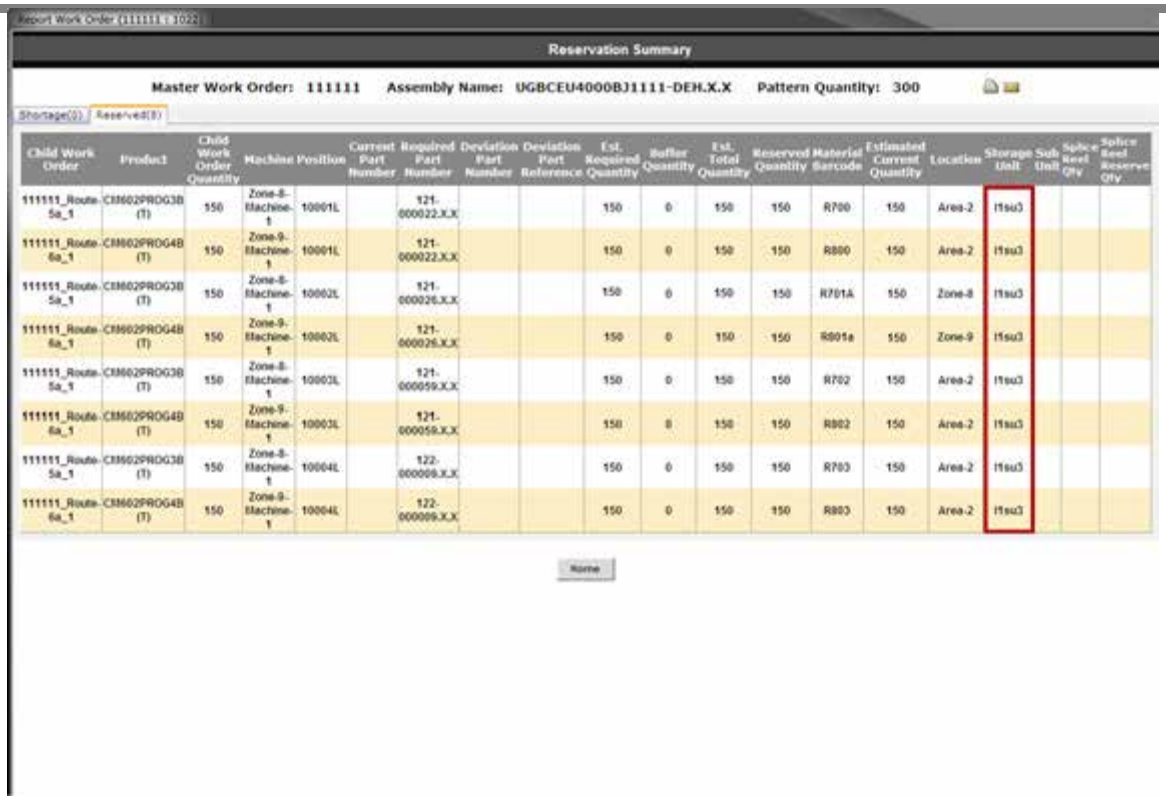
17	After reservation of reels for a sub work order and before material pick, status of sub work order is indicated in the DPS screen by following color scheme:



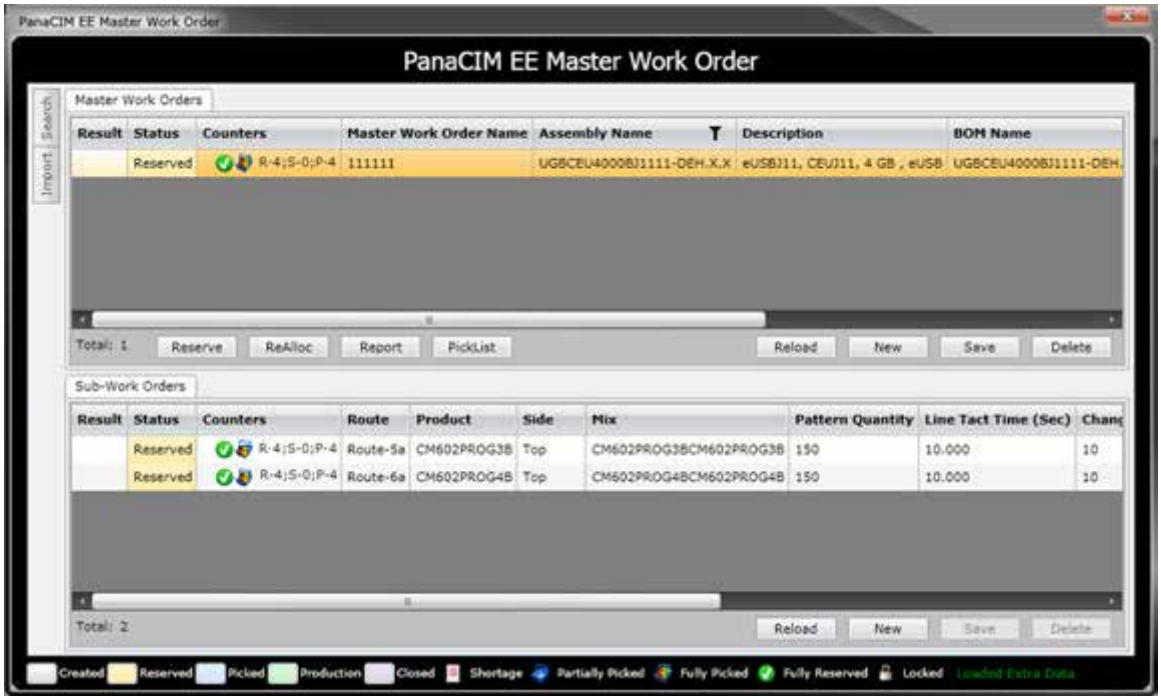
State in which all the reserved reels for sub work order are picked, is also indicated in DPS as shown below:



Note:
After all the reserved reels for a master work order are picked, **Storage Unit** column is populated with the name of storage unit into which picked material is placed. This is shown below:

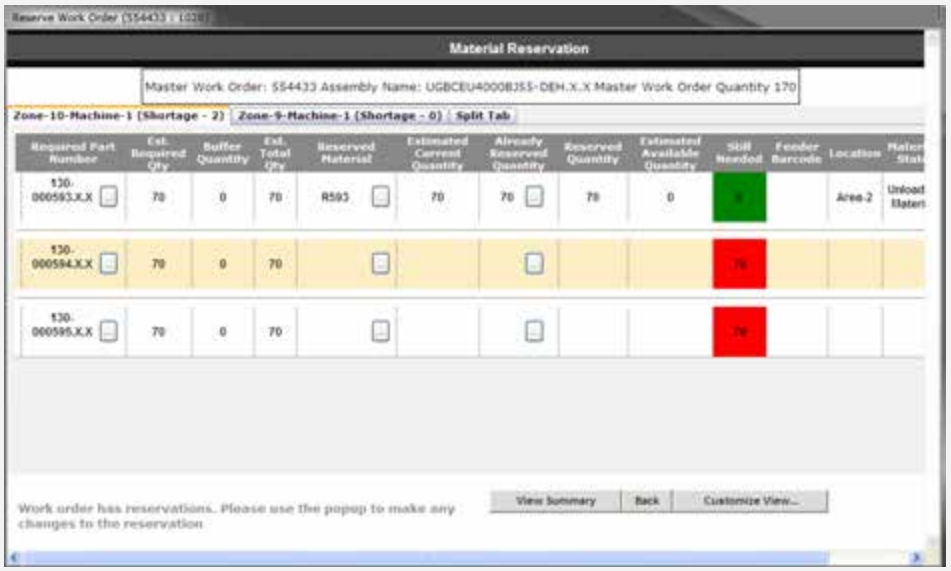
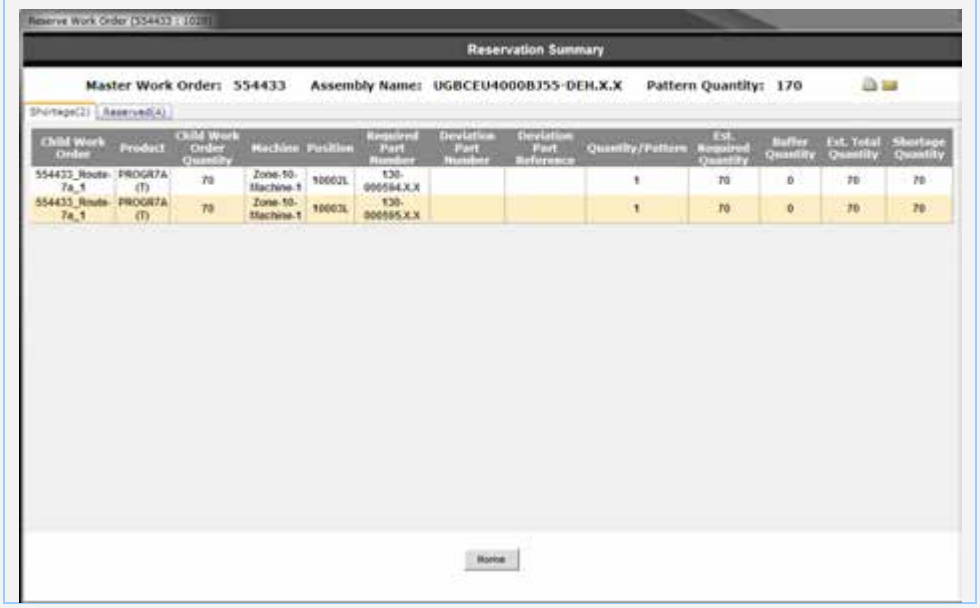


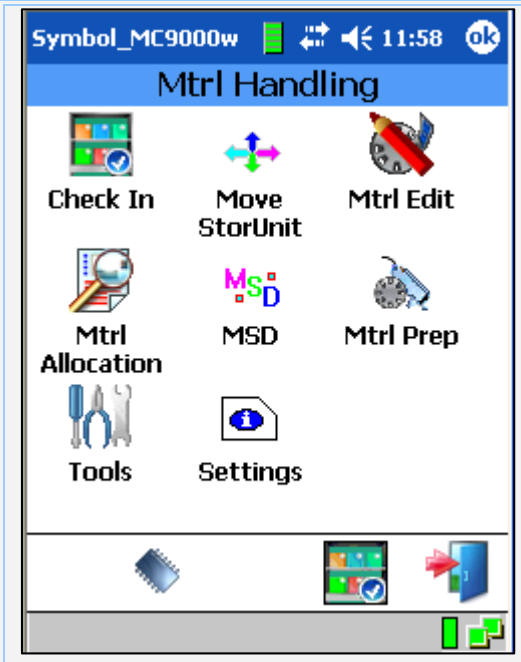
PanaCIM EE Master Work Order screen also flags the Master Work Order and its child work orders with status **"Fully Picked"**. This is shown below:



Fulfilling Shortage for a Work Order

In case reels existing in a material areas doesn't have sufficient part quantity required to complete production for a work order, material is ordered from the warehouse to backfill this shortage. Once the material arrives from the warehouse, operators perform Material Fulfill operation using the mobile scanner. The procedure for performing Material Fulfill operation is shown below:

Step	Description
1	Consider that there is shortage for a master work order for which no reels exist on the factory floor. Shortage is reflected in **Material Reservation** and **Reservation Summary** screens as shown below:  
2	Launch the Material handling screen (shown below) from the Mobile Scanner.



3

Select 

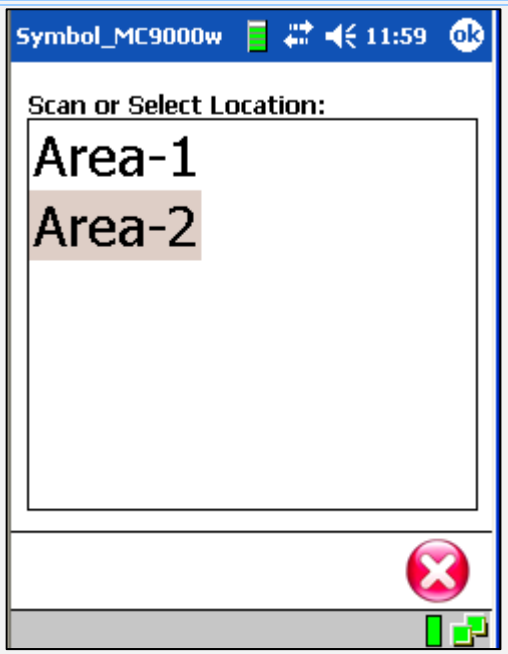
. After doing so, following screen appears:

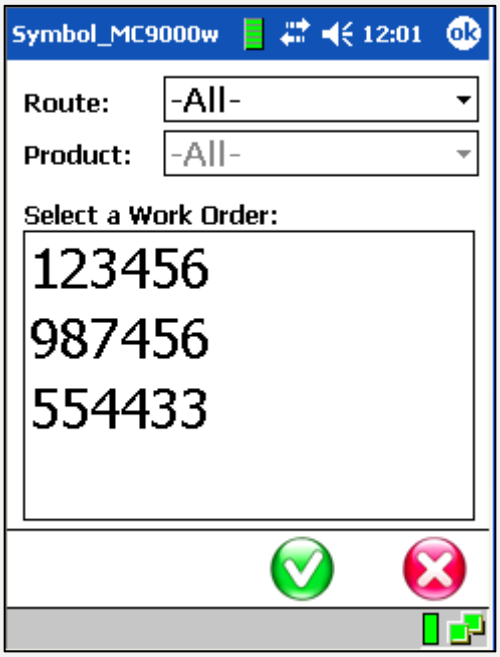


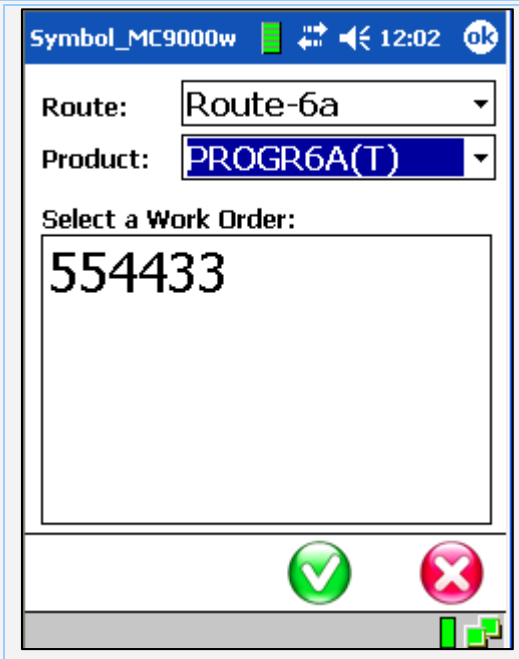
4

Select 

. After doing so, following screen appears listing material areas.

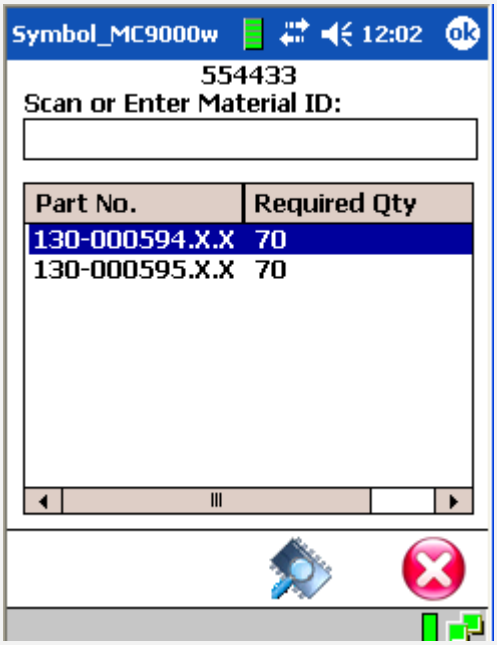


5	Scan or select a location into which material received from warehouse is checked in. After doing so, following screen appears listing the name of master work order for which shortage exists. 
6	Select the route name and product name from the **Route** and **Product** drop-down lists respectively. This lists the name of master work order whose production will run in the selected route.



7

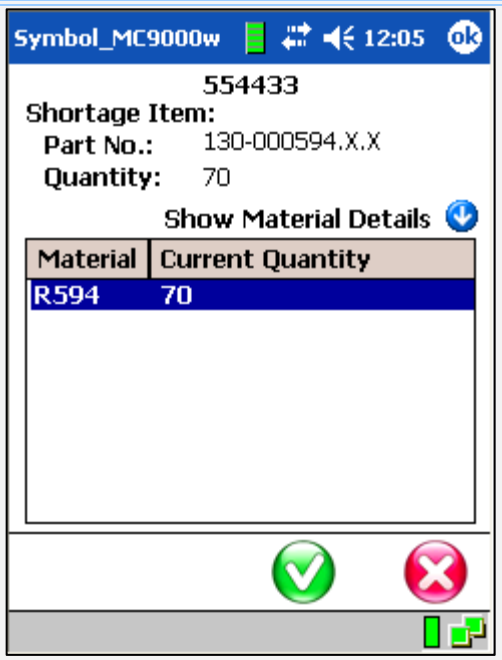
Select the name of the master work order in step 6 and click . After doing so, following screen appears listing parts that have shortages for the selected master work order.



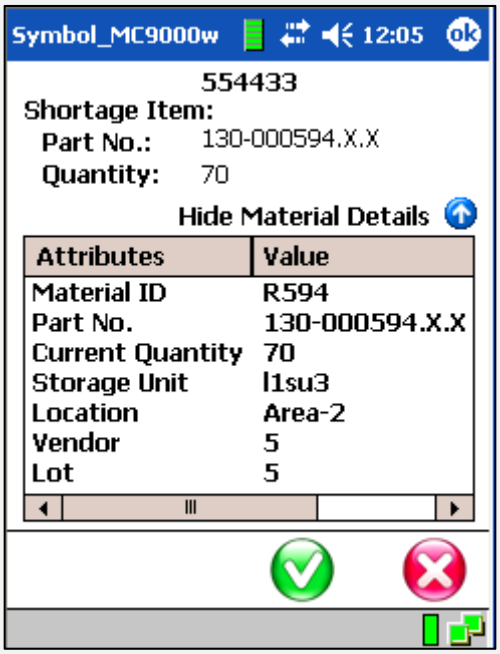
As shown in the screen above, part numbers 130-000594.X.X and 130-000595.X.X are short of required quantity. Quantity that needs to be backfilled by way of ordering material from the warehouse in shown in the **Required Qty** column.

8

Select a part number from the screen shown in step 7 and click to know reel barcodes of the reel containing this selected part. After clicking , following screen appears:

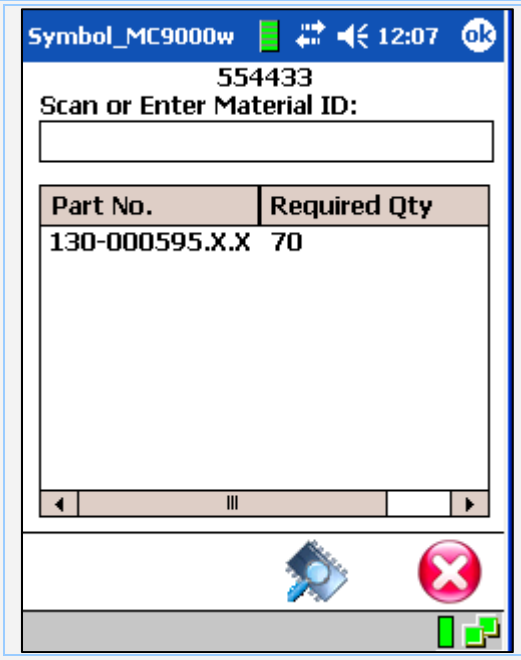


Click  to view reel details of the selected reel. After clicking  , screen layout changes to following:

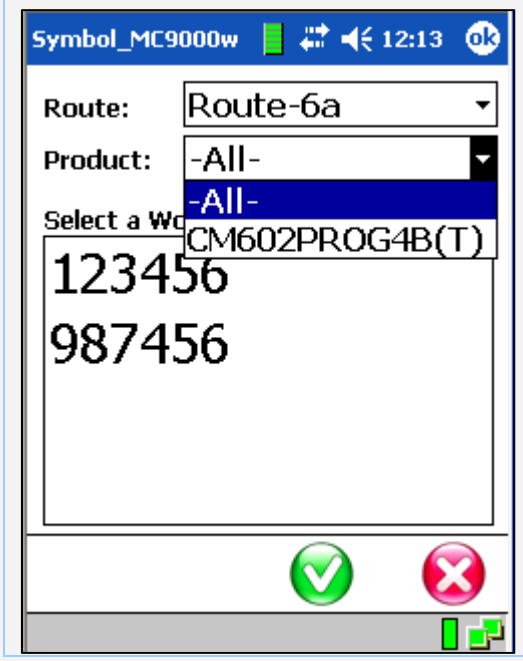


As shown in the screen above, part quantity required for part 130-000594.X.X is 70 and the reel with barcode R594 fulfills this part quantity.

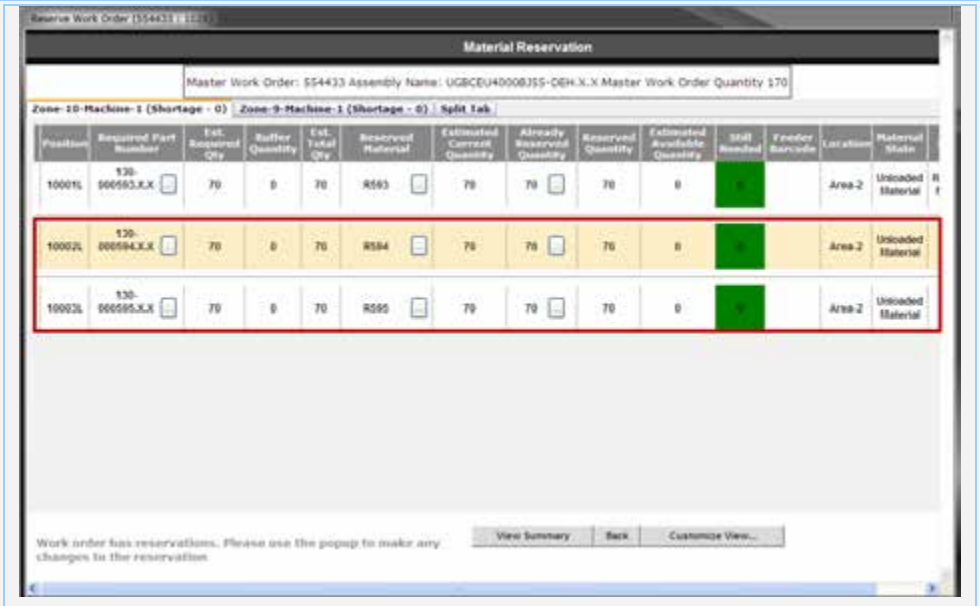
9 Select the reel barcode (as shown in first screen of step 8) and then click  . The previously selected row (shown in step 7) disappears as shown below:

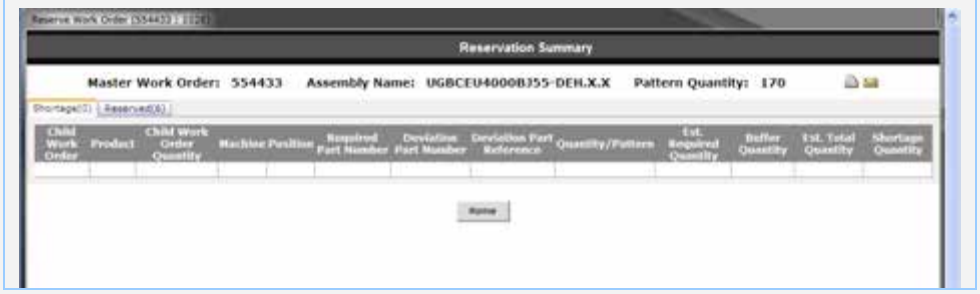


10	Repeat steps 7,8 and 9 to fulfill shortage for each part. Part list keeps on shrinking as shortage quantity is backfilled by way of scanning additional reels that were received from warehouse.
11	After shortages for all the parts is fulfilled, master work order (in this case 554433) disappears from the work order list as shown below:



12	Also, no records are listed in the **Shortage** tab of master work order after all the shortages are fulfilled by completing **Material fulfill** operation from mobile scanner. This is shown below:


Material Reservation
Master Work Order: 554433 Assembly Name: UGBCEU4000BJ55-DEH.X.X Master Work Order Quantity 170
Zone-10-Machine-1 (Shortage - 0)
Zone-9-Machine-1 (Shortage - 0)
Split Tab
Work order has reservations. Please use the popup to make any changes to the reservation
View Summary
Back
Customize View...


Reservation Summary
Master Work Order: 554433
Assembly Name: UGBCEU4000BJ55-DEH.X.X
Pattern Quantity: 170
Shortage(0)
Reserved(6)
Child Work Order
Product
Child Work Order Quantity
Machine Position
Required Part Number
Deviation Part Number
Deviation Part Reference
Quantity/Pattern
Est. Required Quantity
Buffer Quantity
Est. Total Quantity
Shortage Quantity
Home

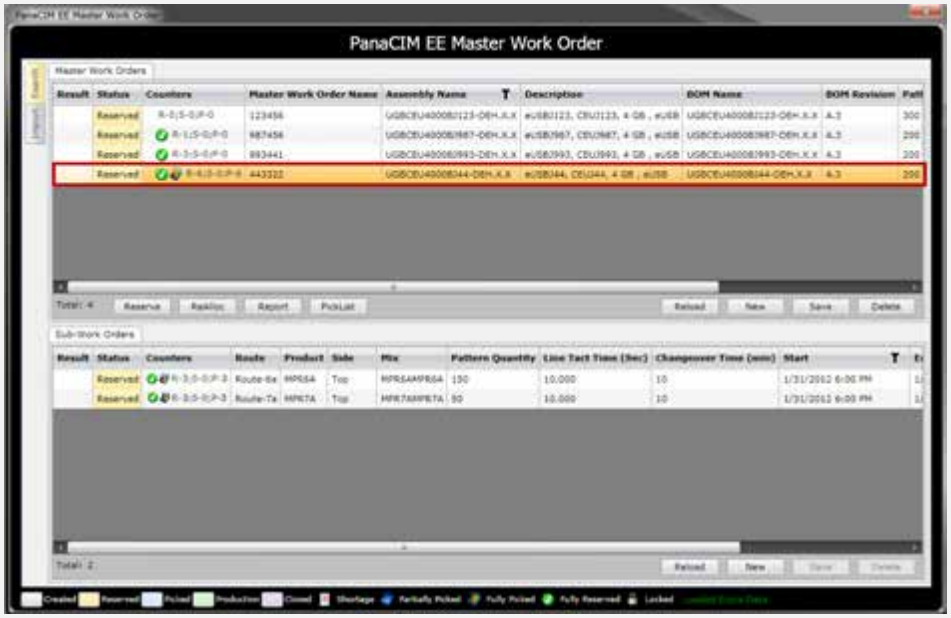
■ Master Work Order Production

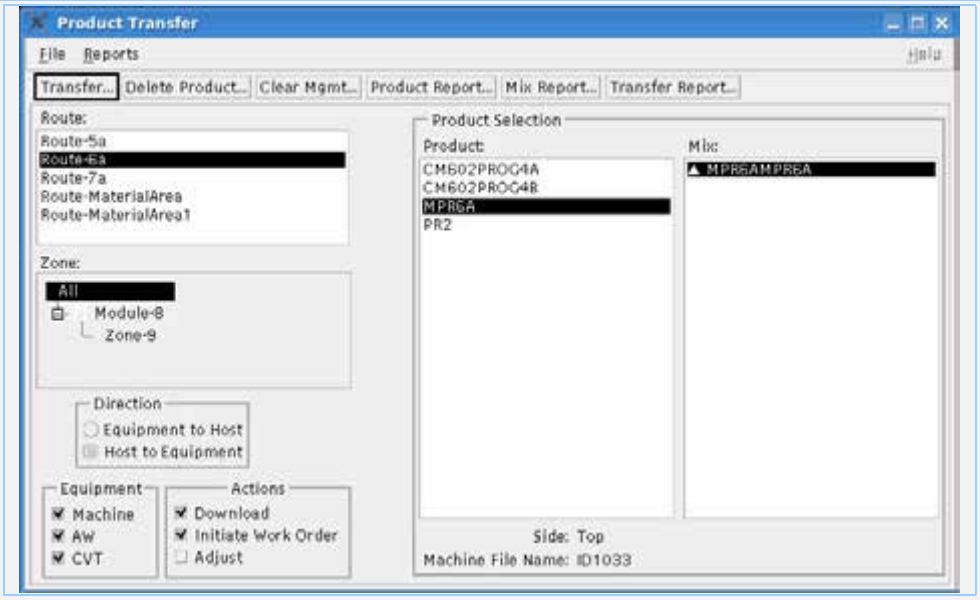
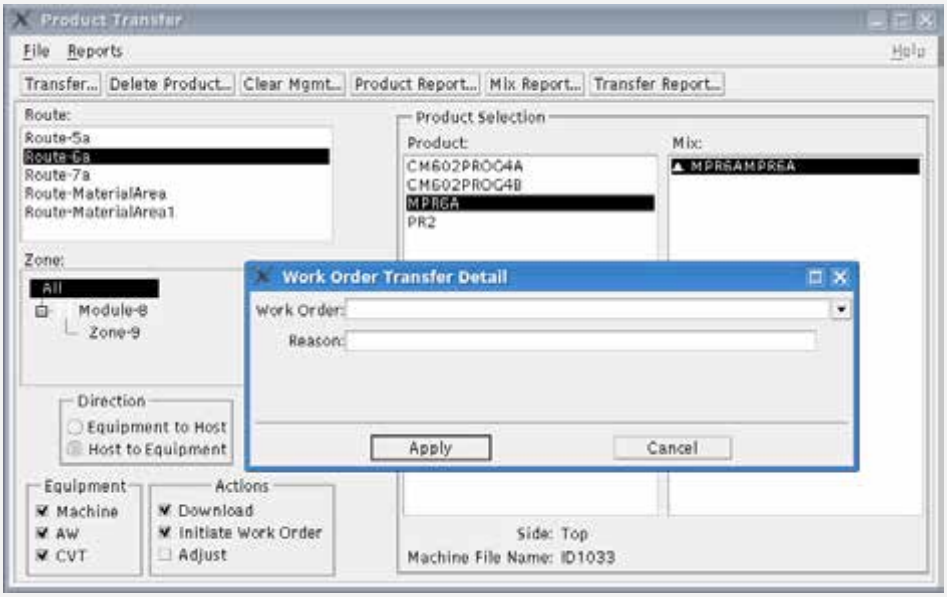
Step	Description
1	Production starts when storage unit (completed kit) is moved to production staging area.
2	User transfers the program corresponding to a child work order to machines in the route. This step is completed until all the production programs for each of the child work order are transferred to its PanaCIM route. This activates the Master work order for production.
3	Production operators load material from the staging area to the feeders.
4	Production operators perform material verification operation with the scanner by scanning/entering material ID, part number and other part attributes like vendor, lot etc.
5	Location of material from Staging area to current feeder location is updated in PanaCIM.
6	Production is started. After production is started, material ID remaining quantity is updated every 5 minutes in PanaCIM database.

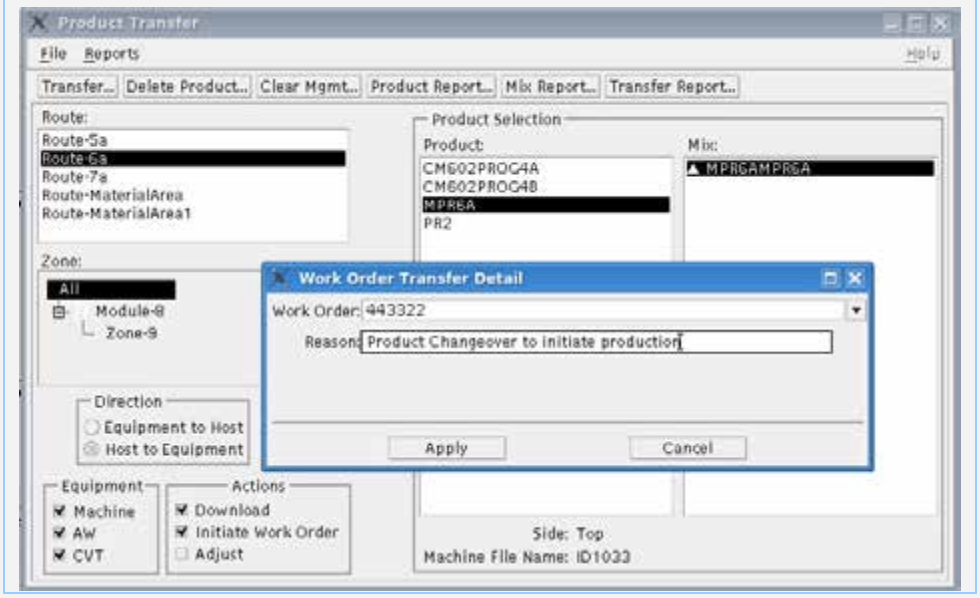
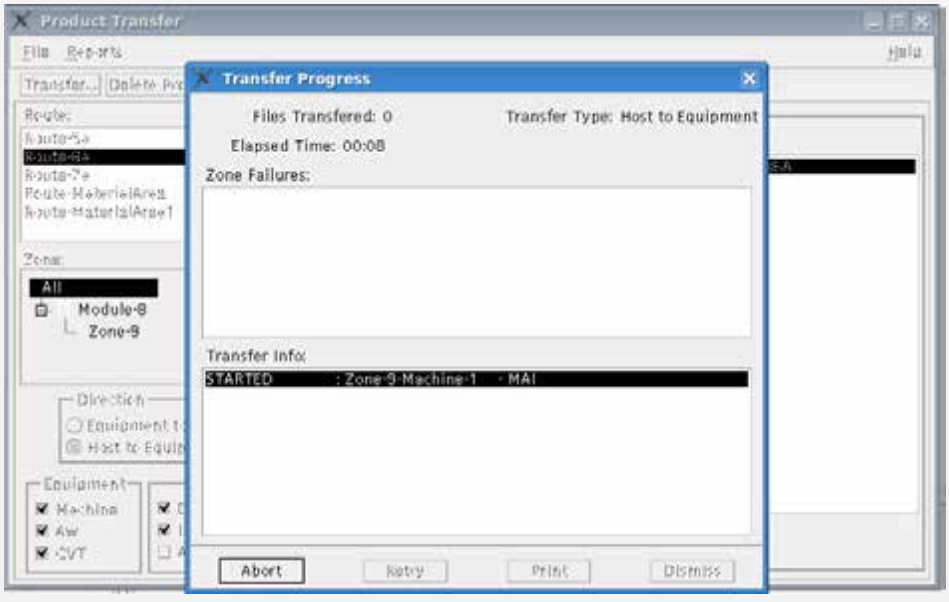
■ Master Work Order Production (CM Limited)

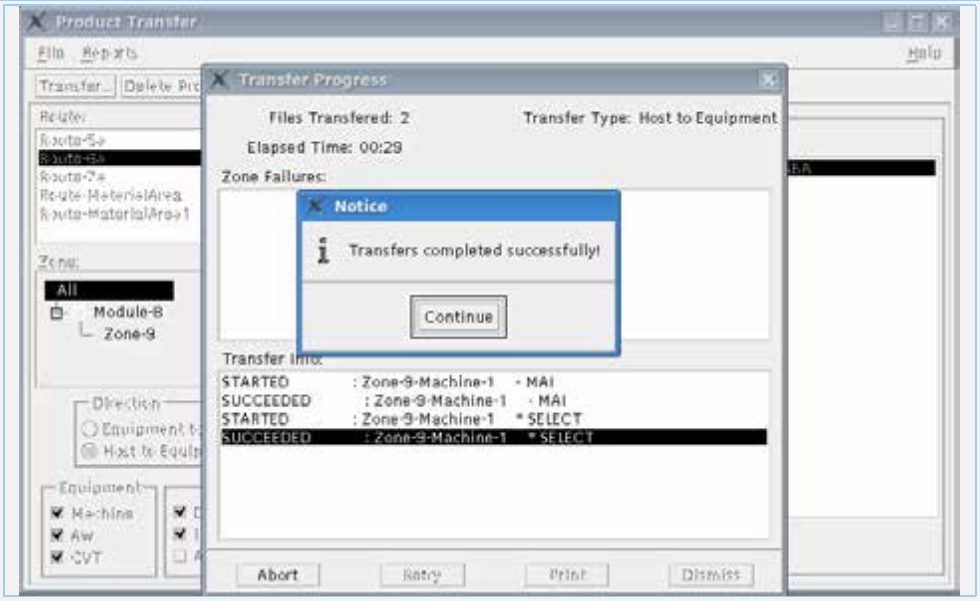
For CM series machines, applicable product program needs to be transferred to the machine after the production start time for sub work order has elapsed so that the PanaCIM may recognize that the production according to the work order (master work order) has started.

Refer to the action table below for the steps involved in putting master work order to production:

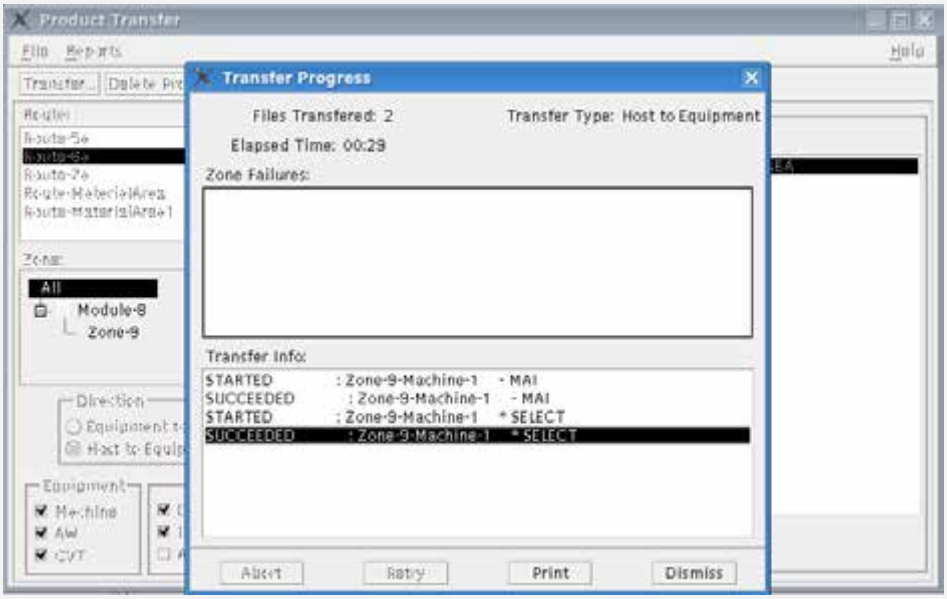
Step	Procedure
1	Consider a Master work order (in this case 443322) that is next for production. 
2	Click  in the PanaCIM launcher bar to open the **Product Transfer** window shown below:

	
3	Select the name of the route on which production needs to be initiated from the **Route** drop-down list. After selecting route name, name of the products imported into that route are shown in the product list.
4	Select a Product from the **Product** list for which sub work order was created in the **PanaCIM EE Master Work Order** screen.
5	Make the following selections: Direction: **Host to Equipment** selected Equipment: **Machine**, **AW** and **CVT** checked Actions: **Download**, **Initiate Work Order** checked
6	Click the Transfer... button to start production on the selected route corresponding to a sub work order of a Master work order.
7	After Transfer... button is clicked, **Work Order Transfer Detail** dialog appears as shown below: 

8	Do the following in the **Work Order Transfer Detail** dialog. (i) Select the name of the Master work order name from the **Work Order** drop-down list. (ii) Enter a reason for Product Transfer in the **Reason** textbox. (iii) Click the **Apply** button. 
9	After clicking the **Apply** button in the **Work Order Transfer Detail** dialog, **Transfer Progress** dialog appears indicating the progress of file transfer from host to equipment. 
10	After MAI file is completely transferred to the equipment in the route, a message appears as indicated below:

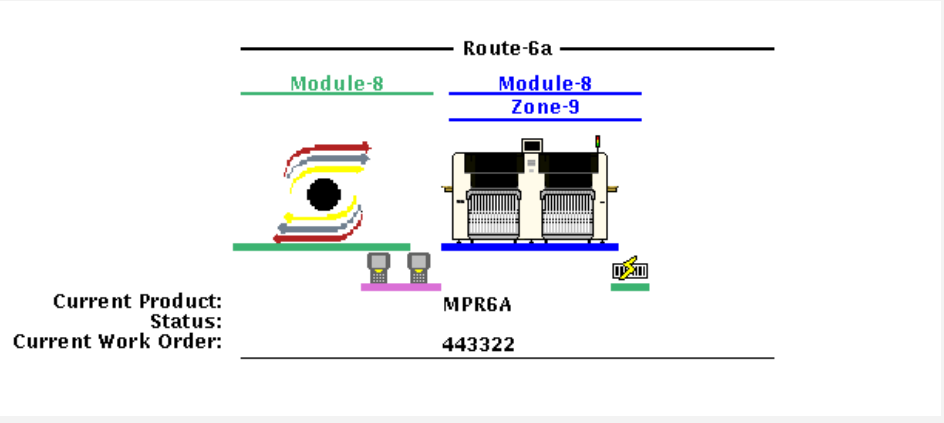


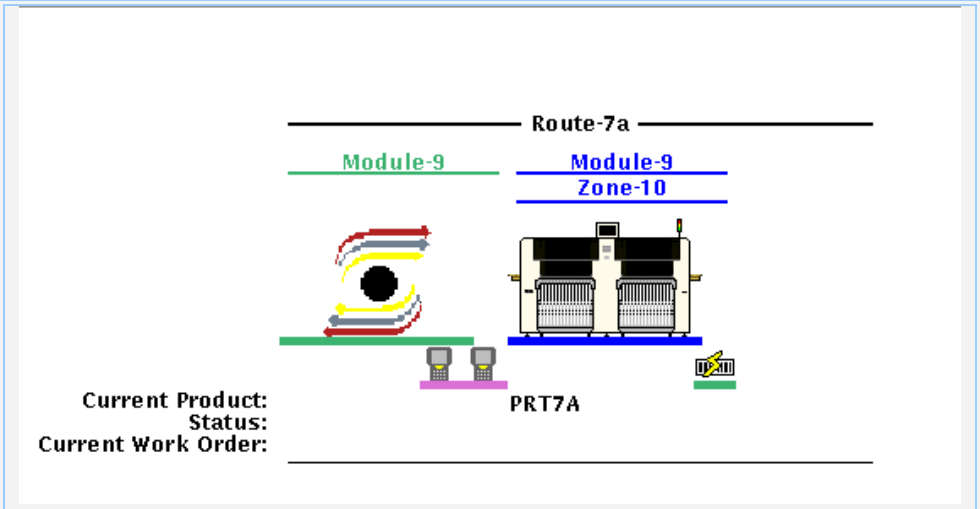
11 Click the **Continue** button to close the message box.

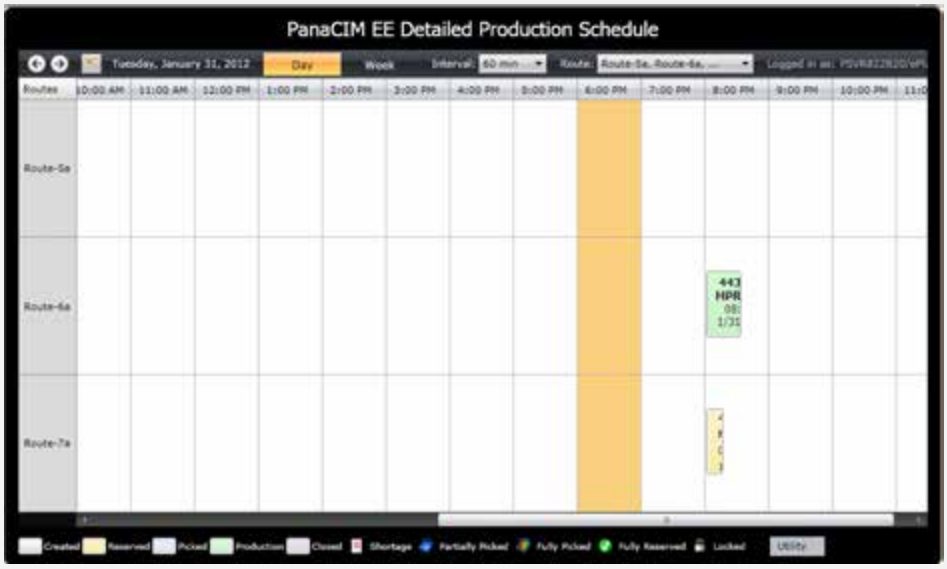


Now close the **Transfer Progress** dialog, since the file has been successfully transferred as shown in the screen above.

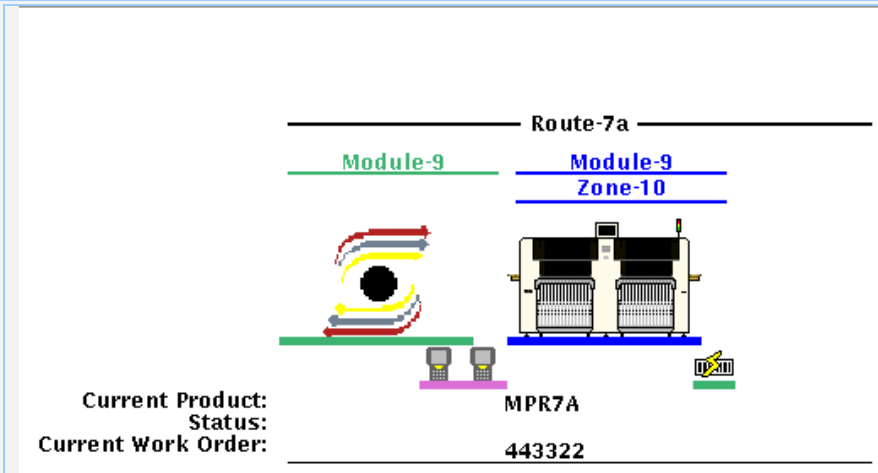
12 After production is started for first sub work order on a route, state of Route Editor and PanaCIM EE DPS screen is the one shown below:

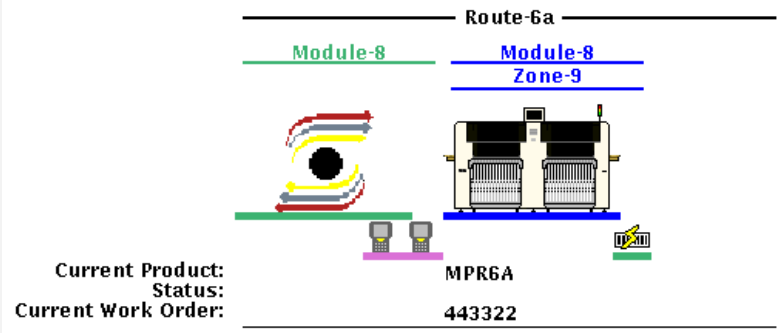






13	At this stage, when production of sub work order is running on a route (in this case Route-6a), you can initiate production on another route for another sub work order of same master work order by following steps 3 to 11.
14	Repeat Steps 3 to 11 to start production for each sub work order of Master work order. When production is started for second sub work order of master work order (in this case 443322), state of **Route Editor** screen is the one shown below:

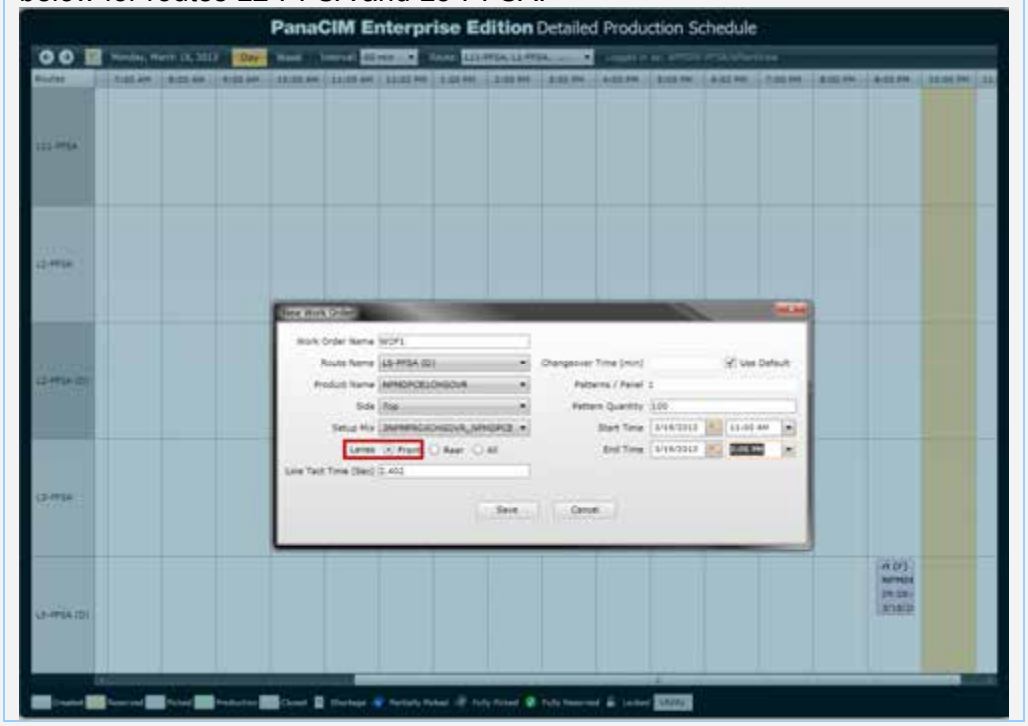

Route-7a
Module-9
Module-9
Zone-10
Current Product:
MPR7A
Status:
Current Work Order:
443322

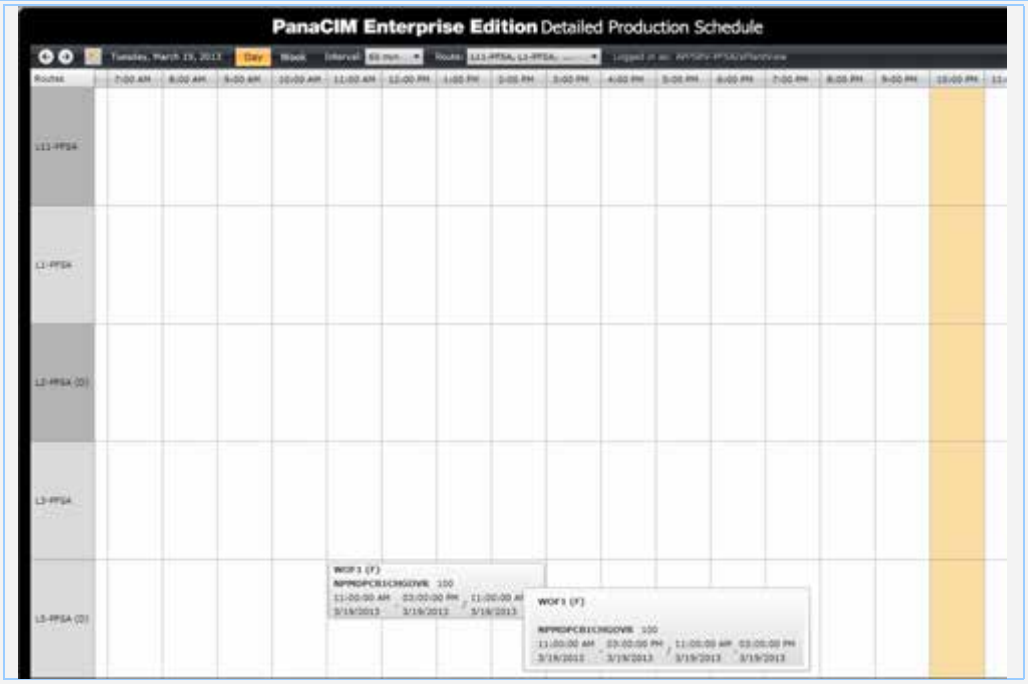

Route-6a
Module-8
Module-8
Zone-9
Current Product:
MPR6A
Status:
Current Work Order:
443322

3.3.14 NPM Dual Lane WO Support

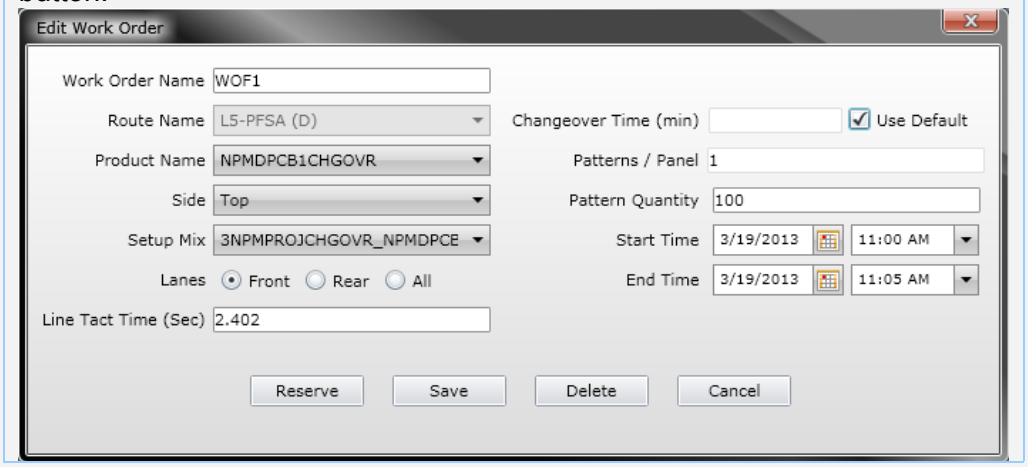
■ Creating Work Orders for NPM Machine

● Front Lane

Step	Procedure
1	Right-click a route containing NPM Dual Lane machine and select New Work Order. After doing so, New Work Order window opens as shown below: **Note:** NPM Dual lane routes have **(D)** suffixed after them in DPS as shown below for routes L2-PFSA and L5-PFSA. 
2	Choose the lane for which work order needs to be created as shown above. For Product in **Shared** mode, you can choose one of the following three options: • **Front** – Work order will run only on **Front** Lane of Dual Lane NPM machine • **Rear** – Work order will run only on **Rear** Lane of Dual Lane NPM machine • **All** – Work order can run on both lanes of Dual Lane NPM machine
3	Click the **Save** button after entering the work order details in the **New Work Order** dialog. Work Order appears in DPS as shown below:



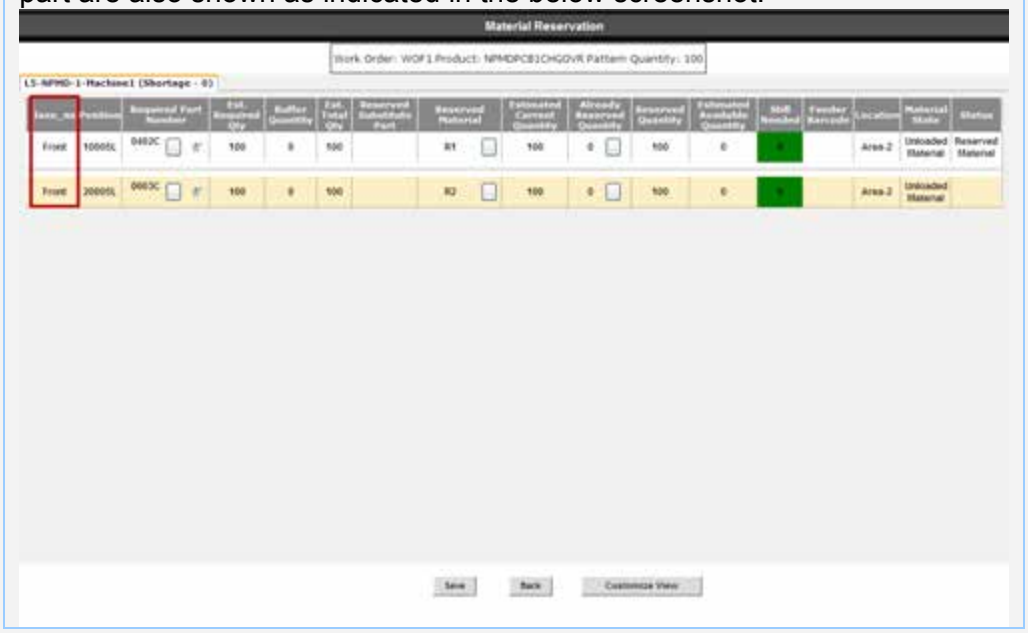
4 Once the work order is created, you can double-click on it to launch **Edit Work Order** dialog. Using **Edit Work Order** dialog, existing work order details can be edited by making the required changes and clicking the **Save** button.



5 To reserve a work order corresponding to a particular lane, click the **Reserve** button. Upon doing so, **Material Reservation** page appears as shown below:



6 Choose one or more selected locations and click the **Generate** button. Upon doing so, following screen appears showing the parts in the product that will run in the designated lane (identified by **lane_no**). Reels reserved for each part are also shown as indicated in the below screenshot.

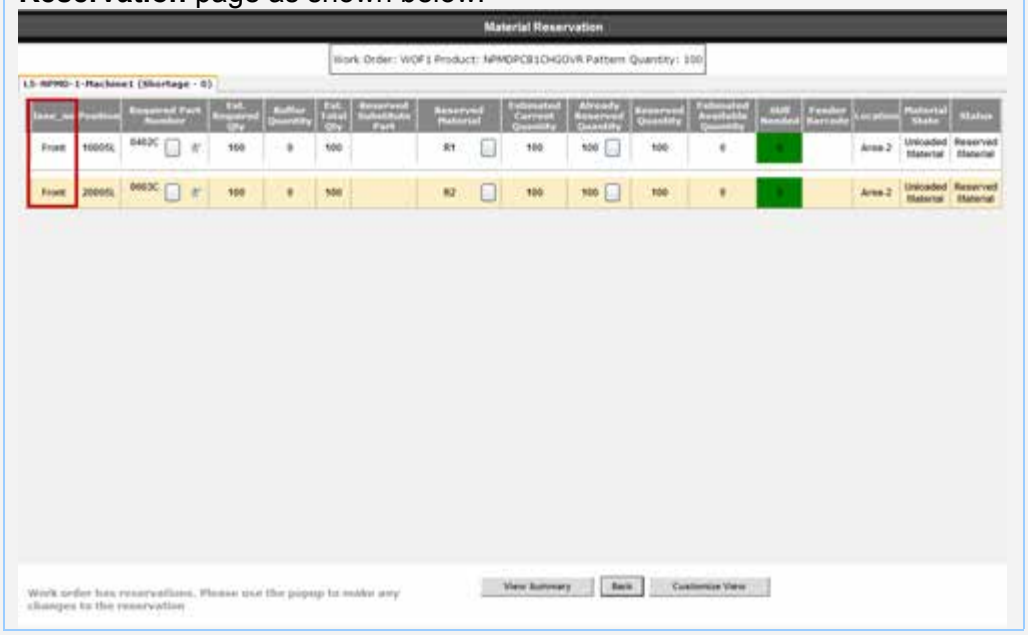


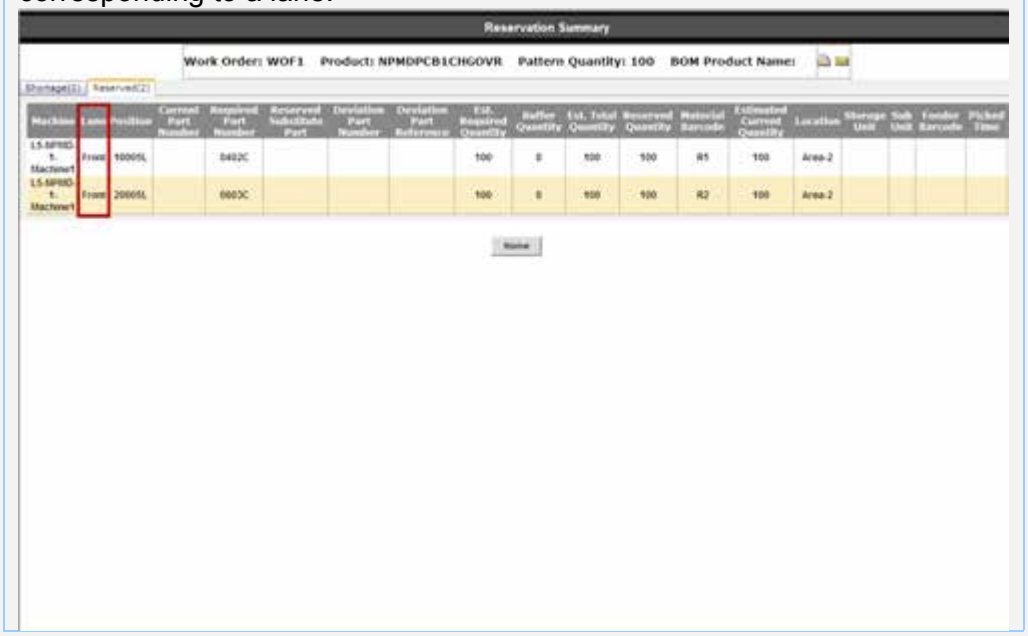
7 Click the **Save** button. Upon doing so, a message appears indicating that reservations are successful provided reels fulfill the part quantity required for production.

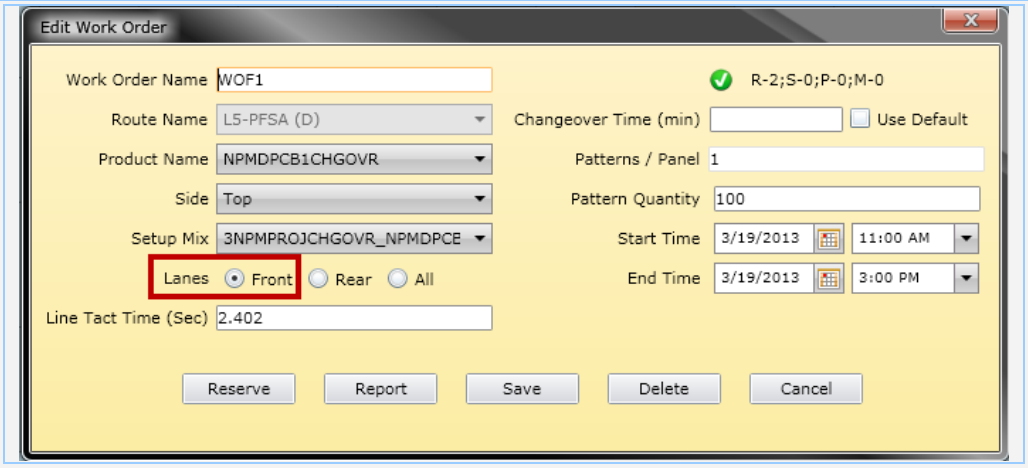


8 Click the **OK** button in the **Reserved Successfully** message to close the dialog. Now **View Summary** button appears at the bottom of the **Material**

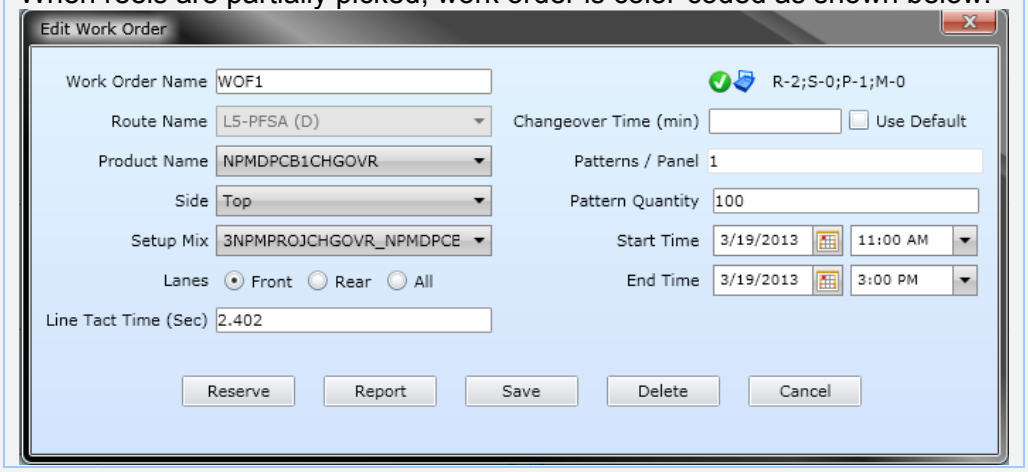
Reservation page as shown below:



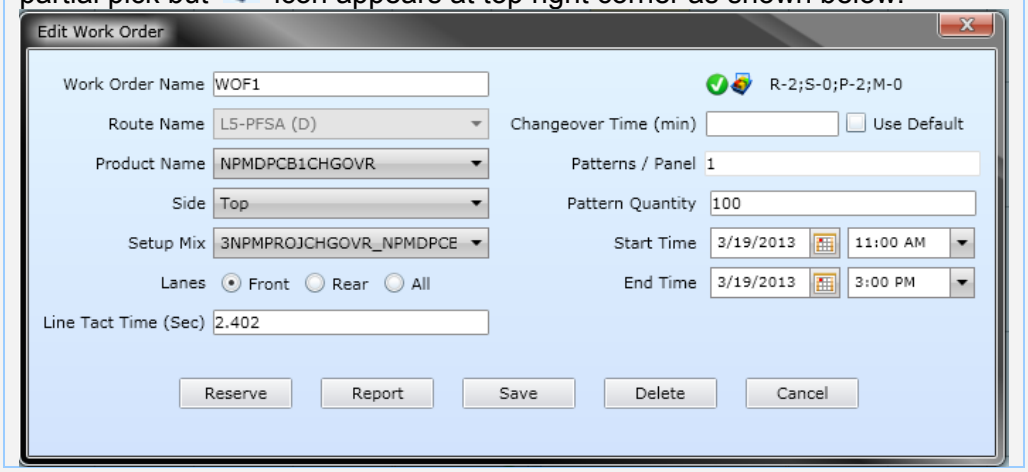
9	Upon clicking the **View Summary** button, **Reservation Summary** page appears listing the reservations that have been done for a work order corresponding to a lane. 
10	Click the **Home** button to return to DPS main page. Double-click the work order to launch Edit Work Order dialog. Both in the DPS as well as **Edit Work Order** dialog, **Reserved** work order is color-coded as shown below:



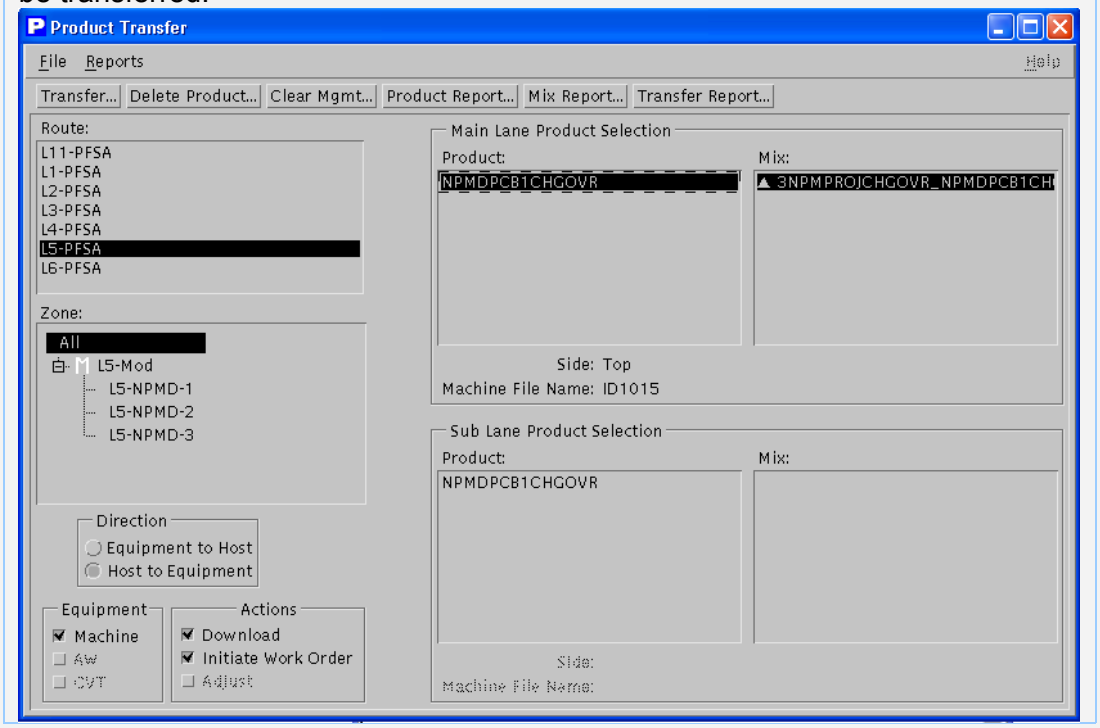
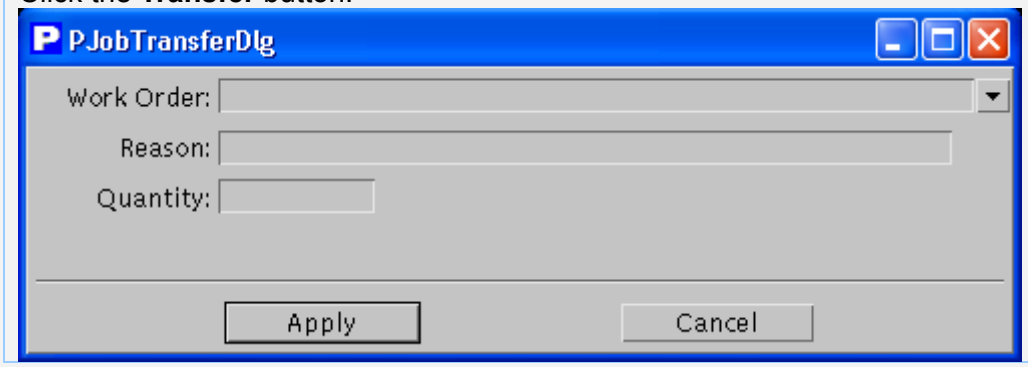
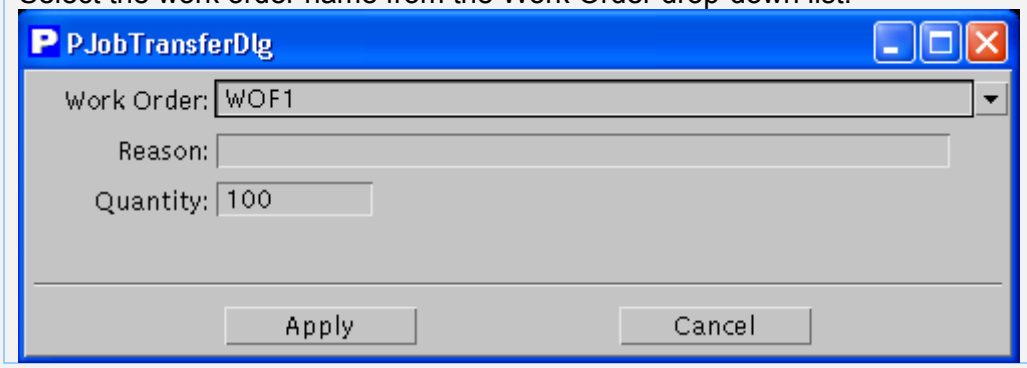
11 When reels are partially picked, work order is color-coded as shown below:

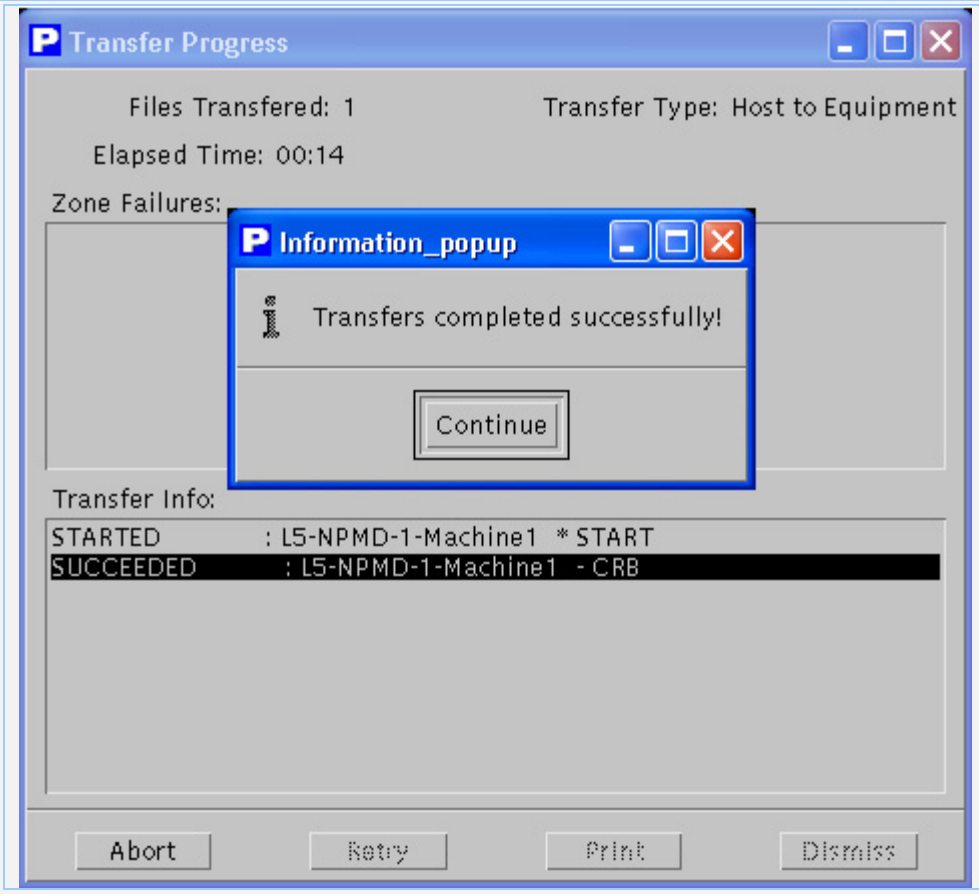


12 When all the reels are fully picked, color code remains the same as for partial pick but icon appears at top right corner as shown below:

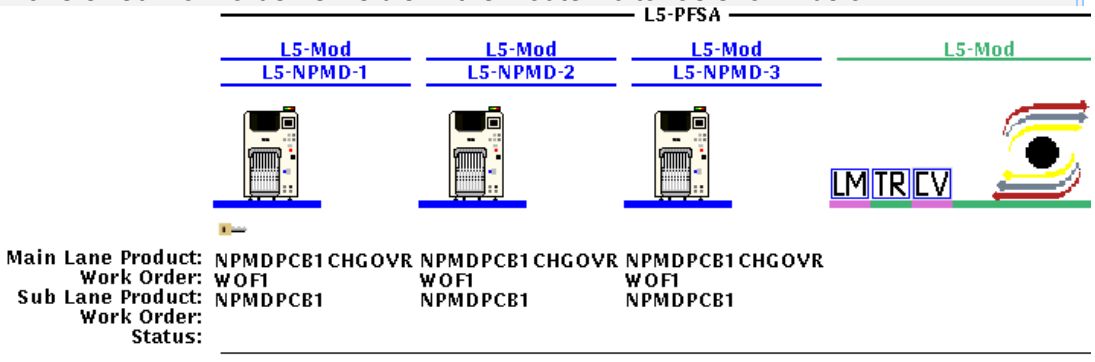


■ Initiating Work Order Production via Product Transfer

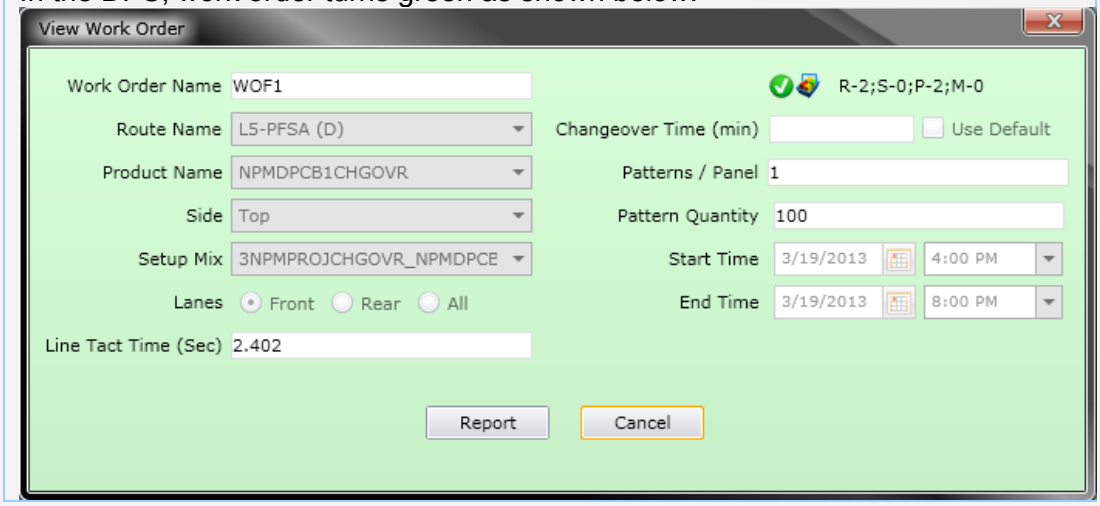
Step	Procedure
1	Launch **Product Transfer** dialog and select route name to which product needs to be transferred. 
2	Click the **Transfer** button. 
3	Select the work order name from the Work Order drop-down list. 
4	Click the **Apply** button to transfer the work order to the machine. Once Transfer is completed, a message **"Transfers completed successfully!"** appears as shown below:



5 Transferred work order is visible in the Route Editor as shown below.



6 In the DPS, work order turns green as shown below:



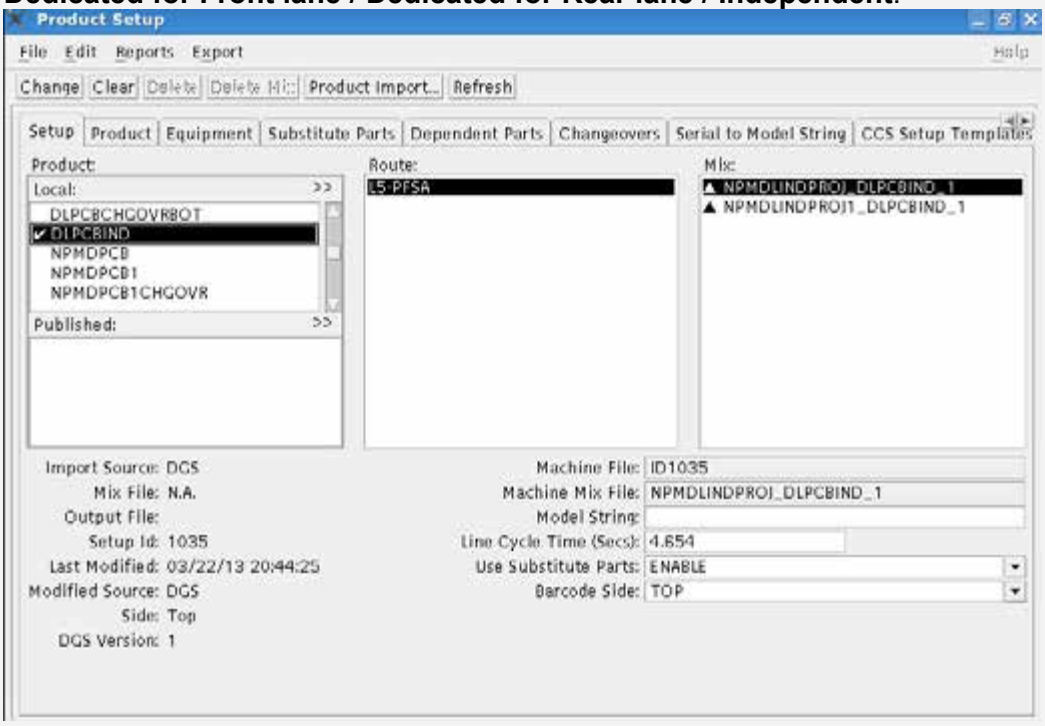
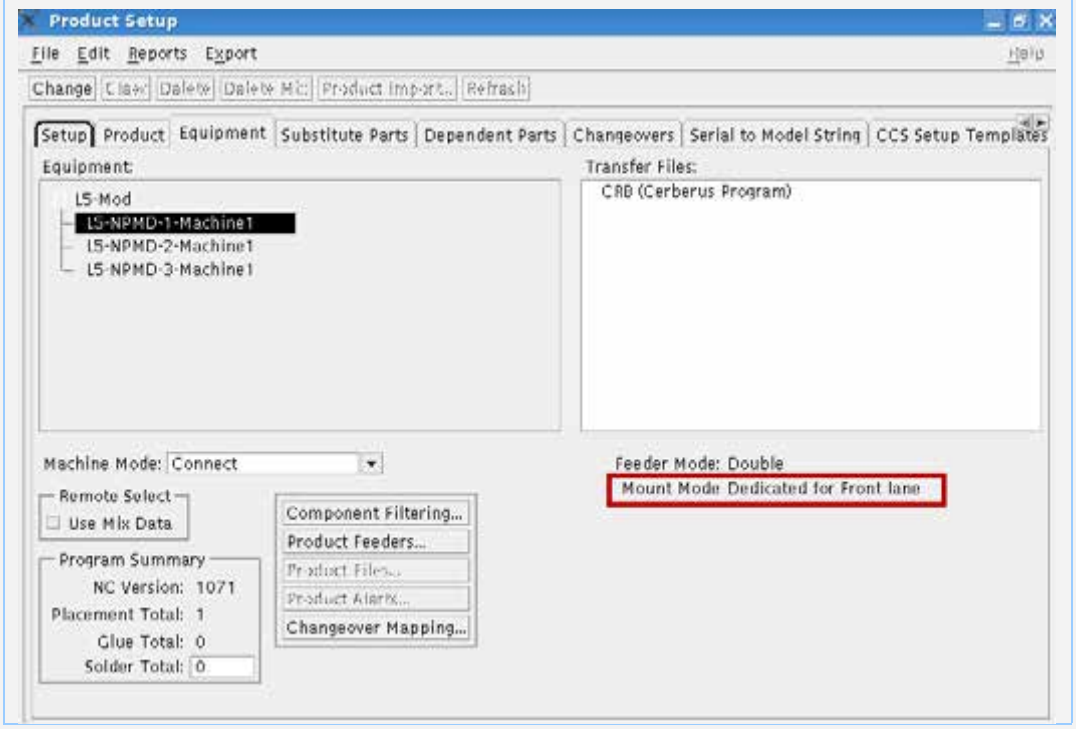
- **WO Creation Scenarios on NPM Dual Lanes**

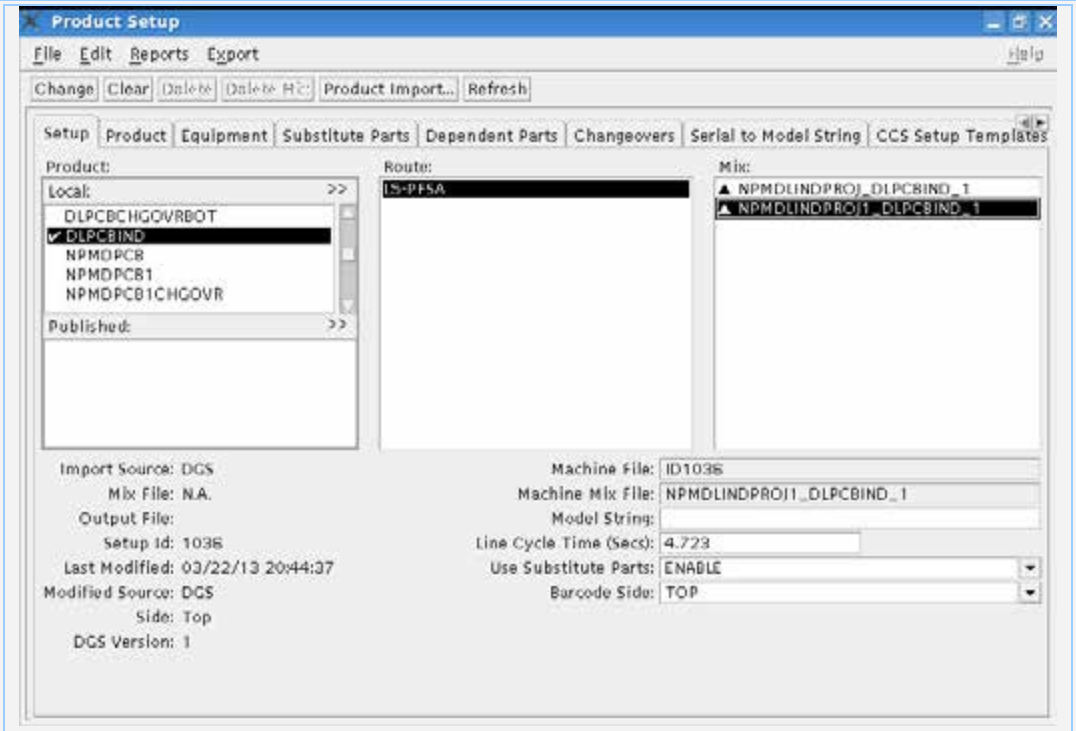
Route	Lane	Work Order	Product	Side	Qty	Result
Same	Same	Same ID	Same	Different	Same	OK: User can assign the same WO to run on same Lane of the Route for same Product AS LONG AS the Side is different.
Same	Different	Same ID	Same	Different	Same	OK: User can assign the same WO to run on different Lane of the same Route for same Product AS LONG AS the Side is different.
Same	Same	Same ID	Same	Same	Same	NG: Work Order must be unique for same Product, Side, Qty combination on same Lane of the same Route.
Same	Same	Different	Same	Same	Same	OK: User can assign same Product, Side, and Qty to same Lane on the same Route AS LONG AS the WO is different.
Same	Different	Different	Same	Same	Different	OK: User can assign same Product, Side, and Qty to different Lanes of the Same Route AS LONG AS the WO is different. PMD currently closes the WO.
Different	NA	Same ID	Same	Different	Same	OK: User can assign the same WO to run on different Routes for same Product AS LONG AS the Side is different.

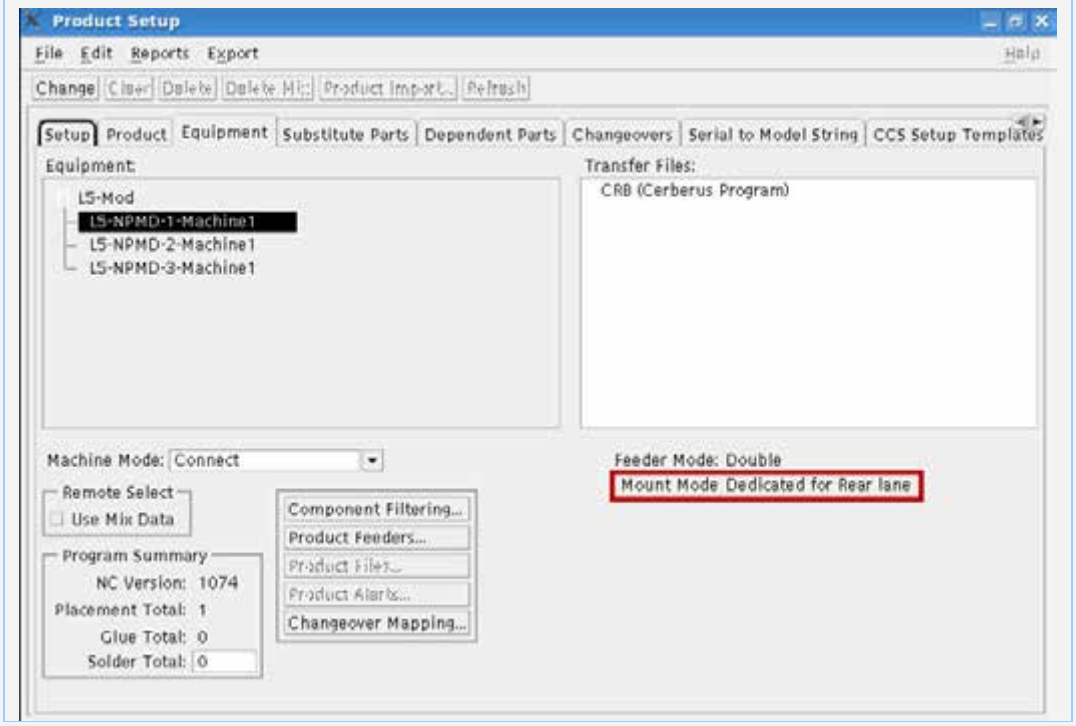
Different	NA	Different	Same	Same	Different	OK: User can assign same Product, Side, and Qty to different Routes AS LONG AS the WO is different.
Different	NA	Same ID	Same	Same	Same	NG: Work Order must be unique for same Product, Side, Qty combination on different Routes.

- **Different Lane rest all conditions are same – Shared vs Independent Mode**

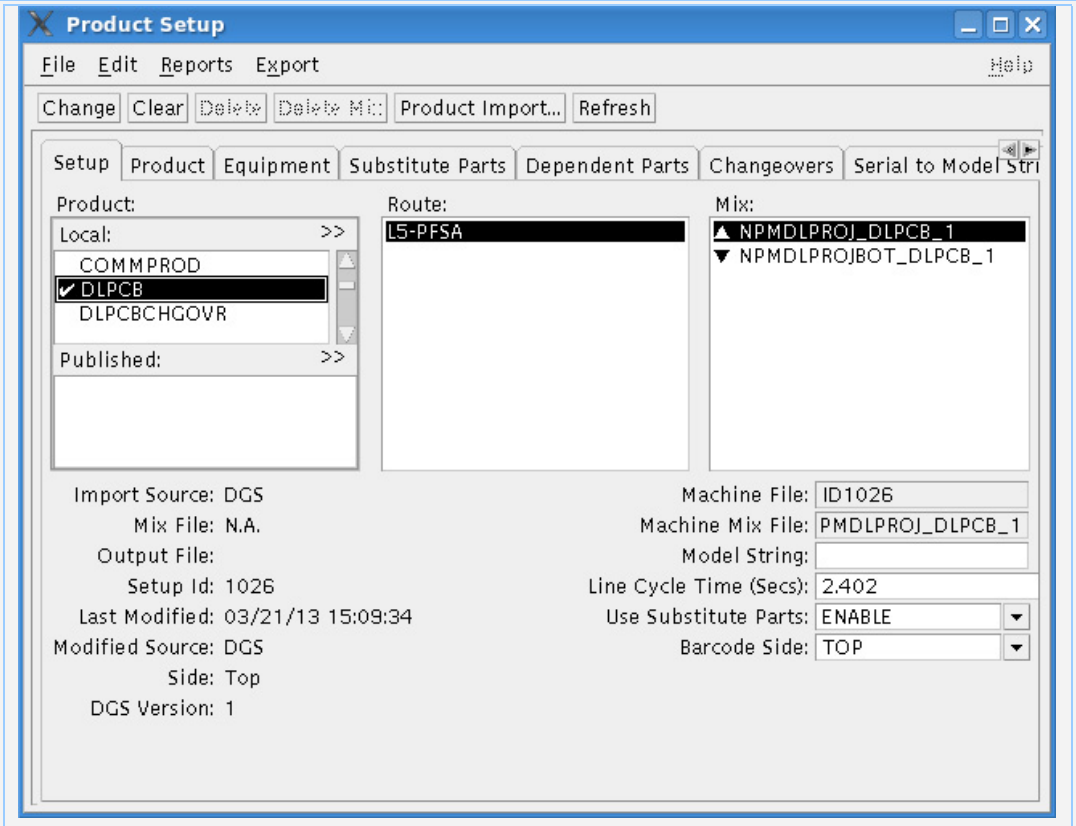
Viewing Product Mode – Shared or Independent

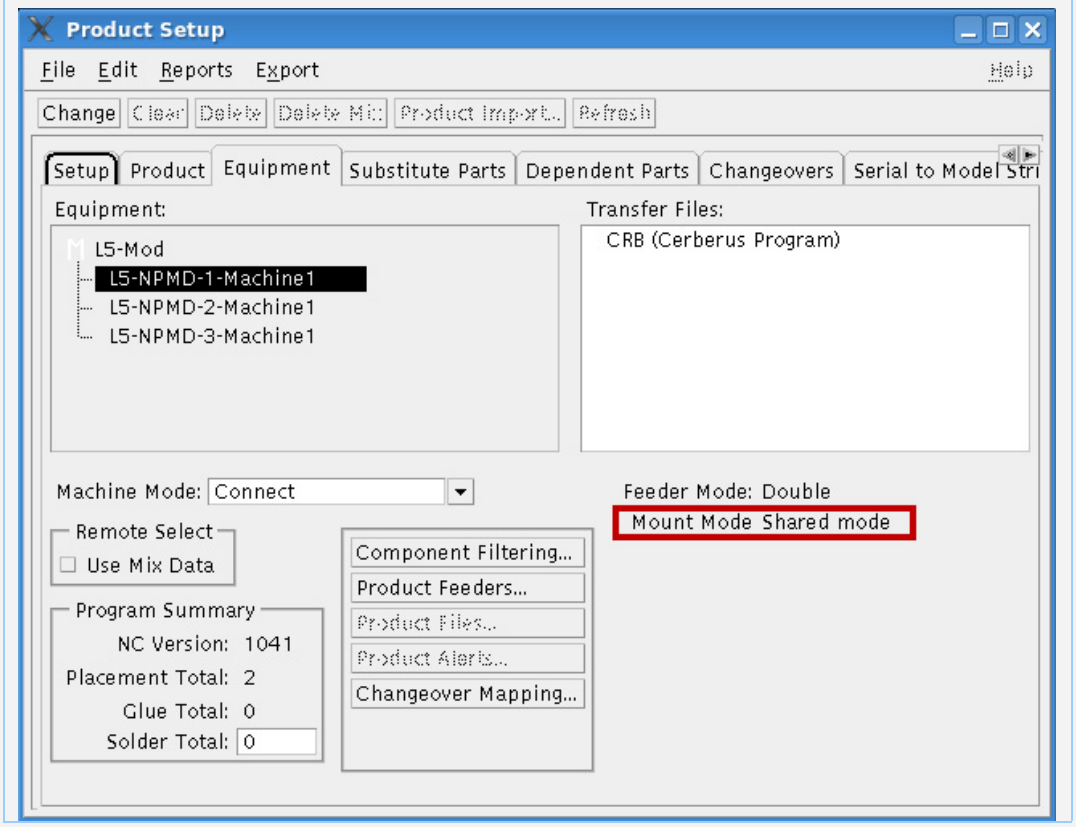
Step	Description
1	Select a Product, Route and Mix in the **Setup** tab and then navigate to **Equipment** tab. In the **Equipment** tab, you will see Mount Mode as **Shared / Dedicated for Front lane / Dedicated for Rear lane / Independent**.  
2	Select same top side product but belonging to different mix and navigate to the **Equipment** tab. In this case, it shows Mount Mode as **Dedicated for Rear lane**.





3	Select a different product and choose its corresponding Route and Mix as shown below. Navigate to **Equipment** tab and click a machine. In this case, Mount Mode is **Shared mode**.

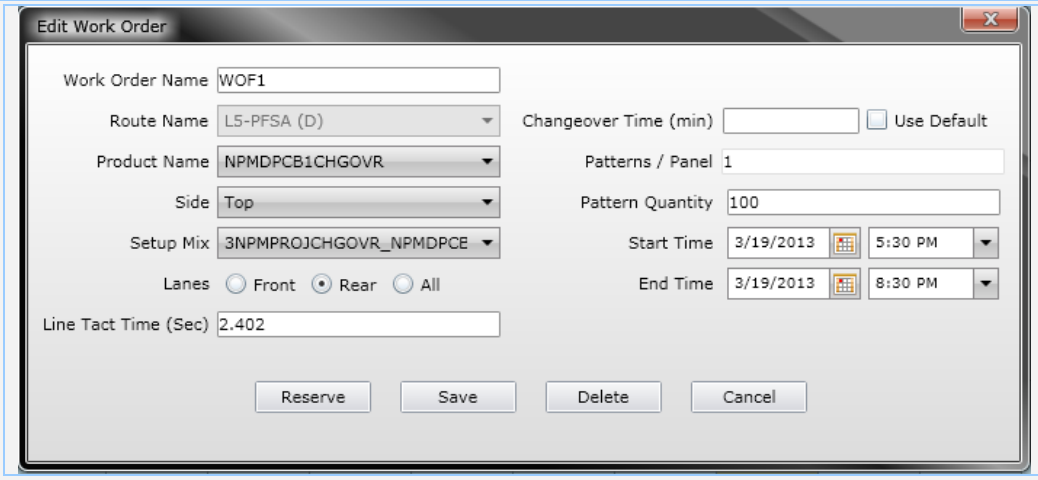

Product Setup
File Edit Reports Export Help
Change Clear Delete Delete Mix Product Import... Refresh
Setup Product Equipment Substitute Parts Dependent Parts Changeovers Serial to Model Stri
Product:
Local: >>
COMMPROD
DLPCB
DLPCBCHGOVR
Published: >>
Route:
L5-PFSA
Mix:
NPMDLPROJ_DLPCB_1
NPMDLPROJBOT_DLPCB_1
Import Source: DGS
Mix File: N.A.
Output File:
Setup Id: 1026
Last Modified: 03/21/13 15:09:34
Modified Source: DGS
Side: Top
DGS Version: 1
Machine File: ID1026
Machine Mix File: PMDLPROJ_DLPCB_1
Model String:
Line Cycle Time (Secs): 2.402
Use Substitute Parts: ENABLE
Barcode Side: TOP


Product Setup
File Edit Reports Export Help
Change Clear Delete Delete Mix Product Import... Refresh
Setup Product Equipment Substitute Parts Dependent Parts Changeovers Serial to Model Stri
Equipment:
L5-Mod
L5-NPMD-1-Machine1
L5-NPMD-2-Machine1
L5-NPMD-3-Machine1
Transfer Files:
CRB (Cerberus Program)
Machine Mode: Connect
Feeder Mode: Double
Mount Mode Shared mode
Remote Select
Use Mix Data
Program Summary
NC Version: 1041
Placement Total: 2
Glue Total: 0
Solder Total: 0
Component Filtering...
Product Feeders...
Product Files...
Product Alerts...
Changeover Mapping...

Shared Mode

Work order can be successfully created when Route Name, Product Name, Side and Pattern Quantity are same. Only thing different is the lane. Product recipe is created in shared mode in DGS. Although work order with same name can be successfully created for each lane, production can only be started for one work order. Any attempt to start production for work order on a lane, when same work order is already running on other lane is not possible.

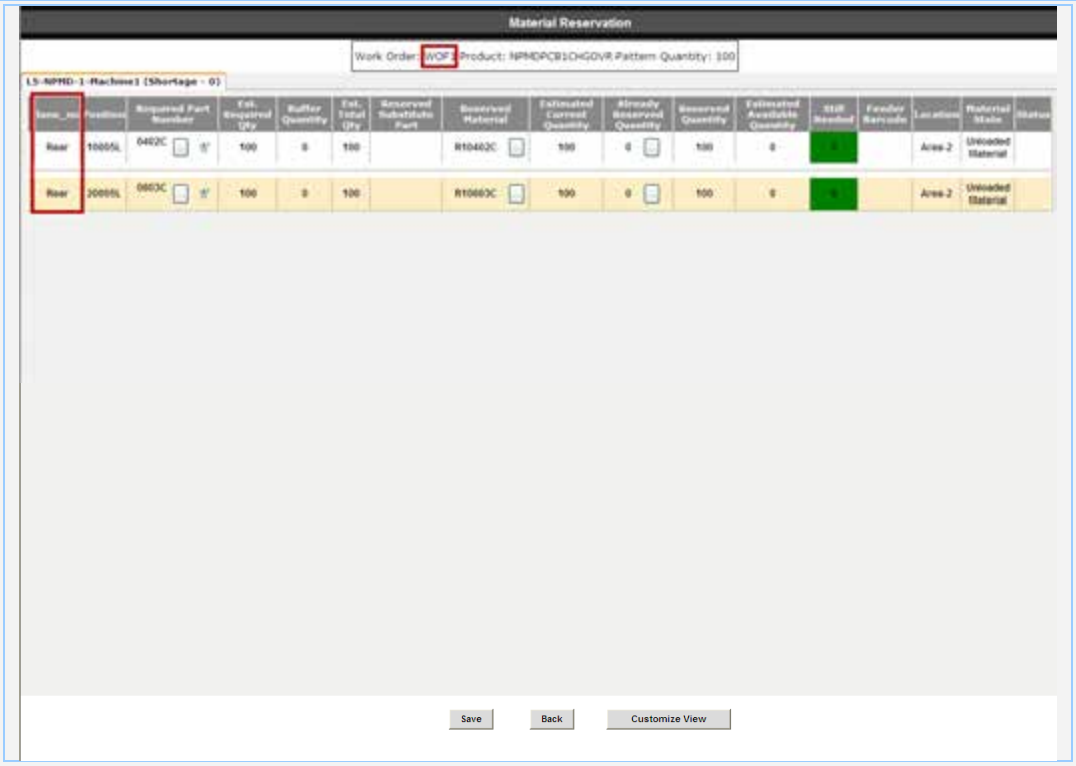
Step	Description
1	Same work order can be created for dual lane NPM machine provided lane is different. As shown below, work order name is same for both the lanes i.e., WOF1. In the DPS screen, Work Order is labeled **WOF1 (F)** for front lane and **WOF1 (R)** for rear lane. 
2	As shown above, work order for front lane is in production while the work order for rear lane has just been created. Double-clicking the work order for rear lane displays the Edit Work Order dialog. For both these work orders, following things are same: • Work Order Name • Route Name • Product Name • Side • Quantity Only thing different is lane.



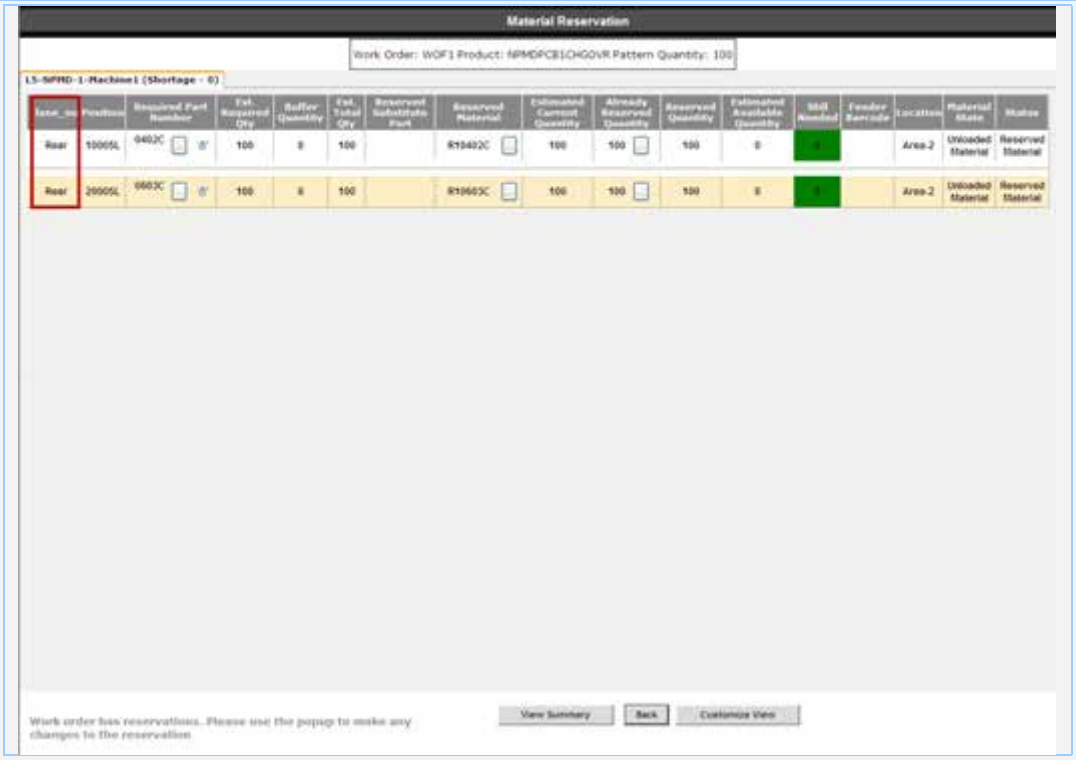
3 Click the **Reserve** button. Upon doing so, following screen appears showing the locations where reels to be reserved for the work order are to be searched.

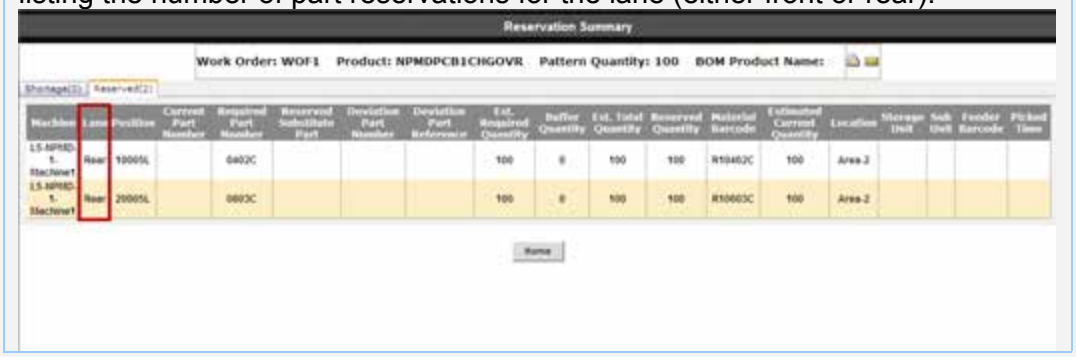


4 Click the **Generate** button. Upon doing so, following screen appears showing the lane (Front or rear) along with part reservations for that lane.



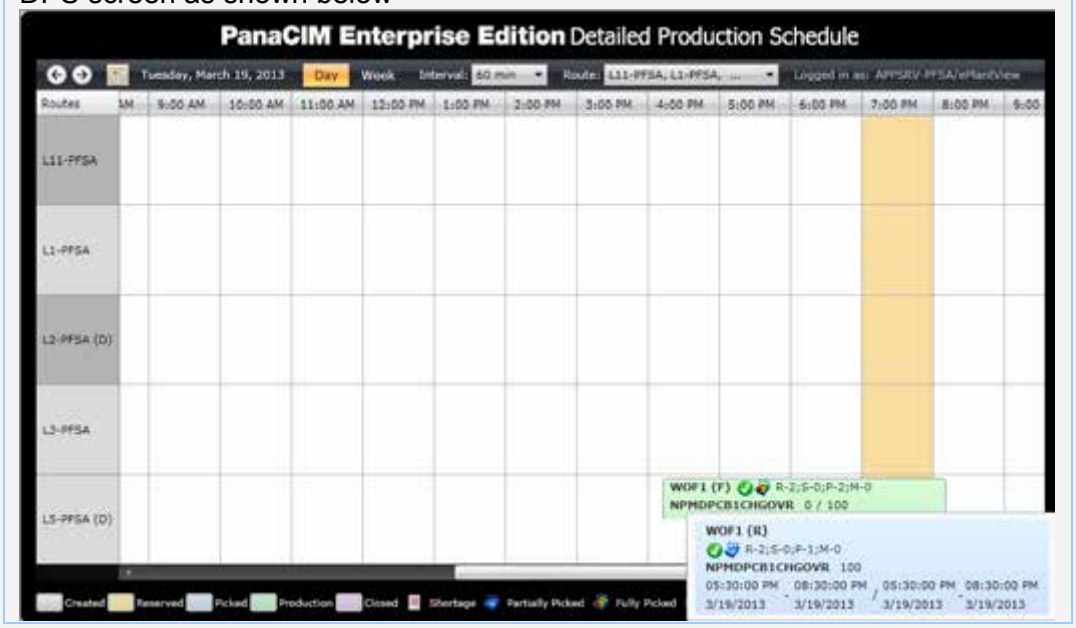
5	Click the **Save** button. Upon doing so, following message appears stating that Part Reservations are successful provided reels have adequate part quantity required by the work order. 
6	After clicking the **OK** button, **Status** column shows that material is reserved for the lane (either front or rear) as shown below:



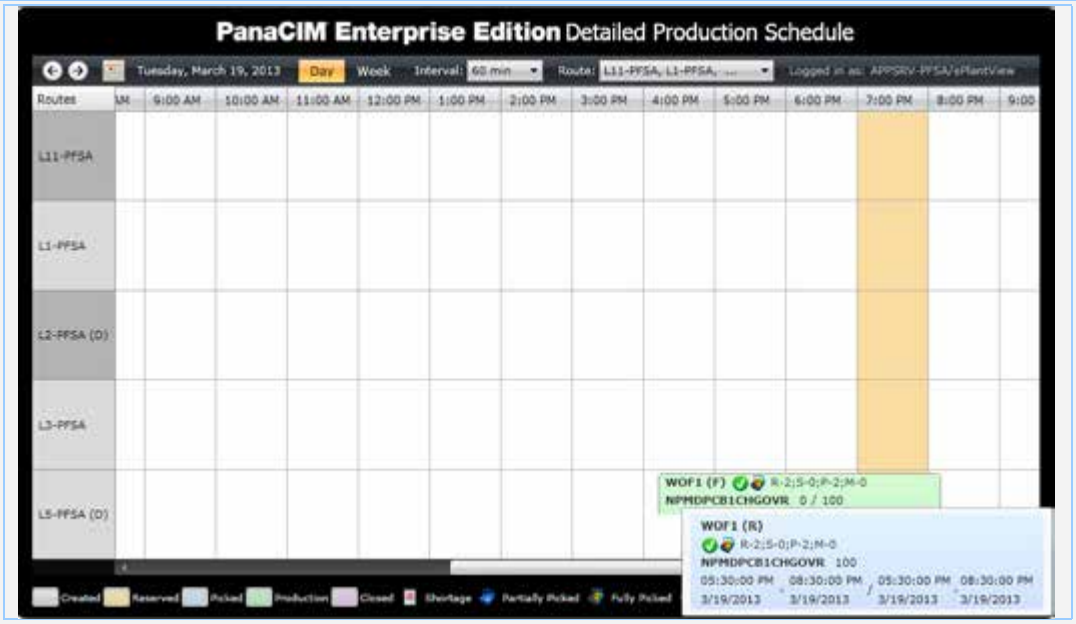
7	Click the **View Summary** button. Upon doing so, following screen appears listing the number of part reservations for the lane (either front or rear). 
8	Click the **Home** button. You are taken to DPS screen where the Work order for the rear lane appears as reserved as shown below:



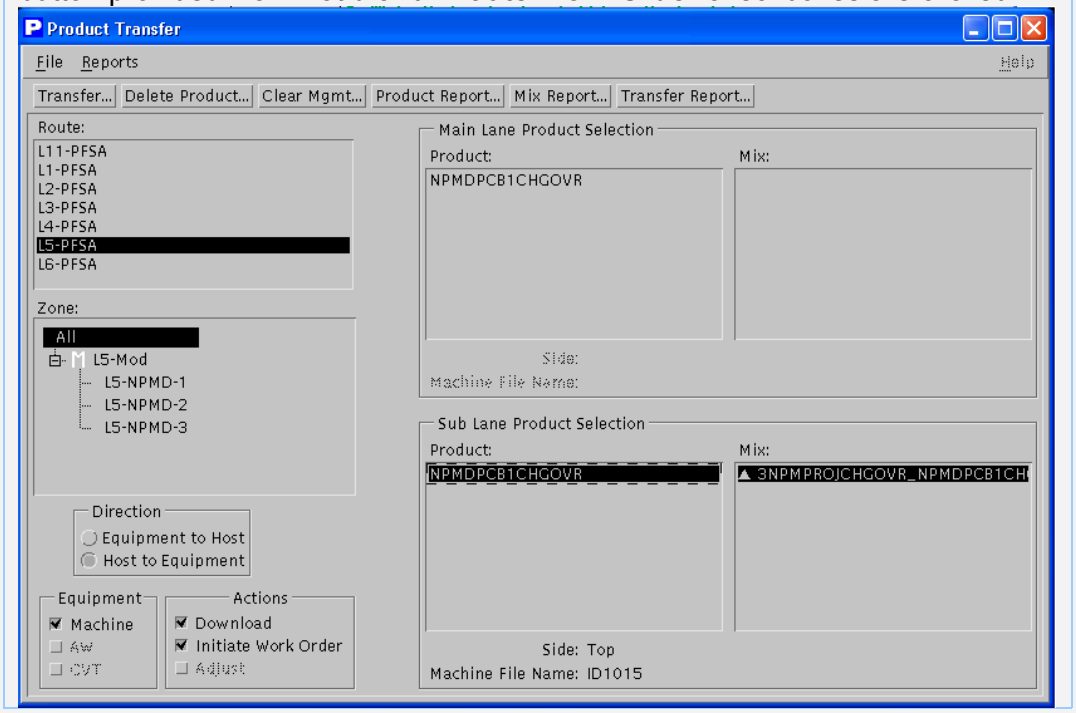
9	After Partial Pick of few reels of rear lane work order, the state is reflected in the DPS screen as shown below



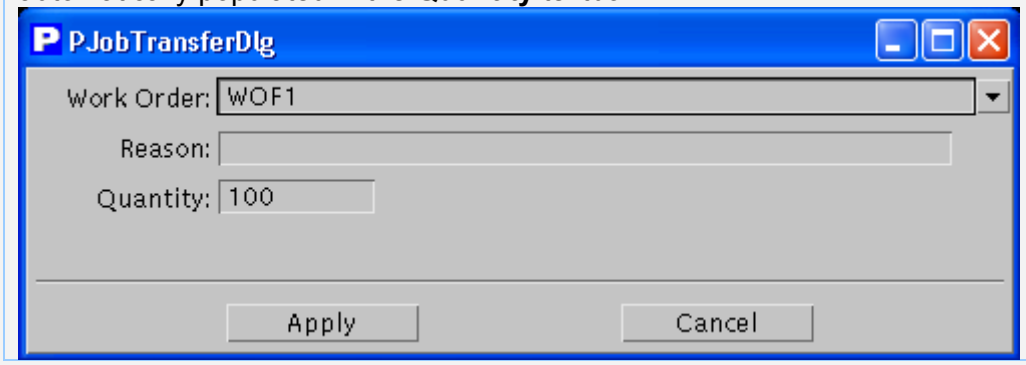
10	Once all the reels of rear lane work order are picked, the status is reflected on the DPS screen.



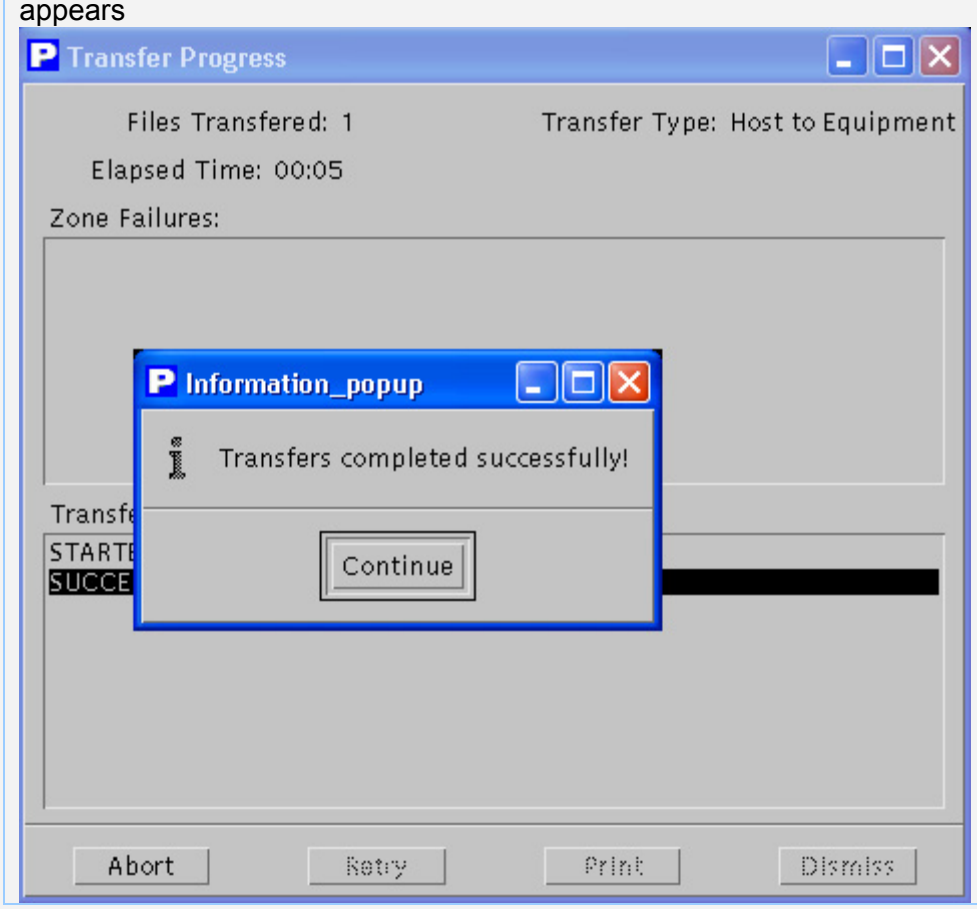
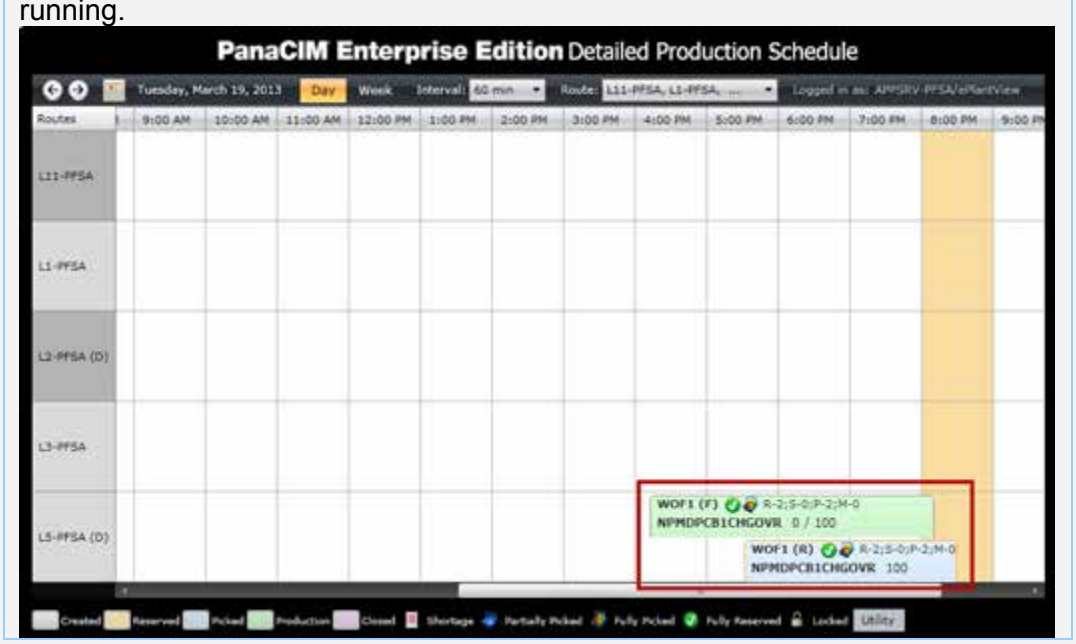
11 WO for rear lane can be put into production using product transfer application. Select the route followed by product in the rear lane and click the **Transfer** button provided **Download** and **Initiate Work Order** checkboxes are clicked.

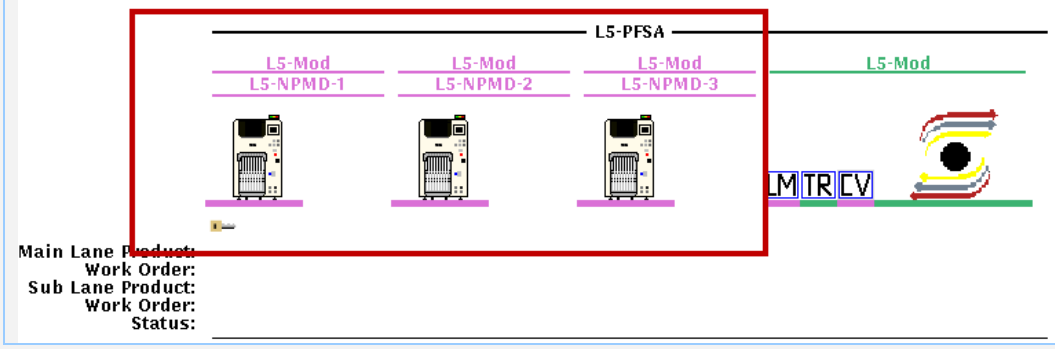


12 Upon doing so, following dialog appears in which the work order to be transferred is chosen. After selecting the work order, corresponding quantity is automatically populated in the **Quantity** textbox.

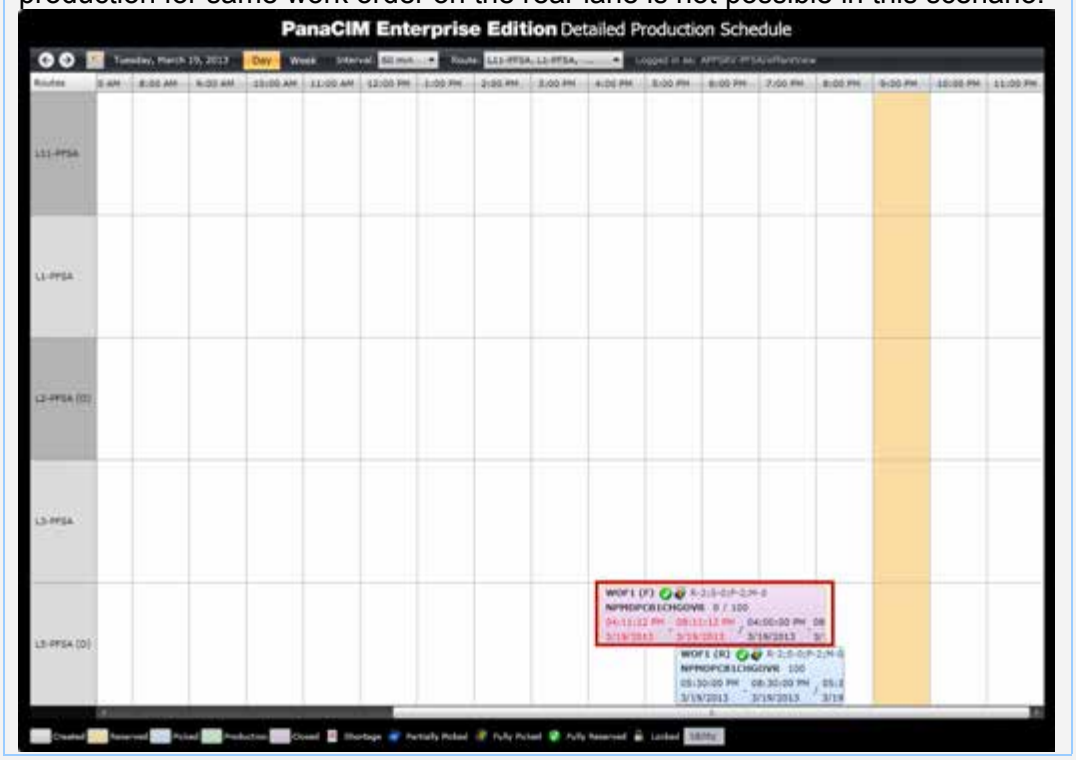


13 Enter a reason and click the **Apply** button. Upon doing so, following message

	appears 
14	Production not started because front lane work order with the same name is still running. 
15	When attempt is made to start production for product in rear lane, LNB interface on CIMC crashes as shown below. **Conclusion:** When front lane work order is in production, you cannot put rear lane work order into production when work order name, route name, product name and side are same for shared mode.

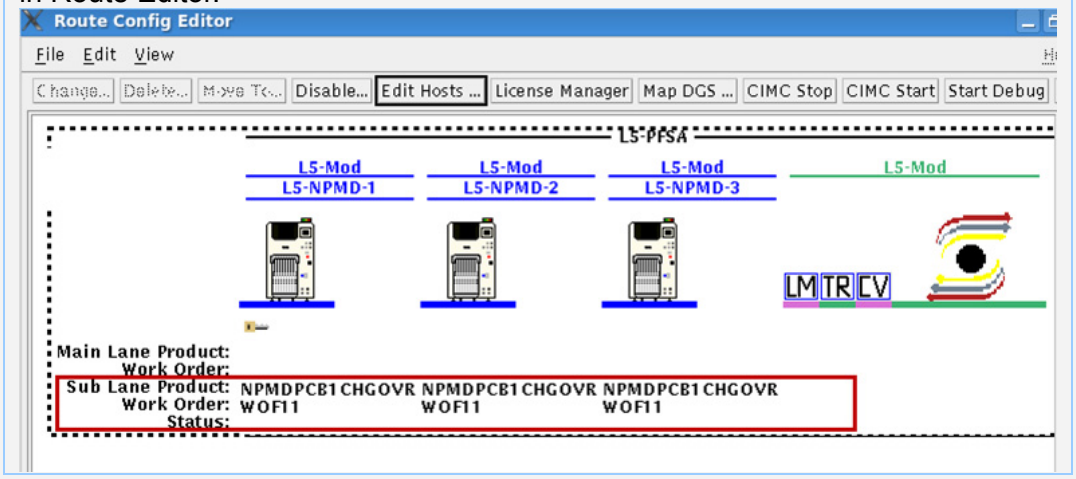


16	Even on closing the work order on the first lane, and then attempting to start the production for same work order on the rear lane is not possible in this scenario.
17	Upon changing the work order name (in this case from WOF1 to WOF11), production for the work order is started after initiating product transfer as shown below:



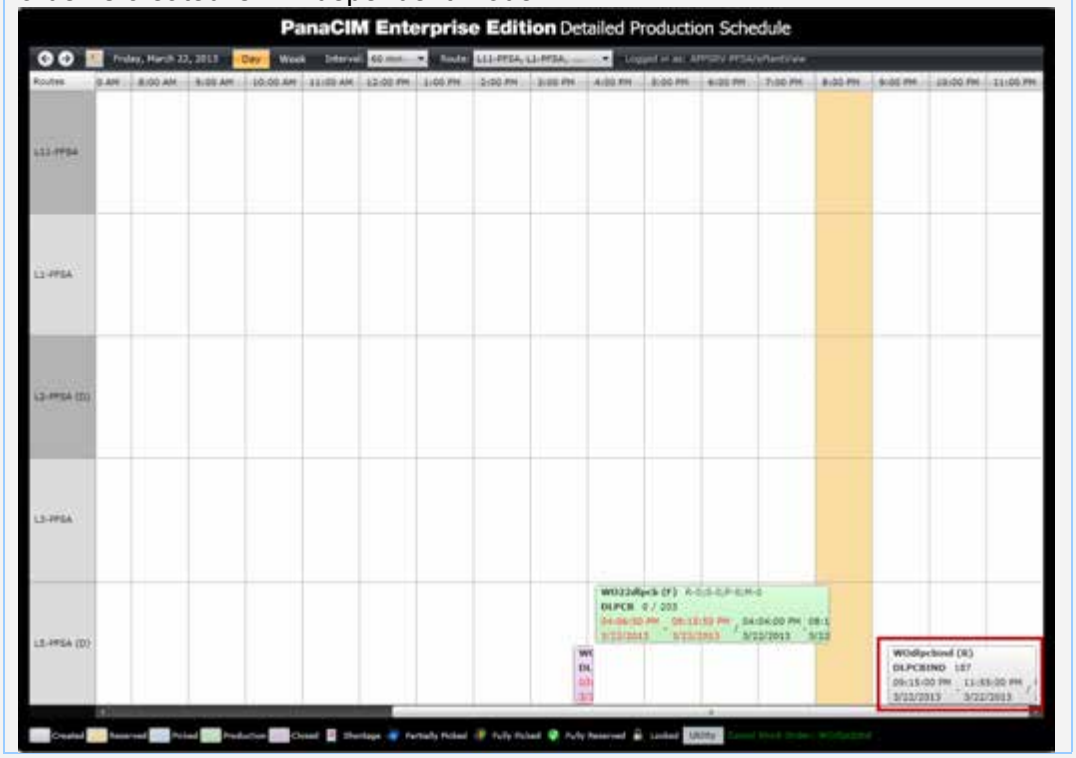
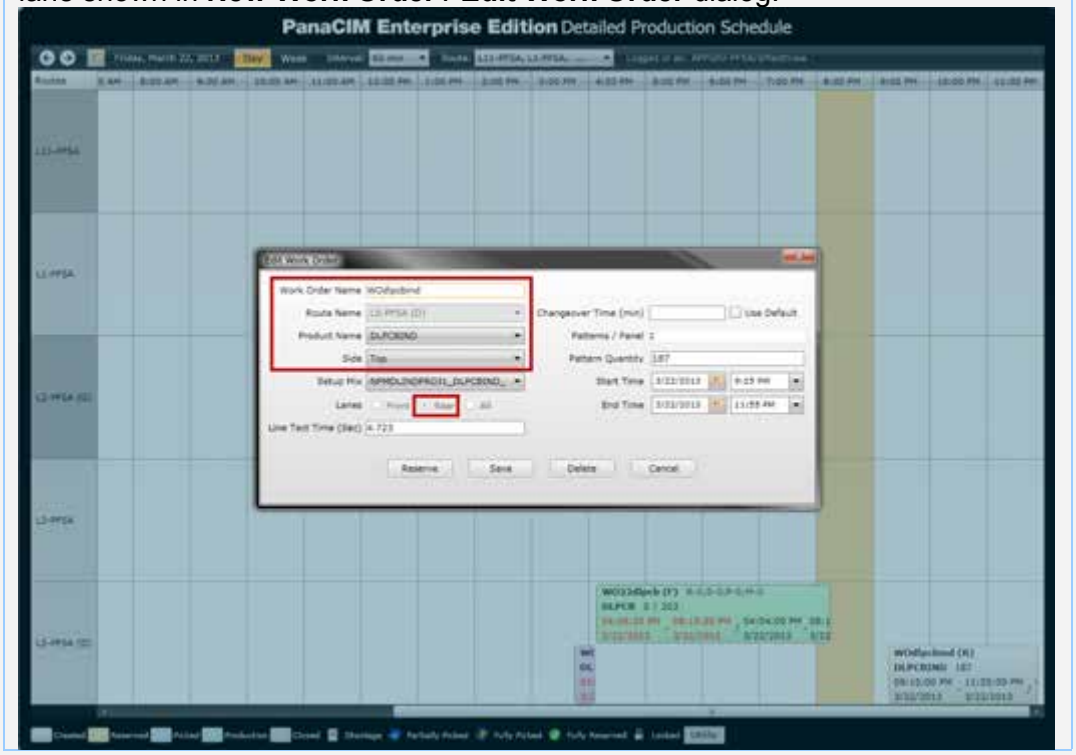


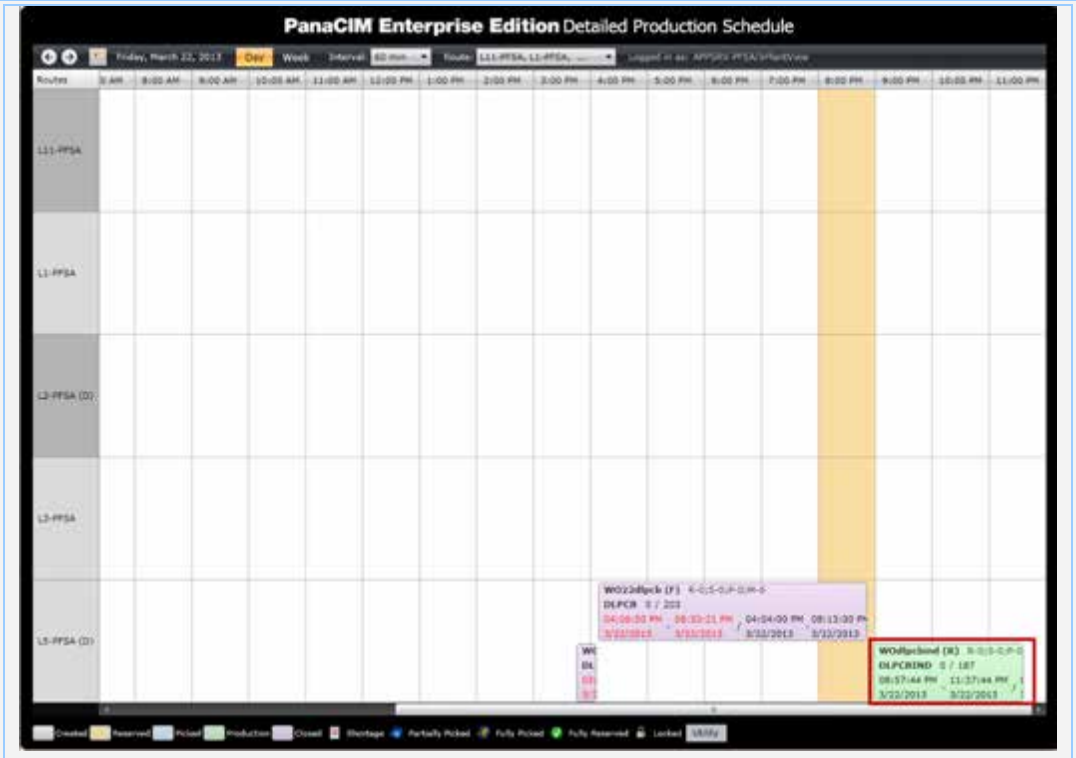
18	Status that work order is in production (in this case WOF11) is correctly reflected in Route Editor.

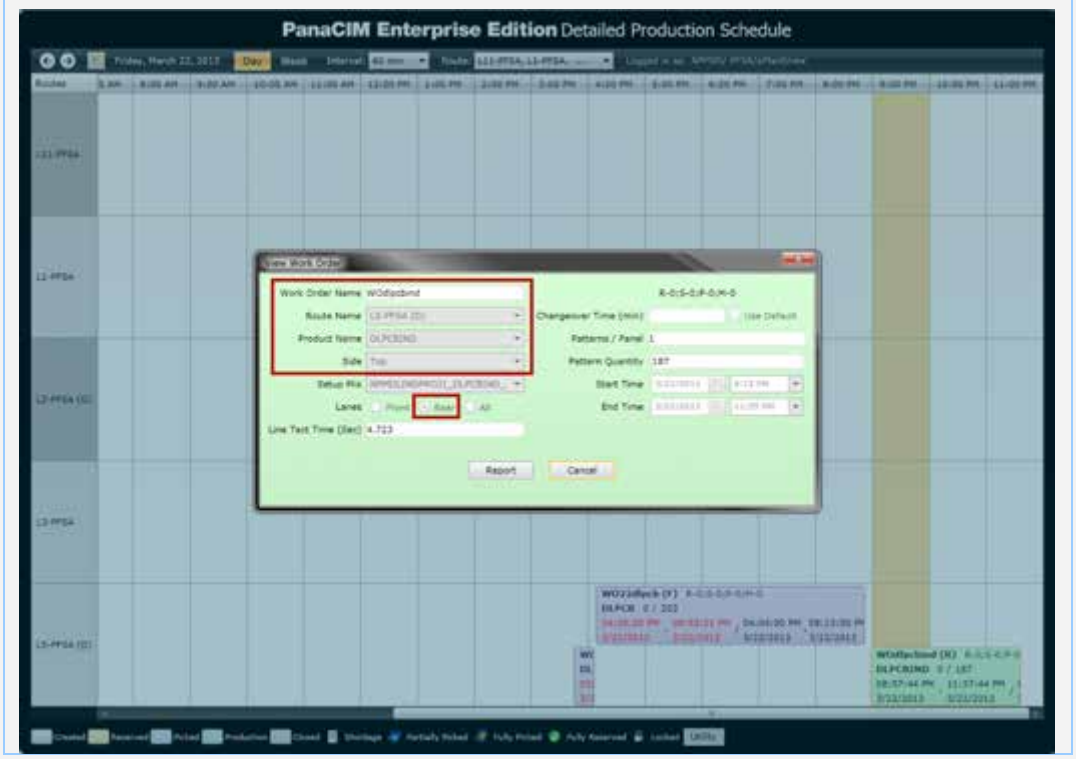


Independent Mode
Work order can be successfully created when Route Name, Product Name, Side and Pattern Quantity are same. Only thing different is the lane. Product recipe is created in **independent** mode in DGS. Production can be simultaneously started for both the work orders in their respective lanes when following attributes are same:

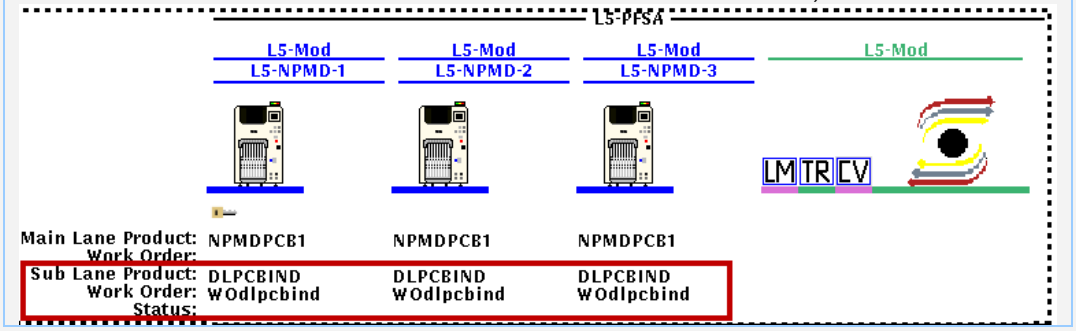
- Work Order Name
- Route Name
- Product Name
- Side
- Quantity

Step	Description
1	Work order is created for rear lane as shown below. The product for which work order is created is in independent mode. 
2	Double-clicking the work order for rear lane displays the Edit Work Order dialog as shown below. **Note:** In shared mode, user can choose a lane as Front, Rear or all while in independent mode once a product is chosen in Edit Work Order dialog in DPS, corresponding lane is already set as front or rear. User cannot edit the selected lane shown in **New Work Order / Edit Work Order** dialog. 
3	Work Order is put into production as shown below:

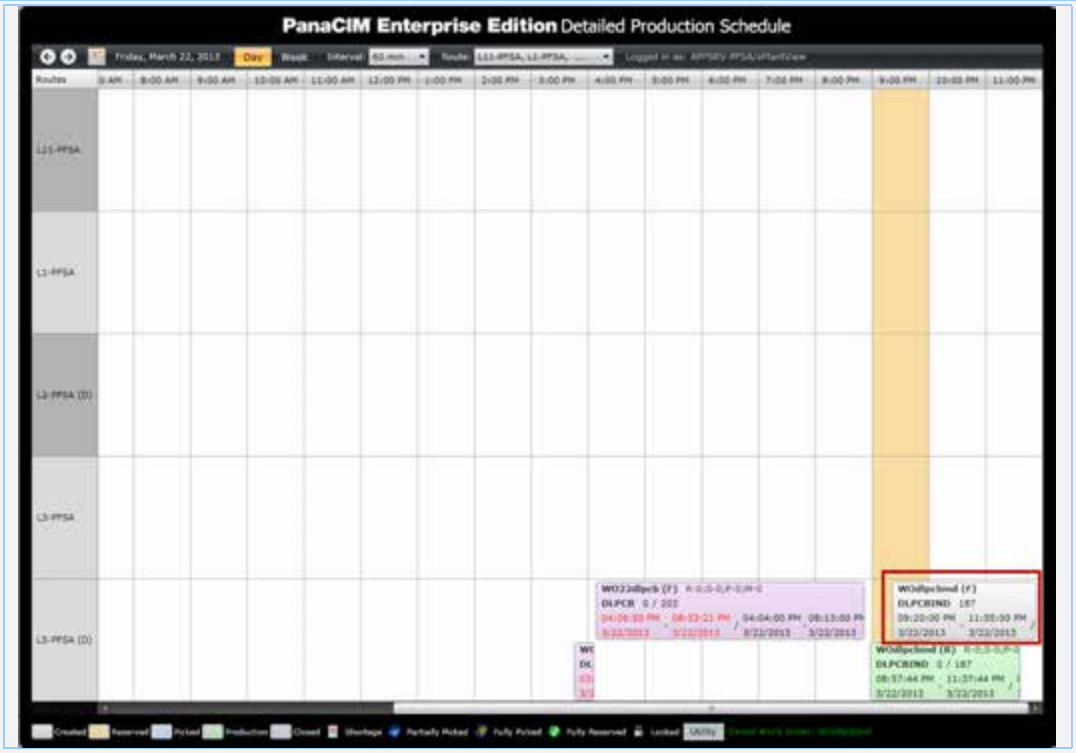


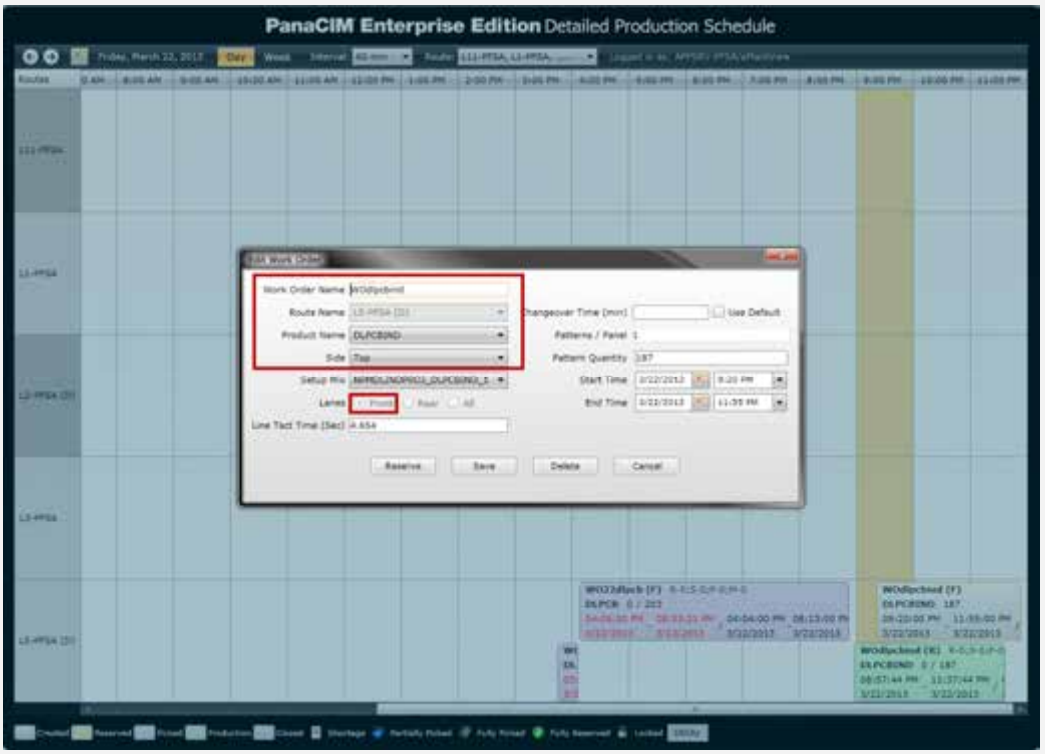


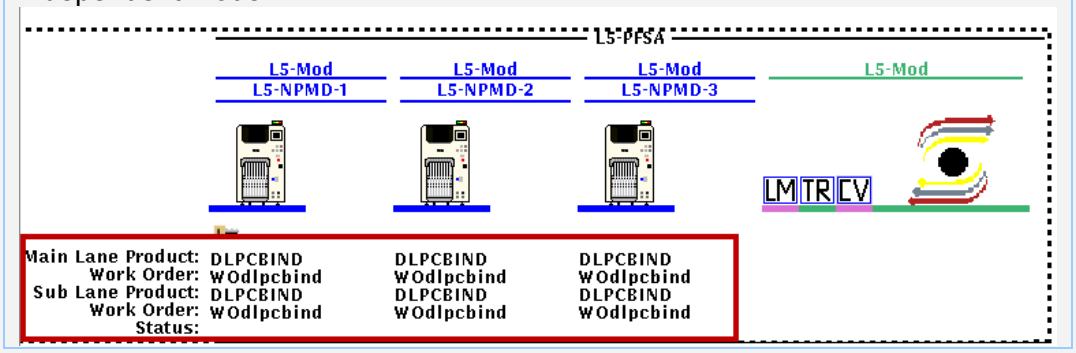
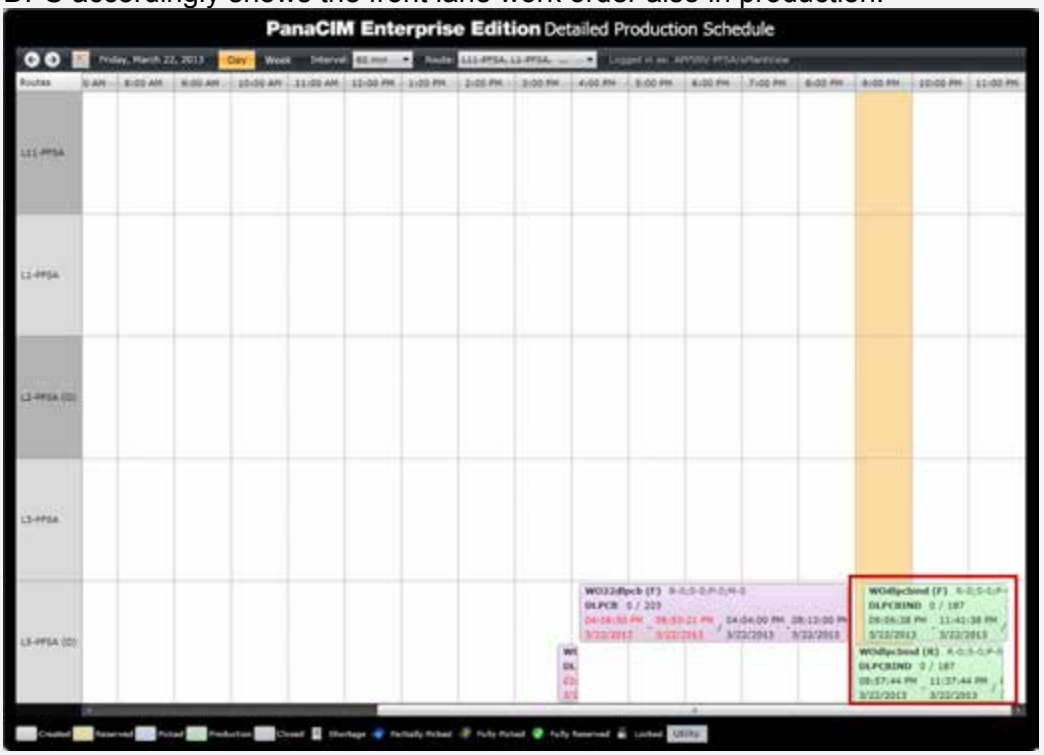
4 Work order is reflected in the route status on the sub lane i.e., rear lane.

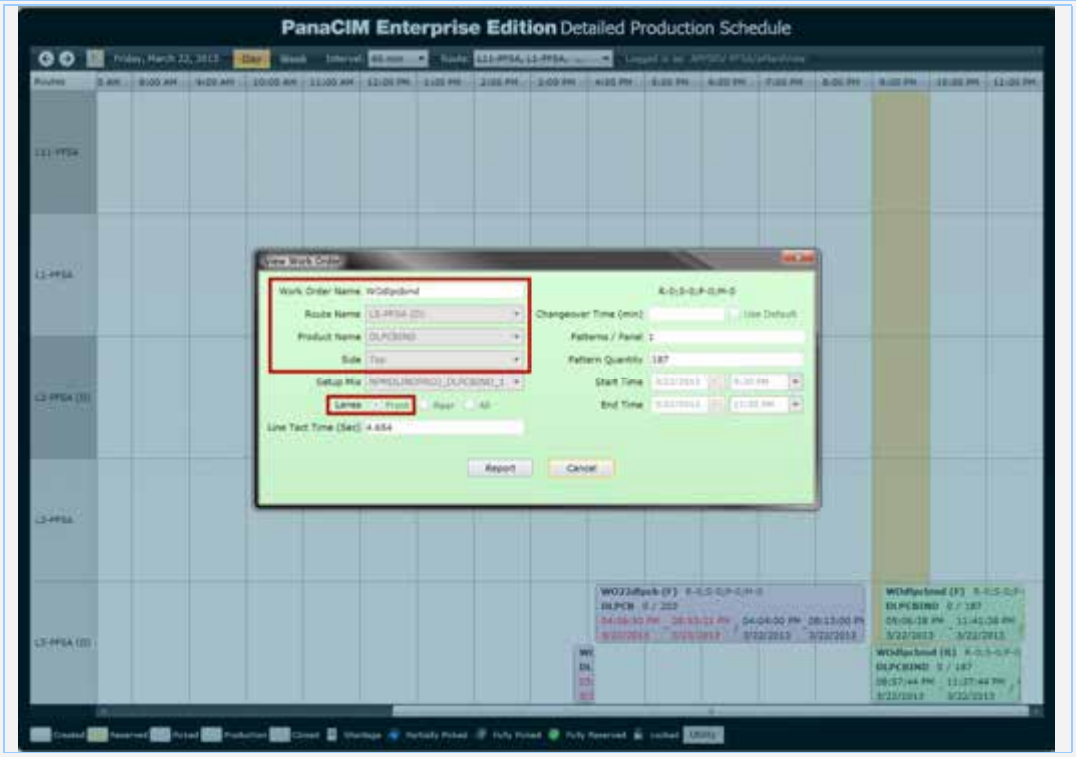


5 Create one more work order for front lane.

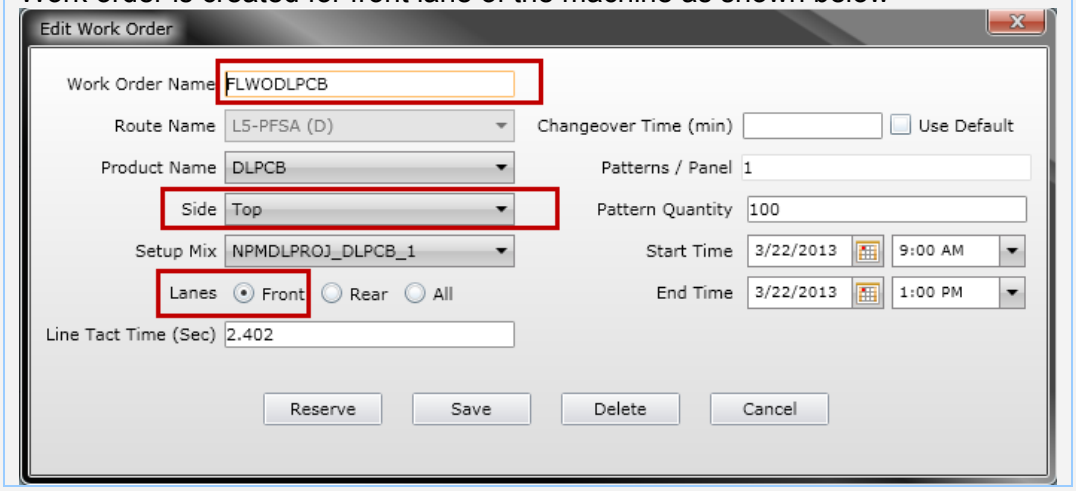
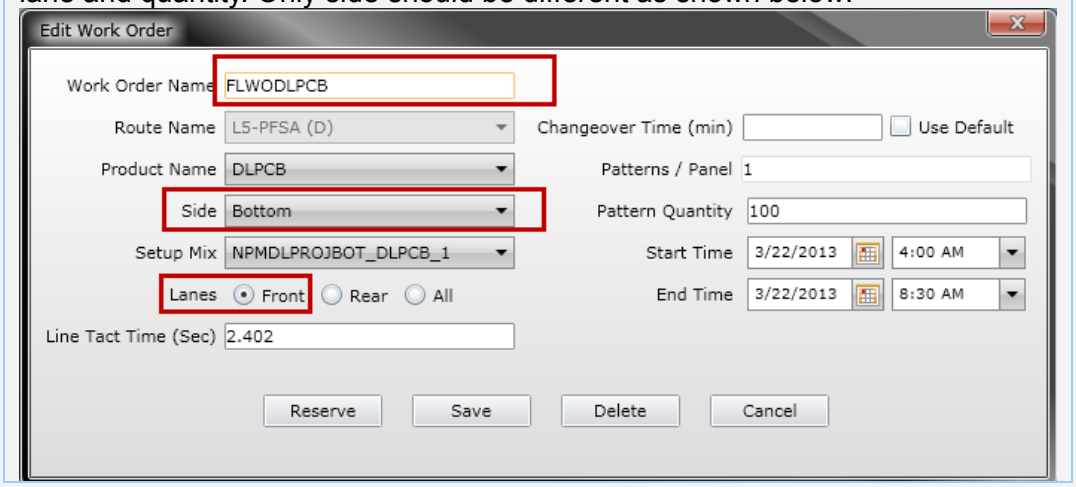


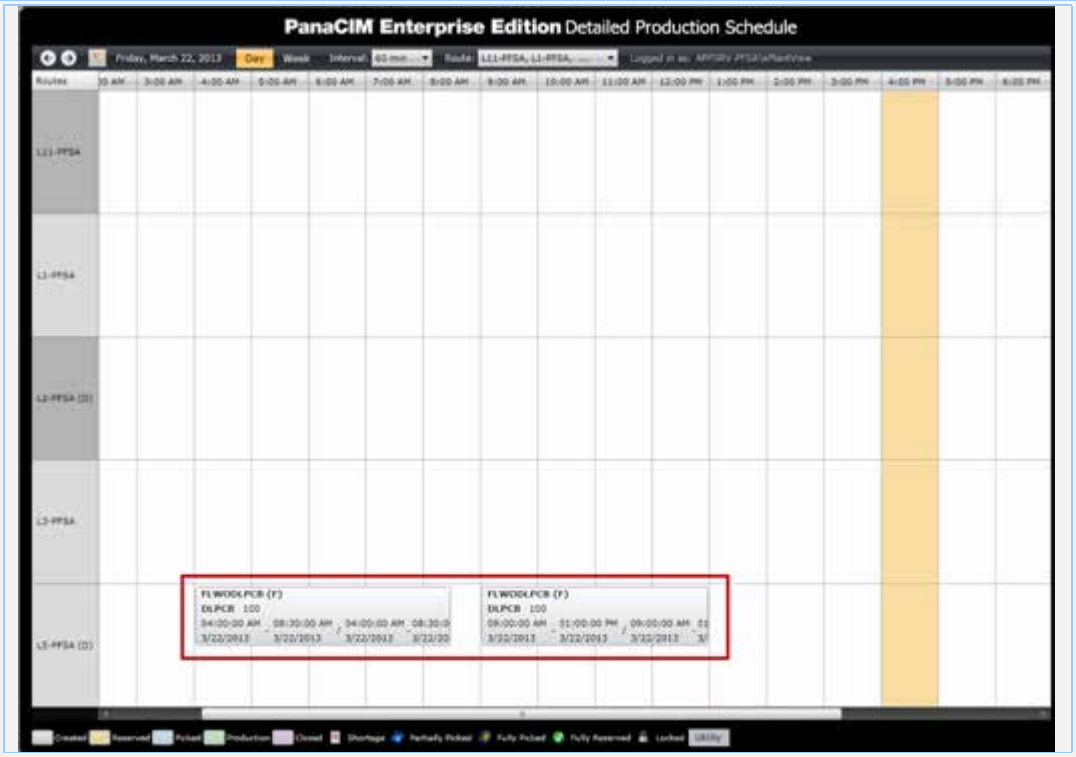
6	Work order details for front lane are as shown below: Work order name, Route Name, Product Name, Side and quantity for front lane work order is same as rear lane work order. 
7	Front lane work order can be successfully put into production when rear lane work order is also in production. Now, LNB interface on CIMC is still up. **Conclusion:** When rear lane work order is in production, you can put front lane work order into production when work order name, route name, product name and side are same provided product imported from DGS was created in

	independent mode. 
8	DPS accordingly shows the front lane work order also in production. 

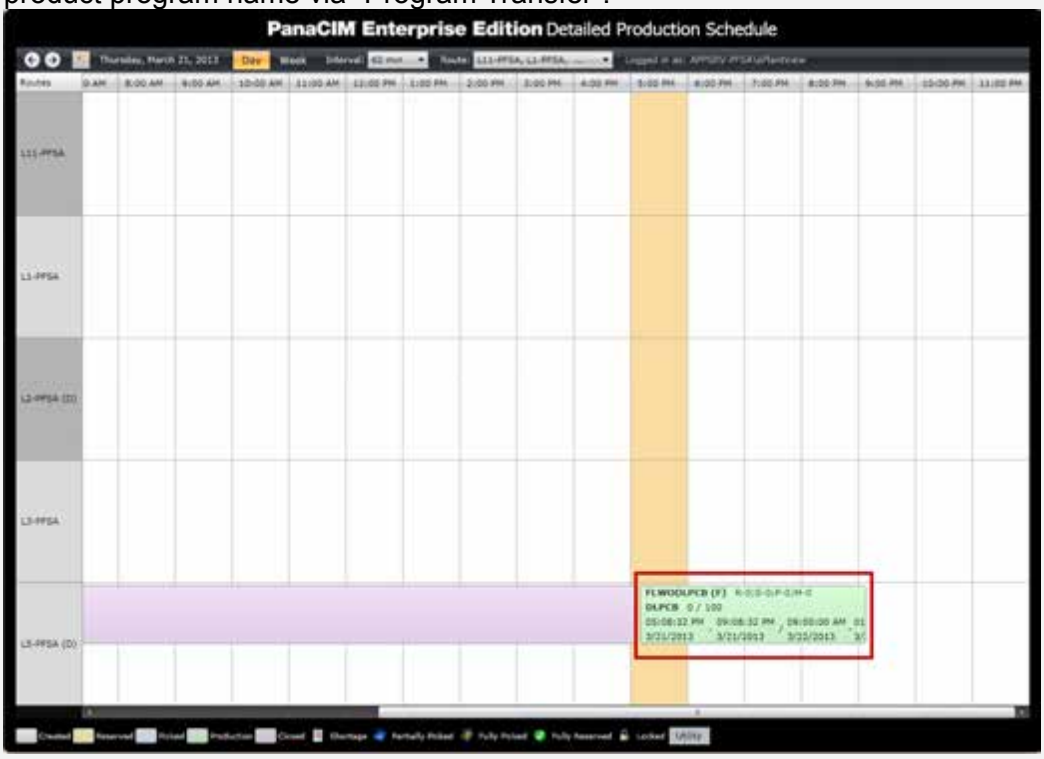
	
9	Independent mode status is reflected in the **Equipment** tab of **Product Setup** dialog. Top side product dedicated for front lane.

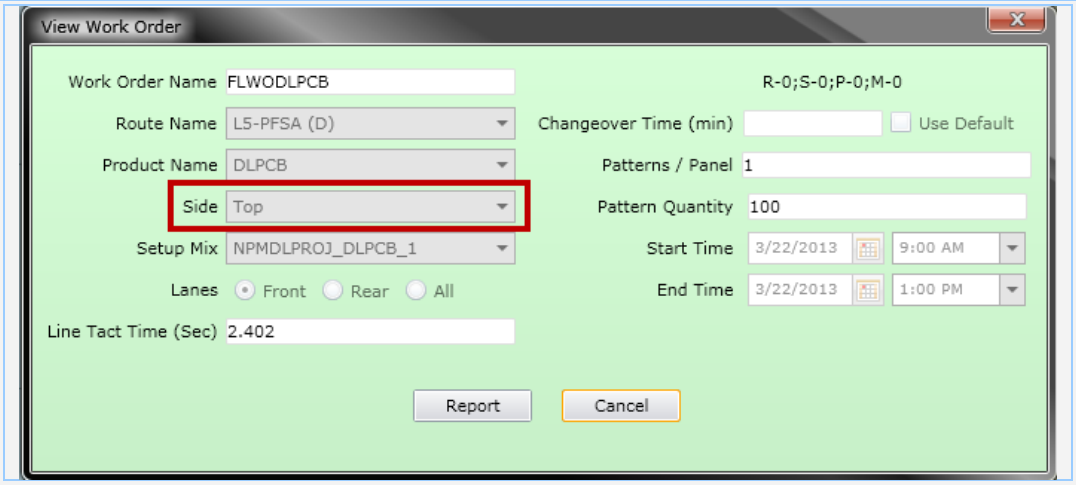
- **Side different rest all conditions same**

Step	Description
1	Work order is created for front lane of the machine as shown below 
2	Another work order with the same name can be created for same route, product, lane and quantity. Only side should be different as shown below: 
3	Both these work orders are visible in DPS as shown below. They will go through their regular cycle of reservation, pick, production and close and each of these statuses is represented in DPS with their corresponding color codes. <Change of work order status (color codes)> Registered (gray) → Material reserved (orange) → Picking/Picked (light blue) → In production (light green) → Closed (pink)



4	Work order is put into production for top side of same product. To start production, transfer the work order name for the top along with the product program name via “Program Transfer”.

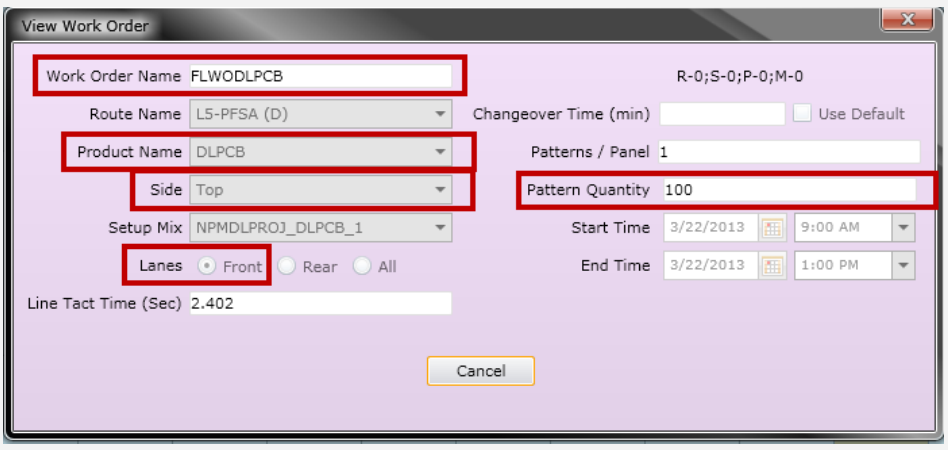


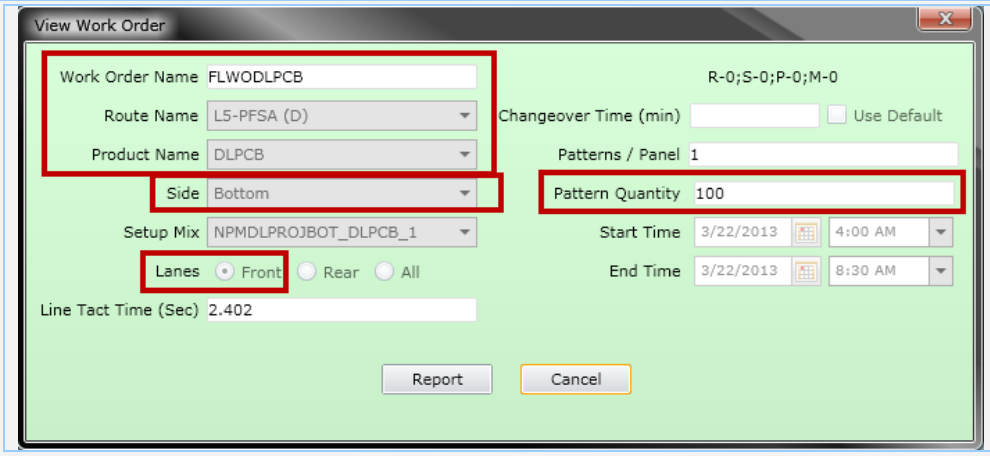


5

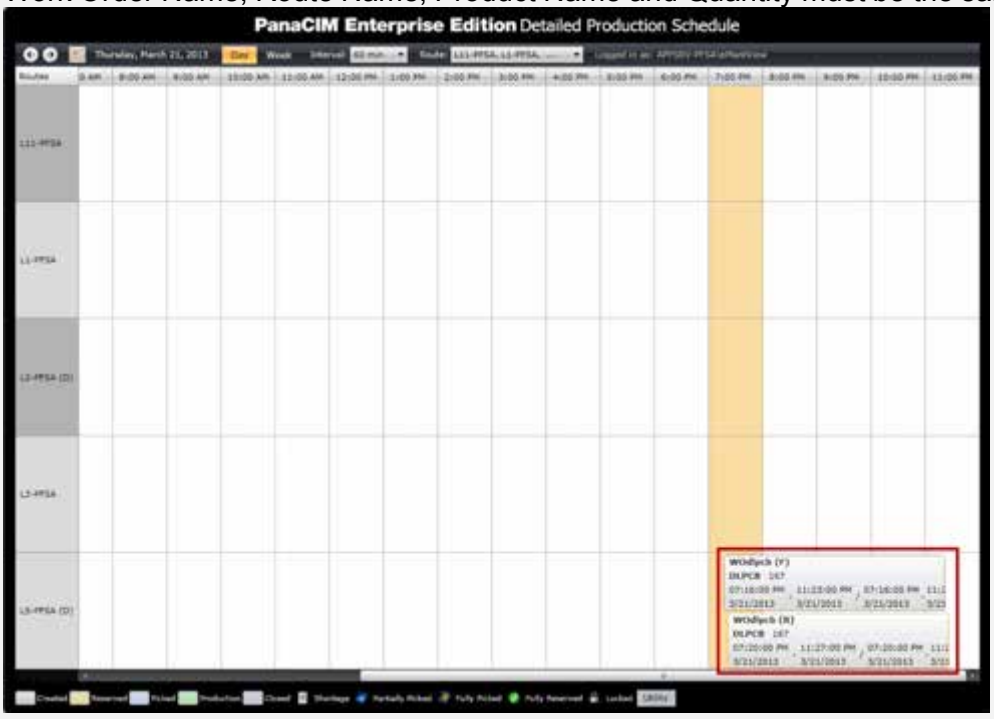
When work order for bottom side created for front lane (whose name is same as top side work order) is put to production, top side work order is closed and bottom side work order comes into production as shown below.
To start production, transfer the work order name for the bottom along with the product program name via “Program Transfer”.

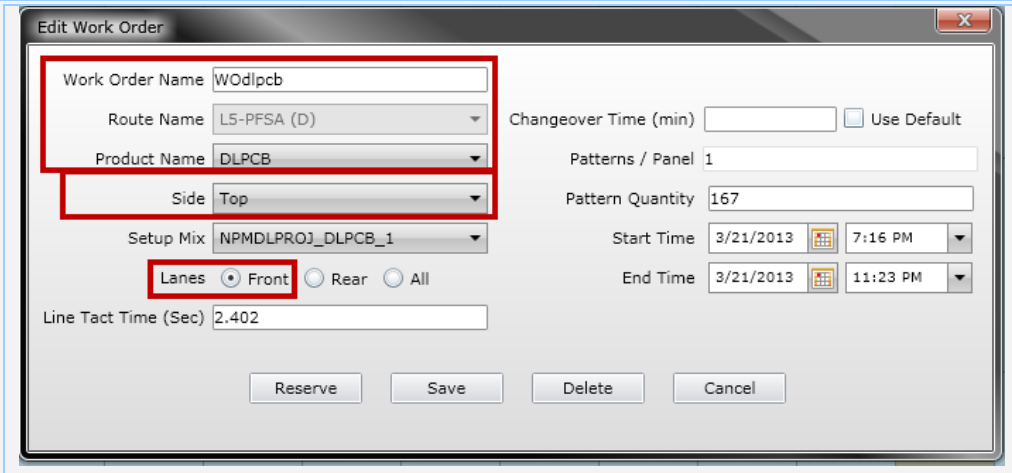


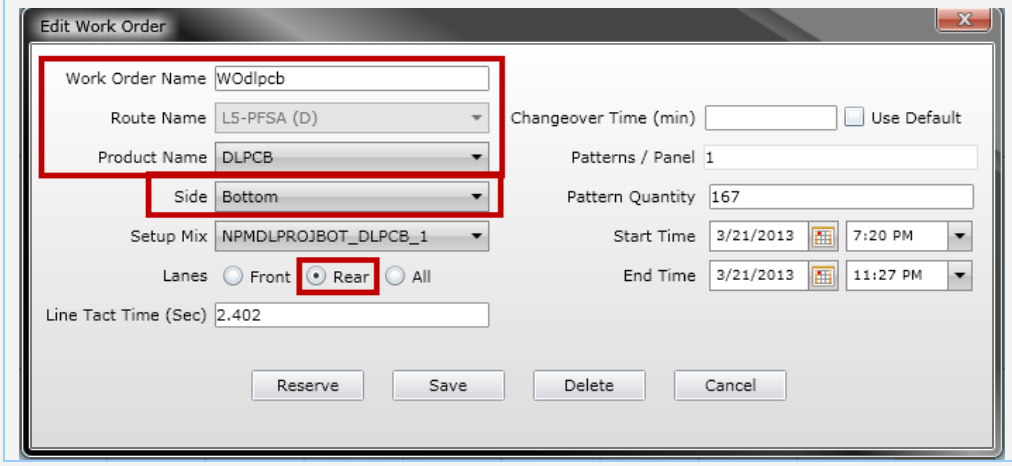




- **Side and lane is different rest is same**

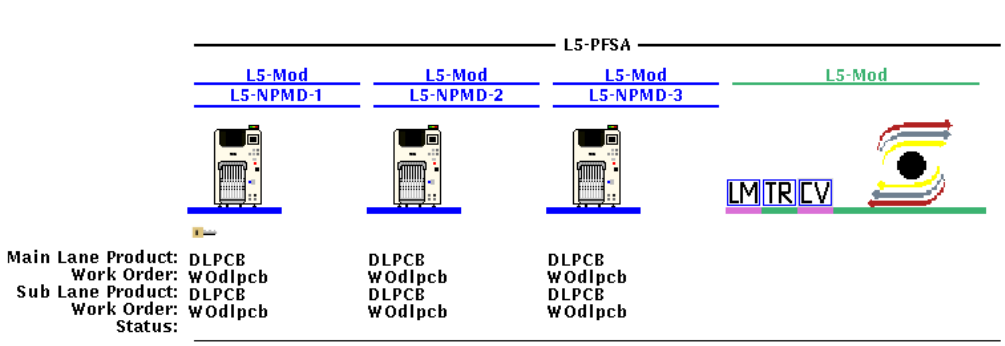
Step	Description
1	Work order is created for front lane and rear lane of the machine as shown below. ・Front lane: Side - **Top** ・Rear lane: Side - **Bottom** Work Order Name, Route Name, Product Name and Quantity must be the same. 



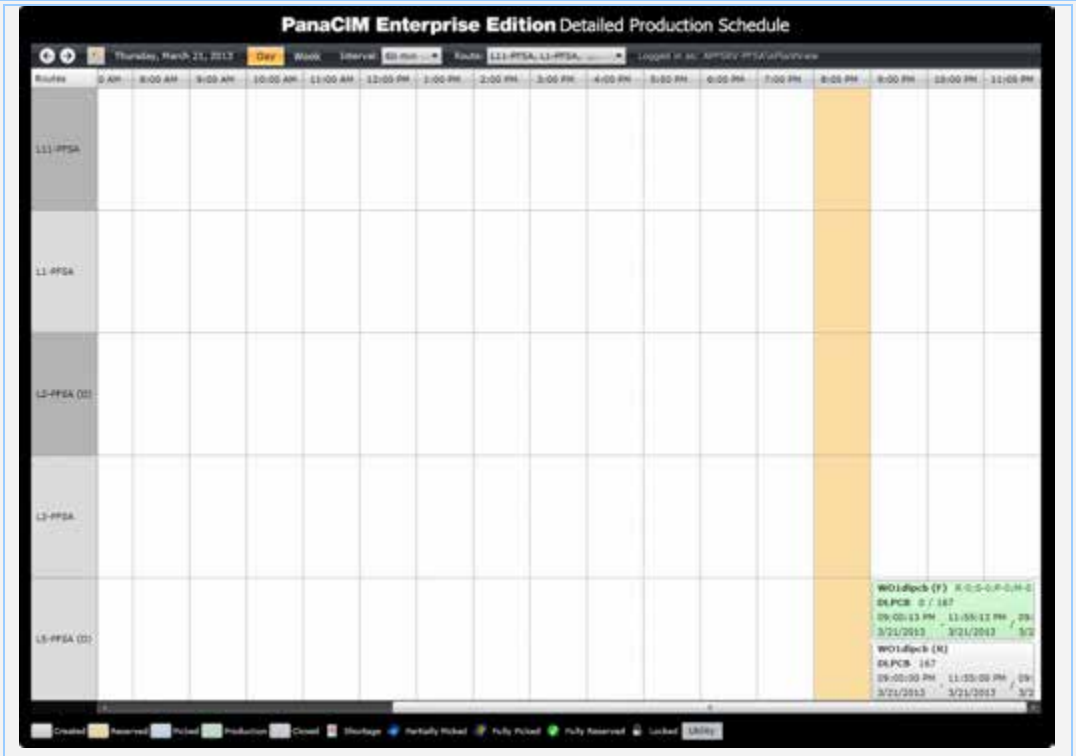


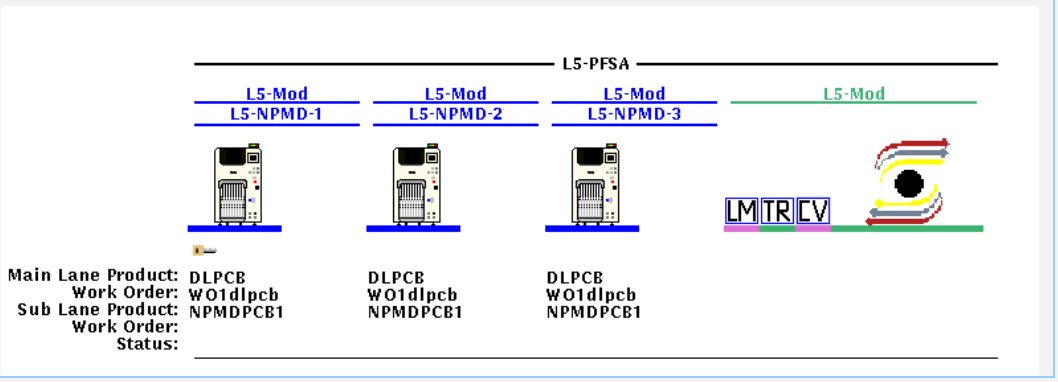
2	As shown above, same work order is created for same route, product name and pattern quantity. Only thing different is product side and lane. They will go through their regular cycle of reservation, pick, production and close and each of these statuses is represented in DPS with their corresponding color codes. <Change of work order status (color codes)> Registered (gray) → Material reserved (orange) → Picking/Picked (light blue) → In production (light green) → Closed (pink)
3	Production can be started for both front lane and rear lane work orders simultaneously corresponding to top side and bottom side products



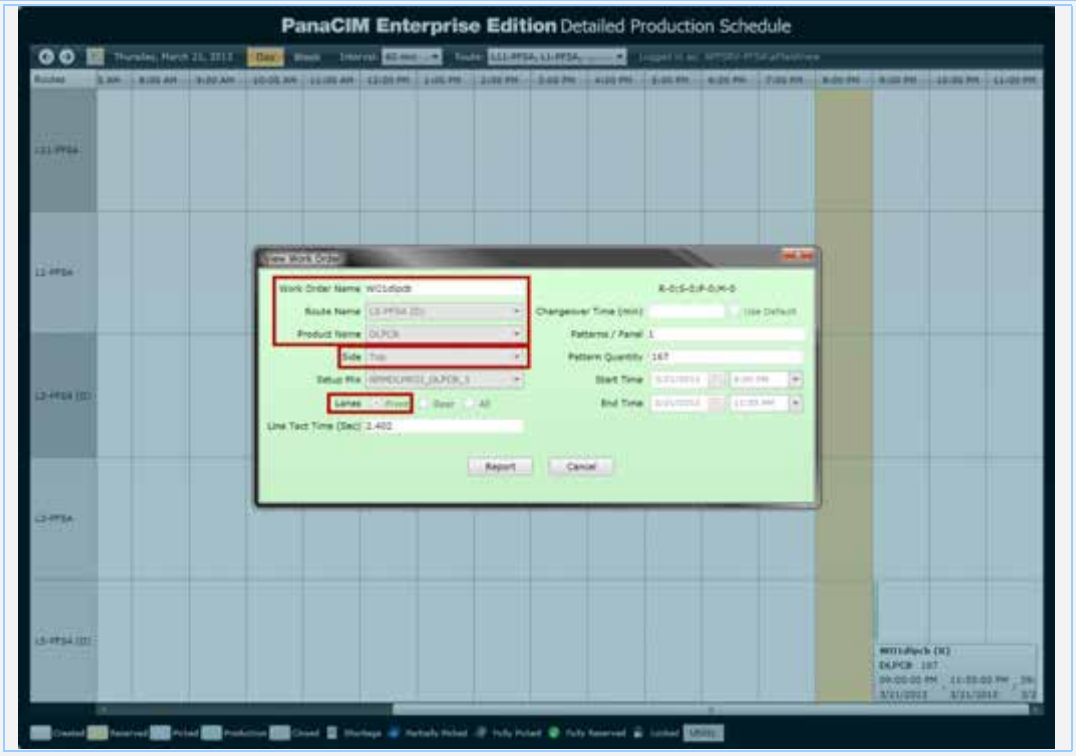


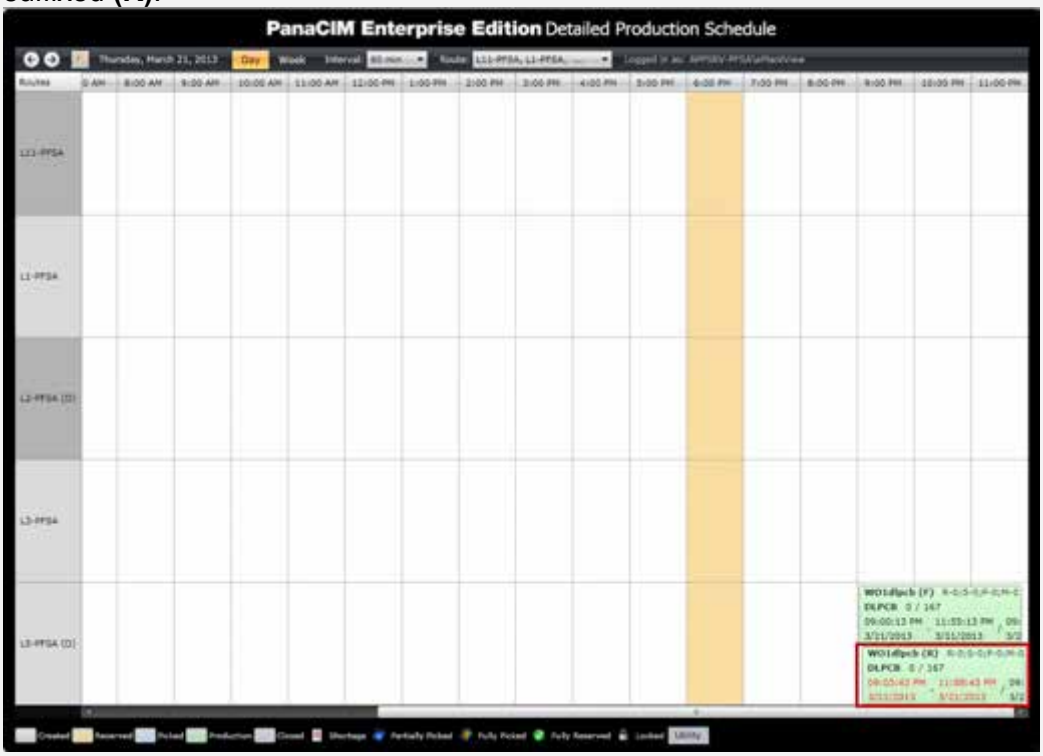
4	Production can be started for one of the lanes only and this is shown below:

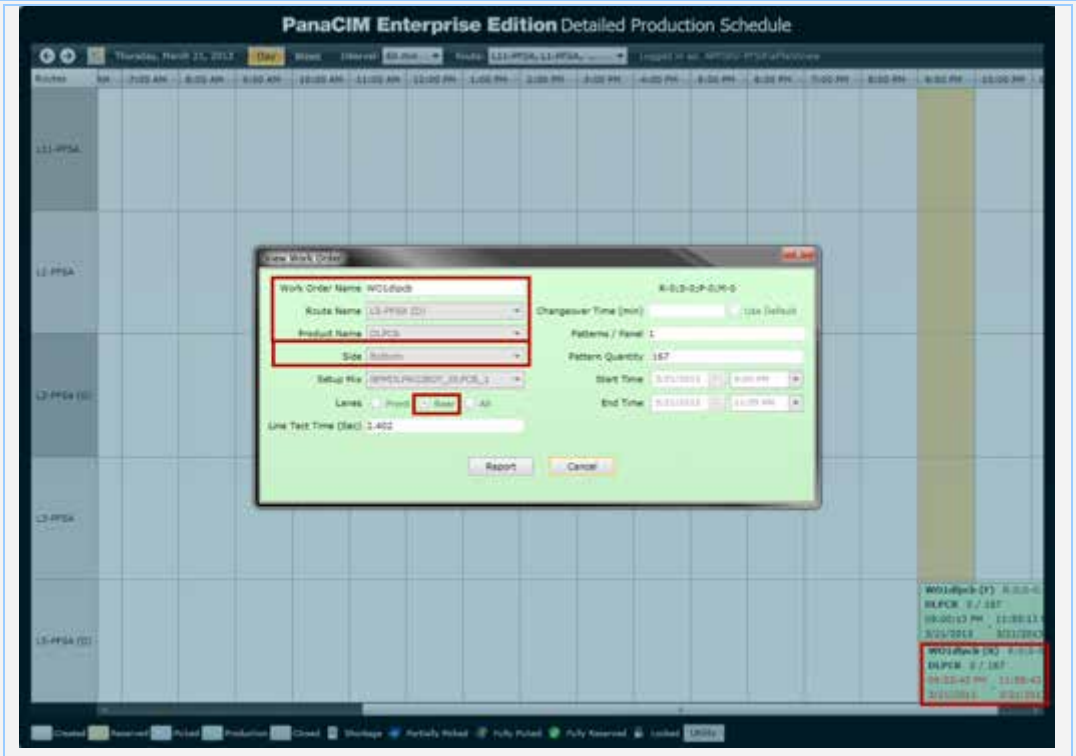




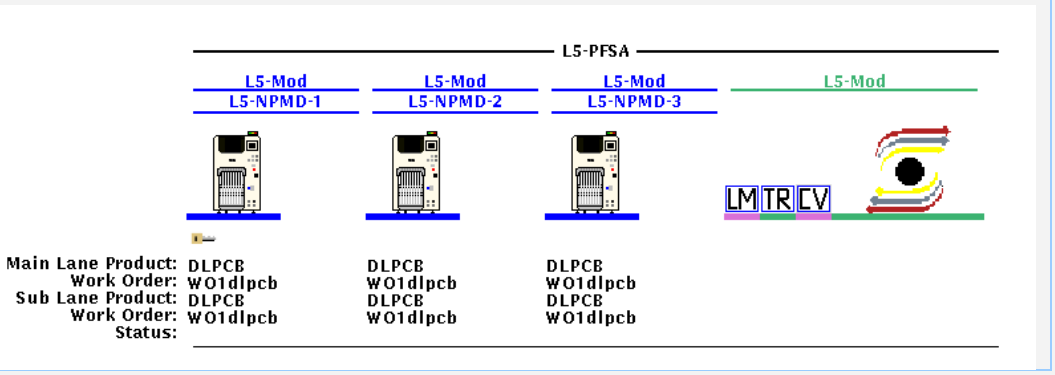
5	As shown above, production is started for top side product in front lane for work order named **WO1dlpcb**.



6	Later on production can be started for rear lane bottom side product whose work order name is same as that of front lane top side product. **Note:** Front lane work order is suffixed by **(F)** while rear lane work order is suffixed **(R)**. 

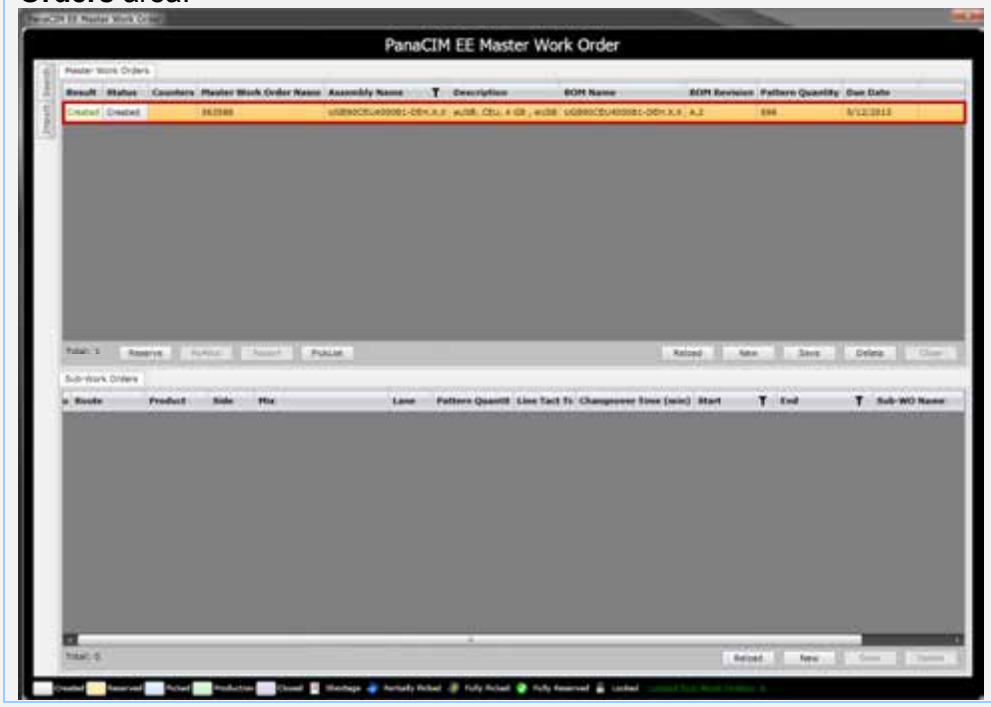


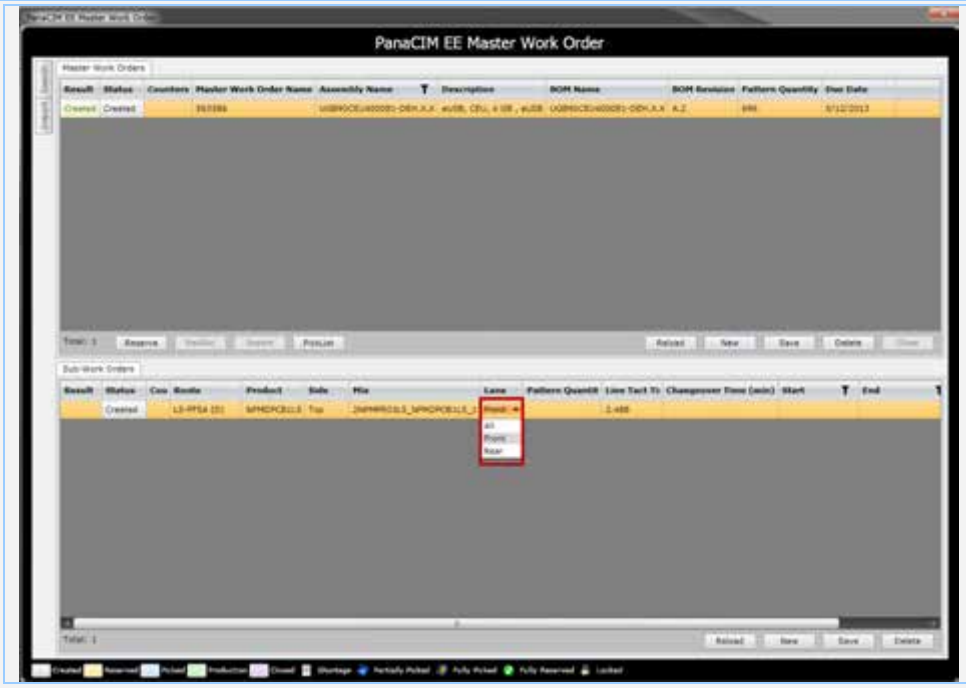
7 In the Route Status and Route Editor, work order is now shown in running in rear lane.

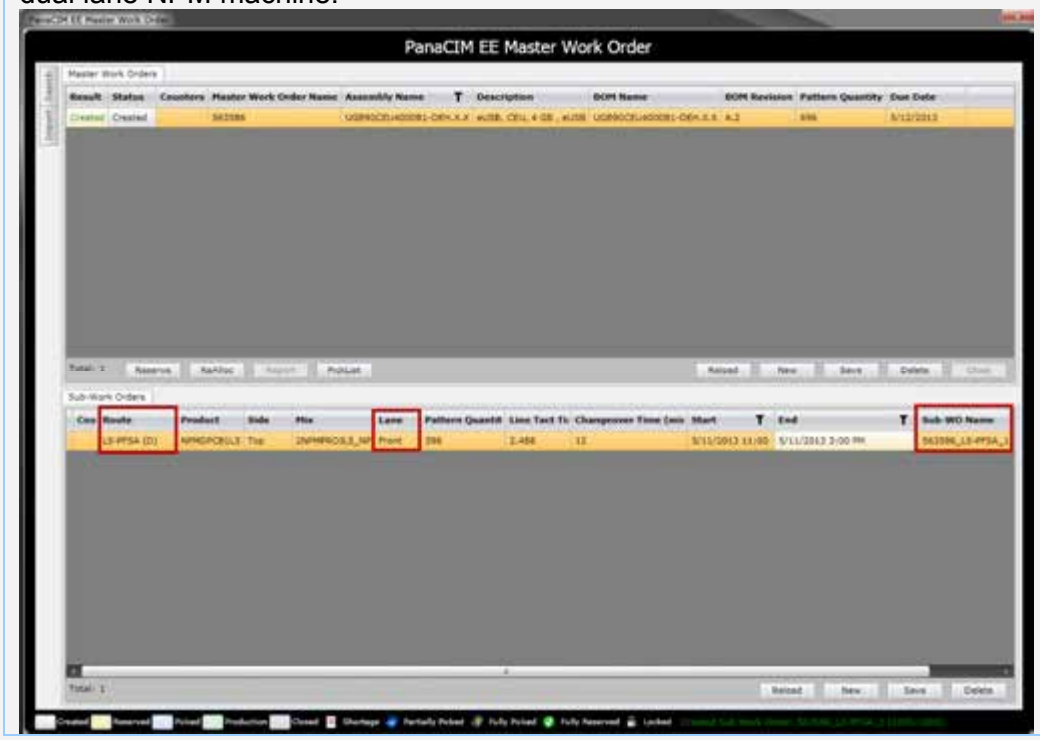


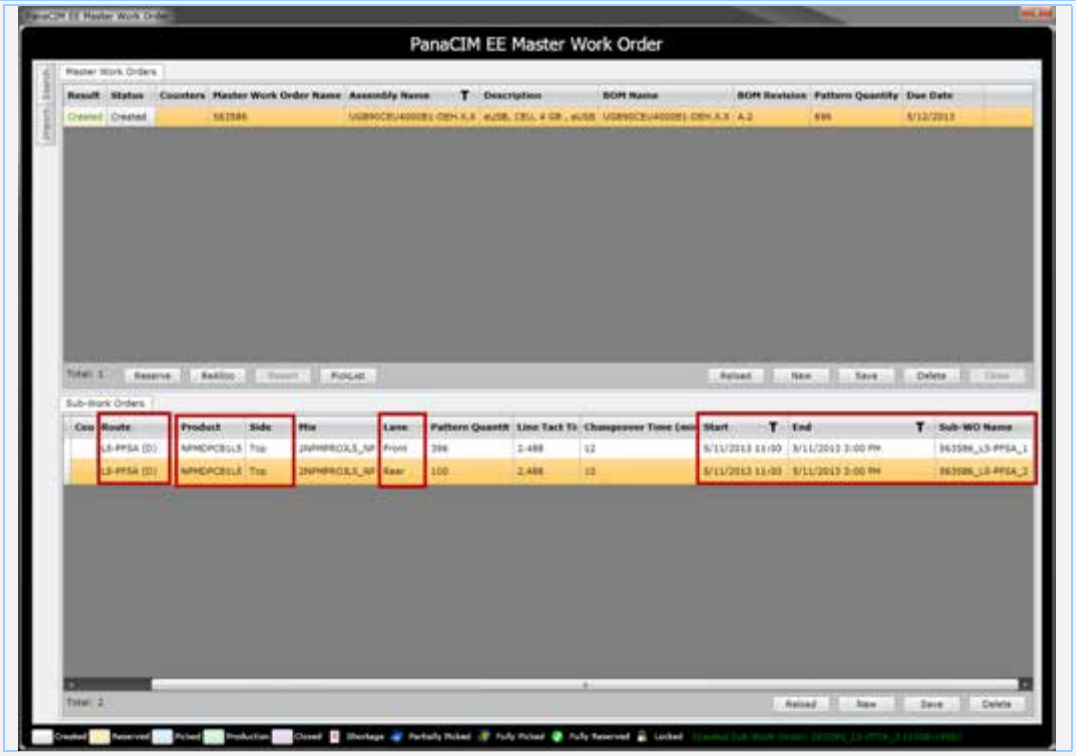
■ Creating and Reserving Master Work Order for Dual Lane NPM Machines

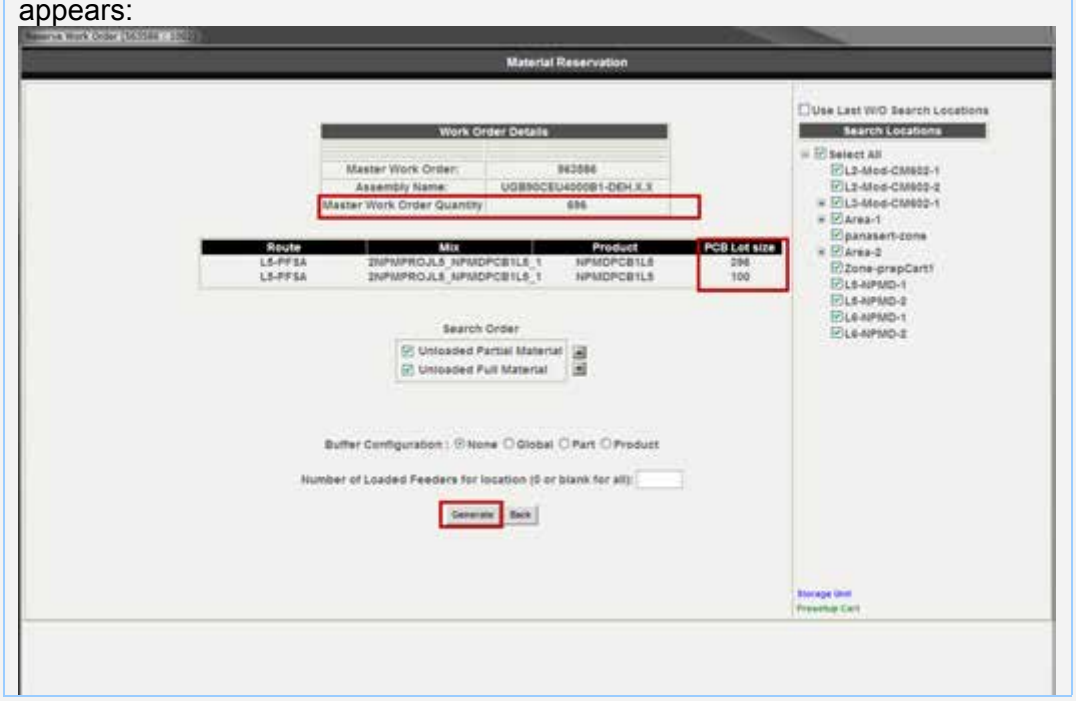
Single master work order can be created for both the lanes of dual lane NPM machines. However, reservation is to be done for complete quantity of sub work order. Following table shows the procedure for creating sub work orders for each lane of NPM machine when master work order is created.

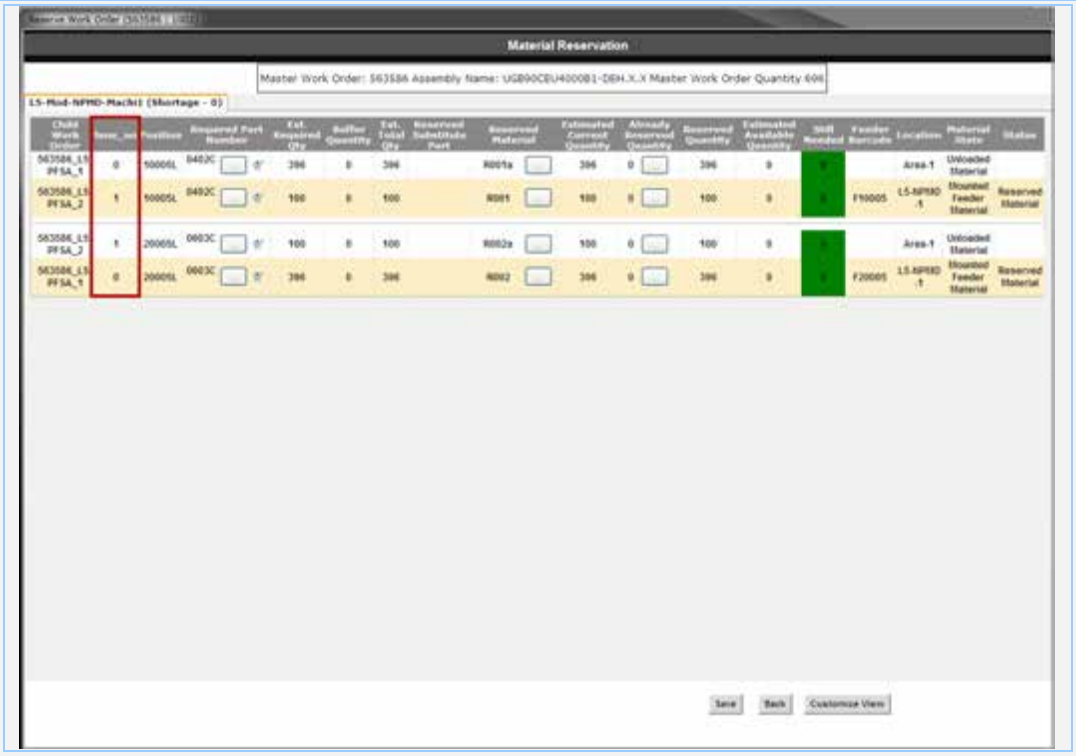
Step	Description
1	Create a master work order by clicking the **New** button in the **Master Work Orders** area. 
2	Once the master work order is created, click the **New** button in the **Sub-Work Orders** area to add a line. With the added sub work order, select or specify the following items, and then click the **Save** button in the sub-work orders area to create a sub work order for a lane. Choose a route containing dual lane NPM machines. Choose the lane as **All**, **Front** or **Rear**. Enter other important information like Pattern Quantity, Changeover Time, Start time and End time. In DPS, dual lane NPM route has **(D)** suffixed at the end of route name.



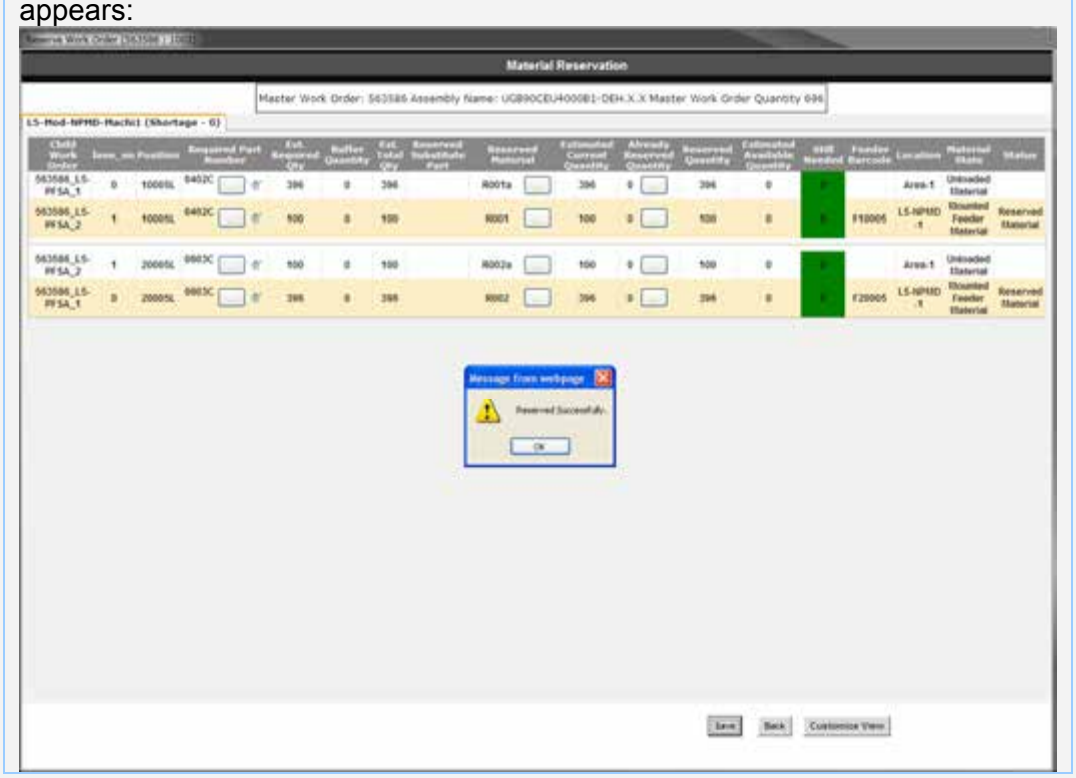
3	Once **Save** button is clicked, sub work order is created for a selected lane of dual lane NPM machine. 
4	Click the **New** button again to create sub work order for same route and same product but for a different lane.



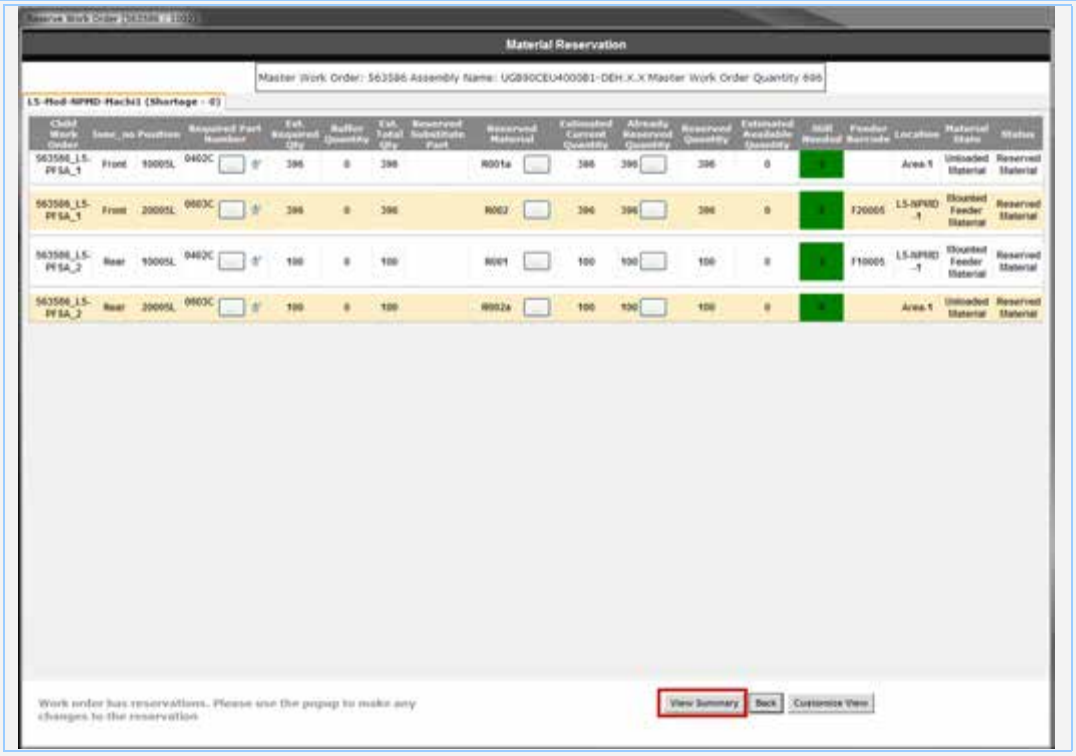
5	Click the **Reserve** button. After clicking the **Reserve** button, following screen appears: 
6	As shown above, reservation is to be done for complete quantity of sub work order and not the master work order. Click the **Generate** button. After clicking the **Generate** button, following screen appears showing reservations for each lane and each sub work order. 0 refers to front lane and 1 refers to rear lane



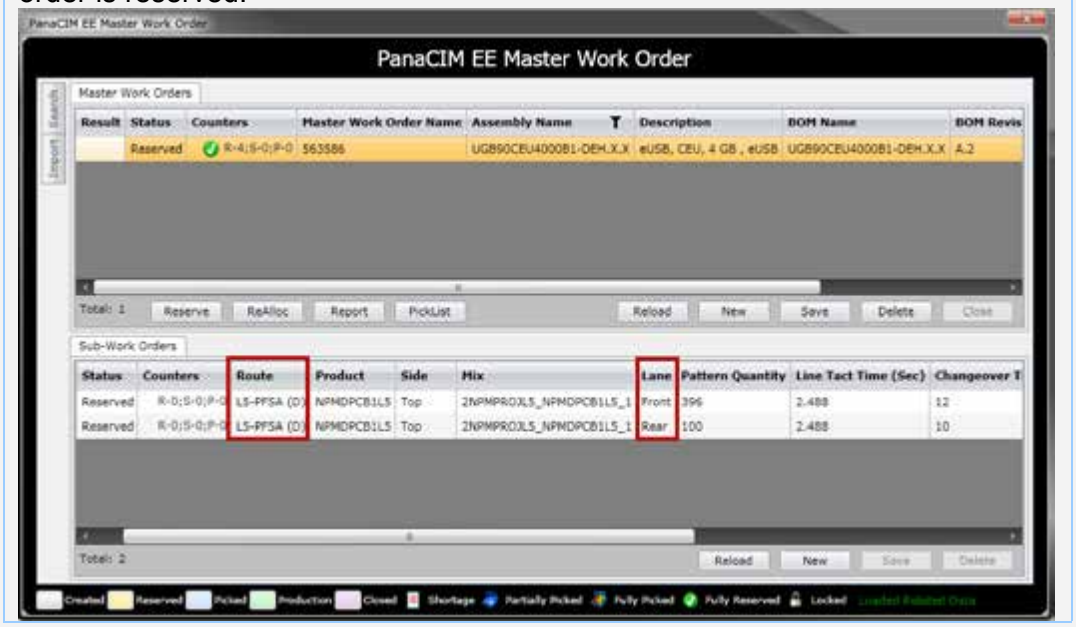
7 Click the **Save** button. Upon doing so, “Reserved Successfully” message appears:



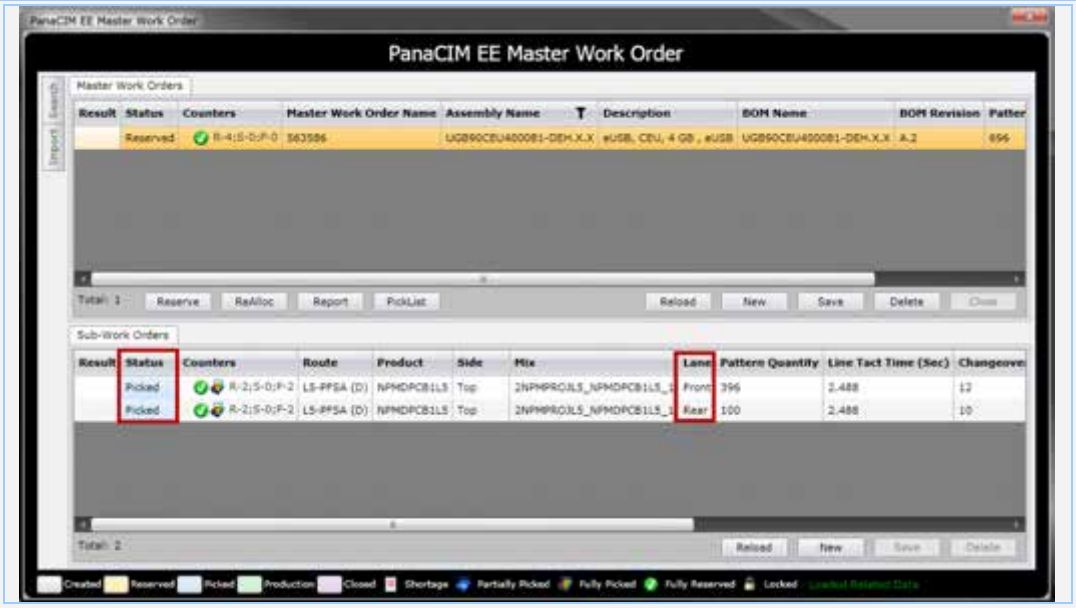
8 Once the “Reserved Successfully” message is closed, option exists to display the reservation summary by clicking on the **View Summary** button.



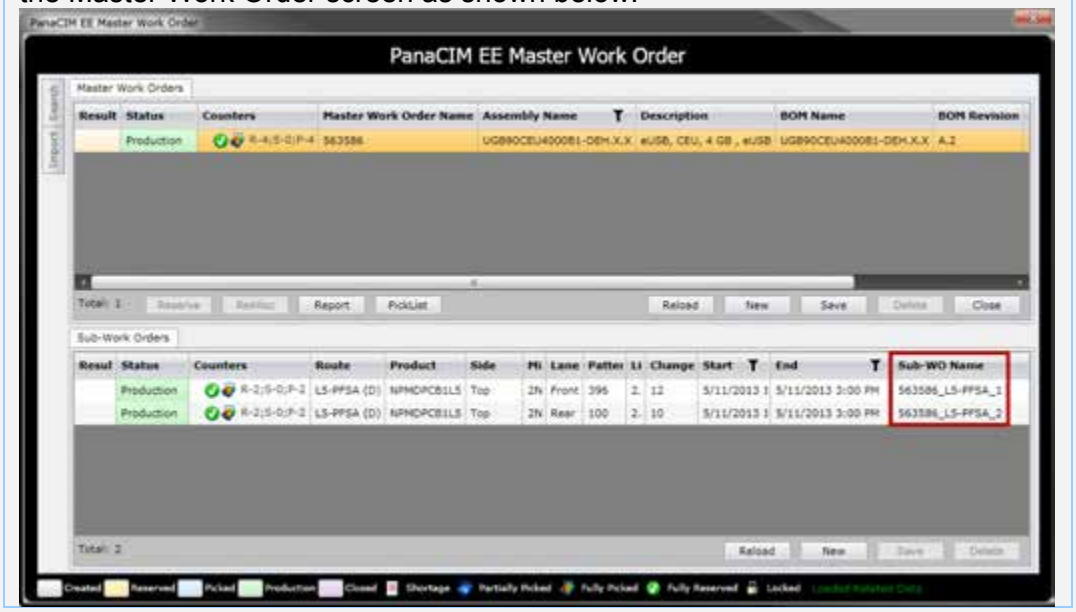
9	In the PanaCIM EE Master Work Order screen, you will see that dual lane work order is reserved.



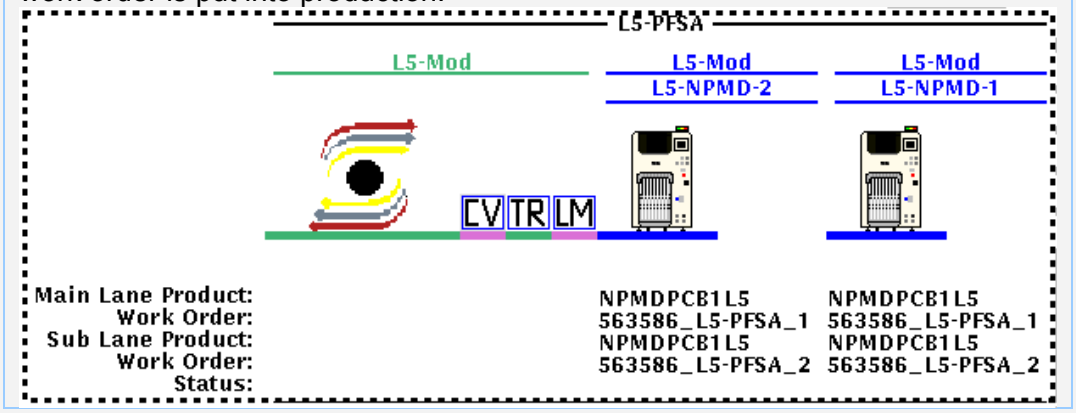
10	Once dual lane work order is picked, its status is reflected in the Master Work Order screen.



11 Once the dual lane work order is put into production, its status will be reflected in the Master Work Order screen as shown below:



12 Route Editor will show each sub work order name for each lane when master work order is put into production.



3.3.15 Report

In PanaCIM Material Control-Material Allocation, you can create various reports. The reports can be created from the PanaCIM Web application which starts up by accessing the PanaCIM application server from browser (Internet Explorer 11 or later).

For details of the PanaCIM Web application, refer to PanaCIM-EE [Basic Operations (Web application)].



■ List of reports

The table below is the list of available reports.

- Material Control-Material Allocation

Report	Description
Material Allocation Shortage Report	Shows detailed information regarding materials which are short of required quantity in upcoming work order.
Material Allocation Reservation Report	Shows detailed information on materials existing in the production zones or supporting areas which have been reserved in the upcoming order.
Material Allocation Setup Report	Shows details of current work order and some important information from previous work order when Product Changeover is initiated.
Material Allocation Production Order History Report	Shows the entire history of work orders created for a particular route and their associated data pertaining to material reservation quantity, material shortage quantity and quantity of materials picked.
Allocatable Component Report	Shows a list of materials allocated to work orders.

■ Filter criteria

For filter criterial of a report, refer to Basic Operations (Web application 3.2.2 [Filter Criteria].

■ Operation procedre

For how to output a report, refer to refer to Basic Operations (Web application) 3.3.1 [Operation Procedure].

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4. FUNCTION - MATERIAL CONTROL - MSD

This chapter explains a setup and operation of PanaCIM-EE / Material Control-MSD function.

4.1 Overview

4.1.1 Preparation

Before configuring the MSD option for use, please ensure that the necessary pre-work has been completed:

- MSD Components labeled with MSD Level
- MSD Components labeled with original bag seal date
- Dry Box entity defined

■ MSD Level(s) and Floor Life

The value (1,2,2a,3,4,5,5a) should be marked on a component's package for maximum allowable floor life time. ***Note**: Level 1 is currently not supported.*

Level	Floor Life (out of bag) at factory / Ambient ≤30°C/60% RH or as stated
1	Unlimited at ≤30°C/85% RH
2	1 year
2a	4 weeks
3	168 hours
4	72 hours
5	48 hours
5a	24 hours
6	Must be dried before use. After dried, components must be used within the time limit specified.

Note: It is to be noted that for MSD level 4 and above, the clock never stops. There is no way of stopping the timer for MSD level 4 and above. This is in accordance to the industry MSD standards

■ Original Bag Seal Date

The date of the original sealing of the package by the vendor. The bag seal date should be clearly marked on each package.

■ Dry Box Definition

The use of MSD components in a factory implies that dry boxes are also used to store these MSD components, in or out of their Moisture Barrier Bags (MBB) when not in use for production. MSD Dry Boxes within the factory are defined by expanding on the base Material Control model for Storage Units. The MSD option of Material Control introduces a special dedication type flag to the basic Storage Unit entity. The dedication type of "Dry Box" is used to represent a MSD Dry Box in the Material Control Factory View. Once the Dry Box entity is defined, MSD component material can be tracked in and out of the Storage Unit and to start/stop the current floor life counter.

■ Baking Unit Definition

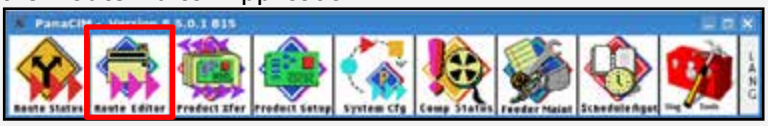
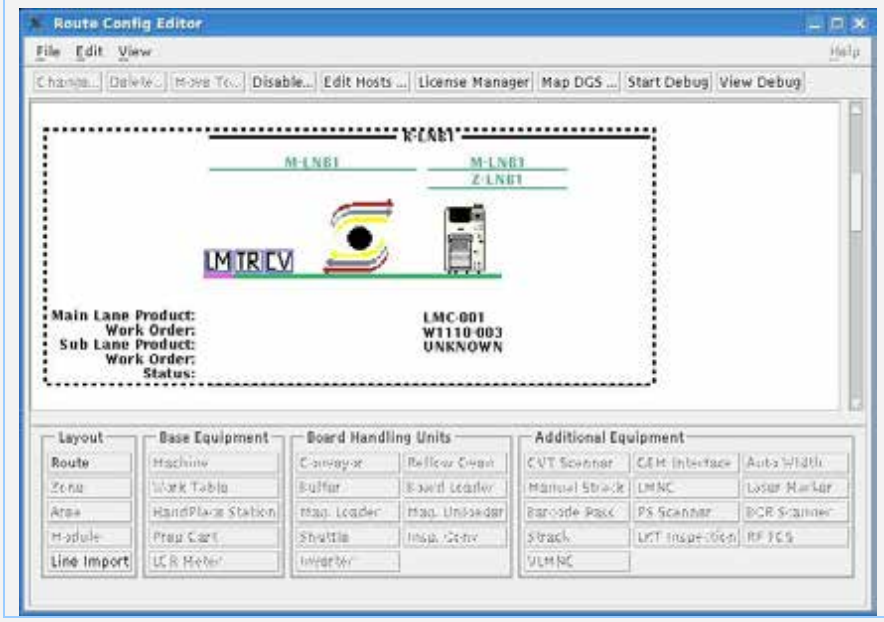
This is used to reset the floor life time of MSD component material. MSD component material can be tracked in and out of the Storage Unit defined as a baking unit, the floor life time and baking count.

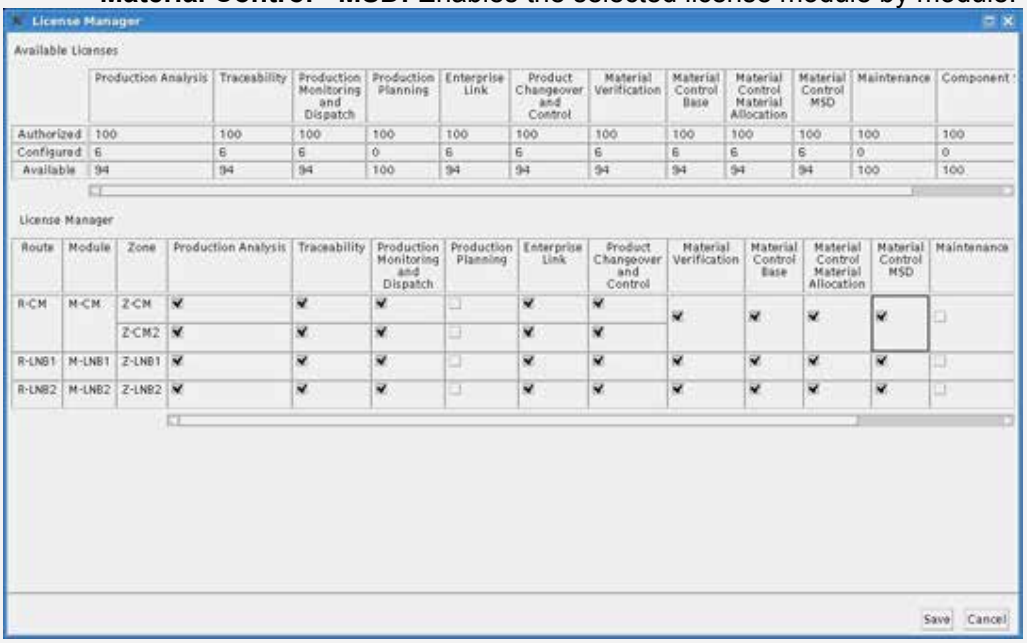
4.2 Settings

4.2.1 Route Editor

- **License Configuration**

  Before using the PanaCIM/Material Control - MSD, the license needs to be enabled via the Route Editor of the PanaCIM controller server.

Step	Procedure
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.

4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Material Control - Base, place a check mark in the following checkboxes: • **Material Verification:** Enables the selected license module by module. • **Material Control - Base:** Enables the selected license module by module. • **Material Control - MSD:** Enables the selected license module by module.  **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After configuring the license, click **Save** and exit the License Manager.

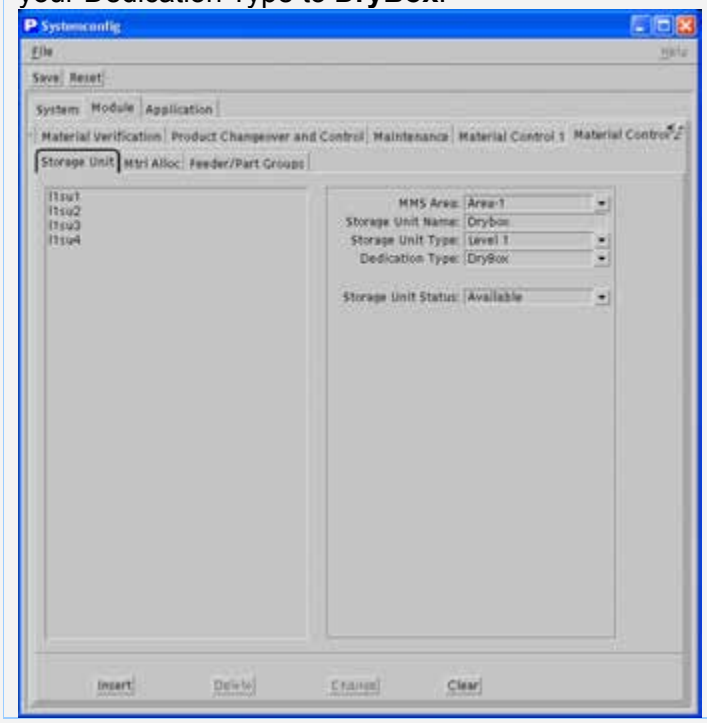
4.2.2 System Config

This section describes any necessary configurations in order to use the Material Control – MSD functionality.

Application	Overview
System Configuration	The System Configuration Application is used to configure the site-dependent aspects of the PanaCIM system. Use the System Configuration Application to configure the following settings for the Material Control - MSD Module: • Configure Dry Box Storage • Configure Baking Unit • Pre-Define MSD Parts
Route Editor	Route Editor Settings are specific to a particular production line. The settings can be observed while the object is running, but the object *must be disabled or stopped* in order to make any changes. Use the Route Editor Application to configure the following settings for the Material Control - MSD: • Configure Licenses

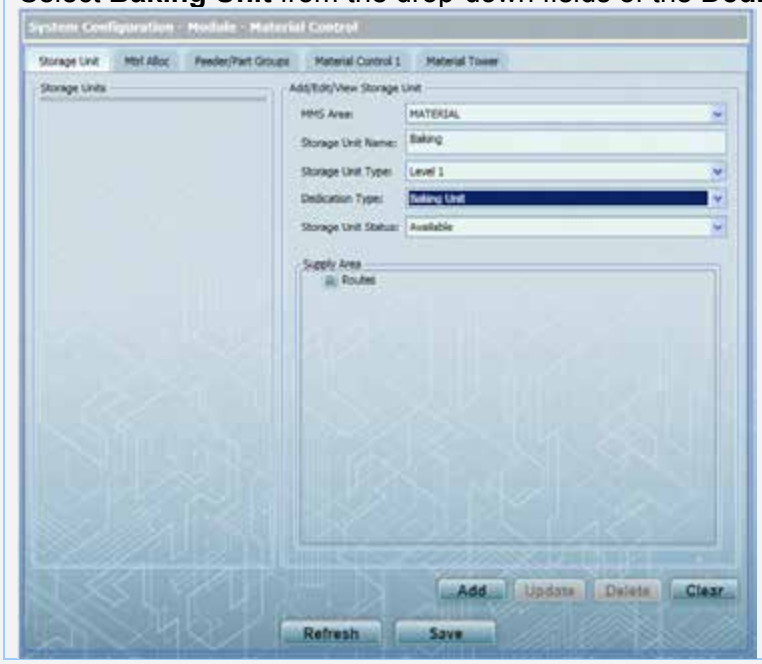
■ Configure Dry Box Storage

Refer to the action table below to configure Dry Box Storage areas using the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Select **Module > Material Control 2 > Storage Unit**.
3	Configure your settings using the drop-down fields ensuring you set your Dedication Type to **DryBox**. 
4	Click **Insert**.
5	Click **Save**.

■ Configure Baking Unit

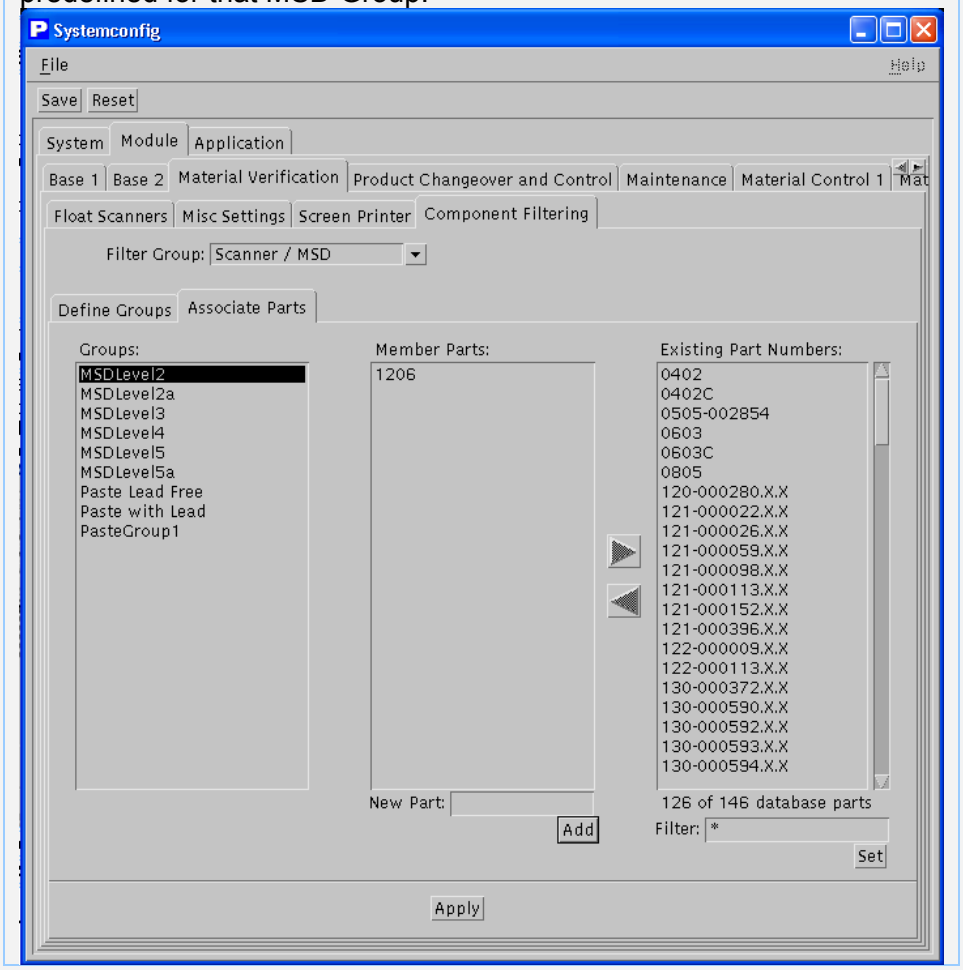
Refer to the action table below to congigure Backing Unit using the System Configuration application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM console.
2	Select **Module > Material Control > Storage Unit**.
3	Select **Baking Unit** from the drop-down fields of the **Dedication Type**. 
4	Click **Add**.
5	Click **Save**.

■ Pre-Define MSD Parts

You have the ability to pre-define parts rather than define-at-use by setting the following in System Configuration.

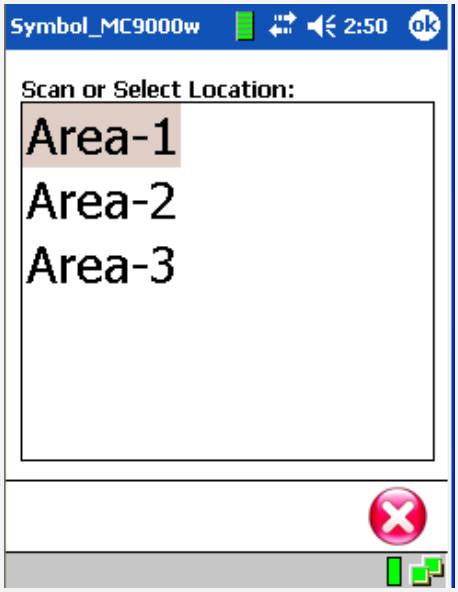
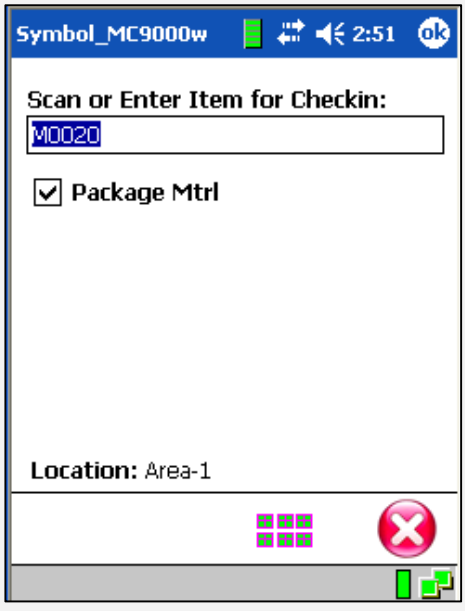
Refer to the action table below to Pre-Define MSD Parts using the System Configuration Application.

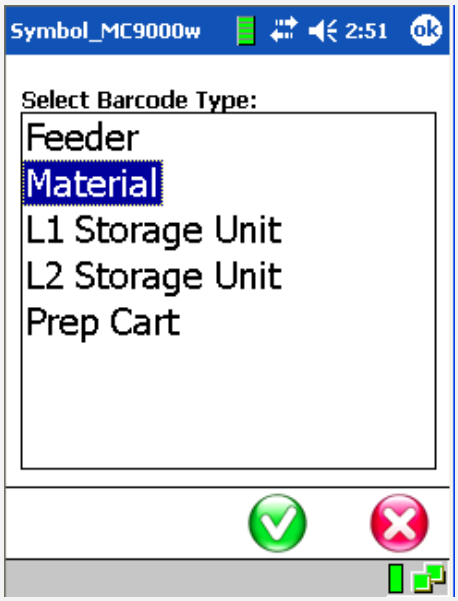
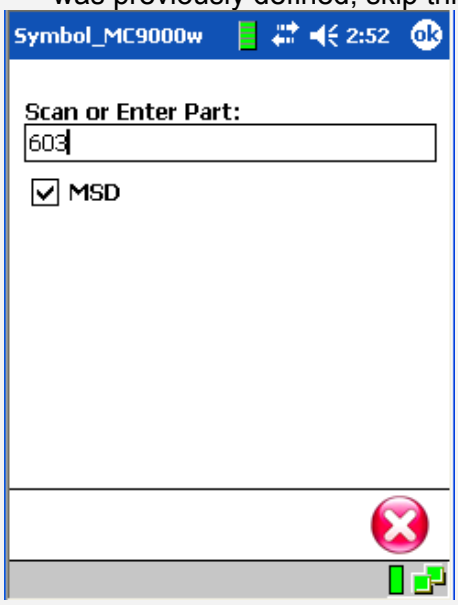
Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Select **Module** > **Material Verification** > **Component Filtering** > **Associate Parts**.
3	The Group(s) list includes groups for each MSD Level. Select an MSD Group from the groups list. Next, select an Existing Part Number(s) and click the arrow to move it to the Member Parts list. This part(s) is now predefined for that MSD Group. 
4	Click **Apply**.
5	Click **Save**.

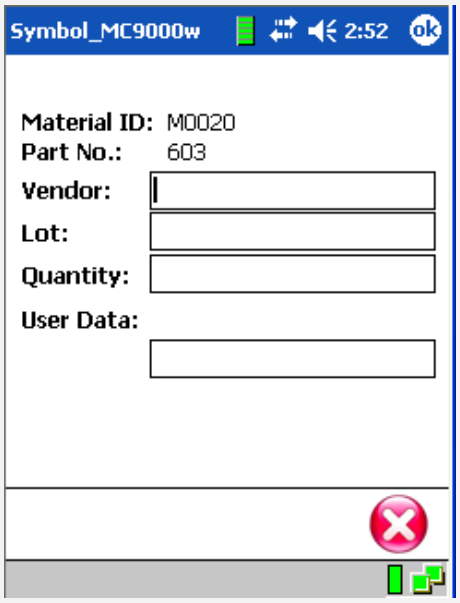
4.3 Operation

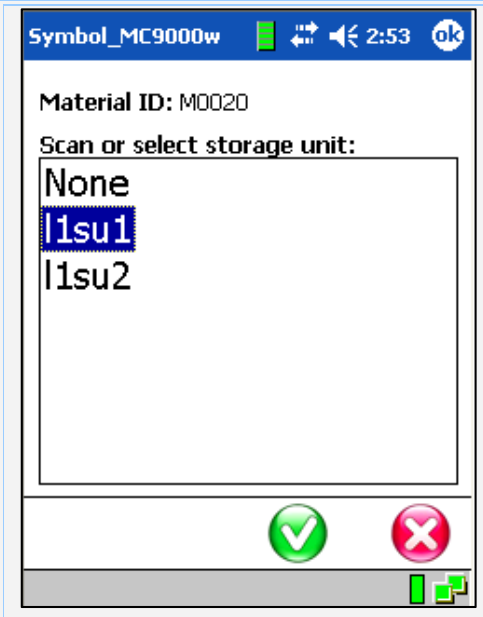
4.3.1 Material Handling - Check In

Refer to the action table below to check MSD components into the Material Area using the Mobile Scanner. This operation applies to dry box / bakcing unit / normal material area. It is to be noted that for MSD components, user can optionally package the material immediately upon check in.

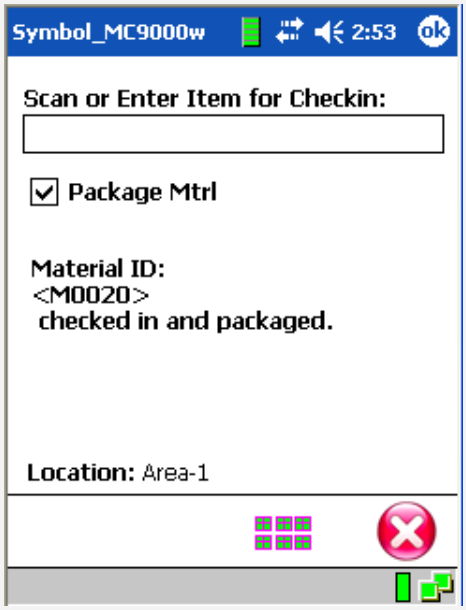
Step	Procedure
1	Select  .
2	Scan or select the **Material Area**. 
3	Scan or enter the Material barcode.  As seen above, **Package Mtrl** checkbox can be checked to allow packaging the MSD component, immediately upon check in. This saves the effort of explicitly performing a packaging operation upon check in.

	By default, **Package Mtrl** option is unchecked but once checked on, it will remain checked until user explicitly unchecks it.
4	After entering the item for check in, following screen appears to choose the material type irrespective of whether **Package Mtrl** option is checked or not:  **Note:** Above screen appears, when operators enter the item for checkin. When operator scans the item for check in, scanner already knows the scanned barcode type and hence the above screen to select barcode type will not appear in this case. Select **Material** to check-in reel containing MSD parts.
5	If the Material ID was not found in the database the scanner will prompt you for the Part. a. Scan or enter the **Part Number**. b. Select the **MSD** checkbox to define the Material item as MSD. If it was previously defined, skip this step.  Check the **MSD** checkbox after scanning or entering part number to

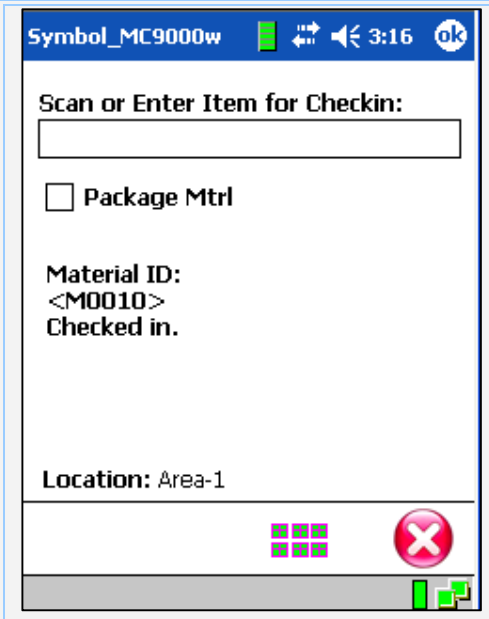
	confirm that the part being checked in is an MSD part.
6	Enter the **Part Details**. 
7	You will be prompted to enter MSD attributes if one of the following is true: • you scanned an item that has been predefined as an MSD item but has one or more MSD attributes undefined. • you scanned a new Material item. Enter the MSD attributes.  **Note:** If the part number entered in step 5 is already associated with MSD group via **Associate Parts** tab within **Component filtering** tab of **System Config** dialog, MSD level cannot be changed by clicking the down arrow.
8	After specifying the MSD part attributes (as shown in step 7), following screen appears to scan or select storage unit



9 After scanning / selecting the drybox where MSD reel is to be checked in and clicking  , following screen appears containing the message that the MSD reel is checked in to dry box and packaged.

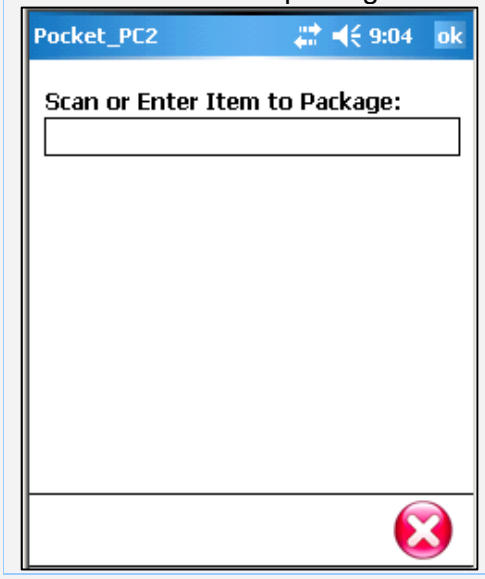
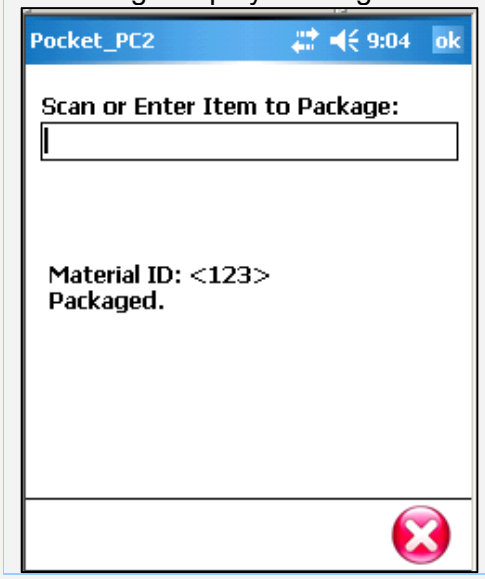


If the **Package Mtrl** option is unchecked in step 2, material is only checked in as shown below:


Symbol_MC9000w
3:16
ok
Scan or Enter Item for Checkin:
Package Mtrl
Material ID:
<M0010>
Checked in.
Location: Area-1

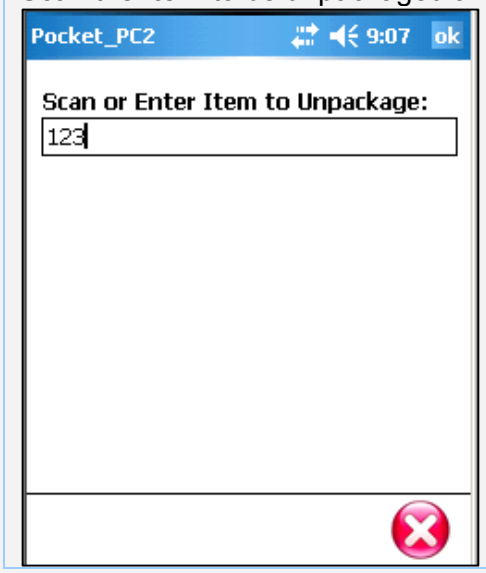
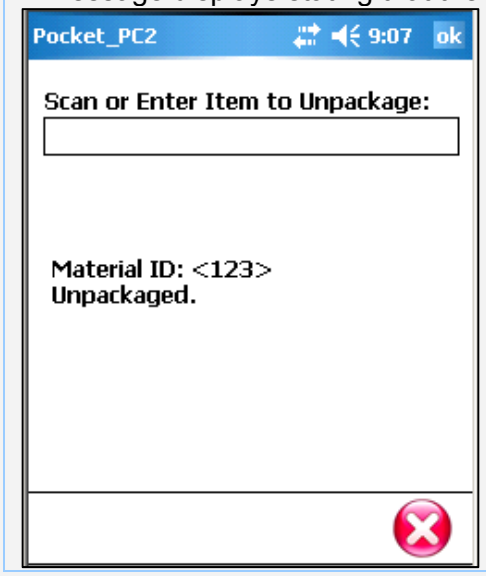
4.3.2 Material Handling - Package Material

Refer to the action table below to stop the floor life counter for an MSD component, and then perform a package operation for this component using the Mobile Scanner.

Step	Procedure
1	Select  .
2	Select  .
3	Scan the **Item** to be packaged and click **Enter**. 
4	A message displays stating that the Material ID has been packaged. 

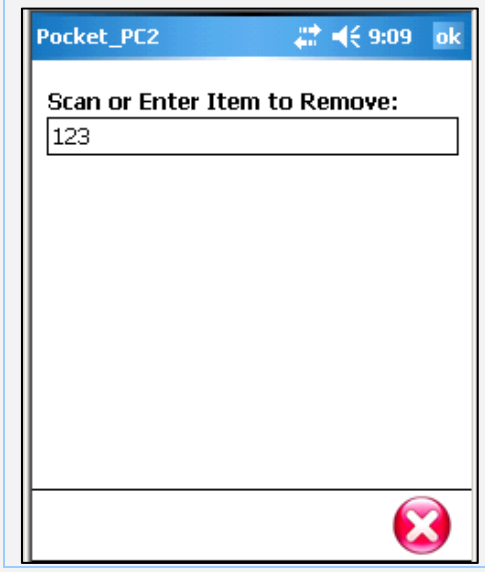
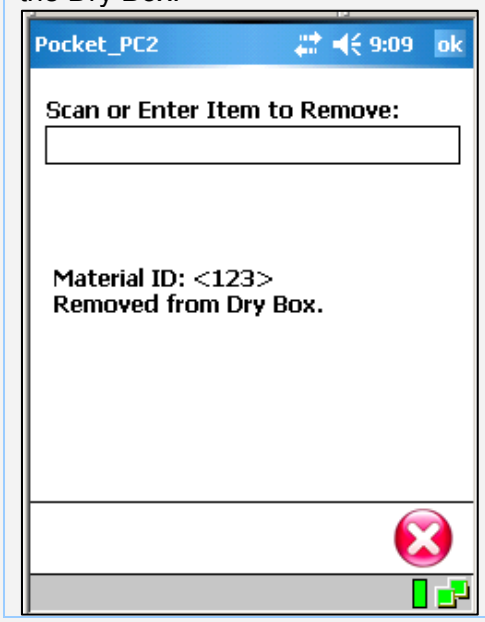
4.3.3 Material Handling - UnPackage Material

Refer to the action table below to start/resume the floor life counter for an MSD component after breaking the seal/re-seal of a package using the Mobile Scanner.

Step	Procedure
1	Select  .
2	Select  .
3	Scan the **Item** to be unpackaged and click **Enter**. 
4	A message displays stating that the Material ID has been unpackaged. 

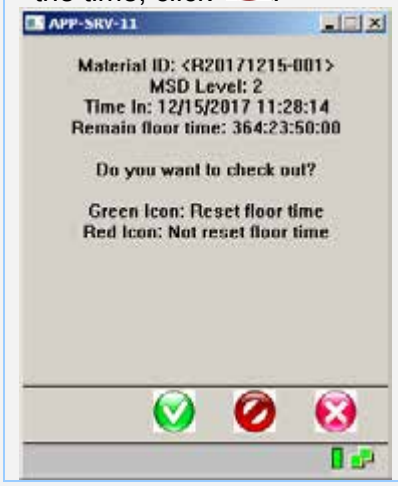
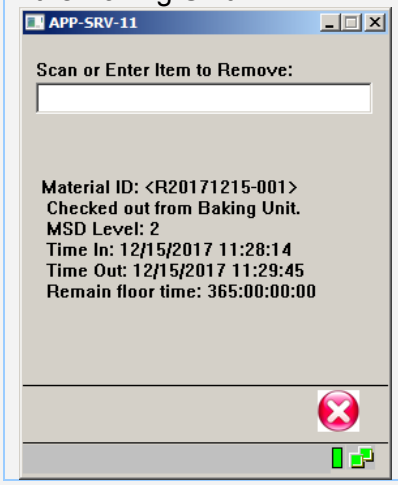
4.3.4 Material Handling - Removing an Item from the Dry Box

Refer to the action table below to remove an item from the Dry Box using the Mobile Scanner.

Step	Procedure
1	Select  ..
2	Select  .
3	Scan or Enter the item to **Remove**. 
4	A message displays stating that the Material ID has been removed from the Dry Box. 

4.3.5 Material Handling - Removing an Item from Baking Unit

Refer to the action table below to remove an item from the Baking Unit using the Mobile Scanner.

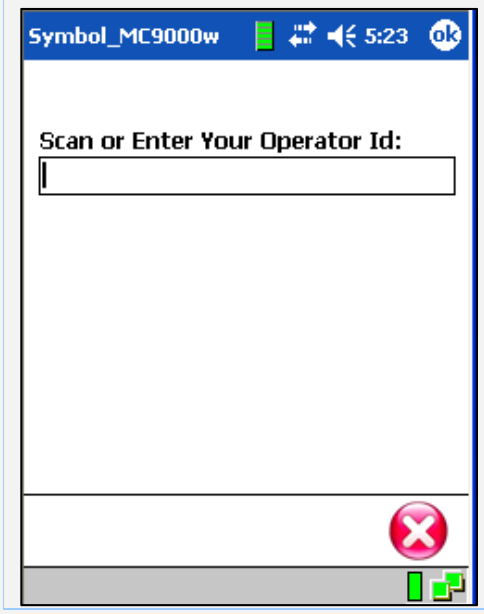
Step	Procedure
1	Select  .
2	Select the baking unit removal icon  .
3	Scan or Enter a material ID.
4	To reset the floor life time, click  and to remove without resetting the time, click  . 
5	A message displays stating that the material ID has been removed from the Baking Unit. 

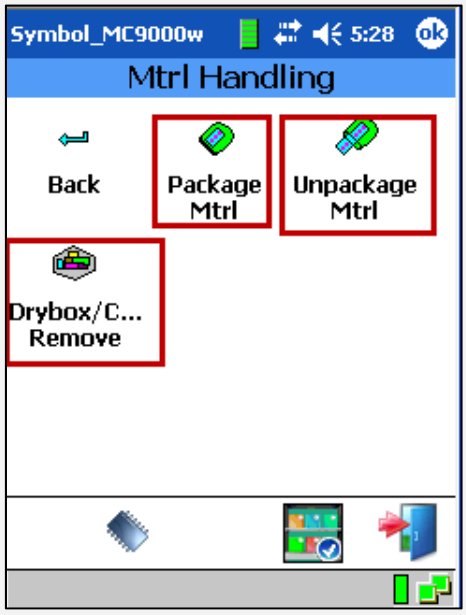
4.3.6 Material Handling - Functions for SPD/SPG/SPV/SP70

The Package/Unpackage Mtrl functions for MSD reels also apply to **“Paste”** reels. The floor time for a **“Paste Level”** reel defined in the define group tab of the system configuration dialog starts reducing by when the scanner is used or **“Package Mtrl”**, **“Drybox/Cooler Remove”** or **“Baking Unit”** function is used during mounting **“Paste”** reels.

The floor time for a **“Paste Level”** reel remains the same (stop reducing) once it has been checked into a **“Drybox”** type storage unit created from **system_config->Module->Material Control 2->Storage Unit** or uses **“Package Mtrl”** to seal the paste parts.

Refer to the action table below to view Material Handling functions for **“Paste”** reels.

Step	Procedure
1	Login using operator ID that has privileges to perform material handling operations. 

2	**Material Handling** screen appears as shown below: 
3	Click  to display the following screen.  **Note:** For details of operations of “Package”, “Unpackage”, “Drybox/Cooler Rmove”, and “Baking Unit”, refer to the step above. It is same as information defined in MSD parts reel.

4.3.7 Production - CM/DT

When using the Mobile Scanner in Production  mode, if an MSD component is expired, a warning message will be displayed during any scan operation of that component. Once an MSD component expires, a Production Lock is triggered. Clearing this lock condition will release the lock.

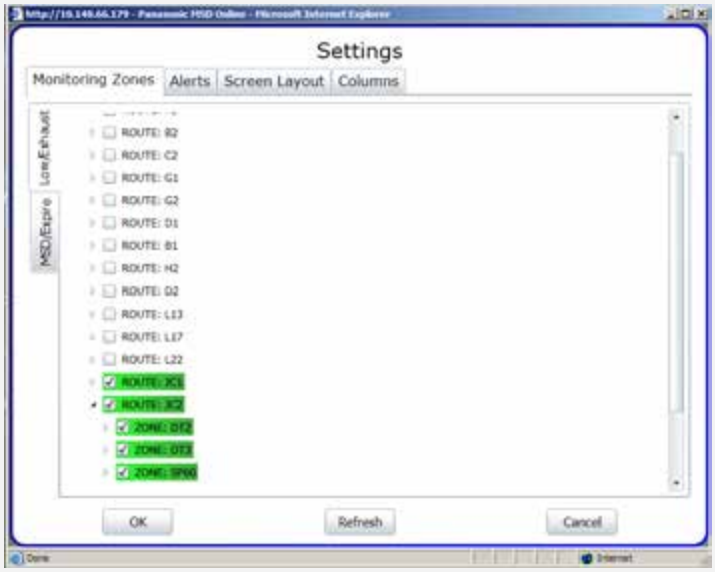
4.3.8 Production - Monitor MSD Components

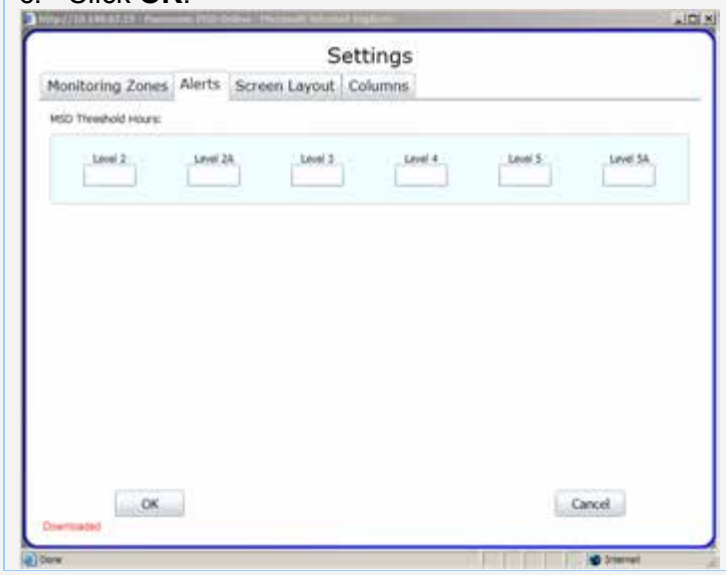
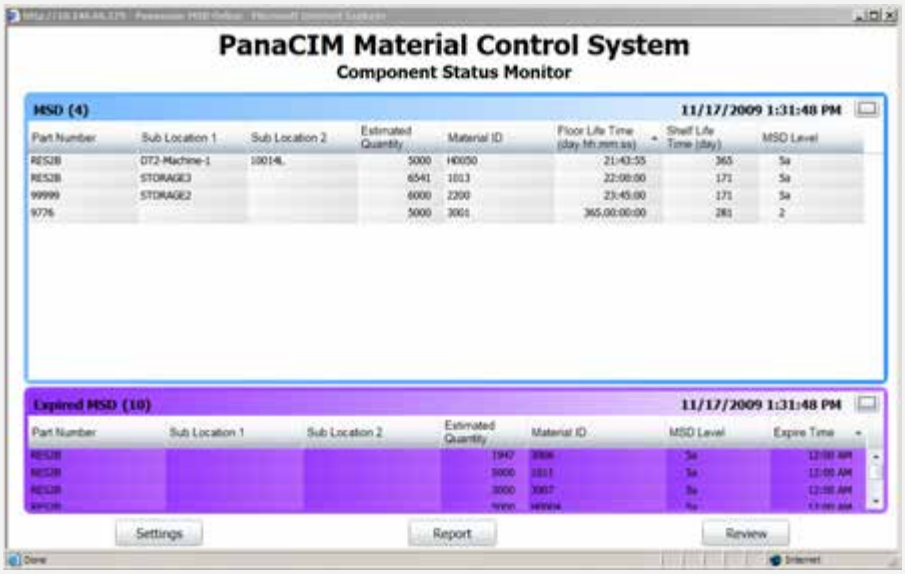
Monitor floor life status for MSD components using the Component Status Monitor Utility. You can configure single or multiple views to monitor the floor life status for a user selected list of locations. Locations include Production Zones and/or Supporting Areas.

- Each MSD component is displayed with its Material ID, location, sub location(s), and floor life remaining time.
- The PanaCIM system automatically updates the status every minute.
- Any MSD components that have passed the pre-defined threshold for remaining floor life time are highlighted for the user.
- Any MSD components that have expired are highlighted in a different color and grouped in a separate section.

Note: Once an MSD component expires, the machine will lock.

Refer to the action table below for guidance in using the Component Status Monitor.

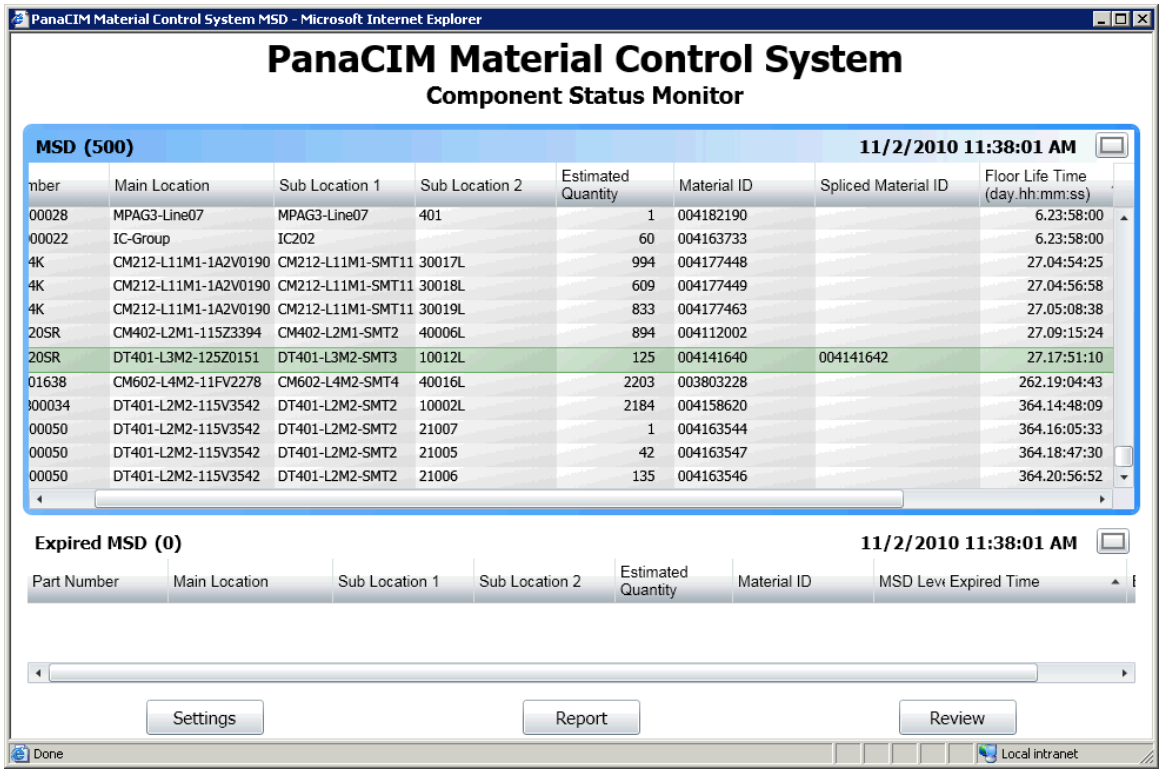
Step	Procedure
1	Open a web browser and enter the appropriate URL in to the **Address** field. a. using the server: http://<localhost>/eplantview/ b. using a PC other than the server: http://<webservername>/eplantview
2	Enter your **User name** and **Password**.
3	Click **OK**.
4	Select the **Material Control** Application and click the **Component Status Monitor** Utility.
5	• Select the **Monitoring Zones/MSD/Expire** tabs, select specific **routes** and/or **zones**. • Optionally, adjust your monitor display by configuring the following tabs: • **Screen Layout:** Adjust your desired screen layout. • **Columns:** Choose the information to be displayed and in what order this information will display. • Click **OK**. 

6	a. Select the **Alerts tab**. b. Enter **MSD Threshold Hours**. c. Click **OK**. 
7	The Component Status Monitor will refresh once each minute.  Expired MSDs will be shown in purple.

4.3.9 Production - Splice Reel Management in the Context of MSD Operations

The MSD Monitoring display has been updated to display the Material ID of any reel that is spliced to the Material ID of MSD reel being monitored / expired.

The image below illustrates the Spliced Material ID of any MSD parts.

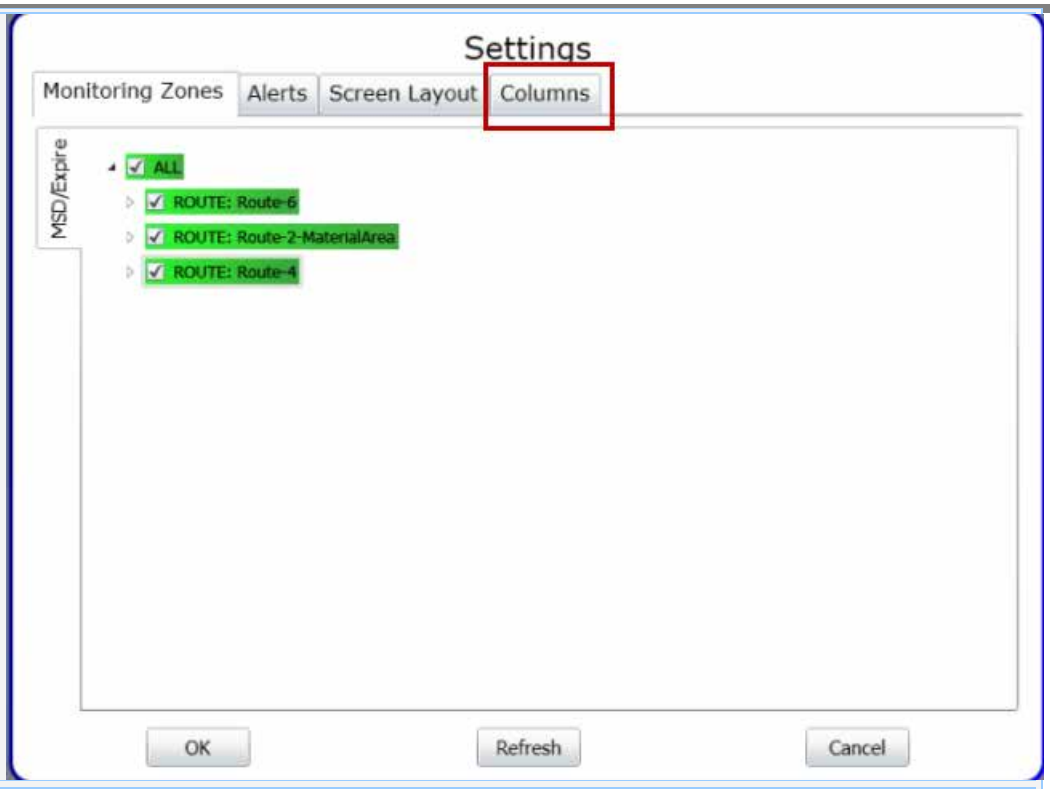


Note:
a) To launch the Component Status Monitor, click the **Material Control** on left hand side of web page and then click **Component Status Monitor**.

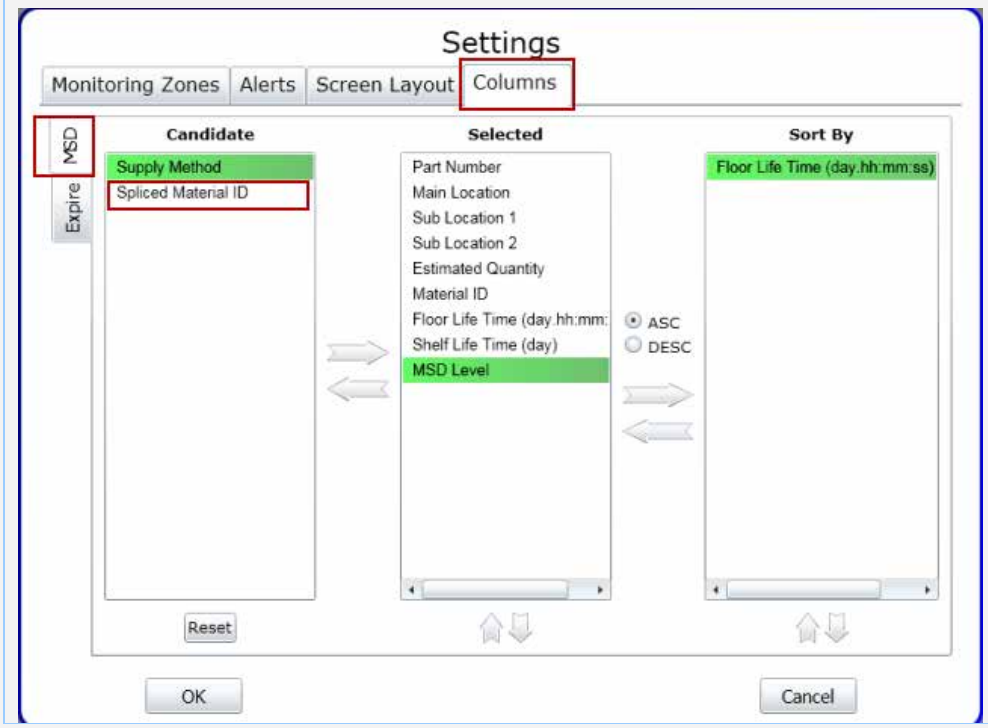
b) The Spliced Material ID column does not display by default. To display this column you must configure your settings.

Refer to the action table below for viewing the Spliced Material ID column.

Step	Procedure
1	Click the **Settings** button in the PanaCIM Material Control System. Following screen appears:

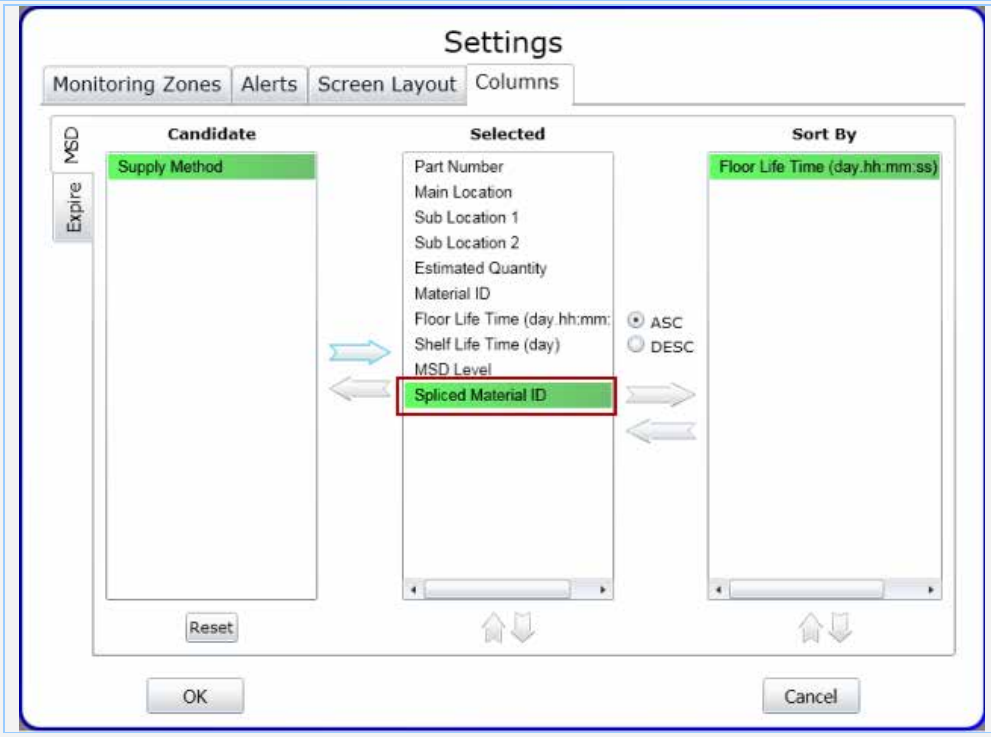


2 Activate the **Columns** tab. Following screen appears:

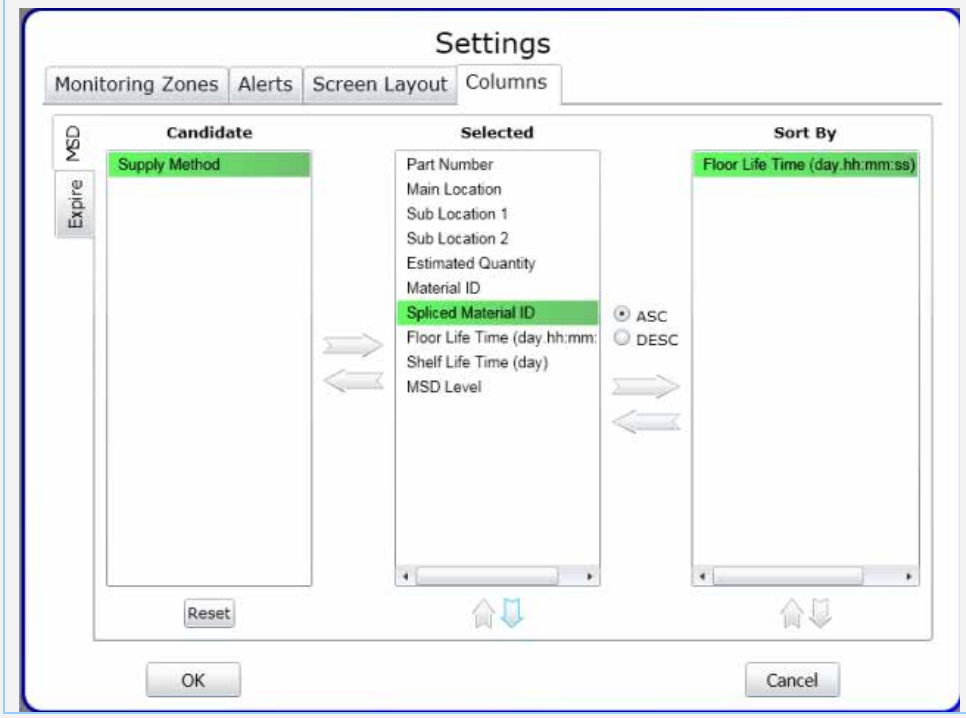


3 Activate the **MSD** tab on the top left corner of the screen.

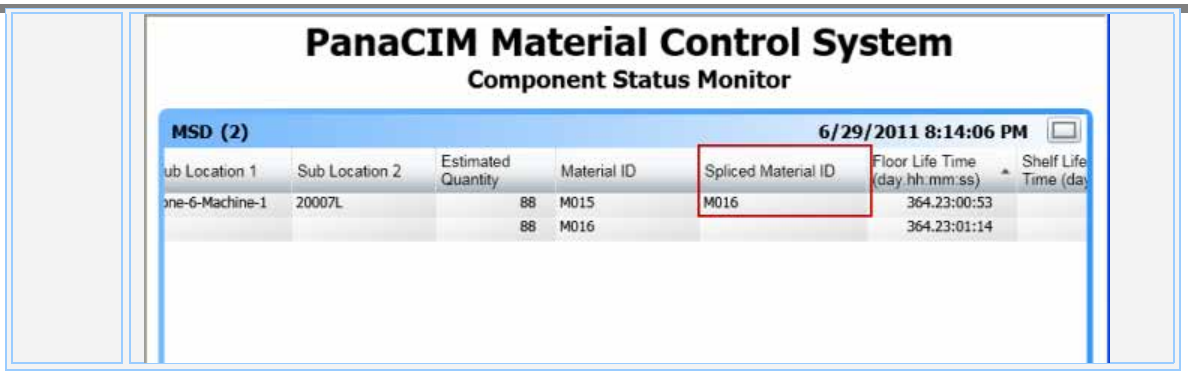
4 Select **Spliced Material ID** underneath the **Candidate** list and click  . **Spliced Material ID** column is now populated in the **Selected** list as shown below:



5 Click the button to position the **Spliced Material ID** column at the desired position. Refer the state of the **Settings** screen when the **Spliced Material ID** column is placed adjacent to the **Material ID** column.



6 Click the **OK** button. Now the **Component Status Monitor** application displays the material ID of the spliced reel for a MSD reel as shown below:



Following table contains actions that will be performed on both the Main reel and spliced reel when following Material Handling operations are performed for either of MSD reel.

Transaction	Description	Actions
Package	This operation performs a package of MSD Material	Both reels will be packaged and applicable MSD timers stopped
Un-Package	This operation performs an unpackage of MSD Material	Both reels will be unpackaged and applicable MSD timers started
Check into Dry box	A reel is placed into the Dry Box	Both reels will be placed and applicable MSD timers stopped
Remove from Dry box	A reel is removed from the Dry box	Both reels will be removed and applicable MSD timers started
Check into Baking Unit	A reel is placed into the Baking Unit	Both reels will be placed and applicable MSD timers stopped
Remove from BakingUnit	A reel is removed from the BakingUnit.	Both reels will be removed and applicable MSD timers started When the floor life time is reset, it is reset to the default value based on the MSD level.

4.3.10 Report

In PanaCIM Material Control- MSD, you can create various reports. The reports can be created from the PanaCIM Web application which starts up from the browser (Internet Explorer 11 or later) by accessing the PanaCIM application server.

For details of the PanaCIM Web application, refer to PanaCIM-EE [Basic Operations (Web application)].



■ List of reports

The table below is the list of available reports.

- Material Control- MSD

Report	Description
MSD Transaction Report	Shows detailed tracking of the elapsed floor life time for a specified MSD component per location and per operation.
Latest MSD Report	Shows the current status for MSD components As MSD component has to be utilized for production within a specified time frame, current floor life time and current shelf life time need to be known and this report provides accurate values of these two attributes.
Expired MSD Report	Shows a list of expired MDS parts.
MSD Part List	Shows the current MSD part and its MSD level.
Baking Operation History	Shows the history of check into or out from the Baking Unit.

■ Filter criteria

For filter criterial of a report, refer to Basic Operations (Web application 3.2.2 [Filter Criteria].

■ Operation procedre

For how to output a report, refer to refer to Basic Operations (Web application) 3.3.1 [Operation Procedure].

5. APPENDIX

5.1 Create Notification Actions

The System Configuration Application can be used to configure notification actions. Notification actions are simple scripts or Applications that interpret and format the Environment Data available to the notification action based on the notification type.

There are some system actions and formatting scripts that are distributed as part of the PanaCIM installation. These are generic actions that can be easily configured for use. User actions can be created by modifying any of the system actions or by creating a new action. User actions should always have a unique name.

The following table describes the notification types available and their descriptions.

Type	Description
Component Low Level I	Invokes the configured notification whenever a component falls below the defined Alert Level I limit.
Component Low Level II	Invokes the configured notification whenever a component falls below the defined Alert Level I Limit.
Component Exhaust	Invokes the configured notification whenever a component enters the exhaust status.
Component Picked	Invokes the configured notification whenever a component is either Picked or Un-Picked.

5.2 Discarding Registered Material and Reuse of Material ID

For the purpose of restricting the operation for verification of used materials, certain materials can be put into the “Discard” state by performing a setup or scanner operation. If you verify a material being in the “Discard” state, a message will be diaplyed, saying “The material cannot be used” and restricting verification of the material.



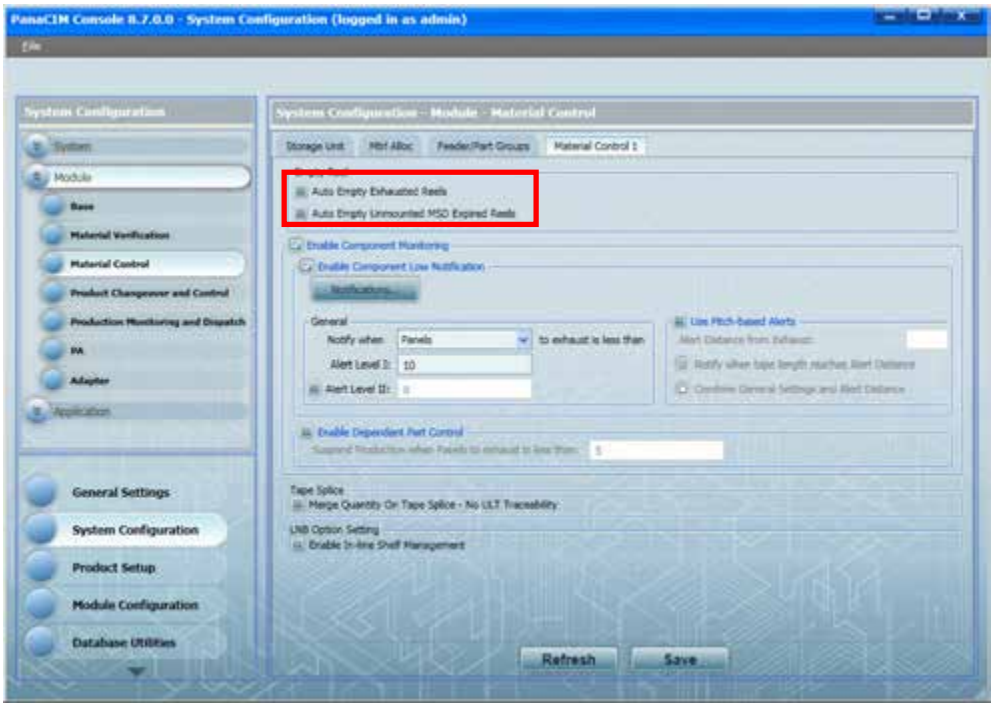
Normally, registered material information is not deleted from the database; however, information of a material that is in the “Discard” state will be deleted from the database after a certain period of time has paased. The material whose information was deleted from the database can be registered as a new material (You cannot reregister the same material unless it is deleted from the database).

■ Discarding material

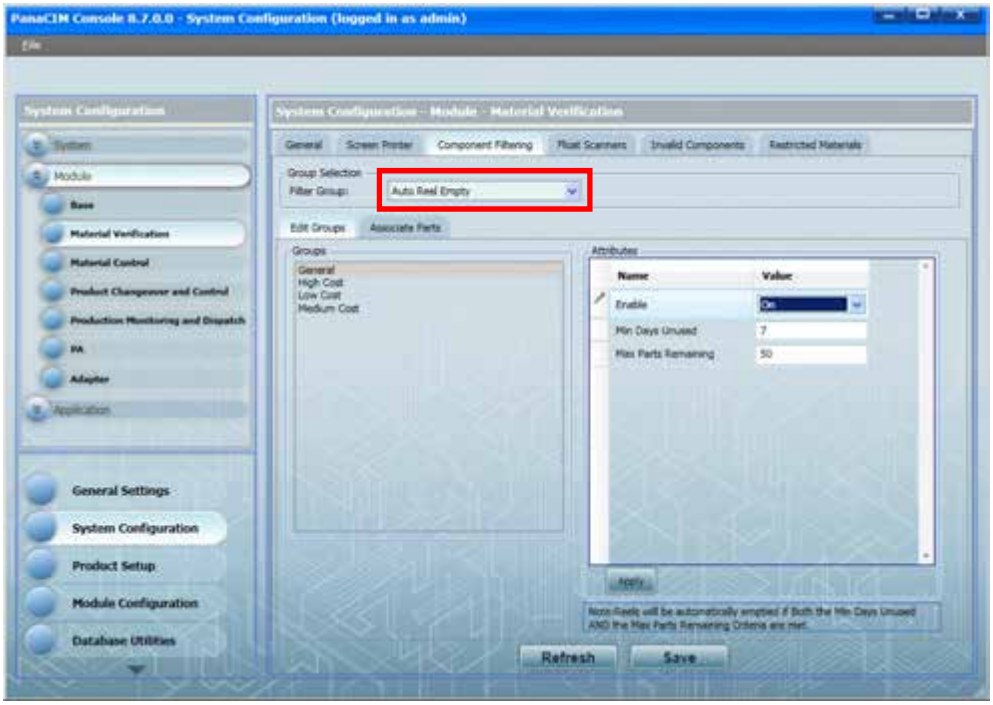
There are the following two methods to put a material into the “Discard” state.

① Enable the setting of “Auto Empty Exhausted Reels.”
Go to "System Configuration - Module – Material Control" screen on the PanaCIM console and enable the setting for ”Auto Empty Exhausted Reels” in the "Material Control 1" tab to put in the ”Discard” state a material removed after exhaust verification or a yet-to-be-transferred material on which reel transfer occurred due to detection of black tape.

② Enable the setting of “Auto Empty Unmounted MSD Expired Reels.”
Go to "System Configuration - Module – Material Control" screen on the PanaCIM console and enable the setting of “Auto Empty Unmounted MSD Expired Reels” in the "Material Control 1" tab to put a material, of which the temperature control component floor life time has reached 0, into the “Discard” state.

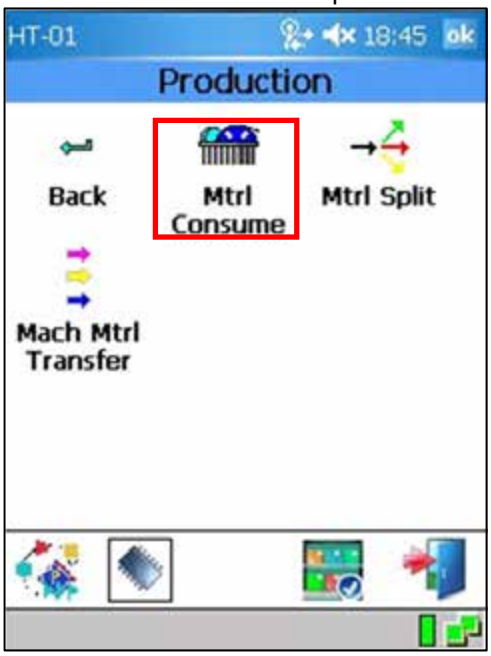


③ Enable the setting of “Auto Reel Empty.”
Go to "System Configuration - Module – Material Verification" screen of the PanaCIM console and enable the setting of “Auto Reel Emtpy” in the “Component Filtering” tab to put a material satisfying the conditions of “Min Days Unused” and “Max Parts Remaining” into the “Discard” state.



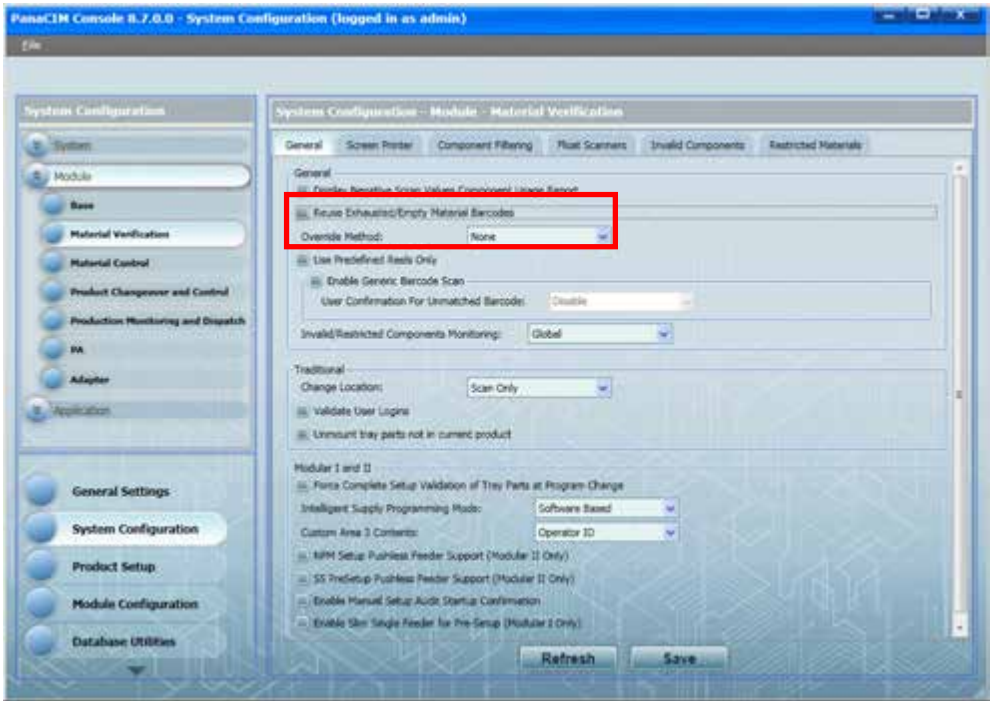
If you choose ‘General’ in Groups, Auto Reel Empty is applied to all the registered materials, and if you choose ‘High Cost’, ‘Low Cost’ or ‘Medium Cost’ in Groups, Auto Reel Empty is applied only to the materials registered with each group.
The input range of “Min Days Unused” is “2 to 99999,” and the input range of “Max Parts Remaining” is “0 to 99999.”

④ Performing material consume operation
Go to "Material Handling - Mtrl Edit - Mtrl Consume" on the scanner and put the material, on which a material consume operation was performed, into the “Discard” state.



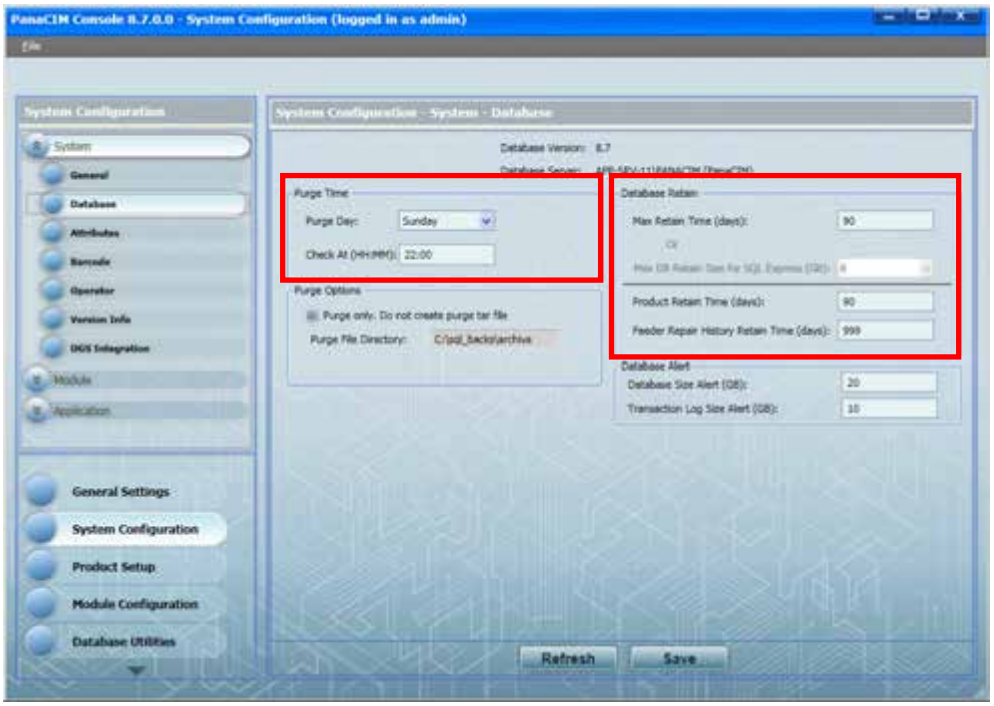
■ Reuse of exhausted material

If the setting of “Reuse Exhausted/Empty Material Barcodes” in the “General” tab on the "System Configuration - Module – Material Verification" screen of the PanaCIM console is enabled, you can verify a material that is in the “Discard” state after restoring the material to the pre-discard state (it is not handled as a new material).



■ Deleting discard material information from the database

Information of the material that is in the “Discard” state will be deleted from the database at the timing of the Purge Time (Purge Day and Check At) when the Max Retain Time in the Database Retain field, set on the "System Configuration - System - Database" screen of the PanaCIM console, is expired.The material whose information was deleted from the database can be used as a new material.



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